

Introduction to Modern Economic Growth: Parts 5-9

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Preface

This book is intended to serve two purposes:

- (1) First and foremost, this is a book about economic growth and long-run economic development. The process of economic growth and the sources of differences in economic performance across nations are some of the most interesting, important and challenging areas in modern social science. The primary purpose of this book is to introduce graduate students to these major questions and to the theoretical tools necessary for studying them. The book therefore strives to provide students with a strong background in dynamic economic analysis, since only such a background will enable a serious study of economic growth and economic development. It also tries to provide a clear discussion of the broad empirical patterns and historical processes underlying the current state of the world economy. This is motivated by my belief that to understand why some countries grow and some fail to do so, economists have to move beyond the mechanics of models and pose questions about the fundamental causes of economic growth.
- (2) In a somewhat different capacity, this book is also a graduate-level introduction to modern macroeconomics and dynamic economic analysis. It is sometimes commented that, unlike basic microeconomic theory, there is no core of current macroeconomic theory that is shared by all economists. This is not entirely true. While there is disagreement among macroeconomists about how to approach short-run macroeconomic phenomena and what the boundaries of macroeconomics should be, there is broad agreement about the workhorse models of dynamic macroeconomic analysis. These include the Solow growth model, the neoclassical growth model, the overlapping-generations model and models of technological change and technology adoption. Since these are all models of economic growth, a thorough treatment of modern economic growth can also provide (and perhaps should provide) an introduction to this core material of modern macroeconomics. Although there are several good graduate-level macroeconomic textbooks, they typically spend relatively little time on the basic core material and do not develop the links between modern macroeconomic analysis and economic dynamics on the one hand and general equilibrium theory on the other. In contrast, the current book does not cover any of the short-run topics in macroeconomics, but provides a thorough and rigorous introduction to what I view to be the core of macroeconomics. Therefore, the second purpose of the book is to provide a graduate-level introduction to modern macroeconomics.

The selection of topics is designed to strike a balance between the two purposes of the book. Chapters 1, 3 and 4 introduce many of the salient features of the process of economic growth and the sources of cross-country differences in economic performance. Even though these chapters cannot do justice to the large literature on economic growth empirics, they provide a sufficient background for students to appreciate the set of issues that are central to the study of economic growth and also a platform for further study of this large literature.

Chapters 5-7 provide the conceptual and the mathematical foundations of modern macroeconomic analysis. Chapter 5 provides the microfoundations for much of the rest of the book (and for much of modern macroeconomics), while Chapters 6 and 7 provide a quick but relatively rigorous introduction to dynamic optimization. Most books on macroeconomics or economic growth use either continuous time or discrete time exclusively. I believe that a serious study of both economic growth and modern macroeconomics requires the student (and the researcher) to be able to go between discrete and continuous time, and choose whichever one is more convenient or appropriate for the set of questions at hand. Therefore, I have deviated from this standard practice and included both continuous-time and discrete-time material throughout the book.

Chapters 2, 8, 9 and 10 introduce the basic workhorse models of modern macroeconomics and traditional economic growth, while Chapter 11 presents the first generation models of sustained (endogenous) economic growth. Chapters 12-15 cover models of technological progress, which are an essential part of any modern economic growth course.

Chapter 16 generalizes the tools introduced in Chapter 6 to stochastic environments. Using these tools, Chapter 17 presents a number of models of stochastic growth, most notably, the neoclassical growth model under uncertainty, which is the foundation of much of modern macroeconomics (though it is often left out of economic growth courses). The canonical Real Business Cycle model is presented as an application. This chapter also covers another major workhorse model of modern macroeconomics, the incomplete markets model of Bewley. Finally, this chapter also presents a number of other approaches to modeling the interaction between incomplete markets and economic growth and shows how models of stochastic growth can be useful in understanding how economies transition from stagnation or slow growth to an equilibrium with sustained growth.

Chapters 18-21 cover a range of topics that are sometimes left out of economic growth textbooks. These include models of technology adoption, technology diffusion, the interaction between international trade and technology, the process of structural change, the demographic transition, the possibility of poverty traps, the effects of inequality on economic growth and the interaction between financial and economic development. These topics are important for creating a bridge between the empirical patterns we observe in practice and the theory. Most traditional growth models consider a single economy in isolation and often after it has already embarked upon a process of steady economic growth. A study of models that incorporate cross-country interdependences, structural change and the possibility of takeoffs will enable us to link core topics of development economics, such as structural change, poverty traps or the demographic transition, to the theory of economic growth.

Finally, Chapters 22 and 23 consider another topic often omitted from macroeconomics and economic growth textbooks; political economy. This is motivated by my belief that the study of economic growth would be seriously hampered if we failed to ask questions about the fundamental causes of why countries differ in their economic performances. These questions inexorably bring us to differences in economic policies and institutions across nations. Political economy enables us to develop models to understand why economic policies and institutions differ across countries and must therefore be an integral part of the study of economic growth.

A few words on the philosophy and organization of the book might also be useful for students and teachers. The underlying philosophy of the book is that all the results that are stated should be proved or at least explained in detail. This implies a somewhat different organization than existing books. Most textbooks in economics do not provide proofs for

many of the results that are stated or invoked, and mathematical tools that are essential for the analysis are often taken for granted or developed in appendices. In contrast, I have strived to provide simple proofs of almost all results stated in this book. It turns out that once unnecessary generality is removed, most results can be stated and proved in a way that is easily accessible to graduate students. In fact, I believe that even somewhat long proofs are much easier to understand than general statements made without proof, which leave the reader wondering about why these statements are true.

I hope that the style I have chosen not only makes the book self-contained, but also gives the students an opportunity to develop a thorough understanding of the material. In line with this philosophy, I present the basic mathematical tools necessary for the development of the main material within the body of the text. My own experience suggests that a “linear” progression, where the necessary mathematical tools are introduced when needed, makes it easier for the students to follow and appreciate the material. Consequently, analysis of stability of dynamical systems, dynamic programming in discrete time and optimal control in continuous time are all introduced within the main body of the text. This should both help the students appreciate the foundations of the theory of economic growth and also provide them with an introduction to the main tools of dynamic economic analysis, which are increasingly used in every subdiscipline of economics. Throughout, when some material is technically more difficult and can be skipped without loss of continuity, it is marked with a “*”. Only material that is tangentially related to the main results in the text or those that should be familiar to most graduate students are left for the Mathematical Appendices.

I have also included a large number of exercises. Students can only gain a thorough understanding of the material by working through the exercises. The exercises that are somewhat more difficult are also marked with a “*”.

This book can be used in a number of different ways. First, it can be used in a one-quarter or one-semester course on economic growth. Such a course might start with Chapters 1-4, then depending on the nature of the course, use Chapters 5-7 either for a thorough study of the general equilibrium and dynamic optimization foundations of growth theory or only for reference. Chapters 8-11 cover the traditional growth theory and Chapters 12-15 provide the basics of endogenous growth theory. Depending on time and interest, any selection of Chapters 16-23 can be used for the last part of such a course.

Second, the book can be used for a one-quarter first-year graduate-level course in macroeconomics. In this case, Chapter 1 is optional. Chapters 3, 5-7, 8-11 and 16 and 17 would be the core of such a course. The same material could also be covered in a one-semester course, but in this case, it could be supplemented either with some of the later chapters or with material from one of the leading graduate-level macroeconomic textbooks on short-run macroeconomics, fiscal policy, asset pricing, or other topics in dynamic macroeconomics.

Third, the book can be used for an advanced (second-year) course in economic growth or economic development. An advanced course on growth or development could use Chapters 1-11 as background and then focus on selected chapters from Chapters 12-23.

Finally, since the book is self-contained, I also hope that it can be used for self-study.

Acknowledgments. This book grew out of the first graduate-level introduction to macroeconomics course I have taught at MIT. Parts of the book have also been taught as part of a second-year graduate macroeconomics course. I would like to thank the students who attended these lectures and made comments that have improved the manuscript. I owe a special thanks to Monica Martinez-Bravo, Samuel Pienknagura, Lucia Tian Tian and

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Please note that this is a preliminary draft of the book manuscript. The draft certainly contains mistakes. Comments and suggestions for corrections are welcome.

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Part 5

Stochastic Growth

This part of the book focuses on stochastic growth models and provides a brief introduction to basic tools of stochastic dynamic optimization. Stochastic growth models are useful for two related reasons. First, a range of interesting growth problems involve either aggregate uncertainty or nontrivial individual level uncertainty interacting with investment decisions and the growth process. Some of these models will be discussed in Chapter 17. Second, the stochastic neoclassical growth model has a wide range of applications in macroeconomics and in other areas of dynamic economic analysis. Various aspects of the stochastic neoclassical growth model will be discussed in the next two chapters. The study of stochastic models requires us to extend the dynamic optimization tools of Chapters 6 and 7 to an environment in which either returns or constraints are uncertain (governed by probability distributions).⁴ Unfortunately, dynamic optimization under uncertainty is considerably harder than the nonstochastic optimization. The generalization of continuous-time methods to stochastic optimization requires fairly advanced tools from measure theory and stochastic differential equations. While continuous-time stochastic optimization methods are very powerful, they are not used widely in macroeconomics and economic growth, so I have decided to focus on discrete-time stochastic models. Thus the next chapter will include the most straightforward generalization of the discrete-time dynamic programming techniques presented in Chapter 6 to stochastic environments. A fully rigorous development of stochastic dynamic programming also requires further mathematical investment than is typically necessary in most macroeconomics and economic growth courses. To avoid a heavy dose of new mathematical tools, in particular a lengthy detour into measure theory at this stage of the book, the next chapter develops the basics of stochastic dynamic programming without measure theory. I will then include a few pointers about how the results in this chapter can be extended and made more rigorous.

⁴Throughout, I do not draw a distinction between risk and uncertainty along the lines of the work by Frank Knight, who identified risk with situations in which there is a known probability distribution of events and uncertainty with situations in which such a probability distribution cannot be specified. While “Knightian uncertainty” may be important in a range of situations, given the set of models being studied here, there is little cost of following the standard practice of using the word “uncertainty” interchangeably with “risk”.

Stochastic Dynamic Programming

This chapter provides an introduction to basic stochastic dynamic programming. To avoid the use of measure theory in the main body of the text, I first focus on economies in which stochastic variables take finitely many values. This will enable us to use Markov chains, instead of general Markov processes, to represent uncertainty. Since many commonly-used stochastic processes, such as those based on normal or uniform distributions, fall outside this class, I will then indicate how the results can be generalized to situations in which stochastic variables can be represented by continuous, or mixture of continuous and discrete, random variables. Throughout my purpose is to provide a basic understanding of the tools of stochastic dynamic programming and how they can be used in dynamic macroeconomic models. For this reason, I will make a number of judicious choices rather than attempting to provide the most general results. Throughout, I focus on stationary problems, that is, the equivalents of Problems A1 and A2 in Chapter 6. Analogs of Theorems 6.11 and 6.12, which applied to nonstationary optimization problems under certainty, can be proved using exactly the same arguments in the stochastic case and I omit these results to save space.

16.1. Dynamic Programming with Expectations

I use a notation similar to that in Chapter 6. Let us first introduce the *stochastic* (random) variable $z(t) \in \mathcal{Z} \equiv \{z_1, \dots, z_N\}$. Note that the set $\mathcal{Z}$ is finite and thus compact, which will simplify the analysis considerably. Let the instantaneous payoff at time t be $U(x(t), x(t+1), z(t))$, where $x(t) \in X \subset \mathbb{R}^K$ for some $K \geq 1$ and $U : X \times X \times \mathcal{Z} \rightarrow \mathbb{R}$. This extends the payoff function in Chapter 6, which took the form $U(x(t), x(t+1))$, by making payoffs directly a function of the stochastic variable $z(t)$. As usual, returns will be discounted by some discount factor $\beta \in (0, 1)$. The initial value $x(0)$ is given. $x(t)$ again denotes the *state variables* (state vector) and $x(t+1)$ the *control variables* (control vector) at time t .

An additional difference from Problem A1 in Chapter 6 is that the constraint on $x(t+1)$ is no longer of the form $x(t+1) \in G(x(t))$. Instead, the constraint also incorporates the stochastic variable $z(t)$ and is written as

$$x(t+1) \in G(x(t), z(t)),$$

where again $G(x, z)$ is a set-valued mapping or a correspondence

$$G : X \times \mathcal{Z} \rightrightarrows X.$$

Suppose that the stochastic variable $z(t)$ follows a (first-order) *Markov chain*.¹ The important property implied by the Markov chain assumption is that the current-value of $z(t)$ only depends on its last period value, $z(t-1)$. Mathematically, this can be expressed as

$$\Pr[z(t) = z_j \mid z(0), \dots, z(t-1)] \equiv \Pr[z(t) = z_j \mid z(t-1)].$$

The simplest example of an economic model with uncertainty represented by a Markov chain would be one in which the stochastic variable takes finitely many values and is independently distributed over time. In this case, clearly, $\Pr[z(t) = z_j \mid z(0), \dots, z(t-1)] = \Pr[z(t) = z_j]$ and the Markov property is trivially satisfied. More generally, however, Markov chains enable us to model economic environments in which stochastic shocks are correlated over time. Markov chains are widely used in the theory of probability, in research in stochastic processes and in various areas of dynamic economic analysis. While the theory of Markov chains is relatively straightforward, not much of this theory is necessary for the basic treatment of stochastic dynamic programming here.

The Markov property not only simplifies the mathematical structure of economic models but also allows us to use relatively simple notation for the probability distribution of the random variable $z(t)$. We can also represent a Markov chain as

$$\Pr[z(t) = z_j \mid z(t-1) = z_{j'}] \equiv q_{jj'},$$

for any $j, j' = 1, \dots, N$, where $q_{jj'} \geq 0$ for all j, j' and

$$\sum_{j=1}^N q_{jj'} = 1 \text{ for each } j' = 1, \dots, N.$$

Here $q_{jj'}$ is also referred to as a *transition probability*, meaning the probability of the stochastic state z transitioning from $z_{j'}$ to z_j . I will make use of this notation in some of the proofs in the next section.

To see how this particular way of introducing stochastic elements into dynamic optimization is useful in economic problems, let us start with a simple example, which is also useful for introducing some additional notation.

EXAMPLE 16.1. Recall the optimal growth problem, where the objective is to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c(t)).$$

As usual, $c(t)$ denotes per capita consumption at time t and $u(\cdot)$ is the instantaneous utility function. The maximand in this problem differs from those studied so far only because of the presence of the expectations operator, $\mathbb{E}_0$, which stands for expectations conditional on information available at time $t = 0$. Expectations are necessary here because the future values

¹I adopt the standard terminology that $z(t)$ follows a Markov chain when it takes finitely (or countably) many values and that it follows a general Markov process when it has a continuous distribution or a mixture of the continuous and discrete distribution.

of consumption per capita is stochastic (as they will depend on the realization of future z 's). In particular, suppose that the production function (per capita) takes the form

$$y(t) = f(k(t), z(t)),$$

where $k(t)$ again denotes the capital-labor ratio and $z(t) \in \mathcal{Z} \equiv \{z_1, \dots, z_N\}$ represents a stochastic variable that affects how much output will be produced with a given amount of inputs. The most natural interpretation of $z(t)$ in this context is as a stochastic TFP term, so one might be tempted to write $y(t) = z(t)f(k(t))$ and in the next chapter I will sometimes impose this form, but there is no mathematical or economic gain from doing so here. Consequently, the constraint facing the maximization problem at time t takes the form

$$(16.1) \quad k(t+1) = f(k(t), z(t)) + (1 - \delta)k(t) - c(t),$$

$k(t) \geq 0$ and given $k(0)$, with δ again representing the depreciation rate. This formulation implies that at the time consumption $c(t)$ is chosen, the random variable $z(t)$ has been realized, thus $c(t)$ is a random variable depending on the realization of $z(t)$. In fact, more generally, $c(t)$ may depend on the entire history of the random variables. For this reason, let us define

$$z^t \equiv (z(0), z(1), \dots, z(t))$$

as the *history* of variable $z(t)$ up to date t . Let $\mathcal{Z}^t \equiv \mathcal{Z} \times \dots \times \mathcal{Z}$ (the t -times product), so that $z^t \in \mathcal{Z}^t$. For given $k(0)$, the level of consumption at time t can be most generally written as

$$c(t) = \tilde{c}[z^t],$$

which simply states that consumption at time t will be a function of the entire sequence of random variables observed up to that point. Clearly, consumption at time t cannot depend on future realizations of the random variable—those values have not been realized yet. Therefore, a consumption plan that depends on future realizations of the stochastic variable z would not be feasible. A function of the form $c(t) = \tilde{c}[z^t]$ is thus natural. Nevertheless, not all functions $\tilde{c}[z^t]$ could be admissible as feasible plans, because they may violate the resource constraint. I return shortly to additional restrictions to ensure feasibility. There is also no point in making consumption a function of the history of capital stocks at this stage, since those are endogenously determined by the choice of past consumption levels and by the realization of past stochastic variables. (When we turn to the recursive formulation of this problem, we will write consumption as a function of the current capital stock and the current-value of the stochastic variable). Let $x(t) = k(t)$, so that

$$\begin{aligned} x(t+1) &= k(t+1) \\ &= f(k(t), z(t)) + (1 - \delta)k(t) - \tilde{c}[z^t] \\ &\equiv \tilde{k}[z^t], \end{aligned}$$

where the second line simply uses the resource constraint with equality and the third line defines the function $\tilde{k} [z^t]$. With this notation, feasibility is easier to express, since

$$k(t+1) \equiv \tilde{k} [z^t]$$

by definition depends only on the history of the stochastic shocks up to time t and not on $z(t+1)$. In addition, feasibility requires that the function $\tilde{k} [\cdot]$ satisfies

$$\tilde{k} [z^t] \leq f(\tilde{k} [z^{t-1}], z(t)) + (1 - \delta) \tilde{k} [z^{t-1}] \text{ for all } z^{t-1} \in \mathcal{Z}^{t-1} \text{ and } z(t) \in \mathcal{Z}.$$

The maximization problem can then be expressed as

$$\max_{\{\tilde{c}[z^t], \tilde{k}[z^t]\}_{t=0}^{\infty}} \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t u(\tilde{c} [z^t]) \middle| z(0) \right]$$

subject to the constraint

$$\tilde{k} [z^t] \leq f(\tilde{k} [z^{t-1}], z(t)) + (1 - \delta) \tilde{k} [z^{t-1}] - \tilde{c} [z^t] \text{ for all } z^{t-1} \in \mathcal{Z}^{t-1} \text{ and } z(t) \in \mathcal{Z},$$

and starting with the initial conditions $\tilde{k} [z^{-1}] = k(0)$ and $z(0)$. This maximization problem can also be written using the instantaneous payoff function $U(x(t), x(t+1), z(t))$ introduced above. In this case, the maximization problem would take the form

$$\max_{\{\tilde{k}[z^t]\}_{t=0}^{\infty}} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t U(\tilde{k} [z^{t-1}], \tilde{k} [z^t], z(t)),$$

where now $U(x(t), x(t+1), z(t)) = u(f(k(t), z(t)) - k(t+1) + (1 - \delta)k(t))$. Notice the timing convention here: $\tilde{k} [z^{t-1}]$ is the value of the capital stock at time t , which is inherited from the investments at time $t-1$ and thus depends on the history of stochastic shocks up to time $t-1$, z^{t-1} , whereas $\tilde{k} [z^t]$ is the choice of capital stock for next period made at time t given the history of stochastic shocks up to time t , z^t .

This example can also be used to give us a first glimpse of how to express the same maximization problem recursively. Since $z(t)$ follows a Markov chain, the current-value of $z(t)$ contains both the information about the available resources for consumption and future capital stock and the information regarding the stochastic distribution of $z(t+1)$. Thus we might naturally expect the policy function determining the capital stock at the next date to take the form

$$(16.2) \quad k(t+1) = \pi(k(t), z(t)).$$

With the same reasoning, the recursive characterization would naturally take the form

$$(16.3) \quad V(k, z) = \sup_{y \in [0, f(k, z) + (1 - \delta)k]} \{u(f(k, z) + (1 - \delta)k - y) + \beta \mathbb{E}[V(y, z') | z]\},$$

where $\mathbb{E}[\cdot | z]$ denotes the expectation conditional on the current-value of z and incorporates the fact that the random variable z is a Markov chain. Let us suppose that this program has a solution, meaning that there exists a feasible plan that achieves the value $V(k, z)$ starting with capital-labor ratio k and stochastic variable z . Then, the set of the next date's capital

stock that achieve this maximum value can be represented by a correspondence $\Pi(k, z) \subset X$ for each $k \in \mathbb{R}_+$ and $z \in \mathcal{Z}$. For any $\pi(k, z) \in \Pi(k, z)$,

$$V(k, z) = u(f(k, z) + (1 - \delta)k - \pi(k, z)) + \beta \mathbb{E}[V(\pi(k, z), z') \mid z].$$

When the correspondence $\Pi(k, z)$ is single valued, then $\pi(k, z)$ would be uniquely defined and the optimal choice of next period's capital stock can be represented as in (16.2).

Example 16.1 already indicates how a stochastic optimization problem can be written in a sequential form and also gives us a hint about how to express such a problem recursively. I now do this more systematically. Let a *plan* be denoted by $\tilde{x}[z^t]$. This plan specifies the value of the vector $x \in \mathbb{R}^K$ for time $t + 1$, i.e., $x(t + 1) = \tilde{x}[z^t]$, for any $z^t \in \mathcal{Z}^t$. Using the same notation as in Chapter 6, the sequence problem takes the form

Problem B1 :

$$\begin{aligned} V^*(x(0), z(0)) &= \sup_{\{\tilde{x}[z^t]\}_{t=-1}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(\tilde{x}[z^{t-1}], \tilde{x}[z^t], z(t)) \\ &\text{subject to} \\ \tilde{x}[z^t] &\in G(\tilde{x}[z^{t-1}], z(t)), \quad \text{for all } t \geq 0 \\ \tilde{x}[z^{-1}] &= x(0) \text{ given,} \end{aligned}$$

where expectations at time $t = 0$, denoted by $\mathbb{E}_0$, are conditioned on the realization of the initial value $z(0)$ and is over the possible infinite sequences of $(z(1), z(2), z(3), \dots)$. For this reason, throughout the symbols $\mathbb{E}_0$ and $\mathbb{E}[\cdot \mid z(0)]$ will be used interchangeably. In this problem, as in the rest of this and the next chapter, I also adopt the convention that $\tilde{x}[z^{-1}] = x(0)$ and write the maximization problem with respect to the sequence $\{\tilde{x}[z^t]\}_{t=-1}^{\infty}$ (which starts at $t = -1$ and the value $\tilde{x}[z^{-1}] = x(0)$ is introduced as an additional constraint). The function V^* is conditioned on $x(0) \in \mathbb{R}^K$, since this is the initial value of the vector x , taken as given, and also on $z(0)$, since the choice of $x(1)$ is made after $z(0)$ is observed (and the expectations are also conditioned on $z(0)$). Finally, the first constraint in Problem B1 ensures that the sequence $\{\tilde{x}[z^t]\}_{t=-1}^{\infty}$ is feasible.

Similar to eq. (16.3) in Example 16.1, the functional equation corresponding to the recursive formulation of this problem can be written as:

Problem B2 :

$$(16.4) \quad V(x, z) = \sup_{y \in G(x, z)} \{U(x, y, z) + \beta \mathbb{E}[V(y, z') \mid z]\}, \text{ for all } x \in X \text{ and } z \in \mathcal{Z}$$

where $V : X \times \mathcal{Z} \rightarrow \mathbb{R}$ is a real-valued function and $y \in G(x, z)$ represents the constraint on next period's state vector as a function of the realization of the stochastic variable z . Problem B2 is a direct generalization of the Bellman equation in Problem A2 of Chapter 6 to a stochastic dynamic programming setup. One can also write Problem B2 as

$$V(x, z) = \sup_{y \in G(x, z)} \left\{ U(x, y, z) + \beta \int V(y, z') Q(z, dz') \right\}, \text{ for all } x \in X \text{ and } z \in \mathcal{Z},$$

where $\int f(z') Q(z_0, dz')$ denotes the Lebesgue integral of the function f with respect to the Markov process for z given last period's value of z as z_0 . This notation is useful in emphasizing that an expectation is nothing but a Lebesgue integral (and thus contains regular summation as a special case). Remembering the equivalence between expectations and integrals is important both for a proper appreciation of the theory and also for recognizing where some of the difficulties in the use of stochastic methods may lie.² There is typically little gain in rigor or insight in using the explicit Lebesgue integral instead of the expectation and I will not do so unless absolutely necessary.

As in Chapter 6, we would like to establish conditions under which the solutions to Problems B1 and B2 coincide. Let us first introduce the set of feasible *plans* starting with an initial value $x(t)$ and a value of the stochastic variable $z(t)$ as

$$\Phi(x(t), z(t)) = \{\{\tilde{x}[z^s]\}_{s=t-1}^\infty : \tilde{x}[z^{s-1}] = x(t) \text{ and } \tilde{x}[z^s] \in G(\tilde{x}[z^{s-1}], z(s)) \text{ for } s = t, t+1, \dots\}.$$

We denote a generic element of $\Phi(x(0), z(0))$ by $\mathbf{x} \equiv \{\tilde{x}[z^t]\}_{t=-1}^\infty$. In contrast to Chapter 6, the elements of $\Phi(x(0), z(0))$ are not infinite sequences of vectors in $\mathbb{R}^K$, but infinite sequences of feasible plans $\tilde{x}[z^t]$ that assign a value $x \in \mathbb{R}^K$ for any history $z^t \in \mathcal{Z}^t$ for any $t = 0, 1, \dots$. We are interested in (i) when the solution $V(x, z)$ to the Problem B2 coincides with the solution $V^*(x, z)$; and (ii) when the set of maximizing plans $\Pi(x, z) \subset \Phi(x, z)$ also generates an optimal feasible plan for Problem B1 (presuming that both problems have feasible plans attaining their supremums). Recall that the set of maximizing plans $\Pi(x, z)$ is defined such that for any $\pi(x, z) \in \Pi(x, z)$,

$$(16.5) \quad V(x, z) = U(x, \pi(x, z), z) + \beta \mathbb{E}[V(\pi(x, z), z') | z].$$

Let us now introduce analogs of Assumption 6.1-6.5 from Chapter 6 and the appropriate generalizations of Theorems 6.1-6.6.

ASSUMPTION 16.1. $G(x, z)$ is nonempty for all $x \in X$ and $z \in \mathcal{Z}$. Moreover, for all $x(0) \in X$, $z(0) \in \mathcal{Z}$, and $\mathbf{x} \in \Phi(x(0), z(0))$, $\lim_{n \rightarrow \infty} \mathbb{E}[\sum_{t=0}^n \beta^t U(\tilde{x}[z^{t-1}], \tilde{x}[z^t], z(t)) | z(0)]$ exists and is finite.

ASSUMPTION 16.2. X is a compact subset of $\mathbb{R}^K$, G is nonempty, compact-valued and continuous. Moreover, let $\mathbf{X}_G = \{(x, y, z) \in X \times X \times \mathcal{Z} : y \in G(x, z)\}$ and suppose that $U : \mathbf{X}_G \rightarrow \mathbb{R}$ is continuous.

Observe that Assumption 16.1 only imposes the compactness of X , since $\mathcal{Z}$ is already compact in view of the fact that it consists of a finite number of elements. Moreover, the continuity of U in (x, y, z) is equivalent to its continuity in (x, y) , since $\mathcal{Z}$ is a finite set, so we can endow it with the discrete topology, so that continuity is automatically guaranteed (see Fact A.11 in Appendix Chapter A). As in Chapter 6, these assumptions enable us to establish a number of useful results about the equivalence between Problems B1 and B2

²In particular, potential difficulties arise when one needs to exchange limits and expectations; in contrast, there is no problem in differentiating a functional under the integral or the expectations sign as long as the integrand is differentiable.

and the solution to the dynamic optimization problems specified above. I state these results without proof here, and provide some of the proofs in Section 16.2 and leave the rest to exercises.

Our first result is a generalization of Theorem 6.1 from Chapter 6.

THEOREM 16.1. (*Equivalence of Values*) Suppose Assumptions 16.1 holds. Then, for any $x \in X$ and any $z \in \mathcal{Z}$, any $V^*(x, z)$ that is a solution to Problem B1 is also a solution to Problem B2. Moreover, any solution $V(x, z)$ to Problem B2 is also a solution to Problem B1, so that $V^*(x, z) = V(x, z)$ for any $x \in X$ and any $z \in \mathcal{Z}$.

The next theorem establishes the *principle of optimality* for stochastic problems. As in Chapter 6, the principle of optimality enables us to break the returns from an optimal plan into two parts, the current return and the continuation return, which now corresponds to expected returns.

THEOREM 16.2. (*Principle of Optimality*) Suppose Assumptions 16.1 holds. For $x(0) \in X$ and $z(0) \in \mathcal{Z}$, let $\mathbf{x}^* \equiv \{\tilde{x}^*[z^t]\}_{t=-1}^\infty \in \Phi(x(0), z(0))$ be a feasible plan that attains $V^*(x(0), z(0))$ in Problem B1. Then,

$$(16.6) \quad V^*(\tilde{x}^*[z^{t-1}], z(t)) = U(\tilde{x}^*[z^{t-1}], \tilde{x}^*[z^t], z(t)) + \beta \mathbb{E}[V^*(\tilde{x}^*[z^t], z(t+1)) \mid z(t)]$$

for $t = 0, 1, \dots$

Moreover, if any $\mathbf{x}^* \in \Phi(x(0), z(0))$ satisfies (16.6), then it attains the optimal value in Problem B1.

The next result establishes the uniqueness of the value function and existence of solutions.

THEOREM 16.3. (*Existence of Solutions*) Suppose that Assumptions 16.1 and 16.2 hold. Then, the unique function $V : X \times \mathcal{Z} \rightarrow \mathbb{R}$ that satisfies (16.4) is continuous and bounded in x for each $z \in \mathcal{Z}$. Moreover, an optimal plan $\mathbf{x}^* \in \Phi(x(0), z(0))$ exists for any $x(0) \in X$ and any $z(0) \in \mathcal{Z}$.

The remaining results, as their analogs in Chapter 6 use further assumptions to establish concavity, monotonicity and the differentiability of the value function.

ASSUMPTION 16.3. U is concave; for any $\alpha \in (0, 1)$ and any $(x, y, z), (x', y', z) \in \mathbf{X}_G$, we have

$$U(\alpha x + (1 - \alpha)x', \alpha y + (1 - \alpha)y', z) \geq \alpha U(x, y, z) + (1 - \alpha)U(x', y', z).$$

Moreover if $x \neq x'$,

$$U(\alpha x + (1 - \alpha)x', \alpha y + (1 - \alpha)y', z) > \alpha U(x, y, z) + (1 - \alpha)U(x', y', z).$$

In addition, $G(x, z)$ is convex in x ; for any $z \in \mathcal{Z}$, any $\alpha \in [0, 1]$, and any $x, x', y, y' \in X$ such that $y \in G(x, z)$ and $y' \in G(x', z)$, we have

$$\alpha y + (1 - \alpha)y' \in G(\alpha x + (1 - \alpha)x', z).$$

ASSUMPTION 16.4. For each $y \in X$ and $z \in \mathcal{Z}$, $U(\cdot, y, z)$ is strictly increasing in its first K arguments, and G is monotone in x in the sense that $x \leq x'$ implies $G(x, z) \subset G(x', z)$ for each $z \in \mathcal{Z}$.

ASSUMPTION 16.5. $U(x, y, z)$ is continuously differentiable in x in the interior of its domain $\mathbf{X}_G$.

THEOREM 16.4. (**Concavity of the Value Function**) Suppose that Assumptions 16.1, 16.2 and 16.3 hold. Then, the unique function V that satisfies (16.4) is strictly concave in x for each $z \in \mathcal{Z}$. Moreover, the optimal plan can be expressed as $\tilde{x}^*[z^t] = \pi(x^*(t), z(t))$, where the policy function $\pi : X \times \mathcal{Z} \rightarrow X$ is continuous in x for each $z \in \mathcal{Z}$.

THEOREM 16.5. (**Monotonicity of the Value Function I**) Suppose that Assumptions 16.1, 16.2 and 16.4 hold and let $V : X \times \mathcal{Z} \rightarrow \mathbb{R}$ be the unique solution to (16.4). Then, for each $z \in \mathcal{Z}$, V is strictly increasing in x .

THEOREM 16.6. (**Differentiability of the Value Function**) Suppose that Assumptions 16.1, 16.2, 16.3 and 16.5 hold. Let π be the policy function defined above and assume that $x' \in \text{Int}X$ and $\pi(x', z) \in \text{Int}G(x', z)$ at $z \in \mathcal{Z}$, then $V(x, z)$ is continuously differentiable at (x', z) , with derivative given by

$$(16.7) \quad D_x V(x', z) = D_x U(x', \pi(x', z), z).$$

These theorems have exact analogs in Chapter 6. Since the value function now also depends on the stochastic variable z , an additional monotonicity result can also be obtained. For this, let us introduce the following additional assumption:

ASSUMPTION 16.6. (i) G is monotone in z in the sense that $z \leq z'$ implies $G(x, z) \subset G(x, z')$ for each any $x \in X$ and $z, z' \in \mathcal{Z}$ such that $z \leq z'$.

(ii) For each $(x, y, z) \in \mathbf{X}_G$, $U(x, y, z)$ is strictly increasing in z .

(iii) The Markov chain for z is monotone in the sense that for any nondecreasing function $f : \mathcal{Z} \rightarrow \mathbb{R}$, $\mathbb{E}[f(z') | z]$ is also nondecreasing in z .

To interpret the last part of this assumption, suppose that $z_j \leq z_{j'}$ whenever $j < j'$. Then, this condition will be satisfied if and only if, for any $\bar{j} = 1, \dots, N$ and any $j'' > j'$, $\sum_{j=\bar{j}}^N q_{jj''} \geq \sum_{j=\bar{j}}^N q_{jj'}$ (see Exercise 16.1).

THEOREM 16.7. (**Monotonicity of the Value Function II**) Suppose that Assumptions 16.1, 16.2 and 16.6 hold and let $V : X \times \mathcal{Z} \rightarrow \mathbb{R}$ be the unique solution to (16.4). Then, for each $x \in X$, V is strictly increasing in z .

16.2. Proofs of the Stochastic Dynamic Programming Theorems*

This section provides proofs for the main theorems provided in the previous section, Theorems 16.1-16.3. The proofs for theorems 16.5-16.7 are straightforward in view of the proofs of corresponding theorems in Chapter 6 and are left as exercises.

First, for any feasible $\mathbf{x} \equiv \{\tilde{x}[z^t]\}_{t=-1}^\infty$, and any initial conditions $x(0) \in X$ and $z(0) \in \mathcal{Z}$, define

$$\bar{U}(\mathbf{x}, z(0)) \equiv \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t U(\tilde{x}[z^{t-1}], \tilde{x}[z^t], z(t)) \mid z(0) \right]$$

and note that for any $x(0) \in X$ and $z(0) \in \mathcal{Z}$,

$$V^*(x(0), z(0)) = \sup_{\mathbf{x} \in \Phi(x(0), z(0))} \bar{U}(\mathbf{x}, z(0)).$$

In view of Assumption 16.1, which ensures that all values are bounded, it follows that V^* must satisfy

$$(16.8) \quad V^*(x(0), z(0)) \geq \bar{U}(\mathbf{x}, z(0)) \text{ for all } \mathbf{x} \in \Phi(x(0), z(0))$$

and

$$(16.9) \quad \text{for any } \varepsilon > 0, \text{ there exists } \mathbf{x}' \in \Phi(x(0), z(0)) \text{ s.t. } V^*(x(0), z(0)) \leq \bar{U}(\mathbf{x}', z(0)) + \varepsilon$$

The conditions for V to be a solution to Problem B2 are similar. For any $x(0) \in X$ and $z(0) \in \mathcal{Z}$,

$$(16.10) \quad V(x(0), z(0)) \geq U(x(0), y, z) + \beta \mathbb{E}[V(y, z(1)) \mid z(0)], \quad \text{all } y \in G(x(0), z(0)),$$

and

$$(16.11) \quad \begin{aligned} &\text{for any } \varepsilon > 0, \text{ there exists } y' \in G(x(0), z(0)) \\ &\text{s.t. } V(x(0), z(0)) \leq U(x(0), y', z(0)) + \beta \mathbb{E}[V(y, z(1)) \mid z(0)] + \varepsilon. \end{aligned}$$

The following lemma is a straightforward generalization of Lemma 6.1 in 6

LEMMA 16.1. *Suppose that Assumption 16.1 holds. Then, for any $x(0) \in X$, any $z(0) \in \mathcal{Z}$, any $\mathbf{x} \equiv \{\tilde{x}[z^t]\}_{t=-1}^\infty \in \Phi(x(0), z(0))$,*

$$\bar{U}(\mathbf{x}, z(0)) = U(x(0), \tilde{x}[z^0], z(0)) + \beta \mathbb{E}[\bar{U}(\{\tilde{x}[z^t]\}_{t=0}^\infty, z(1)) \mid z(0)].$$

PROOF. See Exercise 16.2. □

PROOF OF THEOREM 16.1. If $\beta = 0$, Problems B1 and B2 are identical, thus the result follows immediately. Suppose that $\beta > 0$ and take an arbitrary $x(0) \in X$ and an arbitrary $z(0) \in \mathcal{Z}$, and suppose that $V^*(x(0), z(0))$ is a solution to Problem B1. Then, (16.9) implies that for each $\varepsilon > 0$ and each $z(1) = z_j$ ($j = 1, 2, \dots, N$), there exists $(x(0), \mathbf{x}_\varepsilon^j) \in \Phi(x(0), z(0))$ such that

$$V^*(x(1), z_j) \leq \bar{U}(\mathbf{x}_\varepsilon^j, z_j) + \varepsilon.$$

Therefore,

$$\begin{aligned}
 \mathbb{E}[V^*(x(1), z(1)) \mid z(0)] &= \sum_{j=1}^N q_{jj'} V^*(x(1), z_j) \\
 &\leq \sum_{j=1}^N q_{jj'} \bar{U}(\mathbf{x}_\varepsilon^j, z_j) + \varepsilon \\
 &= \mathbb{E}[\bar{U}(\mathbf{x}_\varepsilon^j, z_j) \mid z(0)] + \varepsilon,
 \end{aligned}$$

where j' is defined by $z(0) = z_{j'}$, and the second line exploits the fact that $\sum_{j=1}^N q_{jj'} = 1$, while the third line uses the definition of the conditional expectation $\mathbb{E}[\cdot \mid z(0)]$. Next, since $(x(0), \mathbf{x}') \in \Phi(x(0), z(0))$, (16.8), Lemma 16.1 and the previous string of inequalities yield

$$\begin{aligned}
 V^*(x(0), z(0)) &\geq U(x(0), \tilde{x}'[z^0], z(0)) + \beta \mathbb{E}[\bar{U}(\mathbf{x}_\varepsilon^j, z_j) \mid z(0)], \\
 &\geq U(x(0), \tilde{x}'[z^0], z(0)) + \beta \mathbb{E}[V^*(x(1), z(1)) \mid z(0)] - \beta \varepsilon.
 \end{aligned}$$

Since the last inequality is true for any $\varepsilon > 0$, it follows that the function V^* satisfies (16.10).

Next, take an arbitrary $\varepsilon > 0$. By (16.9), there exists $\mathbf{x}'_\varepsilon = (x(0), \tilde{x}'_\varepsilon[z^0], \tilde{x}'_\varepsilon[z^1], \dots) \in \Phi(x(0), z(0))$ such that

$$\bar{U}(\mathbf{x}'_\varepsilon, z(0)) \geq V^*(x(0), z(0)) - \varepsilon.$$

Then, condition (16.8) implies that for any $z(1) \in \mathcal{Z}$, $V^*(\tilde{x}'_\varepsilon[z^0], z(1)) \geq \bar{U}(\{\tilde{x}[z^t]\}_{t=0}^\infty, z(1))$ for $\mathbf{x}''_\varepsilon = (\tilde{x}'_\varepsilon[z^0], \tilde{x}'_\varepsilon[z^1], \dots) \in \Phi(\tilde{x}'_\varepsilon[z^0], z(1))$ (recall the definition of $\mathbf{x}'_\varepsilon$). Then, from Lemma 16.1, for any $\varepsilon > 0$,

$$\begin{aligned}
 V^*(x(0), z(0)) - \varepsilon &\leq U(x(0), \tilde{x}'_\varepsilon[z^0], z(0)) + \beta \mathbb{E}[\bar{U}(\{\tilde{x}[z^t]\}_{t=0}^\infty, z(1)) \mid z(0)] \\
 &\leq U(x(0), \tilde{x}'_\varepsilon[z^0], z(0)) + \beta \mathbb{E}[V^*(\tilde{x}'_\varepsilon[z^0], z(1)) \mid z(0)],
 \end{aligned}$$

so that V^* also satisfies (16.11). This establishes that any solution to Problem B1 satisfies (16.10) and (16.11), and is thus a solution to Problem B2.

To establish the converse, note that (16.10) implies that for any $\tilde{x}[z^0] \in G(x(0), z(0))$,

$$V(x(0), z(0)) \geq U(x(0), \tilde{x}[z^0], z(0)) + \beta \mathbb{E}[V(\tilde{x}[z^0], z(1)) \mid z(0)].$$

Now substituting recursively for $V(\tilde{x}[z^0], z(1))$, $V(\tilde{x}[z^1], z(2))$ and so on, and taking expectations,

$$V(x(0), z(0)) \geq \mathbb{E}\left[\sum_{t=0}^n U(\tilde{x}[z^{t-1}], \tilde{x}[z^t], z(t)) \mid z(0)\right] + \beta^{n+1} \mathbb{E}[V(\tilde{x}[z^n], z(n+1)) \mid z(0)].$$

By definition $\lim_{n \rightarrow \infty} \mathbb{E}[\sum_{t=0}^n U(\tilde{x}[z^{t-1}], \tilde{x}[z^t], z(t)) \mid z(0)] = \bar{U}(\mathbf{x}, z(0))$ and by Assumption 16.1 $\lim_{n \rightarrow \infty} \beta^{n+1} \mathbb{E}[V(\tilde{x}[z^n], z(n+1)) \mid z(0)] = \lim_{n \rightarrow \infty} \mathbb{E}[\lim_{m \rightarrow \infty} \sum_{t=n}^m U(\tilde{x}[z^{t-1}], \tilde{x}[z^t], z(t)) \mid z(0)] = 0$, so that (16.8) is verified.

Next, let $\varepsilon > 0$. From (16.11), for any $\varepsilon' = \varepsilon(1 - \beta) > 0$ there exists $\tilde{x}_\varepsilon[z^0] \in G(x(0), z(0))$ such that

$$V(x(0), z(0)) \leq U(x(0), \tilde{x}_\varepsilon[z^0], z(0)) + \beta \mathbb{E}[V(\tilde{x}_\varepsilon[z^0], z(1)) \mid z(0)] + \varepsilon'.$$

Let $\tilde{x}_\varepsilon[z^t] \in G(\tilde{x}_\varepsilon[z^{t-1}], z(t))$, with $\tilde{x}_\varepsilon[z^{-1}] = x(0)$, and define $\mathbf{x}_\varepsilon \equiv (x(0), \tilde{x}_\varepsilon[z^0], \tilde{x}_\varepsilon[z^1], \tilde{x}_\varepsilon[z^2], \dots)$. Again substituting recursively for $V(\tilde{x}_\varepsilon[z^1])$, $V(\tilde{x}_\varepsilon[z^2])$, and so on, and taking expectations,

$$\begin{aligned} V(x(0), z(0)) &\leq \mathbb{E} \left[\sum_{t=0}^n U(\tilde{x}_\varepsilon[z^{t-1}], \tilde{x}_\varepsilon[z^t], z(t)) \mid z(0) \right] \\ &\quad + \beta^{n+1} \mathbb{E}[V(\tilde{x}_\varepsilon[z^n], z(n+1)) \mid z(0)] + \varepsilon' + \varepsilon'\beta + \dots + \varepsilon'\beta^n \\ &\leq \bar{\mathbf{U}}(\mathbf{x}_\varepsilon, z(0)) + \varepsilon, \end{aligned}$$

where the last step follows using the fact that $\varepsilon = \varepsilon' \sum_{t=0}^{\infty} \beta^t$ and that as $\lim_{n \rightarrow \infty} \mathbb{E}[\sum_{t=0}^n U(\tilde{x}_\varepsilon[z^{t-1}], \tilde{x}_\varepsilon[z^t], z(t)) \mid z(0)] = \bar{\mathbf{U}}(\mathbf{x}_\varepsilon, z(0))$. This establishes that V satisfies (16.9) and completes the proof. $\square$

PROOF OF THEOREM 16.2. Suppose that $\mathbf{x}^* \equiv (x(0), \tilde{x}^*[z^0], \tilde{x}^*[z^1], \tilde{x}^*[z^2], \dots) \in \Phi(x(0), z(0))$ is a feasible plan attaining the solution to Problem B1. Let $\mathbf{x}_t^* \equiv (\tilde{x}^*[z^{t-1}], \tilde{x}^*[z^t], \tilde{x}^*[z^{t+1}], \dots)$ be the continuation of this plan from time t .

We first show that for any $t \geq 0$, $\mathbf{x}_t^*$ attains the supremum starting from $\tilde{x}^*[z^{t-1}]$ and any $z(t) \in \mathcal{Z}$, that is,

$$(16.12) \quad \bar{\mathbf{U}}(\mathbf{x}_t^*, z(t)) = V^*(\tilde{x}^*[z^{t-1}], z(t)).$$

The proof is by induction. The hypothesis is trivially satisfied for $t = 0$ since, by definition, $\mathbf{x}_0^* = \mathbf{x}^*$ attains $V^*(x(0), z(0))$.

Next suppose that the statement is true for t , so that $\mathbf{x}_t^*$ attains the supremum starting from $\tilde{x}^*[z^{t-1}]$ and any $z(t) \in \mathcal{Z}$, or equivalently (16.12) holds for t and for $z(t) \in \mathcal{Z}$. Now using this relationship we will establish that (16.12) holds and $\mathbf{x}_{t+1}^*$ attains the supremum starting from $\tilde{x}^*[z^t]$ and any $z(t+1) \in \mathcal{Z}$. Equation (16.12) implies that

$$\begin{aligned} (16.13) \quad V^*(\tilde{x}^*[z^{t-1}], z(t)) &= \bar{\mathbf{U}}(\mathbf{x}_t^*, z(t)) \\ &= U(\tilde{x}^*[z^{t-1}], \tilde{x}^*[z^t], z(t)) \\ &\quad + \beta \mathbb{E}[\bar{\mathbf{U}}(\mathbf{x}_{t+1}^*, z(t+1)) \mid z(t)]. \end{aligned}$$

Let $\mathbf{x}_{t+1} = (\tilde{x}^*[z^t], \tilde{x}^*[z^{t+1}], \dots) \in \Phi(\tilde{x}^*[z^t], z(t+1))$ be any feasible plan starting with state vector $\tilde{x}^*[z^t]$ and stochastic variable $z(t+1)$. By definition,

$\mathbf{x}_t = (\tilde{x}^*[z^{t-1}], \mathbf{x}_{t+1}) \in \Phi(\tilde{x}^*[z^{t-1}], z(t))$. Since, by the induction hypothesis, $V^*(\tilde{x}^*[z^{t-1}], z(t))$ is the supremum starting with $\tilde{x}^*[z^{t-1}]$ and $z(t)$,

$$\begin{aligned} V^*(\tilde{x}^*[z^{t-1}], z(t)) &\geq \bar{\mathbf{U}}(\mathbf{x}_t, z(t)) \\ &= U(\tilde{x}^*[z^{t-1}], \tilde{x}^*[z^t], z(t)) + \beta \mathbb{E}[\bar{\mathbf{U}}(\mathbf{x}_{t+1}, z(t+1)) \mid z(t)] \end{aligned}$$

for any $\mathbf{x}_{t+1}$. Combining this inequality with (16.13),

$$\begin{aligned} (16.14) \quad \mathbb{E}[V^*(\tilde{x}^*[z^t], z(t+1)) \mid z(t)] &= \mathbb{E}[\bar{\mathbf{U}}(\mathbf{x}_{t+1}^*, z(t+1)) \mid z(t)] \\ &\geq \mathbb{E}[\bar{\mathbf{U}}(\mathbf{x}_{t+1}, z(t+1)) \mid z(t)] \end{aligned}$$

for all $\mathbf{x}_{t+1} \in \Phi(\tilde{x}^*[z^t], z(t+1))$. Next, we complete the proof that $\mathbf{x}_{t+1}^*$ attains the supremum starting from $\tilde{x}^*[z^t]$ and any $z(t) \in \mathcal{Z}$ and eq. (16.12) holds starting from $\tilde{x}^*[z^t]$ and any $z(t) \in \mathcal{Z}$. Suppose, to obtain a contradiction, that this is not the case. Then, there exists $\hat{\mathbf{x}}_{t+1} \in \Phi(\tilde{x}^*[z^t], z(t+1))$ for some $z(t+1) = \hat{z}$ such that

$$\bar{\mathbf{U}}(\mathbf{x}_{t+1}^*, \hat{z}) < \bar{\mathbf{U}}(\hat{\mathbf{x}}_{t+1}, \hat{z}).$$

Then, construct the sequence $\hat{\mathbf{x}}_{t+1}^* = \mathbf{x}_{t+1}^*$ if $z(t) \neq \hat{z}$ and $\hat{\mathbf{x}}_{t+1}^* = \hat{\mathbf{x}}_{t+1}$ if $z(t) = \hat{z}$. Since $\mathbf{x}_{t+1}^* \in \Phi(\tilde{x}^*[z^t], \hat{z})$ and $\hat{\mathbf{x}}_{t+1} \in \Phi(\tilde{x}^*[z^t], \hat{z})$, we also have $\hat{\mathbf{x}}_{t+1}^* \in \Phi(\tilde{x}^*[z^t], \hat{z})$. Then, without loss of generality taking $\hat{z} = z_1$,

$$\begin{aligned} \mathbb{E} [\bar{\mathbf{U}}(\hat{\mathbf{x}}_{t+1}^*, z(t+1)) \mid z(t)] &= \sum_{j=1}^N q_{jj'} \bar{\mathbf{U}}(\hat{\mathbf{x}}_{t+1}^*, z_j) \\ &= q_{1j'} \bar{\mathbf{U}}(\hat{\mathbf{x}}_{t+1}, z_j) + \sum_{j=2}^N q_{jj'} \bar{\mathbf{U}}(\mathbf{x}_{t+1}^*, z_j) \\ &> q_{1j'} \bar{\mathbf{U}}(\mathbf{x}_{t+1}^*, z_j) + \sum_{j=2}^N q_{jj'} \bar{\mathbf{U}}(\mathbf{x}_{t+1}^*, z_j) \\ &= \mathbb{E} [\bar{\mathbf{U}}(\mathbf{x}_{t+1}^*, z(t+1)) \mid z(t)], \end{aligned}$$

contradicting (16.14) and completing the induction step, which establishes that $\mathbf{x}_{t+1}^*$ attains the supremum starting from $\tilde{x}^*[z^t]$ and any $z(t+1) \in \mathcal{Z}$.

Equation (16.12) then implies that

$$\begin{aligned} V^*(\tilde{x}^*[z^{t-1}], z(t)) &= \bar{\mathbf{U}}(\mathbf{x}_t^*, z(t)) \\ &= U(\tilde{x}^*[z^{t-1}], \tilde{x}^*[z^t], z(t)) + \beta \mathbb{E} [\bar{\mathbf{U}}(\mathbf{x}_{t+1}^*, z(t+1)) \mid z(t)] \\ &= U(\tilde{x}^*[z^{t-1}], \tilde{x}^*[z^t], z(t)) + \beta \mathbb{E} [V^*(\tilde{x}^*(z^t), z(t+1)) \mid z(t)], \end{aligned}$$

establishing (16.6) and thus completing the proof of the first part of the theorem.

Now suppose that (16.6) holds for $\mathbf{x}^* \in \Phi(x(0), z(0))$. Then, substituting repeatedly for $\mathbf{x}^*$,

$$V^*(x(0), z(0)) = \sum_{t=0}^n \beta^t U(\tilde{x}^*[z^{t-1}], \tilde{x}^*[z^t], z(t)) + \beta^{n+1} \mathbb{E} [V^*(\tilde{x}^*(z^n), z(n+1)) \mid z(0)].$$

In view of the fact that V^* is bounded, $\lim_{n \rightarrow \infty} \beta^{n+1} \mathbb{E} [V^*(\tilde{x}^*(z^n), z(n+1)) \mid z(0)] = 0$ and thus

$$\begin{aligned} \bar{\mathbf{U}}(\mathbf{x}^*, z(0)) &= \lim_{n \rightarrow \infty} \sum_{t=0}^n \beta^t U(\tilde{x}^*[z^{t-1}], \tilde{x}^*[z^t], z(t)) \\ &= V^*(x(0), z(0)), \end{aligned}$$

thus $\mathbf{x}^*$ attains the optimal value in Problem B1. This completes the proof of the second part of the theorem. $\square$

I now provide one possible proof of Theorem 16.3, working with the value function V in Problem B2. An alternative proof working directly with Problem B1 is developed in Exercise 16.3.

PROOF OF THEOREM 16.3. Consider Problem B2. In view of Assumptions 16.1 and 16.2, there exists some $M < \infty$, such that $|U(x, y, z)| < M$ for all $(x, y, z) \in \mathbf{X}_G$. This immediately implies that $|V^*(x, z)| \leq M/(1 - \beta)$, all $x \in X$ and all $z \in \mathcal{Z}$. Consequently, consider the function $V^*(\cdot, \cdot) \in \mathbf{C}(X \times \mathcal{Z})$, where $\mathbf{C}(X \times \mathcal{Z})$ denotes the set of continuous functions defined on $X \times \mathcal{Z}$, where X is endowed with the sup norm, $\|f\| = \sup_{x \in X} |f(x)|$ and $\mathcal{Z}$ is endowed with the discrete topology (recall Fact A.11 in Appendix Chapter A). Moreover, all functions in $\mathbf{C}(X \times \mathcal{Z})$ are bounded because they are continuous and both X and $\mathcal{Z}$ are compact.

Now define the operator T as

$$(16.15) \quad TV(x, z) = \max_{y \in G(x, z)} \{U(x, y, z) + \beta \mathbb{E}[V(y, z') | z]\}.$$

Suppose that $V(x, z)$ is continuous and bounded. Then, $\mathbb{E}[V(y, z') | z]$ is also continuous and bounded, since it is simply given by

$$\mathbb{E}[V(y, z') | z] \equiv \sum_{j=1}^N q_{jj'} V(y, z_j),$$

with j' defined such that $z = z_{j'}$. Moreover, $U(x, y, z)$ is also continuous and bounded over $\mathbf{X}_G$.

A fixed point of the operator T , $V(x, z) = TV(x, z)$, will then be a solution to Problem B2 for given $z \in \mathcal{Z}$. We first prove that such a fixed point (solution) exists.

First note that the maximization problem on the right-hand side of (16.15) is one of maximizing a continuous function over a compact set, and by Weierstrass's Theorem, it has a solution. Consequently, the operator T is well defined and maps the space of continuous bounded functions over the set $X \times \mathcal{Z}$, $\mathbf{C}(X \times \mathcal{Z})$, into itself. It can be verified straightforwardly that T also satisfies Blackwell's sufficient conditions for a contraction (Theorem 6.9 from Chapter 6). Therefore, applying Theorem 6.7, a unique fixed point $V \in \mathbf{C}(X \times \mathcal{Z})$ to (16.15) exists and this is also the unique solution to Problem B2.

Now consider the maximization in Problem B2. Since U and V are continuous and $G(x, z)$ is compact-valued, we can apply Weierstrass's Theorem, Theorem A.9, once more to conclude that $y \in G(x, z)$ achieving the maximum exists. This defines the set of maximizers $\Pi(x, z) \subset \Phi(x, z)$ for Problem B2. Let $\mathbf{x}^* \equiv (x(0), \tilde{x}^*[z^0], \tilde{x}^*[z^1], \tilde{x}^*[z^2], \dots) \in \Phi(x(0), z(0))$ with $\tilde{x}^*[z^t] \in \Pi(\tilde{x}^*[z^{t-1}], z(t))$ for all $t \geq 0$ and each $z(t) \in \mathcal{Z}$. Then, from Theorems 16.1 and 16.2, $\mathbf{x}^*$ is also an optimal plan for Problem B1. $\square$

Finally, the proofs of Theorems 16.4-16.6 are similar to those of Theorems 6.4-6.6 from Chapter 6, and are left as exercises (see Exercises 16.4-16.6). The proof of Theorem 16.7 is similar to 16.5 and is left to Exercise 16.7.

16.3. Stochastic Euler Equations

In Chapter 6, Euler equations and transversality conditions played a central role. In the present context, instead of the standard Euler equations, we have to work with stochastic Euler equations. While this is not conceptually any more involved than the standard Euler equations, stochastic Euler equations are not always easy to manipulate. Sometimes, as in the permanent income hypothesis model studied in Section 16.5, the stochastic Euler equation itself may contain enough economics to be useful. In other instances, our interest will be with the characterization of optimal plans. Although this is typically a non-trivial task, the combination of stochastic Euler equations and the appropriate transversality condition can sometimes be used to determine certain qualitative features of optimal plans.

Let us follow the treatment in Chapter 6 and also build on the results from Section 16.1. Let us use $*$'s to denote optimal values and D for gradients. Then, using Assumption 16.5 and Theorem 16.6, we can write the necessary conditions for an interior optimal plan as

$$(16.16) \quad D_y U(x, y^*, z) + \beta \mathbb{E} [D_x V(y^*, z') \mid z] = 0,$$

where $x \in \mathbb{R}^K$ is the current-value of the state vector, $z \in \mathcal{Z}$ is the current-value of the stochastic variable, and $D_x V(y^*, z')$ denotes the gradient of the value function evaluated at next period's state vector y^* . Now using the stochastic equivalent of the Envelope Theorem for dynamic programming and differentiating (16.5) with respect to the state vector, x :

$$(16.17) \quad D_x V(x, z) = D_x U(x, y^*, z).$$

Here there are no expectations, since this equation is conditioned on the realization of $z \in \mathcal{Z}$. Note that y^* here is a shorthand for $\pi(x, z)$. Now using this notation and combining these two equations, we obtain the canonical form of the stochastic Euler equation

$$D_y U(x, \pi(x, z), z) + \beta \mathbb{E} [D_x U(\pi(x, z), \pi(\pi(x, z), z'), z') \mid z] = 0,$$

where, as in Chapter 6, $D_x U$ represents the gradient vector of U with respect to its first K arguments, and $D_y U$ represents its gradient with respect to the second set of K arguments. Writing this equation in the notation more congruent with the sequence version of the problem, the stochastic Euler equation takes the form

$$(16.18) \quad D_y U(\tilde{x}^*[z^{t-1}], \tilde{x}^*[z^t], z(t)) + \beta \mathbb{E} [D_x U(\tilde{x}^*[z^t], \tilde{x}^*[z^{t+1}], z(t+1)) \mid z(t)] = 0,$$

for $z^{t-1} \in \mathcal{Z}^{t-1}$.

How do we write the transversality condition in this case? The transversality condition essentially requires the discounted marginal return from the state variable to tend to zero as the planning horizon goes to infinity. In a stochastic environment, we clearly have to look at expected returns. The question is what information to condition upon. It turns out that is sufficient to condition on the information available at date $t = 0$, that is, on $z(0) \in \mathcal{Z}$.

Consequently, the transversality condition associated with this stochastic Euler equation takes the form

$$(16.19) \quad \lim_{t \rightarrow \infty} \beta^t \mathbb{E} [D_x U(\tilde{x}^* [z^{t-1}], \tilde{x}^* [z^t], z(s+t)) \cdot \tilde{x}^* [z^{t-1}] \mid z(0)] = 0,$$

given $z(0) \in \mathcal{Z}$.

The next theorem generalizes Theorem 6.10 from Chapter 6 to an environment with uncertainty. In particular, it shows that the transversality condition together with the transformed Euler equations in (16.18) are both necessary and sufficient to characterize an optimal solution to Problem A1 and therefore to Problem A2.

THEOREM 16.8. (*Euler Equations and the Transversality Condition*) *Let $X \subset \mathbb{R}_+^K$ and suppose that Assumptions 16.1-16.5 hold. Then, the sequence of feasible plans $\{\tilde{x}^* [z^t]\}_{t=-1}^\infty$, with $\tilde{x}^* [z^t] \in \text{Int}G(\tilde{x}^* [z^{t-1}], z(t))$ for each $z(t) \in \mathcal{Z}$ and each $t = 0, 1, \dots$, is optimal for Problem B1 given $x(0)$ and $z(0) \in \mathcal{Z}$ if and only if it satisfies (16.18) and (16.19).*

PROOF. (Sufficiency) Consider an arbitrary $x(0) \in X$ and $z(0) \in \mathcal{Z}$, and let $\mathbf{x}^* \equiv \{\tilde{x}^* [z^t]\}_{t=-1}^\infty \in \Phi(x(0), z(0))$ be a feasible plan satisfying (16.18) and (16.19). For any $\mathbf{x} \equiv \{\tilde{x} [z^t]\}_{t=-1}^\infty \in \Phi(x(0), z(0))$, and any $z^\infty \in \mathcal{Z}^\infty$ define

$$\Delta_{\mathbf{x}}(z^\infty) \equiv \lim_{T \rightarrow \infty} \sup \sum_{t=0}^T \beta^t [U(\tilde{x}^* [z^{t-1}], \tilde{x}^* [z^t], z(t)) - U(\tilde{x} [z^{t-1}], \tilde{x} [z^t], z(t))]$$

as the difference of the *realized* objective function between the feasible sequences $\mathbf{x}^*$ and $\mathbf{x}$.

From Assumptions 16.2 and 16.5, U is continuous, concave, and differentiable, so that for any $z^\infty \in \mathcal{Z}^\infty$ and any $\mathbf{x} \in \Phi(x(0), z(0))$

$$\begin{aligned} \Delta_{\mathbf{x}}(z^\infty) &\geq \lim_{T \rightarrow \infty} \sup \sum_{t=0}^T \beta^t [D_x U(\tilde{x}^* [z^{t-1}], \tilde{x}^* [z^t], z(t)) \cdot (\tilde{x}^* [z^{t-1}] - \tilde{x} [z^{t-1}]) \\ &\quad + D_y U(\tilde{x}^* [z^{t-1}], \tilde{x}^* [z^t], z(t)) \cdot (\tilde{x}^* [z^t] - \tilde{x} [z^t])]. \end{aligned}$$

Since this is true for any $z^\infty \in \mathcal{Z}^\infty$, we can take expectations on both sides to obtain

$$\begin{aligned} &\mathbb{E}[\Delta_{\mathbf{x}}(z^\infty) \mid z(0)] \\ &\geq \lim_{T \rightarrow \infty} \sup \mathbb{E} \left[\sum_{t=0}^T \beta^t [D_x U(\tilde{x}^* [z^{t-1}], \tilde{x}^* [z^t], z(t)) \cdot (\tilde{x}^* [z^{t-1}] - \tilde{x} [z^{t-1}]) \mid z(s)] \right] \\ &\quad + \lim_{T \rightarrow \infty} \sup \mathbb{E} \left[\sum_{t=0}^T \beta^t D_y U(\tilde{x}^* [z^{t-1}], \tilde{x}^* [z^t], z(t)) \cdot (\tilde{x}^* [z^t] - \tilde{x} [z^t]) \mid z(s)] \right] \end{aligned}$$

for $z(0) \in \mathcal{Z}$. Rearranging the previous expression,

$$\begin{aligned} \mathbb{E}[\Delta_{\mathbf{x}}(z^\infty) \mid z(0)] &\geq \\ \lim_{T \rightarrow \infty} \sup \mathbb{E} &\left[\sum_{t=0}^T \beta^t D_y U(\tilde{x}^*[z^{t-1}], \tilde{x}^*[z^t], z(t)) \cdot (\tilde{x}^*[z^t] - \tilde{x}[z^t]) \mid z(s) \right] \\ &+ \lim_{T \rightarrow \infty} \sup \mathbb{E} \left[\sum_{t=0}^T \beta^{t+1} D_x U(\tilde{x}^*[z^t], \tilde{x}^*[z^{t+1}], z(t+1)) \cdot (\tilde{x}^*[z^t] - \tilde{x}[z^t]) \mid z(s) \right] \\ &- \lim_{T \rightarrow \infty} \inf \mathbb{E} [\beta^{T+1} D_x U(\tilde{x}^*[z^T], \tilde{x}^*[z^{T+1}], z(T+1)) \cdot \tilde{x}^*[z^T] \mid z(s)] \\ &+ \lim_{T \rightarrow \infty} \sup \mathbb{E} [\beta^{T+1} D_x U(\tilde{x}[z^T], \tilde{x}[z^{T+1}], z(T+1)) \cdot \tilde{x}[z^T] \mid z(s)]. \end{aligned}$$

Since $\mathbf{x}^* \equiv \{\tilde{x}^*[z^t]\}_{t=-1}^\infty$ satisfies (16.18), the terms in first and second lines are all equal to zero. Moreover, since $\mathbf{x}^* \equiv \{\tilde{x}^*[z^t]\}_{t=-1}^\infty$ satisfies (16.19), the third line is also equal to zero. Finally, since U is increasing in x , $D_x U \geq 0$, and $x \geq 0$, the fourth line is nonnegative, establishing that $\mathbb{E}[\Delta_{\mathbf{x}}(z^\infty) \mid z(0)] \geq 0$ for any $\mathbf{x} \in \Phi(x(0), z(0))$. Consequently, $\mathbf{x}^*$ yields higher value than any feasible $\mathbf{x} \in \Phi(x(0), z(0))$, and is therefore optimal.

(Necessity) The proof of necessity mirrors the necessity part of Theorem 6.10. In particular, again consider a feasible plan $\mathbf{x} \in \Phi(x(0), z(0))$ such that $\tilde{x}[z^t] = \tilde{x}^*[z^t] + \varepsilon a[z^t]$ for some variation $a[z^t] \in \mathbb{R}^K$ for each $z^t \in \mathcal{Z}^t$ and a real number ε chosen to be sufficiently small (which is feasible since $\mathbf{x}^* \equiv \{\tilde{x}^*[z^t]\}_{t=-1}^\infty$ is interior). This immediately establishes the necessity of the stochastic Euler equations, (16.18). Next choosing a feasible plan $\tilde{x}[z^t] = (1 - \varepsilon)\tilde{x}^*[z^t]$ and using (16.18) gives

$$\begin{aligned} \mathbb{E}[\Delta_{\mathbf{x}}(z^\infty) \mid z(0)] &= \\ -\varepsilon \lim_{T \rightarrow \infty} \inf \mathbb{E} &[\beta^{T+1} D_x U(\tilde{x}^*[z^T], \tilde{x}^*[z^{T+1}], z(T+1)) \cdot \tilde{x}^*[z^T] \mid z(s)] + \lim_{T \rightarrow \infty} \sum_{t=0}^T \beta^t o(\varepsilon). \end{aligned}$$

If (16.19) is violated, the first term can be made negative and if so, it remains negative as $\varepsilon \rightarrow 0$. This contradicts the fact that $\mathbf{x}^* \equiv \{\tilde{x}^*[z^t]\}_{t=-1}^\infty$ is an optimal plan, establishing the necessity part of the theorem. $\square$

16.4. Generalization to Markov Processes*

What happens if z does not take on finitely many values? For example, z may be represented by a general Markov process, taking values in a compact metric space. The simplest example would be a one-dimensional stochastic variable $z(t)$ given by the process $z(t) = \rho z(t-1) + \sigma \varepsilon(t)$, where $\varepsilon(t)$ has a standard normal distribution. At some level, most of the results we care about generalize to such cases. At another level, however, greater care needs to be taken in formulating these problems both in the sequence form of Problem B1 and in the recursive form of Problem B2. The main difficulty in this case arises in ensuring that there exist appropriately defined feasible plans, which now need to be “measurable” with respect to the information set available at the time. Unfortunately, to state the appropriate theorems in a rigorous manner requires a lengthy detour into measure theory. Instead, I will

assume that both $\mathcal{Z}$ and X are compact and that the function $\tilde{x}[z^t]$ introduced in Section 16.1 is “well-defined”—in particular, finite-valued and measurable. Under these assumptions and again representing all integrals with the expectations, I state the main theorems for stochastic dynamic programming with general Markov processes without proof.

Let us first define $\mathcal{Z}$ as a compact subset of $\mathbb{R}$, which includes $\mathcal{Z}$ consisting of finite number of elements and $\mathcal{Z}$ corresponding to an interval as special cases. Let $z(t) \in \mathcal{Z}$ represent the uncertainty in this environment, and suppose that its probability distribution can be represented as a Markov process, i.e.,

$$\Pr[z(t) \mid z(0), \dots, z(t-1)] \equiv \Pr[z(t) \mid z(t-1)].$$

Let us also use the notation $z^t \equiv (z(0), z(1), \dots, z(t))$ to represent the history of the realizations of the stochastic variable. The objective function and the constraint sets are represented as in Section 16.1, so that $\tilde{x}[z^t]$ again denotes a *feasible plan*. Let the set of feasible plans after history z^t be denoted by $\Phi(\tilde{x}[z^{t-1}], z(t))$. The set of feasible plans starting with $z(0)$ is then $\Phi(x(0), z(0))$. Also whenever there exists a function V that is a solution to Problem B2, let us define $\Pi(x, z) \subset \Phi(x, z)$ such that any $\pi(x, z) \in \Pi(x, z)$ satisfies

$$V(x, z) = U(x, \pi(x, z), z) + \beta \mathbb{E}[V(\pi(x, y), z') \mid z].$$

Finally, to state the appropriate theorems, let us refer to the same assumptions as in Section 16.1, except that these assumptions now require the relevant functions to be *measurable* in the appropriate sense and the correspondence $\Phi(x(t), z^t)$ to always admit a *measurable selection* for all $x(t) \in X$ and $z^t \in \mathcal{Z}^t$. For this reason, I will refer to these assumptions with a * (i.e., instead of Assumption 16.2, I refer to Assumption 16.2*).

THEOREM 16.9. (*Existence of Solutions*) Suppose that $\Phi(x(0), z(0))$ is nonempty for all $z(0) \in \mathcal{Z}$ and all $x(0) \in X$. Suppose also that for any $\mathbf{x} \in \Phi(x(0), z(0))$, $\mathbb{E}[\sum_{t=0}^{\infty} \beta^t U(\tilde{x}[z^{t-1}], \tilde{x}[z^t], z(t)) \mid z(0)]$ is well-defined and finite-valued. Then, any solution $V(x, z)$ to Problem B2 coincides with the solution $V^*(x, z)$ to Problem B1. Moreover, if $\Pi(x, z)$ is nonempty for all $(x, z) \in X \times \mathcal{Z}$, then any $\pi(x, z) \in \Pi(x, z)$ achieves $V^*(x, z)$.

Notice that this theorem already imposes stronger requirements than Assumption 16.1 and hence there is no need to impose the equivalent of Assumption 16.1.

THEOREM 16.10. (*Continuity of Value Functions*) Suppose the hypotheses in Theorem 16.9 are satisfied and Assumption 16.2* holds. Then, there exists a unique function $V : X \times \mathcal{Z} \rightarrow \mathbb{R}$ that satisfies (16.4). Moreover, V is continuous and bounded. Finally, an optimal plan $\mathbf{x}^* \in \Phi(x(0), z(0))$ exists for any $x(0) \in X$ and any $z(0) \in \mathcal{Z}$.

THEOREM 16.11. (*Concavity of Value Functions*) Suppose the hypotheses in Theorem 16.9 are satisfied and Assumptions 16.2* and 16.3* hold. Then, the unique function V that satisfies (16.4) is strictly concave in x for each $z \in \mathcal{Z}$. Moreover, the optimal plan can be expressed as $\tilde{x}^*[z^t] = \pi(x(t), z(t))$, where the policy function $\pi : X \times \mathcal{Z} \rightarrow X$ is continuous in x for each $z \in \mathcal{Z}$.

THEOREM 16.12. (*Monotonicity of Value Functions*) Suppose the hypotheses in Theorem 16.9 are satisfied and Assumptions 16.2* and 16.4* hold. Then, the unique value function $V : X \times \mathcal{Z} \rightarrow \mathbb{R}$ that satisfies (16.4) is strictly increasing in x for each $z \in \mathcal{Z}$.

THEOREM 16.13. (*Differentiability of Value Functions*) Suppose the hypotheses in Theorem 16.9 are satisfied and Assumptions 16.2*, 16.3* and 16.5* hold. Let π be the policy function defined above and assume that $x' \in \text{Int}X$ and $\pi(x', z) \in \text{Int}G(x', z)$ for each $z \in \mathcal{Z}$, then $V(x, z)$ is continuously differentiable at x' , with derivative given by

$$D_x V(x', z) = D_x U(x', \pi(x', z), z).$$

Given the hypotheses of Theorem 16.9, the proofs of these theorems are not difficult, though they are long and require a little care. Somewhat more general versions of these theorems can be found in Stokey, Lucas and Prescott (1989, Chapter 9), who also develop the necessary measure theory and some of the theory of general Markov processes to state more rigorous and complete versions of these theorems.

Finally, note also that Theorem 16.8 applies exactly in this case, since the statement or the proof of this theorem did not make use of the fact that z followed a Markov chain as opposed to a general Markov process.

16.5. Applications of Stochastic Dynamic Programming

I now present a number of applications of the methods of stochastic dynamic programming. Some of the most important applications, related to stochastic growth and growth with incomplete markets, are left for next chapter. In each application, I try to point out how formulating the problem recursively and using stochastic dynamic programming methods simplify the analysis.

16.5.1. The Permanent Income Hypothesis. One of the most important applications of stochastic dynamic optimization is to the consumption smoothing problem of the consumer facing an uncertain income stream. This problem was first discussed by Irving Fisher (1930) and then received its first systematic analysis in Milton Friedman's classic book on consumption theory (1956). With Robert Hall's (1978) seminal paper on dynamic consumption behavior, it became one of the most celebrated macroeconomic models. Here I present a simple version of this problem with linear-quadratic preferences and characterize the solution using the sequence formulation of the problem and also stochastic dynamic programming.

Consider a consumer maximizing discounted lifetime utility

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c(t)),$$

with $c(t) \geq 0$ as usual denoting consumption. To start with, assume that $u(\cdot)$ is strictly increasing, continuously differentiable and concave and denote its derivative by $u'(\cdot)$.

The consumer can borrow and lend freely at a constant interest rate $r > 0$, thus his lifetime budget constraint takes the form

$$(16.20) \quad \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} c(t) \leq \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} w(t) + a(0),$$

where $a(0)$ denotes his initial assets and $w(t)$ is his labor income. Suppose that $w(t)$ is random and takes values from the set $\mathcal{W} \equiv \{w_1, \dots, w_N\}$. This corresponds to potential labor income fluctuations due to aggregate or idiosyncratic shocks facing the individual. To simplify the analysis, let us suppose that $w(t)$ is distributed independently over time and the probability that $w(t) = w_j$ is q_j (naturally with $\sum_{j=1}^N q_j = 1$). Consequently, the lifetime budget constraint (16.20) has to be interpreted as a stochastic constraint. We therefore require this constraint to hold *almost surely*. This implies that the constraint has to hold with probability 1. The reader may wonder why this particular concept from measure theory has crept into our discussion, since $w(t)$ still takes finitely many values. The reason is that even when $w(t)$ takes only finitely many values, the probability distribution for the infinite sequence of random variables $w^\infty \equiv (w(0), w(1), \dots)$ is equivalent to a continuous probability distribution. Nevertheless, for our purposes this is also a technicality and not much more than the requirement that the lifetime budget constraint (16.20) should hold almost surely is necessary for our analysis.

Leaving technicalities aside, the fact that the lifetime budget constraint is stochastic has important economic implications. In particular, although I have not introduced an explicit borrowing constraint, the fact that the lifetime budget constraint must hold with probability 1 imposes *endogenous borrowing constraints*. For example, suppose that $w_1 = 0$ and $q_1 > 0$ (so that this state corresponds to unemployment and zero labor income). Then, there is a positive probability that the individual will receive zero income for any sequence of periods of length $T < \infty$. Then, if the individual ever chooses a negative asset holding, $a(t) < 0$, there will be a positive probability of violating his lifetime budget constraint, even if he were to choose zero consumption in all future periods. Therefore, there is an endogenous borrowing constraint, which takes the form

$$a(t) \geq - \sum_{s=0}^{\infty} \frac{1}{(1+r)^s} w_1 \equiv -b_1,$$

with w_1 denoting the minimum value of w within the set $\mathcal{W}$ and the last relationship defining b_1 .

Let us first solve this problem treating it as a sequence problem, that is, the problem of choosing a sequence of feasible plans $\{\tilde{c}[w^t]\}_{t=0}^{\infty}$. This can be done simply by forming a Lagrangian. Even though there is a single lifetime budget constraint (16.20), it would be incorrect to treat the problem as if there were a unique Lagrange multiplier λ . This is because consumption plans are made conditional on the realizations of events up to a certain date. In particular, consumption at time t will be conditioned on the history of shocks up to that date, $w^t \equiv (w(0), w(1), \dots, w(t))$, and in fact I used the notation $\tilde{c}[w^t]$ to emphasize

that consumption at date t is a mapping from the history of income realizations, w^t . At that point, since there is also more information about how much the individual has earned and how much he has spent, it is also natural to think that the Lagrange multiplier, which represents the marginal utility of money, is also a random variable and can depend only on the realizations of the shocks up to date t , w^t . I therefore write this multiplier as $\tilde{\lambda}[w^t]$.

The first-order conditions for this problem immediately give

$$(16.21) \quad \beta^t u'(\tilde{c}[w^t]) = \frac{1}{(1+r)^t} \tilde{\lambda}[w^t],$$

which requires the (discounted) marginal utility of consumption after history w^t to be equated to the (discounted) marginal utility of income after history w^t , $\tilde{\lambda}[w^t]$. While economically interpretable, this first-order condition is not particularly useful unless we know the law of motion of the marginal utility of income, $\tilde{\lambda}[w^t]$. This law of motion is not straightforward to derive with this formulation. An alternative formulation of the sequence problem, where prices for all possible claims to consumption contingent on any realization of history are introduced, is much more tractable and gives similar results to the recursive approach below. I will introduce this contingent-claims formulation in the analysis of the competitive equilibrium of the neoclassical growth model under uncertainty in the next chapter.

Instead, let us formulate the same problem recursively, which will enable sharper results. Using the tools of this chapter, let us write this problem recursively. First, instead of the lifetime budget constraint, the flow budget constraint of the individual can be written as

$$a' = (1+r)(a+w-c),$$

where a' refers to next period's asset holdings. Conversely, this implies $c = a+w-(1+r)^{-1}a'$. Then, the value function of the individual, conditioned on current asset holding a and current realization of the income shock w , can be written as

$$V(a, w) = \max_{a' \in [-b_1, (1+r)(a+w)]} \left\{ u\left(a+w-(1+r)^{-1}a'\right) + \beta \mathbb{E}V(a', w') \right\},$$

where I have made use of the fact that w is distributed independently across periods, so the expectation of the continuation value is not conditioned on the current realization of w . Now as in Example 6.5 in Chapter 6, where we studied the nonstochastic version of this problem, we need to restrict the set of feasible asset levels to be able to apply Theorems 16.1-16.6 from Section 16.1. In particular, let us take $\bar{a} \equiv a(0) + w_N/r$, where w_N is the highest level of labor income. We can then impose that $a(t) \in [0, \bar{a}]$ and then again verify the conditions under which this has no effect on the solution (in particular the condition for $a(t)$ to be always in the interior of the set, see Exercise 16.11).

The first-order condition for the maximization problem gives

$$(16.22) \quad \frac{1}{1+r} u'(c(t)) = \beta \mathbb{E}_t \frac{\partial V(a(t+1), w(t+1))}{\partial a},$$

where $\mathbb{E}_t$ denotes the expectations given the information at time t . Noting that $\partial V(a', w')/\partial a$ is also the marginal utility of income, this equation is very similar to (16.21). The additional

mileage now comes from the Envelope condition from Theorem 16.6, which implies that

$$\frac{\partial V(a(t), w(t))}{\partial a} = u'(c(t)).$$

Combining this equation with (16.22), we obtain the famous stochastic Euler equation of stochastic permanent income hypothesis:

$$(16.23) \quad u'(c(t)) = \beta(1+r) \mathbb{E}_t u'(c(t+1)).$$

The notable feature here is that on the right-hand side we have the expectation of the marginal utility of consumption at date $t+1$. We thus have a simple stochastic Euler equation.

This equation becomes even simpler and perhaps more insightful, when the utility function is quadratic, for example, taking the form

$$u(c) = \phi c - \frac{1}{2} c^2,$$

with ϕ sufficiently large that in the relevant range $u(\cdot)$ is increasing in c . Using this quadratic form with (16.23), we obtain Hall's famous stochastic equation that

$$(16.24) \quad c(t) = (1 - \kappa) \phi + \kappa \mathbb{E}_t c(t+1),$$

where $\kappa \equiv \beta(1+r)$. A striking prediction of this equation is that variables, such as current or past income, should not predict future consumption growth. A large empirical literature investigates whether or not this is the case in aggregate or individual data, focusing on *excess sensitivity* tests. If future consumption growth depends on current income, this is interpreted as evidence for excess sensitivity, rejecting (16.24). This rejection is often considered as evidence in favor of credit constraints, which prevent individuals from freely borrowing and lending (subject to the endogenous borrowing constraint derived above). Nevertheless, excess sensitivity can also emerge when the utility function is not quadratic (see, for example, Zeldes, 1989, Caballero, 1990).

Equation (16.24) takes an even simpler form when $\beta = (1+r)^{-1}$, that is, when the discount factor is the inverse of the gross interest rate. In this case, $\kappa = 1$ and $c(t) = \mathbb{E}_t c(t+1)$ or $\mathbb{E}_t \Delta c(t+1) = 0$, so that the expected value of future consumption should be the same as today's consumption. This last property is sometimes referred to as the "martingale property," since a random variable $z(t)$ is a *martingale* with respect to some information set Ω_t if $\mathbb{E}[z(t+1) | \Omega_t] = z(t)$. It is a *submartingale*, if $\mathbb{E}[z(t+1) | \Omega_t] \geq z(t)$ and *supermartingale*, if $\mathbb{E}[z(t+1) | \Omega_t] \leq z(t)$. Thus whether consumption is a martingale, submartingales or supermartingale depends on the interest rate relative to the discount factor. Exercises 16.8 and 16.11 further discuss the implications of this equation.

16.5.2. Search for Ideas. This subsection provides another example of an economic problem where dynamic programming techniques are very useful. This example also provides us with an alternative and complementary way of thinking about the endogeneity of technology to that offered by the models presented in Part 4.

Consider the problem of a single entrepreneur, with risk-neutral objective function

$$\sum_{t=0}^{\infty} \beta^t c(t).$$

This entrepreneur's consumption is given by the income he generates in that period (there is no saving or borrowing). The entrepreneur can produce income equal to

$$y(t) = a'(t)$$

at time t , where $a'(t)$ is the quality of the technique he has available for production.³ At $t = 0$, the entrepreneur starts with $a(0) = 0$. From then on, at each date, he can either engage in production using one of the techniques he has already discovered, or spend that period searching for a new technique. Let us assume that each period in which he engages in such a search, he gets an independent draw from a time-invariant distribution function $H(a)$ defined over a bounded interval $[0, \bar{a}]$.

Therefore, the decision of the entrepreneur at each date is whether to search for a new technique or to produce with one of the techniques he has discovered so far. The consumption decision of the entrepreneur is trivial, since there is no saving or borrowing, and he has to consume his current income, $c(t) = y(t)$.

This problem introduces a slightly different perspective on some of the ideas already discussed in the book. In particular, as in the endogenous technological change models studied so far, the entrepreneur has a non-trivial choice which affects the technology available to him; by searching more, which is a costly activity in terms of foregone production, he can potentially improve the set of techniques available to him. Moreover, this economic decision is related to the tradeoffs in the standard models of technological progress and technology adoption; whether to produce with what he has available today or make an "investment" in one more round of search with the hope of discovering something better. This type of economic tradeoff is complementary to the incentives to invest in new technology in the models of endogenous technology.

For now, our main objective is to demonstrate how dynamic programming techniques can be used to analyze this problem. Let us first try to write the maximization problem facing the entrepreneur as a sequence problem. Let us begin with the class of decision rules of the agent. In particular, let $\mathbf{a}^t \in \mathbf{A}^t \equiv [0, \bar{a}]^t$ be a sequence of techniques observed by the entrepreneur over the past t periods, with $a(s) = 0$, if at time s , the entrepreneur engaged in production, and write $\mathbf{a}^t = (a(0), \dots, a(t))$. Then, a decision rule for this individual would be

$$q(t) : \mathbf{A}^t \rightarrow \{a(t)\} \cup \{\text{search}\},$$

which denotes the action of the agent at time t , which is either to produce with the current technique he has discovered, $a(t)$, or to choose $q(t) = \text{"search"}$ and spend that period searching for or researching a new technique. Let $\mathcal{P}_t$ be the set of functions from $\mathbf{A}^t$ into

³The use of a here for the quality of ideas, rather than as asset holdings of individual before, should cause no confusion.

$a(t) \cup \{\text{search}\}$, and $\mathcal{P}^\infty$ the set of infinite sequences of such functions. The most general way of expressing the problem of the individual would be as follows. Let $\mathbb{E}$ be the expectations operator. Then, the individual's problem is

$$\max_{\{q(t)\}_{t=0}^\infty \in \mathcal{P}^\infty} \mathbb{E} \sum_{t=0}^{\infty} \beta^t c(t)$$

subject to $c(t) = 0$ if $q(t) = \text{"search"}$ and $c(t) = a'$ if $q(t) = a'$ for $a(s) = a'$ for some $s \leq t$. Naturally, written in this way, the problem looks complicated, even daunting. The point of writing it in this way is to show that in certain classes of models, the dynamic programming formulation will be quite tractable even when the sequence problem may look quite complicated.

To demonstrate this, high now write this optimization problem recursively using dynamic programming techniques. Let us simplify the formulation of the recursive form of this problem by making two observations (which will both be proved in Exercise 16.12). First, because the problem is stationary we can discard all of the techniques that the individual has sampled except the last one and thus write the problem simply conditioning on the last period's stochastic state. In particular, denote the value of an agent who has just sampled a technique $a \in [0, \bar{a}]$ by $V(a)$. Second, we suppose that once the individual starts producing at some technique a' , he will continue to do so forever, instead of going back to searching again at some future date. This is also intuitive due to the stationarity of the problem; if the individual is willing to accept production at technique a' rather than searching more at time t , he would also do so at time $t+1$. This last observation implies that if the individual accepts production at some technique a' at date t , he will consume $c(s) = a'$ for all $s \geq t$. Consequently, we obtain the value of accepting technique a' as

$$V^{\text{accept}}(a') = \frac{a'}{1 - \beta}.$$

Therefore, we can write

$$\begin{aligned} V(a') &= \max_{q \in \{0,1\}} q V^{\text{accept}}(a') + (1 - q) \beta \mathbb{E} V \\ &= \max \{ V^{\text{accept}}(a'), \beta \mathbb{E} V \} \\ (16.25) \quad &= \max \left\{ \frac{a'}{1 - \beta}, \beta \mathbb{E} V \right\}, \end{aligned}$$

where q is the acceptance decision, with $q = 1$ corresponding to acceptance, and

$$(16.26) \quad \mathbb{E} V = \int_0^{\bar{a}} V(a) dH(a)$$

is the expected continuation value of not producing at the available techniques. The expression in (16.25) follows from the fact that the individual will choose whichever option, starting production or continuing to search, gives him higher utility. That the value of continuing to search is given by (16.26) follows by definition. At the next date, the individual will have

value $V(a)$ as given by (16.25) when he draws a from the distribution $H(a)$, and thus integrating over this expression gives $\mathbb{E}V$. The integral is written as a Lebesgue integral, since $H(a)$ may not have a continuous density.

A SLIGHT DIGRESSION*. Even though the special structure of the search problem enables a direct solution, it is also useful to see that optimal policies can be derived by applying the techniques developed in Section 6.4 in Chapter 6. For this, combine the two previous equations and write

$$\begin{aligned} (16.27) \quad V(a') &= \max \left\{ \frac{a'}{1-\beta}, \beta \int_0^{\bar{a}} V(a) dH(a) \right\}, \\ &= TV(a'), \end{aligned}$$

where the second line defines the mapping T . Now (16.27) is in a form to which we can apply the above theorems. Blackwell's Sufficiency Theorem (Theorem 6.9) applies directly and implies that T is a contraction since it is monotone and satisfies discounting.

Next, let $V \in \mathbf{C}([0, \bar{a}])$, i.e., the set of real-valued continuous (hence bounded) functions defined over the set $[0, \bar{a}]$, which is a complete metric space with the sup norm. Then, the Contraction Mapping Theorem, Theorem 6.7 from Chapter 6, immediately implies that a unique value function $V(a)$ exists in this space. Thus the dynamic programming formulation of the sequential search problem immediately leads to the existence of an optimal solution (and thus optimal strategies, which will be characterized below).

Moreover, Theorem 6.8 also applies by taking S' to be the space of nondecreasing continuous functions over $[0, \bar{a}]$, which is a closed subspace of $\mathbf{C}([0, \bar{a}])$. Therefore, $V(a)$ is nondecreasing. In fact, using Theorem 6.8 we could also prove that $V(a)$ is piecewise linear with first a flat portion and then an increasing portion. Let the space of such functions be S'' , which is another subspace of $\mathbf{C}([0, \bar{a}])$, but is not closed. Nevertheless, now the second part of Theorem 6.8 applies, since starting with any nondecreasing function $V(a)$, $TV(a)$ will be a piecewise linear function starting with a flat portion. Therefore, the theorem implies that the unique fixed point, $V(a)$, must have this property too. ■

The digression above used Theorem 6.8 from Chapter 6 to argue that $V(a)$ would take a piecewise linear form. In fact, in this case, this property can also be deduced directly from (16.27), since $V(a)$ is a maximum of two functions, one of them flat and the other one linear. Therefore $V(a)$ must be piecewise linear, with first a flat portion.

Our next task is to determine the optimal policy using the recursive formulation of Problem B2. The fact that $V(a)$ is linear (and strictly increasing) after a flat portion immediately tells us that the optimal policy will take a *cutoff rule*, meaning that there will exist a cutoff technology level R such that all techniques above R are accepted and production starts, while those $a < R$ are turned down and the entrepreneur continues to search. This cutoff rule property follows because $V(a)$ is strictly increasing after some level, thus if some technology a' is accepted, all technologies with $a > a'$ will also be accepted.

Moreover, this cutoff rule must satisfy the following equation

$$(16.28) \quad \frac{R}{1-\beta} = \int_0^{\bar{a}} \beta V(a) dH(a),$$

so that the individual is just indifferent between accepting the technology $a = R$ and waiting for one more period. Next we also have that since $a < R$ are turned down, for all $a < R$

$$\begin{aligned} V(a) &= \beta \int_0^{\bar{a}} V(a) dH(a) \\ &= \frac{R}{1-\beta}, \end{aligned}$$

and for all $a \geq R$,

$$V(a) = \frac{a}{1-\beta}.$$

Using these observations,

$$\int_0^{\bar{a}} V(a) dH(a) = \frac{RH(R)}{1-\beta} + \int_{a \geq R} \frac{a}{1-\beta} dH(a).$$

Combining this equation with (16.28),

$$(16.29) \quad \frac{R}{1-\beta} = \beta \left[\frac{RH(R)}{1-\beta} + \int_{a \geq R} \frac{a}{1-\beta} dH(a) \right].$$

Manipulating this equation,

$$R = \frac{\beta}{1-\beta H(R)} \int_R^{\bar{a}} a dH(a),$$

which is a convenient way of expressing the cutoff rule R . Equation (16.29) can also be expressed in an alternative and somewhat more intuitive way. To do this, rewrite this equation as

$$\frac{R}{1-\beta} = \beta \left[\int_{a < R} \frac{R}{1-\beta} dH(a) + \int_{a \geq R} \frac{a}{1-\beta} dH(a) \right].$$

Now subtracting $\beta R / (1-\beta) = \beta R \int_{a < R} dH(a) / (1-\beta) + \beta R \int_{a \geq R} dH(a) / (1-\beta)$ from both sides,

$$(16.30) \quad R = \frac{\beta}{1-\beta} \left[\int_R^{\bar{a}} (a - R) dH(a) \right],$$

which is an important way of characterizing the cutoff rule. The left-hand side is best understood as the cost of foregoing production with a technology of R , while the right-hand side is the expected benefit of one more round of search. At the cutoff threshold, these two terms have to be equal, since the entrepreneur is indifferent between starting production and continuing search.

Let us now define the right-hand side of eq. (16.30), the expected benefit of one more search, as

$$\gamma(R) \equiv \frac{\beta}{1-\beta} \left[\int_R^{\bar{a}} (a - R) dH(a) \right].$$

Suppose also that H has a continuous density, denoted by h . Then,

$$\begin{aligned}\gamma'(R) &= -\frac{\beta}{1-\beta}(R - R)h(R) - \frac{\beta}{1-\beta}\left[\int_R^{\bar{a}} dH(a)\right] \\ &= -\frac{\beta}{1-\beta}[1 - H(R)] < 0\end{aligned}$$

This implies that eq. (16.30) has a unique solution. It can be easily verified that a higher β , by making the entrepreneur more patient, increases the cutoff threshold R .

16.5.3. Other Applications. There are numerous other applications of stochastic dynamic programming. In addition to the three growth models studied in the next chapter, the following are noteworthy.

- (1) *Asset Pricing*: following Lucas (1978), we can consider an economy in which a set of identical agents trade claims on stochastic returns of a set of given assets (“trees”). Each agent solves a consumption smoothing problem similar to that in subsection 16.5.1, with the major difference that he or she can now save in assets with stochastic returns rather than (or in addition to) at a constant interest rate. Market clearing will be achieved when the total supply of assets is equal to total demand. This implies that in equilibrium the prices have to be such that each agent is happy to hold the appropriate amount of claims on the returns from these assets. Given the marginal utility of consumption derived from the recursive formulation, these assets can be priced. Exercise 16.14 considers this case.
- (2) *Investment under Uncertainty*: the model of investment under adjustment costs discussed in Section 7.8 of Chapter 7 has much wider application in macroeconomics and industrial organization once augmented by the possibility that firms are uncertain about future demand and/or productivity. Exercise 16.15 considers this case.
- (3) *Optimal Stopping Problems*: the search model discussed in the previous subsection is an example of an optimal stopping problem. More general optimal stopping problems can also be set up and analyzed as stochastic dynamic programming problems. Exercise 16.16 considers an example of such a stopping problem.

16.6. Taking Stock

The material in this section is technical in nature and is useful for its applications more than for its own sake. At the level at which it has been presented here, it has widespread applications in macroeconomics and economic growth. The stochastic neoclassical growth model, which we will see in the next chapter, makes heavy use of the methods developed here and is the workhorse model of modern macroeconomics.

In addition to presenting the basic tools of stochastic dynamic programming, this chapter has presented two important economic models. The first, the stochastic permanent income hypothesis model, is one of the most famous macroeconomic models and has led both to a large theoretical and empirical literature. The early empirical literature focused on excess sensitivity tests as discussed in subsection 16.5.1 using aggregate data. The more recent

literature focuses on micro and panel data in order to derive sharper results about the behavior of individual consumption.

The other substantial model introduced in this chapter is the search for ideas model in subsection 16.5.2, which is adapted from McCall's (1978) labor market search model. McCall's model is the basis of much of the modern equilibrium theory of unemployment. While the model here has been cast in terms of searching for ideas, the reader can easily adapt it to unemployment and use it as an introduction to equilibrium unemployment theory (see Exercise 16.13). In addition, some of the other applications, mentioned above and treated in exercises, including the asset pricing model based on Lucas (1978) and the model of investment under uncertainty, are widely used models in other areas of macroeconomics.

16.7. References and Literature

Most of the references from Chapter 6 are relevant for stochastic dynamic programming as well. The reader may want to look at Howard (1960), Blackwell (1965) and Puterman (1994), for advanced treatments. The most complete treatment of discounted stochastic dynamic programming problems with economic applications is in Stokey, Lucas and Prescott (1989). This chapter covered almost the same material as Stokey, Lucas and Prescott, though at a slightly less technical level. In particular, I presented all the major results of stochastic dynamic programming without introducing measure theory and general Markov processes. A thorough study of stochastic dynamic programming requires a nontrivial investment in these methods. The reader should consult Stokey, Lucas and Prescott (1989, Chapters 8-13), who present a more measure-theoretic approach and develop the necessary material on Markov processes.

The reader may also wish to consult Rudin (1976) or the very lively and readable treatment in Williams (1991) for some of the basic definitions and results in measure theory used in the discussion of Markov processes (though none of this material is not necessary for the main results presented in this book). These references also provide a formal definition of the Lebesgue integral, which I used informally a number of times throughout the text. A slightly more advanced but excellent treatment of measure theory is contained in Royden (1994). Williams (1991) also contains an excellent introductory treatment of martingales, which were referred to in Section 16.5.

Futia (1982) presents a compact and excellent treatment of Markov processes and their applications to stochastic dynamic models. More advanced and complete treatments of Markov processes can be found in Ethier and Kurtz (1985) or Gikhman and Skorohod (1974).

A more detailed and advanced treatment of the necessity and sufficiency of stochastic transversality conditions, stated and proved in Theorem, can be found in Zilcha (1978) and Kamaihigashi (2003).

The best survey of work on consumption is still Deaton (1991). A survey of recent work can be found in Browning and Crossley (2001). Exercise 16.11 is based on Chamberlain

and Wilson (2000) and the reader is referred to this paper for some of the subtle mathematical issues that arise in determining the limiting behavior of the stochastic consumption distribution when the discount factor is equal to the inverse of the gross interest rate. The search for ideas example in subsection 16.5.2 is adapted from McCall's (1978) labor market search model. Kortum (1994) provides the first search-theoretical model of technology choice that I am aware of. Kortum's model is significantly more advanced, but also more insightful than the model presented in subsection 16.5.2. Ljungqvist and Sargent (2005) contains an excellent exposition of the basic McCall model. Pissarides (2001) and Rogerson, Shimer and Wright (2004) provide excellent surveys of recent work in search theory applied to labor market problems.

16.8. Exercises

EXERCISE 16.1. Show that Assumption 16.6 (iii) is satisfied if and only if for any $j'' > j'$ and any $\bar{j} = 1, \dots, N$, $\sum_{j=\bar{j}}^N q_{jj''} \geq \sum_{j=\bar{j}}^N q_{jj'}$. What does this imply about the relationship between the conditional distribution of z given $z_{j''}$ and given $z_{j'}$?

EXERCISE 16.2. * Prove Lemma 16.1.

EXERCISE 16.3. * This exercise develops an alternative proof of Theorem 16.3.

- (1) Choose the appropriate topology on $\mathcal{Z}$ so that U is continuous on $X \times X \times \mathcal{Z}$.
- (2) Use Theorem A.12 in Appendix Chapter A to show that the objective function in Problem B1 is continuous in the product topology and Theorem A.13 and Lemma A.2 to show that the constraint set is compact, and use Theorems A.9 and A.16 to show that $V^*(x(0), z(0))$ is well-defined, continuous and bounded over $X \times \mathcal{Z}$.
- (3) Finally, deduce the same results for $V(x(0), z(0))$ by applying Theorem 16.1.

EXERCISE 16.4. * Prove Theorem 16.4.

EXERCISE 16.5. * Prove Theorem 16.5.

EXERCISE 16.6. * Prove Theorem 16.6.

EXERCISE 16.7. * Prove Theorem 16.7.

EXERCISE 16.8. Consider the stochastic permanent income hypothesis model studied in Section 16.5 and suppose that $u(c)$ is not quadratic. Explain the conditions under which the excess sensitivity tests described in that section would fail even when the stochastic Euler equation (16.23) holds. [Hint: you may want to consider the CRRA preferences for concreteness].

- EXERCISE 16.9.
- (1) Consider the stochastic permanent income hypothesis model studied in Section 16.5 and assume that the interest rate r is no longer constant, but is equal to $r(t) > 0$ at time t . Derive the equivalent of (16.23) in this case. Show that excess sensitivity tests can be applied in this case as well.
 - (2) Now suppose that $r(t)$ is a random variable taking one of finitely many values, $r_1, \dots, r_N$, and to simplify the analysis, suppose that the realizations of the interest rate are independent over time. Derive the equivalent of (16.23) in this case. Show that excess sensitivity tests can be applied in this case as well.

EXERCISE 16.10. Consider the stochastic permanent income hypothesis model studied in Section 16.5. Suppose that instead of being distributed independently, $w(t)$ follows a Markov chain. Show that (16.23) still holds. Now suppose that $u(c)$ takes a quadratic form and assume that the econometrician incorrectly believes that $w(t)$ is independently distributed, so that the individual has superior information relative to the econometrician. Show that a regression of consumption growth on past income realizations will still lead to a zero coefficient (thus the excess sensitivity test will not reject). [Hint: make use of the law of iterated expectations, which states that if Ω is an information set that is finer than Ω' and z is a random variable, then $\mathbb{E}[\mathbb{E}[z | \Omega] | \Omega'] = \mathbb{E}[z | \Omega']$.]

EXERCISE 16.11. In the stochastic permanent income hypothesis model studied in Section 16.5, suppose that $c(t) \geq 0$, $u(\cdot)$ is twice continuously differentiable, everywhere strictly concave and strictly increasing, and $u''(\cdot)$ is increasing. Suppose also that $w(t)$ has a non-degenerate probability distribution with lower support equal to 0.

- (1) Show that consumption can never converge to a constant level.
- (2) * Prove that if $u(\cdot)$ takes the CRRA form and $\beta < (1+r)^{-1}$, then there exists some $\bar{a} < \infty$ such that $a(t) \in (0, \bar{a})$ for all t .
- (3) * Prove that when $\beta \leq (1+r)^{-1}$, there does not necessarily exist $\bar{a} < \infty$ such that $a(t) \in (0, \bar{a})$ for all t . [Hint: first suppose $u(\cdot)$ takes the CARA form, consider the case where $\beta = (1+r)^{-1}$, and take the stochastic sequence where $w(t) = w_N$ for an arbitrarily large number of periods, which is a positive probability sequence. Then, generalize this argument to the case where $\beta \leq (1+r)^{-1}$.]
- (4) * Suppose that $u''(\cdot)$ is nondecreasing. Prove that when $\beta \leq (1+r)^{-1}$, marginal utility of consumption follows a (nondegenerate) supermartingale and therefore consumption must converge to infinity. [Hint: note that in this case (16.23) implies $u'(c(t)) \geq \mathbb{E}u'(c(t+1))$ and use this equation to argue that consumption must be increasing “on average”].
- (5) * How is the analysis in part 4 mortified if $u''(\cdot)$ is decreasing?

EXERCISE 16.12. Consider the model of searching for ideas introduced in subsection 16.5.2. Suppose that the entrepreneur can use any of the techniques he has discovered in the past to produce at any point in time. But he can also stop production at any point and go back to searching.

- (1) Prove that if the entrepreneur has turned down production at some technique a' at date t , he will never accept technique a' at date $t+s$, for $s > 0$ (i.e., he will not accept it for any possible realization of events between dates t and $t+s$).
- (2) Prove that if the entrepreneur accepts technique a' at date t , he will continue to produce with this technique for all dates $s \geq t$ rather than stopping production and going back to searching.
- (3) Using 1 and 2, show that the maximization problem of the entrepreneur can be formulated as in the text without loss of any generality.

- (4) Now suppose that when not producing, the entrepreneur receives income b . Write the recursive formulation for this case and show that as b increases, the cutoff threshold R increases.

EXERCISE 16.13. Formulate the problem in subsection 16.5.2 as one of an unemployed worker sampling wages from an exogenously given stationary wage distribution $H(w)$. The objective of the worker is to maximize the net present discounted value of his income stream. Assume that once the worker accepts a job he can work at that wage forever.

- (1) Formulate the dynamic maximization problem of the worker recursively assuming that once the worker finds a job he will never quit.
- (2) Prove that the worker will never quit a job that he has accepted.
- (3) Prove that the worker will use the reservation wage R for deciding what job to accept.
- (4) Calculate the expected duration of unemployment for the worker.
- (5) Show that if the wages in the wage distribution $H(w)$ are offered by firms and all workers are identical, the wage offers of all firms other than those offering $w = R$ are not profit-maximizing. What does this observation imply about the McCall search model?

EXERCISE 16.14. Consider an economy populated by identical households each with preferences given by $\mathbb{E} [\sum_{t=0}^{\infty} \beta^t u(c(t))]$, where $u(\cdot)$ is strictly increasing, strictly concave and twice continuously differentiable. Normalize the measure of agents in the economy to 1. Each household has a claim to a single tree, which delivers $z(t)$ units of consumption good at time t . Assume that $z(t)$ is a random variable taking values from the set $\mathcal{Z} \equiv \{z_1, \dots, z_N\}$ and is distributed according to a Markov chain (all trees have exactly the same output, so there is no gain in diversification). Each household can sell any fraction of its trees or buy fractions of new trees, though he cannot sell trees short (i.e., negative holdings are not allowed). Suppose that the price of a tree when the current realization of $z(t)$ is z is given by the function $p: \mathcal{Z} \rightarrow \mathbb{R}_+$. There are no other assets to transfer resources across periods.

- (1) Show that for a given price function $p(z)$, the flow budget constraint of a representative household can be written as

$$c(t) + p(z(t))x(t+1) \leq [z(t) + p(z(t))]x(t),$$

where $x(t)$ denotes the tree holdings of the household at time t . Interpret this constraint.

- (2) Show that for a given price function $p(z)$, the maximization problem of the representative household subject to the flow budget constraint and the constraint that $c(t) \geq 0$, $x(t) \geq 0$ can be written in a recursive form as follows

$$V(x, z) = \sup_{y \in [0, p(z)^{-1}(z+p(z))x]} \left\{ u((z + p(z))x - p(z)y) + \beta \mathbb{E} [V(y, z') \mid z] \right\}.$$

- (3) Use the results from Section 16.1 to show that $V(x, y)$ has a solution, is increasing in both of its arguments and strictly concave, and is differentiable in x in the interior of its domain.
- (4) Derive the stochastic Euler equations for this maximization problem.
- (5) Now impose market clearing, which implies that $x(t) = 1$ for all t . Explain why this condition is necessary and sufficient for market clearing.
- (6) Under market clearing, derive $p(z)$ the equilibrium prices of trees as a function of the current realization of z .

EXERCISE 16.15. Consider a discrete stochastic version of the investment model from Section 7.8, where a firm maximizes the net present discounted value of its profits, with discount factor given by $(1 + r)^{-1}$ and instantaneous returns given by

$$f(k(t), z(t)) - i(t) - \phi(i(t)).$$

Here $f(k(t), z(t))$ is the revenue or profit of the firm as a function of its capital stock, $k(t)$, and a stochastic variable, representing productivity or demand, $z(t)$. As in Section 7.8, $i(t)$ is investment and $\phi(i(t))$ represents adjustment costs.

- (1) Assume that $z(t)$ has a distribution represented by a Markov chain. Formulate the sequence version of the maximization problem of the firm.
- (2) Formulate the recursive version of the maximization problem of the firm.
- (3) Provide conditions under which the two problems have the same solutions.
- (4) Derive the stochastic Euler equation for the investment decision of the firm and compare the results to those in Section 7.8.

EXERCISE 16.16. Consider a general stopping problem, where the objective of the individual is to maximize $\mathbb{E} [\sum_{t=0}^{\infty} \beta^t u(y(t))]$. Let us assume that the individual faces a stream of random variables represented by $z(t)$ and assume that $z(t)$ follows a Markov chain. At any t , the individual can “stop” the process. Let $y(t) = 0$ while the individual has not stopped and $y(t) = z(s)$ if the individual has stopped the process at some $s \leq t$.

- (1) Formulate the problem of the individual as a stochastic dynamic programming problem and show that there exists some R^* such that the individual will stop the process at time t if $z(t) \geq R^*$.
- (2) Now assume that $z(t)$ has a distribution at time t given by $H(z | \zeta(t))$ and $\zeta(t)$ follows a Markov chain with values in the finite set $\mathcal{Z}$. Formulate the problem of the individual as a stochastic dynamic programming problem. Prove that there exists a function $R^* : \mathcal{Z} \rightarrow \mathbb{R}_+$ such that the individual will stop the process when $z(t) \geq R^*(\zeta(t))$ when the current state is $\zeta(t)$. Explain why the stopping rule is no longer constant. What does this result imply for the job acceptance decisions of unemployed workers studied in Exercise 16.13 when the distribution of wages is different during periods of recession?

CHAPTER 17

Stochastic Growth Models

In this chapter, I present four models of stochastic growth emphasizing different aspects of the interaction between growth and uncertainty. The first is the baseline neoclassical growth model (with complete markets) augmented with stochastic productivity shocks, first studied by Brock and Mirman (1972). This model is not only an important generalization of the baseline neoclassical growth of Chapter 8, but also provides the starting point of the influential *Real Business Cycle* models, which are used extensively for the study of a range of short- and medium-run macroeconomics questions. I present this model and some of its implications in the next three sections. The baseline neoclassical growth model incorporates complete markets in the sense that households and firms can trade using any Arrow-Debreu commodity. In the presence of uncertainty, this implies that a full set of *contingent claims* is traded competitively. For example, an individual can buy an asset that will pay one unit of the final good after a pre-specified history. The presence of complete markets—or the full set of contingent claims—implies that individuals can fully insure themselves against idiosyncratic risks. The source of interesting uncertainty in these models is aggregate shocks. For this reason, the standard neoclassical growth model under uncertainty does not even introduce idiosyncratic shocks (had they been present, they could have been easily diversified away).

This discussion shows the importance of contingent claims in the basic neoclassical model under uncertainty. Moreover, trading in contingent claims is not only sufficient, but it is essentially also necessary for the representative household assumption to hold in environments with uncertainty. This is illustrated in Section 17.4, which considers a model where households cannot use contingent claims and can only trade in riskless bonds. This model, which builds on Bewley’s seminal work in the 1970s and the 1980s, explicitly prevents risk-sharing across households and thus features “incomplete markets”—in particular, one of the most relevant type of market incompleteness for macroeconomic questions, which prevents the sharing or diversification of idiosyncratic risk. Households face a stochastic stream of labor income and can only achieve consumption smoothing via “self-insurance,” that is, by borrowing and lending at a market interest rate. Like the overlapping generations model of Chapter 9, the Bewley model does *not* admit a representative household. The Bewley model is not only important in illustrating the role of contingent claims in models under uncertainty, but also because it is a tractable model for the study of a range of macroeconomic questions related to risk, income fluctuations and policy. Consequently, over the past decade or so, it has become a workhorse model for macroeconomic analysis.

The last two sections, Sections 17.5 and 17.6, turn to stochastic overlapping generations models. The first presents a simple extension of the canonical overlapping generations model that includes stochastic elements.

Section 17.6 shows how stochastic growth models can be useful in understanding the process of takeoff from low growth into sustained growth, which was already discussed in Chapter 1. A notable feature of the long-run experience of many societies is that the early stages of economic development were characterized by slow or no growth in income per capita and by frequent economic crises. The process of takeoff not only led to faster growth, but also to a more steady (less variable) growth process. An investigation of these issues requires a model of stochastic growth. Section 17.6 presents a model that provides a unified framework for the analysis of the variability of economic performance and takeoff. The key feature is the tradeoff between investment in risky activities and safer activities with lower returns. At the early stages of development, societies do not have enough resources to invest in sufficiently many activities to achieve diversification and are thus forced to bear considerable risk. As a way of reducing this risk, they also invest in low-return safe activities, such as a storage or safe technology and low-yield agricultural products. The result is an equilibrium process that features a lengthy period of slow or no growth associated with high levels of variability in economic performance. The growth is truly stochastic and an economy can escape this stage of development and takeoff into sustained growth only when its risky investments are successful for a number of consecutive periods. When this happens, the economy achieves better diversification and also better risk management through more developed financial markets. Better diversification reduces risk and also enables the economy to channel its investments in higher return activities, increasing its productivity and growth rate. Thus this simple model of stochastic growth presents a stylistic account of the process of takeoff from low and variable growth and to sustained and steady growth. The model I will use to illustrate these ideas features both a simple form of stochastic growth and also endogenously incomplete markets. I will therefore use this model to show how some simple ideas from Markov processes can be used to characterize the stochastic equilibrium path of a dynamic economy and also to highlight potential inefficiencies resulting from models with endogenous incomplete markets. Finally, this model will give us a first glimpse of the relationship between financial development and economic growth, a topic that will be discussed more extensively in Chapter 20.

17.1. The Brock-Mirman Model

The first systematic analysis of economic growth with stochastic shocks was undertaken by Brock and Mirman in their 1972 paper. Brock and Mirman focused on the optimal growth problem and solved for the social planner's maximization problem in a dynamic neoclassical environment with uncertainty. Since, with competitive and complete markets, the First and Second Welfare Theorems still hold, the equilibrium growth path is identical to the optimal growth path. Nevertheless, the analysis of equilibrium growth is more involved and also

introduces a number of new concepts. I start with the Brock-Mirman approach and then discuss competitive equilibrium growth under uncertainty in the next section.

The economy is similar to the baseline neoclassical growth model studied in Chapters 6 and 8. It is in discrete time and the aggregate production function is now given by

$$(17.1) \quad Y(t) = F(K(t), L(t), z(t)),$$

where $z(t)$ denotes a stochastic *aggregate* productivity term affecting how productive a given combination of capital and labor will be in producing the unique final good of the economy. Let us suppose that $z(t)$ follows a Markov chain with values in the set $\mathcal{Z} \equiv \{z_1, \dots, z_N\}$. Many applications of the neoclassical growth model under uncertainty also assume that the stochastic shock is a labor-augmenting productivity term, so that the aggregate production function takes the form $Y(t) = F(K(t), z(t)L(t))$, though for the analysis here, we do not need to impose this additional restriction. Suppose that the production function F satisfies Assumptions 1 and 2, and define per capita output and the per capita production function as

$$\begin{aligned} y(t) &\equiv \frac{Y(t)}{L(t)} \\ &\equiv f(k(t), z(t)), \end{aligned}$$

with $k(t) \equiv K(t)/L(t)$ once again corresponding to the capital-labor ratio. A fraction δ of the existing capital stock depreciates at each date. Finally, suppose also that the numbers $z_1, \dots, z_N$ are arranged in ascending order and that $j > j'$ implies $f(k, z_j) > f(k, z_{j'})$ for all $k \in \mathbb{R}_+$. This assumption implies that higher values of the stochastic shock z correspond to greater productivity at *all* capital-labor ratios. In addition, let us assume that $z(t)$ follows a monotone Markov chain (as defined in Assumption 16.6), so that a higher value of z today makes higher values in the future more likely.

On the preference side, the economy admits a representative household with instantaneous utility function $u(c)$ that satisfies the standard assumptions laid out in Assumption 3 in Chapter 8. The representative household supplies one unit of labor inelastically, so that $K(t)$ and $k(t)$ can be used interchangeably (and there is no reason to distinguish total consumption $C(t)$ from per capita consumption $c(t)$). Finally, consumption and saving decisions at time t are made after observing the realization of the stochastic shock for time t , $z(t)$.

The sequence version of the expected utility maximization problem of a social planner in this economy can be written as

$$(17.2) \quad \max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c(t))$$

subject to

$$(17.3) \quad k(t+1) = f(k(t), z(t)) + (1 - \delta)k(t) - c(t) \text{ and } k(t) \geq 0,$$

with given $k(0) > 0$. The resource constraint (17.3) must hold that each state and for each history of the stochastic shock, z_t (and I have not yet introduced the conditioning on the history of these shocks to keep the formulation of the initial problem simple).

To characterize the optimal growth path using the sequence problem we would need to define feasible plans, in particular, the mappings $\tilde{k}[z^t]$ and $\tilde{c}[z^t]$ introduced in the previous chapter, with $z^t \equiv (z(0), \dots, z(t))$ again standing for the history of (aggregate) shocks up to date t . Rather than going through these steps again, let us directly look at the recursive version of this program, which can be written as

$$(17.4) \quad V(k, z) = \max_{k' \in [0, f(k, z) + (1-\delta)k]} \{u(f(k, z) + (1-\delta)k - k') + \beta \mathbb{E}[V(k', z') | z]\},$$

where I used “max” rather than “sup”, since this maximization problem does have a solution. In particular, the main theorems from the previous chapter immediately apply to this problem and yield the following result:

PROPOSITION 17.1. *In the stochastic optimal growth problem described above, the value function $V(k, z)$ is uniquely defined, strictly increasing in both of its arguments, strictly concave in k and differentiable in $k > 0$. Moreover, there exists a uniquely defined policy function $\pi(k, z)$ such that the capital stock at date $t+1$ is given by $k(t+1) = \pi(k(t), z(t))$.*

PROOF. The proof simply involves verifying that Assumptions 16.1-16.6 from the previous chapter are satisfied, so that Theorems 16.1-16.7 can be applied. To do this, first define $\bar{k}$ such that $\bar{k} = f(\bar{k}, z_N) + (1-\delta)\bar{k}$, and show that starting with $k(0)$, the capital-labor ratio will always remain within the compact set $[0, \max\{k(0), \bar{k}\}]$. $\square$

In addition, the following proposition can also be established.

PROPOSITION 17.2. *In the stochastic optimal growth problem described above, the policy function for next period’s capital stock, $\pi(k, z)$, is strictly increasing in both of its arguments.*

PROOF. From Assumption 3, u is differentiable and from Proposition 17.1 V is differentiable in k . Moreover, by the same argument as in the proof of Proposition 17.1, $k \in (0, \bar{k})$, so we are in the interior of the domain of the objective function. Thus, the value function V is differentiable in its first argument, and

$$u'(f(k, z) + (1-\delta)k - k') - \beta \mathbb{E}[V'(k', z') | z] = 0,$$

where V' denotes the derivative of the $V(k, z)$ function with respect to its first argument. Since from Proposition 17.1 V is strictly concave in k , this equation can hold when the level of k or z increases only if k' also increases. For example, an increase in k reduces the first-term (because u is strictly concave), hence an increase in k' is necessary to increase the first term and to reduce the second term (by the concavity of V). The argument for the implications of an increase in z is similar. $\square$

It is also straightforward to derive the stochastic Euler equations corresponding to the neoclassical growth model with uncertainty. For this purpose, let us first define the policy

function for consumption as

$$\pi^c(k, z) \equiv f(k, z) + (1 - \delta)k - \pi(k, z),$$

where $\pi(k, z)$ is the optimal policy function for next date's capital stock determined in Proposition 17.1. Using this notation, the stochastic Euler equation can be written as

$$(17.5) \quad u'(\pi^c(k, z)) = \beta \mathbb{E} \left[(f'(\pi(k, z), z') + (1 - \delta)) u'(\pi^c(\pi(k, z), z')) \mid z \right],$$

where f' denotes the derivative of the per capita production function with respect to the capital-labor ratio, k . In this form, the Euler equation looks complicated. A slightly different way of expressing this equation makes it both simpler and more intuitive:

$$(17.6) \quad u'(c(t)) = \beta \mathbb{E}_t [p(t+1) u'(c(t+1))],$$

where $\mathbb{E}_t$ denotes the expectation conditional on information available at time t and $p(t+1)$ is the stochastic marginal product of capital (including undepreciated capital) at date $t+1$. This form of writing the stochastic Euler equation is also useful for comparison with the competitive equilibrium because $p(t+1)$ corresponds to the stochastic (date $t+1$) dividends paid out by one unit of capital invested at time t . Finally, we can also write the transversality condition associated with the optimal plan as

$$(17.7) \quad \lim_{t \rightarrow \infty} \mathbb{E} [\beta^t (f'(k(t), z(t)) + 1 - \delta) u'(c(t)) k(t) \mid z(0)] = 0$$

given $z(0) \in \mathcal{Z}$, where for notational simplicity I have again used $c(t) = \pi^c(k(t), z(t))$ and $k(t) = \pi(k(t-1), z(t-1))$. It is straightforward to verify that Theorem 16.8 applies to this environment and implies that eq.'s (17.6) and (17.7) are sufficient to characterize the solution to the optimal growth problem specified here.

Although Proposition 17.1 characterizes the form of the value function and policy functions, it has two shortcomings. First, it does not provide us with an analog of the “Turnpike Theorem” of the nonstochastic neoclassical growth model. In particular, it does not characterize the long-run behavior of the neoclassical growth model under uncertainty. Second, while the characterization provides a number of qualitative results about the value and the policy functions, it does not deliver comparative static results.

A full analysis of the long-run behavior of the stochastic growth model would take us too far afield into the analysis of Markov processes. Nevertheless, a few simple observations are useful to appreciate the salient features of the stochastic law of motion of the capital-labor ratio in this model. The capital stock at date $t+1$ is given by the policy function π , thus

$$(17.8) \quad k(t+1) = \pi(k(t), z(t)),$$

which defines a general Markov process, since before the realization of $z(t)$, $k(t+1)$ is a random variable, with its law of motion governed by the last period's value of $k(t)$ and the realization of $z(t)$. If $z(t)$ has a non-degenerate distribution, $k(t)$ does not typically converge to a single value (see Exercise 17.4). Instead, we may hope that it will converge to an *invariant limiting distribution*. It can indeed be verified that this is the case. The Markov process (17.8) defines a sufficiently well-behaved stochastic process that starting with any $k(0)$, it converges

to a unique invariant limiting distribution, meaning that when we look at sufficiently faraway horizons, the distribution of k should be independent of $k(0)$. Moreover, the average value of $k(t)$ in this invariant limiting distribution will be the same as the time average of $\{k(t)\}_{t=0}^T$ as $T \rightarrow \infty$ (so that the stochastic process for the capital stock is “ergodic”). Consequently, a “steady-state” equilibrium now corresponds not to specific values of the capital-labor ratio and output per capita but to an invariant limiting distribution. If the stochastic variable $z(t)$ takes values within a sufficiently small set, this limiting invariant distribution would hover around some particular value, which we may wish to refer to as a *quasi-steady-state* value of the capital-labor ratio, because even though the equilibrium capital-labor ratio may not converge to this value, it will have a tendency to return to a neighborhood thereof. But in general the range of the limiting distribution could be quite wide.

To obtain a better understanding of the behavior of the neoclassical growth model under uncertainty, I next provide a simple example, which allows us to obtain a closed-form solution for the policy function π .

EXAMPLE 17.1. Suppose that $u(c) = \log c$, $F(K, L, z) = zK^\alpha L^{1-\alpha}$ and $\delta = 1$. The stochastic shock z again follows a Markov chain over the set $\mathcal{Z} \equiv \{z_1, \dots, z_N\}$, with transition probabilities denoted by $q_{jj'}$. Let $k \equiv K/L$. The stochastic Euler equation (17.5) implies

$$(17.9) \quad \frac{1}{zk^\alpha - \pi(k, z)} = \beta \mathbb{E} \left[\frac{\alpha z' \pi(k, z)^{\alpha-1}}{z' \pi(k, z)^\alpha - \pi(\pi(k, z), z')} \middle| z \right],$$

which is a relatively simple functional equation in a single function $\pi(\cdot, \cdot)$. Though simple, this functional equation would still be difficult to solve unless we had some idea about what the solution looked like. Here, fortunately, the method of “guessing and verifying” the solution of the functional equation becomes handy. Let us conjecture that

$$\pi(k, z) = B_0 + B_1 z k^\alpha.$$

Substituting this guess into (17.9),

$$(17.10) \quad \frac{1}{(1 - B_1) z k^\alpha - B_0} = \beta \mathbb{E} \left[\frac{\alpha z' (B_0 + B_1 z k^\alpha)^{\alpha-1}}{z' (B_0 + B_1 z k^\alpha)^\alpha - B_0 - B_1 z' (B_0 + B_1 z k^\alpha)^\alpha} \middle| z \right].$$

It is straightforward to check that this equation cannot be satisfied for any $B_0 \neq 0$ (see Exercise 17.5). Thus imposing $B_0 = 0$ and writing out the expectation explicitly with $z = z_{j'}$, this expression becomes

$$\frac{1}{(1 - B_1) z_{j'} k^\alpha} = \beta \sum_{j=1}^N q_{jj'} \frac{\alpha z_j (B_1 z_{j'} k^\alpha)^{\alpha-1}}{z_j (B_1 z_{j'} k^\alpha)^\alpha - B_1 z_j (B_1 z_{j'} k^\alpha)^\alpha}.$$

Simplifying each term within the summation,

$$\frac{1}{(1 - B_1) z_{j'} k^\alpha} = \beta \sum_{j=1}^N q_{jj'} \frac{\alpha}{B_1 (1 - B_1) z_{j'} k^\alpha}.$$

Now taking $z_{j'}$ and k out of the summation and using the fact that, by definition, $\sum_{j=1}^N q_{jj'} = 1$, we can cancel the remaining terms and obtain

$$B_1 = \alpha\beta,$$

so that regardless of the exact Markov chain for z , the optimal policy rule is

$$\pi(k, z) = \alpha\beta z k^\alpha.$$

The reader can verify that this is identical to the result in Example 6.4 in Chapter 6, with z there corresponding to a nonstochastic productivity term. Consequently, in this case the stochastic elements have not changed the form of the optimal policy function. Exercise 17.6 shows that the same result applies when z follows a general Markov process rather than a Markov chain.

Using this example, we can fully analyze the stochastic behavior of the capital-labor ratio and output per capita. In fact, the stochastic behavior of the capital-labor ratio in this economy is identical to that of the overlapping generations model analyzed in Section 17.5 and Figure 17.1 in that section applies exactly to this example. A more detailed discussion of these issues is left to Exercise 17.7. Unfortunately, Example 17.1 is one of the few instances where the neoclassical growth model admits closed-form solutions. In particular, if the depreciation rate of the capital stock δ is not equal to 1, the neoclassical growth model under uncertainty does not admit an explicit form characterization (see Exercise 17.8).

17.2. Equilibrium Growth under Uncertainty

Let us now consider the competitive equilibria of the neoclassical growth model under uncertainty. The environment is identical to that in the previous section and z corresponds to an aggregate productivity shock affecting all production units. We continue to assume that z follows a Markov chain. Defining the Arrow-Debreu commodities in the standard way, so that goods indexed by different realizations of the history z^t correspond to different commodities, this is an economy with a countable infinity of commodities. The Second Welfare Theorem, Theorem 5.7 from Chapter 5, applies and implies that the optimal growth path characterized in the previous section can be decentralized as a competitive equilibrium (see Exercise 17.10). Moreover, since we are focusing on an economy with a representative household, this allocation is a competitive equilibrium without any redistribution of endowments. These observations justify the frequent focus on social planner's problems in analyses of stochastic growth models in the literature.

Here I will briefly discuss the explicit characterization of competitive equilibria of this economy both to show the equivalence between the optimal growth problem more explicitly and the equilibrium growth problem under complete markets and also to introduce a number of important ideas related to the pricing of various contingent claims in competitive equilibrium under uncertainty. The complete markets assumption in this context implies that, in principle, any commodity—including any contingent claim—can be traded competitively.

Nevertheless, as shown by our analysis in Section 5.8 in Chapter 5, in practice there is no need to specify or trade all of these commodities and a subset of the available commodities is sufficient to provide all the necessary trading opportunities to households and firms. The analysis in this section will also show which subsets of commodities or contingent claims are typically sufficient to ensure an equilibrium with complete markets. In particular, I first present the characterization of the competitive equilibrium under uncertainty when the full set of commodities are traded (and all trades take place at time $t = 0$). I will then show how an equivalent characterization of the competitive equilibrium can be obtained with sequential trading and with the help of a smaller set of contingent claims, *the Arrow securities* (recall Section 5.8 in Chapter 5). In both formulations, the key step in the characterization of the equilibrium is the formulation of the appropriate market clearing conditions and the resulting no-arbitrage conditions.

17.2.1. Competitive Equilibrium with Full Set of Commodities. Preferences and technology are as in the previous section. Recall that the economy admits a representative household and that the production side of the economy can be represented by a representative firm (Theorem 5.4). Let us first consider the problem of the representative household. This household will maximize the objective function given by (17.2) subject to the lifetime budget constraint (written from the viewpoint of time $t = 0$).

To write the lifetime budget constraint of the household, let $\mathcal{Z}^t$ be the set of all possible histories of the stochastic variable z^t up to date t and $\mathcal{Z}^\infty$ be the set of infinite histories. With a slight abuse of notation, I will write $z^t \in \mathcal{Z}^\infty$ to denote a possible history of length t . For any z^t , let $p_0[z^t]$ be the price of the unique final good at time t in terms of the final good of date 0 following a history z^t , $c[z^t]$ be the time t consumption of the household following history z^t , and $w_0[z^t]$ be the wage rate and thus total labor earnings of the household, in terms of the final good dated 0 following history z^t . Using this notation, the household's lifetime budget constraint can be written as

$$(17.11) \quad \sum_{t=0}^{\infty} \sum_{z^t \in \mathcal{Z}^\infty} p_0[z^t] c[z^t] \leq \sum_{t=0}^{\infty} \sum_{z^t \in \mathcal{Z}^\infty} w_0[z^t] + k(0).$$

A number of features about this lifetime budget constraint are worth noting.¹ First and most importantly, there are no expectations. This is because this is an economy with complete markets, which implies that the household is making all of his (lifetime) trades in the initial period of the economy $t = 0$ at a well-defined price vector for all Arrow-Debreu commodities. Consequently, the lifetime budget constraint applies in exactly the same way as a static budget constraint in the standard theory of general equilibrium. More explicitly, the household buys claims to different “contingent” consumption bundles. These bundles are contingent in the sense that they are conditioned on the history of the aggregate state

¹Here $c[z^t]$ can be interpreted as a policy mapping from possible histories of the stochastic variable to consumption levels, which was defined as $\tilde{c}[z^t]$ in the previous chapter. I use the simpler expression $c[z^t]$ in this chapter both to simplify notation and also to emphasize the slightly different interpretation of this object in the present context as “contingent claims” on consumption after history z^t placed at date $t = 0$.

variable (stochastic shock) z^t and thus whether they are realized and delivered depends on the realization of the sequence of the stochastic shock. For example, $c[z^t]$ denotes units of final good allocated to consumption at time t if history z^t is realized. If a different history is realized, then this claim will not be exercised. This way of writing the lifetime budget constraint reiterates the importance of thinking in terms of Arrow-Debreu commodities.

Second, with this interpretation the left-hand side is simply the total expenditure of the individual taking the prices of all possible claims, i.e., the entire set of $p_0[z^t]$ s, as given. The right-hand side has a similar interpretation, except that it denotes the labor earnings of the household rather than his expenditures. The last term on the right-hand side is the value of the initial capital stock per capita, which is part of the household's initial wealth. As noted above, the price of the final good at date $t = 0$ is normalized to 1. As in the standard neoclassical growth model, capital is in terms of the final good, thus has a price of 1 as well.²

Finally, the right-hand side of (17.11) could also include profits accruing to the individuals (as in Definition 5.1 in Chapter 5). The fact that the aggregate production function exhibits constant returns to scale combined with the presence of competitive markets implies that equilibrium profits will be equal to 0. This enables us to omit the additional term for profits in the representative household's budget constraint without loss of any generality.

The objective function of the household at time $t = 0$ can also be written somewhat more explicitly than (17.2) as follows:

$$(17.12) \quad \sum_{t=0}^{\infty} \beta^t \sum_{z^t \in \mathcal{Z}^{\infty}} q[z^t | z^0] u(c[z^t]),$$

where $q[z^t | z^0]$ is the probability at time 0 that the history z^t will be realized at time t . I have written this in the form of a conditional probability to create continuity between the models that assume all trades take place at date $t = 0$ and the models with sequential trading. Notice that there is no longer the expectations operator in this objective function. Instead, the explicit summation over all possible events weighted by their probabilities has been introduced.³

For the characterization of the competitive equilibrium from the viewpoint of trading at time $t = 0$, it is most convenient to consider the maximization of (17.12) subject to (17.11)—rather than specifying this problem recursively, which will be the approach adopted in the next subsection. Assuming that an interior solution exists, the first-order conditions of this

²Recall that the initial value of the aggregate stochastic variable $z(0)$ is also taken as given. This is important for enabling us to normalize the price of the initial period capital to 1. As we will see shortly there will be a different price sequence for capital purchases conditional on the realization of the current stochastic variable. Had $z(0)$ not been known, the price of initial capital stock would also have to be conditioned on the value of $z(0)$.

³In fact, more generally we could think of the preferences of the representative household as defined over the entire set of commodities, i.e., as a functional $U(c[z^t]_{z^t \in \mathcal{Z}^{\infty}})$. This emphasizes that the household is maximizing its utility defined over different commodities, which here correspond to consumption goods in different dates and in different states. Equation (17.12) exploits—and emphasizes—the fact that the household has preferences that are additively separable over these different commodities.

problem is

$$(17.13) \quad \beta^t q [z^t | z^0] u' (c [z^t]) = \lambda p_0 [z^t]$$

for all t and all z^t , where λ is the Lagrange multiplier on (17.11) and corresponds to the marginal utility of income at date $t = 0$ (see Exercise 17.11 on why a single multiplier for the lifetime budget constraint is sufficient in this case). Combining this first-order condition for two different date t histories z^t and $\hat{z}^t$,

$$\frac{u' (c [\hat{z}^t])}{u' (c [z^t])} = \frac{p_0 [\hat{z}^t] / q [\hat{z}^t | z^0]}{p_0 [z^t] / q [z^t | z^0]},$$

which shows that the right-hand side is the relative price of consumption claims conditional on histories z^t and $\hat{z}^t$. Combining this first-order condition for histories z^t and z^{t+1} such that $z^{t+1} = (z^t, z(t+1))$, we obtain

$$(17.14) \quad \frac{\beta u' (c [z^{t+1}])}{u' (c [z^t])} = \frac{p_0 [z^{t+1}] / q [z^{t+1} | z^0]}{p_0 [z^t] / q [z^t | z^0]},$$

so that the right-hand side now corresponds to the contingent interest rate between date t and $t + 1$ conditional on z^t (and contingent on the realization of z^{t+1}). While these expressions are intuitive, they cannot be used to characterize equilibrium consumption or investment sequences until we know more about the prices $p_0 [z^t]$. We will be able to derive these prices from the profit maximization problem of the representative firm.

Let us consider the value of the firm at date $t = 0$. To do this, let us define one more price sequence, $R_0 [z^t]$ corresponding to the price of one unit of capital after the state has been revealed as z^t and also denote the capital and labor employment levels of the representative firm after history z^t by $K^e [z^t]$ and $L [z^t]$. Two points are noteworthy here. First, $R_0 [\cdot]$ here refers to the *price* of capital goods, not the rental price (whereas in the deterministic growth models R was the rental price of capital). Second, I have introduced the additional superscript “ e ” for the capital to distinguish the capital employed by the firm after history z^t from the capital that is saved by the households after history z^t .

The value of the firm can then be written as

$$\sum_{t=0}^{\infty} \beta^t \sum_{z^t \in \mathcal{Z}^{\infty}} \{ p_0 [z^t] (F (K^e [z^t], L [z^t], z(t)) + (1 - \delta) K^e [z^t]) - R_0 [z^t] K^e [z^t] - w_0 [z^t] L [z^t] \},$$

where recall that $R_0 [z^t]$ is the price of capital after history z^t and $w_0 [z^t]$ is the wage rate conditional on history z^t . Profit maximization by the firm implies

$$\begin{aligned} p_0 [z^t] \left(\frac{\partial F (K^e [z^t], L [z^t], z(t))}{\partial K^e} + (1 - \delta) \right) &= R_0 [z^t] \\ p_0 [z^t] \frac{\partial F (K^e [z^t], L [z^t], z(t))}{\partial L} &= w_0 [z^t]. \end{aligned}$$

Using constant returns to scale and expressing everything in per capita terms, these first-order conditions can be written as

$$(17.15) \quad \begin{aligned} p_0 [z^t] (f' (k^e [z^t], z(t)) + (1 - \delta)) &= R_0 [z^t] \\ p_0 [z^t] (f (k^e [z^t], z(t)) - k^e [z^t] f' (k^e [z^t], z(t))) &= w_0 [z^t], \end{aligned}$$

where f' denotes the derivative of the per capita production function with respect to the capital-labor ratio, $k^e \equiv K^e/L$. The first equation relates the price of the final good to the price of capital goods and to the marginal productivity of capital, while the second equation determines the wage rate in terms of the price of the final good and the marginal (physical) product of labor. Equation (17.15) can also be interpreted as stating that the price of a unit of capital good after history z^t , $R_0 [z^t]$, is equal to the value of the dividends paid out by this unit of capital inclusive of undepreciated capital, that is, the price of the final good, $p_0 [z^t]$, times the marginal product of capital $f' (k^e [z^t], z(t))$ plus the $(1 - \delta)$ fraction of the capital that is not depreciated and paid back to the holder of the capital good in terms of date $t + 1$ final good. An alternative way of formulating the competitive equilibrium and writing (17.15) is to assume that capital goods are *rented*—not purchased—by firms, thus introducing a rental price sequence for capital goods. Exercise 17.12 shows that this alternative formulation leads to identical results. This is not surprising because, with complete markets, buying one unit of capital today and selling contingent claims on $1 - \delta$ units of capital tomorrow is equivalent to renting. Whether one uses the formulation in which capital goods are purchased or rented by firms is then just a matter of convenience and emphasis.

The key step in the characterization of a competitive equilibrium is the specification of the set of market clearing conditions. For labor, this is straightforward and requires

$$(17.16) \quad L [z^t] = 1 \text{ for all } z^t.$$

To write the market clearing condition for capital, recall that per capita production after history z^t is given by $f (k^e [z^t], z(t)) + (1 - \delta) k^e [z^t]$, and this is divided between consumption $c [z^t]$ and savings $s [z^t]$. The capital used at time $t + 1$ (after history z^{t+1}) must be equal to $s [z^t]$, since this is the amount of capital available at the beginning of date $t + 1$. Savings $s [z^t]$ is therefore the equivalent of capital stock choice of the planner for the next period, $k [z^{t+1}]$, in terms of the terminology in the previous subsection. Market clearing for capital implies that for any $z^{t+1} = (z^t, z(t + 1))$,

$$(17.17) \quad k^e [z^{t+1}] = s [z^t],$$

because the amount of available capital at time t is fixed regardless of the realization of $z(t + 1)$. The capital market clearing condition can then be written as

$$(17.18) \quad c [z^t] + s [z^t] \leq f (s [z^{t-1}], z(t)) + (1 - \delta) s [z^{t-1}]$$

for any $z^{t+1} = (z^t, z(t + 1))$.

The *no arbitrage* conditions that are essential in the characterization of the competitive equilibrium, which will link the price of capital conditional on z^{t+1} ($R_0 [z^{t+1}]$) to the price of the final good at time t ($p_0 [z^t]$), are then directly implied by the capital market clearing

conditions. In particular, consider the following *riskless arbitrage*; the household buys one unit of the final good after history z^t and saves it to be used as capital at time $t + 1$. It simultaneously sells claims on capital goods for each $z^{t+1} = (z^t, z(t+1))$. These combined transactions carry no risk, since the one unit of the final good bought after history z^t will cover the obligation to pay one unit of capital good after any history $z^{t+1} = (z^t, z(t+1))$. Consequently, this transaction should not make or lose money, which implies the no arbitrage condition

$$(17.19) \quad p_0 [z^t] = \sum_{z(t+1) \in \mathcal{Z}} R_0 [(z^t, z(t+1))].$$

A competitive equilibrium is defined in a standard manner as feasible policies determining consumption and capital levels, $\{c[z^t], s[z^t], k^e[z^{t+1}]\}_{z^t \in \mathcal{Z}^t}$, and price sequences, $\{p_0[z^t], R_0[z^t], w_0[z^t]\}_{z^t \in \mathcal{Z}^t}$, such that households maximize utility (i.e., satisfy (17.13)), firms maximize profits (i.e., satisfy (17.15) and (17.19)), and labor and capital markets clear (i.e., (17.16), (17.17), and (17.18) are satisfied).

To characterize the equilibrium path, let us substitute from (17.15) and (17.19) into the first-order condition for consumption given by (17.13) and rearrange to obtain

$$(17.20) \quad u'(c[z^t]) = \sum_{z(t+1) \in \mathcal{Z}} \frac{\lambda p_0 [z^{t+1}]}{\beta^t q[z^t | z^0]} (f'(k[z^{t+1}], z(t+1)) + (1 - \delta)).$$

Next using (17.13) for $t + 1$, we also have

$$\begin{aligned} \beta u'(c[z^{t+1}]) &= \frac{\lambda p_0 [z^{t+1}]}{\beta^t q[z^{t+1} | z^0]} \\ &= \frac{\lambda p_0 [z^{t+1}]}{\beta^t q[z^{t+1} | z^t] q[z^t | z^0]}, \end{aligned}$$

where the second line simply uses the fact that, by the law of iterated expectations, $q[z^{t+1} | z^0] \equiv q[z^{t+1} | z^t] q[z^t | z^0]$. Substituting this into (17.20),

$$\begin{aligned} u'(c[z^t]) &= \beta \sum_{z(t+1) \in \mathcal{Z}} q[z^{t+1} | z^t] (f'(k[z^{t+1}], z(t+1)) + (1 - \delta)) u'(c[z^{t+1}]) \\ (17.21) \quad &= \beta \mathbb{E}[(f'(k[z^{t+1}], z(t+1)) + (1 - \delta)) u'(c[z^{t+1}]) | z^t], \end{aligned}$$

which is identical to (17.6). Since from Theorem 16.8, the stochastic Euler equation, (17.6), and the transversality condition (17.7) are sufficient for optimal growth, the equivalence between optimal and equilibrium growth under uncertainty follows by observing that the lifetime budget constraint, (17.11), and the transversality condition of the representative household imply (17.7). In particular, with an argument similar to that in Section 8.6, we can show that the representative household's optimization problem will ensure that condition (17.7) is satisfied. To establish this claim, let us use (17.21) and the fact that $q[z^t | z^s] = q[z^t | z^{t-1}] q[z^{t-1} | z^s]$ (from the law of iterated expectations) to rewrite (17.7) as

$$\lim_{t \rightarrow \infty} \left[\beta^{t-1} \sum_{z^t \in \mathcal{Z}^t} q[z^{t-1} | z^s] u'(c[z^{t-1}]) k[z^t] \right] = 0.$$

Now using (17.14) between states z^t and z^s , this is equivalent to

$$\lim_{t \rightarrow \infty} \left[\frac{\beta^s u'(c[z^s])}{p_0[z^s]} \sum_{z^t \in \mathcal{Z}^t} p_0[z^{t-1}] k[z^t] \right] = 0,$$

where $u'(c[z^s])$ and $p_0[z^s]$ are taken out of the summation operator, since the summation is conditional on z^s . Finally, substituting from (17.13) and noting that $k[z^t]$ is equal to the asset holdings of the representative household after history z^t , $a[z^t]$,

$$\lambda q[z^s | z^0] \lim_{t \rightarrow \infty} \left[\sum_{z^t \in \mathcal{Z}^t} p[z^{t-1}] a[z^t] \right] = 0.$$

If this condition is violated for some $z^s \in \mathcal{Z}^s$, then the lifetime budget constraint (17.11) cannot hold. Therefore, the representative household's optimization problem ensures that (17.7) must hold. This establishes the following proposition.

PROPOSITION 17.3. *In the above-described economy, optimal and competitive growth paths coincide.*

PROOF. See Exercise 17.13. □

17.2.2. Competitive Equilibrium with Sequential Trading. Complementary insights can be obtained by considering the equilibrium problem in its equivalent form with sequential trading and using the appropriate Arrow securities rather than all trades taking place at the initial date $t = 0$. To do this, we will write the budget constraint of the representative household somewhat differently. First, normalize the price of the final good at each date to 1 (recall the discussion in Section 5.8 in Chapter 5). The $a[z^t]$ s now correspond to (basic) Arrow securities that pay out only in specific states on nature. More explicitly, $a[z^t]$ denotes the assets of the household in terms of the final good at date t conditional on history z^t . We interpret $\{a[z^t]\}_{z^t \in \mathcal{Z}^t}$ as a set of contingent claims that the household has purchased that will pay $a[z^t]$ units of the final good at date t when history z^t is realized. We also denote the price of a claim to one unit of $a[z^t]$ at time $t - 1$ after history z^{t-1} by $\bar{p}[z(t) | z^{t-1}]$, where naturally $z^t = (z^{t-1}, z(t))$. The amount of these claims purchased by the household is denoted by $a[(z^{t-1}, z(t))]$. Consequently, the flow budget constraint of the household can be written as

$$c[z^t] + \sum_{z(t+1) \in \mathcal{Z}} \bar{p}[z(t+1) | z^t] a[(z^t, z(t+1))] \leq w[z^t] + a[z^t],$$

where $w[z^t]$ is the equilibrium wage rate after history z^t in terms of final goods dated t , so the right-hand side is the total amount of resources available to the household after history z^t , which will be spent on consumption $c[z^t]$ and for purchasing contingent claims to final good at the next date, $a[(z^t, z(t+1))]$. The total expenditure on these is equal to $\sum_{z(t+1) \in \mathcal{Z}} \bar{p}[z(t+1) | z^t] a[(z^t, z(t+1))]$.

With this formulation, we can once again write the sequence version of the optimization problem of the household. To save space, let us directly go to the recursive formulation,

leaving the sequence version of the household's problem with sequential trading to Exercise 17.14.

Preparing for the recursive formulation, let a denote the current asset holdings of the household (in terms of the notation above, you can think of this as the realization of the current assets after some history z^t has been realized). Then, the flow budget constraint of the household can be written as

$$c + \sum_{z' \in \mathcal{Z}} \bar{p}[z' | z] a'[z' | z] \leq w + a,$$

where the function $\bar{p}[z' | z]$ summarizes the prices of contingent claims (for next date's state z' given current state z) and $a'[z' | z]$ denotes the corresponding asset holdings. Let $V(a, z)$ be the value function of the household when it holds a units of the final good as assets and the current realization of the stochastic variable is z . The choice variables of the household are contingent asset holdings for the next date, denoted by $a'[z' | z]$, and consumption today denoted by $c[a, z]$. Let us also denote the probability that next period's stochastic variable will be equal to z' conditional on today's value being z by $q[z' | z]$. Then, taking the sequence of equilibrium prices $\bar{p}$ as given, the value function of the representative household can be written as

$$(17.22) \quad V(a, z) = \max_{\{a'[z' | z]\}_{z' \in \mathcal{Z}}} \left\{ u \left(a + w - \sum_{z' \in \mathcal{Z}} \bar{p}[z' | z] a'[z' | z] \right) + \beta \sum_{z' \in \mathcal{Z}} q[z' | z] V(a'[z' | z], z') \right\}.$$

Theorems 16.1-16.7 from the previous chapter can again be applied to this value function (see Exercise 17.15). The first-order condition for current consumption can now be written as

$$\bar{p}[z' | z] u'(c[a, z]) = \beta q[z' | z] \frac{\partial V(a'[z' | z], z')}{\partial a}$$

for any $z' \in \mathcal{Z}$ with $c[a, z]$ denoting the optimal consumption conditional on asset holdings a and stochastic variable z .

The key to the characterization of the equilibrium is again the capital market clearing condition. Since the asset holdings of the representative household must be decided before next period's stochastic shock z' is realized, the capital market clearing requires that

$$a'[z' | z] = a'[z],$$

for all $z' \in \mathcal{Z}$. In other words, in the aggregate, the same amount of assets will be present in all states at the next date. This implies that the first-order condition for consumption can be alternatively written as

$$(17.23) \quad \bar{p}[z' | z] u'(c[a, z]) = \beta q[z' | z] \frac{\partial V(a'[z], z')}{\partial a}.$$

The capital market clearing condition again provides the key *no arbitrage condition*, which now takes the form

$$(17.24) \quad \sum_{z' \in \mathcal{Z}} \bar{p}[z' | z] R[z' | z] = 1,$$

where $R[z' | z]$ is the price of capital goods when the current state is z' and last period's state was z . Intuitively, the cost of one unit of the final good now, which is 1, has to be equal to the return that the individual will obtain by saving this good to be used as capital next period. When tomorrow's state is z' , the gross rate of return in terms of tomorrow's goods is $R[z' | z]$ and the relative price of tomorrow's goods in terms of today's goods is $\bar{p}[z' | z]$. Summing over all possible states z' tomorrow must then have total return of 1 to ensure no arbitrage (see Exercise 17.16). Let us now combine (17.23) with the Envelope condition,

$$\frac{\partial V(a, z)}{\partial a} = u'(c[a, z]),$$

and then multiply both sides of (17.23) by $R[z' | z]$ and sum over all $z' \in \mathcal{Z}$ to obtain the first-order condition of the household as

$$\begin{aligned} u'(c[a, z]) &= \beta \sum_{z' \in \mathcal{Z}} q[z' | z] R[z' | z] u'(c[a', z']). \\ &= \beta \mathbb{E}[R[z' | z] u'(c[a', z']) | z]. \end{aligned}$$

Next the market clearing condition for capital, combined with the fact that the only asset in the economy is capital, implies that

$$a = k.$$

Therefore, this first-order condition can be written as

$$u'(c[k, z]) = \beta \mathbb{E}[R[z' | z] u'(c[k', z']) | z]$$

which is identical to (17.6). A similar argument to that in the previous subsection again establishes that the transversality condition of the optimal growth problem, (17.7), is satisfied in the competitive equilibrium. Consequently, the approach based on sequential trading also leads to a competitive equilibrium allocation identical to the solution to the optimal growth problem.

The analysis in the previous two subsections illustrates the equivalence between the optimal growth problem (under uncertainty) and the competitive equilibrium allocation (with a complete set of markets). Given this equivalence, also implied directly by the Second Welfare Theorem (Theorem 5.7, Chapter 5) and the fact that the former is considerably simpler, much of the literature focuses on the optimal growth problem rather than the explicit characterization of the competitive equilibrium under uncertainty. The (stochastic) equilibrium path of all the real variables is immediately obtained from this optimal growth problem and the various different equilibrium prices are given by the Lagrange multipliers of the same optimal growth problem. In particular, for most purposes the most important prices are the prices of capital goods, i.e., the $R[z' | z]$ s, which are also the relevant intertemporal prices. Equations (17.5) and (17.6) show that these can be easily obtained from the marginal product of capital in the optimal growth path.

17.3. Application: Real Business Cycle Models

One of the most important applications of the neoclassical growth model under uncertainty over the past 25 years has been to the analysis of short- and medium-run fluctuations. The approach, pioneered by Kydland and Prescott's seminal (1982) paper and referred to as the *Real Business Cycle* theory, uses the neoclassical growth model with aggregate productivity shocks in order to provide a framework for the analysis of macroeconomic fluctuations. The Real Business Cycle (RBC) theory has been one of the most active research areas of macroeconomics in the 1990s and also one of the most controversial. On the one hand, its conceptual simplicity and its relative success in matching certain moments of employment, consumption and investment fluctuations for a given (appropriately chosen) sequence of aggregate productivity shocks have attracted a large following. On the other and, the absence of monetary factors and demand shocks, the traditional pillars of Keynesian economics and previous research on macroeconomic fluctuations, has generated a ferocious opposition and much debate on the merits of this theory. The merits of the RBC theory are not relevant for our focus here and would take us too far afield from the key questions of economic growth. Nevertheless, a brief exposition of the canonical RBC model is useful for two purposes. First, it constitutes one of the most important applications of the neoclassical growth model under uncertainty and has become another one of the workhorse models for macroeconomic research over the past 25 years. Second, it illustrates how the introduction of labor supply choices into the neoclassical growth model under uncertainty generates new insights. So far I have assumed, except in Exercise 8.31 in Chapter 8, that labor is supplied inelastically and this choice has enabled us to focus on the first-order issues related to economic growth. Because the issue of labor supply is central to a number of questions in macroeconomics, a brief analysis of the neoclassical growth model with labor supply is also useful.

The economic environment is identical to that in the previous two sections, with the only difference that the instantaneous utility function of the representative household now takes the form

$$u(C, L),$$

where C denotes consumption and L labor supply. I use uppercase letters for consistency with what will come below. I assume that u is jointly concave and differentiable in both of its arguments and strictly increasing in C and strictly decreasing in L . I also assume that L has to lie in some convex compact set $[0, \bar{L}]$.

Given the equivalence between the optimal growth and the equilibrium growth problems, I focus on the optimal growth formulation here and set up the social planner's problem. This problem can be expressed as the maximization of

$$\mathbb{E} \sum_{t=0}^{\infty} \beta^t u(C(t), L(t))$$

subject to the flow resource constraint

$$K(t+1) \leq F(K(t), L(t), z(t)) + (1 - \delta)K(t) - C(t),$$

where $z(t)$ again represents an aggregate productivity shock following a monotone Markov chain.

The social planner's problem can be written recursively as

$$(17.25) \quad V(K, z) = \sup_{\substack{L \in [0, \bar{L}] \\ K' \in [0, F(K, L, z) + (1-\delta)K]}} \{u(F(K, L, z) + (1-\delta)K - K', L) + \beta \mathbb{E}[V(K', z') | z]\}.$$

The following proposition is again a direct consequence of Theorems 16.1-16.7:

PROPOSITION 17.4. *The value function $V(K, z)$ defined in (17.25) is continuous and strictly concave in K , strictly increasing in K and z , and differentiable in $K > 0$. There exist uniquely defined policy functions $\pi^k(K, z)$ and $\pi^l(K, z)$ that determine the level of capital stock chosen for next period and the level of labor supply as a function of the current capital stock K and the stochastic variable z .*

PROOF. See Exercise 17.17. □

Under the assumption that an interior solution exists, the relevant prices can be obtained from the appropriate multipliers and the standard first-order conditions characterize the form of the equilibrium. In particular, the two key first-order conditions determine the evolution of consumption over time and the equilibrium level of labor supply. Denoting the derivatives of the u function with respect to its first and second arguments by u_c and u_l , the derivatives of the F function by F_k and F_l , and defining the policy function for consumption as

$$\pi^c(K, z) \equiv F(K, \pi^l(K, z), z) + (1-\delta)K - \pi^k(K, z),$$

these take the form

$$(17.26) \quad \begin{aligned} u_c(\pi^c(K, z), \pi^l(K, z)) &= \beta \mathbb{E} \left[R(\pi^k(K, z), z') u_c(\pi^c(\pi^k(K, z), z'), \pi^l(\pi^k(K, z), z')) | z \right], \\ w(K, z) u_c(\pi^c(K, z), \pi^l(K, z)) &= -u_l(\pi^c(K, z), \pi^l(K, z)). \end{aligned}$$

where

$$\begin{aligned} R(K, z) &= F_k(K, z) + (1-\delta) \\ w(K, z) &= F_l(K, z) \end{aligned}$$

denote the gross rate of return to capital and the equilibrium wage rate. Notice that the first condition in (17.26) is essentially identical to (17.5), whereas the second is a static condition determining the level of equilibrium (or optimal) labor supply. The second condition does not feature expectations, since it is conditional on the current-value of the capital stock, K , and the current realization of the aggregate productivity variable, z .

Why is this framework useful for the analysis of macroeconomic fluctuations? The answer lies in the fact that estimates of total factor productivity, along the lines described in Chapter 3, indicate that it is procyclical—that is, it fluctuates considerably and higher in periods during which output is above trend and unemployment is low. So let us think of a period

in which z is low. Clearly, if there is no offsetting change in labor supply, output will be low, so we can think of this period as a “recession”. Moreover, under standard assumptions, the wage rate $w(K, z)$ and equilibrium labor supply will decline.⁴ Thus there will be low employment as well as low output. Thus a negative productivity shock generates two of the important characteristics of a recessionary period. In addition, if the Markov chain (or more generally the Markov process) governing the behavior of z exhibits persistence, output will be low the following period as well, so output and employment will exhibit *persistent fluctuations*. Finally, provided that the aggregate production function $F(K, L, z)$ takes a form such that low output is also associated with low marginal product of capital, the expectation of future low output will typically reduce savings and thus future levels of capital stock (though this does also depend on the form of the utility function, which regulates the desire for consumption smoothing and the balance between income and substitution effects).

This brief discussion suggests that the neoclassical growth model under uncertainty with labor supply choices and with aggregate productivity shocks may generate some of the major qualitative features of macroeconomic fluctuations. The RBC literature also argues that this model, under suitable assumptions, generates the major *quantitative* features of the business cycles, such as the correlations between output, investment, and employment. A large part of the debate on the merits of the RBC model focused on: (i) whether the model did indeed match these moments in the data; (ii) whether these were the right empirical objects to look at (for example, as opposed to persistence in employment or output at different frequencies); and (iii) whether a framework in which the driving force of fluctuations is exogenous changes in aggregate productivity is sidestepping the more interesting question of why there are shocks that cause recessions and booms. It is fair to say that, while the RBC debate is not as active today as it was in the 1990s, there has not been a complete agreement on these questions. In the meantime, many extensions of the standard RBC model have improved over the bare-bones version presented here.

The model presented here considered the neoclassical growth model without exogenous technological progress. Exercise 17.18 introduces exogenous technological progress into this model and shows that the analysis is essentially unchanged. The next example considers a very simple case of the RBC model that can be solved in closed form (though the price of doing so is that some of the interesting features of the model are lost).

EXAMPLE 17.2. Consider an example economy similar to that studied in Example 17.1. In particular, suppose that $u(C, L) = \log C - \gamma L$, $F(K, L, z) = zK^\alpha L^{1-\alpha}$, and $\delta = 1$. Productivity z follows a monotone Markov chain over the set $\mathcal{Z} \equiv \{z_1, \dots, z_N\}$, with transition

⁴It is useful to note, however, that there is no agreement as to whether wages are indeed procyclical. Average wages do not seem to be procyclical over the business cycle, but this may be because of “selection bias,” the fact that the composition of the labor force changes over the business cycle as those who lose their jobs during recessions are typically different from the “average” worker. Depending on how one corrects for this potential source of bias, wages appear to be either mildly procyclical or acyclical. See, for example, Bils (1985), Barsky, Parker and Solon (1994) and Abraham and Haltiwanger (1995).

probabilities denoted by $q_{jj'}$. As in that example, let us conjecture that

$$\pi^k(K, z) = BzK^\alpha L^{1-\alpha}.$$

Then, with these functional forms, the stochastic Euler equation for consumption, (17.26), implies

$$\frac{1}{(1-B)zK^\alpha L^{1-\alpha}} = \beta \mathbb{E} \left[\frac{\alpha z' (BzK^\alpha L^{1-\alpha})^{-(1-\alpha)} (L')^{1-\alpha}}{(1-B)z' (BzK^\alpha L^{1-\alpha})^\alpha (L')^{1-\alpha}} \middle| z \right],$$

where L' denotes next period's labor supply. Canceling constants within the expectations and taking terms that do not involve z' out of the expectations operator this equation simplifies to

$$\frac{1}{zK^\alpha L^{1-\alpha}} = \beta \mathbb{E} \left[\alpha (BzK^\alpha L^{1-\alpha})^{-1} \middle| z \right],$$

which yields

$$B = \alpha\beta.$$

The resulting policy function for the capital stock is therefore

$$\pi^k(K, z) = \alpha\beta zK^\alpha L^{1-\alpha},$$

which is identical to that in Example 17.1. Next, the first-order condition for labor implies

$$\frac{(1-\alpha)zK^\alpha L^{-\alpha}}{(1-B)zK^\alpha L^{1-\alpha}} = \gamma.$$

The resulting policy function for labor can be obtained as

$$\pi^l(K, z) = \frac{(1-\alpha)}{\gamma(1-\alpha\beta)},$$

which implies that labor supply is constant. This is because with the preferences as specified here, the income and the substitution effects cancel out, thus the increase in wages induced by a change in aggregate productivity has no effect on labor supply. Exercise 17.19 shows that the same result obtains whenever the utility function takes the form of $U(C, L) = \log C + h(L)$ for some decreasing and concave function h . Overall, this simple version of the RBC model with a closed-form solution therefore generates positive covariation between output and investment, but does not lead to labor fluctuations.

17.4. Growth with Incomplete Markets: The Bewley Model

I now turn to a fundamentally different model of economic growth, where the economy does not admit a representative household and idiosyncratic risks are not diversified. This model was first introduced and studied by Truman Bewley (1977, 1980, 1986). It has subsequently been revived, extended and applied to a variety of new questions including the structure of optimal fiscal policy, business cycle fluctuations and asset pricing in Aiyagari (1994, 1997) and Krusell and Smith (1998, 1999). Many economists believe that, to a first approximation, such a structure provides a better approximation to reality than the complete markets neoclassical growth model. Unfortunately, this model, which I will refer to as the Bewley economy, is considerably more complicated than the baseline neoclassical growth model.

As we will discuss below, however, the assumption that there is no insurance for individual income fluctuations—except through “self-insurance,” that is, the process of accumulating assets to be used in a rainy day—is extreme and may limit the applications of the current model in the growth context.

The economy is populated by a continuum 1 of households and the set of households is denoted by $\mathcal{H}$. Each household has preferences given by (17.2) and supplies labor inelastically. Suppose also that the second derivative of this utility function, $u''(\cdot)$, is increasing. Differently from the baseline neoclassical growth model, the efficiency units that each household supplies vary over time. In particular, each household $h \in \mathcal{H}$ has a labor endowment of $z^h(t)$ at time t , where $z^h(t)$ is an independent draw from the set $\mathcal{Z} \equiv [z_{\min}, z_{\max}]$, where $0 < z_{\min} < z_{\max} < \infty$, so that the minimum labor endowment is $z_{\min}$. Suppose that the labor endowment of each household is identically and independently distributed with distribution function $G(z)$ defined over $[z_{\min}, z_{\max}]$.

The production side of the economy is the same as in the canonical neoclassical growth model under certainty and is represented by an aggregate production function satisfying Assumptions 1 and 2, as in (17.1). The only difference is that $L(t)$ is now the sum (integral) of the heterogeneous labor endowments of all the agents and is written as

$$L(t) = \int_{h \in \mathcal{H}} z^h(t) dh.$$

Appealing to a law of large numbers type argument, $L(t)$ is constant at each date and can be normalized to 1. Thus output per capita in the economy can be expressed as

$$y(t) = f(k(t)),$$

with $k(t) = K(t)$. Notice that there is no longer any aggregate productivity shock. The only source of uncertainty is at the individual level (i.e., it is idiosyncratic). Consequently, while individual households will experience fluctuations in their labor income and consumption, we can imagine a *stationary* equilibrium in which aggregates are constant over time. Throughout this section I focus on such a stationary equilibrium. In particular, in a stationary equilibrium the wage rate, w , and the gross rate of return on capital R will be constant (though of course their levels will be determined endogenously to ensure equilibrium). Let us first take these prices as given and look at the behavior of a typical household $h \in \mathcal{H}$ (I am using the language “typical” household, since, though not representative, this household faces an identical problem to all other households in the economy). This household will solve the following maximization problem: maximize (17.2) subject to the flow budget constraint

$$a^h(t+1) \leq Ra^h(t) + wz^h(t) - c^h(t)$$

for all t , where $a^h(t)$ is the asset holding of household $h \in \mathcal{H}$ at time t . Consumption cannot be negative, so $c^h(t) \geq 0$. In addition, though we do not impose any exogenous borrowing constraints, with the same reasoning as in the model of the permanent income hypothesis in subsection 16.5.1 in the previous chapter, the requirement that the household should satisfy

its lifetime budget constraint in all histories imposes the endogenous borrowing constraint

$$(17.27) \quad \begin{aligned} a^h(t) &\geq -\frac{z_{\min}}{R-1} \\ &\equiv -b, \end{aligned}$$

for all t (see Exercise 17.20). We can then write the maximization problem of household $h \in \mathcal{H}$ recursively as

$$(17.28) \quad V^h(a, z) = \max_{a' \in [-b, Ra+wz]} \left\{ u(Ra + wz - a') + \beta \mathbb{E} \left[V^h(a', z') \mid z \right] \right\}.$$

Standard dynamic programming arguments then establish the following proposition:

PROPOSITION 17.5. *The value function $V^h(a, z)$ defined in (17.28) is uniquely defined, continuous and strictly concave in a , strictly increasing in a and z , and differentiable in $a \in (-b, Ra + wz)$. Moreover, the policy function that determines next period's asset holding $\pi(a, z)$ is uniquely defined and continuous in a .*

PROOF. See Exercise 17.21. □

Moreover, as in Proposition 17.2:

PROPOSITION 17.6. *The policy function $\pi(a, z)$ derived in Proposition 17.5 is strictly increasing in a and z .*

PROOF. See Exercise 17.22. □

The total amount of capital stock in the economy can be obtained by aggregating the asset holdings of all households in the economy, thus in a stationary equilibrium,

$$\begin{aligned} k(t+1) &= \int_{h \in \mathcal{H}} a^h(t) dh \\ &= \int_{h \in \mathcal{H}} \pi(a^h(t), z^h(t)) dh. \end{aligned}$$

This equation integrates over all households taking their asset holdings and the realization of their stochastic shock as given. It states that both the average of current asset holdings and also the average of tomorrow's asset holdings must be equal by the definition of a stationary equilibrium. To understand this condition, recall that as in the neoclassical growth model, the policy function $a' = \pi(a, z)$ defines a Markov process. Under fairly weak assumptions this Markov process will admit a unique invariant distribution. If this were not the case, the economy could have multiple stationary equilibria or even there might be problems of non-existence. For our purposes here, we can ignore this complication and assume the existence of a unique invariant distribution, which is denoted by $\Gamma(a)$, so that the stationary equilibrium capital-labor ratio is given by

$$k^* = \int \int \pi(a, z) d\Gamma(a) dG(z),$$

which uses the fact that z is distributed identically and independently across households and over time.

Next turning to the production side, factor prices are the same as the neoclassical growth model under certainty, that is,

$$\begin{aligned} R &= f'(k^*) + (1 - \delta) \\ w &= f(k^*) - k^* f'(k^*). \end{aligned}$$

Recall from Chapters 6 and 8 that the neoclassical growth model with complete markets and no uncertainty implies that there exists a unique steady state in which $\beta R = 1$, that is,

$$(17.29) \quad f'(k^{**}) = \beta^{-1} - (1 - \delta),$$

where k^{**} refers to the capital-labor ratio of the neoclassical growth model under certainty.

Perhaps the most interesting implication of the Bewley economy is that this is no longer true. In particular:

PROPOSITION 17.7. *In any stationary equilibrium of the Bewley economy, the stationary equilibrium capital-labor ratio k^* is such that*

$$(17.30) \quad f'(k^*) < \beta^{-1} - (1 - \delta)$$

and

$$(17.31) \quad k^* > k^{**},$$

where k^{**} is the capital-labor ratio of the neoclassical growth model under certainty.

PROOF. (Sketch) Suppose $f'(k^*) \geq \beta^{-1} - (1 - \delta)$. Then, the result in Exercise 16.11 from the previous chapter implies that each household's expected consumption is strictly increasing. This implies that average consumption in the population, which is deterministic, is strictly increasing and would tend to infinity. This is not possible in view of Assumption 2, which implies that aggregate resources must always be finite. This establishes (17.30). Given this result, (17.31) immediately follows from (17.29) and from the strict concavity of $f(\cdot)$ (Assumption 1). $\square$

Intuitively, the interest rate in the incomplete markets economy is “depressed” relative to the neoclassical growth model with certainty because each household has an additional self-insurance, or *precautionary*, incentive to save. These additional savings increase the capital-labor ratio and reduce the equilibrium interest rate. Interestingly, therefore, the Bewley economy, like the overlapping generations model of Chapter 9, leads to a higher capital intensity of production than the standard neoclassical growth model. Observe that in both cases, the lack of a representative household plays an important role in this result.

While the Bewley model is an important workhorse for macroeconomic analysis, two of its features may be viewed as potential shortcomings. First, as already remarked in the context of the overlapping generations model, the source of inefficiency coming from overaccumulation of capital is unlikely to be important for explaining income per capita differences across countries. Thus the Bewley model is not interesting because of the greater capital-labor ratio that it generates. Instead, it is important as an illustration of how an economy might

exhibit a stationary equilibrium in which aggregates are constant while individual households have uncertain and fluctuating consumption and income profiles. It also emphasizes issues of individual risk in the context of a relatively familiar neoclassical growth setup. Issues of individual risk bearing are important in the context of economic development as shown in Section 17.6 below and also in Chapter 21. Second, the incomplete markets assumption in this model may be extreme. In practice, when their incomes are low, household may receive transfers, either because they have entered into some form of private insurance or because of government-provided social insurance. Instead, the current model exogenously assumes that there are no insurance possibilities. Much more satisfactory would be models in which the lack of insurance opportunities are derived from microfoundations (for example, from moral hazard or adverse selection) or models in which the set of active markets is determined endogenously. While models of limited insurance due to moral hazard or adverse selection are beyond the scope of this book, I will return to an economic growth model with endogenously incomplete markets in Section 17.6.

17.5. The Overlapping Generations Model with Uncertainty

Let us now briefly consider a stochastic version of the canonical overlapping generations model from Section 9.3 in Chapter 9. Time is discrete and runs to infinity. Each individual lives for two periods. Suppose as in Section 9.3 that the utility of a household in generation t is given by

$$(17.32) \quad U(t) = \log c_1(t) + \beta \log c_2(t+1).$$

There is a constant rate of population growth equal to n , so that

$$(17.33) \quad L(t) = (1+n)^t L(0),$$

where $L(0)$ is the size of the first generation. As in Section 9.3, the aggregate production technology is Cobb-Douglas but now also includes an aggregate stochastic shock z , which is assumed to follow a Markov process. Consequently, total output at time t is given by

$$Y(t) = z(t) K(t)^\alpha L(t)^{1-\alpha}.$$

Expressing this in per capita terms,

$$y(t) = z(t) k(t)^\alpha.$$

To simplify the notation, suppose also that capital depreciates fully ($\delta = 1$). Factor prices clearly only depend on the current-values of z and the capital-labor ratio k :

$$(17.34) \quad \begin{aligned} R(k, z) &= \alpha z k^{\alpha-1} \\ w(k, z) &= (1 - \alpha) z k^\alpha. \end{aligned}$$

The consumption Euler equation for an individual of generation t takes the form

$$\begin{aligned} \frac{c_2(t+1)}{c_1(t)} &= \beta R(t+1) \\ &= \beta R(k, z), \end{aligned}$$

with $R(k, z)$ given by (17.34). The total amount of savings at time t is then given by $s(t) = s(k(t), z(t))$ such that

$$(17.35) \quad s(k, z) = \frac{\beta}{1 + \beta} w(k, z),$$

which, as in the canonical overlapping generations model of Section 9.3 in Chapter 9 and also as in the baseline Solow growth model of Chapter 2, corresponds to a constant savings rate, now equal to $\beta / (1 + \beta)$.

Combining (17.35) with (17.33) and the fact that $\delta = 1$, next date's capital stock $k(t + 1)$ can be written as

$$(17.36) \quad \begin{aligned} k(t + 1) &= \pi(k, z) \\ &= \frac{s(k, z)}{(1 + n)} \\ &= \frac{\beta(1 - \alpha) z k^\alpha}{(1 + n)(1 + \beta)}. \end{aligned}$$

Clearly, if $z = \bar{z}$, this equation would have a unique steady state with capital-labor ratio given by

$$(17.37) \quad k^* = \left[\frac{\beta(1 - \alpha) \bar{z}}{(1 + n)(1 + \beta)} \right]^{\frac{1}{1 - \alpha}}.$$

However, when z has a non-degenerate distribution, (17.36) defines a stochastic first-order difference equation. As in the neoclassical growth model under uncertainty, the long-run equilibrium of this model will correspond to an invariant distribution of the capital stock. In this particular case, however, since (17.36) is very tractable, we can obtain more insights about the behavior of the economy with a diagrammatic analysis.

Suppose that z is distributed between $[z_{\min}, z_{\max}]$, then the behavior of the economy can be analyzed by plotting *the stochastic correspondence* associated with (17.36), which is done in Figure 17.1. The stochastic correspondence plots the entire range of possible values of $k(t + 1)$ for a given value of $k(t)$. The upper thick curve corresponds to the realization of $z_{\max}$, while the lower thick curve corresponds to the realization of $z_{\min}$. The dashed curve in the middle is for $z = \bar{z}$. Observe that the curves for both $z_{\min}$ and $z_{\max}$ start above the 45° line, which is a consequence of the Inada condition implied by the Cobb-Douglas technology—the marginal product of capital is arbitrarily large when the capital stock is close to zero. The stochastic correspondence enables a simple analysis of the dynamics of certain stochastic models. For example, Figure 17.1 plots a particular sample path of capital-labor ratio in this economy, where starting with $k(0)$, the economy first receives a fairly favorable productivity shock moving to $k(1)$. Following this, there is another moderately favorable productivity realization and the capital-labor ratio increases to $k(2)$. In the following period, however, the realization of the stochastic variable is quite bad and the capital-labor ratio and thus output per capita decline sharply. This figure illustrates the type of dynamics that can emerge. Similar methods will be used in the next section in a somewhat richer model.

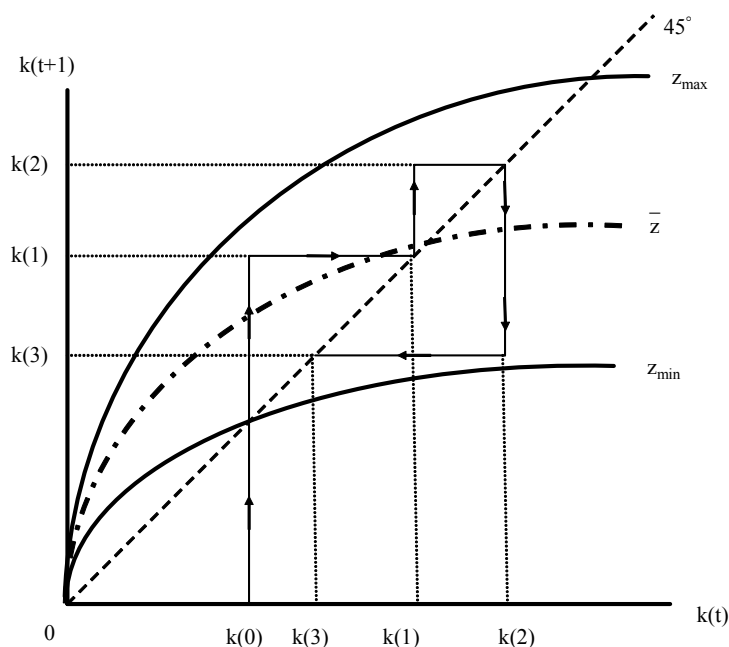


FIGURE 17.1. This figure shows the stochastic correspondence of the overlapping generations model. Values for next period's capital-labor ratio within the two curves marked $z_{\min}$ and $z_{\max}$ are possible. The path $k(0) \rightarrow k(1) \rightarrow k(2) \rightarrow k(3)$ illustrates a particular realization of the stochastic path of the equilibrium capital-labor ratio.

Another noteworthy feature of this model is that, together with the stochastic Solow model discussed in Exercise 17.3 and the specific form of the neoclassical growth model in Example 17.1, it provides a much simpler model of stochastic growth than the neoclassical growth under uncertainty. In the overlapping generations model and the Solow model, this is because saving decisions are “myopic” and remain unaffected by the distribution of stochastic shocks or even their realizations. Thus for the analysis of a range of macroeconomic questions, these more myopic models or the simple neoclassical model of Example 17.1 might provide a tractable attractive alternative to the full neoclassical growth model under uncertainty.

17.6. Risk, Diversification and Growth

In this section, I present a stochastic model of long-run growth based on Acemoglu and Zilibotti (1997). This model is useful for two distinct purposes. First, because it is simpler than the baseline neoclassical growth model under uncertainty, it will provide a complete characterization of the stochastic dynamics of growth and show how simple ideas from the theory of Markov processes can be used in the context of the study of economic growth. Second and more important, this model will enable the analysis of a number of important topics in the theory of long-run growth. In particular, I have so far focused on models with

balanced growth and relatively well-behaved transitional dynamics. The experience of economic growth over the past few thousand years has been much less “orderly” than implied by these models, however. Until about 200 years ago, growth in income per capita was relatively rare. Sustained growth in income per capita is a relatively recent phenomenon. Before this “takeoff” into sustained growth, societies experienced periods of growth followed by large slumps and crises. Acemoglu and Zilibotti (1997), Imbs and Wacziarg (2003), and Koren and Tenreyro (2007) document that even today richer countries have much more stable growth performances than less developed economies, which suffer from much higher variability in their growth rates. In many ways, this pattern of relatively risky growth and low productivity followed by a process of capital-deepening, financial development and better risk management is a major characteristic of the history of economic growth. The famous economic historian Fernand Braudel describes the start of economic growth in Western Europe as follows:

“The advance occurred very slowly over a long period and was broken by sharp recessions. The right road was reached and thereafter never abandoned, only during the eighteenth century, and then only by a few privileged countries. Thus, before 1750 or even 1800 the march of progress could still be affected by unexpected events, even disasters.” F. Braudel (1973, p. xi).

In the model I present here, these patterns arise endogenously because the extent to which the economy can diversify risks by investing in imperfectly correlated activities is limited by the amount of capital it possesses. As the amount of capital increases, the economy achieves better diversification and faces fewer risks. The resulting equilibrium process thus generates greater variability and risk at the early stages of development and these risks are significantly reduced after the economy manages to “take off” into sustained growth. Moreover, the desire of individual households to avoid risk makes them invest in lower return less risky activities during the early stages of development, thus the growth rate of the economy is also endogenously limited during this pre-takeoff stage. In addition, in this model, economic development goes hand-in-hand with financial development as greater availability of capital enables better risk sharing through asset markets. Finally, because the model is one of endogenously incomplete markets, it also enables us to show that price-taking behavior by itself is not sufficient to guarantee Pareto optimality, and the form of inefficiency of the equilibrium in this economy will be interesting both on substantive and methodological grounds.

17.6.1. The Environment. Consider an overlapping generations economy, where each generation lives for two periods. There is no population growth and the size of each generation is normalized to 1. The production sector consists of two sectors. The first sector produces final goods with the Cobb-Douglas production function

$$(17.38) \quad Y(t) = K(t)^\alpha L(t)^{1-\alpha},$$

where as usual $L(t)$ is total labor and $K(t)$ is the total capital stock available at time t . Capital depreciates fully after use ($\delta = 1$ in terms of our previous notation).

The second sector transforms savings at time $t - 1$ into capital to be used for production at time t . This sector consists of a continuum $[0, 1]$ of intermediates, and stochastic elements only affect this sector. In particular, let us represent possible states of nature also with the unit interval and assume that intermediate sector $j \in [0, 1]$ pays a positive return only in state j and nothing in any other state. This formulation implies that investing in a sector is equivalent to buying a (Basic) Arrow security that only pays in one state of nature. Since there is a continuum of sectors, the probability that a single sector will have positive payoff is 0, but if a household invests in some subset J of $[0, 1]$, then there will be positive returns with probability equal to the (Lebesgue) measure on the set J . Thus each intermediate sector is a risky activity, but a household (and in particular, the representative household in the economy) can diversify risks by investing in multiple sectors. In particular, if one were to invest in all of the sectors, then one would receive positive returns with probability 1. What makes the economic interactions in this model non-trivial is that investing in all sectors may not be possible at every date because of potential *nonconvexities*. More specifically, let us assume that each sector has a *minimum size requirement*, denoted by $M(j)$ and positive returns will be realized only if aggregate investment in that sector exceeds $M(j)$.

In light of this description, let $I(j, t)$ be the aggregate investment in intermediate sector j at time t . This investment will generate date $t + 1$ capital equal to $QI(j, t)$ if state j is realized and $I(j, t) \geq M(j)$, and nothing otherwise. Thus aggregate investment in the intermediate sector exceeding the minimum size requirement is *necessary* for any positive returns.

In addition to the risky intermediate sectors, there is also a safe intermediate sector which transforms one unit of savings at date t into q units of date $t + 1$ capital. The crucial assumption is that

$$(17.39) \quad q < Q,$$

so that the safe option is also less productive.

The requirement that $I(j, t) \geq M(j)$ combined with the fact that the amount of capital obtained from savings $I(j, t)$ in state j is equal to $QI(j, t)$ implies that all intermediate sectors have linear technologies, but only after the minimum size requirement, $M(j)$, is met. For any $I(j, t) < M(j)$, the output is equal to 0. In order to simplify the exposition and the computations, let us adopt a simple distribution of minimum size requirements by intermediate sector:

$$(17.40) \quad M(j) = \max \left\{ 0, \frac{D}{(1 - \gamma)}(j - \gamma) \right\}.$$

This equation implies that intermediate sectors $j \leq \gamma$ have no minimum size requirement, so aggregate investments of any size can be made in these sectors. For the remaining sectors, the minimum size requirement increases linearly. Figure 17.2 shows the minimum size requirements with the thick line. This figure will be used to illustrate the determination of the set of open sectors once the equilibrium investments are specified.

It is worth noting that there are three important features introduced so far.

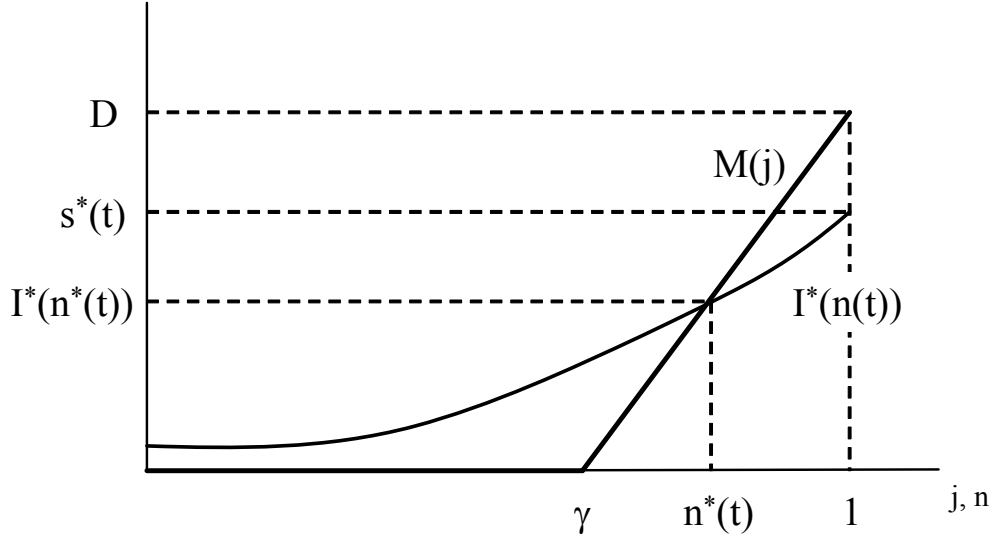


FIGURE 17.2. Minimum size requirements, $M(j)$, of different sectors and demand for assets, $I^*(n)$.

- (1) Risky investments have a higher expected return than the safe investment, which is captured by the assumption that $Q > q$.
- (2) The output of the risky investments (of the intermediate sectors) are imperfectly correlated so that there is safety in numbers.
- (3) The mathematical formulation here implies a simple relationship between investments and returns. As already hinted above, if a household holds a portfolio consisting of an equi-proportional investment I in all sectors $j \in \bar{J} \subseteq [0, 1]$, and the (Lebesgue) measure of the set $\bar{J}$ is p , then the portfolio pays the return QI with probability p , and nothing with probability $1 - p$.

The first two features imply that if the aggregate production set of this economy had been convex, for example because $D = 0$, all households would invest an equal amount in all intermediate sectors and manage to diversify all risks without sacrificing any of the high returns. However, in the presence of nonconvexities, as captured by the minimum size requirements, there is a tradeoff between insurance and high productivity.

Let us next turn to the preferences of households. Recall that each generation has size normalized to 1. Consider a household from a generation born at time t . The preferences of this household are given by

$$(17.41) \quad \mathbb{E}_t U(c_1(t), c_2(t+1)) = \log(c_1(t)) + \beta \int_0^1 \log c_2(j, t+1) dj,$$

where $c_1(t)$ is the consumption of final goods in the first period of this household's life (which is at time t) and c_2 refers to consumption in the second period of this household. $\mathbb{E}_t$ refers

to the expectations operator, which is necessary because second period consumption is risky. This is spelled out on the right-hand side of (17.41), with $c_2(j, t + 1)$ denoting consumption in state j at time t . The integral replaces the expectation using the fact that all states are equally likely. As in the canonical overlapping generations model, each household has 1 unit of labor when young and no labor endowment when old. Thus the total supply of labor in the economy is 1. Moreover, in the second period of their lives, each individual (household) consumes the return from their savings. For future reference, the set of young households at time t is denoted by $\mathcal{H}_t$ and Figure 17.3 depicts the life cycle and the various decisions of a typical household, emphasizing that uncertainty affects the return on their savings and thus the amount of capital they will have in the second period of their lives.

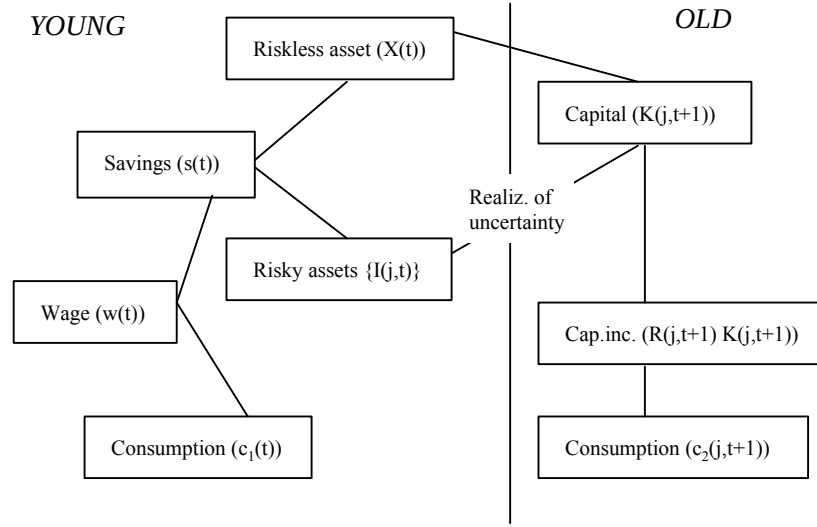


FIGURE 17.3. Life cycle of a typical household.

The aggregate capital stock depends on the realization of the state of nature, which determines how much of the investments in different intermediate sectors at time t is turned into capital. The capital stock at time $t + 1$ therefore depends on the realization of the state of nature as well as the composition of investment of young households. In particular, in state j , the aggregate stock of capital is

$$K(j, t + 1) = \int_{h \in \mathcal{H}_t} (qX^h(t) + QI^h(j, t)) dh$$

where $I^h(j, t)$ is the amount of savings invested by (young) household $h \in \mathcal{H}_t$ in sector j at time t , and $X^h(t)$ is the amount invested in the safe intermediate sector.

Since the capital stock is potentially random, so will be output and factor prices. In particular, both labor and capital are assumed to be traded in competitive markets, so the equilibrium factor prices will be given by their marginal products. Since the total capital

stock in state j at time $t + 1$ is $K(j, t + 1)$ and the total supply of labor is equal to 1, these prices are given by

$$\begin{aligned}
 w(j, t + 1) &= (1 - \alpha)K(j, t + 1)^\alpha \\
 (17.42) \qquad &= (1 - \alpha) \left(\int_{h \in \mathcal{H}_t} (qX^h(t) + QI^h(j, t)) dh \right)^\alpha.
 \end{aligned}$$

and

$$\begin{aligned}
 R(j, t + 1) &= \alpha K(j, t + 1)^{\alpha-1} \\
 (17.43) \qquad &= \alpha \left(\int_{h \in \mathcal{H}_t} (qX^h(t) + QI^h(j, t)) dh \right)^{\alpha-1}.
 \end{aligned}$$

To complete the description of the environment, let us also specify the market structure of the intermediate sector. Suppose that households make investments in different intermediate sectors through *financial intermediaries*. There is free entry into financial intermediation (either by a large number of firms or by the households themselves). Any intermediary can form costlessly and mediate funds for a particular sector, that is, it can collect funds, invest them in a particular intermediate sector and provides the corresponding Arrow securities to its investors. The important requirement is that, to be able to invest, any financial intermediary should raise enough funds to cover the minimum size requirement. For now, suppose that each financial intermediary can operate only a single sector, which rules out the formation of a grand financial intermediary managing all investments.⁵ I will return to this issue in subsection 17.6.5. We denote the price charged for a security associated with intermediate sector j at time t by $P(j, t)$. Although the decentralized equilibrium in this economy will be defined and analyzed in the next section, we can already make a few useful observations about the prices for the securities, the $P(j, t)$ s. Clearly $P(j, t) < 1$ is not possible, since one unit of the security requires one unit of the final good, so $P(j, t) < 1$ would lose money. What about $P(j, t) > 1$? This is also ruled out by free entry. Imagine that a particular intermediary offers security j at some price $P(j, t) > 1$ and raises enough funds so that the total investment in this intermediate sector $I(j, t)$ is greater than the minimum size requirement $M(j)$. But in that case, some other intermediary can also enter, offer a lower price for the security, and attract all the funds that were otherwise received by the first intermediary. This argument shows that $P(j, t) > 1$ is not possible either, so that equilibrium behavior will force $P(j, t) = 1$ for all securities that are being supplied. However, we will see that securities for not all intermediate sectors will be supplied in equilibrium.

⁵To simplify the notation and the argument, I am sacrificing mathematical rigor here. Since there is a continuum of sectors, all (equilibrium) statements should be accompanied with the qualifier “*almost everywhere*”. This implies that investment in a single sector (or in fact a countable subset of the $[0, 1]$ sectors) may deviate from optimality. In addition, a fully rigorous analysis would require each financial intermediary to deal with a set of intermediate sectors of measure ε and then consider the limit $\varepsilon \rightarrow 0$. Throughout, I will ignore these qualifications and impose that investment in each sector needs to be consistent with equilibrium and assume that each intermediary controls a single sector.

17.6.2. Equilibrium. I now characterize the equilibrium of the economy described in the previous subsection. Recall the two observations from the previous paragraph. First, not all intermediate sectors will be *open* at each date, meaning that there will be securities for only a subset of the intermediate sectors at any date. Let the set of intermediate sectors that are open at date t be denoted by $J(t)$. Second, by the argument at the end of the previous subsection, for any $j \in J(t)$ free entry implies that $P(j, t) = 1$. These two observations enable us to write the problem of a representative household $h \in \mathcal{H}_t$ taking prices and the set of available securities at time t as given. This problem takes the following form:

$$(17.44) \quad \max_{s(t), X(t), [I(j, t)]_{0 \leq j \leq 1}} \log c(t) + \beta \int_0^1 \log c(j, t+1) dj,$$

subject to:

$$(17.45) \quad X(t) + \int_0^1 I(j, t) dj = s(t),$$

$$(17.46) \quad c(j, t+1) = R(j, t+1) (qX(t) + QI(j, t)),$$

$$(17.47) \quad I(j, t) = 0, \quad \forall j \notin J(t),$$

$$(17.48) \quad c(t) + s(t) \leq w(t),$$

where I have suppressed the superscript h to simplify the notation. Here eq. (17.44) is the expected utility (objective function) of the representative household. Equations (17.45)-(17.48) are the constraints on this maximization problem. The first one, (17.45), requires that the investment in the safe sector and the sum of the investments in all other securities are equal to the total savings of the household, $s(t)$. Equation (17.46) expresses consumption in state j at time $t+1$. Two features are worth noting. First, recall households supply labor only when young and consume capital income when old. This implies that second period consumption for the household is equal to its capital holdings times the rate of return to capital, $R(j, t+1)$ given by (17.43). This rate of return is conditioned on state j (at time $t+1$) since the amount of capital and thus the marginal product of capital will differ across states. Second, the amount of capital available to the household is equal to what it receives from the safe investment, $qX(t)$, plus the return from the Arrow security for state j , $QI(j, t)$. Equation (17.47) encapsulates a major constraint on household behavior: it emphasizes that the household cannot invest in any security that is not being supplied in the market. In particular, recall that $I(j, t) \geq 0$ is necessary for an intermediate sector to be open and this may not be possible for all sectors in $[0, 1]$, so some subset of the sectors in $[0, 1]$ may not be open and thus there will not be securities for the sectors that are *not* traded in equilibrium. Naturally, the household cannot invest in these non-traded securities and the constraint (17.47) ensures this. Finally, (17.48) requires the sum of consumption and savings to be less than or equal to the income of the household, which only consists of its wage income, given by (17.42).

We are now in a position to define an equilibrium. A *static equilibrium* is an equilibrium for time t , taking the amount of capital available at time t , $K(t)$, and thus the wage $w(t)$ as given. A time t tuple

$\langle s^*(t), X^*(t), [I^*(j, t)]_{0 \leq j \leq J^*(t)}, J^*(t), [P^*(j, t)]_{0 \leq j \leq J^*(t)}, w^*(j, t), R^*(j, t) \rangle$ is a static equilibrium if $s^*(t), X^*(t), [I^*(j, t)]_{0 \leq j \leq 1}$ solve the maximization of (17.44) subject to (17.45)-(17.48) for given $[P^*(j, t)]_{0 \leq j \leq J^*(t)}, J^*(t), w^*(j, t)$ and $R^*(j, t)$; $w^*(j, t)$ and $R^*(j, t)$ are given by (17.42) and by (17.43); and $J^*(t)$ and $[P^*(j, t)]_{0 \leq j \leq J^*(t)}$ are such that for all $j \in J^*(t)$, $P^*(j, t) = 1$ and the set $J^*(t)$ is determined by free entry in the sense that if some $j' \notin J^*(t)$ were offered for a price $P(j', t) \geq 1$, then the solution to the modified maximization problem (17.44) subject to (17.45)-(17.48) would involve $I(j', t) < M(j)$; in other words, there is no more room for one more intermediate sector to open and attract sufficient funds to cover the minimum size requirement.

A *dynamic equilibrium* is a sequence of static equilibria linked to each other through (17.42) given the realization of the state $j(t)$ at each $t = 1, 2, \dots$

Because preferences in (17.44) are logarithmic, the saving rate of all households will be constant as in the canonical overlapping generations model. Consequently, the following saving rule applies regardless of the risk-return tradeoff:

$$(17.49) \quad s^*(t) \equiv s^*(w(t)) = \frac{\beta}{1 + \beta} w(t).$$

Given this result, a household's optimization problem can be broken into two parts: first, the amount of savings is determined, and then an optimal portfolio is chosen. This decomposition of the optimization problem is particularly useful because of two observations (with proofs left as exercises):

- (1) For any $j, j' \in J(t)$, $I^*(j, t) = I^*(j', t)$. Intuitively, since each household is facing the same price for all of the traded *symmetric* Arrow securities, it prefers to purchase an equal amount of each and thus achieving a *balanced portfolio* (see Exercise 17.24).
- (2) The set of open projects at time t will take the form $J^*(t) = [0, n^*(t)]$ for some $n(t) \in [0, 1]$. Intuitively, when only a subset of projects can be opened in equilibrium, intermediate sectors with small minimum size requirements will be opened before those with greater minimum size requirements. Consequently, if an intermediate sector j^* is open, all sectors $j \leq j^*$ must also be open (see Exercise 17.25).

The previous two observations also imply that we can divide the states of nature at time t into two sets: states in $[0, n(t)]$ that are “good” in the sense that the society is lucky and its risky investments have delivered positive returns, and states in $(n(t), 1]$ that are “bad” in the sense that the society is unlucky and its risky investments have zero returns. Clearly, the rate of return to capital (and the wage rate) will take different values in these two sets of states. Let us denote the rate of return to capital when a good state is realized by $R^G(t+1)$ and when a bad state is realized by $R^B(t+1)$ —these returns are dated $t+1$, because they are paid out at time $t+1$. In light of this structure, the maximization problem of a representative

household can be written in the much simpler form:

$$(17.50) \quad \max_{X(t), I(t)} n^*(t) \log [R^G(t+1)(qX(t) + QI(t))] + (1 - n^*(t)) \log [R^B(t+1)qX(t)],$$

subject to:

$$(17.51) \quad X(t) + n^*(t)I(t) \leq s^*(t)$$

where $n^*(t)$, $R^G(t+1)$ and $R^B(t+1)$ are taken as given by the representative household, and $s^*(t)$ is given by (17.49). Clearly, from (17.43)

$$R^B(t+1) = \alpha(qX(t))^{\alpha-1}$$

is the marginal product of capital in the “bad” state, when the realized state is $j > n^*(t)$ and no risky investment pays off, and

$$R^G(t) = \alpha(qX(t) + QI(t))^{\alpha-1}$$

applies in the “good state”, i.e. when the realized state is $j \leq n^*(t)$.

Straightforward maximization of (17.50) subject to (17.51) yields the unique solution to the household’s problem as:

$$(17.52) \quad X^*(t) = \frac{(1 - n^*(t))Q}{Q - qn^*(t)} s^*(t),$$

and

$$(17.53) \quad I^*(j, t) = \begin{cases} I^*(n^*(t)) \equiv \frac{Q-q}{Q-qn^*(t)} s^*(t), & \text{for } j \leq n^*(t) \\ 0 & \text{for } j > n^*(t) \end{cases}.$$

Notably, eq. (17.53) implies that the demand for *each asset* (or investment in each intermediate sector) grows as the measure of open sectors increases—that is, $I^*(n)$ is strictly increasing in n . This is because when more securities are available, the risk-diversification opportunities improve and consumers become willing to reduce their investments in the safe asset and increase their investments in risky projects. This represents an important economic force. What holds back investment in the higher productivity sectors in this economy is the fact that they are riskier than the safe sector. But since there is “safety in numbers” (that is, a first-order benefit from diversification), when there are more financial assets (more open sectors), each household is willing to invest more in risky assets in total. This complementarity between the set of traded assets and investments will play an important role in the dynamics of economic development below.

Equations (17.49), (17.52) and (17.53) completely characterize the utility-maximizing behavior of the representative household given the set of intermediate sectors that are active. To completely characterize the equilibrium, we need to find the set of sectors that are active. This is equivalent to finding a threshold sector $n^*(t)$ such that all $j \leq n^*(t)$ can meet their minimum size requirements while no additional sector can enter and raise enough funds to meet its minimum size requirements. Diagrammatically, this can be done by plotting the level of investment for each sector in a balanced portfolio, $I^*(n^*(t))$ given by (17.53), together

with the minimum size requirement, $M(j)$ given by (17.40). The first curve can be loosely interpreted as the “demand for assets” in the financial market and the curve for (17.40) can be thought of as corresponding to the “supply of assets”. The two curves and their intersection is plotted in Figure 17.2. The figure shows a unique intersection between the two curves. However, because both curves are upward-sloping, more than one intersection is possible in general. It can be verified that the condition $Q \geq (2 - \gamma)q$ is sufficient to ensure a unique intersection (see Exercise 17.26). If this condition is violated, there might be multiple solutions, corresponding to multiple equilibria. These equilibria would involve different number of active sectors. When there are only a few active sectors, households invest a large fraction of their resources in the safe asset, and in equilibrium only a few risky sectors can be operated. In contrast, when there is a significant number of active risky sectors, each household invests a large fraction of its resources in risky assets. This enables more sectors to be open and creates better risk diversification for all households. When such multiple equilibria exist, the equilibrium with more active sectors gives higher ex ante utility to all households. While interesting for illustrating the forces at work, one would expect that financial intermediaries might be successful in avoiding this type of coordination failures. Motivated by this reasoning, let us focus on the part of the parameter space where $Q \geq (2 - \gamma)q$. In that case, the static equilibrium is uniquely defined and the following proposition summarizes this equilibrium.

PROPOSITION 17.8. *Suppose that $Q \geq (2 - \gamma)q$ and that $K(t)$ is given. Then, there exists a unique time t equilibrium in which all sectors $j \leq n^*(t) = n^*[K(t)]$ are open and those $j > n^*[K(t)]$ are shut, where*

$$(17.54) \quad n^*[K(t)] = \frac{(Q + q\gamma) - \{(Q + q\gamma)^2 - 4q[D^{-1}(Q - q)(1 - \gamma)\Gamma K(t)^\alpha + \gamma Q]\}^{1/2}}{2q}$$

if $K(t) \leq D^{1/\alpha}\Gamma^{-1/\alpha}$, and $n^*[K(t)] = 1$ if $K(t) > D^{1/\alpha}\Gamma^{-1/\alpha}$ with Γ defined as $\Gamma \equiv (1 - \alpha)\beta(1 + \beta)^{-1}$. In this equilibrium,

$$s^*(t) = \frac{\beta}{1 + \beta}(1 - \alpha)K(t),$$

and $X^*(t)$ and $I^*(j, t)$ are given by (17.52) and (17.53) with $n^*(t) = n^*[K(t)]$.

PROOF. See Exercise 17.27. □

An important feature is that the equilibrium threshold sector $n^*[K]$ is increasing in K . When there is more capital, the economy is able to open more intermediate sectors. This again contributes to the complementarity in the behavior mentioned above, since eq. (17.53), in turn, implies that when there are more open sectors, investment in each sector will increase.

17.6.3. Equilibrium Dynamics. Let us next turn to the characterization of equilibrium dynamics. Given the static equilibrium in Proposition 17.8, it is straightforward to characterize the full stochastic equilibrium process. The law of motion for the capital stock, $K(t)$, will be given by a simple Markov process. Recall that investments in risky sectors

will be successful with probability $n^*[K(t)]$ when the capital stock is $K(t)$, and it will be unsuccessful with the complementary probability, $1 - n^*[K(t)]$. This implies the following stochastic law of motion for the capital stock:

$$(17.55) \quad K(t+1) = \begin{cases} \frac{q(1-n^*[K(t)])}{Q-qn^*[K(t)]} Q\Gamma K(t)^\alpha & \text{with probability } 1 - n^*[K(t)] \\ Q\Gamma K(t)^\alpha & \text{with probability } n^*[K(t)] \end{cases}$$

where $n^*[K(t)]$ is given by eq. (17.54) and recall that $\Gamma \equiv (1 - \alpha)\beta(1 + \beta)^{-1}$. Notice that the first line of (17.55) is always less than the second line, which reflects the fact that the second line refers to the case in which the investments in the intermediate sectors have been successful.

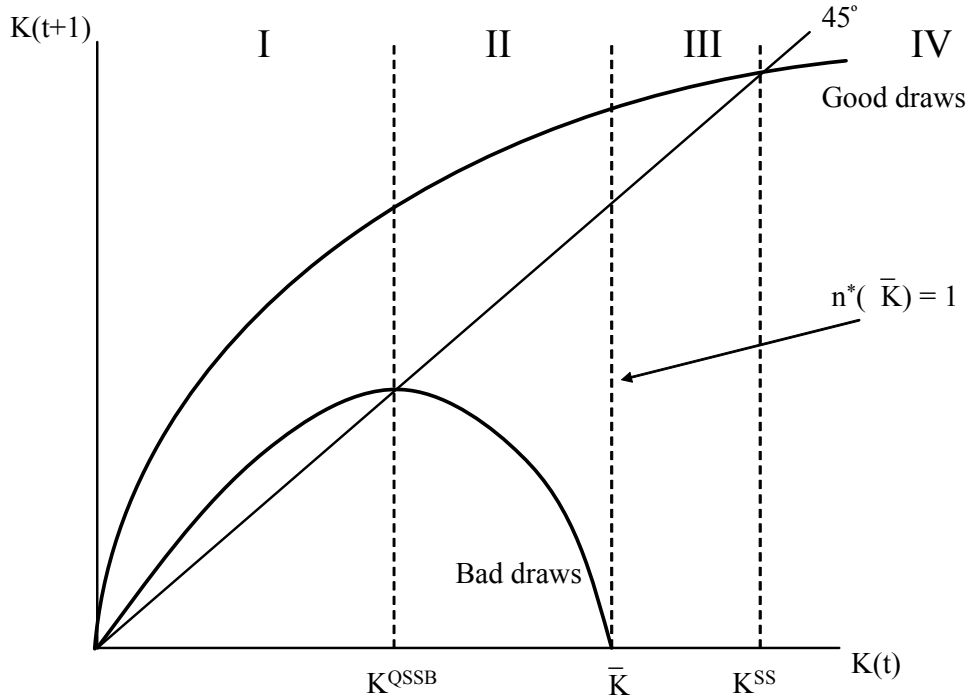


FIGURE 17.4. The stochastic correspondence of the capital stock.

Equation (17.55) is a particularly simple Markov process, since given $K(t)$, $K(t+1)$ can only take two values. However, it is a Markov process not a Markov chain, since for different values of $K(t)$, the possible values of $K(t+1)$ belong to the entire $\mathbb{R}_+$. A diagrammatic analysis of this Markov process is particularly illuminating. Consider Figure 17.4, which plots the stochastic correspondence of the Markov process in (17.55) and is thus similar to Figure 17.1 in the previous section. The main difference is that in Figure 17.1, any value between the two curves for $z_{\min}$ and $z_{\max}$ were possible. In contrast, here, only values exactly on the two curves plotted in the figure are possible. The first curve corresponds to $Q\Gamma K(t)^\alpha$. This is the value of the capital stock that would result if households followed

their equilibrium investment strategies given in (17.52) and (17.53), and at each date, the economy turned out to be lucky, so that their investments always had positive return. The second, inverse U-shaped, curve corresponds to $q(1 - n^*[K(t)])Q\Gamma K(t)^\alpha / (Q - qn^*[K(t)])$ and thus applies if the economy is unlucky at each date. Both curves start above the 45° line near zero for the same reason as that given for the similar pattern in Figure 17.1 (because the aggregate production function (17.38) satisfies the Inada conditions). The economy will be on the upper curve with probability $n^*[K(t)]$ and on the lower curve with probability $1 - n^*[K(t)]$. This implies that not only do the probabilities of success and failure change with the aggregate capital stock but so does average productivity. To quantify this variation in average productivity, let us define *expected total factor productivity* (TFP) conditional on the proportion of intermediate sectors that are open:

$$(17.56) \quad \sigma^e(n^*[K(t)]) = (1 - n^*[K(t)]) \frac{q(1 - n^*[K(t)])}{Q - qn^*[K(t)]} Q + n^*[K(t)] Q.$$

Straightforward differentiation establishes that $\sigma^e(n^*[K(t)])$ is strictly increasing in $n^*[K(t)]$. This implies that as the economy develops and manages to open more intermediate sectors, its productivity increases endogenously. Since $n^*[K]$ is increasing in K , this implies that average productivity is increasing in the capital stock of the economy.

PROPOSITION 17.9. *The expected total factor productivity of the economy $\sigma^e(n^*[K])$ is increasing in n^* and thus increasing in K .*

Inspection of Figure 17.4 also suggests that the following two levels of capital stock are special and useful in the analysis.

- (i): K^{QSSB} refers to the “quasi steady state” of an economy which *always has unlucky draws*. An economy would converge towards this quasi steady state if it follows the optimal investments characterized above, but the sectors chosen by the households never have positive payoff due to bad luck.
- (ii): K^{QSSG} refers to the “quasi steady state” of an economy which *always receives good news*, meaning that it is always on the upper curve in Figure 17.4.

These two capital stock levels are plotted in the figure and are also easy to compute as:

$$(17.57) \quad K^{QSSB} = \left[\frac{q(1 - n^*[K^{QSSB}])}{Q - qn^*[K^{QSSB}]} Q \Gamma \right]^{\frac{1}{1-\alpha}} \quad \text{and} \quad K^{QSSG} = (Q\Gamma)^{\frac{1}{1-\alpha}}.$$

The form of K^{QSSG} is particularly noteworthy, since it refers to the case in which the economy never faces any risk and thus acts very much like a standard neoclassical growth model. In particular, if, in equilibrium, $n^*[K^{QSSG}] = 1$, then K^{QSSG} becomes a proper steady state and the economy would stay at this level of capital stock once it reaches it. This is because once the economy accumulates sufficient capital to open all intermediate sectors, it would eliminate all risk and would always be on the upper curve in Figure 17.4.

Equations (17.54) and (17.57) show that the condition for this good steady state to exist, that is, for $n^*[K^{QSSG}] = 1$, is that the saving level corresponding to K^{QSSG} be sufficient to ensure a balanced portfolio of investments, of at least D , in all the intermediate sectors. It is straightforward to show that the following condition is sufficient to ensure this

$$(17.58) \quad D < \Gamma^{\frac{1}{1-\alpha}} Q^{\frac{\alpha}{1-\alpha}}.$$

Thus when (17.58) is satisfied, the good quasi steady state will indeed generate sufficient capital to open all sectors and eliminate all the risk, thus becoming a proper steady state. In this case, we denote K^{QSSG} by K^{SS} . Under the assumption that (17.58) is satisfied, Figure 17.4 draws $n^*[K^{SS}]$. Now returning to this figure, we can get a better sense of the stochastic dynamics of this equilibrium. The figure divides the range of capital stocks into four regions. In region I, the capital stock is low enough so that both the curves conditional on good draws and bad draws are above the 45° line, so that in this range the economy will grow regardless of whether it experiences good or bad productivity realizations. Next comes region II, which in many ways is the most interesting one. Here the economy grows if it receives positive shocks but suffers a crisis if its investments are unsuccessful. Between these two regions lies the bad quasi steady state K^{QSSB} . The figure justifies the terminology of calling this level of capital stock a “quasi steady state,” since when $K < K^{QSSB}$, the economy will definitely grow towards K^{QSSB} . When $K > K^{QSSB}$, the economy may grow or contract. Nevertheless, as noted above, because $n^*[K]$ is increasing in K , in the right-hand side neighborhood of K^{QSSB} , the economy has the highest probability of contracting (recall that to the left of K^{QSSB} , negative shocks do not lead to a contraction).

For most parameter values, the economy tends to spend a long time in region II. Acemoglu and Zilibotti (1997) provide examples where the number of periods in which the economy is in regions I and II could be arbitrarily large. However, if the economy were to receive a sequence of good news, it would ultimately exit from region II and enter region III. The level of capital stock that divides these two regions, $\bar{K}$, is defined such that $n^*[\bar{K}] = 1$. This means that once the economy reaches the capital stock of $\bar{K}$, it has enough capital to open all the sectors. Consequently, in region III all risk is diversified and the dynamics are exactly the same as those of the canonical overlapping generations model without uncertainty. Finally, starting anywhere in region III, the economy travels towards the steady state K^{SS} , which stands between regions III and IV. Region IV, on the other hand, has so much capital that even with the positive shocks, the economy will contract. Naturally, unless it starts there, the economy will never enter region IV.

This discussion, combined with Figure 17.4, gives a fairly complete characterization of the stochastic equilibrium growth path. In particular, an economy that starts with a low enough capital stock will first experience some growth, but then spend a long time fluctuating between successful periods and periods of severe crises. Eventually, a string of good news will take the economy to a level of capital stock such that much (here all) of the risks can be diversified. At this level, we can think of the economy as achieving *takeoff* as in Rostow’s account discussed

in Chapter 1. The economy is experiencing a takeoff in two senses. First, after takeoff it successfully diversifies all risk, so that growth from this point onwards progresses steadily rather than being subject to significant fluctuations as in region II. Second, Proposition 17.9 implies that the aggregate (labor and total factor) productivity will increase after this level of capital. Thus takeoff comes with a decline in the fluctuations in economic activity and an increase in productivity.

In addition, as the economy develops by accumulating more capital, it achieves both higher productivity and better diversification, and manages its risks better. This takes the form of more sectors being open, which equivalently corresponds to more financial intermediaries being active. Thus in this model financial and economic development go hand-in-hand. In this respect, it is important to emphasize that in the model it is neither economic growth that causes financial development nor financial development that causes economic growth. Both are determined jointly and affect each other along the equilibrium path.

A natural question is whether the economy will necessarily reach region III and then region IV. The next proposition answers this question.

PROPOSITION 17.10. *Suppose that condition (17.58) holds, then the stochastic process $\{K(t)\}_{t=1}^{\infty}$ converges to the point K^{SS} with probability 1.*

PROOF. See Exercise 17.28. □

This proposition establishes that the variability of growth in the economy will *eventually* decline (and in fact disappear). But one might wish to know whether the amplitudes of economic fluctuations are systematically related to the level of the capital stock or output in the economy. This is particularly relevant, since, as already discussed, both cross-sectional and time-series comparisons suggest that poorer nations suffer from greater economic variability. To answer this question, the natural variable to look at is the conditional variance of TFP (whose expected value was defined in (17.56) above). Define $\sigma(n^*[K(t)])$ as a random variable that takes the values $q(1 - n^*[K(t)])Q / (Q - qn^*[K(t)])$ and Q with respective probabilities $(1 - n^*[K(t)])$ and $n^*[K(t)]$. The expectation of this random variable is $\sigma^e(n^*[K(t)])$ as defined in (17.56). Then, taking logs, we can rewrite (17.55) as

$$(17.59) \quad \Delta \log(K(t+1)) = \log \Gamma - (1 - \alpha) \log(K(t)) + \log(\sigma(n^*[K(t)])) .$$

It is clear from this equation that capital (and output) growth volatility, after removing the deterministic “convergence effects” due to the standard neoclassical effects, are determined by the stochastic component σ . Denoting the (conditional) variance of $\sigma(n^*[K(t)])$ given $K(t)$ by V_n , we can state the following proposition.

PROPOSITION 17.11. *Let $\mathcal{V}_n \equiv \text{Var}(\sigma(n^*) | n^*) = n^*(1 - n^*)[Q(Q - q) / (Q - qn^*)]^2$. Then,*

- *If $\gamma \geq Q / (2Q - q)$, then $\partial \mathcal{V}_n / \partial K \leq 0$ for all $K \geq 0$.*

- If $\gamma < Q/(2Q - q)$, then there exists $\tilde{K}$ defined such that $n^*(\tilde{K}) = Q/(2Q - q) < 1$ and.

$$\begin{aligned}\frac{\partial \mathcal{V}_n}{\partial K} &\leq 0 \text{ for all } K \geq \tilde{K} \\ \frac{\partial \mathcal{V}_n}{\partial K_t} &> 0 \text{ for all } K < \tilde{K}.\end{aligned}$$

PROOF. See Exercise 17.29. $\square$

The behavior of the variability of growth in this proposition results from the counteracting effects of two forces; first, as the economy develops, more savings are invested in risky assets; and second, as more sectors are opened, idiosyncratic risks are better diversified. The proposition shows that if $\gamma \geq Q/(2Q - q)$, the second effect always dominates and thus the richer economies are less risky. If $\gamma < Q/(2Q - q)$, then the first effect dominates for sufficiently low levels of capital stock, but once the capital stock reaches a critical threshold, $\tilde{K}$, the second effect again dominates. Thus except for sufficiently low levels of capital, the variability of the growth rate is everywhere decreasing in the income (or capital) level of the economy.

17.6.4. Efficiency. The previous subsection completely characterized the stochastic equilibrium of the economy. Is this equilibrium Pareto efficient? Since all households are price takers, it may be conjectured that the answer to this question must be yes. In this subsection I show that this is not the case. Though at first surprising, this result will turn out to be intuitive and interesting. First, it results from an economically meaningful *pecuniary externality*. Second, it makes sense from the viewpoint of the theory of general equilibrium; though all households are price takers, this is not an Arrow-Debreu economy because the set of traded commodities is determined endogenously by a zero profit condition. To illustrate these issues in the most transparent way I ignore any potential source of intertemporal inefficiency (which, we know from Chapter 9, may arise in overlapping generations economies). For this reason, the analysis of efficiency takes a particular level of savings $s(t)$, or equivalently the current level of the capital stock $K(t)$, as given and looks at whether the way in which savings are allocated across different sectors of the economy is (constrained) efficient. This can be done by considering the social planner's problem. This problem can be written as:

$$(17.60) \quad \max_{n(t), X(t), [I(j,t)]_{0 \leq j \leq n(t)}} \int_0^{n(t)} \log(qX(t) + QI(j,t))dj + (1 - n(t)) \log(qX(t))$$

subject to

$$X(t) + \int_0^{n(t)} I(j,t) dj \leq s(t).$$

More specifically, the social planner chooses the set of sectors that are active, which is denoted by $[0, n(t)]$, the amount that will be invested in the safe sector $X(t)$ and the allocation of funds among the other sectors denoted by $[I(j,t)]_{0 \leq j \leq n(t)}$. In principle, the social planner could have chosen the set of active sectors not to be an interval of the form $[0, n(t)]$, but the

same argument as in Exercise 17.25 ensures that there is no loss of generality in imposing this form. The constraint makes sure that the sum of investments in the safe and the risky sectors is less than the amount of savings available to the planner. The main difference between this program and the maximization problem of the representative household, (17.44), is that the social planner also chooses $n(t)$, while the representative household took the set of available assets as given. The social planner's allocation (and thus the Pareto optimal allocation) is given by the solution to this maximization problem. The next proposition characterizes the solution.

PROPOSITION 17.12. *Let $n^*[K(t)]$ be given by (17.54), and $s(t)$ and $K(t)$ denote current level of savings and capital stock available to the social planner. Then, the unique solution to the maximization problem in (17.60) is as follows:*

- For all $s(t) < D$, the set of active sectors is given by $[0, n^S[K(t)]]$, where $n^S[K(t)] > n^*[K(t)]$. The amount of investment in the safe sector is given by $X^S(t)$, where $X^S(t) < X^*(K(t))$. Finally, there exists a sector $j^*(t) \in (0, n^S[K(t)])$ such that the portfolio of risky sectors for each household takes the form

$$(17.61) \quad \begin{aligned} I^S(j, t) &= M(j^*) > M(j) && \text{for } j < j^*(t) \\ I^S(j, t) &= M(j) && \text{for } j \in [j^*(t), n^S[K(t)]] \\ I^S(j, t) &= 0 && \text{for } j > n^S[K(t)] \end{aligned}$$
- For all $s(t) \geq D$, $n^S[K(t)] = n^*[K(t)] = 1$ and $I^S(j, t) = s(t)$ for all $j \in [0, 1]$.

PROOF. See Exercise 17.30. □

This proposition implies that, when the economy has not achieved full diversification, the social planner will open more sectors than the decentralized equilibrium. She will finance these additional sectors by deviating from the *balanced portfolio*, which was always a feature of the equilibrium allocation. In other words, she will invest less in the sectors without the minimum size requirement. The Pareto optimal allocation of funds is shown in Figure 17.5.

The deviation from the balanced portfolio implies that the social planner is implicitly *cross-subsidizing* the sectors with high a minimum size requirements at the expense of sectors with low or no minimum size requirements. This is because, starting with a balanced portfolio, opening a few more sectors always benefits all households, who will be able to achieve better risk diversification. The only way the social planner can achieve this is by implicitly taxing sectors that have low or no minimum size requirements (so that they have lower investments) and subsidizing the marginal sectors with high minimum size requirements.

Why does the decentralized equilibrium not achieve the same allocation? There are two complementary ways of providing the intuition for this. The first is that a marginal dollar of investment by a household in a high minimum size requirements sector creates a *pecuniary externality*, because this investment makes it possible for the sector to be active and to provide better risk diversification possibilities to all the other households. However, each household, taking equilibrium prices as given, ignores this pecuniary externality and tends

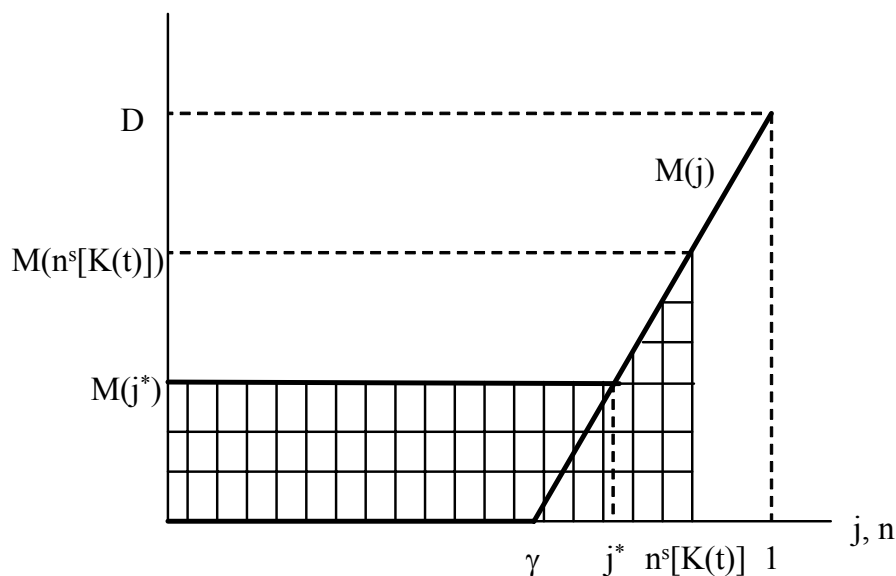


FIGURE 17.5. The efficient portfolio allocation.

to underinvest in marginal sectors with high minimum size requirements. Thus the source of inefficiency is that each household ignores its impact on others' diversification opportunities. The second intuition for this result is related. Because households take the set of prices as given and in equilibrium $P(j, t) = 1$ for all active sectors, they will always hold a balanced portfolio. However, the Pareto optimal allocation involves cross-subsidization across sectors in a non-balanced portfolio. Market prices do not induce the households to hold the right portfolio.

At this point, the reader may wonder why the First Welfare Theorem does not apply in this environment (especially since all households are price takers). The reason why the First Welfare Theorem does not apply is that the decentralized equilibrium here does not correspond to an Arrow-Debreu equilibrium. In particular, this is an equilibrium for an economy with endogenously incomplete markets, where the set of active markets is determined by zero profit (free-entry) condition. All commodities that are traded in equilibrium are priced competitively, but there is no "competitive pricing" for commodities that are not traded. Instead, in an Arrow-Debreu equilibrium, all commodities, even those that are not traded in equilibrium, are priced and in fact a potential commodity would not be traded in equilibrium only if its price were equal to zero and at zero prices generated excess supply. In this sense, the equilibrium characterized here is not an Arrow-Debreu equilibrium. In fact, it can be verified that such an equilibrium does not exist in this economy because of the nonconvexity in the production set. Instead, the equilibrium concept used here is a more natural competitive equilibrium notion; it requires that all commodities that are traded in equilibrium are priced competitively and then determines the set of traded commodities by

a free-entry condition. Some additional discussion of this equilibrium concept is provided in the References and Literature section below

17.6.5. Inefficiency with Alternative Market Structures. Would the market failure in portfolio choices be overcome if some financial institution could coordinate households' investment decisions? Imagine that rather than all households acting in isolation and ignoring their impact on each others' decisions, funds are intermediated through a financial coalition-intermediary. This intermediary can collect all the savings and offer to each saver a *complex* security (as different from an Arrow security) that pays $QI^S(j, t) + qX^S(t)$ in each state j , where $I^S(j, t)$ and $X^S(t)$ are as in the optimal portfolio. Holding this security would make each consumer better-off compared to the equilibrium.

Although from this discussion it may appear that the inefficiency identified here may not be robust to the formation of more complex financial institutions, this is not the case. The remarkable result is that unless some rather strong assumptions are made about the set of contracts that a financial intermediary can offer, equilibrium allocations resulting from competition among intermediaries will be identical to the equilibrium allocation in Proposition 17.8. A full analysis of this issue is beyond the scope of the current book, but a brief discussion gives the flavor. Let us model more complex financial intermediaries as "intermediary-coalitions," that is, as a set of households who join their savings together and invest in a particular portfolio intermediate sectors. Such coalitions may be organized by a specific household, and if it is profitable for other households to join the coalition, the organizer of the coalition can charge a premium (or a joining fee) thus making profits. Let us assume that there is free entry into financial intermediation or coalition-building, so that any household can attempt to exploit profit opportunities if there are any. Let us also impose some structure on how the timing of financial intermediation works and also how individual households can participate into different coalitions. In particular, let us adopt the following assumptions.

- (1) Coalitions maximize a weighted utility of their members at all points in time. In particular, a coalition cannot commit to a path of action that will be against the interests of its members in the continuation game.
- (2) Coalitions cannot exclude other households (or coalitions) from investing in a particular project.

Acemoglu and Zilibotti (1997) then prove the following result.

PROPOSITION 17.13. *The set of equilibria of the financial intermediation game described above is always nonempty and all equilibria have exactly the same structure as those characterized in subsection 17.6.2 and Proposition 17.8.*

I will not provide a proof of this proposition, since a formal statement of the proposition and the proof require additional notation. But the intuition is straightforward: as shown in Proposition 17.12, the Pareto optimal allocation involves a non-balanced portfolio and cross-subsidization across different sectors. This implies that the shadow price of investing in some

sectors should be higher than in others, even though the cost of investing in each sector is equal to 1 (in terms of date t final goods). These differences in shadow prices will then support a non-balanced portfolio. Recall also that it is the sectors with no minimum size requirement or low minimum size requirements that are being implicitly taxed in this allocation. This kind of cross-subsidization is difficult to sustain in equilibrium because each household would deviate towards slightly reducing its investments in coalitions/intermediaries that engage in cross-subsidization and undertake investments on the side to move its portfolio towards a balanced one (by investing in no minimum size or low minimum size sectors). At the end, only allocations without cross-subsidization can survive as equilibria, and those are identical to equilibria characterized in subsection 17.6.2.

The most important implication of this result is that even with unrestricted financial intermediaries or coalitions, the inefficiency resulting from endogenously incomplete markets cannot be prevented. The key economic force is that each household creates a positive pecuniary externality by holding a non-balanced portfolio but in a decentralized equilibrium each household wishes to and can easily move towards a balanced portfolio, undermining efforts to sustain the efficient allocation.

17.7. Taking Stock

This section presented a number of different models of stochastic growth. My selection of topics was geared towards achieving two objectives. First, I introduced a number of workhorse models of macroeconomics, such as the neoclassical growth model under uncertainty and the basic Bewley model. These models are not only useful for the analysis of economic growth but also have a wide range of applications in the macroeconomics literature.

Second, the model in Section 17.6 demonstrated how stochastic models can significantly enrich the analysis of economic growth and economic development. In particular, this model showed how a simple extension of our standard models can generate an equilibrium path in which economies spend a long time with low productivity and suffer frequent crises. They take off into sustained and steady growth once they receive a sequence of favorable realizations. The takeoff process not only reduces volatility and increases growth, but is also associated with better management of risk and greater financial development. Though stylistic, this model provides a good approximation to the economic development process that much of Western Europe underwent over the past 700 years or so. It also emphasizes the possibility that luck may have played an important role in the timing of takeoff and perhaps even in determining which countries were early industrializers. Therefore, this model provides an attractive formalization of the luck hypothesis discussed in Chapter 4. Nevertheless, underlying the equilibrium in this model is a set of market institutions that enable households to trade in competitive markets and to invest freely. Thus my interpretation would be that the current model shows how random elements and luck can matter for the timing of takeoff among countries that satisfy the major prerequisites for takeoff. This could account for some of the current-day cross-country income differences and may provide important insights about

the beginning of the process of sustained growth. However, institutional factors—whether those prerequisites are satisfied—are more important for understanding why some parts of the world did not takeoff during the 19th century and have not yet embarked on a path of sustained and steady growth. These are topics that will be discussed in the rest of the book.

It is also worth noting that the model in Section 17.6 introduces a number of important ideas related to incomplete markets. The Bewley model presented in Section 17.4 is a prototypical incomplete markets model, and as with most incomplete markets models in the literature, it takes the set of markets that are open as given. In contrast, the model in Section 17.6 is a model with endogenously incomplete markets. The analysis showed that the fact that the set of markets that are open (the set of sectors that are active) is determined in equilibrium with a free-entry condition can lead to a novel type of Pareto inefficiency due to pecuniary externalities (even though all households take prices as given). Although this type of Pareto inefficiency is different from those highlighted so far, there are some important parallels between the fact that an insufficient number of markets are open in this model and too few intermediate goods being produced in the baseline endogenous technological change model of Chapter 13.

17.8. References and Literature

The neoclassical growth model under uncertainty, presented in Section 17.1, was first analyzed by Brock and Mirman (1972). Because the analysis of the social planner's problem is considerably easier than the study of equilibrium growth under uncertainty, most analyses in the literature look at the social planner's problem and then appeal to the Second Welfare Theorem. Stokey, Lucas and Prescott (1989) provide an example of this approach. An analysis of the full stochastic dynamics of this model requires a more detailed discussion of the general theory of Markov processes. Space restrictions preclude me from presenting these tools. The necessary material can be found in Stokey, Lucas and Prescott (1989, Chapters 8, 11, 12 and 13) or the reader can look at Futia (1982) for a more compact treatment. More advanced and complete treatments are presented in Ethier and Kurtz (1985) or Gikhman and Skorohod (1974). The tools in Stokey, Lucas and Prescott (1989) are sufficient to prove that the optimal path of capital-labor ratio in the neoclassical growth model under uncertainty converges to a unique invariant distribution and they can also be used to prove the existence of a stationary equilibrium in the Bewley economy.

The first systematic analysis of competitive equilibrium under uncertainty is provided in Lucas and Prescott (1971) and Mehra and Prescott (1979). Sargent and Ljungqvist (2004, Chapter 12) provides a good textbook treatment. The material in Section 17.2 is similar to that in Sargent and Ljungqvist, but is more detailed and provides a few additional results.

The Real Business Cycle literature is enormous and the treatment in Section 17.3 only scratches the surface. The classic papers in this literature are Kydland and Prescott (1982) and Long and Plosser (1983). Sargent and Ljungqvist (2004) again provides a good introduction. The collection of papers in Cooley and Prescott (1995) is an excellent starting point,

emphasizing the achievements of the RBC literature and providing a range of tools for theoretical as well as quantitative analysis using recursive competitive models. Blanchard and Fischer (1989) discusses the critiques of the RBC approach. The interested reader is also referred to the exchange between Edward Prescott and Lawrence Summers (Prescott, 1986, and Summers, 1986) and to the review of the more recent literature in King and Rebelo (1999).

Section 17.4 presents the incomplete markets model first introduced by Truman Bewley (1977, 1980, 1986). This model has become one of the workhorse models of macroeconomics and has been used for analysis of business cycle dynamics, income distribution, optimal fiscal policy, monetary policy and asset pricing. A more modern treatment is provided in Aiyagari (1994), though the published version of the paper does not contain any of the mathematical analysis. The reader is referred to Bewley (1977, 1980) and to the working paper version, Aiyagari (1993), for more details on some of the propositions stated in Section 17.4 as well as a proof of existence of a stationary equilibrium, which I did not provide in the text. Krusell and Smith (1998, 1999, 2005), among others, have used this model for business cycle analysis and have also provided new quantitative tools for the study of incomplete market economies.

Section 17.5 is a simple stochastic extension of the baseline overlapping generations model. I am not aware of any similar treatment, though none of the material in this section is new or difficult.

Section 17.6 builds on Acemoglu and Zilibotti (1997) and more details on some of the results stated in that section are provided in Acemoglu and Zilibotti (1997). Evidence on the relationship between economic development and volatility is provided in Acemoglu and Zilibotti (1997), Imbs and Wacziarg (2003) and Koren and Tenreyro (2007). Ramey and Ramey (1994) also provide related evidence, though they emphasize the effect of volatility on growth using cross-country regression analysis rather than the theoretical interactions between growth and volatility. As noted above, the concept of decentralized equilibrium used in this model is not Arrow-Debreu. Instead it imposes price-taking behavior in all open markets and determines the set of open markets via a free-entry condition. This type of equilibrium concept is commonly used in general equilibrium theory, for example, Hart (1980), Makowski (1980), and Allen and Gale (1991). Koren and Tenreyro (2007) present a generalization of the Acemoglu and Zilibotti (1997) model. Acemoglu and Zilibotti (1997) also contain an analysis of international capital flows in a similar framework and this analysis is further extended in Martin and Rey (2002).

17.9. Exercises

EXERCISE 17.1. Proposition 17.2 shows that $k(t+1)$ is increasing in $k(t)$ and $z(t)$. Provide sufficient conditions such that $c(t)$ is also increasing in these variables.

EXERCISE 17.2. Consider the neoclassical growth model under uncertainty analyzed in Section 17.1 and assume that $z(t)$ is realized after $c(t)$ and $k(t+1)$ are chosen.

- (1) Show that if $z(t)$ is distributed independently across periods, the choice of capital stock and consumption in this economy is *identical* to that in a neoclassical growth model under certainty with a modified production function. Explain this result.
- (2) Now suppose that $z(t)$ is not distributed independently across periods, Establish the equivalent of Proposition 17.1 for this case. How does the behavior in this economy differ from the canonical version of the neoclassical growth model under uncertainty in Section 17.1.

EXERCISE 17.3. Consider the same production structure as in Sections 17.1 and 17.2 but assume that regardless of the level of the capital stock and the realization of the stochastic variable, each household saves a constant fraction s of its income. Characterize the stochastic laws of motion of this economy. How does behavior in this economy differ from that in the canonical neoclassical growth model under uncertainty.

EXERCISE 17.4. Consider the neoclassical growth model under uncertainty studied in Section 17.1.

- (1) Provide conditions under which $\pi(k, z)$ is strictly increasing in both of its arguments.
- (2) Show that when this is the case, the capital-labor ratio can never converge to a constant unless z has a degenerate distribution (that is, it always takes the same value).

EXERCISE 17.5. Consider Example 17.1.

- (1) Prove that eq. (17.10) cannot be satisfied for any $B_0 \neq 0$.
- (2) Conjecture that the value function for this example takes the form $V(k, z) = B_2 + B_3 \log k + B_4 \log z$. Verify this guess and compute the parameters B_2 , B_3 and B_4 .

EXERCISE 17.6. Show that the policy function in Example 17.1, $\pi(k, z) = \beta \alpha z k^\alpha$, applies when z follows a general Markov process rather than a Markov chain. [Hint: instead of the summation, replace the expectations sign with an appropriately defined (Lebesgue) integral and cancel terms under the integral sign].

- EXERCISE 17.7. (1) Consider the economy analyzed in Example 17.1 with $0 < z_1 < z_N < \infty$. Characterize the limiting invariant distribution of the capital-labor ratio and show that the stochastic correspondence of the capital stock can be represented by Figure 17.1 in Section 17.5. Use this figure to show that the capital-labor ratio, k , will always grow when it is sufficiently small and always decline when it is sufficiently large.
- (2) Next consider the special case where z takes two values z_h and z_l , with each value persisting with probability $q > 1/2$ and switching to the other value with probability $1 - q$. Show that as $q \rightarrow 1$, the behavior of the capital-labor ratio converges to the equilibrium behavior of the same object in the neoclassical growth model under certainty.

EXERCISE 17.8. Consider the economy studied in Example 17.1, but suppose that $\delta < 1$. Show that in this case there does not exist a closed-form expression for the policy function $\pi(k, z)$.

EXERCISE 17.9. Consider an extended version of the neoclassical growth model under uncertainty such that the instantaneous utility function of the representative household is $u(c, b)$, where b is a random variable following a Markov chain.

- (1) Setup and analyze the optimal growth problem in this economy. Show that the optimal consumption sequence satisfies a modified stochastic Euler equation.
- (2) Prove that Theorem 5.7 can be applied to this economy and the optimum growth path can be decentralized as a competitive growth path.

EXERCISE 17.10. Write the maximization problem of the social planner explicitly as a sequence problem, with output, capital and labor following different histories interpreted as different Arrow-Debreu commodities. Using this formulation, show that the conditions of Theorem 5.7 are satisfied, so that the optimal growth path can be decentralized as an equilibrium growth path.

EXERCISE 17.11. Explain why in subsection 16.5.1 in the previous chapter, the Lagrange multiplier $\tilde{\lambda}[y^t]$ was conditioned on the entire history of labor income realizations, while in the formulation of the competitive equilibrium with a full set of Arrow-Debreu commodities (contingent claims) in Section 17.2, there is a single multiplier λ associated with the lifetime budget constraint.

EXERCISE 17.12. Consider the model of competitive equilibrium in Section 17.2. Repeat the analysis of the competitive equilibrium of the neoclassical growth model under uncertainty by assuming that instead of a price for buying and selling capital goods in each state (at the price sequence in terms of date 0 final good given by $R_0[z^t]$), there is a market for renting capital goods. Let the rental price of capital goods in terms of date 0 final good be $r_0[z^t]$ when the sequence of stochastic variables is z^t . Characterize the competitive equilibrium and show that it is equivalent to that obtained in Section 17.2. Explain why the two formulations give identical results.

EXERCISE 17.13. Prove Proposition 17.3. [Hint: use Theorem 16.8 together with (17.6) and (17.21) and then show that the lifetime budget constraint of the representative household, (17.11), implies (17.7)].

EXERCISE 17.14. Characterize the competitive equilibrium path of the neoclassical growth model under uncertainty analyzed in Section 17.2 using sequential trading, but the sequence problem formulation rather than the recursive formulation.

EXERCISE 17.15. Show that Theorems 16.1-16.7 can be applied to $V(a, z)$ defined in (17.22) and establish that $V(a, z)$ is continuous, strictly increasing in both of its arguments, concave and differentiable in a .

EXERCISE 17.16. Derive eq. (17.24).

EXERCISE 17.17. Prove Proposition 17.4.

EXERCISE 17.18. Consider the RBC model presented in Section 17.3 and suppose that the production function takes the form $F(K, zAL)$, with both z and A corresponding to labor-augmenting technological productivity terms. Suppose that z follows a Markov chain and $A(t+1) = (1+g)A(t)$ is an exogenous and deterministic productivity growth process. Setup the social planner's problem in this case. [Hint: use a transformation of variables to make the recursive equation stationary]. What restrictions do we need to impose on $U(C, L)$ to ensure that the optimal growth path corresponds to a "balanced growth path," where labor supply does not (with probability 1) go to zero or infinity?

EXERCISE 17.19. In Example 17.2, suppose that the utility function of the representative household is $u(C, L) = \log C + h(L)$, where $h(\cdot)$ is a continuous, decreasing and concave function. Show that the equilibrium level of labor supply is constant and independent of the level of capital stock and the realization of the productivity shock.

EXERCISE 17.20. Explain why in the Bewley-Aiyagari model of Section 17.4, the budget constraint of the household must hold along all sample paths, and compare the resulting constraint, (17.27) to (17.11) in Section 17.2.

EXERCISE 17.21. Prove Proposition 17.5.

EXERCISE 17.22. Prove Proposition 17.6.

EXERCISE 17.23. What would happen if, instead of the logarithmic preferences (17.41), the utility function of the representative household in Section 17.6 took the more general form $u(c_1(t)) + \mathbb{E}_t u(c_2(t+1))$? Could the growth rate of the economy be higher when $u(\cdot)$ becomes more concave? [Hint: distinguish between the effect of $u(\cdot)$ on asset allocation given the level of savings and its effect on the total amount of savings].

EXERCISE 17.24. In the model of Section 17.6, prove that the maximization problem of the representative household implies that for any $j, j' \in J(t)$, $I^*(j, t) = I^*(j', t)$.

EXERCISE 17.25. In the model of Section 17.6, prove that if an intermediate sector $j^* \in J(t)$, then all sectors $j \leq j^*$ are also in $J(t)$.

EXERCISE 17.26. In the model of Section 17.6, prove that the condition $Q \geq (2 - \gamma)q$ is sufficient to ensure that there is a unique intersection between the curves for (17.40) and (17.53) in Figure 17.2.

EXERCISE 17.27. Prove Proposition 17.8. In particular, show that (i) if the equilibrium $n < n^*[K]$, then there exists a profitable deviation for a financial intermediary to offer securities based on a previously-unavailable sector and make positive profits; and (ii) if $n > n^*[K]$, feasibility is violated.

EXERCISE 17.28. (1) Prove Proposition 17.10.

(2) Suppose that condition (17.58) is not satisfied. Does the stochastic process $\{K(t)\}_{t=0}^\infty$ converge? Does it converge to a point?

EXERCISE 17.29. Prove Proposition 17.11.

EXERCISE 17.30. Prove Proposition 17.12. [Hint: setup the Lagrangian for the social planner and show that when all sectors cannot be opened, then the social planner will not choose a balanced portfolio.]

EXERCISE 17.31. * Consider the following two-period economy similar to the environment described in Section 17.6. There are I financial intermediaries who compete a la Bertrand without using any resources. They invest funds on behalf of households in any of the projects of this economy. There are N projects/assets in this economy indexed by $j = 1, 2, \dots, N$. Asset j requires a minimum size investment $M(j)$ and without loss of generality rank the projects in ascending order of minimum size.

There is continuum of consumers with measure normalized to 1, each with the utility function $u(c) + \mathbb{E}v(c')$, where c is consumption today, c' is consumption tomorrow, so that $\mathbb{E}v(c')$ denotes expected utility from tomorrow's consumption. Each consumer has total resources equal to w and decides how much to consume and how much to save and then how to allocate his savings. Assume that $u(\cdot)$ and $v(\cdot)$ are strictly concave and increasing. Funds today are turned into consumption tomorrow by financial intermediaries. Alternatively, funds can also be invested in a safe linear technology with rate of return q . Let the investment in asset j be $K(j)$, then if $K(j) \geq M(j)$, then asset j has probability π of paying out $Qk(j)$ such that $\pi Q > q$ (thus the safe technology is less productive). On the other hand if $K(j) < M(j)$, the pay-out is equal to zero.

- (1) Denote the “share price” of \$1 invested in project j , which pays out $\$Q$ with probability π and zero otherwise, by $p(j)$. Show that financial competition ensures that if $K(j), K(j') > 0$, then $p(j) = p(j') = 1$.
- (2) Now assume that the returns of each project is *independently drawn*, that is, the probability that asset j pays out Q is equal to π independent of the realization of the returns of other projects. Show that $K(j) = K(j')$ for all j and j' .
- (3) Characterize the decentralized equilibrium of this economy.
- (4) Show that when some projects are inactive, the decentralized equilibrium may be constrained Pareto inefficient. Explain why the decentralized equilibrium may be constrained efficient in some cases even though some projects are inactive.
- (5) Characterize the efficient allocation.
- (6) Can you establish the inefficiency of the decentralized equilibrium without independence?
- (7) Informally discuss what happens if $M(j)$ is not a minimum size requirement but a fixed cost (such that average costs are falling). [Hint: there are two cases to distinguish; (1) linear prices; (2) price discrimination].

Part 6

Technology Diffusion, Trade and Interdependences

One of the most major shortcomings of the models presented so far is that each country is treated as an “isolated island,” not interacting with the rest of the countries in the world. This is problematic for at least two reasons. The first is related to the technological interdependences across countries and the second to international trade (in commodities and in assets). In this part of the book, I will investigate the implications of technological and trade interdependences on the process of economic growth.

The models presented so far treat technology either as exogenous or as endogenously generated within the boundaries of the economy in question. We have already seen how allowing for endogeneity of technology provides new and important insights about the process of growth. But should we think of the potential technology differences between Portugal and Nigeria as resulting from lower R&D in Nigeria? The answer to this question is most probably no. Nigeria, like most less-developed or developing countries, imports many of its technologies from the rest of the world. The same is likely to be the case for Portugal despite its substantially more developed economy. This suggests that a framework in which *frontier* technologies in the world are produced in the United States or other advanced economies and then copied or adopted by other “follower” countries provides a better approximation to reality. Therefore, to understand technology differences between advanced and developing economies, we should not only, or not even primarily, focus on differential rates of endogenous technology generation in these economies, but on their decisions concerning *technology adoption* and *efficient technology use*.

While the exogenous growth models of Chapters 2 and 8 have this feature, they too have important shortcomings. First, technology is entirely exogenous, so interesting economic decisions only concern investment in physical capital. There is a conceptually and empirically compelling sense in which technology is different from physical capital (and also from human capital), so we would like to understand sources of differences in technology arising endogenously across countries. Thus the recognition that technology adoption from the world frontier matters is not the same as accepting that the Solow or the neoclassical growth model are the best vehicle for studying cross-country income differences. Second, while the emphasis on technology adoption makes the process of growth resemble the exogenous growth models of Chapters 2 and 8, technological advances at the world level are unlikely to be “manna from heaven”. Instead, economic growth at the world level either results from the interaction of the adoption and R&D decisions of all countries or perhaps from the innovations by frontier economies. This implies that models in which the growth rate at the world level is endogenous and interacts (and coexists) with technology adoption may provide a better approximation to reality and a better framework for the analysis of the mechanics of economic growth. In addition to technology adoption, other interactions across countries, such as international trade, may also play the same role of allowing for endogenous growth at the world level together with growth in each specific country that depends on technological and other developments at the world level.

In Chapter 18, I start with models of technology adoption and investigate the factors affecting the speed and nature of technology adoption. In addition to factors slowing down technology diffusion and the importance of barriers against new technologies, I will discuss the role of whether technologies from the world frontier are *appropriate* for the needs of less-developed countries. Recall also that “technology differences” not only reflect differences in techniques used in production, but also differences in the organization of production affecting the efficiency with which existing factors of production are utilized. A satisfactory theory of technology differences among countries must therefore pay attention to barriers to technology adoption and to potential inefficiencies in the organization of production, leading to apparent technology differences across countries. In Chapter 18, I also provide a simple model of inefficient technology adoption resulting from contracting problems among firms.

The second major element missing from our analysis so far, international trade and international capital flows, will be addressed in Chapter 19. International trade in commodities and assets link the economic fortunes of the countries in the world as well. For example, economies with low capital-labor ratios may be able to borrow internationally, which would change equilibrium dynamics. Similarly and perhaps more importantly, less productive countries that export certain goods to the world economy will be linked with other economies because of changes in relative prices—because of changes in their “terms of trade”. This type of terms-of-trade effects may also work towards creating a framework in which, while the world economy grows endogenously, the growth rates of each country is linked to those of others through trading relationships. Finally, I will emphasize the connections between international trade and technology adoption, in particular, emphasizing how trade and the “international product cycle” facilitate technology diffusion.

Throughout the rest of the book, including this part, I will be somewhat less comprehensive than in the previous chapters. In particular, to economize on space I will be more selective in the range of models covered, focusing on the models which I believe provide the main insights in an economical fashion. I will leave some of the alternative models that also relate to the economic issues under consideration to the discussion of the literature at the end or to exercises. In addition, I will make somewhat greater use of simplifying assumptions and I will leave the proofs of results that are similar to those provided so far and the relaxation of some of the simplifying assumptions to exercises.

CHAPTER 18

Diffusion of Technology

In many ways, the problem of innovation ought to be harder to model than the problem of technology adoption. Nevertheless, the literature on economic growth and development has made more progress on models of innovation, such as those we discussed in Chapters 13-15, than on models of technology diffusion. This is in part because the process of technology adoption involves many challenging features. First, even within a single country, we observe considerable differences in the technologies used by different firms in the same narrowly-defined industry. Second and relatedly, it is difficult to explain how in the globalized world which we live in some countries may fail to import and use technologies that would significantly increase their productivity. In this chapter, I begin the study of these questions. Since potential barriers to technology adoption are intimately linked to the analysis of the political economy of growth, I will return to some of these themes in Part 8 of the book. For now the emphasis will be on how technological interdependencies change the mechanics of economic growth and can thus enrich our understanding of the potential sources of cross-country income differences and economic growth over time.

I will first provide a brief overview of some of the empirical patterns pertaining to technology adoption and diffusion within countries and industries, and how this appears to be important for within-industry productivity differences. I will then turn to a benchmark model of world equilibrium with technology diffusion, which will provide a reduced-form model for analyzing the slow diffusion of technological know-how across countries. I will then enrich this model by incorporating investments in R&D and technology adoption. Next, I will discuss issues of appropriate technology, and finally, I will turn to the impact of contractual imperfections on technology adoption decisions. Throughout this chapter, the only interaction among the countries in the world will be through technological exchange, and there will be no international trade in assets or in commodities.

18.1. Productivity Differences and Technology

Let us first start with a brief overview of productivity and technology differences within countries. This overview will help us place the cross-country differences in productivity and technology into perspective. The most important lesson from the within-country studies is that productivity and technology differences are ubiquitous even across firms within narrow sectors in the same country.

18.1.1. Productivity and Technology Differences within Narrow Sectors. A large literature uses longitudinal micro-data (often for the manufacturing sector) to study labor and total factor productivity differences across plants within narrow sectors (for example three-digit or four-digit manufacturing sectors). For our focus, the most important pattern that emerges from these studies is that, even within a narrow sector of the US economy, there is a significant amount of productivity differences across plants, with an approximately two or three-fold difference between the top and the bottom of the distribution (see, for example the survey in Bartelsman and Doms, 2000, for a summary of various studies and estimates). In addition, these productivity differences appear to be highly persistent (e.g., Bailey, Hulten and Campbell, 1992).

There is little consensus on what the causes of these differences are. Many studies find a correlation between plant productivity and plant or firm size, various measures of technology (in particular IT technology), capital intensity, the skill level of the workforce and management practices (e.g., Davis and Haltiwanger, 1991, Doms, Dunn and Troske, 1997, Black and Lynch, 2004). Nevertheless, since all of these features are choice variables for firms, these correlations cannot be taken to be causal. Thus to a large extent the determinants of productivity differences across plants are still unknown. In this light, it should not appear as a surprise that there is no consensus on the determinants of cross-country differences in productivity.

Nevertheless, the existing evidence suggests that technology differences are an important factor, at least as an approximate cause, for productivity differences. For example, Doms, Dunn and Troske (1997) and Haltiwanger, Lane and Spletzer (1999) document significant technology differences across plants within narrow sectors. Interestingly, as emphasized by Doms, Dunn and Troske (1997) and Caselli and Coleman (2001), a key determinant of technology adoption decisions seems to be the skill level of the workforce of the plant (often proxied by the share of non-production workers), though adoption of new technology does not typically lead to a significant change in the skill level of the employees of the plant. These results suggest that, consistent with some of the models discussed in Chapters 10 and 15, differences in the availability of skills and skilled workers might be an important determinant of technology adoption (and development) decisions. I will return to the role of skills in productivity and technology differences in the cross-country context below.

The distribution of productivity differences across firms appears to be related to the entry of new and more productive plants (and the exit of less productive plants). For example, consistent with the basic Schumpeterian models of economic growth discussed in Chapter 14, Bartelsman and Doms (2000) and Foster, Haltiwanger and Krizan (2000) document that entry of new plants has an important contribution to industry productivity growth. Nevertheless, entry and exit appear to account for only about 25% of average TFP growth, with the remaining productivity improvements accounted for by continuing plants. This suggests that models in which firms continually invest in technology and productivity (for example such as the model of step-by-step innovation in Section 14.4 in Chapter 14) may be important for

understanding the productivity differences across firms and plants and also for the study of cross-country productivity differences.

18.1.2. Diffusion of New Technologies. A key implication of the sectoral studies is that, despite our presumption that technology and know-how are freely available and can be adopted easily, there are considerable technology and productivity differences among firms operating under similar circumstances. Nevertheless, cross-sectional distributions of productivity and technology are not stationary. In particular, new and more productive technologies, once they arrive on the scene, diffuse and over time are adopted by more firms and plants. The literature on technology diffusion studies this process of adoption of new technologies. As one might expect, there are parallels between the issue of technology diffusion across countries and slow technology diffusion across firms. It is therefore important to briefly overview the main findings all the technology diffusion literature.

The classic paper in this area is Griliches's (1957) study of the adoption of hybrid corn in the US. Griliches showed that the more productive hybrid corn diffused only slowly in the US agriculture and that this diffusion was affected by the local economic conditions of different areas. Consistent with the theoretical models presented so far, his study found evidence that the likelihood of adoption was related to the productivity contribution of the hybrid corn in a particular area, the market size and the skill level of the area. The importance of these factors has been found in other studies as well (see, for example, Mansfield, 1998, for a survey). Another important result of Griliches's study was to uncover the famous S-shape of diffusion, whereby a particular technology first spreads slowly and then once it reaches a critical level of adoption, it starts spreading much more rapidly. Finally, once a large fraction of the target population adopts the technology, the rate of adoption again declines. The overall pattern thus approximates an S curve or a logistic function. Jovanovic and Lach (1989), among others, show how this type of diffusion process can emerge as an equilibrium of an industry model with knowledge spillovers.

The important lesson for our focus here is that productivity and technology differences are not only present across countries, but also within countries. Moreover, even within countries better technologies do not immediately get adopted by all firms. In light of these patterns, the presence of significant productivity and technology differences across countries should not be entirely surprising. Nevertheless, the causes of within-country and cross-country productivity and technology differences might be different, and despite the presence of within-country differences, the significant cross-country differences do pose a puzzle that requires investigation. For example, within-country productivity differences might be due to differences in managerial (entrepreneurial) ability or related to the success of the match between the manager and the technology (or the product). These types of explanations would be unlikely to account for why almost all firms in many less developed countries are much less productive than the typical firms in the United States or other advanced economies. Motivated by the evidence briefly surveyed here, I will discuss both models in which technology diffuses slowly

across countries and also models in which productivity differences may remain even when instantaneous technology diffusion and adoption are possible.

18.2. A Benchmark Model of Technology Diffusion

18.2.1. A Model of Exogenous Growth. In the spirit of providing the main insights with the simplest possible models, let us return to the Solow growth model of Chapter 2. In particular, suppose that the world economy consists of J countries, indexed $j = 1, \dots, J$, each with access to an aggregate production function for producing the unique final good of the world economy,

$$Y_j(t) = F(K_j(t), A_j(t) L_j(t)),$$

where $Y_j(t)$ is the output of this unique final good in country j at time t , and $K_j(t)$ and $L_j(t)$ are the capital stock and labor supply. Finally, $A_j(t)$ is the technology of this economy, which is both country-specific and time-varying. In line with the result in Theorem 2.6 in Chapter 2, technological change has already been assumed to be purely labor-augmenting (Harrod-neutral) form. The aggregate production function F is assumed to satisfy the standard neoclassical assumptions, that is, Assumptions 1 and 2 from Chapter 2. In particular, recall that these assumptions imply that F exhibits constant returns to scale. Throughout this chapter and the next, whenever the world economy consists of J countries, I assume that J is large enough so that each country is “small” relative to the rest of the world and thus it ignores its effect on world aggregates.¹

Using our usual approach, income per capita in country j at time t is

$$\begin{aligned} y_j(t) &\equiv \frac{Y_j(t)}{L_j(t)} \\ &= A_j(t) F\left(\frac{K_j(t)}{A_j(t) L_j(t)}, 1\right) \\ &\equiv A_j(t) f(k_j(t)), \end{aligned}$$

where

$$k_j(t) \equiv \frac{K_j(t)}{A_j(t) L_j(t)}$$

is the effective capital-labor ratio of country j at time t .

Suppose that time is continuous, that there is population growth at the constant rate $n_j \geq 0$ in country j , and that there is an exogenous saving rate equal to $s_j \in (0, 1)$ in country j and a depreciation rate of $\delta \geq 0$ for capital, so that the law of motion of capital for each country is given by

$$(18.1) \quad \dot{k}_j(t) = s_j f(k_j(t)) - (n_j + g_j(t) + \delta) k_j(t),$$

¹This can be thought of in two ways. Either we can think of J as a large finite number, or consider the limit where $J \rightarrow \infty$.

Alternatively, I could have assumed that there is a continuum rather than a countable number of countries. None of the results in this and the next chapter depend on whether the number of countries are taken to be a continuum or a countable set, and throughout I work with a countable, in fact, finite, number of countries to simplify the exposition.

where

$$(18.2) \quad g_j(t) \equiv \frac{\dot{A}_j(t)}{A_j(t)}$$

is the (endogenously-determined) growth rate of technology of country j at time t (see Exercise 18.1). The endogenously given initial conditions are $k_j(0) > 0$ and $A_j(0) > 0$ for each $j = 1, \dots, J$.

To start with, technology diffusion is modeled in a reduced-form way. Let us assume that the world's technology frontier, denoted by $A(t)$, grows exogenously at the constant rate

$$g \equiv \frac{\dot{A}(t)}{A(t)} > 0,$$

with an initial condition $A(0) > 0$. We refer to $A(t)$ as the world technology or sometimes as the “world technology frontier”. It encapsulates the maximal knowledge that any country can have, so that $A_j(t) \leq A(t)$ for all j and t . Moreover, each country's technology progresses as a result of *absorbing* the world's technological knowledge. In particular, let us posit the following law of motion for each country's technology:

$$(18.3) \quad \dot{A}_j(t) = \sigma_j (A(t) - A_j(t)) + \lambda_j A_j(t),$$

where $\sigma_j \in (0, \infty)$ and $\lambda_j \in [0, g)$ for each $j = 1, \dots, J$ (see Exercise 18.2). Equation (18.3) implies that each country absorbs world technology at some exogenous rate σ_j . I refer to this parameter as the *technology absorption* rate. In practice, absorption corresponds both to straightforward adoption of existing technologies and to *adaptation* of existing blueprints to the conditions prevailing in a specific country, so that they can be used with the other technologies and practices in place. This parameter will vary across countries because of differences in their human capital or other investments (see below) and also because of institutional or policy barriers affecting technology adoption. This parameter multiplies the difference $A(t) - A_j(t)$, since it is this difference that remains to be absorbed by the country in question—if $A(t) = A_j(t)$, there is nothing to absorb from the world technology frontier. Though natural, this formulation has important economic consequences. In particular, it implies that countries that are relatively “backward” in the sense of having a low $A_j(t)$ compared to the frontier, will tend to grow faster, because they have more technology to absorb or more room for *catch-up*. This potential advantage of relatively backward economies will play an important role in ensuring a stable world income distribution across countries. It also formalizes an idea of going back to Gerschenkron's (1962) essay *Economic Backwardness in Historical Perspective*. Gerschenkron argued that rapid catch-up by relatively backward countries was important for understanding cross-country growth patterns. He also suggested that the organization of production in the process of catching up is (or should) be different than the organization of production appropriate for frontier economies. I will return to this theme in Chapter 20.

Equation (18.3) also implies that technological progress can happen “locally” as well, that is, building upon the knowledge stock of country j , $A_j(t)$. The parameter λ_j captures

the speed at which this happens. This equation therefore contains the two major forms of technological progress that a particular country can experience; absorption from the world technology frontier and local technological advances. Its functional form is adopted for simplicity.

Notice that (18.3) already sidesteps one of the major issues raised at the beginning of this chapter: it posits that despite the level of globalization the world has reached and the relatively free-flow of information among individuals across the globe, the process of technology transfer between countries is a slow one. The assumption that $\sigma_j < \infty$ imposes this feature. In particular, since $\sigma_j < \infty$, $A_j(t) < A(t)$ will imply that $A_j(t + \Delta t) < A(t + \Delta t)$, at least for $\Delta t > 0$ and sufficiently small. Consequently, countries that have access to only a subset of the production techniques (blueprints) available in the world will not immediately acquire all of the knowledge that they do not currently have access to.

To proceed with the analysis of this model, let us define

$$a_j(t) \equiv \frac{A_j(t)}{A(t)}$$

as an inverse measure of the proportional technology gap between country j in the world or alternatively as an inverse measure of country j 's *distance to the frontier* (distance to the world technology frontier), we can then write the above equation as (see Exercise 18.3):

$$(18.4) \quad \dot{a}_j(t) = \sigma_j - (\sigma_j + g - \lambda_j) a_j(t).$$

Clearly, the initial conditions $A(0) > 0$ and $A_j(0) > 0$ give a unique initial condition for the differential equation for a_j , $a_j(0) \equiv A_j(0)/A(0) > 0$.

Given the description of the environment above, the dynamics of the world income per capita levels and technology are determined by $2J$ differential equations. For each j , one of (18.1) and one of (18.4) applies. These equations characterize the steady-state distribution of technology and income per capita in the world economy and its transitional dynamics. What makes the analysis of this world equilibrium relatively straightforward is the *block recursiveness* of the system of differential equations governing the behavior of income per capita and technology across countries. The law of motion of (18.4) for country j only depends on $a_j(t)$, so it can be solved without reference to the law of motion of $k_j(t)$ and to the law of motion of $\{k_{j'}(t), a_{j'}(t)\}_{j' \neq j}$. Once (18.4) is solved, then (18.1) becomes a first-order nonautonomous differential equation in a single variable. The fact that it is nonautonomous is a consequence of the fact that it has $g_j(t)$ on the right-hand side, which can be determined as

$$g_j(t) = \frac{\dot{a}_j(t)}{a_j(t)} + g.$$

Once we solve for the law of motion of $a_j(t)$, this is simply a function of time, making (18.1) a simple nonautonomous differential equation.

Let us start the analysis with the steady-state world equilibrium. A *world equilibrium* is defined as an allocation $\left\{ [k_j(t), a_j(t)]_{t \geq 0} \right\}_{j=1}^J$ such that (18.1) and (18.4) are satisfied for

each $j = 1, \dots, J$ and for all t , starting with the initial conditions $\{k_j(0), a_j(0)\}_{j=1}^J$. A *steady-state world equilibrium* is then defined as a steady state of this equilibrium process, that is, an equilibrium with $\dot{k}_j(t) = \dot{a}_j(t) = 0$ for each $j = 1, \dots, J$. The “steady-state equilibria” studied in this chapter will exhibit constant growth, so I could have alternatively referred to them as *balanced growth path equilibria*. Throughout I will use the term steady-state equilibrium for consistency.

Now imposing these steady-state conditions, we obtain:

PROPOSITION 18.1. *In the above-described model, there exists a unique steady-state world equilibrium in which income per capita in all countries grows at the same rate $g > 0$. Moreover, for each $j = 1, \dots, J$,*

$$(18.5) \quad a_j^* = \frac{\sigma_j}{\sigma_j + g - \lambda_j},$$

and k_j^* is uniquely determined by

$$s_j \frac{f(k_j^*)}{k_j^*} = n_j + g + \delta.$$

The steady-state world equilibrium $\{k_j^*, a_j^*\}_{j=1}^J$ is globally stable in the sense that starting with any strictly positive initial values $\{k_j(0), a_j(0)\}_{j=1}^J$, the equilibrium path $\{k_j(t), a_j(t)\}_{j=1}^J$ converges to $\{k_j^*, a_j^*\}_{j=1}^J$.

PROOF. (**Sketch**) First solve (18.1) and (18.4) for each $j = 1, \dots, J$ imposing the steady-state condition that $\dot{k}_j(t) = \dot{a}_j(t) = 0$. This yields a unique solution, establishing the uniqueness of the steady-state equilibrium. Then, standard arguments show that the steady state a_j^* of the differential equation for $a_j(t)$ is globally stable. Next using this result, the global stability of the steady state of the differential equation for $k_j(t)$ follows straightforwardly. Exercise 18.4 asks you to complete the details of this proof. $\square$

A number of features about this world equilibrium are noteworthy. First, there is a unique steady-state world equilibrium and it is globally stable. This enables us to perform simple comparative static and comparative dynamic exercises (see Exercise 18.5). Second and most importantly, despite differences in saving rates and technology absorption rates across countries, income per capita in all economies grows at the same rate equal to the growth rate of the world technology frontier, g . Why is this? The technology adoption equation, (18.3), provides the answer to this question; the rate of technology diffusion (absorption) is higher when the gap between the world technology frontier and the technology level of a particular country is greater. Thus there is a force pulling backward economies towards the technology frontier, and in steady state this force is powerful enough to ensure that all countries grow at the same rate.

Does this imply that all countries will converge to the same level of income per capita? The answer is clearly no. Differences in saving rates and absorption rates translate into *level*

differences (instead of growth rate differences) across countries. For example, a society with a low level of σ_j will initially grow less than others, until it is sufficiently behind the world technology frontier. At this point, it will also grow at the world rate, g . This discussion illustrates that it is precisely the endogenous technology gap between a country and the world frontier that ensures growth at the rate g for all countries. Thus societies that are unsuccessful in absorbing world technologies, those that impose barriers slowing technology diffusion (those with low σ_j) and those that are not sufficiently innovative in developing their own local technologies (those with low λ_j) will be poorer. Moreover, as in the baseline Solow model, those with low saving rates will also be poorer. These results are summarized in the following proposition.

PROPOSITION 18.2. *Steady-state income per capita level of country j can be written as $y_j^*(t) = \exp(gt) y_j^*$, where y_j^* is increasing in σ_j, λ_j and s_j and decreasing in n_j and δ . It does not depend on $\sigma_{j'}, \lambda_{j'}, s_{j'}$ and $n_{j'}$ for any $j' \neq j$.*

PROOF. See Exercise 18.7. □

A particularly convenient—but also restrictive—feature of the equilibrium studied here is that even though there is technology diffusion and interdependence in this world equilibrium, there is no interaction among countries. Each country's steady-state income per capita (and in fact path of income per capita) only depends on the behavior of the world technology frontier and its own parameters. Later in this chapter, we will see models in which there is more interaction between the decisions of individual countries.

18.2.2. Consumer Optimization. It is straightforward to incorporate consumer optimization into this benchmark model of technology transfer. In particular, let us now suppose that each country admits a representative household with preferences at time $t = 0$ given by

$$(18.6) \quad U_j = \int_0^\infty \exp(-(\rho - n_j)t) \left[\frac{\tilde{c}_j(t)^{1-\theta} - 1}{1-\theta} \right] dt,$$

where $\tilde{c}_j(t) \equiv C_j(t)/L_j(t)$ is per capita consumption in country j at time t and I imposed that all countries have the same time discount rate, ρ . This latter feature is to simplify the discussion in the text, and Exercise 18.9 generalizes the results in this subsection to a world economy with different discount rates. This is an important generalization, since it highlights that a stable world income distribution does not depend on equal discount rates or asymptotically equal saving rates across countries.

As in the neoclassical growth model, the flow resource constraint facing the representative household can be written as

$$\dot{k}_j(t) = f(k_j(t)) - c_j(t) - (n_j + g_j(t) + \delta) k_j(t),$$

where $c_j(t) \equiv \tilde{c}_j(t)/A_j(t) \equiv C_j(t)/A_j(t)L_j(t)$ is consumption normalized by effective units of labor. This equation now replaces (18.1) as the law of motion of effective capital-labor ratio of country j .

The world equilibrium and the steady-state world equilibrium are defined in a similar fashion, except that instead of a constant saving rate consumption processes must now maximize the utility of the representative household in each country subject to their resource constraint. An analysis similar to that in Chapter 8 leads to the following proposition:

PROPOSITION 18.3. *Consider the above-described model with consumer optimization with preferences given by (18.6) and suppose that $\rho - n_j > (1 - \theta)g$. Then, there exists a unique steady-state world equilibrium where for each $j = 1, \dots, J$, a_j^* is given by (18.5) and k_j^* is uniquely determined by*

$$f'(k_j^*) = \rho + \delta + \theta g,$$

and consumption per capita in each country grows at the rate $g > 0$.

Moreover, the steady-state world equilibrium is globally saddle-path stable in the sense that starting with any strictly positive initial values $\{k_j(0), a_j(0)\}_{j=1}^J$, initial consumption to effective labor ratios are $\{c_j(0)\}_{j=1}^J$ and the equilibrium path $\{k_j(t), a_j(t), c_j(t)\}_{j=1}^J$ converges to $\{k_j^, a_j^*, c_j^*\}_{j=1}^J$, where c_j^* is the steady-state consumption to effective labor ratio in economy j .*

PROOF. (Sketch) We can first show that a_j^* can be determined from the differential equation in (18.4) without reference to any other variables and satisfies (18.5). The consumer Euler equations and the dynamics of capital accumulation are the same as in the baseline neoclassical growth model, taking into account that in steady state $g_j(t) = g$. To complete the proof of the proposition, we need to show the stability of a_j^* , and then taking into account the behavior of $g_j(t)$, we must establish the saddle path stability of k_j^* using the same type of analysis as in Chapter 8—which is slightly more complicated here because the differential equation for capital accumulation is not autonomous. You are asked to complete these details in Exercise 18.8. $\square$

This proposition shows that all of the qualitative results of the benchmark model of technology diffusion apply regardless of whether we assume constant saving rates or dynamic consumer maximization (as long as we ensure that the growth rate is not so high as to violate the transversality condition). Naturally, an equilibrium now corresponds not only to processes of $\{k_j(t), a_j(t)\}$ but also includes the time path of consumption per unit of effective labor, $c_j(t)$. Consequently, the appropriate notion of stability is saddle-path stability, which the equilibrium in Proposition 18.3 satisfies.

18.2.3. The Role of Human Capital in Technology Diffusion. The model presented above is in part inspired by a classic short paper by Richard Nelson and Edmund Phelps (1966), which was already discussed in Section 10.8 of Chapter 10. Recall that the Becker-Mincer view emphasizes how human capital increases the productivity of the labor hours supplied by an individual. While this approach allows the effect of human capital to be different in different tasks, in most applications it is presumed that greater human capital

translates into higher productivity in all or most tasks, with the set of productive tasks typically taken as given. In contrast, Nelson and Phelps (and Ted Schultz) emphasize the role of human capital in facilitating the adoption of new technologies and adaptation to changing environments.

In terms of the model described above, the simplest way of capturing this argument is to posit that the parameter σ_j is a function of the human capital of the workforce. The greater is the human capital of the workforce, the higher is the absorption capacity of the economy. If so, high human capital societies will be richer because, as shown in Proposition 18.2, economies with higher σ_j have higher steady-state levels of income.

While this modification leaves the mathematical exposition of the model unchanged, the implications for how we view growth experiences of societies with different levels of human capital are potentially quite different than in the Becker-Mincer approach (or at the very least, than in the simplest version of the Becker-Mincer approach). The latter approach suggests that we can approximate the role of human capital in economic development by carefully accounting for its role in the aggregate production function. This, in turn, can be done by estimating individual returns to schooling and returns to other dimensions of human capital in the labor market. The Nelson-Phelps-Schultz view, on the other hand, suggests that even if the contribution of human capital to productivity in regular activities may be limited, lack of human capital will seriously slow down the process of technology diffusion. The model here shows how, in a dynamically evolving world economy, this effect can lead to a lower level of income per capita for the country in question.

18.2.4. Barriers to Technology Adoption. As discussed in Chapter 8, one of the main criticisms against the neoclassical growth model has been its inability to generate quantitatively large cross-country income per capita differences. Most economists view this as related to the fact that the basic neoclassical growth model does not provide an explanation for “technology differences”. The model in this section presents a reduced-form model of technology differences across countries, and thus enables us to enrich the neoclassical growth model and the Solow model to incorporate technology differences. Nevertheless, such a theory will be useful only to the extent that the key parameters such as σ_j and λ_j can be mapped to reality. The previous subsection discussed ideas linking the parameter σ_j to human capital. An alternative, emphasized by Parente and Prescott (1994), is to link σ_j to barriers to technology adoption. Parente and Prescott construct a variant of the neoclassical growth model in which investments affect technology absorption, and countries differ in terms of the “barriers” that they place on the path of firms in this process. In terms of the reduced-form model here, the Parente-Prescott mechanism can be captured by interpreting σ_j as a function of property rights institutions or other institutional or policy features.

This perspective is useful as it gives us a concrete way of thinking of the reasons why σ_j may vary across countries. Nevertheless, it is still unsatisfactory in two important respects. First, exactly how these institutions affect technology adoption is left as a blackbox.

Second and more importantly, why some societies choose to create barriers against technology adoption while others do not is left unexplained. The models that combine technology diffusion with endogenous technology decisions, which will be presented in the next section, make some progress on the first point. In fact, Parente and Prescott constructed a model in which firms undertook investments to acquire technologies from the world technology frontier. Nevertheless, their model is closer to the neoclassical growth model than to the endogenous technology models presented in Part 4 of the book, because it does not feature investments in the creation or adoption of new technologies that can be identified with R&D decisions. Since the endogenous technological change models are more widely used and offer richer insights about the nature of technology, I introduce endogenous technology adoption decisions in the context of these models. The question of why some societies block technology adoption will be the topic of Part 8 below.

18.3. Technology Diffusion and Endogenous Growth

In the previous section, technology diffusion took place “exogenously,” in the sense that firms did not engage in R&D or investment-type activities in order to improve their technologies. In this section, I introduce these types of purposeful activities directed at improving technology. The material in this section therefore complements the models of technology diffusion of the previous section in the same way that endogenous technological change models complemented (and advanced upon) the neoclassical framework with exogenous technology. The section is separated into two parts. In the first, the world growth rate will be taken as exogenous, while it will be endogenized in the second part.

18.3.1. Exogenous World Growth Rate. To keep the exposition as brief as possible, I will use the baseline endogenous technological change model with expanding machine variety and lab-equipment specification as in Section 13.1 of Chapter 13 and I will frequently refer to the analysis there. Clearly, different versions of the endogenous technological change models could be used for the same purposes.

The aggregate production function of economy $j = 1, \dots, J$ at time t is

$$(18.7) \quad Y_j(t) = \frac{1}{1-\beta} \left[\int_0^{N_j(t)} x_j(\nu, t)^{1-\beta} d\nu \right] L_j^\beta,$$

where L_j is the aggregate labor input, which is assumed to be constant over time, $N_j(t)$ denotes the different number of varieties of machines available to country j at time t , and $x_j(\nu, t)$ is the total amount of machine type ν used at time t . Suppose again that x 's depreciate fully after use. As in Chapter 13, each variety in economy j is owned by a technology monopolist, which will sell machines embodying this technology at the profit maximizing (rental) price $\chi_j(\nu, t)$. This monopolist can produce each unit of the machine at a cost of $\psi \equiv 1 - \beta$ units on the final good, where this normalization is again introduced to simplify the expressions.

Since there is no international trade, firms in country j can only use technologies supplied by technology monopolists in their country. This assumption introduces the potential differences in the knowledge stock available to different countries.

Each country admits a representative household with the same preferences as in (18.6), except that there is no population growth, that is, $n_j = 0$ for all j . New varieties are again produced by investment, and thus the resource constraint for each country at each point in time is

$$(18.8) \quad C_j(t) + X_j(t) + \zeta_j Z_j(t) \leq Y_j(t),$$

where $X_j(t)$ is investment or spending on inputs at time t and $Z_j(t)$ is expenditure on technology adoption at time t , which may take the form of R&D or other expenditures, such as the purchase or rental of machines embodying new technologies. The parameter ζ_j is introduced as a potential source of differences in the cost of technology adoption across countries, which may result from institutional barriers against innovation as emphasized by Parente and Prescott (1994), from subsidies to R&D and to technology, or from other tax policies. As discussed in Section 8.10 in Chapter 8, many authors identify this parameter with tax distortions on investment-type activities and often proxy it with the relative price of investment to consumption goods. In the next chapter, we will see when this might be valid and when it might be misleading.

The main difference from the environment in Chapter 13 is in the *innovation possibilities frontier*,

$$(18.9) \quad \dot{N}_j(t) = \eta_j \left(\frac{N(t)}{N_j(t)} \right)^\phi Z_j(t),$$

where $\eta_j > 0$ for all j , and $\phi > 0$ and is common to all economies. This form of the innovation possibilities frontier captures the same basic idea as (18.3) in the previous section, but what matters is not the absolute gap in technology, but the proportional gap. This functional form is again adopted for simplicity. I assume that each economy starts with some initial technology stock $N_j(0) > 0$. Finally, as noted above, the world technology frontier of varieties expands at an exogenous rate $g > 0$, that is,

$$(18.10) \quad \dot{N}(t) = gN(t).$$

The analysis in Chapter 13 implies that the flow profits of a technology monopolist at time t in economy j is given by

$$\pi_j(t) = \beta L_j.$$

Suppose a steady-state (BGP) equilibrium exists in which the interest rate is constant at some level $r_j^* > 0$. Then, the net present discounted value of a new machine is

$$V_j^* = \frac{\beta L_j}{r_j^*}.$$

If the steady state involves the same rate of growth in each country, then $N_j(t)$ will also grow at the rate g , so that $N_j(t)/N(t)$ will remain constant, say at some level ν_j^* . In that

case, an additional unit of technology spending will create benefits equal to $\eta_j (\nu_j^*)^{-\phi} V_j^*$ counterbalanced against the cost of ζ_j . Free entry (with positive activity) then requires

$$(18.11) \quad \nu_j^* = \left(\frac{\eta_j \beta L_j}{\zeta_j r^*} \right)^{1/\phi},$$

where I have also used the fact that given the preferences (18.6), equal growth rates across countries imply that the interest rate will be the same in all countries (and in fact it will be equal to $r^* = \rho + \theta g$).

Since a higher ν_j implies that country j is technologically more advanced and thus richer than others, eq. (18.11) shows that societies with better innovation possibilities frontiers, as captured by the parameter η_j , and those with lower cost of R&D, corresponding to lower ζ_j , will be more advanced and richer. This equation also incorporates a scale effect as in the standard endogenous technological change models, so a country with a greater labor force will also be richer. This is for the same reason as a greater labor force leads to faster growth in the baseline endogenous technological change model: a greater labor force creates more demand for machines, making R&D more profitable.

This analysis leads to the following proposition:

PROPOSITION 18.4. *Consider the model with endogenous technology adoption described in this section. Suppose that $\rho > (1 - \theta)g$. Then, there exists a unique steady-state world equilibrium in which relative technology levels are given by (18.11) and all countries grow at the same rate $g > 0$.*

Moreover, this steady-state equilibrium is globally saddle-path stable, in the sense that starting with any strictly positive vector of initial conditions $N(0)$ and $(N_1(0), \dots, N_J(0))$, the equilibrium path of $(N_1(t), \dots, N_J(t))$ converges to $(\nu_1^ N(t), \dots, \nu_J^* N(t))$.*

PROOF. (Sketch) First show that the specified steady-state equilibrium is the only steady state equilibrium in which all countries grow at the same rate. Then, consider the value function of technology monopolist in each country as in Chapter 13 and show that the number of varieties in each countries must asymptotically grow at the rate g . Exercise 18.11 asks you to complete this proof. $\square$

This result and the preceding analysis therefore show that endogenizing investments in technology adoption leads to an equilibrium pattern similar to that in the previous section. The main difference is that we can now pinpoint the factors that affect the rates of technology adoption and relate them to the profit incentives of firms. An explicit model of technology decisions also allows us to investigate how differences in the cost of investing in technology might affect cross-country differences in technology and income (see Exercise 18.12).

18.3.2. Endogenous Growth in the World. The model in the previous subsection was simplified by the fact that the world growth rate was exogenous. A more satisfactory model would derive the world growth rate from the technology adoption and R&D activities of each country. Such models are typically more involved, because the degree of interaction

among countries in the world equilibrium is now considerably greater. In addition, a certain amount of care needs to be taken so that the world economy grows at a constant endogenous rate, while there are still forces that ensure relatively similar growth rates across countries. Naturally, one may also wish to construct models in which countries grow at permanently different long run rates (see, for example, Exercise 13.8 in Chapter 13). The evidence in Chapter 1 suggests that such long-run growth differences are present when we look at the past 200 or 500 years, but there are more limited sustained growth rates differences over the past 60 years or so (implying only small changes in the postwar world income distribution). Thus whether one wants to have long-run growth rate differences across countries is a modeling choice—it partly depends on whether one thinks of a model with a long transition leading to the large income differences, or wishes to approximate the past 200 or 500 years as corresponding to “steady-state behavior”. Since such growth rates differences emerge straightforwardly in many models (including all of the endogenous technology models so far, see again Exercise 13.8), in this subsection I focus on forces that will keep countries growing at similar rates in the presence of endogenous technological change at the world level.

The main difference from the model in the previous subsection is that the world growth equation, (18.10), which specified exogenous world growth at the rate g , is now replaced with an equation that links the improvements in the world technology to technological improvements in each country. In particular, let us adopt the simplest way of aggregating the technologies of different countries, which is by taking their arithmetic average:

$$(18.12) \quad N(t) = \frac{1}{J} \sum_{j=1}^J N_j(t).$$

With this new equation, $N(t)$ no longer corresponds to the “world technology frontier”. Instead, it represents average technology in the world, and as long as there are some differences across countries, it will naturally be the case that $N_j(t) > N(t)$ for at least some j . Nevertheless, having the world technology correspond to an average of the technology of each country is a natural generalization of the ideas presented so far in this chapter. One disadvantage of this formulation is that it implies that the contribution of each country to the world technology is the same. Exercise 18.18 discusses alternative ways of aggregating individual country technologies into a world technology term and shows that the qualitative results here do not depend on the specification in (18.12). Besides eq. (18.12), all the other equations from the previous subsection continue to hold.

The main result of this section is that the pattern of cross-country growth will be similar to that in the previous subsection, but now the growth rate of the world economy, g , will be endogenous, resulting from the investments in technologies made by firms in each country. In particular, suppose that there exists a steady-state world equilibrium in which each country grows at the rate g . Then, (18.12) implies that the world technology index, $N(t)$, will also grow at the same rate g . Now, as in the previous subsection, the net present discounted value of a new machine in country j is $\beta L_j / r^*$, and the no-arbitrage condition in R&D investments

implies that, for given g , each country j 's relative technology, ν_j^* , should satisfy (18.11). However, now dividing both sides of eq. (18.12) by $N(t)$ implies that the steady-state world equilibrium must satisfy:

$$(18.13) \quad \begin{aligned} \frac{1}{J} \sum_{j=1}^J \nu_j^* &= 1 \\ \frac{1}{J} \sum_{j=1}^J \left(\frac{\eta_j \beta L_j}{\zeta_j (\rho + \theta g)} \right)^{1/\phi} &= 1, \end{aligned}$$

where the second line uses the definition of ν_j^* from (18.11) and substitutes for the common interest rate r^* as a function of the world growth rate. The only unknown in eq. (18.13) is g . Moreover, the left-hand side is clearly strictly decreasing in g , so this equation can be satisfied for at most one value of g , say g^* . A well-behaved world equilibrium would require the growth rates to be positive and not so high as to violate the transversality condition. The following condition is necessary and sufficient for the world growth rate to be positive:

$$(18.14) \quad \frac{1}{J} \sum_{j=1}^J \left(\frac{\eta_j \beta L_j}{\zeta_j \rho} \right)^{1/\phi} > 1.$$

Moreover, by usual arguments, when this condition is satisfied, there will exist a unique $g^* > 0$ that will satisfy (18.13) (if this condition were violated, (18.13) would not hold, and we would have $g = 0$ as the world growth rate). Therefore, the following proposition follows:

PROPOSITION 18.5. *Suppose that (18.14) holds and that the solution g^* to (18.13) satisfies $\rho > (1 - \theta)g^*$. Then, there exists a unique steady-state world equilibrium in which growth at the world level is given by g^* and all countries grow at this common rate. This growth rate is endogenous and is determined by the technologies and policies of each country. In particular, a higher η_j or L_j or a lower ζ_j for any country $j = 1, \dots, J$ increases the world growth rate.*

PROOF. See Exercise 18.15. □

A number of features about this equilibrium are noteworthy. First, taking the world growth rate as given, the structure of the equilibrium is very similar to that in Proposition 18.4. Thus the fact that all countries grow at the same rate and that differences in the innovation possibilities frontier, η_j , the size the labor force, L_j , and the extent of potential distortions in technology investments, ζ_j , translate into level differences across countries has exactly the same intuition as in that proposition. What is more interesting is that essentially the same model as in the previous subsection now gives us an “endogenous” growth rate for the world economy. In particular, while growth for each country appears “exogenous” in the sense that, each country accumulates towards a world-determined growth rate, the growth rate of the world economy is endogenous and results from the investments of the firms in each country. As such, the current model provides a more satisfactory framework for the analysis of the process of world growth than both the purely exogenous growth models and the purely endogenous growth models. In the current model, technological progress and

economic growth are the outcome of investments by all countries in the world, but there are sufficiently powerful forces in the world economy, here working through technological spillovers that pull relatively backward countries towards the world average, ensuring equal long-run growth rates for all countries in the long run. Naturally, equal growth rates are still consistent with quite *large level differences* across countries (see Exercise 18.12).

Proposition 18.5 used a number of simplifying assumptions. First, each country was assumed to have the same discount rate. This was only for simplicity, and Exercise 18.16 considers the case in which countries differ according to their discount rates. Second, the proposition only describes the steady-state equilibrium. Transitional dynamics are now more complicated, since the “block recursiveness” of the dynamical system is lost. The differential equations describing the equilibrium path for all countries need to be analyzed together. Nevertheless, local stability of the steady-state world equilibrium can be established, and this is analyzed in Exercise 18.15.

18.4. Appropriate and Inappropriate Technologies and Productivity Differences

The models presented so far in this chapter explicitly introduced a slow process of technology diffusion from the world stock of knowledge to the set of techniques used in production in each country. This was motivated either by some process of costly (and slow) technology absorption or because of barriers to technology adoption. However, as noted at the beginning of the chapter, in the highly globalized world we live in, where information technology and information flows make a wide range of blueprints easily accessible to most individuals and firms around the world, we should perhaps expect even faster technology transfer across countries. Why does rapid diffusion of ideas not remove all, or at least most, cross-country technology differences? Leaving the discussion of institutional or policy barriers preventing technology diffusion to later, in this section I focus on how “technology” differences and income gaps can remain substantial even with free flow of ideas.

A first important idea is that productivity differences may remain even if all differences in “techniques” disappear, because production is organized differently and the extent of inefficiency in production may vary across countries. A model along these lines will be discussed later in this chapter. Another important idea is that technologies of the world technology frontier may be *inappropriate* to the needs of specific countries, so that importing the most advanced frontier technologies may not guarantee the same level of productivity for all countries. At some level, this idea is both simple and attractive. Clearly, technologies and skills consist of bundles of complementary attributes and these bundles vary across countries, so that there is no guarantee that a new technology that works well given the skills and competences in the United States or Switzerland will also do so in Nigeria or Turkey. Nevertheless, without specifying these attributes that make some technologies work well in certain nations and not in others, this story will have little explanatory power. In this section, I present three versions of this story that may have some theoretical and empirical appeal. First, I discuss how differences in exogenous (e.g., geographic) conditions may make

the same set of technologies differentially productive in different areas. Second, I show how differences in capital intensity across countries may change the appropriateness of different types of technologies. Finally, most of this section will be devoted to the implications of differences in skill supplies across countries for the appropriateness of frontier technologies to developing economies. In this context, I will show how the degree of appropriateness or inappropriateness of technologies may arise endogenously in the world equilibrium and also introduce a model of economic growth where labor has to be allocated across different sectors, which is of independent interest.

18.4.1. Inappropriate Technologies. The idea of inappropriate technologies can be best illustrated by an example on health innovations. Suppose that productivity in country j at time t , $A_j(t)$, is a function of whether there are effective cures against certain diseases affecting their populations. Suppose that there are two different diseases, heart attack and malaria. Countries $j = 1, \dots, J'$ are affected by malaria and not by heart attacks, while $j = J' + 1, \dots, J$ are affected by heart attacks and are unaffected by malaria. If the disease affecting country j has no cure, then productivity in that country given by $A_j(t) = \underline{A}$, while when a cure against this disease is introduced, then $A_j(t) = \bar{A}$. Now imagine that a new cure against heart attacks is discovered and becomes freely available to all countries. Consequently, the productivity in countries $j = J' + 1, \dots, J$ increases from $\underline{A}$ to $\bar{A}$, but productivity in countries $j = 1, \dots, J'$ remains at $\underline{A}$. This simple example thus illustrates how technologies of the world frontier may be “inappropriate” to the needs of some of the countries (in this case, the J' countries affected by malaria). In fact, in this extreme case, a technological advance that is freely available to all countries in the world increases productivity in a subset of the countries and creates cross-country income differences.

Is there any reason to expect that issues of the sort might be important? The answer is both yes and no. Over 90% of the world R&D is carried out in OECD economies. There are therefore natural reasons to expect that new technologies should be optimized for the conditions in OECD countries or should explicitly deal with the problems that these countries are facing. This suggests that an analysis of the implications of appropriate technology is a promising area. Nevertheless, other than the issue of disease prevention, there are not many obvious *fixed* country characteristics that will create this type of “inappropriateness”. Instead, the issue of appropriate technology is much more likely to be important in the context of whether new technologies increasing productivity via process and product innovations will function well at different *factor intensities*. The next two subsections focus on whether technologies developed in advanced economies can be productively used at different capital-labor and skilled-unskilled labor ratios than those for which they have been *designed*.

18.4.2. Capital-Labor Ratios and Inappropriate Technologies. A seminal paper by Atkinson and Stiglitz (1969) entitled “A New View of Technical Change” argued that a useful way of modeling technological change is to view it as shifting isoquants (increasing productivity) at a *given capital-labor ratio*. For example, a firm that is using a specific

machine, say a particular type of tractor, with a single worker, may discover a way to increase the productivity of the worker. This innovation can be used by any other firm employing the same tractor with a single worker. But it would be much less valuable to firms using oxen or less advanced tractors, or even to firms using more advanced tractors. Thus technological changes are localized for specific capital-labor ratios and when used with different capital labor ratios, they do not bring the same benefits. The implications of this observation for cross-country income differences can be quite major. If new technologies are developed for high capital-intensive production processes in OECD countries, they may be of little use to labor-abundant less-developed economies, where most production units will be functioning at lower capital-labor ratios than those in the OECD. This point is developed in the context of a Solow-type growth model by Basu and Weil (1998). I provide a simple version of their argument here.

Suppose that the production technology for all countries in the world is

$$Y = A(k | k') K^{1-\alpha} L^\alpha,$$

so that output per worker becomes

$$y \equiv \frac{Y}{L} = A(k | k') k^{1-\alpha},$$

where $k = K/L$ is the capital-labor ratio of the country in question, and $A(k | k')$ is the (total factor) productivity of technology designed to be used with capital-labor ratio k' when used instead with capital-labor ratio k . I have suppressed the time and country indices to simplify notation.

For example, suppose that

$$A(k | k') = A \min \left\{ 1, \left(\frac{k}{k'} \right)^\gamma \right\}$$

for some $\gamma \in (0, 1)$. That is, when a technology designed for the capital labor ratio k' is used with a lower capital-labor ratio, there is a loss in efficiency.

Now suppose that new technologies are developed in richer economies, which have greater capital-labor ratios. Then, productivity in a less developed country with the capital-labor ratio $k < k'$ will be

$$(18.15) \quad y = A(k | k') k^{1-\alpha} = A k^{1-\alpha+\gamma} (k')^{-\gamma}.$$

An immediate implication of eq. (18.15) is that less-developed countries will be less productive even when they are producing with the same techniques. Moreover this productivity disadvantage will be larger when the gap in the capital intensity of production between these countries and in the technologically advanced economies is greater. Depending on the value of the parameter γ , the implication of this type of inappropriateness might be important for understanding cross-country income differences. With the same arguments as in Chapters 2 and 3, we may want to think of $\alpha \approx 2/3$. In this case, an economy with an eight times higher capital-labor ratio than another would only be twice as rich, when both countries have access to the same technology and there is no issue of inappropriate technologies. But if $\gamma = 2/3$ and

the country with the higher capital-labor ratio is the frontier one setting the level of k' in terms of the function $A(k | k')$, the implied difference would be eightfold rather than the twofold difference implied by the model that overlooked the issue of appropriate technology. Thus inappropriateness of technologies have the potential to increase the implied cross-country income differences, even when all countries have access to the same technologies. Exercise 18.20 provides more details on this model.

18.4.3. Endogenous Technological Change and Appropriate Technology. The Atkinson-Stiglitz and Basu-Weil approach discussed in the previous subsection emphasizes differences in capital intensity between rich and poor economies. The evidence discussed in Section 18.1 suggests that differences in human capital may be particularly important in the adoption of technology. Moreover, the past 30 years have witnessed the introduction of a range of skill-biased technologies both in developed economies and in many developing countries (see Autor, Katz and Krueger, 1998, Acemoglu, 2002b, for general surveys, Berman, Bound and Machin, 1998, for evidence across OECD countries, and Berman and Machin, 2000, for evidence on skill-biased technological change in developing economies). Given this evidence, we may expect a mismatch between the skill requirements of frontier technologies and the available skills of the workers in less-developed countries to be potentially more important than differences in capital intensity. In this subsection, I outline the model introduced in Acemoglu and Zilibotti (2001), which emphasizes the implications of the mismatch between technologies developed in advanced economies and the skills of the workforce of the less-developed countries. Furthermore, this will enable us to use the ideas related to directed technological change developed in Chapter 15 and also provide us with a tractable multi-sector growth model.

The world economy consists of two groups of countries, the North and the South, and as in Chapter 15, two types of workers, skilled and unskilled. There are two differences between the North and the South. First, all R&D and new innovations take place in the North (so that the North approximates the OECD or the US and some of the other advanced economies). The South simply copies technologies developed in the North. Because of lack of intellectual property rights in the South, the main market of new technologies will be Northern firms. Second, the North is more skill-abundant than the South, in particular,

$$H^n/L^n > H^s/L^s,$$

where H^j denotes the number of skilled workers in country j and L^j denotes the number of unskilled workers. We will use $j = n$ or s to denote the North or the South, and assume that there are many Northern and many Southern countries. There is no population growth. Throughout, all countries have access to the same set of technologies, so there will be no issue of slow technology diffusion. All differences in productivity will arise from the potential mismatch between technology and skills.

On the preference side, all economies are assumed to admit a representative household with the standard preferences, for example, as given in (18.6) above with $n_j = 0$ for all

countries, since there is no population growth. The final good in each country is produced as a Cobb-Douglas aggregate of a continuum 1 of intermediate goods, that is,

$$(18.16) \quad Y_j(t) = \exp \left[\int_0^1 \ln y_j(i, t) di \right]$$

where $Y_j(t)$ is the amount of final good in country j at time t , while $y_j(i, t)$ is the output of intermediate i . As usual, total output is spent on consumption, $C_j(t)$, intermediate expenditures, $X_j(t)$, and also in the North, there will be R&D expenditures equal to $Z_j(t)$. The South will not undertake R&D, but can adopt technologies developed in the North.

Let us assume that the technology for producing intermediate i in country j at time t is given as follows:

$$(18.17) \quad y_j(i, t) = \frac{1}{1-\beta} \left[\int_0^{N_L(t)} x_{L,j}(i, \nu, t)^{1-\beta} d\nu \right] [(1-i)l_j(i, t)]^\beta \\ + \frac{1}{1-\beta} \left[\int_0^{N_H(t)} x_{H,j}(i, \nu, t)^{1-\beta} d\nu \right] [i\omega h_j(i, t)]^\beta.$$

A number of features about this intermediate production function is worth noting. First each intermediate can be produced using two alternative technologies, one using skilled workers, the other one using unskilled labor. Here $l_j(i, t)$ is the number of unskilled workers working in intermediate i in country j at time t . $h_j(i, t)$ is defined similarly. Second, skilled and unskilled workers have different productivities in different industries—incorporating a pattern of cross-industry comparative advantage. In particular, the presence of the terms $1-i$ and i in the production function (18.17) implies that skilled workers are relatively more productive in higher indexed intermediates, while unskilled workers have a comparative advantage in lower indexed intermediates. Third, skilled workers also have an absolute advantage, captured by the parameter ω , which is assumed to be greater than 1. Fourth, as in the standard models with machine varieties, $x_{L,j}(i, \nu)$ denotes the quantity of machines of type ν used with unskilled workers, and $x_{H,j}(i, \nu)$ is defined similarly. This part of the production function parallels to those used in Chapter 15. The number of machine varieties available to be used with skilled and unskilled workers differ and are equal to $N_L(t)$ and $N_H(t)$. The important point here is that these quantities are not indexed by j , since all technologies are available to all countries. This implies that we are ignoring the issue of slow diffusion and focusing on differences arising purely from inappropriateness of technology. Finally, as usual, the term $1/(1-\beta)$ is introduced as a convenient normalization.

Let us assume that the final good sectors and the labor markets are competitive. Again as in Chapters 13 and 15, a technology monopolist can produce these machines at marginal cost ψ and supplies the quantities of machines. Let the prices of these machines be denoted by $p_{L,j}^x(\nu, t)$ and $p_{H,j}^x(\nu, t)$ for the two sectors in country j for machine of type ν at time t . Note that these prices do not depend on i , since the machines are not sector specific. Instead, they are skill specific. As in Chapters 13 and 15, profit maximization by the final

good producers leads to the following demands for machines:

$$\begin{aligned} x_{L,j}(i, \nu, t) &= \left[p_j(i, t) ((1-i)l_j(i, t))^\beta / p_{L,j}^x(\nu, t) \right]^{1/\beta}, \\ x_{H,j}(i, \nu, t) &= \left[p_j(i, t) (i\omega h_j(i, t))^\beta / p_{L,j}^x(\nu, t) \right]^{1/\beta}, \end{aligned}$$

where $p_j(i, t)$ is the relative price of intermediate i in country j at time t in terms of the final good (which is set as the numeraire in each country). The technology monopolist in the North will be the firm that invents the new type of machine, so here the analysis is identical to that in Chapters 13 and 15.

What about in the South? To keep the treatment of Northern and Southern economies symmetric, I assume that in each Southern economy a “technology” firm adopts the new technology invented in the North (at no cost) and acts as the monopolist supplier of that machine for the producers in its own country. Moreover, let us assume that the marginal cost of producing machines for this Southern firm is the same as the inventor in the North, equal to ψ .

As usual, the isoelastic demand for machines imply that the profit-maximizing price for the technology monopolists will be a constant markup over marginal cost, and I normalize the cost to $\psi \equiv 1 - \beta$. The symmetry between the North and the South implies that the price of machines and thus the demand for machines will take the same form in all countries. In particular, we obtain output in sector i in any country j as

$$(18.18) \quad y_j(i, t) = \frac{1}{1-\beta} p_j(i, t)^{(1-\beta)/\beta} [N_L(t) (1-i)l_j(i, t) + N_H(t) i\omega h_j(i, t)].$$

For each economy, $N_L(t)$ and $N_H(t)$ are the state variables. Given these state variables the equilibrium is straightforward to characterize. In particular, the following proposition determines the structure of equilibrium in each country.

PROPOSITION 18.6. *In any country j , given the world technologies $N_L(t)$ and $N_H(t)$, there exists a threshold $I_j(t) \in [0, 1]$ such that skilled workers will be employed only in sectors $i > I_j(t)$, that is, for all $i < I_j(t)$, $h_j(i, t) = 0$, and for all $i > I_j(t)$, $l_j(i, t) = 0$.*

Moreover, prices and labor allocations across sectors will be such that: for all $i < I_j(t)$, $p_j(i, t) = P_{L,j}(t) (1-i)^{-\beta}$ and $l_j(i, t) = L_j/I_j(t)$, while for all $i > I_j(t)$, $p_j(i, t) = P_{H,j}(t) i^{-\beta}$ and $h_j(i, t) = H_j/(1-I_j(t))$ where the positive numbers $P_{L,j}(t)$ and $P_{H,j}(t)$ can be interpreted as the price indices for labor-intensive and skill-intensive intermediates.

PROOF. See Exercise 18.21. □

With Proposition 18.6, the characterization of equilibrium given the level a world technologies $N_L(t)$ and $N_H(t)$ is straightforward. In particular, the technology for the final goods sector in (18.16) implies that the price indices in country j at time t must satisfy

$$(18.19) \quad \frac{P_{H,j}(t)}{P_{L,j}(t)} = \left(\frac{N_H(t) \omega H_j / (1 - I_j(t))}{N_L(t) L_j / I_j(t)} \right)^{-\beta}.$$

Moreover, the threshold sector $I_j(t)$ in country j at time t is indifferent between using skilled and unskilled workers (and technologies) for production, thus $P_{L,j}(t)(1 - I_j(t))^{-\beta} = P_{H,j}(t)I_j(t)^{-\beta}$. Combining this with (18.19),

$$(18.20) \quad \frac{P_{H,j}(t)}{P_{L,j}(t)} = \left(\frac{N_H(t)}{N_L(t)} \frac{\omega H_j}{L_j} \right)^{-\beta/2},$$

and the equilibrium threshold $I_j(t)$ is uniquely pinned down by

$$(18.21) \quad \frac{I_j(t)}{1 - I_j(t)} = \left(\frac{N_H(t)}{N_L(t)} \frac{\omega H_j}{L_j} \right)^{-1/2}.$$

Combining these two equations, we can also derive the level of total output in economy j as

$$(18.22) \quad Y_j(t) = \exp(-\beta) \left[(N_L(t) L_j)^{1/2} + (N_H(t) \omega H_j)^{1/2} \right]^2,$$

and the skill premium as

$$(18.23) \quad \frac{w_{H,j}(t)}{w_{L,j}(t)} = \omega \left(\frac{N_H(t)}{N_L(t)} \right)^{1/2} \left(\frac{\omega H_j}{L_j} \right)^{-1/2}$$

(see Exercise 18.22). An interesting feature of this characterization, apparent from eq. (18.22) is that the multi-sector model in this section leads to an equilibrium allocation in which the level of output is identical to that given a CES production function within elasticity of substitution equal to 2. In fact, this phenomenon is more general and by changing the pattern of comparative advantage of skilled and unskilled workers in different sectors, one can obtain models with aggregate production functions of any elasticities of substitution.

The characterization of the equilibrium above already shows that the type of technologies, $N_L(t)$ and $N_H(t)$, will impact productivity in economies with different factor proportions differently. For example, consider the extreme case in which $H^s = 0$, so that there are no skilled workers in the South. Then, an increase in $N_H(t)$ will increase productivity in the North, but will have no effect in the South. Naturally, when there are skilled and unskilled workers in both the North and the South, the implications of the changes in these two technologies will not be as extreme, but the general principle will continue to apply: an increase in $N_H(t)$ relative to $N_L(t)$ will benefit the skill-abundant North more than the skill-scarce South. But conversely, an increase in $N_L(t)$ will tend to benefit Southern economies relatively more. Thus the question becomes whether the world technology will be appropriate to the needs of the North or the South. Here the features that new technologies are developed in the North and that there are no intellectual property rights for Northern R&D in the South become important. In particular, these features imply that new technologies will be developed—*designed*—for the needs of the North.

To communicate the main ideas related to the emergence of technologies that are inappropriate to factor proportions in the South, let us adopt the simplest version of the directed technological change model from Chapter 15 (in particular, Section 15.3 with the

lab-equipment specification) and suppose that

$$(18.24) \quad \dot{N}_L(t) = \eta Z_L(t) \text{ and } \dot{N}_H(t) = \eta Z_H(t),$$

which is the same as the innovation possibilities frontier in Section 15.3, except that η_L and η_H have been set equal to each other for simplification. The analysis there, combined with the fact that the relevant market sizes are given by H^n and L^n (because research firms can only sell their technologies to Northern firms) implies that the steady-state (balanced growth) equilibrium must take the following form:

PROPOSITION 18.7. *With the lab-equipment specification of directed technological change as in (18.24) and no intellectual property rights in the South, the unique steady-state equilibrium involves Northern relative prices*

$$\frac{P_H^n}{P_L^n} = \left(\frac{\omega H^n}{L^n} \right)^{-\beta}$$

and world relative technology ratio

$$(18.25) \quad \frac{N_H^*}{N_L^*} = \frac{\omega H^n}{L^n}.$$

Moreover, in the North the threshold sector satisfies

$$\frac{1 - I^{n*}}{I^{n*}} = \frac{\omega H^n}{L^n}$$

and the skill premium is

$$\frac{w_H^{n*}}{w_L^{n*}} = \omega.$$

This steady-state equilibrium is globally saddle path stable.

PROOF. **(Sketch)** Equation (18.18) immediately implies that, given $N_L(t)$ and $N_H(t)$ and the prices of skilled and unskilled workers, relative profitability on employing skilled workers is strictly increasing in $i \in [0, 1]$. This implies that there must exist a threshold $I_j(t)$ as specified in the proposition. The Cobb-Douglas specification in (18.16) implies an allocation of labor across intermediates, and the corresponding relationship between the prices of intermediates using skilled labor and those using unskilled labor, so that expenditures on different intermediates are equalized. You are asked to complete the details of this argument, derive the expression for the threshold and the skill premium, and also establish the stability of the equilibrium in Exercise 18.23. $\square$

To understand the implications of directed technological change for equilibrium relative technologies N_L and N_H , let us next introduce three simple concepts. The first is *net output* in country j defined as

$$NY_j \equiv Y_j - X_j,$$

that is, output minus the spending on intermediates. The second and the third are *income per capita* and *income per effective unit of labor* in different countries, defined as

$$y_j \equiv \frac{Y_j}{L_j + H_j} \text{ and } y_j^{eff} \equiv \frac{Y_j}{L_j + \omega H_j}.$$

All of these quantities are functions of labor supplies and of relative technologies, in particular of N_H/N_L . These dependences are suppressed to simplify notation.

The next result shows that the steady-state technologies N_L^* and N_H^* are indeed “appropriate” for the conditions (factor proportions) in the North, and that this creates endogenous income differences between the North and the South.

PROPOSITION 18.8. *Consider the above-described model. Then:*

- (1) *The steady-state equilibrium technology ratio N_H^*/N_L^* is such that, given a constant level of for given $N_H + N_L$, it achieves the unique maximum of net output in the North, NY^n , as a function of relative technology N_H/N_L .*
- (2) *At the steady-state equilibrium technology ratio N_H^*/N_L^* , $y_n > y_s$ and $y_n^{eff} > y_s^{eff}$.*

PROOF. See Exercise 18.24. □

This proposition establishes two important results. First, the steady-state equilibrium technology is indeed appropriate for the needs of the North. This is intuitive, since research firms are innovating targeting the Northern markets (in particular the relative supply of skills in the North). Moreover, the statement that there is a unique maximum of NY_n (given the total amount of “technology” $N_H + N_L$) also implies that net output in the South, NY_s , given by a similar expression, will *not* be maximized by N_H^*/N_L^* . This is the essence of the second result contained in this proposition: because technologies are developed in the North (in practice, corresponding loosely to the OECD) and are designed for the needs (factor proportions) of Northern economies, they are inappropriate for the needs of the South. As a result, income per capita and income per effective units of labor in the North will be higher than in the South. Thus the process of directed technological change, combined with import of frontier technologies to less-developed economies, creates an advantage for the more advanced economies and acts as a force towards greater cross-country inequality. Therefore, the issue of potential mismatch between the technologies of the world frontier and the skills of less-developed countries creates a force towards large income per capita differences among these countries. Acemoglu and Zilibotti (2001) show that this source of cross-country income differences can be quite substantial in practice. Therefore, inappropriateness of technologies of the world to the needs of the less-developed countries, especially the potential mismatch between technology and skill, can create significant income differences.

18.5. Contracting Institutions and Technology Adoption

An important determinant of differences in technology and technology adoption are institutional differences across societies. I have already noted how the parameter σ_j in the model of Section 18.2 can be interpreted as varying across countries because of differences in policies and institutions erecting barriers against technology adoption. Naturally, an approach that links σ_j to such “technology barriers” is rather reduced-form and is most useful in providing a perspective in discussions. To make further progress, we need more micro-founded models of why there are barriers to technology adoptions and how these barriers affect technology

choices. The reasons why certain groups may want to erect barriers against the introduction of new technologies will be discussed in detail in Part 8 below. In Part 7, I discuss other factors affecting the efficiency of the organization of production, which can also be loosely related to “technology choices”. However, before turning to these models, it is useful to show how differences in the ability to write contracts between firms and their suppliers (or firms and their workers) may have first-order effect on technology adoption decisions. I now briefly discuss a model of endogenous technology adoption, which again builds on the framework developed in Chapter 13. The purpose of this model is to illustrate how contractual difficulties can lead to important technological differences across countries and to emphasize the other side of the issue of technology adoption, that is, how the conditions in the adopting country affect the use of these technologies by firms. The model I present is a slight simplification of that by Acemoglu, Antras and Helpman (2007). The main focus is how differences in contracting institutions across countries will affect relationships between producers and suppliers and thus change the profitability of technology adoption. I will also use this model to illustrate how analysis of contracting problems (in this instance between firms) can be easily incorporated into the types of models studied so far.

18.5.1. Description of the Environment. For simplicity, consider a static world and focus on a single country. There exists a continuum of final goods $q(z)$, with $z \in [0, M]$, where M represents the number (measure) of final goods (I use M here, since N will denote technology choice). All consumers have identical preferences,

$$(18.26) \quad u = \left(\int_0^M q(\nu)^\beta d\nu \right)^{1/\beta} - \psi e, \quad 0 < \beta < 1,$$

where e is the total effort exerted by this individual, with ψ representing the cost of effort in terms of real consumption. The parameter $\beta \in (0, 1)$ determines the elasticity of demand and implies that the elasticity of substitution between final goods, $1/(1 - \beta)$, is greater than 1. These preferences imply the demand function

$$q(\nu) = \left[\frac{p(\nu)}{p^I} \right]^{-1/(1-\beta)} \frac{A}{p^I},$$

for each producer $\nu \in [0, M]$, where $p(\nu)$ is the price of good ν , A is the aggregate spending level, and

$$p^I \equiv \left[\int_0^M p(\nu)^{-\beta/(1-\beta)} d\nu \right]^{-(1-\beta)/\beta}$$

is the ideal price index, which is taken as the numeraire, so that $p^I = 1$. This implies that each final good producer will face a demand function of the form $q = Ap^{-1/(1-\beta)}$, where q denotes quantity and p denotes price, and I have dropped the conditioning on z , since I will focus on the decisions of a single firm. The resulting revenue function for the firm can therefore be written as

$$(18.27) \quad R = A^{1-\beta} q^\beta.$$

Production depends on the technology choice of the firm, which is denoted by $N \in \mathbb{R}_+$. More advanced technologies involve a greater range of intermediate goods (inputs), supplied by different suppliers. The transactions between the producer and the suppliers will necessitate contracting relationships. For each $j \in [0, N]$, let $X(j)$ be the quantity of intermediate input j . The production function of the representative firm takes the standard CES form

$$(18.28) \quad q = N^{\kappa+1-1/\alpha} \left[\int_0^N X(j)^\alpha dj \right]^{1/\alpha},$$

where $\alpha \in (0, 1)$, so that the elasticity of substitution between inputs, $\varepsilon \equiv 1/(1 - \alpha)$, is always greater than one. In addition, $\kappa > 0$. The standard specification of the CES aggregator would not involve the term $N^{\kappa+1-1/\alpha}$ (that is, it would implicitly set $\kappa = 1/\alpha - 1$). In that case, as in Section 12.4 in Chapter 12, when $X(j) = X$, total output is $q = N^{1/\alpha} X$, and both the elasticity of substitution between inputs and the elasticity of output to changes in technology, N , would be governed by the same parameter, α . By introducing the term $N^{\kappa+1-1/\alpha}$ in front of the integral, we are separating these two elasticities.

There is a large number of profit-maximizing suppliers that can produce the necessary intermediate goods. Suppose that each supplier has the same outside option $w_0 > 0$. For now, let us take w_0 as given and also assume that each intermediate input needs to be produced by a different supplier with whom the firm needs to contract (see Exercise 18.31 on endogenizing this outside option). A supplier assigned to the production of an intermediate input needs to undertake relationship-specific investments in a unit measure of (symmetric) activities. The marginal cost of investment for each activity is ψ as specified in (18.26). The production function of intermediate inputs is Cobb-Douglas and symmetric in the activities,

$$(18.29) \quad X(j) = \exp \left[\int_0^1 \ln x(i, j) di \right],$$

where $x(i, j)$ is the level of investment in activity i performed by the supplier of input j . This formulation will allow a tractable parameterization of contractual incompleteness, whereby a subset of the investments necessary for production will be nonverifiable and thus noncontractible. Finally, let us assume that adopting a technology N involves costs $\Gamma(N)$, and impose the following two restrictions on $\Gamma(N)$:

- (i) For all $N > 0$, $\Gamma(N)$ is twice differentiable, with $\Gamma'(N) > 0$ and $\Gamma''(N) > 0$.
- (ii) For all $N > 0$, $N\Gamma''(N) / [\Gamma'(N) + w_0] > [\beta(\kappa + 1) - 1] / (1 - \beta)$.

These restrictions are standard. In particular, they introduce enough convexity to ensure interior solutions.

The relationship between the producer and its suppliers requires contracts to ensure that the suppliers deliver the required inputs. Let the payment to supplier j consist of two parts: an ex ante payment $\tau(j) \in \mathbb{R}$ before the investments, the $x(i, j)$'s, take place, and a payment $s(j)$ after the investments. Then, the payoff to supplier j , also taking account of her outside

option, is

$$(18.30) \quad \pi_x(j) = \max \left\{ \tau(j) + s(j) - \int_0^1 \psi x(i, j) di, w_0 \right\}.$$

Similarly, the payoff to the firm is

$$(18.31) \quad \pi = R - \int_0^N [\tau(j) + s(j)] dj - \Gamma(N),$$

where R is revenue and the other two terms on the right-hand side represent costs. Substituting (18.28) and (18.29) into (18.27), revenue can be expressed as

$$(18.32) \quad R = A^{1-\beta} N^{\beta(\kappa+1-1/\alpha)} \left[\int_0^N \left(\exp \left(\int_0^1 \ln x(i, j) di \right) \right)^\alpha dj \right]^{\beta/\alpha}.$$

18.5.2. Equilibrium under Complete Contracts. As a benchmark, consider the “idealized” case of complete contracts, where the firm has full control over all investments and pays each supplier her outside option. Conceptually, complete contracts correspond to the case in which markets are complete, and intermediates of different qualities can be bought and sold in a quasi-competitive fashion. Almost all of the models presented so far in this book have assumed complete contracts (the exception being the model in Section 10.6 in Chapter 10). While this is a good approximation for many commodities, complete contracts (or the corresponding complete markets) may not always capture the essence of the interaction between firms and their suppliers, especially when contracting institutions are somewhat imperfect, so that using courts or other legal sanctions against firms that breach their contractual agreements might be costly.

To prepare for our treatment below of technology adoption under incomplete contracts, consider a game where the firm chooses a technology level N and makes a contract offer $\left[\{x(i, j)\}_{i \in [0,1]}, \{s(j), \tau(j)\} \right]$ for every input $j \in [0, N]$. If a supplier accepts this contract for input j , she is obliged to supply $\{x(i, j)\}_{i \in [0,1]}$ as stipulated in the contract in exchange for the payments $\{s(j), \tau(j)\}$. A *subgame perfect equilibrium* of this game is a strategy combination for the firm and the suppliers such that suppliers maximize (18.30) and the firm maximizes (18.31). An equilibrium can be alternatively represented as a solution to the following maximization problem:

$$(18.33) \quad \max_{N, \{x(i, j)\}_{i,j}, \{s(j), \tau(j)\}_j} R - \int_0^N [\tau(j) + s(j)] dj - \Gamma(N)$$

subject to (18.32) and the suppliers’ *participation constraint*,

$$(18.34) \quad s(j) + \tau(j) - \psi \int_0^1 x(i, j) di \geq w_0 \text{ for all } j \in [0, N].$$

Since the firm has no reason to provide rents to the suppliers, it chooses payments $s(j)$ and $\tau(j)$ that satisfy (18.34) with equality. Moreover, with complete contracts, $\tau(j)$ and $s(j)$ are perfect substitutes, so only the sum $s(j) + \tau(j)$ matters and is determined in equilibrium—this will not be the case when contracts are incomplete.

Moreover, since the firm's objective function, (18.33), is (jointly) concave in the investment levels $x(i, j)$ and these investments are all equally costly, the firm chooses the same investment level x for all activities in all intermediate inputs. Now, substituting for (18.34) in (18.33), we obtain the following simpler unconstrained maximization problem for the firm:

$$(18.35) \quad \max_{N, x} A^{1-\beta} N^{\beta(\kappa+1)} x^\beta - \psi Nx - \Gamma(N) - w_0 N.$$

The first-order conditions of this problem imply:

$$(18.36) \quad (N^*)^{\frac{\beta(\kappa+1)-1}{1-\beta}} A \kappa \beta^{1/(1-\beta)} \psi^{-\beta/(1-\beta)} = \Gamma'(N^*) + w_0,$$

$$(18.37) \quad x^* = \frac{\Gamma'(N^*) + w_0}{\kappa \psi}.$$

Equations (18.36) and (18.37) can be solved recursively. The restrictions on the function Γ above ensure that eq. (18.36) has a unique solution for N^* , which, together with (18.37), yields a unique solution for x^* .

When all the investment levels are identical and equal to x , output is $q = N^{\kappa+1}x$. Since a total of $NX = Nx$ inputs are used in the production process, a natural measure of productivity is output divided by total input use, $P = N^\kappa$. In the case of complete contracts this productivity level is $P^* = (N^*)^\kappa$, which is increasing in the level of technology. Summarizing this analysis:

PROPOSITION 18.9. *Consider the above-described model, take A as given and suppose that there are complete contracts. Then, there exists a unique equilibrium with technology and investment levels $N^* > 0$ and $x^* > 0$ given by (18.36) and (18.37). Furthermore, this equilibrium satisfies:*

$$\frac{\partial N^*}{\partial A} > 0, \quad \frac{\partial x^*}{\partial A} \geq 0, \quad \frac{\partial N^*}{\partial \alpha} = \frac{\partial x^*}{\partial \alpha} = 0.$$

PROOF. See Exercise 18.27. □

In the case of complete contracts, the size of the market, which corresponds to A and from the viewpoint of the individual firm is exogenous, has a positive effect on investments by suppliers of intermediate inputs and productivity, because a greater market size makes both suppliers' and the producer's investments more productive. The other noteworthy implication of this proposition is that under complete contracts, the level of technology and thus productivity do not depend on the elasticity of substitution between intermediate inputs, $1/(1-\alpha)$.

18.5.3. Equilibrium under Incomplete Contracts. Let us next consider the same environment under incomplete contracts. We model the imperfection of the contracting institutions by assuming that there exists a $\mu \in [0, 1]$ such that, for every intermediate input j , investments in activities $0 \leq i \leq \mu$ are observable and verifiable and therefore contractible, while investments in activities $\mu < i \leq 1$ are not contractible. Consequently, a contract stipulates investment levels $x(i, j)$ for the μ contractible activities, but does not specify the investment levels in the remaining $1 - \mu$ noncontractible activities. Instead, suppliers choose

their investments in noncontractible activities in anticipation of the ex post distribution of revenue, and may decide to withhold their services in these activities from the firm. Economies with weak contracting institutions will have a low μ , thus will feature only a small set of tasks that are contractible, whereas more developed contracting institutions will correspond to high levels of μ .

The ex post distribution of revenues in activities that are not ex ante contractible will be determined by multilateral bargaining between the firm and its suppliers. The exact bargaining protocol will determine investment incentives of suppliers and the profitability of investment for the firm. Below we will use the *Shapley value* as a natural solution concept for this multilateral bargaining game. First, consider the timing of events:

- The firm adopts a technology N and offers a contract $[\{x_c(i, j)\}_{i=0}^\mu, \tau(j)]$ for every intermediate input $j \in [0, N]$, where $x_c(i, j)$ is an investment level in a contractible activity and $\tau(j)$ is an upfront payment to supplier j . The payment $\tau(j)$ can be positive or negative.
- Potential suppliers decide whether to apply for the contracts. Then, the firm chooses N suppliers, one for each intermediate input j .
- All suppliers $j \in [0, N]$ simultaneously choose investment levels $x(i, j)$ for all $i \in [0, 1]$. In the contractible activities $i \in [0, \mu]$ the suppliers will invest $x(i, j) = x_c(i, j)$.
- The suppliers and the firm bargain over the division of revenue, and at this stage, suppliers can withhold their services in noncontractible activities.
- Output is produced and sold, and the revenue R is distributed according to the bargaining agreement.

Let us now characterize a *symmetric subgame perfect equilibrium* (SSPE) of this game, where bargaining outcomes in all subgames are determined by Shapley values.

Behavior along the SSPE can be described by a tuple $\{\tilde{N}, \tilde{x}_c, \tilde{x}_n, \tilde{\tau}\}$ in which $\tilde{N}$ represents the level of technology, $\tilde{x}_c$ the investment in contractible activities, $\tilde{x}_n$ the investment in noncontractible activities, and $\tilde{\tau}$ the upfront payment to every supplier. That is, for every $j \in [0, \tilde{N}]$ the upfront payment is $\tau(j) = \tilde{\tau}$, and the investment levels are $x(i, j) = \tilde{x}_c$ for $i \in [0, \mu]$ and $x(i, j) = \tilde{x}_n$ for $i \in (\mu, 1]$. With a slight abuse of terminology, I denote the SSPE by $\{\tilde{N}, \tilde{x}_c, \tilde{x}_n\}$.

As is typically the case in extensive form complete information games, the SSPE can be characterized by backward induction. First, consider the penultimate stage of the game, with N as the level of technology, x_c as the level of investment in contractible activities. Suppose also that each supplier other than j has chosen a level of investment in noncontractible activities equal to $x_n(-j)$ (these are all the same, because we are constructing a symmetric equilibrium), while the investment level in every noncontractible activity by supplier j is $x_n(j)$. Given these investments, the suppliers and the firm will engage in multilateral bargaining. Denote the return to supplier j resulting from this bargaining by $\bar{s}_x[N, x_c, x_n(-j), x_n(j)]$. The optimal investment by supplier j implies that $x_n(j)$ must be

chosen to maximize $\bar{s}_x [N, x_c, x_n(-j), x_n(j)]$ minus the cost of investment in noncontractible activities, $(1 - \mu) \psi x_n(j)$. In a symmetric equilibrium, $x_n(j) = x_n(-j)$, or in other words, x_n needs to be a fixed-point given by:

$$(18.38) \quad x_n \in \arg \max_{x_n(j)} \bar{s}_x [N, x_c, x_n, x_n(j)] - (1 - \mu) \psi x_n(j).$$

Equation (18.38) can be thought of as an “incentive compatibility constraint,” with the additional symmetry requirement. While this equation is written with “ $\in$ ” to allow for the fact that there may be more than one maximizers of the expression on the right-hand side, the structure of the current model ensures that there will be a unique maximizer, thus “ $\in$ ” can be replaced with “ $=$ ”.

In a symmetric equilibrium with technology N , with investment in contractible activities given by x_c and with investment in noncontractible activities equal to x_n , the revenue of the firm is given by $R = A^{1-\beta} (N^{\kappa+1} x_c^\mu x_n^{1-\mu})^\beta$. Moreover, let $s_x(N, x_c, x_n) = \bar{s}_x(N, x_c, x_n, x_n)$, then the Shapley value of the firm is obtained as a residual:

$$s_q(N, x_c, x_n) = A^{1-\beta} (N^{\kappa+1} x_c^\mu x_n^{1-\mu})^\beta - N s_x(N, x_c, x_n).$$

Now consider the stage in which the firm chooses N suppliers from a pool of applicants. If suppliers expect to receive less than their outside option, w_0 , this pool is empty. Therefore, for production to take place, the final-good producer has to offer a contract that satisfies the participation constraint of suppliers under incomplete contracts, that is,

$$(18.39) \quad \bar{s}_x(N, x_c, x_n, x_n) + \tau \geq \mu \psi x_c + (1 - \mu) \psi x_n + w_0 \quad \text{for } x_n \text{ that satisfies (18.38).}$$

In other words, given N and (x_c, τ) , each supplier $j \in [0, N]$ should expect her Shapley value plus the upfront payment to cover the cost of investment in contractible and noncontractible activities and the value of her outside option.

The maximization problem of the firm can then be written as:

$$\max_{N, x_c, x_n, \tau} s_q(N, x_c, x_n) - N\tau - \Gamma(N)$$

subject to (18.38) and (18.39).

With no restrictions on τ , the participation constraint (18.39) will be satisfied with equality; otherwise the firm could reduce τ without violating (18.39) and increase its profits. Therefore, the upfront payment τ can be solved out from this constraint and substituted into the firm’s objective function. This yields the simpler maximization problem,

$$(18.40) \quad \max_{N, x_c, x_n} s_q(N, x_c, x_n) + N [\bar{s}_x(N, x_c, x_n, x_n) - \mu \psi x_c - (1 - \mu) \psi x_n] - \Gamma(N) - w_0 N,$$

subject to (18.38).

The SSPE $\{\tilde{N}, \tilde{x}_c, \tilde{x}_n\}$ solves this problem, and the corresponding upfront payment satisfies

$$(18.41) \quad \tilde{\tau} = \mu \psi \tilde{x}_c + (1 - \mu) \psi \tilde{x}_n + w_0 - \bar{s}_x(\tilde{N}, \tilde{x}_c, \tilde{x}_n, \tilde{x}_n).$$

The key issue in the presence of incomplete contracts is that the payments from the firm to its suppliers will be determined ex post through bargaining rather than through contractual

arrangements. As noted above, different bargaining protocols between suppliers and the producer will lead to somewhat different results. In the current context, the most natural choice appears to be the Shapley value, since it provides a plausible and tractable division rule for multilateral bargaining problems. The derivation of this formula is not essential for the results here, thus it is included for completeness at the end of this section. The next proposition provides the form of this bargaining solution.

PROPOSITION 18.10. *Suppose that supplier j invests $x_n(j)$ in her noncontractible activities, all the other suppliers invest $x_n(-j)$ in their noncontractible activities, every supplier invests x_c in her contractible activities, and the level of technology is N . Then, the Shapley value of supplier j is*

$$(18.42) \quad \bar{s}_x[N, x_c, x_n(-j), x_n(j)] = (1 - \gamma) A^{1-\beta} \left[\frac{x_n(j)}{x_n(-j)} \right]^{(1-\mu)\alpha} x_c^{\beta\mu} x_n(-j)^{\beta(1-\mu)} N^{\beta(\kappa+1)-1},$$

where

$$(18.43) \quad \gamma \equiv \frac{\alpha}{\alpha + \beta}.$$

PROOF. See subsection 18.5.4. □

A number of features of (18.42) are worth noting. First, the derived parameter $\gamma \equiv \alpha/(\alpha + \beta)$ represents the bargaining power of the firm; it is increasing in α and decreasing in β . A higher elasticity of substitution between intermediate inputs—a higher α —raises the firm's bargaining power, because it makes every supplier *less essential* in production and therefore raises the share of revenue appropriated by the firm. In contrast, a higher elasticity of demand for the final good—higher β —reduces the firm's bargaining power, because, for any coalition, it reduces the marginal contribution of the firm to the coalition's payoff as a fraction of revenue.

Second, in equilibrium, all suppliers invest equally in all the noncontractible activities, that is, $x_n(j) = x_n(-j) = x_n$, and so

$$(18.44) \quad \begin{aligned} s_x(N, x_c, x_n) &= \bar{s}_x(N, x_c, x_n, x_n) = (1 - \gamma) A^{1-\beta} x_c^{\beta\mu} x_n^{\beta(1-\mu)} N^{\beta(\kappa+1)-1} \\ &= (1 - \gamma) \frac{R}{N}, \end{aligned}$$

where $R = A^{1-\beta} x_c^{\beta\mu} x_n^{\beta(1-\mu)} N^{\beta(\kappa+1)}$ is the total revenue of the firm. Thus, the joint Shapley value of the suppliers, $N s_x(N, x_c, x_n)$, equals the fraction $1 - \gamma$ of the revenue, and the firm receives the remaining fraction γ , so that

$$(18.45) \quad \begin{aligned} s_q(N, x_c, x_n) &= \gamma A^{1-\beta} x_c^{\beta\mu} x_n^{\beta(1-\mu)} N^{\beta(\kappa+1)} \\ &= \gamma R. \end{aligned}$$

This is a relatively simple rule for the division of revenue between the firm and its suppliers.

Finally, when α is smaller, $\bar{s}_x[N, x_c, x_n(-j), x_n(j)]$ is more concave with respect to $x_n(j)$, because greater complementarity between the intermediate inputs implies that a given

change in the relative employment of two inputs has a larger impact on their relative marginal products. The impact of α on the concavity of $\bar{s}_x(\cdot)$ will play an important role in the following results. The parameter β , on the other hand, affects the concavity of revenue in output (see (18.27)), but has no effect on the concavity of $\bar{s}_x$, because with a continuum of suppliers, a single supplier has an infinitesimal effect on output.

To characterize a SSPE, let us first derive the incentive compatibility constraint using (18.38) and (18.42):

$$x_n = \arg \max_{x_n(j)} (1 - \gamma) A^{1-\beta} \left[\frac{x_n(j)}{x_n} \right]^{(1-\mu)\alpha} x_c^{\beta\mu} x_n^{\beta(1-\mu)} N^{\beta(\kappa+1)-1} - \psi(1 - \mu) x_n(j).$$

Relative to the producer's first-best (complete contracts) characterized above, there are two differences. First, the term $(1 - \gamma)$ implies that the supplier is not the full residual claimant of the return from her investment in noncontractible activities and thus underinvests in these activities. Second, as discussed above, multilateral bargaining distorts the perceived concavity of the private return relative to the social return. Using the first-order condition of this problem and solving for the fixed point by substituting $x_n(j) = x_n$ yields a unique x_n :

$$(18.46) \quad x_n = \bar{x}_n(N, x_c) \equiv \left[\alpha(1 - \gamma) \psi^{-1} x_c^{\beta\mu} A^{1-\beta} N^{\beta(\kappa+1)-1} \right]^{1/[1-\beta(1-\mu)]}.$$

This equation implies that investments in noncontractible activities are increasing in α . This follows from the fact that $\alpha(1 - \gamma) = \alpha\beta/(\alpha + \beta)$ is increasing in α . The economics of this relationship is the outcome of two opposing forces. The share of the suppliers in revenue, $(1 - \gamma)$, is decreasing in α , because greater substitution between the intermediate inputs reduces the suppliers' ex post bargaining power. But a greater level of α also reduces the concavity of $\bar{s}_x(\cdot)$ in x_n , increasing the marginal reward from investing further in noncontractible activities. Because the latter effect dominates, x_n is increasing in α .

Another interesting feature is that contractible and noncontractible activities are complements, and in particular, $\bar{x}_n(N, x_c)$ is increasing in x_c . Finally, the effect of N on x_n is ambiguous, since investment in noncontractible activities declines with the level of technology when $\beta(\kappa + 1) < 1$ and increases with N when $\beta(\kappa + 1) > 1$. This is because an increase in N has two opposite effects on a supplier's incentives to invest; a greater number of inputs increases the marginal product of investment due to the "love-for-variety" embodied in the technology, but at the same time, the bargaining share of a supplier, $(1 - \gamma)/N$, declines with N . For large values of κ , the former effect dominates, while for small values of κ , the latter dominates.

Now, using (18.44), (18.45) and (18.46), the firm's optimization problem (18.40) can be expressed as the maximization of

$$(18.47) \quad A^{1-\beta} \left[x_c^\mu \bar{x}_n(N, x_c)^{1-\mu} \right]^\beta N^{\beta(\kappa+1)} - \psi N \mu x_c - \psi N (1 - \mu) \bar{x}_n(N, x_c) - \Gamma(N) - w_0 N$$

with respect to N and x_c , where $\bar{x}_n(N, x_c)$ is defined in (18.46). Substituting (18.46) into (18.47) and differentiating with respect to N and x_c results in two first-order conditions,

which yield a unique solution $(\tilde{N}, \tilde{x}_c)$ to (18.47):

$$(18.48) \quad \left[\tilde{N}^{\frac{\beta(\kappa+1)-1}{1-\beta}} A \kappa \beta^{\frac{1}{1-\beta}} \psi^{-\frac{\beta}{1-\beta}} \right] \left[\frac{1 - \alpha(1 - \gamma)(1 - \mu)}{1 - \beta(1 - \mu)} \right]^{\frac{1-\beta(1-\mu)}{1-\beta}} \\ \times [\beta^{-1} \alpha(1 - \gamma)]^{\frac{\beta(1-\mu)}{1-\beta}} \\ = \Gamma'(\tilde{N}) + w_0,$$

$$(18.49) \quad \tilde{x}_c = \frac{\Gamma'(\tilde{N}) + w_0}{\kappa \psi}.$$

As in the complete contracts case, these two conditions determine the equilibrium recursively. First, (18.48) gives $\tilde{N}$, and then given $\tilde{N}$, (18.49) yields $\tilde{x}_c$. Moreover, using (18.46), (18.48) and (18.49) gives the level of investment in noncontractible activities as

$$(18.50) \quad \tilde{x}_n = \frac{\alpha(1 - \gamma)[1 - \beta(1 - \mu)]}{\beta[1 - \alpha(1 - \gamma)(1 - \mu)]} \left(\frac{\Gamma'(\tilde{N}) + w_0}{\kappa \psi} \right).$$

Comparing (18.37) to (18.49), we see that for a given N the implied level of investment in contractible activities under incomplete contracts, $\tilde{x}_c$, is identical to the investment level in contractible activities under complete contracts, x^* . This highlights the fact that differences in investments in contractible activities between these economic environments only result from differences in technology adoption. In fact, comparing (18.36) with (18.48), we see that $\tilde{N}$ and N^* differ only because of the two bracketed terms on the left-hand side of (18.48). These represent the distortions created by bargaining between the firm and its suppliers. Intuitively, technology adoption is distorted because incomplete contracts reduce investments in noncontractible activities below the level of investment in contractible activities and this “underinvestment” reduces the profitability of technologies with high N . As $\mu \rightarrow 1$ (and contractual imperfections disappear), both of these bracketed terms on the left-hand side of (18.48) go to 1 and $(\tilde{N}, \tilde{x}_c) \rightarrow (N^*, x^*)$.

I next provide a number of comparative static results on the SSPE under incomplete contracts and compare the incomplete contracts equilibrium to the equilibrium under complete contracts. The comparative static results are facilitated by the block-recursive structure of the equilibrium; any change in A , μ or α that increases the left-hand side of (18.48) also increase $\tilde{N}$, and the effect on $\tilde{x}_c$ and $\tilde{x}_n$ can then be obtained from (18.49) and (18.50). The main results are provided in the next proposition:

PROPOSITION 18.11. *Consider the above-described model with incomplete contracts and suppose that the restrictions on Γ hold. Then, there exists a unique SSPE under incomplete contracts, $\{\tilde{N}, \tilde{x}_c, \tilde{x}_n\}$, characterized by (18.48), (18.49) and (18.50). Furthermore, $\{\tilde{N}, \tilde{x}_c, \tilde{x}_n\}$ satisfies $\tilde{N}, \tilde{x}_c, \tilde{x}_n > 0$,*

$$\tilde{x}_n < \tilde{x}_c,$$

$$\begin{aligned}
 \frac{\partial \tilde{N}}{\partial A} &> 0, \quad \frac{\partial \tilde{x}_c}{\partial A} \geq 0, \quad \frac{\partial \tilde{x}_n}{\partial A} \geq 0, \\
 \frac{\partial \tilde{N}}{\partial \mu} &> 0, \quad \frac{\partial \tilde{x}_c}{\partial \mu} \geq 0, \quad \frac{\partial (\tilde{x}_n/\tilde{x}_c)}{\partial \mu} > 0, \\
 \frac{\partial \tilde{N}}{\partial \alpha} &> 0, \quad \frac{\partial \tilde{x}_c}{\partial \alpha} \geq 0, \quad \frac{\partial (\tilde{x}_n/\tilde{x}_c)}{\partial \alpha} > 0.
 \end{aligned}$$

PROOF. See Exercise 18.28. □

This proposition states that suppliers invest less in noncontractible activities than in contractible activities. In particular:

$$(18.51) \quad \frac{\tilde{x}_n}{\tilde{x}_c} = \frac{\alpha(1-\gamma)[1-\beta(1-\mu)]}{\beta[1-\alpha(1-\gamma)(1-\mu)]} < 1,$$

which follows from eq.'s (18.49) and (18.50) and from the fact that $\alpha(1-\gamma) = \alpha\beta/(\alpha+\beta) < \beta$ (recall (18.43)). This is intuitive: the producer firm is the full residual claimant of the return to investments in contractible activities and it dictates these investments in the contract. In contrast, investments in noncontractible activities are decided by the suppliers, who are not the full residual claimants of the returns generated by these investments (recall (18.44)) and thus underinvest in these activities.

In addition, the level of technology and investments in both contractible and noncontractible activities are increasing in the size of the market, in the fraction of contractible activities (quality of contracting institutions), and in the elasticity of substitution between intermediate inputs. The impact of the size of the market is intuitive; a greater A makes production more profitable and thus increases investments and equilibrium technology. Better contracting institutions, on the other hand, imply that a greater fraction of activities receive the higher investment level $\tilde{x}_c$ rather than $\tilde{x}_n < \tilde{x}_c$. This makes the choice of a more advanced technology more profitable. A higher N , in turn, increases the profitability of further investments in $\tilde{x}_c$ and $\tilde{x}_n$. Better contracting institutions also close the (proportional) gap between $\tilde{x}_c$ and $\tilde{x}_n$ because with a higher fraction of contractible activities, the marginal return to investment in noncontractible activities is also higher.

A higher α (lower complementarity between intermediate inputs) also increases technology choices and investments. The reason is related to the discussion in the previous subsection where it was shown that a higher α reduces the share of each supplier but also makes $\bar{s}_x(\cdot)$ less concave. Because the latter effect dominates, a lower degree of complementarity increases supplier investments and makes the adoption of more advanced technologies more profitable.

One of the main implications of this analysis is that contractual frictions (here captured by the incomplete contracts equilibrium) cause underinvestment in quality, and thus discourage technology adoption and reduce productivity. This is summarized in the next proposition. Note that productivity under incomplete contracts is $\tilde{P} = \tilde{N}^\kappa$, while productivity under complete contracts is $P^* = (N^*)^\kappa$.

PROPOSITION 18.12. Let $\{\tilde{N}, \tilde{x}_c, \tilde{x}_n\}$ be the unique SSPE with incomplete contracts and let $\{N^*, x^*\}$ be the unique equilibrium with complete contracts. Then

$$\tilde{N} < N^* \text{ and } \tilde{x}_n < \tilde{x}_c < x^*.$$

PROOF. See Exercise 18.29. □

Since incomplete contracts lead to the choice of less advanced (lower N) technologies, they also reduce productivity and investments in contractible and noncontractible activities. Acemoglu, Antras and Helpman (2007) also show that the technology and income differences resulting from relatively modest differences in contracting institutions can be quite large. Therefore, the link between contracting institutions and technology adoption provides us with a theoretical mechanism that might generate significant technology differences across countries.

18.5.4. The Shapley Value and the Proof of Proposition 18.10 *. The concept of Shapley values, first proposed by Shapley (1953) has both intuitive and game theoretic appeal. In a bargaining game with a finite number of players, each player's Shapley value is the average of her contributions to all coalitions that consist of players ordered below her in all feasible permutations. More explicitly, in a game with $T+1$ players, let $g = \{g(0), g(1), \dots, g(T)\}$ be a permutation of $0, 1, 2, \dots, T$, where player 0 is the firm and players $1, 2, \dots, T$ are the suppliers, and let $z_g^j = \{j' \mid g(j) > g(j')\}$ be the set of players ordered below j in the permutation g . Let us denote the set of all feasible permutations by G , the set of all subsets of $T+1$ players by S , and the value of a coalition consisting of a subset of the $T+1$ players by $v : S \rightarrow \mathbb{R}$. Then, the Shapley value of player j is

$$s_j = \frac{1}{(T+1)!} \sum_{g \in G} [v(z_g^j \cup j) - v(z_g^j)].$$

Let us now derive the *asymptotic* Shapley value proposed by Aumann and Shapley (1974), which involves considering the limit of this expression as the number of players goes to infinity. Let there be T suppliers each one controlling a range $\xi = N/T$ of the continuum of intermediate inputs. Due to symmetry, all suppliers provide an amount x_c of contractible activities. As for the noncontractible activities, consider a situation in which a supplier j supplies an amount $x_n(j)$ per noncontractible activity, while the $T-1$ remaining suppliers supply the same amount $x_n(-j)$ (note that we are again appealing to symmetry).

To compute the Shapley value for this particular supplier j , we need to determine the marginal contribution of this supplier to a given coalition of agents. A coalition of n suppliers and the firm yields a sales revenue of

$$F_{IN}(n, N; \xi) = A^{1-\beta} N^{\beta(\kappa+1-1/\alpha)} x_c^{\beta\mu} \left[(n-1) \xi x_n(-j)^{(1-\mu)\alpha} + \xi x_n(j)^{(1-\mu)\alpha} \right]^{\beta/\alpha},$$

when the supplier j is in the coalition, and a sales revenue

$$F_{OUT}(n, N; \xi) = A^{1-\beta} N^{\beta(\kappa+1-1/\alpha)} x_c^{\beta\mu} \left[n \xi x_n(-j)^{(1-\mu)\alpha} \right]^{\beta/\alpha}$$

when supplier j is *not* in the coalition. Notice that even when $n < N$, the term $N^{\beta(\kappa+1-1/\alpha)}$ remains in front, because it represents a feature of the technology affecting output independent of the amount and quality of the inputs provided by the suppliers. On the other hand, productivity suffers because the term in square brackets is lower.

The Shapley value of player j is then

$$(18.52) \quad s_j = \frac{1}{(T+1)!} \sum_{g \in G} [v(z_g^j \cup j) - v(z_g^j)].$$

The fraction of permutations in which $g(j) = i$ is $1/(T+1)$ for every i . If $g(j) = 0$, then $v(z_g^j \cup j) = v(z_g^j) = 0$, because in this event the firm is necessarily ordered *after* j . If $g(j) = 1$ then the firm is ordered before j with probability $1/T$ and after j with probability $1-1/T$. In the former case $v(z_g^j \cup j) = F_{IN}(1, N; \xi)$, while in the latter case $v(z_g^j \cup j) = 0$. Therefore the conditional expected value of $v(z_g^j \cup j)$, given $g(j) = 1$, is $F_{IN}(1, N; \xi)/T$. By similar reasoning, the conditional expected value of $v(z_g^j)$ is $F_{OUT}(0, N; \xi)/T$. Repeating the same argument for $g(j) = i$, $i > 1$, the conditional expected value of $v(z_g^j \cup j)$, given $g(j) = i$, is $iF_{IN}(i, N; \xi)/T$, and the conditional expected value of $v(z_g^j)$ is $iF_{OUT}(i-1, N; \xi)/T$. It then follows from (18.52) that

$$\begin{aligned} s_j &= \frac{1}{(T+1)T} \sum_{i=1}^T i [F_{IN}(i, N; \xi) - F_{OUT}(i-1, N; \xi)] \\ &= \frac{1}{(N+\xi)N} \sum_{i=1}^T i \xi [F_{IN}(i, N; \xi) - F_{OUT}(i-1, N; \xi)] \xi. \end{aligned}$$

Substituting for the expressions of F_{IN} and F_{OUT} ,

$$\begin{aligned} s_j &= \frac{A^{1-\beta} N^{\beta(\kappa+1-1/\alpha)} x_c^{\beta\mu}}{(N+\xi)N} \sum_{i=1}^T i \xi \left\{ i \xi x_n(-j)^{(1-\mu)\alpha} + \xi \left[x_n(j)^{(1-\mu)\alpha} - x_n(-j)^{(1-\mu)\alpha} \right] \right\}^{\beta/\alpha} \xi \\ &\quad - \frac{A^{1-\beta} N^{\beta(\kappa+1-1/\alpha)} x_c^{\beta\mu}}{(N+\xi)N} \sum_{i=1}^T i \xi \left[i \xi x_n(-j)^{(1-\mu)\alpha} - \xi x_n(-j)^{(1-\mu)\alpha} \right]^{\beta/\alpha} \xi. \end{aligned}$$

Now using a first-order Taylor expansion (see Theorem A.23 in Appendix Chapter A), we obtain

$$s_j = \frac{A^{1-\beta} N^{\beta(\kappa+1-1/\alpha)} x_c^{\beta\mu} (\beta/\alpha) \xi x_n(j)^{(1-\mu)\alpha}}{(N+\xi)N} \sum_{i=1}^T (i\xi) \left[i \xi x_n(-j)^{(1-\mu)\alpha} \right]^{(\beta-\alpha)/\alpha} \xi + o(\xi),$$

where $o(\xi)$ represents terms such that $\lim_{\xi \rightarrow 0} o(\xi)/\xi = 0$. Rearranging this expression and dividing by $o(\xi)$,

$$\frac{s_j}{\xi} = \frac{A^{1-\beta} N^{\beta(\kappa+1-1/\alpha)} (\beta/\alpha) \left[\frac{x_n(j)}{x_n(-j)} \right]^{(1-\mu)\alpha} x_c^{\beta\mu} x_n(-j)^{\beta(1-\mu)}}{(N+\xi)N} \sum_{i=1}^T (i\xi)^{\beta/\alpha} \xi + \frac{o(\xi)}{\xi}.$$

Now taking the limit as $T \rightarrow \infty$, which is also equivalent to the limit $\xi = N/T \rightarrow 0$, $\lim_{\xi \rightarrow 0} o(\xi)/\xi = 0$, gives the Riemann integral (recall Section B.2 in Appendix Chapter B):

$$\lim_{T \rightarrow \infty} \left(\frac{s_j}{\xi} \right) = \frac{A^{1-\beta} N^{\beta(\kappa+1-1/\alpha)} (\beta/\alpha) \left[\frac{x_n(j)}{x_n(-j)} \right]^{(1-\mu)\alpha} x_c^{\beta\mu} x_n (-j)^{\beta(1-\mu)}}{N^2} \int_0^N z^{\beta/\alpha} dz.$$

Solving this integral yields

$$\lim_{T \rightarrow \infty} (s_j/\xi) = (1 - \gamma) A^{1-\beta} \left[\frac{x_n(j)}{x_n(-j)} \right]^{(1-\mu)\alpha} x_c^{\beta\mu} x_n (-j)^{\beta(1-\mu)} N^{\beta(\kappa+1)-1},$$

with $\gamma \equiv \alpha/(\alpha + \beta)$. This corresponds to eq. (18.42) and completes the proof of the proposition.

18.6. Taking Stock

This chapter presented models of technology differences across societies. While the baseline endogenous growth models, such as those studied in Part 4, are useful in understanding the incentives of research firms to create new technologies and can generate different rates of technological change across different economies, two factors suggest that a somewhat different perspective is necessary for understanding technology differences across nations. First, technology and productivity differences do not only exist across nations, but are ubiquitous within countries. Even within narrowly-defined sectors, there are substantial productivity differences across firms and only a small portion of these differences can be attributed to differences in capital intensity of production. This within-country pattern suggests that technology adoption and use decisions of firms are complex and new technologies only diffuse slowly across firms. This pattern gives us some clues about potential sources of productivity and technology differences across nations and suggests that a somewhat slow process of technology diffusion across countries may not be unreasonable. Second, while the United States or Japan can be thought of as creating their own technologies via the process of research and development, most countries in the world are technology importers rather than technological leaders. This is not to deny that some firms in these societies do engage in R&D nor to imply that a number of important technologies, most notably those related to the Green Revolution, have been invented in developing countries. These exceptions notwithstanding, adoption of existing frontier technologies appears more important for most firms in developing countries than the creation of entirely new technologies. This perspective also suggests that a detailed analysis of technology diffusion and technology adoption decisions is necessary for obtaining a good understanding of productivity and technology differences across countries.

A number of important lessons have emerged from our study in this chapter.

1. We can make considerable progress in understanding technology and productivity differences across nations by positing a slow process of technology transfer across countries. Namely, in light of the within-country evidence, which suggests that even within narrowly-defined sectors in the same country different technologies can survive side-by-side for long periods of time, it seems reasonable to assume that technologically backward economies

will only slowly catch up to those at the frontier. Such an approach enables us to have a tractable model of technology differences across countries. An important element of models of technology diffusion is that they create a built-in advantage for countries (or firms) that are relatively behind; since there is a larger gap for them to close, it is relatively easier for them to close it. This catch-up advantage for backward economies ensures that models of slow technology diffusion will lead to differences in income levels, not necessarily in growth rates. In other words, the canonical model of technology diffusion implies that countries that create barriers against technology diffusion or those that are slow in adopting new technologies for other reasons will be poor, but they will eventually converge to the growth rate of the frontier economies. Thus a study of technology diffusion enables us to develop a model of world income distribution, whereby the position of each country in the world income distribution is determined by their ability to absorb new technologies from the world frontier. This theoretical machinery is also useful in enabling us to build a framework in which, while each country may act as a neoclassical exogenous growth economy, importing its technology from the world frontier, the entire world behaves as an endogenous growth economy, with its growth rate determined by the investment in R&D decisions of all the firms in the world. This class of models becomes useful when we wish to think of the joint process of world growth and world income distribution across countries. They also emphasize that much is being lost in terms of insights when we focus our attention on the baseline neoclassical growth model in which each country is treated as an “isolated island,” not interacting with others in the world. Technological interdependences across countries implies that we should often consider the world equilibrium, not simply the equilibrium of each country on its own.

2. While slow diffusion of existing technologies across countries is reasonable, in the globalized world we live in today it is becoming increasingly easier for firms to adopt technologies that have already been tried and implemented in other parts of the world. Once we allow a relatively rapid diffusion of technologies, does there remain any reason for technology or productivity differences across countries (beyond differences in physical and human capital)? The second part of the chapter has argued that the answer to this question is also yes and is related to the “appropriateness” of technologies. A given technology will not have the same impact on the productivity in all economies, because it may be a better match to the conditions or to the factor proportions of some countries than of others. Part of this chapter was devoted to explaining how the issue of appropriate technologies can play a role in different contexts. In this age of pervasive skill-biased technologies, a particularly important channel of appropriateness is the potential match between technologies developed at the world frontier and the skills of the adopting country’s workforce. A potential technology-skill mismatch can lead to large endogenous productivity differences. If the types of technologies developed at the world frontier were random, the possibility of the technology-skill mismatch creating a significant gap between rich and poor nations would be a mere possibility, no more. However, there are reasons to suspect that technology-skill mismatch may be more important, because of the organization of the world technology market. Two features are important here. First,

the majority of frontier technologies are developed in a few rich countries. Second, the lack of effective intellectual property rights enforcement implies that technology firms in rich countries target the needs of their own domestic market. This creates a powerful force towards new technologies that are appropriate to (“designed for”) the needs of the rich countries, and thus are typically inappropriate to the factor proportions of developing nations. In particular, new technologies will be “too skill-biased” to be effectively used in developing countries. This source of inappropriateness of technologies can create a large endogenous technology and income gap among nations.

3. Productivity differences do not simply stem from differences in the use of different techniques of production, but also because production is organized differently around the world. A key reason for such differences is institutions and policies in place in different parts of the world. The last part of the chapter showed how contracting institutions, affecting what types of contracts firms can write with their suppliers, can have an important effect on their technology adoption decisions and thus on cross-country differences on productivity. Contracting institutions are only one of many potential organizational differences across countries that might impact equilibrium productivity. My purpose in presenting these ideas in this chapter is to emphasize the importance of endogenous productivity differences resulting from differences in the organization of production. We will see more on this when we turn to the relationship between the process of economic growth and the process of economic development in Part 7 of the book.

18.7. References and Literature

The large literature documenting productivity and technology differences across firms and the patterns of technology diffusion were discussed in Section 18.1 and the relevant references can be found there. The simple model of technology diffusion presented in Section 18.2 is inspired by Gerschenkron (1962) essay and by Nelson and Phelps’s (1966) seminal paper, though I am not aware of a paper that presents a simple general equilibrium treatment similar to that in Section 18.2. Ideas similar to those of Nelson and Phelps were also developed independently by Schultz (1967), who went further than Nelson and Phelps in showing how these ideas could be applied in a variety of different settings, especially in the context of technology adoption in agriculture. The Nelson-Phelps approach, which was discussed in greater detail in Chapter 10, has been important in a number of recent papers. Benhabib and Spiegel (1994) reinterpret and modify Barro-style growth regressions in light of Nelson-Phelps’s view of human capital. Aghion and Howitt (1998) also provide a similar reinterpretation of growth regressions. Caselli (1999), Greenwood and Yorukoglu (1997), Galor and Moav (2001) and Aghion, Howitt and Violente (2001) provide models inspired by the Nelson-Phelps-Schultz view of human capital and applied to understanding the recent increase in the returns to skills and the United States and other OECD economies. In Acemoglu (2002b), I provide a critique of these explanations of the rise in wage inequality.

The model in Section 18.3 is inspired by Howitt (2000), but is different in a number of important respects. First, Howitt uses a model of Schumpeterian growth rather than the baseline expanding input variety model used here. This difference is not important, and the choice here was motivated to simplify the exposition. Second, Howitt uses a model without scale effects. Since our interest here is not with scale effects, the added complication necessary to remove scale effects was deemed unnecessary. Finally, there are more widespread technological externalities in Howitt's model. Thus in many ways, the model in Section 18.3 is a much simplified version of Howitt's model, but it involves all the necessary ingredients for a benchmark model of endogenous growth at the world level.

The ideas of appropriate technology discussed in Section 18.4 have a long pedigree. Many development economists in the 1960s realized the importance of the issues of appropriate technology. The classic work here is Stewart (1977), though similar ideas were also discussed in Salter (1966) and David (1974). A classic treatment was provided Atkinson and Stiglitz (1969), who suggested a simple and powerful formalization of how technological change can be localized and thus may not be easy transfer from one productive units to another (or from one country to another). Atkinson and Stiglitz's idea is incorporated into a growth model by Basu and Weil (1998), which was the basis of one of the models in Section 18.4. The last part of this section draws on Acemoglu and Zilibotti (2001), where we developed a model of appropriate technologies due to skill differences across countries and combine it with directed technological change to show how there will be a bias towards technologies inappropriate to the needs of poorer nations. That paper also provided evidence that these effects could be quantitatively large and patterns of sectoral differences are consistent with the importance of this type of technology-skill mismatch. In Acemoglu (2002b), I showed that technology-skill mismatch applies in a more general model of directed technological change than the one in Acemoglu and Zilibotti (2001) discussed here (see Exercise 18.26).

Finally, the model presented in Section 18.5 draws upon Acemoglu, Antras and Helpman (2007). A number of other models also generate endogenous productivity or technology differences across countries as a result of differences in the organization of production. Some of these will be discussed in Chapter 21.

18.8. Exercises

EXERCISE 18.1. Derive eq. (18.1).

EXERCISE 18.2. Show that if the restriction that $\lambda_j \in [0, g)$ in Section 18.2 is relaxed, the requirement that $A_j(t) \leq A(t)$ can be violated.

EXERCISE 18.3. Derive eq. (18.4).

EXERCISE 18.4. Complete the proof of Proposition 18.1.

EXERCISE 18.5. Derive the effect of an increase in λ_j on the law of motion of $a_j(t)$ and $k_j(t)$. How does this differ from the effect of an increase in σ_j ? Explain why these two parameters have different effects on technology and capital stock dynamics.

EXERCISE 18.6. In the model of Section 18.2, show that if $g = 0$, then all countries converge to the same level of technology. Explain carefully why $g > 0$ leads to steady-state technology differences, while these differences disappear when $g = 0$.

EXERCISE 18.7. (1) Set up the world equilibrium problem in subsection 18.2.2 as one in which the Second Welfare Theorem holds within each country. Under this assumption, carefully define an equilibrium path. Explain the significance of this assumption.

(2) Now set up the world equilibrium problem without appealing to the Second Welfare Theorem. Explain why the mathematical problem is identical to that in part 1 of this exercise.

(3) Prove Proposition 18.2.

EXERCISE 18.8. (1) Why is the condition $\rho - n_j > (1 - \theta)g$ necessary in Proposition 18.3?

(2) Complete the proof of Proposition 18.3.

EXERCISE 18.9. In the model of Section 18.2 with consumer optimization, suppose that preferences in country j are given by

$$U_j = \int_0^\infty \exp(-(\rho_j - n_j)t) \left[(c_j(t)^{1-\theta} - 1) / (1 - \theta) \right] dt,$$

where the ρ_j 's differs across countries.

(1) Show that a unique steady-state world equilibrium still exists and all countries grow at the rate g .

(2) Provide an intuition for why countries grow at the same rate despite different rates of discounting.

(3) Show that this steady-state equilibrium is globally saddle-path stable.

EXERCISE 18.10. * Consider the model of Section 18.2 with F corresponding to the production function of an individual firm j (with a slight abuse of notation) and (18.3) corresponding to the law of motion of the technology of the firm, with $\sigma_j = \sigma(h_j)$, where h_j is the average human capital of the workers of firm j and σ is a strictly increasing and differentiable function. To simplify the discussion, suppose that each firm employs a single worker (which is without loss of any generality given constant returns to scale).

(1) Derive the wage of the worker of human capital h_j . [Hint: this consists of the workers value of marginal product in production plus the increase in the productivity of the firm because of the improvement in the firm's technology due to the higher human capital of the worker].

(2) Show that an increase in g (at any point t) will increase worker wages. Derive the implications of changes in g on the returns to human capital. Contrast an increase in the returns to human capital driven by an increase in g with those discussed in Chapter 15.

EXERCISE 18.11. Complete the proof of Proposition 18.4.

EXERCISE 18.12. Consider the model in subsection 18.3.1 and suppose that all countries have the same labor force size $L_j = 1$ and the same $\eta_j = \eta$, and only differ in terms of their ζ_j 's. Imagine that the range of ζ_j 's is the same as used in the quantitative evaluation of the neoclassical growth model in Chapter 8.

- (1) Evaluate the impact of these differences in ζ_j 's on cross-country technology and income differences for different values of ϕ .
- (2) What value of ϕ is necessary so that a fourfold difference in ζ_j 's translates into a thirtyfold difference in income per capita?
- (3) How would you interpret the economic significance of such a value of ϕ ? Would this be a satisfactory model of cross-country technology and income differences? If yes, explain why it is more attractive than the neoclassical model and other alternatives we have seen so far. If not, suggest what important features are missing and how they might be introduced.

EXERCISE 18.13. * Consider the model in subsection 18.3.1. Suppose that preferences are given by $U_j = \int_0^\infty \exp(-\rho_j t) \left[(c_j(t)^{1-\theta} - 1) / (1-\theta) \right] dt$, where ρ_j differs across countries. Show that an equivalent of Proposition 18.4, with a unique globally saddle-path stable world equilibrium where all countries grow at the same rate, applies.

EXERCISE 18.14. Show that (18.14) is necessary and sufficient for a positive world growth rate in the model of subsection 18.3.2. Write down the conditions that characterize the world equilibrium when this condition is not satisfied.

EXERCISE 18.15. Prove Proposition 18.5.

EXERCISE 18.16. * Analyze the local dynamics of the steady-state world equilibrium in Proposition 18.5. [Hint: linearize the system of differential equations around the steady state].

EXERCISE 18.17. * Consider Proposition 18.5 with the discount rates, the ρ_j 's differing across countries. Prove that a unique steady-state world equilibrium, with all countries growing at the same rate, still exists.

EXERCISE 18.18. In the model of subsection 18.3.2, replace eq. (18.12) with

$$N(t) = G(N_1(t), \dots, N_J(t)),$$

where G is increasing in all of its arguments and homogeneous of degree 1.

- (1) Generalize the results in Proposition 18.5 to this case and derive an equation that determines the world growth rate implicitly.
- (2) Derive an explicit equation for the world growth rate for the specific case in which $N(t) = \max_j N_j(t)$.

EXERCISE 18.19. In the model of subsection 18.3.2, there is a strong scale effect.

- (1) Show that if population grows at some constant rate $n > 0$ in each country, there will not exist a steady-state equilibrium.
- (2) Construct a variation of this model along the lines of the semi-endogenous growth models of Section 13.3 in Chapter 13, where this strong scale effect is removed and

there is long run growth at a constant rate (when population grows at the rate $n > 0$ in each country). [Hint: modify eq. (18.9), so that $\dot{N}_j(t) = \eta_j N(t)^\phi N_j(t)^{-\tilde{\phi}} Z_j(t)$, where $\tilde{\phi} > \phi$].

- (3) Provide a full characterization of the steady-state world equilibrium in this case.

EXERCISE 18.20. Consider the model in subsection 18.4.2. Suppose that the world consists of two countries with constant and equal populations, and constant savings rates $s_1 > s_2$. Suppose that the production function in each country is given by (18.15) with k' corresponding to the highest capital-labor ratio in any country experienced until then. There is no technological progress and both countries start with the same capital-labor ratio.

- (1) Characterize the steady-state world equilibrium (that is, the steady-state capital-labor ratios in both countries).
- (2) Characterize the output per capita dynamics in the two economies. How does an increase in γ affect these dynamics?
- (3) Show that the implied income per capita differences (in steady state) between the two countries are increasing in γ . Interpret this result.
- (4) Do you think this model provides a good/plausible mechanism for generating large income differences across countries? Substantiate your answer with theoretical or empirical arguments.

EXERCISE 18.21. Complete the proof of Proposition 18.6. In particular, explicitly derive the expression for the threshold $I_j(t)$ and the skill premium $w_{H,j}(t)/w_{L,j}(t)$ in country j at time t .

EXERCISE 18.22. Derive the equilibrium expressions (18.20)-(18.23).

EXERCISE 18.23. Prove Proposition 18.7. [Hint: in steady state the profits from owning a skill-complementary and unskilled labor-complementary machine must be equal].

EXERCISE 18.24. Prove Proposition 18.8.

EXERCISE 18.25. Consider the model of appropriate technology in subsection 18.4.3.

- (1) Suppose that now research firms can sell their machines to all producers in the world, including those in the South and can charge the same markup. Derive the steady-state equilibrium under these conditions.
- (2) Comparing your answer in part 1 to the analysis in the text, derive the implications of intellectual property rights enforcement in the South on equilibrium technologies? What are the implications for income per capita differences between the North and the South?
- (3) In view of your answer to 1 and 2 above, could it be the case that Southern economies prefer lack of intellectual property rights enforcement to full intellectual property rights enforcement? [Hint: distinguish between a world in which there is a single Southern country versus one in which there are many].

EXERCISE 18.26. * Instead of the multi-sector model in subsection 18.4.3, suppose that output is given by an aggregate production function of the form

$Y(t) = \left[\gamma Y_L(t)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\gamma) Y_H(t)^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}$ as in Chapter 15, with Y_L and Y_H being produced exactly as in that chapter. Assume, as in subsection 18.4.3, that new technologies are developed in the North for the Northern market only.

- (1) Characterize the steady-state (BGP) equilibrium of this economy. [Hint: use exactly the same analysis as in Chapter 15 and subsection 18.4.3].
- (2) Show that if $\sigma \equiv \varepsilon - (\varepsilon - 1)(1 - \beta)$ is equal to 2, the results are identical to those in subsection 18.4.3.
- (3) Derive the equivalents of Proposition 18.8.
- (4) Do the implications of inappropriate technologies become more or less important when σ increases?

EXERCISE 18.27. Prove Proposition 18.9.

EXERCISE 18.28. Prove Proposition 18.11.

EXERCISE 18.29. Prove Proposition 18.12.

EXERCISE 18.30. * Consider the model of Section 18.5. Suppose that there is a total population of L . Assume that each individual can work as a supplier for one of the M products, or he can work in the process of technology adoption. For this reason, suppose that the cost of technology adoption is given by $\Gamma(N) \equiv w\Gamma_0(N)$, where w is the wage rate, corresponding to the outside option of each supplier.

- (1) Characterize the general equilibrium of the economy by endogenizing A for a given number of products M . In particular, show that in equilibrium the following market clearing condition must be satisfied:

$$M\Gamma_0(N^*) = L,$$

where N^* is the equilibrium technology choice (number of suppliers).

- (2) What is the effect of an increase in μ on N^* ? Explain the result.
- (3) Now suppose that the M products differ according to their elasticity of substitution, in particular, each product has a different α , with the distribution of α 's across products given by a distribution function $G(\alpha)$ with support within the interval $[0, 1]$. Let $N^*(\alpha)$ be the equilibrium technology choice (number of suppliers) for a product with parameter α . Show that the market clearing condition now takes the form:

$$M \int_0^1 \Gamma_0(N^*(\alpha)) dG(\alpha) = L.$$

- (4) What is the effect of an increase in μ on the equilibrium in this case?
- (5) How would you endogenize Q in this model? What types of insights would this generate?

EXERCISE 18.31. Consider the model of Section 18.5. What types of organizational forms might emerge when contracting institutions are imperfect (that is, when μ is very low)? In particular, discuss how vertical integration and repeated interactions between suppliers and

producers might change the results discussed in that section. How would you model each of these?

CHAPTER 19

Trade and Growth

The previous chapter discussed how technological linkages across countries and technology adoption decisions lead to a pattern of interdependent growth across countries. This chapter studies world equilibria with international trade in financial assets or commodities. I start with growth in economies that can borrow and lend internationally, and discuss how this affects cross-country income differences and growth dynamics. I then turn to the growth implications of international trade in commodities.

Our first task is to construct models of world equilibria, which feature both international trade in commodities (or intermediate goods) and economic growth. The exact interactions between trade and growth depend on the nature of trade that countries engage in. I will try to provide an overview of these different interactions. I start with a model in which trade is of the Heckscher-Ohlin type, that is, it originates only because of differences in factor abundance across countries, and growth is driven by capital accumulation. Then, I will turn to a model of Ricardian type, where trade is driven by technological comparative advantage. The main difference between these two approaches concerns whether the prices of the goods that a country supplies to the world are affected by its own production and accumulation decisions. These models will shed new light on the patterns of interdependences across countries, for example, showing that growth in one country cannot be analyzed in isolation from the growth experiences of other nations in the world.

Our second task is to turn to a central question of the literature on trade and growth: whether international trade encourages economic growth. The answer to this question also depends on exactly how trade is modeled, as well as on what the source of economic growth is (in particular learning-by-doing versus innovation). Throughout, the emphasis will be on the importance of considering the world equilibrium rather than the equilibrium of a closed economy in isolation.

19.1. Growth and Financial Capital Flows

In a globalized economy, if the rates of return to capital differ across countries, we would expect capital to flow towards areas where its rate of return is higher. This simple observation has a number of important implications for growth theory. First, it implies a very different pattern of economic growth in a financially integrated world. Our first task in this section is to illustrate the implications of international capital flows for economic growth and show how they significantly change transitional dynamics in the basic neoclassical growth model. Our second task is to highlight what new lessons can be derived from the analysis of economic

growth in the presence of international capital flows. The presence of international capital flows raises a number of puzzles, most notably, the one emphasized by Lucas (1990): “Why Does Capital Not Flow from Rich to Poor Countries?”. This simple question helps us think about a range of important issues in economic growth and economic development. While a model of free flow of capital around the world is a good starting point, the existing evidence is not entirely consistent with such free flows. In particular, free flows of capital lead to a pattern of growth that appears counterfactual. Moreover, a large literature in international finance, starting with Feldstein and Horioka (1980), points out that there is much less *net* flows of capital from countries with high saving rates towards those with lower saving rates than a theory of frictionless international capital markets would suggest. In the next section, I briefly discuss why capital flows across countries may be hampered and what the implications of this are for cross-country growth dynamics.

19.1.1. A World Equilibrium with Free Financial Flows. Consider a world economy consisting of J countries, indexed $j = 1, \dots, J$, each with access to an aggregate production function for producing a unique final good:

$$Y_j(t) = F(K_j(t), A_j(t)L_j(t)),$$

where $Y_j(t)$ is the output of this unique final good in country j at time t , and $K_j(t)$ and $L_j(t)$ are the capital stock and labor supply, $A_j(t)$ is again the country-specific Harrod-neutral technology term. The production function F satisfies Assumptions 1 and 2 from Chapter 2. As in the previous chapter, each country is “small” and ignores its effects on world aggregates. Throughout the section technological change occurs at a constant rate across countries, though there may be level differences in technology, that is,

$$A_j(t) = A_j \exp(gt),$$

where g is the common growth rate of technology in the world.

Suppose that each country admits a representative household with the standard preferences at time $t = 0$ given by

$$(19.1) \quad U_j = \int_0^\infty \exp(-(\rho - n)t) \left[\frac{\tilde{c}_j(t)^{1-\theta} - 1}{1-\theta} \right] dt,$$

where $\tilde{c}_j(t)$ is per capita consumption in country j at time t and I have imposed that all countries have the same time discount rate, ρ , and the same population growth rate n . Moreover, let us assume that all countries start with the same population at time $t = 0$, which, without loss of any generality, is normalized to 1, so that $L_j(0) = 1$ for all $j = 1, \dots, J$, and

$$L_j(t) = L(t) = \exp(nt),$$

for all j . In addition, Assumption 4 from Chapter 8 continues to be satisfied, so that $\rho - n > (1 - \theta)g$.

The key feature of this economy is the presence of international borrowing and lending. Consistent with the permanent income hypothesis for individual consumption decisions, borrowing and lending will allow a smoother consumption profile for households (in particular for the representative household) in each country. But since the desire for a smoother consumption profile was one of the main reasons why the capital stock did not adjust immediately to its steady-state (or BGP) value, the opportunities for international financial transactions will influence the dynamics of capital accumulation and growth.

More specifically, let $B_j(t) \in \mathbb{R}$ denote the net borrowing of country j from the world at time t . Let $r(t)$ denote the world interest rate. Free capital flows imply that this interest rate is independent of which country is borrowing and whether a country is borrowing or lending to others. Moreover, consistent with our assumption that each country is small relative to the world, all countries are price takers in the international financial markets, so they can borrow or lend as much as they like at this interest rate. Consequently, the flow resource constraint facing the representative household in each country will be somewhat different from that in subsection 18.2.2 and can be written as

$$(19.2) \quad \dot{k}_j(t) = f(k_j(t)) - c_j(t) + b_j(t) - (n + g + \delta)k_j(t),$$

where, as usual, $k_j(t) \equiv K_j(t)/A_j(t)L_j(t)$ is the effective capital-labor ratio in country j at time t , $c_j(t) \equiv C_j(t)/A_j(t)L_j(t)$ is the ratio of consumption to effective labor, and

$$y_j(t) \equiv \frac{Y_j(t)}{L_j(t)} \equiv A_j(t)f(k_j(t))$$

is income per capita, while

$$b_j(t) \equiv \frac{B_j(t)}{A_j(t)L_j(t)}$$

denotes the net borrowing normalized by effective labor. The most important feature of eq. (19.2) is that, in contrast to all other resource constraints so far, it does not require domestic consumption and investment to be equal to domestic production. Instead, there are potential transfers of resources from the rest of the world, $B_j(t)$, which can be used for consumption or investment. Conversely, the country may be transferring resources to the rest of the world, so that it consumes and invests less than its production. Naturally, once we allow for international borrowing and lending, we must ensure that each country, thus each representative household, satisfies an international budget constraint. For this purpose, let $\mathcal{A}_j(t)$ denote the international asset position of country j at time t . If $\mathcal{A}_j(t)$ is positive, the country is a net lender and has positive claims on output produced in other countries, while if it is negative, the country is a net borrower. The flow international budget constraint for country j at time t can then be written as:

$$(19.3) \quad \dot{\mathcal{A}}_j(t) = r(t)\mathcal{A}_j(t) - B_j(t),$$

which simply states that the country earns the world interest rate, $r(t)$, on its existing asset position $\mathcal{A}(t)$ (or accumulates further debt if the latter is negative) and in addition receives transfers $B(t)$ from the rest of the world (or makes transfers to the rest of the world when

$B(t)$ is negative). If transfers from the rest of the world exceed the interest earned on current assets, the asset position of the country deteriorates, that is, $\dot{A}_j(t) < 0$. The no-Ponzi game condition (for example, from Chapter 8) now applies to the representative household in each country, and thus indirectly applies to the international asset position of a country and requires that

$$\lim_{t \rightarrow \infty} A_j(t) \exp \left(- \int_0^t r(s) ds \right) = 0$$

for each $j = 1, \dots, J$. In writing this equation, I incorporated that each country faces the world interest rate, $r(t)$, at all points in time. The intuition for this expression is the same as the no-Ponzi game condition, (8.15) in Chapter 8.

As with the other variables, it is convenient to express the net asset position of the country in terms of effective labor units, so let us define

$$\mathbf{a}_j(t) \equiv \frac{A_j(t)}{A_j(t) L_j(t)},$$

which implies that (19.3) can be rewritten as

$$(19.4) \quad \dot{\mathbf{a}}_j(t) = (r(t) - g - n) \mathbf{a}_j(t) - b_j(t)$$

and the no-Ponzi game condition becomes

$$(19.5) \quad \lim_{t \rightarrow \infty} \mathbf{a}_j(t) \exp \left(- \int_0^t (r(s) - g - n_j) ds \right) = 0.$$

Naturally, the amount of borrowing and lending in the world has to balance out. This implies the world capital market clearing condition

$$\sum_{j=1}^J B_j(t) = 0$$

must hold at all times t . Now dividing and multiplying each term by $A_j(t) L_j(t)$, and recalling that $A_j(t) = A_j \exp(gt)$ and $L_j(t) = L(t)$ for all j , the world capital market clearing condition can be written as:

$$(19.6) \quad \sum_{j=1}^J A_j b_j(t) = 0$$

for all $t \geq 0$.

With access to international capital markets, the problem of the representative household in each country can be written as maximizing (19.1) subject to (19.2), (19.4) and (19.5).

A world equilibrium is now defined as processes of normalized consumption levels, capital stocks and asset positions for each country, $\left\{ [k_j(t), c_j(t), \mathbf{a}_j(t)]_{t \geq 0} \right\}_{j=1}^J$, and a time path of world interest rates, $[r(t)]_{t \geq 0}$, such that each country's allocation maximizes the utility of the representative household in each country, and the world financial market clears, that is, (19.6) is satisfied. A steady-state world equilibrium is defined as a world equilibrium in which $k_j(t)$ and $c_j(t)$ are constant and output in each country grows at a constant rate. As

in previous chapters, we could alternatively refer to this allocation as a BGP rather than a steady-state equilibrium.

The equilibrium of this world economy with free financial flows is quite straightforward to characterize. It is useful to first present a number of simple intermediate results to emphasize a number of important economic ideas.

PROPOSITION 19.1. *In the world equilibrium of the economy with free flows of capital,*

$$k_j(t) = k(t) = f'^{-1}(r(t) + \delta) \text{ for all } j = 1, \dots, J,$$

where $f'^{-1}(\cdot)$ is the inverse function of $f'(\cdot)$ and $r(t)$ is the world interest rate.

PROOF. See Exercise 19.1. □

With free flows of capital, each firm in each country will stop renting capital only when its marginal product is equal to the opportunity cost, which is given by the world rental rate (the world interest rate plus the depreciation rate). Consequently, effective capital-labor ratios are equalized across countries. Note, however, that this does *not* imply equalization of capital-labor ratios. To the extent that two countries j and j' have different levels of productivity, $A_j(t)$ and $A_{j'}(t) \neq A_j(t)$, their capital-labor ratios are not, and *should not*, be equalized. This is an important point to which we will return below.

The next proposition focuses on the steady-state world equilibrium.

PROPOSITION 19.2. *Suppose that Assumption 4 is satisfied. Then, in the world economy with free flows of capital, there exists a unique steady-state world equilibrium in which output, capital and consumption per capita in all countries grow at the rate g and effective capital-labor ratios are given by*

$$k_j^* = k^* = f'^{-1}(\rho + \delta + \theta g) \text{ for all } j = 1, \dots, J.$$

Moreover, in the steady-state equilibrium,

$$\lim_{t \rightarrow \infty} \dot{\mathcal{A}}_j(t) = 0 \text{ for all } j = 1, \dots, J.$$

PROOF. See Exercise 19.2. □

At some level this result is very intuitive: with free capital flows, the world economy is *integrated*. This integrated world economy has a unique steady-state equilibrium similar to that in the standard neoclassical growth model. This steady-state equilibrium not only determines the effective capital-labor ratio and its growth rate, but also the distribution of the available capital across different countries in the world economy. Even though this proposition is intuitive, its proof requires some care, to ensure that no country runs a Ponzi scheme and that this implies the *normalized* asset position of each country (and each household within each country), $\mathbf{a}_j(t)$ for each j , must asymptote to a constant. This last feature is no longer the case when the model is extended so that countries differ according to their discount rates (see Exercise 19.2).

Let us next consider the transitional dynamics of the world economy. The analysis of transitional dynamics is simplified by the fact that the world behaves as an integrated economy rather than an independent collection of economies (see Exercise 19.2). Consequently, the following result is straightforward:

PROPOSITION 19.3. *In the world equilibrium of the economy with free flows of capital, there exists a unique equilibrium path $\left\{ [k_j(t), c_j(t), a_j(t)]_{t \geq 0} \right\}_{j=1}^J$ that converges to the steady-state world equilibrium. Along this equilibrium path, $k_j(t)/k_{j'}(t) = 1$ and $c_j(t)/c_{j'}(t) = \text{constant}$ for any two countries j and j' .*

PROOF. See Exercise 19.3. □

Intuitively, the integrated world economy acts as if it has a single neoclassical aggregate production function, thus the characterization of the dynamic equilibrium path and of transitional dynamics from Chapter 8 applies. In addition, Proposition 19.1 implies that $k_j(t)/k_{j'}(t)$ is constant and the standard Euler equations imply that $c_j(t)/c_{j'}(t)$ is constant. Therefore, both production and consumption in each economy grow in tandem.

The following is an important corollary to Proposition 19.3:

COROLLARY 19.1. *Consider the world economy with free flows of capital. Suppose that at time t , a fraction λ of the capital stock of country j is destroyed. Then, capital flows immediately to this country ($\dot{a}_j(t) \rightarrow -\infty$) to ensure that $k_j(t')/k_{j'}(t') = 1$ for all $t' \geq t$ and for all $j' \neq j$.*

PROOF. This is a direct implication of Propositions 19.1 and 19.3. The latter implies that there exists a unique globally stable equilibrium, while the former implies that for all t , $k_j(t)/k_{j'}(t) = 1$. This is only possible by an immediate inflow of capital into country j . □

This result implies that in the world economy with free flows of capital, there are only transitional dynamics for the aggregate world economy, but no transitional dynamics separately for each country (in particular, $k_j(t)/k_{j'}(t) = 1$ for all t and any j and j'). This is intuitive, since international capital flows will ensure that each country has the same effective capital-labor ratio, thus dynamics resulting from slow capital accumulation are removed. This corollary therefore implies that any theory emphasizing the role of transitional dynamics in explaining the evolution of cross-country income differences must implicitly limit the extent or the speed of international capital flows. The evidence on this point is mixed. While the amount of gross capital flows in the world economy is large, the “Feldstein-Horioka puzzle,” which will be discussed below, still remains a puzzle—in particular, countries that save more also tend to invest more rather than lending this money internationally. One reason for this might be the potential risk of sovereign default by countries that borrow significant amounts from the world financial markets. Exercise 19.4 investigates this issue.

Although the implications of this corollary for cross-country patterns of divergence can be debated, its implications for cross-regional convergence are clear; cross-regional patterns

of convergence cannot be related to slow capital accumulation as in the baseline neoclassical growth model. Exercise 19.5 asks you to apply this corollary to investigations of income convergence across US regions and states.

19.2. Why Doesn't Capital Flow from Rich to Poor Countries?

The model studied in the previous section provides us with a framework to answer the question posed above and in the title of this section. In the basic Solow and neoclassical growth models, a key source of cross-country income differences is capital-labor ratios. For example, if we consider a world economy in which all countries have access to the same technology and there are no human capital differences, the *only* reason why one country would be richer than another is differences in capital-labor ratios. But if two countries with the same production possibilities set differ in terms of their capital-labor ratios, then the rate of return to capital will be lower in the richer economy and there will be incentives for capital to flow from rich to poor countries. I now discuss reasons why capital may not flow from societies with higher capital-labor ratios to those with greater capital scarcity.

19.2.1. Capital Flows under Perfect International Capital Markets. One potential answer to the question posed above is provided by the analysis in the previous section. With perfect international capital markets, capital flows will equalize effective capital-labor ratios. But this does *not* imply equalization of capital-labor ratios. This result, which follows directly from the analysis in the previous section, is stated in the next proposition. Note that this result does not give a complete answer to our question, since it takes productivity differences across countries as given. Nevertheless, it explains how, given these productivity differences, there is no compelling reason to expect capital to flow from rich to poor countries.

PROPOSITION 19.4. *Consider a world economy with identical neoclassical preferences across countries and free flows of capital. Suppose that countries differ according to their productivities, the A_j 's. Then, there exists a unique steady state equilibrium in which capital-labor ratios differ across countries (in particular, effective capital-labor ratios, the k_j 's, are equalized), and there are no capital flows across countries.*

PROOF. See Exercise 19.7. □

This proposition states that there is no reason to expect capital flows when countries differ according to their productivities. The more productive countries will have higher capital-labor ratios. To the extent that two countries j and j' have different levels of productivity, $A_j(t)$ and $A_{j'}(t) > A_j(t)$, their capital-labor ratios should not be equalized, instead, country j' should have a higher capital-labor ratio than j . Consequently, capital need not flow from rich to poor countries, because rich countries are more “productive”. This is in fact similar to the explanation suggested in Lucas (1990), except that Lucas also linked differences in A_j 's to differences in human capital and in particular to human capital externalities. Instead, Proposition 19.4 emphasizes that any sources of differences in A_j 's will generate this pattern.

The reader would be right to object at this point that this is only a “proximate” answer to the question, since it provides no reason for why productivity differs across countries. This objection is largely correct. Nevertheless, this proposition is still useful, since it suggests a range of explanations for the lack of capital flows from rich to poor countries that do not depend on the details of the world financial system, but instead focus on productivity differences across countries. We have already made some progress in understanding the potential sources of productivity differences across countries, and as we make more progress, we will start having better answers to the question of why capital does not flow from rich to poor countries (in fact, why it might sometimes flow from poor to rich countries).

19.2.2. Capital Flows under Imperfect International Financial Markets. It is also useful to note that there are other reasons, besides Proposition 19.4, why capital may not flow from rich to poor countries. In particular, it may be the case that the rate of return to capital is higher in poor countries, but financial market frictions or issues of sovereign risk may prevent such flows. For example, lenders might worry that a country that has a negative asset position might declare international bankruptcy and not repay its debts. Alternatively, domestic financial problems in developing countries (which will be discussed in Chapter 21) may prevent or slow down the flows of capital from rich to poor countries. For whatever reason, if the international financial markets are not perfect and capital cannot flow freely from rich to poor countries, we may expect large differences in the return to capital across countries.

Existing evidence on this topic is mixed. Three different types of evidence are relevant. First, a number of studies, including Trefler’s (1993) important work discussed in Chapter 3 and recent work by Caselli and Feyrer (2007), suggest that differences in the return to capital across countries are relatively limited. These estimates are directly relevant to the question of whether there are significant differences in the returns to capital across countries, but they are computed under a variety of assumptions (in Trefler’s case, they rely on data on factor contents of trade and make a variety of assumptions on the impact of trade on factor prices; Caselli and Feyrer, on the other hand, require comparable and accurate measures of quality-adjusted differences in capital stocks across countries).

Second and somewhat in contrast to the aggregate results, a number of papers exploiting microdata, for example, summarized in Banerjee and Duflo (2005), suggest that the rate of return for additional investment in some firms in less-developed countries could be as high as 100%. Nonetheless, this evidence, even if taken at face value, does not suggest that there will be strong incentives for capital to flow from rich to poor countries, since it may be generated by within-country credit market imperfections. In particular, it may be that the rate of return is very high for a range of credit-rationed firms, but various incentive problems make it impossible for domestic or foreign financial institutions to lend to these firms on profitable terms. If these developing economies were to receive an infusion of additional foreign capital, the rate of return would not be given by the rate of return of credit-rationed firms, but by the rate of return of unconstrained firms, which is presumably much lower. Consequently,

the incentives for capital to flow from rich to poor countries may be quite weak as suggested by Proposition 19.4.

Finally, directly related to the issue of the flow of capital across countries is the evidence related to the Feldstein-Horioka puzzle. In an influential paper, Martin Feldstein and Charles Horioka (1980) pointed out a striking fact: differences in savings and investment rates across countries are highly correlated. In particular, Feldstein and Horioka used various different samples to run a regression of the form:

$$\Delta \left(\frac{I_j(t)}{Y_j(t)} \right) = \alpha_0 + \alpha_1 \Delta \left(\frac{S_j(t)}{Y_j(t)} \right),$$

where $\Delta(I_j(t)/Y_j(t))$ is the change in the investment to GDP ratio of country j between some prior date and date t , and $\Delta(S_j(t)/Y_j(t))$ is the change in the savings to GDP ratio. Imagine that savings to GDP ratio varies across countries and over time because of “shocks” to the saving rate or other reasons. In a world with free capital flows, we would expect these changes in savings to have no effect on investment, thus we should estimate a coefficient of $\alpha_1 \approx 0$. In contrast, Feldstein and Horioka estimated a coefficient close to 1 (around 0.9) for OECD economies. Similar results have been found for other samples of countries, though other studies, most notably Taylor (1994), argue that including additional controls removes the puzzle. Feldstein and Horioka and much of the literature that has followed them has interpreted the positive correlation between investment and savings as evidence against free capital flows. Naturally, in practice there are a number of econometric issues one needs to worry about before one can reach a precise conclusion. For example, Exercise 19.6 shows how correlation between investment and savings can arise without imperfections in international financial markets (because the major difference across countries is in investment opportunities). Nevertheless, the Feldstein-Horioka puzzle suggests that issues of sovereign risk might be important in practice and may create barriers to the free flow of capital across countries. Models incorporating endogenous sources of sovereign risk together with the process of economic growth could be an interesting area for future research.

19.3. Economic Growth in a Heckscher-Ohlin World

We have so far focused on the growth implications of trade in financial assets, which enables countries to change the time profile of their consumption. Perhaps more important is international trade in commodities, which allows countries to exploit their comparative advantages (resulting from technology or differences in factor proportions). I now turn to a simple model of growth in a world consisting of countries that trade in commodities. This model builds on work by Ventura (1997), who constructed a tractable model of world equilibrium based on the *Heckscher-Ohlin* model of trade.

The Heckscher-Ohlin model is the benchmark approach to international trade. It posits that countries have access to the same (or similar) technologies, and the main source of trade is differences in factor proportions—that some countries have more capital relative to labor than others or more human capital than others, and so on. Clearly, an analysis of these issues

necessitates the specification of models in which there are multiple commodities used either in consumption or as intermediates in the production. For the sake of concreteness (without loss of generality), I pursue the second alternative, which also creates continuity with the models in Chapters 13 and 15.

Suppose that each country has access to an aggregate production function of the following form:

$$(19.7) \quad Y_j(t) = F(X_j^K(t), X_j^L(t)),$$

where $Y_j(t)$ is final output in country j at time t , and $X_j^L(t)$ and $X_j^K(t)$ are respectively labor- and capital-intensive intermediates (inputs). I use the letter X to denote these inputs, since they refer to the amounts of these inputs *used in production* rather than the amount of inputs *produced* in country j . In the presence of international trade these two quantities will typically differ. F denotes a constant returns to scale production function and again satisfies Assumptions 1 and 2 from Chapter 2 (except that it is defined over two intermediate inputs rather than labor and capital). Notice that Assumption 2 also incorporates the Inada conditions, which will play an important role in the analysis below. The production of the final good is competitive.

The theory of international trade is a well-developed and rich area of economics, and provides useful results on the structure of production and trade. Here my purpose is not to review these results, but to illustrate the implications of Heckscher-Ohlin type international trade for economic growth. Therefore, I adopt the simplest possible setting, which involves each intermediate input being produced by one factor. In particular,

$$(19.8) \quad Y_j^L(t) = A_j L_j(t)$$

and

$$(19.9) \quad Y_j^K(t) = K_j(t),$$

where the use of Y instead of X here emphasizes that these quantities refer to the local production, not the use, of these intermediates. Also, as usual, $L_j(t)$ is total labor input in country j at time t , supplied inelastically, and $K_j(t)$ is the total capital stock of the country. One feature about these intermediate production functions is worth noting: there are potential productivity differences across countries in the production of the labor-intensive good, but not in the production of the capital-intensive good. This is the same assumption as the one adopted in Ventura (1997). Exercise 19.10 shows the implications of allowing differences in the productivity of the capital-intensive sector as well. For now, it suffices to note that this assumption makes it possible to derive a well-behaved world equilibrium, and it is in the spirit of allowing only labor-augmenting technological progress in the basic neoclassical model. Moreover, this assumption is not entirely unreasonable, since we may think of differences in A_j 's as reflecting differences in the human capital embodied in labor. Notice also

that there is no technological progress. This is again to simplify the exposition, and Exercise 19.12 extends the model in this section to incorporate labor-augmenting technological progress.

Throughout the rest of this chapter, I assume that there is free international trade in commodities—in intermediate goods. This is an extreme assumption, since trading internationally involves costs and many analyses of international trade incorporate the physical costs of transportation and tariffs. The main insights for economic growth do not depend on whether or not there are such costs, so I simplify the analysis by assuming costless international trade. The most important implication of this assumption is that the prices of traded commodities, here the intermediate goods, are the same in all countries and are equal to their “world prices”. Then, the world supply and demand for these commodities will determine these prices. In particular, the world prices of the labor-intensive and the capital-intensive intermediates at time t are denoted by $p^L(t)$ and $p^K(t)$, respectively. Both of these prices are in terms of the final good in the world market, which is taken as numeraire, with price normalized to 1.¹

Given the production technologies in (19.8) and (19.9), this immediately implies that the wage rate and the rental rate of capital in country j at time t are given by

$$\begin{aligned}w_j(t) &= A_j p^L(t) \\ R_j(t) &= p^K(t).\end{aligned}$$

These two equations summarize the most important economic insights of the model studied here. Factor prices shape the incentives to accumulate capital in the neoclassical growth model and are typically determined by the capital-labor ratio (recall Chapter 8). The specific structure here, in contrast, implies that these factor prices are determined by world prices. In particular, since capital is used only in the production of the capital-intensive intermediate and there is free trade in intermediates, the rental rate of capital in each country is given by the world price of the capital-intensive intermediate. A similar reasoning applies to the wage rate, with the only difference that, because of cross-country differences in the productivity of labor, wage rates are not equalized; instead it is the effective wage rates, the $w_j(t)/A_j$'s, that are equalized. Let us follow Treffer (1993) in referring to this pattern as *conditional factor price equalization* across countries, meaning that, once we take into account intrinsic productivity differences of factors, there is equalization of factor prices across countries. Conditional factor price equalization is weaker than the celebrated factor price equalization of international trade theory, which would require $w_j(t)$'s to be equalized across countries. Instead, $w_j(t)/A_j$'s are equalized.

¹In this model, there is no loss of generality in assuming that the price of the final good is normalized to 1 in each country even if there is no trade in final good. This is because all goods are traded and there are no differences in costs of living (purchasing power parity) across countries. This will no longer be in the models studied in the next section. I take no position on whether there is trade in the final good, but, as specified below, there is no international lending and borrowing.

In this model, equalization of factor prices (or conditional factor prices) is an immediate consequence of free trade in goods, since each factor is only used in the production of a single traded intermediate. Nevertheless, factor price equalization results are considerably more general than the specialized structure here might suggest. In particular, factor price equalization or conditional factor price equalization results apply in general international trade models without trading frictions under fairly weak assumptions. Intuitively, trading commodities is a way of trading factors; if there is sufficient trade in commodities—especially sufficient trade in commodities with different factor intensities—then countries that are more abundant in one factor will sell enough of the goods embedding that factor to equalize factor prices across countries. In the jargon of international trade theory, with free trade of commodities, there exists a *cone of diversification*, such that when factor proportions of different countries are within this cone, there will be (conditional) factor price equalization. Our extreme assumption that labor is used in the production of the labor-intensive intermediate and capital is used in the production of the capital-intensive intermediate is useful as it ensures that the cone of diversification is large enough to include any possible configuration of the distribution of capital and labor stocks across countries.

The reader may also wonder why conditional factor price equalization is important. Its main importance for us is that when there is conditional factor price equalization, factor prices in each country are *entirely independent* of its capital stock and labor (provided that the country in question is “small” relative to the rest of the world; recall footnote 1 in the previous chapter). Thus each country will be taking intermediate prices, and consequently factor prices, as given when it makes its allocation and accumulation decisions. In fact, the distinguishing feature of the model analyzed in this section is this independence of factor prices from accumulation decisions, which is, in turn, a direct implication of a world of Heckscher-Ohlin trade.

In addition, as in previous chapters, capital depreciates at an exponential rate δ in each country, so that the interest rate is

$$\begin{aligned} r_j(t) &= R_j(t) - \delta \\ (19.10) \qquad &= p^K(t) - \delta. \end{aligned}$$

Let us next specify the resource constraint. While there is free international trade in commodities, there is no international trade in assets. Thus we are abstracting from the issues of international lending and borrowing discussed in the previous two sections. This will enable us to isolate the effects of international trade in the simplest possible way. Lack of international lending and borrowing implies that at every date, each country must run a *balanced international trade*. In terms of the variables introduced so far, this implies the following trade balance equation:

$$(19.11) \qquad p^K(t) [X_j^K(t) - Y_j^K(t)] + p^L(t) [X_j^L(t) - Y_j^L(t)] = 0,$$

for all j and all t . This equation is intuitive; it requires that for each country (at each date) the value of their net sales of the capital-intensive good should be made up by their net purchases

of the labor-intensive good. For example, if $X_j^K(t) - Y_j^K(t) < 0$, so that the country is a net supplier of the capital-intensive good (that is, it uses less of the capital-intensive good in its final good sector than it produces), then it must be a net purchaser of the labor-intensive good, that is, $X_j^L(t) - Y_j^L(t) > 0$.

In addition to this trade balance equation, there is the usual resource constraint affecting each country, which takes the form

$$(19.12) \quad \dot{K}_j(t) = F(X_j^K(t), X_j^L(t)) - C_j(t) - \delta K_j(t),$$

for all j and t . In addition, world market clearing requires

$$(19.13) \quad \sum_{j=1}^J X_j^L(t) = \sum_{j=1}^J Y_j^L(t) \text{ and } \sum_{j=1}^J X_j^K(t) = \sum_{j=1}^J Y_j^K(t) \text{ for all } t.$$

The important feature in this equation is that both the consumption good and the capital good are produced with the same technology—one unit of the final good can be transformed into one unit of consumption good or one unit of capital or the investment good. In the next section, we will see how different factor intensities of consumption and capital goods can be allowed in models of international trade and growth. But for now, the simpler setup with the consumption and investment goods having the same factor intensities is sufficient for our purposes.

Finally, on the preference side, I again assume that each country admits a representative household with standard preferences

$$(19.14) \quad U_j = \int_0^\infty \exp(-(\rho - n)t) \frac{c_j(t)^{1-\theta} - 1}{1-\theta} dt,$$

where $c_j(t) \equiv C_j(t) / L_j(t)$ is per capita consumption in country j at time t and all countries have the same time discount rate, ρ , and also the same rate of population growth. Without loss of any generality, let us assume that all the decisions within each country is made by the representative household of that country and that $\rho > n$ to ensure positive discounting and finite lifetime utilities (see Chapter 8, in particular, Assumption 4'). Finally, to simplify the analysis let us also suppose that $L_j(0) = L$ for each $j = 1, 2, \dots, J$, which, combined with the common population growth assumption, implies that

$$(19.15) \quad L_j(t) = L(t) \text{ for each } j = 1, 2, \dots, J.$$

Exercise 19.11 generalizes the results here to the case in which population levels vary across countries.

With a reasoning similar to that in Chapter 8, a key object is the ratio of “capital-like” intermediates relative to “labor-like” intermediates in production. For this reason, let us define

$$x_j(t) \equiv \frac{X_j^K(t)}{X_j^L(t)},$$

so that

$$\begin{aligned}
 Y_j(t) &= F(X_j^K(t), X_j^L(t)) \\
 &= X_j^L(t) F\left(\frac{X_j^K(t)}{X_j^L(t)}, 1\right) \\
 (19.16) \quad &\equiv X_j^L(t) f(x_j(t)),
 \end{aligned}$$

where the third line defines the function $f(\cdot)$ in the usual way exploiting the constant returns to scale nature of the function F . I refer to $x_j(t)$ as the capital intermediate intensity of country j . Finally, $k_j(t) \equiv K_j(t)/L_j(t)$ is again the capital labor ratio in country j at time t .

A *world equilibrium* can be expressed as processes of consumption, capital accumulation and capital intermediate intensity decision for each country and world prices, $\left[\{c_j(t), k_j(t), x_j(t)\}_{j=1}^J, p^K(t), p^L(t)\right]_{t \geq 0}$, such that $[c_j(t), k_j(t), x_j(t)]_{t \geq 0}$ maximize the utility of the representative household in country j subject to (19.11) and (19.12) given world prices, $[p^K(t), p^L(t)]_{t \geq 0}$, and world prices are such that world markets clear, that is, the equations in (19.13) hold. A *steady-state world equilibrium* is defined similarly as an equilibrium in which all of these quantities are constant.

Let us start with a straightforward result about the allocation of production around the world:

PROPOSITION 19.5. *Consider the above-described model. In any world equilibrium,*

$$x_j(t) = x_{j'}(t) = \frac{\sum_{j=1}^J k_j(t)}{\sum_{j=1}^J A_j} \text{ for any } j \text{ and } j' \text{ and any } t.$$

PROOF. Given world prices at time t , the representative household in each country maximizes $F(X_j^L(t), X_j^K(t))$ subject to (19.11). Denoting the derivatives of this function by F_L and F_K , this implies

$$\frac{F_K(X_j^K(t), X_j^L(t))}{F_L(X_j^K(t), X_j^L(t))} = \frac{p^K(t)}{p^L(t)} \text{ for any } j \text{ and any } t.$$

Using the definition in (19.16) and the linear homogeneity of F , this can be written as

$$\frac{f'(x_j(t))}{f(x_j(t)) - x_j(t) f'(x_j(t))} = \frac{p^K(t)}{p^L(t)} \text{ for any } j \text{ and any } t,$$

where the left-hand side is strictly decreasing in $x_j(t)$, thus defines a unique $x_j(t)$ given the world price ratio. Since $x_j(t)$'s are equal across countries, they must all be equal to the ratio of capital-intensive intermediates to labor-intensive intermediates in the world, so that

$$x_j(t) = \frac{\sum_{j=1}^J K_j(t)}{\sum_{j=1}^J A_j L_j(t)}$$

for $j = 1, \dots, J$. Using the fact that $k_j(t) = K_j(t)/L_j(t) = K_j(t)/L(t)$ (because of (19.15)) completes the proof of the proposition. $\square$

This proposition implies that regardless of differences in capital-labor ratios across countries, the ratio of capital-intensive to labor-intensive intermediates in production will be equalized across countries. The equalization of the use of the ratio of capital-intensive to labor-intensive intermediates in the production of the final good enables us to aggregate the production and capital stocks of different countries to obtain the behavior of world aggregates. In particular, let $c(t)$ be the average consumption per capita in the world and $k(t)$ be the average capital-labor ratio in the world, given by

$$c(t) \equiv \frac{1}{J} \sum_{j=1}^J c_j(t) \text{ and } k(t) \equiv \frac{1}{J} \sum_{j=1}^J k_j(t).$$

The next proposition shows that world aggregates follow laws of motion very similar to those in the standard neoclassical closed economy.

PROPOSITION 19.6. *Consider the above-described model. Then, in any world equilibrium, the world averages follow the laws of motion given by*

$$\begin{aligned} \frac{\dot{c}(t)}{c(t)} &= \frac{1}{\theta} \left(f' \left(\frac{k(t)}{A} \right) - \delta - \rho \right) \\ \dot{k}(t) &= Af \left(\frac{k(t)}{A} \right) - c(t) - (n + \delta) k(t), \end{aligned}$$

where $r(t) = p^K(t)$ is the world interest rate at time t , and

$$A = \frac{1}{J} \sum_{j=1}^J A_j$$

is average labor productivity.

PROOF. Using (19.11), (19.12) and Proposition 19.5, the law of motion of the capital stock of country j can be written as

$$\dot{K}_j(t) = p^K(t) K_j(t) + p^L(t) A_j L(t) - C_j(t) - \delta K_j(t).$$

Now define $K(t) \equiv \frac{1}{J} \sum_{j=1}^J K_j(t)$, sum over $j = 1, \dots, J$, and use the definitions of $p^K(t)$ and $p^L(t)$, Proposition 19.5 and the linear homogeneity of F (together with Theorem 2.1) to obtain

$$\sum_{j=1}^J \dot{K}_j(t) = F \left(\sum_{j=1}^J K_j(t), \sum_{j=1}^J A_j L(t) \right) - \sum_{j=1}^J C_j(t) - \delta \sum_{j=1}^J K_j(t).$$

Dividing both sides by $JL(t)$ and using Theorem 2.1 once more,

$$\frac{\dot{K}(t)}{L(t)} = Af \left(\frac{K(t)}{AL(t)} \right) - c(t) - \delta \frac{K(t)}{L(t)}.$$

Now using the definition of $k(t)$ gives the second differential equation.

To obtain the differential equation for $c(t)$, aggregate the Euler equation for the representative household in each county, $\dot{c}_j(t)/c_j(t) = (r(t) - \rho)/\theta$, for each j . This completes the proof of the proposition. $\square$

The result in this proposition is not surprising. With (conditional) factor price equalization, the world behaves as an integrated economy, and thus obeys the two key differential equations of the neoclassical model. Now using the previous two propositions, we can characterize the form of the steady-state world equilibrium.

PROPOSITION 19.7. *Consider the above-described model. There exists a unique steady-state equilibrium whereby*

$$(19.17) \quad f'(x_j^*) = f'\left(\frac{k^*}{A}\right) = \rho + \delta \text{ for all } j,$$

where

$$(19.18) \quad x_j^* = x^* = \frac{\sum_{j=1}^J K_j(t)}{L(t) \sum_{j=1}^J A_j} \text{ and } k^* = \frac{\sum_{j=1}^J K_j(t)}{JL(t)}.$$

Moreover

$$(19.19) \quad p^{K^*} = \rho + \delta.$$

PROOF. The proof follows from Proposition 19.6. The Inada conditions in Assumption 2 rule out sustained growth. Therefore, world average consumption must remain constant in steady state, and the interest rate must satisfy $r^* = p^{K^*} - \delta = \rho$. Propositions 19.5 and 19.6 then yield (19.17) and (19.18). $\square$

Proposition 19.7 shows that the steady-state world equilibrium takes a very simple form, with the ratio of capital-intensive to labor-intensive intermediates pinned down purely by the aggregate production function F (or its transform, f) and by the ratio of total capital to total labor in the world. The reason why steady-state production structure is determined by world supplies of capital and labor is simple: in the presence of (conditional) factor price equalization, the world economy is effectively integrated. We have already seen in the previous two sections how capital flows can make the world become integrated. The analysis in this section shows that Heckscher-Ohlin trade also leads to the same result (as long as it guarantees conditional factor price equalization).

While the structure of the steady-state equilibrium is rather straightforward, transitional dynamics in this world economy are somewhat more involved. In fact, the behavior of individual economies can be quite rich and complicated. Nevertheless, the fact that world averages obey the equations of the neoclassical growth model ensures that the steady-state world equilibrium is globally stable.

PROPOSITION 19.8. *Consider the above-described economy. The steady-state equilibrium characterized in Proposition 19.7 is globally saddle-path stable.*

PROOF. With the arguments in the proof of Proposition 19.6, for any process of world prices $[p^L(t), p^K(t)]_{t \geq 0}$, the problem of the representative household in each country j at any time t satisfies the differential equations:

$$\frac{\dot{c}_j(t)}{c_j(t)} = \frac{1}{\theta} (p^K(t) - \delta - \rho)$$

$$\dot{k}_j(t) = [p^K(t) - (n + \delta)] k_j(t) + p^L(t) A_j - c_j(t).$$

Standard arguments from Chapter 8 applied to world averages in Proposition 19.6 imply that world averages converge to the unique world steady state equilibrium and $[p^K(t)]_{t \geq 0}$ converges to $\rho + \delta$. This immediately implies that consumption per capita and capital-labor ratio of each country also converges to their steady-state values. With $p^{K*} = \rho + \delta$, the convergence is necessarily to the unique steady-state world equilibrium. $\square$

The analysis so far showed that a world economy consisting of a collection of economies engaged in Heckscher-Ohlin trade generates a pattern of growth similar to that in Chapter 8, with each country converging to a unique steady state. There is one important difference, however. As in the model with international borrowing and lending in the previous section, the nature of the transitional dynamics is very different from the closed-economy neoclassical growth models. Here, despite the absence of international capital flows, the rate of return to capital is equalized across countries. Thus there are no transitional dynamics because a country with a higher rate of return to capital is accumulating capital faster than the rest. This model therefore also emphasizes the potential pitfalls of using the closed-economy growth model for the analysis of output and capital dynamics across countries and regions.

Nevertheless, the results on transitional dynamics are perhaps the less interesting implications of the current model. One of my main objectives in this chapter is to illustrate how the presence of international trade changes the conclusions of closed economy growth models. The current framework already points out how this can happen. Notice that while the world economy has a standard neoclassical technology satisfying Assumptions 1 and 2, each country faces an “AK” technology, since it can accumulate as much capital as it wishes without running into diminishing returns. In particular, for every additional unit of capital at time t , a country receives a return of $p^K(t)$, which is independent of its own capital stock. So how is it that the world does not generate endogenous growth? The answer is that while each country faces an AK technology, and thus can accumulate when the price of capital-intensive intermediates is high, accumulation by all countries drives down the price of capital-intensive intermediate goods to a level that is consistent with steady state. In other words, the price of capital-intensive intermediates will adjust to ensure the steady-state equilibrium where capital, output and consumption per capita are constant (see the proof of Proposition 19.8). While this process describes the long-run dynamics, it also opens the door for a very different type of short-run (or “medium-run”) dynamics, especially for countries that have different saving rates than others.

To illustrate this possibility in the simplest possible way, consider the following thought experiment. Let us start with the world economy in steady state and suppose that one of the countries experiences a decline in its discount rate from ρ to $\rho' < \rho$. What will happen? The answer is provided in the next proposition.

PROPOSITION 19.9. *Consider the above-described model. Suppose J is arbitrarily large and the world starts in steady state at time $t = 0$, then the discount rate of country 1 declines*

to $\rho' < \rho$. After this change, there exists some $T > 0$ such that for all $t \in [0, T)$, country one grows at the rate

$$g_1 = \frac{\dot{c}_1(t)}{c_1(t)} = \frac{1}{\theta} (\rho - \rho').$$

PROOF. In steady state, Proposition 19.8 and eq. (19.19) imply that $p^{K*} = \rho + \delta$. As long as country 1 is small (which will be the case during some interval $[0, T)$), it faces this price as the return on capital. This implies that the country's dynamics will be identical to those of the AK economy in Chapter 11, Section 11.1, with $A = \rho > \rho'$, and the result that the growth rate is constant follows from the analysis there. $\square$

Essentially, given conditional factor-price equalization, each country faces an AK technology, thus can accumulate capital and grow without running into diminishing returns. The price of capital-intensive intermediates and thus the rate of return to capital is pinned down by the discount rate of other countries in the world, so that country 1, with its lower discount rate, will have an incentive to save faster than the rest of the world and can achieve positive growth of income per capita (while the rest of the world has constant income per capita).

Therefore, the model of economic growth with Heckscher-Ohlin trade can easily rationalize bouts of rapid growth (“growth miracles”) by the countries that change their policies or their savings rates (or discount rates). Ventura (1997) suggests this model as a potential explanation for why, starting in the 1970s, East Asian tigers may have grown rapidly without running into diminishing returns. Since in the 1970s and 1980s East Asian economies were indeed more open to international trade than many other developing economies and have accumulated capital rapidly (e.g., Young, 1992, 1995, Vogel, 2006), this explanation is quite plausible. It shows how international trade can temporarily prevent the diminishing returns to capital that would set in because of rapid accumulation and enable sustained growth at higher rates.

Nevertheless, such behavior cannot go on forever. This follows from Assumption 2 above, which implies that world output cannot grow in the long run. So how is Proposition 19.9 consistent with this? The answer is that this proposition describes behavior in the “medium run”. This is the reason why the statement of the proposition is for $t \in [0, T)$. At some point, country 1 will become so large relative to the rest of the world that it will essentially own almost all of the capital of the world. At that point or in fact even before this point is reached, country 1 can no longer be considered a “small” country; its capital accumulation will have a major impact on the relative price of the capital-intensive intermediate. Consequently, the rate of return on capital will eventually fall so that accumulation by this country comes to an end. Naturally, an alternative path of adjustment could take place if, at some future date, the discount rate of country 1 increases back to ρ , so that the world economy again settles into a steady state.

The important lesson from this discussion is that while the current model can generate growth miracles, these can only apply in the “medium run”. The fact that growth miracles can happen only in the medium run highlights another important feature of the current

model. Exercise 19.9 shows that the current model does *not* admit a steady-state equilibrium (or even a well-defined distribution of world income) when discount rates differ across countries. In other words, the well behaved world equilibrium in the world income distribution that emerges from this model relies on the knife-edge case in which all countries have the same discount rate (and also the same productivity of the capital-intensive intermediates, see Exercise 19.10). This feature is not only a shortcoming of the current model, but more generally a shortcoming of all Heckscher-Ohlines approach to trade and growth. In the traditional Heckscher-Ohlin model there is no comparative advantage coming from technology, so that each country is either small and takes world prices as given, or becomes sufficiently large to influence world prices for all commodities. This seems an unappealing feature on both empirical and intuitive grounds; while it is plausible that countries take prices of the goods that they import as given, they often influence the world prices of at least some of the goods that they export (such as copper for Chile, Microsoft windows for the United States or Lamborghinis for Italy). The next section will show that models with *Ricardian* features avoid these unappealing implications and provide a richer and more tractable framework for the analysis of the interaction between international trade and economic growth.

19.4. Trade, Specialization and the World Income Distribution

In this section, I present a model of the world economy in which countries trade intermediate goods, because of Ricardian features—productivity or technology differences. In particular, each country will specialize in the production of a subset of the available goods in the world economy and will supply those to the world economy. Consequently, each country will affect the prices of the goods that it supplies to the world. Put differently, each country's terms of trade will be endogenous and will depend on the rate at which it accumulates capital. We will see that such a model is more flexible than the one discussed in the previous section, since it can allow for differences in discount rates (and saving rates) and also enables us to perform a richer set of comparative static results. The model economy presented here builds on Acemoglu and Ventura (2002). I will start with a simplified version of this model, which features physical capital as the only factor of production. I will then present the full model in which both physical capital and labor are used to produce consumption and investment goods.

In addition to the nature of trade (Heckscher-Ohlin versus Ricardian), another major difference between the model in this section and the previous one will be that now, as in Section 18.3 in the previous chapter, the world economy will exhibit endogenous growth, with the growth rate determined by the investment decisions of all countries. Despite endogenous growth at the world level, international trade (without any technological spillovers) will create sufficient interactions to ensure a common long-run growth rate for all countries. Therefore, the current model will show how international trade, like technological spillovers, will create a powerful force limiting the extent to which divergence can occur across countries.

19.4.1. Basics. Consider a world economy consisting of a large number J of “small” countries, again indexed by $j = 1, \dots, J$. There is a continuum of intermediate products indexed by $\nu \in [0, N]$, and two final products that are used for consumption and investment. There is free trade in intermediate goods and no trade in final products or assets. Lack of trade in consumption and investment goods enables us to focus on trade in intermediates. Lack of trade in assets again rules out international borrowing and lending.

Countries differ in their technology, savings and economic policies. For example, country j will be defined by its characteristics (μ_j, ρ_j, ζ_j) , where μ is an indicator of how advanced the technology of the country is, ρ is its rate of time preference, and ζ is a measure the effect of policies and institutions on the incentives to invest. All of these characteristics potentially vary across countries with a given distribution, but are constant over time. In addition, I assume that each country has a population normalized to 1 and there is no population growth.

All countries admit a representative household with utility function:

$$(19.20) \quad \int_0^\infty \exp(-\rho_j t) \ln C_j(t) dt,$$

where $C_j(t)$ is consumption of country j date t . Preferences are logarithmic and thus more specialized than the typical CRRA preferences used so far (for example, in terms of the preferences in (19.1), they involve $\theta \rightarrow 1$). Logarithmic preferences enable us to simplify the exposition without any substantive loss of generality. Note, however, that the preferences in (19.20) are significantly more flexible than those in the previous section because they allow the discount rates, the ρ_j 's, to differ across countries. We also assume that country j starts with a capital stock of $K_j(0) > 0$ at time $t = 0$.

The budget constraint of the representative household in country j at time t is

$$(19.21) \quad \begin{aligned} p_j^I(t) \dot{K}_j(t) + p_j^C(t) C_j(t) &= Y_j(t) \\ &= r_j(t) K_j(t) + w_j(t), \end{aligned}$$

where $p_j^I(t)$ and $p_j^C(t)$ are the prices of the investment and consumption goods in country j at date t (in terms of the numeraire, which will be the ideal price index of traded intermediates; see below). Despite international trade in intermediates, because consumption and investment goods are not traded, their prices might differ across countries. As usual, $K_j(t)$ is the capital stock of country j at time t , $r_j(t)$ is the rental rate of capital, which may also differ across countries, and $w_j(t)$ is the wage rate. Equation (19.21) requires investment plus consumption expenditures to be equal to total income and also imposes that there is no depreciation. This assumption is adopted to reduce notation. The more important feature is that investment, $\dot{K}_j(t)$, is multiplied by $p_j^I(t)$, while consumption is multiplied by $p_j^C(t)$. This reflects the fact that investment and consumption goods will have different production technologies and thus their prices will differ. In this respect, the model in this section is closely related to that in Section 11.3 in Chapter 11. The second equality in (19.21) specifies that total output is equal to capital income plus labor income— $r_j(t)$ is the rental rate of capital, $K_j(t)$ is the

total capital holdings in country j and $w_j(t)$ denotes total labor earnings, since population is normalized to 1.

As noted above, our focus here is with Ricardian models which feature specialization. I introduce specialization in the simplest possible way; the N intermediates available in the world economy are partitioned across the J countries, such that each intermediate can only be produced by one country. This assumption, which is often referred to as the *Armington* preferences or technology in the international trade literature, ensures that while each country is small in import markets, it will affect its own terms of trades by the amount of the goods it exports. Denoting the measure of goods produced by country j by μ_j , our assumption implies that

$$(19.22) \quad \sum_{j=1}^J \mu_j = N.$$

It follows from this equation that a higher level of μ_j implies that country j has the technology to produce a larger variety of intermediates, so we can interpret μ as an indicator of how advanced the technology of the country is. I assume that all firms within each country have access to the technology to produce these intermediates, which ensures that all intermediates are produced competitively.

Moreover, let us assume that in each country the production technology of intermediates is such that one unit of capital produces one unit of any of the intermediates that the country is capable of producing and that there is free entry into the production of intermediates. This assumption implies that the prices of all intermediates produced in country j at time t are given by

$$(19.23) \quad p_j(t) = r_j(t),$$

where recall that $r_j(t)$ is the rental rate of return in country j at time t .

19.4.2. The AK Model. Before presenting the full model, it is convenient to start with a simplified version, where capital is the only factor of production. Consequently, in terms of eq. (19.21), $w_j(t) = 0$, and

$$Y_j(t) = r_j(t) K_j(t).$$

Suppose that both consumption and investment goods are produced using domestic capital as well as a bundle of all the intermediate goods in the world (which are all traded freely). In particular, the production function for consumption goods in country j is:

$$(19.24) \quad C_j(t) = \chi K_j^C(t)^{1-\tau} \left(\int_0^N x_j^C(t, \nu)^{\frac{\varepsilon-1}{\varepsilon}} d\nu \right)^{\frac{\tau\varepsilon}{\varepsilon-1}}.$$

A number of features are worth noting. First, K_j^C denotes domestic capital used in the consumption goods sector and enters the production function with exponent $1-\tau$. Intuitively, this term corresponds to the services of the domestic capital stock used in the production of consumption goods. It represents the “non-traded” component of the production process, which depends on the services provided by non-traded goods using the capital available in

the country. Since there is no international trade in assets, it must be the domestic capital stock that is used in providing these non-traded services, and if a country has a relatively low capital stock, the relative price of capital will be high and less of it will be used in producing consumption goods (and investment goods; see below). Second, the term in parentheses represents the bundle of intermediates purchased from the world economy. In particular, $x_j^C(t, \nu)$ is the quantity of intermediate good ν purchased and used in the production of consumption goods in country j at time t . The expression implies that it is the CES aggregate of all the intermediates, with an elasticity of substitution ε , that matters in the production of consumption goods. I assume that

$$\varepsilon > 1,$$

which avoids the counterfactual and counterintuitive pattern of “immiserizing growth” (see Exercise 19.23). The use of CES aggregates is familiar by now and enables us to have tractable structure. The expression also makes it clear that there is a continuum N of intermediates (given by eq. (19.22) above). Notice that this CES aggregator has an exponent τ , which ensures that the production function for consumption goods exhibits constant returns to scale. The parameter τ is the elasticity of the production function of consumption goods with respect to traded intermediates and will also be the share of trade in GDP for all countries in this world economy (see Exercise 19.15). Finally, χ is a constant introduced for normalization (see Exercise 19.13).

The production function for investment goods in country j is:

$$(19.25) \quad I_j(t) = \zeta_j^{-1} \chi K_j^I(t)^{1-\tau} \left(\int_0^N x_j^I(t, \nu)^{\frac{\varepsilon-1}{\varepsilon}} d\nu \right)^{\frac{\tau\varepsilon}{\varepsilon-1}},$$

which is identical to that for consumption goods, except for the presence of the term ζ_j . This allows differential levels of productivity, due to technology or policy, in the production of investment goods across countries. The assumption that these differences are in the investment good sector rather than in the production of consumption goods is consistent with results on the relative prices of investment goods discussed previously, which suggested that in poorer economies investment goods are relatively more expensive. In terms of the production functions specified here, we may want to think of greater distortions as corresponding to higher levels of ζ_j , since we will see that higher ζ_j will reduce output and increase the relative price of investment goods. We will see below that in the full model with both capital and labor, the relative price of investment goods are determined in equilibrium and also depend on technology and discount rates.

Market clearing for capital naturally requires

$$(19.26) \quad K_j^C(t) + K_j^I(t) + K_j^\mu(t) \leq K_j(t),$$

where $K_j^\mu(t)$ capital used in the production of intermediates and $K_j(t)$ is the total capital stock of country j at time t .

The reader can also see why this model is referred to as the “AK version”; the production of both consumption and investment goods uses capital and intermediates that are directly

produced from capital. Thus a doubling of the world capital stock will double the output of all intermediates and of consumption and investment goods.

While we can directly work with the production functions for consumption and investment goods, (19.24) and (19.25), as in many trade models, it is simpler to work with *unit cost functions*, which express the cost of producing one unit of consumption and investment goods in terms of the numeraire (which will be chosen as the ideal price index for intermediates, see eq. (19.32) below). Exercise 19.13 shows that the production functions (19.24) and (19.25) are equivalent to the unit cost functions for consumption and production given by

$$(19.27) \quad B_j^C \left(r_j(t), [p(t, \nu)]_{\nu \in [0, N]} \right) = r_j(t)^{1-\tau} \left[\left(\int_0^N p(t, \nu)^{1-\varepsilon} d\nu \right)^{\frac{\tau}{1-\varepsilon}} \right],$$

$$(19.28) \quad B_j^I \left(r_j(t), [p(t, \nu)]_{\nu \in [0, N]} \right) = \zeta_j r_j(t)^{1-\tau} \left[\left(\int_0^N p(t, \nu)^{1-\varepsilon} d\nu \right)^{\frac{\tau}{1-\varepsilon}} \right],$$

where $p(t, \nu)$ is the price of the intermediate ν at time t and the constant χ in (19.24) and (19.25) is chosen appropriately (see Exercise 19.13). Notice that these prices are not indexed by j , since there is free trade in intermediates and thus all countries face the same intermediate prices. The specification using the unit cost functions simplifies the analysis.

A world equilibrium is defined in the usual fashion, as processes of prices, capital stock levels and consumption levels for each country, such that all markets clear and the representative household in each country maximizes its utility given the processes for prices. Namely, an equilibrium is represented by

$$\left[\{p_j^C(t), p_j^I(t), r_j(t), K_j(t), C_j(t)\}_{j=1}^J, [p(t, \nu)]_{\nu \in [0, N]} \right]_{t \geq 0}.$$

Notice that while the prices of consumption and investment goods and the return to capital are country specific, the prices of intermediates are not. A steady-state world equilibrium is also defined in the usual fashion, in particular, requiring that all prices are constant (as before, this “steady-state” equilibrium will involve balanced growth).

The characterization of the world equilibrium in this case is made relatively simple by the *AK* technology (and the logarithmic preferences). In particular, the maximization of the representative household, that is, the maximization of (19.20) subject to (19.21) for each j yields the following first-order conditions

$$(19.29) \quad \frac{r_j(t) + \dot{p}_j^I(t)}{p_j^I(t)} - \frac{\dot{p}_j^C(t)}{p_j^C(t)} = \rho_j + \frac{\dot{C}_j(t)}{C_j(t)}$$

for each j and t , and the transversality condition:

$$(19.30) \quad \lim_{t \rightarrow \infty} \exp(-\rho_j t) \frac{p_j^I(t) K_j(t)}{p_j^C(t) C_j(t)} = 0,$$

for each j (see Exercise 19.14).

Equation (19.29) is the Euler equation. This equation might first appear slightly different from the standard Euler equations, but the reader will see that it is identical to the Euler

equations implied by the two-sector model in Section 11.3 in Chapter 11 (recall, in particular, eq. (11.31)). The difference from the standard Euler equations stems from the fact that we now have potentially different technologies for producing consumption and investment goods, thus individuals that delay consumption have to take into account the change in the relative price of consumption versus investment goods—which explains the presence of the term $\dot{p}_j^I(t)/p_j^I(t) - \dot{p}_j^C(t)/p_j^C(t)$. In this light, it is clear that this equation simply requires the (net) rate of return to capital to be equal to the rate of time preference plus the slope of the consumption path.

Equation (19.30) is the transversality condition. Integrating the budget constraint and using the Euler and transversality conditions, we obtain a particularly simple consumption function,

$$(19.31) \quad p_j^C(t) C_j(t) = \rho_j p_j^I(t) K_j(t),$$

which can be interpreted as individuals spending a fraction ρ_j of their wealth on consumption at every instant (recall that in this simplified model, there is no labor income and $p_j^I(t) K_j(t)$ is consumer wealth at current prices).

The analysis so far has therefore characterized the prices of intermediates and the behavior of the consumption and capital stock for each country. Let us next determine the prices of consumption and investment goods and the relative prices of intermediates in the world economy. As a first step towards this, I define the numeraire for this world economy as the ideal price index for the basket of all the (traded) intermediates. Since the intermediates always appear in the CES form, the corresponding ideal price index is simply

$$(19.32) \quad \begin{aligned} 1 &= \left[\int_0^N p(t, \nu)^{1-\varepsilon} d\nu \right]^{\frac{1}{1-\varepsilon}} \\ &= \sum_{j=1}^J \mu_j p_j(t)^{1-\varepsilon}. \end{aligned}$$

Here the first equation defines the ideal price index, while the second uses the fact that country j produces μ_j intermediates, and each of these intermediates have the same price $p_j(t) = r_j(t)$ as given by (19.23) above.

This choice of numeraire has another convenient implication. Our assumption that each country is small implies that each exports practically all of its production of intermediates and imports the ideal basket of intermediates from the world economy. Consequently, $p_j(t) = r_j(t)$ is not only the price of intermediates produced by country j , but also its *terms of trade*—defined as the price of the exports of a country divided by the price of its imports.

Next, using the price normalization in (19.32), eq.'s (19.27) and (19.28) imply that the equilibrium prices of consumption and investment goods in country j at time t are given by

$$(19.33) \quad p_j^C(t) = r_j(t)^{1-\tau} \text{ and } p_j^I(t) = \zeta_j r_j(t)^{1-\tau}.$$

This completes the characterization of all the prices in terms of the rate of return to capital. To compute the rate of return to capital, we need to impose market clearing for capital in

each country. In addition, we also have a trade balance equation for each country. However, by Walras's Law, one of these equations is redundant. It turns out to be more convenient to use the trade balance equation, which can be written as

$$(19.34) \quad Y_j(t) = \mu_j r_j(t)^{1-\varepsilon} Y(t),$$

where $Y(t) \equiv \sum_{j=1}^J Y_j(t)$ is total world income at time t . To see why this equation ensures balanced trade, note that each country spends a fraction τ of its income on intermediates, and since each country is small, this implies a fraction τ of its income being spent on imports. At the same time, the rest of the world spends a fraction $\tau \mu_j p_j(t)^{1-\varepsilon}$ of its total income on intermediates produced by country j (this follows because of the CES aggregator over intermediates combined with the observations that $p_j(t)$ is the relative price of country j 's intermediates and there are μ_j of them). Noting that total world income is $Y(t)$ and that $p_j(t) = r_j(t)$, we obtain (19.34). Exercise 19.15 asks you to derive this equation from the capital market clearing equation, (19.26), thus verifying the use of the Walras's Law.

The equations derived so far, in particular (19.23), (19.31), (19.33) and (19.34) together with the resource constraint, (19.21), characterize the world equilibrium fully.

Let us start by describing the state of the world economy, which can simply be represented by the distribution of capital stocks across the J economies (these are the only endogenous state variables). Their law of motion is obtained simply by combining (19.21), (19.31) and (19.33) on the one hand, and (19.21) and (19.34) on the other. In particular, for each j and t , the law of motion of the capital stock is described by the following pair of differential equations:

$$(19.35) \quad \frac{\dot{K}_j(t)}{K_j(t)} = \frac{r_j(t)^\tau}{\zeta_j} - \rho_j,$$

$$(19.36) \quad r_j(t) K_j(t) = \mu_j r_j(t)^{1-\varepsilon} \sum_{i=1}^J r_i(t) K_i(t).$$

These two equations completely characterize the world equilibrium. Starting with a cross section of capital stocks at time t , $\{K_j(t)\}_{j=1}^J$, (19.36) gives the cross section of terms of trade and interest rates, $\{r_j(t)\}_{j=1}^J$. Given this cross section of interest rates, (19.35) describes exactly how the cross section of capital stocks will evolve.

The simplicity of these laws of motion are noteworthy. The first, (19.35), determines the evolution of the capital stock of each country simply as a function of their own parameters, ζ_j , the distortions on the investment good producing sector, and ρ_j , the discount rate, as well as the equilibrium rental rate. The second, (19.36), expresses the rental rate for each country as a function of the rental rates and capital stocks of other countries.

These two equations immediately establish the following important result:

PROPOSITION 19.10. *There exists a unique steady-state world equilibrium where*

$$(19.37) \quad \frac{\dot{K}_j(t)}{K_j(t)} = \frac{\dot{Y}_j(t)}{Y_j(t)} = g^*$$

for $j = 1, \dots, J$, and the world steady-state growth rate g^* is the unique solution to equation

$$(19.38) \quad \sum_{j=1}^J \mu_j [\zeta_j (\rho_j + g^*)]^{(1-\epsilon)/\tau} = 1.$$

The steady-state rental rate of capital and the terms of trade in country j are given by

$$(19.39) \quad r_j^* = p_j^* = [\zeta_j (\rho_j + g^*)]^{1/\tau}.$$

This unique steady-state equilibrium is globally saddle-path stable.

PROOF. (Sketch) By definition, a steady-state equilibrium must have constant prices, thus constant r_j^* 's. This implies that in any steady state, for each $j = 1, \dots, J$, $\dot{K}_j(t)/K_j(t)$ must grow at some constant rate g_j . Suppose these rates are not equal for two countries j and j' . Taking the ratio of eq. (19.36) for these two countries yields a contradiction, establishing that $\dot{K}_j(t)/K_j(t)$ is constant for all countries. Equation (19.34) then implies that all countries also grow at this common rate, say g^* . Given this common growth rate, (19.35) immediately implies (19.39). Substituting this back into (19.36) gives (19.38). Since these equations are all uniquely determined and (19.38) is strictly decreasing in g^* , thus has a unique solution, the steady-state world equilibrium is unique.

To establish global stability, it suffices to note that (19.36) implies that $r_j(t)$ is decreasing in $K_j(t)$. Thus whenever a country has a high capital stock relative to the world, it has a lower rate of return on capital, which from (19.35) slows down the process of capital accumulation in that country. This process ensures that the world economy, and all economies, move towards the unique steady-state world equilibrium. Exercise 19.16 asks you to provide a formal proof of stability. $\square$

The results summarized in this proposition are quite remarkable. First, despite the high degree of interaction among the various economies, there exists a unique globally stable steady-state world equilibrium. Second, this equilibrium takes a relatively simple form. Third and most important, in this equilibrium all countries grow at the same rate g^* . This third feature is quite surprising, since each economy has access to a AK technology, thus without any international trade, each country would grow at a different rate (for example, those with lower ζ_j 's or ρ_j 's would have higher long-run growth rates). The process of international trade acts as a powerful force keeping countries together, ensuring that in the long run they will all grow at the same rate. In other words, international trade, together with specialization, leads to a stable world income distribution.

Why is this? The answer is related to the terms-of-trade effects encapsulated in eq. (19.36). To understand the implications of this equation, consider the special case where all countries have the same technology parameter, that is, $\mu_j = \mu$ for all j . Suppose also that a particular country, say country j , has lower ζ_j and ρ_j than the rest of the world. Then, (19.35) implies that this country will tend to accumulate more capital than others. But (19.36) makes it clear that this cannot go on forever and country j , by virtue of being richer than the world average, will also have a lower rate of return on capital. This lower rate of

return will ultimately compensate the greater incentive to accumulate in country j , so that capital accumulation in this country converges to the same rate as in the rest of the world.

Intuitively, while each country is “small” relative to the world, it has market power in the goods that it supplies to the world. When it exports more of a particular good, the price of that good declines, so that world consumers should wish to consume the greater amount of this good that is being supplied in the world market. This implies that when a country accumulates faster than the rest of the world, and thus increases the supply of its exports relative to the supplies of other countries exports, it will face *worsening terms of trades*. This negative terms of trade effect will reduce the income of the country that is accumulating faster. However, more important than this level effect is the dynamic effects of the changes in terms of trades. Recall that eq. (19.23) links the rate of return to capital to the terms of trade faced by the country. When a country experiences a worsening in its terms of trade, it also experiences a decline in the rate of return to capital and in the interest rate that the households face. This slows down its rate of capital accumulation, ensuring that in the steady-state equilibrium all countries accumulate and grow at the same rate.

Therefore, this model shows how pure trade linkages are sufficient to ensure that countries that would otherwise grow at different rates pull each other towards a common growth rate and the result is a stable world income distribution. Naturally, growth at a common rate does not imply that countries with different characteristics will have the same level of income. Exactly as in models of technological interdependences in the previous chapter, countries with better characteristics (higher μ_j and lower ζ_j and ρ_j) will grow at the same rate as the rest of the world, but will be *richer* than other countries. This is most clearly shown by the following equation, which summarizes the world income distribution. Let $y_j^* \equiv Y_j(t)/Y(t)$ the relative income of country j in steady state. Then, eq.’s (19.34) and (19.39) yield

$$(19.40) \quad y_j^* = \mu_j [\zeta_j (\rho_j + g^*)]^{(1-\varepsilon)/\tau}.$$

This equation shows that countries with better technology (high μ_j), lower distortions (low ζ_j) and lower discount rates (low ρ_j) will be relatively richer. Equation (19.40) also highlights that the elasticity of income with respect to ζ_j and ρ_j depends on the elasticity of substitution between the intermediates, ε , and the degree of openness, τ . When ε is high and τ is relatively low, small differences in ζ_j ’s and ρ_j ’s can lead to very large differences in income across countries. This observation is interesting for another reason; recall from Chapters 2 and 3 that the Solow growth model generates a similar equation linking the world income distribution to differences in savings rates and technology. In particular, recall that in a world with a Cobb-Douglas aggregate production function and no human capital differences, the Solow model implies that

$$(19.41) \quad y_j^* = A_j \left(\frac{s_j}{g^*} \right)^{\alpha/(1-\alpha)},$$

where A_j is the relative labor-augmenting productivity of country j , s_j is its savings rate, g^* is again the world growth rate and α is the exponent of capital in the Cobb-Douglas production

function, which is also equal to the share of capital in national income. Equation (19.40) shows that the implications of the world economy with trade are very similar, except that (i) the role of the labor-augmenting technologies is played by the technological capabilities of the country, which determine the range of goods in which it has a comparative advantage; (ii) the role of the saving rate is played by the discount rate ρ_j and the policy parameter affecting the distortions on the production of investment goods, ζ_j ; (iii) instead of the share of capital in national income, the elasticity of substitution between intermediates and the degree of trade openness affects how spread out the world income distribution is. Exercise 19.17 develops these points further.

19.4.3. The General Model. The model presented in the previous subsection has a number of striking implications. The most important is that despite the possibility of endogenous growth at the country level, world relative prices adjust in such a way as to keep the world income distribution stable. Consequently, differences in preferences and technology across countries translate into differences in income levels along a stable income distribution, rather than into differences in permanent growth rates. However, the reader may wonder how general this result is. The result was derived in the context of a collection of *AK* economies. In this subsection, I show that the results generalize to an economy in which both capital and labor are used. To maintain the tractability of the model of the previous subsection, and in fact in order to obtain almost identical equations to those from the previous subsection, I make use of the structure of production first used by Rebelo (1991), which we encountered in Section 11.3 in Chapter 11, where the production of investment goods only uses capital, while the production of consumption goods uses both capital and labor. While the exact mathematical derivations here depend on these specific assumptions, the general insights do not.

More specifically, preferences, demographics, the production functions for intermediates and the production function for investment goods are as same as in the previous subsection. The main difference is that the production function for consumption goods has now changed to

$$C_j(t) = \chi K_j^C(t)^{(1-\tau)\gamma} (L_j(t))^{(1-\tau)(1-\gamma)} \left(\int_0^N x_j^C(t, \nu)^{\frac{\varepsilon-1}{\varepsilon}} d\nu \right)^{\frac{\tau\varepsilon}{\varepsilon-1}}$$

for some $\gamma \in (0, 1)$, and χ is again a normalizing constant and $L_j(t)$ is the total labor supply of country j at time t . All of this labor supply is used in the production of the consumption good, since neither the production of intermediates nor the production of the investment good use labor. The labor endowment in the economy is supplied inelastically by the representative household, and without loss of any generality, let us normalize $L_j(t) = 1$. This implies that in terms of (19.21), $w_j(t)$ stands both for the wage rate per unit of labor and for total labor income. The associated unit cost function for the consumption good is

$$B_j^C(w_j(t), r_j(t), [p(t, \nu)]_{\nu \in [0, N]}) = w_j(t)^{(1-\tau)(1-\gamma)} r_j(t)^{(1-\tau)\gamma} \left[\left(\int_0^N p(t, \nu)^{1-\varepsilon} d\nu \right)^{\frac{\tau}{1-\varepsilon}} \right].$$

Using the same price normalization, that is, (19.32), intermediate prices are still given by (19.23) and the price of the investment good in country j at time t is

$$p_j^I(t) = \zeta_j r_j(t)^{1-\tau}.$$

The price of the consumption good is obtained, with a similar reasoning, as

$$(19.42) \quad p_j^C(t) = w_j(t)^{(1-\tau)(1-\gamma)} r_j(t)^{(1-\tau)\gamma}.$$

The maximization problem of the representative household in each country is essentially unchanged, except for the stream of labor income that the household receives. This maximization problem again leads to the necessary and sufficient conditions given by (19.29) and (19.30). Combining these two equations, we again obtain that consumption expenditure is given as the fraction of the lifetime wealth of the household, which now consists of the value of capital plus the discounted value of future labor earnings (see Exercise 19.18):

$$(19.43) \quad p_j^C(t) C_j(t) = \rho_j \left(p_j^I(t) K_j(t) + \int_t^\infty \exp \left(- \int_t^z \frac{r_j(s) + \dot{p}_j^I(s)}{p_j^I(s)} ds \right) w(z) dz \right).$$

It is also straightforward to show that (19.34) still gives the necessary trade balance equation for each country.

The final condition we need to impose is market clearing for labor. Recall that labor demand comes only from the consumption goods sector, and given the Cobb-Douglas assumption, this demand is $(1-\gamma)(1-\tau)$ times consumption expenditure, $p_j^C C_j$, divided by the wage rate, w_j . So the market clearing condition for labor in country j at time t is:

$$(19.44) \quad 1 = (1-\gamma)(1-\tau) \frac{p_j^C(t) C_j(t)}{w_j(t)}.$$

Because (19.44) implies that labor income, $w_j(t)$, is always proportional to consumption expenditure, the optimal consumption rule, (19.43), can be simplified to the following convenient equation:

$$(19.45) \quad p_j^C(t) C_j(t) = \frac{\rho_j}{1 - (1-\gamma)(1-\tau)} p_j^I(t) K_j(t).$$

In other words, households again consume a constant fraction of the value of the capital stock, but this fraction now depends not only on their discount rate, ρ_j , but also on the technology parameters, τ and γ . In light of this derivation, the following two propositions are straightforward:

PROPOSITION 19.11. *In the general model with labor, the world equilibrium is characterized by (19.35) for each j and t , as well as two additional equations*

$$(19.46) \quad r_j(t) K_j(t) + w_j(t) = \mu_j r_j(t)^{1-\varepsilon} \sum_{i=1}^J [r_i(t) K_i(t) + w_i(t)], \text{ and}$$

$$(19.47) \quad \frac{w_j(t)}{r_j(t) K_j(t) + w_j(t)} = \frac{(1-\gamma)(1-\tau) \rho_j}{[\gamma + (1-\gamma)\tau] \zeta_j^{-1} r_j(t) + (1-\gamma)(1-\tau) \rho_j}.$$

PROOF. See Exercise 19.19. □

The derivation and the intuition for this result follow those in the previous subsection. For a given cross section of capital stocks, eq.'s (19.46) and (19.47) determine the cross section of rental rates and wage rates, and given the cross-sectional rental rates, (19.35) determines the evolution of the distribution of capital stocks in the world economy.

The next proposition shows that the structure of the world equilibrium is essentially identical to that in the previous subsection.

PROPOSITION 19.12. *There exists a unique steady-state world equilibrium. In this equilibrium, capital stock and output in each country grows at the constant rate g^* as in (19.37) above, and the world steady-state growth rate g^* is the unique solution to (19.38). This unique steady-state equilibrium is globally stable.*

PROOF. See Exercise 19.20. □

This proposition implies that the results regarding the stable income distribution continue to apply in this more general model. Moreover, eq. (19.40) still gives the world income distribution in the steady-state world equilibrium.

This more general model does not simply replicate the results of the simpler AK model, however. One important implication of this more general model concerns the relative prices of investment and consumption goods. As discussed previously, the empirical evidence strongly suggests that the price of investment goods relative to consumption goods is greater in poor countries. Many models adopt a reduced-form approach to this empirical regularity and argue that it must be due to frictions affecting the investment sector in poor economies. However, only models that allow for trade and different production functions for consumption and investment goods can be truly useful for understanding the sources of differences in these relative prices. The current model, which incorporates these features, naturally generates this pattern of relative prices. The equilibrium derivation above immediately implies the following relative price in each country:

$$\frac{p_j^I(t)}{p_j^C(t)} = \zeta_j \left(\frac{r_j(t)}{w_j(t)} \right)^{(1-\gamma)(1-\tau)},$$

so that the relative price of investment goods will be higher in countries that have high ζ_j and low wages. The first part of this result, that countries with high ζ_j 's (high distortions on investment good sectors) have higher relative prices of investment goods, is natural and consistent with the presumption in the literature. However, eq. (19.47) above shows that countries with worse technology (low μ_j) and higher discount rates (high ρ_j) will also have lower wages and, via this channel, they will have higher relative prices of investment goods. Therefore, the current model not only provides us with a tractable framework for the analysis of international trade of economic growth, and how trade acts as a force stabilizing the world income distribution, but it also generates a cross section of the relative prices of investment and consumption goods that is consistent with the patterns we observe in the data. Furthermore, it highlights that the relative price of investment goods may vary across countries for

reasons different from distortions on the investment sector, so that considerable care is necessary when using the observed variation in these relative prices in the context of one-sector and/or closed-economy models as the previous literature has done.

In concluding this section, let us return to a comparison of the economic forces emphasized here with those of Section 19.3. Recall that in the model of the previous section, each country takes the world product and factor prices as given, and then accumulates without running into diminishing returns to scale. In contrast, the model in this section has emphasized how capital accumulation by a country will increase the world supply of goods in which it specializes, thus creating powerful terms-of-trade effects. These terms-of-trade effects are the reason why the long-run world income distribution is stable and the fast-growing countries tend to increase the growth rate of the rest of the world. Can the approaches in these two sections be reconciled? I believe the answer is yes. One way to reconcile these two approaches is to view them as applying at different stages of development and for different kinds of goods. Imagine, for example, a world in which some goods are “standardized” and can be produced in any country. When a country is producing these goods, it does not face terms-of-trade effects and can accumulate without running into diminishing returns to capital. As discussed in the previous section, this might be a good approximation to the situation experienced by the East Asian tigers in the 1970s and 80s, when they specialized in medium-tech goods (e.g., Vogel, 2006). However, as countries become richer, they also produce and consume more specialized goods. These goods often come in differentiated varieties and thus a greater supply of any one of these goods will create terms-of-trade effects. Consequently, if a country is in the stage of development where it produces more of the specialized goods, further capital accumulation will run into diminishing returns because of terms-of-trade effects. An interesting research area is to construct models combining these two forces and determining when one becomes more important than the other.

19.5. Trade, Technology Diffusion and the Product Cycle

The previous chapter highlighted the importance of technology diffusion in understanding cross-country income differences. But this was done in the context of a world consisting of a collection of closed economies. The presence of international trade enriches the process of technology diffusion, since it introduces the possibility of the “international product cycle,” whereby technology diffusion goes hand-in-hand with certain products previously produced by technologically advanced economies migrating to less-developed nations. The idea of the international product cycle was first suggested by Vernon (1966). Here I present a simple model originally developed by Krugman (1979), which provides a formalization of these ideas. The main use of the model presented here is that, thanks to its simplicity, it has many applications in the study of various different issues in macroeconomics, international trade and economic development.

19.5.1. The International Division of Labor. Consider the world economy consisting of two sets of countries, the North and the South. For the analysis in this section, it

does not matter whether there is one Northern and one Southern country, or many countries within each group. There is free international trade, without any trading costs.

All individuals in all countries have the same CES preferences with love-for-variety defined over a consumption index. This consumption index for country $j \in \{n, s\}$ at time t is

$$(19.48) \quad C_j(t) = \left(\int_0^{N(t)} c_j(t, z)^{\frac{\varepsilon-1}{\varepsilon}} dz \right)^{\frac{\varepsilon}{\varepsilon-1}}$$

where $c_j(t, z)$ is the consumption of the z th good in country $j \in \{n, s\}$ at time t , $N(t)$ is the total number of goods in the world economy at time t that will be determined endogenously and traded freely, and $\varepsilon > 1$ is the elasticity of substitution between these goods. Naturally, without the free-trade assumption, the range of goods consumed by households in country j would not be $N(t)$, but a subset of these goods to which they have access to.

Each country admits a representative household with dynamic preferences defined over streams of consumption $C_j(t)$. For our purposes here, there is no need to specify what these dynamic preferences are, but for concreteness, the reader may want to assume that these are given by the standard CRRA preferences as in (19.1).

The key assumption of the model is that goods fall into two categories: new goods are just invented in the North and can only be produced there; old goods have been invented in the past and their production technology has been imitated by the South, so they can be produced both in the South and in the North.

The technology of production is simple: one worker produces one unit of any good to which the country in which he is located has access to. Workers in the North have access to all goods, but workers in the South only have access to “old goods”. It is important to emphasize that when producing old goods, Northern workers have no productive advantage. Their only advantage (and the only difference in technology) arises because they have access to a larger set of goods.

Suppose that the total labor supply in the North is L^n at all times and the labor supply in the South is L^s . All labor is supplied inelastically. An equilibrium is defined in the usual way as processes of prices for all goods and allocation of labor across goods.

This description of the environment immediately implies that there can be two types of equilibria.

- (1) *Equalization equilibrium*: in this type of equilibrium, there are sufficiently few new goods that both workers in the South and the North will produce some of the old goods. We will see below that in this type of equilibrium both new goods and old goods will command the same price, and incomes in the North and South will be the same. This justifies the label “equalization equilibrium”.
- (2) *Specialization equilibrium*: in this type of equilibrium the South specializes in the production of old goods, while the Northern producers specializes in the production of new goods.

Let us start by studying the international division of labor for a given set of new and old goods, $N^n(t)$ and $N^o(t)$, where naturally the total number of goods is $N(t) = N^n(t) + N^o(t)$. Since the North has access to all goods, while the South only has access to old goods, the ratio $N^n(t)/N^o(t)$ (or $N(t)/N^o(t)$) can be interpreted as a measure of the technology gap between the North and the South.

To start with, let us suppose that the world is in a specialization equilibrium. Clearly, the prices of all new goods will be equal and the prices of all old goods will also be equalized. Denote these two sets of prices by $p^n(t)$ and $p^o(t)$. Let the wage rate in the North be $w^n(t)$ and that in the South $w^s(t)$. A specialization equilibrium then implies that

$$(19.49) \quad \begin{aligned} p^n(t) &= w^n(t) \\ p^o(t) &= w^s(t). \end{aligned}$$

It must be the case that $w^n(t) \geq w^s(t)$, since otherwise Northern workers would prefer to produce old goods. Thus a specialization equilibrium can exist only if when all old goods are produced in the South, the implied equilibrium wage rate in the South is lower than that in the North. To find out when this will be so is straightforward. The CES preferences specified in (19.48) imply that utility maximization requires the ratio of the consumption of new and old goods to satisfy

$$(19.50) \quad \frac{c^n(t)}{c^o(t)} = \left(\frac{p^n(t)}{p^o(t)} \right)^{-\varepsilon}.$$

Specialization implies that all of the labor force of the South is used to produce old goods, while all of the labor force of the North is employed in the production of new goods. Therefore,

$$(19.51) \quad c^n(t) = \frac{L^n}{N^n(t)} \text{ and } c^o(t) = \frac{L^s}{N^o(t)}.$$

Combining the previous three equations yields the following simple relationship between relative wages and labor supplies and technology,

$$(19.52) \quad \frac{w^n(t)}{w^s(t)} \equiv \omega(t) = \left(\frac{N^n(t)}{N^o(t)} \frac{L^s}{L^n} \right)^{1/\varepsilon}.$$

Notice that the right-hand side of (19.52) consists of pre-determined (or constant) quantities at time t . Thus they determine a unique relative wage between the North and the South. A specialization equilibrium will exist only if this ratio $\omega(t)$ is greater than or equal to 1. If it happens to be less than 1, then a specialization equilibrium does not exist; instead, the equilibrium will take the form of an equalization equilibrium. In this equalization equilibrium, wages in the North and the South are equalized, and some of the old goods are produced in the North. In particular, suppose that $\omega(t)$ as defined by (19.52) is strictly less than 1. Then, there exists a unique equilibrium, which takes the form of an equalization equilibrium, where new goods and old goods all command the same price, and are consumed in the same

quantity. Therefore,

$$c^n(t) = \frac{\phi L^n}{N^n(t)} \text{ and } c^o(t) = \frac{L^s + (1 - \phi) L^n}{N^o(t)},$$

where $\phi \in (0, 1)$ is chosen such that $c^n(t) = c^o(t)$. We know that such a $\phi \in (0, 1)$ exists, since $\omega(t) < 1$, which implies that $c^n(t) > c^o(t)$ at $\phi = 1$.

The characterization of the equilibrium is shown diagrammatically in Figure 19.1. This figure shows that there is a downward sloping relationship between the relative supply of labor in the North, L^n/L^s , and the earnings premium in the North, $\omega \equiv w^n/w^s$. It also shows that when $L^n/L^s = N^n(t)/N^o(t)$, the relationship becomes flat at $w^n/w^s = 1$, because in this case the relative supply of labor in the North is sufficiently large that we enter the region of equalization equilibrium.

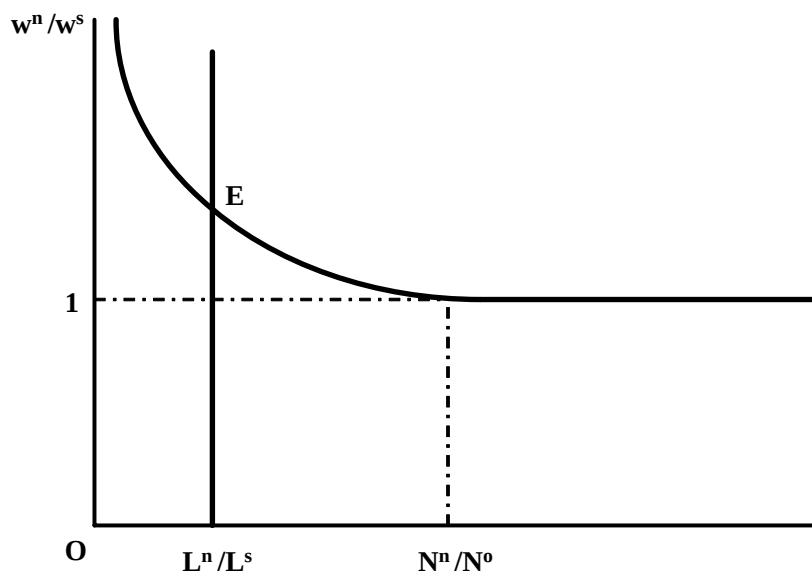


FIGURE 19.1. Determination on the relative wages in the North and the South in the basic product cycle model.

An interesting implication of this equilibrium is that even when there is a technology gap between the North and the South, Northern and Southern incomes may be equalized. There will only be an income gap between the North and the South when the technology gap is relatively large or when the labor supply in the South, L^s , is sufficiently large. This last feature is particularly interesting in the context of the current wave of globalization, which has involved the incorporation of India and China into the world economy as potential low-cost producers of “old” goods.

While we may think that the case with a sufficiently large technology gap and sufficiently large L^s , which leads to a positive income gap between the North and the South is more realistic, the possibility that such a gap may not exist is of theoretical interest and helps

us understand the impact of the international division of labor on cross-country income differences. The possibility that incomes in the North and the South are equalized may appear surprising at first, but the intuition is straightforward. International trade ensures that the Southern consumers have access to goods that their country does not have the technology to produce. Consequently, despite the fact that the South is technologically behind the North, it may achieve the same consumption bundle and the same level of income. This discussion therefore suggests that international trade is a powerful force limiting the extent of cross-country income inequality (for example, resulting from technological differences). This is typically the case, but perhaps surprisingly, not always so. Exercise 19.28 goes through the implications of trade on cross-country income differences and shows that even in the context of the current model, it can sometimes lead to a larger gap of income between rich and poor countries.

19.5.2. Product Cycles and Technology Transfer. The characterization of the equilibrium in the previous subsection was for a given number of new and old goods. Our interest in this model originates because its relative simplicity enables us to endogenize the number of new and old goods, and generates a pattern of product cycle across countries. Here I will follow Krugman (1979) and endogenize the number of new and old goods using a model of exogenous technological change. Exercise 19.27 considers a version of this model with endogenous creation of new products.

In particular, let us suppose that new goods are created in the North according to the following simple differential equation

$$\dot{N}(t) = \eta N(t),$$

with some initial condition $N(0) > 0$ and innovation parameter $\eta > 0$. Goods invented in the North can be imitated by the South. As in the models of technology diffusion in the previous chapter, this process is assumed to be slow and to follow the differential equation

$$\dot{N}^o(t) = \iota N^n(t),$$

where $\iota > 0$ is the imitation parameter, and this differential equation has a motivation similar to the technology diffusion equations in the previous chapter, and captures the idea that the South can only imitate from the set of goods that have not so far been imitated (of which there is a total of $N^n(t)$ at time t). Also, as specified above, $N(t) = N^n(t) + N^o(t)$. Combining these equations, we obtain a unique globally stable steady-state ratio of new to old goods given by

$$(19.53) \quad \frac{N^n(t)}{N^o(t)} = \frac{\eta}{\iota}.$$

This equation is intuitive: the ratio of new to old goods will be high when the rate of innovation in the North, η , is high relative to the rate of imitation from the South, ι . Combining

this equation with (19.52), the equilibrium wage ratio between the North and the South is

$$(19.54) \quad \frac{w^n(t)}{w^s(t)} = \max \left\{ \left(\frac{\eta L^s}{\iota L^n} \right)^{1/\varepsilon}, 1 \right\}.$$

In this expression, when the max operator picks 1, we are in the equalization equilibrium. Otherwise we are in the specialization equilibrium. Since the ratio $w^n(t)/w^s(t)$ also corresponds to the ratio of income between the North and the South, this equation also implies that a high rate of innovation by the North makes the South relatively poor (though not absolutely so), while a higher rate of imitation by the South makes the South relatively richer and the North relatively poorer (see Exercise 19.26). In view of the results from the previous chapter, these results are not surprising.

An important and interesting feature of this steady-state equilibrium is the product cycle. Let us focus on the specialization equilibrium. In this case, new goods are invented in the North and produced there by workers that receive relatively high wages (since in the specialization equilibrium, $w^n(t) > w^s(t)$). After a while, a given new good is imitated by the South, so its production shifts to the South, where labor costs are lower. Thus in this model we witness the international product cycle, starting with production at high labor costs in the North, and then transitioning to a mode of “cheap production” in the South.

An important application of the product cycle model is to the implications of international protection of intellectual property rights (IPR). The rate of imitation ι can also be considered as an inverse measure of the international protection of IPR. Then, as shown in Exercise 19.26, in this baseline model, stronger international IPR protection will always increase the income gap between the North and the South. Interestingly, however, the exercise also shows that it does not always lead to a welfare improvement in the North.

19.6. Trade and Endogenous Technological Change

The effect of trade on growth has attracted much academic and policy attention. Most economists believe that trade promotes growth, and there is both micro and macro evidence consistent with this belief. A number of papers, for example, Dollar (1992) and Sachs and Warner (1995), find a positive correlation between openness to international trade and economic growth. While these studies are not entirely convincing, owing to the typical difficulties of reaching causal conclusions from growth regressions (recall the discussion in Chapter 3), other papers have tried to overcome these difficulties by using instrumental-variables strategies. In this context, a well-known paper by Frankel and Romer (1999) exploits differences in the trade capacity of countries as given by the gravity equations of trade as a source of variation to estimate the effect of trade on long-run income differences. Gravity equations, which are widely used in the empirical trade literature, link the volume of trade between two countries to their geographic and economic characteristics and their interactions (such as size of country, GDP, distance or other characteristics). Frankel and Romer exploit the geographically-determined component of these gravity equations to construct a measure of

“predicted trade” for each country and use this as an instrument for actual trade openness. Using this strategy, they show that greater trade is associated with higher income per capita (thus with greater long-run growth). In addition, recent microeconomic evidence by Bernard and Jensen (1997), Bernard, Eaton, Jensen and Kortum (2004) and others show that firms that engage in exporting are typically more productive, which might be partly due to “learning-by-exporting,” though at least some part of this correlation is likely to be due to selection (Melitz, 2003). Similarly, firms in developing countries that import machinery from more advanced economies appear to be more productive (e.g., Goldberg and Pavnik, 2007). Nevertheless, a number of economists are skeptical of the growth effects of trade. Rodrik (1997) and Rodriguez and Rodrik (1999) argue that the empirical evidence that trade promotes growth is not entirely compelling. On the theoretical side, a number of authors, for example, Matsuyama (1992) and Young (1993), have presented models in which international trade can slow down growth in some countries.

In this and the next section, I investigate some of the simplest models that link trade to growth in order to investigate the potential impacts of international trade on economic growth. I start with a model illustrating how trade opening may change the pace of endogenous technological change. This model is inspired by Grossman and Helpman (1991b), who investigate many different interactions between international trade and endogenous technological change. Briefly, the model consists of two independent economies that can be approximated by the baseline endogenous technological change model with expanding input varieties as in Chapter 13. In fact, the model is identical to the lab-equipment specification in Section 13.1. The advantage of this model is that there are no knowledge spillovers, thus we do not have to make some potentially problematic assumption about knowledge spillovers occurring at the same time as trade opening.² We will look at these two economies first without any international trade and then with costless international trade, and then compare the equilibrium growth rates under these two scenarios. Naturally, a smoother transition, in which trade costs decline slowly, is more realistic in practice, but the sharp thought experiment of moving from autarky to full trade integration is sufficient for us to obtain the main insights concerning the effect of international trade on technological progress.

Given the analysis in Section 13.1 of Chapter 13, there is no need to repeat the analysis here. It suffices to say that we consider two economies, say 1 and 2, with identical technologies, identical preferences, and identical labor forces normalized to 1 (and no population growth). Preferences and technologies are also the same as those specified in Section 13.1. Consequently, a slight variation on Proposition 13.1 in that section immediately implies the following result:

²If, instead of the lab-equipment specification, we were to use the specification with knowledge spillovers and the two countries produced different sets of inputs, it would be necessary to make additional modeling assumptions. For example, we would need to decide whether and how much the inputs produced in the foreign country increase the productivity of R&D in the home country before and after trade opening. Exercise 19.31 shows that assumptions concerning how the extent of knowledge spillovers change with trade opening influence the conclusions regarding the effect of trade on growth.

PROPOSITION 19.13. *Suppose that condition*

$$(19.55) \quad \eta\beta > \rho \text{ and } 2(1 - \theta)\eta\beta < \rho$$

holds. Then, in autarky there exists a unique equilibrium in which starting from any level of technology, both countries innovate and grow at the same rate

$$(19.56) \quad g^A = \frac{1}{\theta}(\eta\beta - \rho).$$

PROOF. See Exercise 19.29. □

Next, let us analyze what happens when these two economies start trading. The exact implications of trade will depend on whether, before trade opening, the two countries were producing some of the same inputs or not (recall that there is a continuum of available inputs that can be produced). To the extent that they were producing some of the same inputs, the static gains from trade will be limited. If, on the other hand, the two countries were producing different inputs, there will be larger static gains. However, our interest here is with the dynamic effects of trade opening, that is, the effects of trade opening on economic growth. Once again, the analysis from Chapter 13 immediately leads to the following result:

PROPOSITION 19.14. *Suppose that condition (19.55) holds. Then, after trade opening, the world economy and both countries produce new technologies and grow at the rate*

$$g^T = \frac{1}{\theta}(2\eta\beta - \rho) > g^A,$$

where g^A is the autarky growth rate given by (19.56).

PROOF. See Exercise 19.30. □

This proposition shows that opening to international trade encourages technological change and increases the growth rate of the world economy. The reason is simple; international trade enables each input producer to access a larger market, and this makes inventing new inputs more profitable. This greater profitability translates into a higher rate of innovation and more rapid growth.

The main effect captured in this simple model is reasonably robust. Grossman and Helpman (1991b) provide a number of extensions and also richer models of international trade (for example with multiple factors). The economic force, a version of the market size effect, leading to the innovation gains from trade is also reasonably robust. Nevertheless, a number of caveats are necessary. First, as Exercise 19.31 shows, if the R&D sector competes with production, there will be powerful offsetting effects, because trade will also increase the demand for production workers. In this case, the qualitative result in this section, that trade opening increases the rate of technological progress, generally applies, but it is also possible to construct versions of this baseline model, where this effect is entirely offset. Exercise 19.31 also provides an example of this type of extreme offset, which should be borne in mind as a useful caveat. Second, Exercise 19.32 shows that if the full scale effect is removed and we focus on an economy with semi-endogenous growth (as the model studied in Section 13.3 in Chapter 13), trade opening will increase innovation temporarily, but not in the long run.

19.7. Learning-by-Doing, Trade and Growth

The previous section showed how international trade can increase economic growth in all countries in the world by encouraging faster technological progress. In addition to this effect of trade on growth working via technological change, the “static gains” from trade are well recognized and understood. By improving the allocation of resources in the world economy, these static gains can also encourage economic growth. Nevertheless, as mentioned in the previous section, many commentators and some economists remain skeptical of the positive growth effects of international trade. A popular argument, often used to justify infant industry protection, is that the static gains from trade come at the cost of dynamic gains, because international trade induces some countries to specialize in industries with relatively low growth potential. In this section, I will outline a simple model with this feature. Richer models that also lead to similar conclusions have been presented by, among others, Matsuyama (1992), Young (1993) and Galor and Mountford (2006). There are also more subtle arguments for why trade may have negative effects on growth based on institutional differences across countries, which are discussed at the end of this chapter. My purpose here is to use the simplest model to illustrate the potential negative effects of trade—and also show when they may not apply. As in the models by Matsuyama and Young, the mechanism for potential dynamic losses from trade (for some countries) will be the presence of learning-by-doing externalities in some sectors.

Consider a world economy consisting of two blocks of countries, the North and the South, and suppose that each block consists of many identical countries. The thought experiment is a move from autarky to full international trade integration between these two blocks. To simplify the exposition and to focus on the main ideas, let us assume that all countries are “almost identical”. In particular, each country has a total labor force of 1, and labor can be used to produce one of two intermediate goods with the production functions

$$Y_j^1(t) = A_j(t) L_j^1(t) \text{ and } Y_j^2(t) = L_j^2(t),$$

with the labor market clearing condition

$$L_j^1(t) + L_j^2(t) \leq 1$$

for $j \in \{n, s\}$ denoting a Northern or Southern country. Moreover, let us assume that the total number of Northern and Southern countries are equal, and denote the total number of countries in the world by $2J$.

The final good is produced as a CES aggregate of these two intermediates. Once again distinguishing between the production of intermediates and their use in the final good sector, we write this as

$$Y_j(t) = \left[\gamma X_j^1(t)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\gamma) X_j^2(t)^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}},$$

where ε is the elasticity of substitution between the two intermediates and suppose that $\varepsilon > 1$. The case of $\varepsilon = 1$ (where the production function becomes Cobb-Douglas) is also of interest,

and I will treat this case separately. Moreover, to simplify the algebra and the exposition below, I set $\gamma = 1/2$.

Learning-by-doing is modeled as follows:

$$(19.57) \quad \frac{\dot{A}_j(t)}{A_j(t)} = \eta L_j^1(t),$$

so that when more workers are employed in sector 1, the technology of sector 1 improves. There are no learning-by doing opportunities in sector 2. Thus one might think of sector 1 as manufacturing or some high-tech sector, while sector 2 may correspond to agriculture or to relatively low-tech sectors (though whether there are greater opportunities for learning-by-doing in manufacturing than in agriculture is quite debatable). As in Romer' (1986a) model of growth through externalities (recall Chapter 11), each producer ignores the positive externality that it creates on the future productivity of sector 1 by its production decisions today.

The only difference between the North and the South is a small "comparative advantage" for the North in the production of sector 1. In particular, I assume that

$$(19.58) \quad A_n(0) = 1 \text{ and } A_s(0) = 1 - \delta,$$

where $\delta > 0$ and it will be taken to be a small number.

Given this structure, the equilibrium both without international trade and with international trade are relatively straightforward to characterize. The key in both cases is that the value of the marginal product of labor (the "wage rates") in the two sectors have to be equalized or only one of the two sectors will be active. Let us start with the closed economy and suppose that both sectors have to be active at t . This implies that the marginal products have to be equalized in the two sectors, thus

$$(19.59) \quad p_j^1(t) A_j(t) = p_j^2(t),$$

where $p_j^1(t)$ and $p_j^2(t)$ denote the prices of the two intermediates in country j in terms of the final good, and $A_j(t)$ is the level of productivity in sector 1 in country j . Notice that prices are indexed by j , since we are in the closed economy case. Profit-maximization by the final good producers immediately implies that

$$\begin{aligned} \frac{p_j^1(t)}{p_j^2(t)} &= \left(\frac{X_j^1(t)}{X_j^2(t)} \right)^{-\frac{1}{\varepsilon}} \\ &= \left(\frac{A_j(t) L_j^1(t)}{1 - L_j^1(t)} \right)^{-\frac{1}{\varepsilon}}, \end{aligned}$$

where $L_j^1(t)$ denotes the amount of labor allocated to sector 1 in country j at time t , and naturally, the amount of labor allocated to sector 2 is $L_j^2(t) = 1 - L_j^1(t)$. Combining this with (19.59),

$$(19.60) \quad L_j^1(t) = \frac{A_j(t)^{\varepsilon-1}}{1 + A_j(t)^{\varepsilon-1}}.$$

The evolution of the productivity of sector 1 is then given by (19.57).

PROPOSITION 19.15. *Consider the above-described model, and suppose that $\varepsilon > 1$ and $\delta \rightarrow 0$. Then, in the absence of international trade the equilibrium involves the allocation of labor given by (19.60) for all j and t . In particular, $L_j^1(t=0) = 1/2$, and $L_j^1(t)$ monotonically converges to 1. The growth rate of each country $g_j(t)$ converges to $g^* = \eta$.*

*If, on the other hand, $\varepsilon = 1$, then $L_j^1(t) = 1/2$ for all t , and the long-run growth rate of each country is $g^{**} = \eta/2$.*

PROOF. See Exercise 19.33. □

Next consider the same world economy with free international trade starting at time $t = 0$. For each intermediate good, there is now only a single world price, $p^1(t)$ for good 1 and $p^2(t)$ for good 2. With standard arguments, these prices satisfy

$$\begin{aligned} \frac{p^1(t)}{p^2(t)} &= \left(\frac{X_n^1(t) + X_s^1(t)}{X_n^2(t) + X_s^2(t)} \right)^{-\frac{1}{\varepsilon}} \\ &= \left(\frac{A_n(t) L_n^1(t) + A_s(t) L_s^1(t)}{2 - L_n^1(t) - L_s^1(t)} \right)^{-\frac{1}{\varepsilon}}, \end{aligned}$$

where the subscripts n and s denote Northern and Southern countries.

It is straightforward to verify that as a result of the slight comparative advantage introduced in eq. (19.58), at $t = 0$ the marginal product of Northern workers in sector 1 is higher and all of the labor force in the North will be employed in sector 1, while all of the labor force in the South will be employed in sector 2. Moreover, all of sector 1 production will be in Northern countries and all of sector 2 production will be in the South. In all subsequent periods, the productivity of Northern workers in sector 1 is even higher, while the productivity of Southern workers in sector 1 remains stagnant. Consequently:

PROPOSITION 19.16. *Consider the above-described model. Then, with free international trade, the equilibrium is as follows: $L_n^1(t) = 1$ and $L_s^1(t) = 0$ for all t . In this equilibrium,*

$$\frac{\dot{A}_n(t)}{A_n(t)} = \eta \text{ and } \frac{\dot{A}_s(t)}{A_s(t)} = 0.$$

The world economy converges to a growth rate of $g^ = \eta$ in the long run. Throughout, the ratio of income in the North and the South is given by*

$$\frac{Y_n(t)}{Y_s(t)} = A_n(t)^{\frac{\varepsilon-1}{\varepsilon}}.$$

Consequently, if $\varepsilon > 1$, then the North becomes progressively richer relative to the South, so that $\lim_{t \rightarrow \infty} Y_n(t)/Y_s(t) = \infty$. If, instead hand, $\varepsilon = 1$, then the relative incomes of the North and the South remain constant, so that $Y_n(t)/Y_s(t) = \text{constant}$ for all t .

PROOF. See Exercise 19.34. □

This proposition contains the main result on how international trade can harm certain countries when there are learning-by-doing externalities in some sectors. In particular, the

South has a slight comparative disadvantage in sector 1. In the absence of trade, it devotes enough of its resources to that sector and achieves the same growth rate as the North. However, if there is free trade, the South specializes in sector 2 (because of its slight comparative disadvantage in sector 1) and fails to benefit from the learning-by-doing opportunities offered by sector 1. As a result, the South becomes progressively poorer relative to the North. This proposition therefore captures the main critique against international trade coming from models such as Young (1993) and proponents of the infant industry arguments.

However, the proposition also shows some of the shortcomings of these arguments. For example, if $\varepsilon = 1$ (or sufficiently close to 1), specialization in sector 2 does not hurt the South. The reason is closely related to the effects highlighted in Section 19.4: the increase in the productivity of sector 1 in the North creates a negative terms of trade effect against the North. This effect is always present, but when $\varepsilon = 1$, it becomes sufficiently powerful to prevent the impoverishment of the South despite the fact that they have specialized in the sector with the low growth potential. Another caveat is highlighted in Exercise 19.35: in the world economy described here, infant industry protection will *not* help the South. Even if there international trade is prevented for a period of duration $T > 0$ for “protecting” some infant industry, the ultimate outcome will be the same as in Proposition 19.16.

So what are we to make of the results in this section and the general issue of the impact of trade on growth? An immediate answer is that the juxtaposition of the models of this and the previous section suggest that the effect of trade on growth must be an empirical one. Since there are models that highlight both the positive and the negative effects of trade on growth, the debate can be resolved only by empirical work. Nevertheless, the theoretical perspectives are still useful. A couple of issues are particularly worth noting. First, the effect of trade integration on the rate of endogenous technological progress may be limited because of the factors already discussed at the end of the previous section. For example, significant effects are possible only when trade opening does not increase wages in the final good sector competing for workers against the R&D sector (which will be the case when the R&D sector does not compete for workers with the final good sector). Moreover, if the extreme scale effects are removed, trade opening creates a temporary boost in innovation, but does not necessarily change long-run growth rate. Nevertheless, the benefits of the greater market size for firms involved in innovation must be present in any model of endogenous technological change. Taking all of these factors into account, we should expect some inducement to innovation from trade opening. Whether these effects are commensurate with or even greater than the static gains of international trade is much harder to ascertain. It may well be that the static gains from trade are more important than the subsequent innovation gains. On the other side of the tradeoff are the potential costs of trade in terms of inducing specialization of some economies in the wrong sectors. The model in this section illustrates this possibility. Nevertheless, I believe that the potential negative effects of trade on growth because of such “incorrect” specialization are much exaggerated. First, there is no strong evidence that learning-by-doing externalities are important in general and much more important in some

sectors than in others (which is what is necessary for “incorrect” specialization). Second, even if this were the case, in most situations specialization is not perfect, thus some amount of learning-by-doing takes place in all economies. Third and most important, international flows of information, which often accompany trade opening but also exist independently, imply that improvements in productivity in some countries will affect productivity in others that were not initially specializing in those sectors (for example, Korea was initially an importer of cars, and is now a net exporter, its productivity in the automotive sector having increased with technology transfer). Finally, as the main result in this section showed, terms-of-trade effects ameliorate any negative impact of specialization in some countries. All in all, it seems that the theoretical case for worrying about the negative growth implications of trade is very weak.

19.8. Taking Stock

This section had three main objectives. The first was to emphasize the shortcoming of using the closed-economy models for the analysis of the economic growth patterns across countries or regions. We have seen that both international trade in assets (international borrowing and lending) and international trade in commodities change the dynamics and also possibly the long-run implications of the closed-economy neoclassical growth models. For example, international capital flows remove transitional dynamics, because economies that are short of capital do not need to accumulate it slowly, but can borrow in international markets. Naturally, there are limits to how much international borrowing can take place. Countries are sovereign entities, thus it is relatively easy for them to declare bankruptcy once they have borrowed a lot. Consequently, the sovereign borrowing risk might place limits on the ability of countries to use international markets to smooth consumption and to increase their investments rapidly. Even in this case, some amount of international lending will take place and this will have an important effect on the equilibrium dynamics of output and the capital stock. The available evidence shows that the amount of gross capital flows are very large, though the Feldstein-Horioka puzzle, that fluctuations in investment are correlated with fluctuations in savings, shows that there are limits to net international capital flows. An investigation of why, despite the very large size of the gross capital flows, net international capital flows do not play a greater role in international consumption and investment smoothing is an interesting area for future research. While there is some research on this topic in international finance, its implications for economic growth are important and need to be studied.

We have further seen that international trade in commodities also changes the implications of the neoclassical growth model. For example, in the model of economic growth with Heckscher-Ohlin trade in Section 19.3, international trade in goods plays the same role as international lending and borrowing, and significantly changes cross-country output dynamics. Thus even in the absence of international lending and borrowing, the implications of

approaches that model the entire world equilibrium are significantly different from those focusing on closed-economy dynamics. The model of economic growth with Ricardian trade in Section 19.4 also showed that output dynamics are very different in the presence of trade. In that model, there would be no convergence across countries without trade, but international trade, via the terms-of-trade effects it induces, creates a powerful force that links the real incomes of different countries. Consequently, the long-run equilibrium involves a stable world income distribution and the short-run dynamics are very different from the closed-economy models.

The second objective was to highlight how the nature of international trade interacts with the process of economic growth. Sections 19.3 and 19.4 focused on this issue. The model of economic growth with Heckscher-Ohlin trade showed how economic growth increases the effective elasticity of output with respect to capital for each country, because of (conditional) factor price equalization. This is useful in understanding how certain economies, such as East Asian tigers, can grow rapidly for extended periods relying on capital accumulation without running into diminishing returns. However, our analysis also showed that a pure Heckscher-Ohlin model may not be an appropriate framework for the analysis of the interactions across countries. In contrast, the model in Section 19.4 emphasized how Ricardian trade, based on technological comparative advantage, creates a new source of diminishing returns to accumulation for each country based on terms-of-trade effects. As a country accumulates more capital, it starts exporting more of the goods in which it specializes. The result is a worsening of its terms of trade, effectively reducing the rate of return to further capital accumulation. The analysis showed how this force leads to a stable world income distribution, whereby rapidly growing economies pull up the laggards to grow at the same rate as themselves. How are we to reconcile the different implications of the models in Sections 19.3 and 19.4? One possibility is to imagine a world that is a mixture of the models of these two sections. It may be that some goods are “standardized” and can be produced in any country. When producing these goods, there are no terms-of-trade effects. So if a country can grow only by producing these goods, it can escape the standard diminishing returns to capital thanks to international trade. This might be a good approximation to the situation experienced by the East Asian tigers in the 1970s and 80s, when they specialized in medium-tech goods. However, as countries become richer they also produce and consume more specialized goods. These goods often come in differentiated varieties and thus a greater supply of any one of these goods will create terms-of-trade effects. Consequently, if a country is in the stage of development where it produces more of the specialized goods, further capital accumulation will run into diminishing returns through the mechanism highlighted in Section 19.4. Regardless of how the forces emphasized in these two approaches are combined, they both show the importance of modeling the world equilibrium and also the importance of viewing the changes in the rate of return to capital in the context of the international trading relations.

The third objective of this chapter was to investigate the effect of international trade on economic growth. Sections 19.6 and 19.7 illustrated two different approaches, one emphasizing the beneficial effects of trade on growth, the other one the potential negative effects. Both classes of models are useful to have in one's arsenal in the analysis of world equilibrium and economic growth. The usefulness of these models notwithstanding, the impact of international trade on economic growth is ultimately an empirical question, though our theoretical analysis has already highlighted some important mechanisms and also suggested that the negative effects of trade on growth are unlikely to be important. Whether the positive effects of trade on technological progress are quantitatively significant remains an open question. It may well be that static gains of trade are more important than its dynamic gains. Nevertheless, any analysis of international trade must take its implications on economic growth and technological change into account.

19.9. References and Literature

This chapter covered a variety of models. Section 19.1 focused on the implications of international financial flows on economic growth. This topic is discussed in detail in Chapter 3 of Barro and Sala-i-Martin (2004), both with and without limits to financial flows. Obstfeld and Rogoff (1996) Chapters 1 and 2 provide a more detailed analysis of international borrowing and lending. Chapter 6 of Obstfeld and Rogoff provides an excellent introduction to the implications of imperfections in international capital markets. Work that models these imperfections explicitly includes Atkeson (1991), Bulow and Rogoff (1989a, 1989b), Kehoe and Perri (2002) and Aguiar, Amador and Gopinath (2006). The Feldstein-Horioka puzzle, which was also discussed in Section 19.1, is still an active area of research. Obstfeld (1995) and Obstfeld and Taylor (2004) present surveys of much of the research on this topic. Taylor (1994), Baxter and Crucini (1993) and Kraay and Ventura (2002) propose potential resolutions for the Feldstein-Horioka puzzle.

Section 19.2 is motivated by Lucas's classic (1990) article. There is a large literature on why capital does not flow from rich to poor countries. Obstfeld and Taylor (2004) contain a survey of the work in this area. The work by Caselli and Feyrer (2007) discussed above provides a method for estimating cross-country differences in the marginal productive capital and argues that differences in the return to capital are limited. This work supports models that account for the lack of capital flows based on productivity differences, such as the model presented in Section 19.2. Recent work by Chirinko and Mallick (2007) argues that the Caselli and Feyrer (2007) procedure may lead to misleading results because they do not incorporate adjustment costs in investment in their calculations and that once these costs are incorporated, returns to capital differ significantly across countries. See also recent work by Alfaro, Kalemli-Ozcan and Volosovych (2005), which also emphasizes productivity differences and links these to institutional factors which we discussed in Chapter 4.

The rest of the chapter relies on some basic knowledge of international trade theory. Space restrictions preclude a detailed review. The reader is referred to a standard text, for

example, Dixit and Norman (1990). Section 19.3 provides a slight generalization of the model in Ventura (1997) (it considers a general constant returns to scale production function rather than CES production function used in Ventura, 1997), though it omits some of the more detailed characterization of transitional dynamics in the paper. A similar but less rich model was first analyzed by Stiglitz (1971). Stiglitz did not include labor-augmenting productivity differences across nations and assumed exogenous saving rates. Other papers that combine Heckscher-Ohlin trade with models of economic growth include Atkeson and Kehoe (2000) and Cunat and Maffezzoli (2001). Section 19.4 builds on Acemoglu and Ventura (2002). The importance of terms-of-trade effects are well recognized in the theory of international trade (see again Dixit and Norman, 1990), but their growth implications had not previously been recognized. The model presented in Acemoglu and Ventura (2002) is a much simplified Ricardian model, exploiting the structure of preferences first introduced by Armington (1969), but in the production of the final good rather than in preferences. Richer Ricardian models typically build on the seminal article by Dornbusch, Fischer and Samuelson (1977), though this richer setup has not yet been integrated with growth models. Ventura (2005) provides a survey of international trade and economic growth, focusing on the models in Sections 19.3 and 19.4.

The model in Section 19.5 builds on Krugman's (1979) seminal article on the product cycle. As noted in the text, Vernon (1966) was the first to formulate the problem of the international product cycle, emphasizing the economic forces modeled in Krugman (1979) and in Section 19.5 here. Grossman and Helpman (1991b) provide richer models of the product cycle with endogenous technology, similar to the economy discussed in Exercise 19.27. Antras (2006) provides a new perspective on the international product cycle that relies on the importance of incomplete contracts. In his model, contractual problems between Northern producers and Southern subsidiaries constitute a barrier slowing down the transfer of goods to the South. Only after goods become sufficiently "standardized," the contracting problems become less severe and the transfer of production to the south takes place.

There is a large empirical literature on the impact of trade on growth. Many of the best-known papers in this literature were discussed at the beginning of Section 19.6. The rest of Section 19.6 builds on Romer and Rivera Batiz (1991) and Grossman and Helpman (1991b), but uses the formulation from Section 13.1 in Chapter 13. Grossman and Helpman (1991b) assume that R&D requires labor and introduce competition between the R&D sector and the final good sector. In this case, the nature of the knowledge spillovers becomes important for the implications of trade on the pace of endogenous technological progress. Romer and Rivera Batiz (1991) also discuss the implications of the form of the innovation possibilities frontier for the effects of trade on technological change. This point, which is developed in Exercise 19.31, also features in recent work by Atkeson and Burstein (2006). Grossman and Helpman (1991b) also present much richer models with multiple sectors and factor proportion differences across countries, leading to Heckscher-Ohlin type trade. Another potential effect of international trade on technological change would be by influencing the direction of technological change.

This topic is analyzed in detail in Acemoglu (2003b), where I show that trade opening with imperfect intellectual property rights can make new technologies more skill-biased than before trade opening. Similar models are also analyzed in Thoenig and Verdier (2002) and Epifani and Gancia (2006).

Section 19.7 presents a model inspired by Young (2003) and Matsuyama (2002). Lucas (1988) and Galor and Mountford (2006) also present similar models, which feature interaction between specialization and learning-by-doing. Other models where international trade may be costly for some countries rely on differences in the amount of rents generated by different sectors because of imperfections in the labor market or institutional problems. Levchenko (2007) and Nunn (2006) present models in which trade leads to the transfer of rent-creating jobs from countries with weak institutions to those with better institutions and may be harmful to countries with weak institutions.

19.10. Exercises

EXERCISE 19.1. Prove Proposition 19.1.

EXERCISE 19.2. Prove Proposition 19.2. [Hint: use (19.5) together with the fact that consumption and output grow at the same rate in each country to show that in the steady state it is optimal for each country (or each consumer in each country) to choose $\dot{\mathcal{A}}_j(t) \rightarrow 0$.] Consider the world economy with free flows of capital, but assume that each country has a different discount factor ρ_j .

- (1) Prove that 19.1 still holds.
- (2) Show that there does not exist a steady-state equilibrium with $\dot{\mathcal{A}}_j(t) = 0$ for all j . Explain the intuition for this result.
- (3) Characterize the asymptotic equilibrium (the equilibrium path as $t \rightarrow \infty$). Suppose that $\rho_{j'} < \rho_j$ for all $j \neq j'$. Show that the share of the world capital that is used in country j' will tend to 1. What does this imply for the relationship between GDP and GNP across countries.
- (4) Is the form of the asymptotic equilibrium in part 3 of this exercise realistic? If not, explain how you would modify the model to achieve a more realistic world equilibrium in the presence of free capital flows.

EXERCISE 19.3. This exercise asks you to prove Proposition 19.3.

- (1) Show that $c_j(t)/c_{j'}(t)$ is constant for all j and j' .
- (2) Show that given the result in Proposition 19.1, the integrated world equilibrium can be represented by a single aggregate production function. [Hint: use an argument similar to that leading to Proposition 19.6].
- (3) Relate this result and Proposition 19.6 to Theorem 5.4 in Chapter 5. Explain why these “aggregation” results would not hold without free capital flows.
- (4) Given the result in parts 1 and 2, apply an analysis similar to that for the global stability of the equilibrium path in the basic neoclassical growth model to establish

the global stability of the equilibrium path here. Given global stability, prove the uniqueness of the equilibrium path.

EXERCISE 19.4. * Consider a world economy with international capital flows, but suppose that because of sovereign default risk a country cannot borrow more than a fraction $\phi > 0$ of its capital stock. Consequently, in terms of the model in Section 19.1, we have the restriction that

$$b_j(t) \leq \phi k_j(t).$$

- (1) Show that the steady-state equilibrium of the world economy is not affected by this constraint. Explain the intuition for this result carefully.
- (2) Characterize the transitional dynamics of the world economy under this constraint. Show that Corollary 19.1 no longer holds.

EXERCISE 19.5. Barro and Sala-i-Martin (1994, 2004) use growth regressions to look at the patterns of convergence across US regions and states. They find that there is a slow pattern of convergence across regions and states, and they interpret this through the lenses of the neoclassical growth model. Explain why Corollary 19.1 implies that this interpretation is not appropriate. Suggest instead an alternative explanation for why convergence across regions and states might be slow. [Hint: should we expect technology or capital to flow more rapidly across regions?]

EXERCISE 19.6. Consider the baseline AK model studied in Chapter 11, and suppose that countries have the same production technology, but differ according to their discount rates, the ρ_j 's. Show that there will be persistent differences in saving and investment rates across countries that are correlated, even in the presence of free financial flows across countries. Provide a precise intuition for this result. Explain why this model could not account for the Feldstein-Horioka puzzle, which does not refer to the correlation between saving and investment in levels but in differences. Can you extend this model to account for the Feldstein-Horioka puzzle?

EXERCISE 19.7. Prove Proposition 19.4.

EXERCISE 19.8. Show that if in the model of Section 19.3, there are free capital flows across countries, this will have no effect on the equilibrium allocation.

EXERCISE 19.9. Consider the model in Section 19.3 with different discount rates across countries. Prove that there does not exist a steady-state equilibrium.

EXERCISE 19.10. * Consider the model in Section 19.3, but assume that (19.9) is now modified to

$$Y_j^K(t) = B_j K_j(t),$$

where B_j 's potentially differ across countries. Characterize the world equilibrium. What would happen if there were free capital flows in this case?

EXERCISE 19.11. (1) Reformulate and prove the main results in Section 19.3 for the case in which population levels differ across countries. [Hint: instead of $k(t)/A$, relevant prices will not depend on $\sum_{j=1}^J K_j(t) / \sum_{j=1}^J AL_j(t)$].

- (2) What happens if each country has a different rate of population growth, n_j ?

EXERCISE 19.12. *

- (1) Show that all the results in Section 19.3 continue to hold if the CRRA preferences in (19.14) is now modified to an arbitrary strictly increasing, strictly concave utility function $u(c)$.
- (2) Now let us go back to the preferences as in (19.14), but suppose that productivity of labor in each country is given by

$$A_j(t) = A_j \exp(gt).$$

Show that all of the results from the text continue to apply, and in particular, derive the equivalent of Proposition 19.4.

- (3) Finally, let us suppose that F in (19.7) does not satisfy Assumption 2. How does this affect the analysis and the results?

EXERCISE 19.13. Derive the unit cost functions (19.27) and (19.28) from the production functions (19.24) and (19.25). Determine the value of the constant χ .

EXERCISE 19.14. Derive eq.'s (19.29) and (19.30).

EXERCISE 19.15. Consider the model in Section 19.4.

- (1) Derive the trade balance equation, (19.34), from the capital market clearing equation, (19.26).
- (2) Prove that the ratio of imports to GDP at each t is equal to τ .

EXERCISE 19.16. Provide a rigorous proof of the global (saddle-path) stability of the steady-state world equilibrium in Proposition 19.10.

EXERCISE 19.17. (1) Derive eq.'s (19.40) and (19.41).

- (2) Explain the roles of the different parameters in determining cross-country income dispersion.
- (3) Using reasonable parameter values show how the model with international trade can generate much larger differences in income per capita across countries resulting from small parameter differences.

EXERCISE 19.18. Derive eq. (19.43).

EXERCISE 19.19. Prove Proposition 19.11.

EXERCISE 19.20. Prove Proposition 19.12.

EXERCISE 19.21. Consider the steady-state world equilibrium in the model of Section 19.4.

- (1) Show that an increase in τ does not necessarily increase the steady-state world equilibrium growth rate g^* as given by (19.38). Provide an intuition for this result.
- (2) Show that even when τ does not increase growth, it increases world welfare. [Hint: to simplify the answer to this part of the question, you can simply look at steady state welfare].
- (3) Interpret this finding in light of the debate about the effect of trade on growth.
- (4) Provide a sufficient condition for an increase in τ to increase the world growth rate and interpret this condition.

EXERCISE 19.22. * Consider the model of Section 19.4, except that instead of utility maximization by a representative household, assume that each country saves a constant fraction s_j of its income. Show that terms-of-trade effects will be present in equilibrium, but the steady state will be “degenerate,” with the relative prices of goods supplied by the highest saving country going to zero. Explain why exogenous savings versus dynamic utility maximization give different answers in this case.

EXERCISE 19.23. * Consider the model of Section 19.4, but assume that $\varepsilon < 1$. Characterize the equilibrium. Show that in this case countries that have lower discount rates will be relatively poor. Provide a precise intuition for this result. Explain why the assumption that $\varepsilon < 1$ may not be plausible.

EXERCISE 19.24. * Consider the baseline AK model in Section 19.4. Suppose that production and allocation decisions within each country is made by a “country-specific social planner” (who maximizes the utility of the representative household within the country).

- (1) Show that the equilibrium in the text is no longer an equilibrium. Explain why.
- (2) Characterize the equilibrium in this case and show that all of the qualitative results derived in the text apply. In particular, provide generalizations of Propositions 19.11 and 19.12.
- (3) Show that world welfare is lower in this case than in the equilibrium in the text. Explain why.
- (4) Do you find the equilibrium in this exercise or the one in the text more plausible? Justify your answer.

EXERCISE 19.25. * Consider the model with labor in Section 19.4. Suppose that countries can invest in order to create new varieties of products. Suppose that if a particular firm creates such a variety, it becomes the monopolist and can charge a markup equal to the monopoly price to all consumers in the world, until this variety is destroyed endogenously, which happens at the exponential rate $\delta > 0$.

- (1) Show that the optimal monopoly price for a firm in country j at time t is: $p_j(t) = (\varepsilon r_j(t)) / (\varepsilon - 1)$. Interpret this equation.
- (2) Suppose that a new variety can be created by using $1/\eta$ units of labor. Show how this changes the labor market clearing condition and specify the free-entry condition.
- (3) Define a world BGP as an equilibrium in which all countries grow at the same rate. Show that such an equilibrium exists and is uniquely defined. Explain the economic forces that lead to the existence of such a “stable” equilibrium. [Hint: show that in this BGP the number of varieties that each country produces is constant].
- (4) What is the effect of an increase in the discount rate ρ on the number of varieties that a country produces? Interpret this result.
- (5) Discuss informally how the analysis and the results would be modified if new products were produced using a combination of labor and capital?

EXERCISE 19.26. Show that in the model of Section 19.5 an increase in ι will always (weakly) close the relative income gap between the North and the South. Characterize the conditions under which an increase in ι will make the North worse-off (in terms of reducing its real income). Interpret these results.

EXERCISE 19.27. This exercise asks you to endogenize innovation decisions in the model of Section 19.5. Assume that new goods are created by technology firms in the North as in the model in Section 13.4 in Chapter 13, and these firms are monopolist suppliers until the good they have invented is copied by the South. The technology of production is the same as before, and assume that new goods can be produced by using final goods, with the technology $\dot{N}(t) = \eta Z(t)$, where $Z(t)$ is final good spending. Imitation is still exogenous and takes place at the rate ι . Once a good is imitated, it can be produced competitively in the South.

- (1) Show that for a good that is not copied by the South, the price will be

$$p(t, \nu) = \frac{\varepsilon}{\varepsilon - 1} w^n(t).$$

- (2) Characterize the static equilibrium for given levels of $N^n(t)$ and $N^o(t)$.
- (3) Compute the net present value of a new product for a Northern firm. Why does it differ from the expression in Section 13.4?
- (4) Impose the free-entry condition and derive the equilibrium rate of technological change for the world economy. Compute the world growth rate.
- (5) What is the effect of an increase in ι on the equilibrium? Can an increase in ι make the South worse-off? Explain the intuition for this result.

EXERCISE 19.28. Consider a variation of the product cycle model in Section 19.5. Suppose there is no international trade, so that, the number of goods produced and consumed in each country will differ.

- (1) Show that wages and incomes in the North and the South at time t are

$$w^n(t) = N(t)^{\frac{1}{\varepsilon-1}} \text{ and } w^s(t) = N^o(t)^{\frac{1}{\varepsilon-1}}.$$

- (2) Derive a condition for relative income differences to be smaller in this case than in the model with international trade. Provide a precise intuition for why international trade may increase relative income differences
- (3) If trade increases the income differences between the North and the South, does it mean that it reduces welfare in the South? [Hint: if you wish, you can again use the steady-state welfare levels].

EXERCISE 19.29. Prove Proposition 19.13.

EXERCISE 19.30. Prove Proposition 19.14.

EXERCISE 19.31. Consider the model in Section 19.6, but assume that new products are created with the innovation possibilities frontier as in Section 13.2 in Chapter 13. Assume that before trade knowledge spillovers are created by the entire set of available inputs in the world economy, that is, the innovation possibilities frontier is similar to (13.24) in Section

13.2, except that

$$\dot{N}^j(t) = \eta N(t) L_R^j(t)$$

for country j , where $N(t) = N^1(t) + N^2(t)$ and $L_R^j(t)$ is the number workers working in R&D in country j . Consequently, trade opening does not change the structure of knowledge spillovers.

- (1) Show that in this model, trade opening has no effect on the equilibrium growth rate. Provide a precise intuition for this result.
- (2) Next assume that before trade opening the innovation possibilities frontier takes the form $\dot{N}^j(t) = \eta N^j(t) L_R^j(t)$. Show that in this case, trade opening leads to an increase in the equilibrium growth rate as in Proposition 19.14. Explain why the results are different.
- (3) Which of the specifications in 1 and 2 is more plausible? In light of your answer to this question, how do you think trade opening should affect economic growth.

EXERCISE 19.32. Consider the model in Section 19.6, with two differences. First, population grows at the rate n in both countries. Second, the innovation possibilities frontier is given as

$$\dot{N}^j(t) = \eta N(t)^{-\phi} Z^j(t)$$

for country j , where $N(t) = N^1(t) + N^2(t)$. Show that trade opening leads to greater technological progress upon impact, but the long-run growth rate of each country remains unchanged.

EXERCISE 19.33. Prove Proposition 19.15.

EXERCISE 19.34. (1) Prove Proposition 19.16.

- (2) Explain why when $\varepsilon = 1$, specialization in the sector without learning-by-doing does not have an adverse effect on the relative income of the South.
- (3) What are the implications of trade opening on relative incomes if $\varepsilon < 1$?
- (4) Characterize the equilibrium if all economies are closed until time $t = T$ and then open to international trade at time T . What are the implications of this result for infant industry protection.

EXERCISE 19.35. Consider the economy in Section 19.7, but suppose that the South is bigger than the North. In particular, assume that

$$(19.61) \quad (1 - \delta)^{-\varepsilon} < \frac{L^S}{L^N} < 1 + (2 - \delta)(1 - \delta)^{-\varepsilon}.$$

- (1) Show that in this case not all Southern workers will work in sector 2 and there will be some learning-by-doing in the South. Why is (19.61) necessary for this result?
- (2) How does this affect the long-run equilibrium? [Hint: show that the limiting value of L_s^1 is equal to 0]. Why is (19.61) necessary for this result?

Part 7

Economic Development and Economic Growth

In this part of the book, I discuss the relationship between economic development and economic growth. The first question that the reader will rightly ask is why there is (or there should be) a distinction between economic development and economic growth. This question is particularly apt because I have argued in Chapter 1 that societies that are rich—*developed*—today are those that have grown steadily over the past 200 years and those that are poor or *less-developed* are those that have not achieved this type of steady growth. This perspective suggests that economic development and economic growth are essentially the same thing and should be studied together. Nevertheless, there are two reasons, one good and one bad, for drawing a distinction between development and growth. The good reason is that even though economic development and growth are part of the same process, models of growth emphasize different aspects of this process than models of economic development. In particular, the models studied so far focus on either balanced growth or transitional dynamics leading to balanced growth. Even though these transitional dynamics have been analyzed in a number of contexts, our main interest has been to ensure that they take us towards a BGP. Behavior along or near the BGP of a neoclassical or endogenous growth economy provides a good approximation to the behavior of relatively developed societies. But many salient features of economic growth at lower incomes or at earlier stages of development are not easy to map to this “orderly” behavior of balanced growth. In fact, Simon Kuznets and other economists have documented that even in more developed economies, many aspects of the process of economic growth are far from the balanced benchmark implied by the standard neoclassical growth model.

Motivated by these patterns, in his classic book *Modern Economic Growth*, Simon Kuznets defines economic growth as follows:

“We identify the economic growth of nations as a sustained increase in per capita or per worker product, most often accompanied by an increase in population and usually by sweeping structural changes. In modern times these were changes in the industrial structure within which product was turned out and resources employed—away from agriculture toward nonagricultural activities, the process of industrialization; in the distribution of population between the countryside and the cities, the process of urbanization; in the relative economic position of groups within the nation distinguished by employment status, attachment to various industries, level of per capita income, and the like; in the distribution of product by use—among household consumption, capital formation, and the government consumption, and within each of these major categories by further subdivisions; in the allocation of product by its origin within the nation’s boundaries and elsewhere; and so on.” Simon Kuznets (1966).

Although one might debate whether this is the most functional definition of economic growth, it does capture a range of important changes that accompany economic growth in most societies. And yet, the models of economic growth studied so far do not do justice to

the complex process described by Kuznets. They provide a framework for explaining the sustained increase in income per capita or output per worker. But our models do not feature Kuznets's *sweeping structural changes*.

A complementary perspective to Kuznets's vision is provided by early development economists, such as Hirschman, Nurske and Rosenstein-Rodan, who emphasized the importance of potential *market failures* and *poverty traps* in the process of development. If such market failures and poverty traps are an important determinant of economic performance, then we may expect them to be more widespread in less-developed, poorer economies.³ Thus one might expect Kuznets's structural change to be accompanied by a process that involves the organization of production becoming more efficient and the economy moving from the interior of the aggregate production possibilities set towards its frontier. Throughout, I use the term *structural change* to describe the features emphasized by Kuznets, while *structural transformation* refers to changes in the organization and efficiency of production accompanying the process of development.

A useful theoretical perspective might therefore be to consider the early stages of economic development taking place in the midst of—or even via—this type of structural transformation, which includes both the structural changes emphasized by Kuznets and the process of less-developed economies approaching their efficiency frontier. We may then expect this structural transformation to ultimately bring the economy to the neighborhood of balanced growth, where our focus so far has been. If this perspective is indeed useful, then we would like to develop unified models that explain both the structural changes at the early stages of development and the behavior approximated by balanced growth at the later stages.

Some of the models presented so far take steps in this direction. For example, the model of takeoff in Section 17.6 in Chapter 17 captures a specific type of transformation, from volatile, low-productivity growth into sustained, stable growth. In addition, many of the models in Chapter 18 emphasize the difference between frontier economies and technological followers. Nevertheless, I have not offered a framework that can do justice to Kuznets's vision and this is largely because the current growth literature is far from a satisfactory framework that can achieve this objective. In this light, the distinction between economic growth and economic development can be justified by arguing that, in the absence of a unified framework or perhaps precisely *before* we can develop a unified framework, we need to study the two aspects of the long-run growth process separately. Economic growth, according to this division of labor, focuses on balanced growth, the growth behavior of the world economy, and other aspects of the growth process approximating the behavior of relatively developed economies. Economic development, on the other hand, becomes the study of structural transformations, and the efficiency implications of these transformations, at the early stages of development. Models of economic development would then focus on structural changes in

³In fact, these theoretical perspectives may be the justification for referring to relatively poor economies as *underdeveloped* rather than as *developing*. In what follows, unless there is a special reason for using these terms, I stick with the less tainted adjectives “less-developed” or relatively poor.

the production and consumption, on urbanization, on the size and the composition of the population, on the occupational structure, and on changes in living and social arrangements. The study of economic development will thus seek to understand when, why and how these processes take place and whether they contribute to a less-developed economy moving towards the frontier of its production possibilities set. Since, as emphasized by Kuznets, economic growth in relatively developed economies also incorporates important element of structural change, part of our analysis in the context of economic development will also shed light on the nature of economic growth in more advanced nations, for example, by helping us understand why and how relatively balanced growth can often go hand-in-hand with major changes in the sectoral composition of output and employment.

The second—the not-so-satisfactory—reason for the distinction between economic growth and economic development is that there are separate literatures on these two topics, with very different emphases and often different questions. The economic growth literature focuses on the theoretical and empirical questions we have so far addressed in this book. The economic development literature, on the other hand, focuses on empirical analyses of education, poverty, discrimination, women's economic and social status, child outcomes, health, lending relations and agriculture in less-developed economies. Much of this literature is non-theoretical. It documents how economic relationships work in less-developed economies or identifies specific market failures. This literature has provided us with numerous facts that are helpful in understanding the economic relations in less-developed economies and has sometimes acted as a conduit for micro reforms that have improved the lives of the citizens of these less-developed economies. But this literature does not ask questions about the aspects of the process of economic development I have emphasized here—that is, it does not pose the question of *why* some countries are less productive and poorer, and *how and why* these less-developed economies can undergo the process of structural transformation associated with, and necessary for, modern economic growth. This implies that even though the reason for drawing a distinction between economic growth and economic development might be literature-driven, it may still be useful. Moreover, based on this distinction one may attempt to bridge the gap that exists between the distinct development and growth literatures by combining the theoretical tools developed in this book with the wealth of evidence collected by the empirical development literature. Such a combination might ultimately lead to a more satisfactory framework for understanding the process of economic development (though unfortunately space restrictions preclude me from pursuing these issues in detail here).

These two reasons motivate my acceptance of the standard distinction between economic development and economic growth. Although I go along with this standard distinction, throughout I emphasize how it is exactly the same tools that are useful for understanding the process of economic development—the structural transformations emphasized by Kuznets, Hirschman, Nurske and Rosenstein-Rodan—as well as the more orderly process of economic growth. My hope is that this approach will engender both greater efforts to develop a unified theoretical framework useful for understanding the process of development and also

theoretical approaches that can make contact with and benefit from the wealth of evidence collected by the empirical development literature.

I organize this part of the book into two chapters. The first, Chapter 20, will focus on models that take only a minimal departure from the balanced growth approaches studied so far while still shedding some light on the structural changes emphasized by Kuznets. The models in this chapter can thus be viewed as extensions of the neoclassical growth models in Chapters 8 and 11 designed to confront various important empirical patterns that are salient over the development process. However, these models neither do full justice to the process of sweeping structural changes emphasized by Kuznets nor do they capture the complex aspects of the process of economic development associated with the move from the interior of the production possibilities set towards the frontier. The second, Chapter 21, will focus on a number of models that investigate various different aspects of this process, including financial development, the demographic transition, urbanization and other social changes. Furthermore, they highlight the importance of potential market failures that may cause development traps. These models present a range of exciting questions and different modeling approaches, but at the expense of providing less unity. Each model makes a different set of assumptions and the profession is far from a unified framework for the analysis of the major structural transformations involved in the process of development. The purpose of Chapter 21 is not to provide such a unified framework but to introduce the reader to these interesting and important questions. It should also be noted that the division between the two chapters is not perfect. Some of the models of structural transformation studied in Chapter 21 can be seen as closely related to the structural change models in Chapter 20. Moreover, some topics, such as the beginning of industrialization, can be treated both as a process of structural change and also as an outcome of a society solving certain market failures. Thus, there is quite a bit of arbitrariness in the decision of whether a particular topic should be in Chapter 20 or Chapter 21.

Structural Change and Economic Growth

In this chapter, I discuss various different approaches to the analysis of structural change. The next two sections focus on the shift of employment and production from agriculture to manufacturing, and then from manufacturing to services. This is a useful starting point both because changes in the composition of employment and production are an important part of the process of economic development and also because, as emphasized by Kuznets and others, similar changes are present even beneath the façade of balanced modern growth. Consequently, these two sections will focus on demand-side and supply-side reasons why we may expect structural change as an economy becomes richer, but also emphasize how such structural changes can be reconciled with balanced growth. Section 20.3 turns to a related theme. As emphasized in Chapter 1, industrialization appears to be an important element underlying the takeoff that led to modern growth and thus to the large cross-country income differences we witness today. In this section, I present a simple model of industrialization, which again emphasizes the importance of structural change, but also shows how pre-industrial agricultural productivity may be a key determinant of the process of industrialization and takeoff.

20.1. Non-Balanced Growth: The Demand Side

Figure 20.1 provides a summary of some of the major changes in the structure of production that the US economy has undergone over the past 150 years. It shows that the share of US employment in agriculture stood at around 90% of the labor force at the beginning of the 19th century, while only a very small fraction of the US labor force worked in manufacturing and services. By the second half of the 19th century, both manufacturing and services had expanded to over 20% of employment, accompanied by a steep decline in the share of agriculture. Over the past 150 years or so, the share of employment in agriculture has continued to decline and now stands at less than 5%, while over 70% of US workers now work in service industries. The share of manufacturing first increased when the share of agriculture started its decline, but has been on a downward trend over the past 40 years or so and now stands at just over 20%. When we look at consumption shares, the general trends are similar, though the share of consumption expenditures on agricultural products is still substantial because of changes in relative prices and relative productivities (and also partly because of imports of agricultural goods). The changes in the composition of employment in the British economy towards the end of the 18th century are also consistent with the US patterns shown in Figure 20.1 (see, for example, Mokyr, 1989). Consequently, similar patterns are present in all OECD economies. Some of the less-developed economies are still largely agricultural but the time

trend is inexorably towards a smaller share of agriculture. Because of Kuznets's emphasis on structural change and seminal work on the topic, Kongsamut, Rebelo and Xie (2001) refer to these changes in the composition of employment and production as the *Kuznets facts*. They provide a tractable model to reconcile this type of structural change with the *Kaldor facts* emphasized so far, that is, the relative constancy of factor shares and the interest rate.

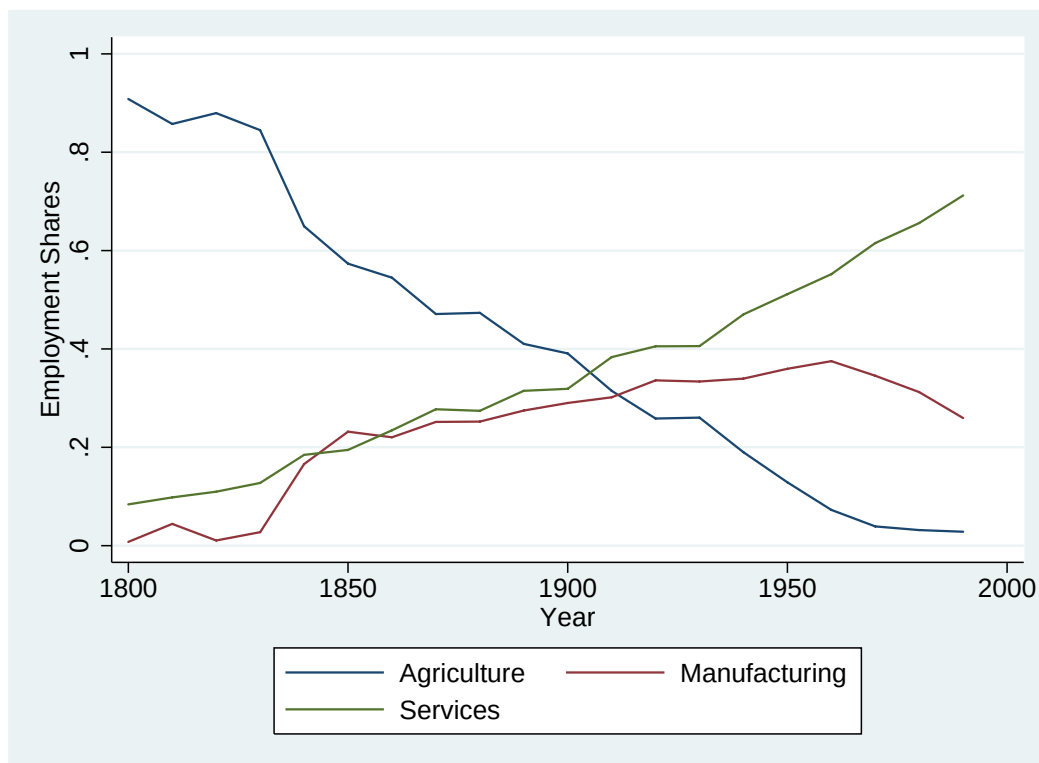


FIGURE 20.1. The share of US employment in agriculture, manufacturing and services, 1800-2000.

Figure 20.1 paints a picture of changes in sectoral employment that includes a significant *non-balanced* component. Consequently, models that depart from Kaldor facts over the early stages of the development process might be useful for understanding broader aspects of structural change. Kongsamut, Rebelo and Xie instead take a more modest departure from the baseline neoclassical growth model and propose a model that can account for a certain degree of non-balanced growth at the sectoral level, while still remaining consistent with the Kaldor facts of aggregate balanced growth. Even though it is designed to match the Kaldor facts regardless of the stage of development, its tractability makes this model a useful starting point for our analysis. Moreover, as emphasized by Kuznets, once we look beneath the aggregate facts of balanced growth, we see structural changes in the composition of employment and production, even in relatively advanced economies. A model consistent with the Kaldor facts provides us with the simplest approach to these types of sectoral changes that appear to be ongoing in most economies.

At the heart of Kongsamut, Rebelo and Xie's approach is the so-called *Engel's Law*, which states that as a household's income increases, the fraction that it spends on food (agricultural products) declines. While calling this observation a law may exaggerate its status, this observation, first made by the 19th-century German statistician Ernst Engel, appears to be a remarkably robust pattern in the data. Kongsamut, Rebelo and Xie extend Engel's Law, by also positing that as a household becomes richer, it will desire not only to spend less on food, but will also wish to spend more on services. In particular, consider the following infinite-horizon economy. Population grows at the exogenous rate $n \geq 0$, so that total labor supply is

$$(20.1) \quad L(t) = \exp(nt) L(0).$$

The economy admits a representative household who supplies labor inelastically and has standard preferences given by

$$(20.2) \quad U(0) \equiv \int_0^\infty \exp(-(\rho - n)t) \frac{c(t)^{1-\theta} - 1}{1-\theta} dt,$$

with $\theta \geq 0$ and $c(t)$ denoting the consumption aggregate produced out of agricultural, manufacturing and service goods. I use the lower case letter here to emphasize that this is consumption per capita. Aggregate consumption itself consists of agricultural, manufacturing and services consumptions, with an aggregator of the form:

$$(20.3) \quad c(t) = (c^A(t) - \gamma^A)^{\eta^A} c^M(t)^{\eta^M} (c^S(t) + \gamma^S)^{\eta^S},$$

where $c^A(t) \in [\gamma^A, \infty)$ denotes per capita agricultural consumption at time t , $c^M(t) \in \mathbb{R}_+$ is manufacturing consumption, and $c^S(t) \in \mathbb{R}_+$ is services consumption, while γ^A , γ^S , η^A , η^M and η^S are positive constants. This general functional form of the aggregator (preferences) is often referred to as the *Stone-Geary* preferences. It is a highly tractable way of introducing income elasticities that are different from one for different subcomponents of consumption, which will enable us to incorporate Engel's Law. In particular, this aggregator implies that there is a minimum or *subsistence* level of agricultural (food) consumption equal to γ^A . The household must consume at least this much food to survive and in fact, consumption and utility are not defined when the household does not consume the minimum amount of food (recall (negative number)^{1- θ} is undefined for $\theta > 0$). After this level of food consumption has been achieved, the household starts to demand other items, in particular, manufactured goods (e.g., textiles and durables) and services (e.g., health, entertainment, wholesale and retail). However, as we will see shortly, the presence of the γ^S term in the aggregator implies that the household will spend a positive amount on services only after certain levels of agricultural and manufacturing consumption have been reached.

Suppose that the economy is closed, thus agricultural, manufacturing and services consumption must be met by domestic production. I follow Kongsamut, Rebelo and Xie and assume the following production functions for the agricultural, manufacturing and service

goods:

$$\begin{aligned}
 (20.4) \quad Y^A(t) &= B^A F(K^A(t), X(t) L^A(t)), \\
 Y^M(t) &= B^M F(K^M(t), X(t) L^M(t)), \\
 Y^S(t) &= B^S F(K^S(t), X(t) L^S(t)),
 \end{aligned}$$

where $Y^j(t)$ for $j \in \{A, M, S\}$ denotes the output of agricultural, manufacturing and services at time t , $K^j(t)$ and $L^j(t)$ for $j \in \{A, M, S\}$ are the levels of capital and labor allocated to the agricultural, manufacturing and services sectors at time t , B^j for $j \in \{A, M, S\}$ is a Hicks-neutral productivity term for the three sectors and finally, $X(t)$ is a labor-augmenting (Harrod-neutral) productivity term affecting all sectors (I use the letter X instead of the standard A to distinguish this from the agricultural good). The function F satisfies the usual neoclassical assumptions, Assumptions 1 and 2, and thus in particular, exhibits constant returns to scale. Two other features in (20.4) are worth noting. First, the production function for all three sectors are identical. Second, the same labor-augmenting technology term affects all three sectors. Both of these features are clearly unrealistic, but they are useful to isolate the demand-side sources of structural change and to contrast them with the supply-side factors that will be discussed in the next section. Furthermore, Exercise 20.7 below show that they can be relaxed to some degree. Let us take the initial population, $L(0) > 0$, and the initial capital stock, $K(0) > 0$, as given, and also assume that there is a constant rate of growth of the labor-augmenting technology term, that is,

$$(20.5) \quad \frac{\dot{X}(t)}{X(t)} = g$$

for all t , with initial condition $X(0) > 0$. To ensure that the transversality condition of the representative household holds, I impose the same assumption as in the basic neoclassical growth model of Chapter 8, Assumption 4 (which, recall, implies that $\rho - n > (1 - \theta)g$).

Market clearing for labor and capital requires

$$(20.6) \quad K^A(t) + K^M(t) + K^S(t) = K(t),$$

and

$$(20.7) \quad L^A(t) + L^M(t) + L^S(t) = L(t),$$

where $K(t)$ and $L(t)$ are the total supplies of capital and labor at time t .

Another key assumption of the Kongsamut, Rebelo and Xie model builds on Rebelo (1991) and imposes that it is the manufacturing good that is used in the production of the investment good. Consequently, market clearing for the manufacturing good takes the form

$$(20.8) \quad \dot{K}(t) + c^M(t) L(t) = Y^M(t),$$

where, for simplicity, I ignore capital depreciation (otherwise there would be an additional term $\delta K(t)$ on the left-hand side). This equation states that the total production of manufacturing goods is distributed between consumption of manufacturing goods and new capital stock, which will be used for the production of agricultural, manufacturing and service goods

in the future. Since the economy admits a representative household, eq.'s (20.4)-(20.8) can also be taken to represent the representative household's budget constraint.

In addition, market clearing for the agricultural and service goods take the standard forms

$$(20.9) \quad c^A(t) L(t) = Y^A(t) \text{ and } c^S(t) L(t) = Y^S(t),$$

where the left-hand sides of both equations are multiplied by $L(t)$ to turn per capita consumption levels into total consumptions.

All markets are competitive. Let us choose the price of the manufacturing good at each date as the numeraire, which leaves us with the prices of agricultural goods, $p^A(t)$, and of services, $p^S(t)$, and factor prices $w(t)$ and $r(t)$. The consumption aggregator (20.3) immediately implies that the prices of agricultural and service goods must satisfy

$$(20.10) \quad \frac{p^A(t) (c^A(t) - \gamma^A)}{\eta^A} = \frac{c^M(t)}{\eta^M},$$

and

$$(20.11) \quad \frac{p^S(t) (c^S(t) + \gamma^S)}{\eta^S} = \frac{c^M(t)}{\eta^M}.$$

Competitive factor markets also imply

$$(20.12) \quad w(t) = \frac{\partial B^M F(K^M(t), X(t) L^M(t))}{\partial L^M},$$

and

$$(20.13) \quad r(t) = \frac{\partial B^M F(K^M(t), X(t) L^M(t))}{\partial K^M},$$

where I could have equivalently used the marginal products from other sectors, with identical results.

A competitive equilibrium is defined in the usual manner as processes of sectoral factor demands, $[K^A(t), K^M(t), K^S(t), L^A(t), L^M(t), L^S(t)]_{t=0}^\infty$, that maximize profits given the processes of the total supplies of capital and labor, $[K(t), L(t)]_{t=0}^\infty$, and the processes of prices, $[p^A(t), p^M(t), w(t), r(t)]_{t=0}^\infty$; prices, $[p^A(t), p^M(t), w(t), r(t)]_{t=0}^\infty$, that satisfy (20.10)-(20.13) given $[K^A(t), K^M(t), K^S(t), L^A(t), L^M(t), L^S(t)]_{t=0}^\infty$; and processes of consumption and capital, $[c^A(t), c^M(t), c^S(t), K(t)]_{t=0}^\infty$, that maximize (20.2) subject to (20.4)-(20.8); and a labor supply process, $[L(t)]_{t=0}^\infty$, that satisfies (20.1). In addition, suppose that

$$(20.14) \quad B^A F(K^A(0), X(0) L^A(0)) > \gamma^A L(0),$$

so that the economy starts with enough capital and technological know-how to produce more than the minimum necessary amount of agricultural consumption

An equilibrium is straightforward to characterize in this economy. Because the production functions of the all three sectors are identical, the following result obtains immediately:

PROPOSITION 20.1. *Suppose (20.14) holds. Then, in any equilibrium, the following conditions are satisfied:*

$$(20.15) \quad \frac{K^A(t)}{X(t)L^A(t)} = \frac{K^M(t)}{X(t)L^M(t)} = \frac{K^S(t)}{X(t)L^S(t)} = \frac{K(t)}{X(t)L(t)} \equiv k(t)$$

for all t , where the last equality defines $k(t)$ as the aggregate effective capital-labor ratio of the economy, and also

$$(20.16) \quad p^A(t) = \frac{B^M}{B^A}$$

and

$$(20.17) \quad p^S(t) = \frac{B^M}{B^S}$$

for all t .

PROOF. See Exercise 20.2. □

The results in this proposition are intuitive. First, the fact that the production functions are identical implies that the capital-labor ratios allocated to the three sectors must be equalized. Second, given (20.15), the equilibrium price relationships (20.16) and (20.17) follow from the fact that the marginal products of capital and labor have to be equalized in all three sectors.

Proposition 20.1 does not make use of the preference side. Next incorporating utility maximization on the side of the representative household, in particular, deriving the standard Euler equation for the representative household and then using eq.'s (20.10)-(20.11), we obtain the following additional equilibrium conditions:

PROPOSITION 20.2. *Suppose (20.14) holds. Then, in any equilibrium,*

$$(20.18) \quad \frac{\dot{c}^M(t)}{c^M(t)} = \frac{1}{\theta} (r(t) - \rho)$$

for all t and moreover, provided that Assumption 4 holds, the transversality condition of the representative household is satisfied. In addition,

$$(20.19) \quad \frac{B^M (c^A(t) - \gamma^A)}{B^A \eta^A} = \frac{c^M(t)}{\eta^M}$$

and

$$(20.20) \quad \frac{B^M (c^S(t) + \gamma^S)}{B^S \eta^S} = \frac{c^M(t)}{\eta^M},$$

for all t .

PROOF. See Exercise 20.3. □

In analogy to the standard models, we may want to define a *BGP* in this economy as an equilibrium path in which output and consumption of all three sectors grow at the same constant rate. The next proposition shows that such a BGP does not exist.

PROPOSITION 20.3. *Suppose that either $\gamma^A > 0$ and/or $\gamma^S > 0$. Then, a BGP does not exist.*

PROOF. See Exercise 20.4. □

This result is not surprising. Since the preferences of the representative household incorporate Engel's Law, the household would always like to change the composition of its consumption, and this will be reflected in a change in the composition of production. Instead of a BGP, let us define a weaker notion of "balanced growth," which I will refer to as a *constant growth path* (CGP). A CGP requires that the rate of growth of aggregate consumption is asymptotically constant.¹ Given the preferences in (20.2), the constant growth rate of consumption implies that the interest rate must also be constant asymptotically. In a CGP, output, consumption and employment in the three sectors may grow at different rates.

PROPOSITION 20.4. *Suppose (20.14) holds. Then, in the above-described economy a CGP exists if and only if*

$$(20.21) \quad \frac{\gamma^A}{B^A} = \frac{\gamma^S}{B^S}.$$

In a CGP $k(t) = k^$ for all t , and moreover, consumption and employment in the three sectors evolve according to*

$$(20.22) \quad \frac{\dot{c}^A(t)}{c^A(t)} = g \frac{c^A(t) - \gamma^A}{c^A(t)}, \quad \frac{\dot{c}^M(t)}{c^M(t)} = g, \quad \text{and} \quad \frac{\dot{c}^S(t)}{c^S(t)} = g \frac{c^S(t) + \gamma^S}{c^S(t)},$$

and

$$\frac{\dot{L}^A(t)}{L^A(t)} = n - g \frac{\gamma^A/L^A(t)}{B^A X(t) F(k^*, 1)}, \quad \frac{\dot{L}^M(t)}{L^M(t)} = n, \quad \text{and} \quad \frac{\dot{L}^S(t)}{L^S(t)} = n + g \frac{\gamma^S/L^S(t)}{B^S X(t) F(k^*, 1)}$$

for all t .

Moreover, in the CGP the share of national income accruing to capital is constant.

PROOF. See Exercise 20.5. □

This model therefore delivers a tractable framework for the analysis of structural change that has potential relevance both for the experience of economies at the early stages of development and also for understanding the patterns of growth of relatively advanced countries. Engel's Law (augmented with the highly income elastic demand for services) generates a demand-side force towards non-balanced growth. In particular, as their incomes grow, consumers wish to spend a greater fraction of their budget on services and a smaller fraction on food (agricultural goods). This makes an equilibrium with fully balanced growth impossible. Instead, different sectors grow at different rates and there is reallocation of labor and capital across sectors. Nevertheless, Proposition 20.4 shows that under condition (20.21) a CGP exists and in this equilibrium, structural change takes place despite the fact that the interest rate and the share of capital in national income are constant. This model therefore delivers

¹Kongsamut, Rebelo and Xie, instead, define the concept of generalized balanced growth path, where the interest rate is constant. Clearly, given the CRRA preferences in (20.2), the two notions are equivalent.

many of the features that are useful for thinking of the long haul of the process of development; in particular, the equilibrium path can be consistent with the Kaldor facts, and there is a continuous process of structural change, whereby the share of agriculture in production and employment declines over the development process and the share of services increases.

On the downside, a number of potential shortcomings of the current model are worth noting. First, one may argue that the process of structural change in this model falls short of the sweeping changes discussed by Kuznets. Although I focused on the CGP, it is straightforward to incorporate transitional dynamics into the model. Exercise 20.6 shows that if the effective capital-labor ratio starts out below its CGP value of k^* in Proposition 20.4, then there will be additional transitional dynamics in this model complementing the structural changes. Nevertheless, even these transitional dynamics probably fall short of the sweeping structural changes emphasized by Kuznets. To some degree, whether or not this is so is a matter of taste and emphasis. The current model also does not incorporate the various different aspects of more fundamental structural transformations that will be discussed in the next chapter, though it was also not meant to incorporate these transformations.

Second, the assumption that all three sectors have the same production function appears restrictive. Nevertheless, this assumption can be relaxed to some degree. Exercise 20.7 discusses how this can be done. Perhaps more important is the assumption that investments for all three sectors use only the manufacturing good. This assumption is similar in nature to the assumption that only capital is used to produce capital (investment) goods in Rebelo's (1991) model (recall Chapter 11). Exercise 20.10 shows that if this assumption is relaxed, it is no longer possible to reconcile the Kuznets and the Kaldor facts in the context of this model.

Third, the model presented here is designed to generate a constant share of employment in manufacturing. Although this pattern is broadly consistent with the US experience over the past 150 years, when we look at even earlier stages of development, almost all employment is in agriculture. This implies that early stages of structural change must also involve an increase in the share of employment in manufacturing. A number of models in the literature generate this pattern by also introducing land as an additional factor of production. Exercise 20.8 provides an example and Section 21.2 will present a model incorporating land as a major factor of production in the context of the study of population dynamics.

Finally and most importantly, the condition necessary for a CGP, (20.21), is a rather "knife-edge" condition. Why should this specific equality between technology and preference parameters hold? In the final analysis, there is no compelling argument that this condition should be satisfied. Nevertheless, even when it fails, it can be argued that the behavior of the model may approximate the structural changes we observe in practice and Exercise 20.9 illustrates this with an example in which sectoral production functions differ, but are all of the Cobb-Douglas form.

20.2. Non-Balanced Growth: The Supply Side

The previous section showed how the process of structural change can be driven by a generalized form of Engel's Law, that is, by the desires of the consumers to change the composition of their consumption as they become richer. An alternative approach to why growth may be non-balanced was first proposed by Baumol's (1967) seminal work. Baumol suggested that "uneven growth" (or what I am referring to here as non-balanced growth) will be a general feature of the growth process because different sectors will grow at different rates owing to different rates of technological progress (for example, technological progress might be faster in manufacturing than in agriculture or services). Although Baumol's original article derived this result only under a variety of special assumptions, the general insight that there might be supply-side forces pushing the economy towards non-balanced growth is considerably more general. Here I review some ideas based on Acemoglu and Guerrieri (2006), who emphasize the supply-side causes of non-balanced growth. Ultimately, both the rich patterns of structural change during the early stages of development and those we witness in more advanced economies today require models that combine supply-side and demand-side factors. Nevertheless, isolating these factors in separate models is both more tractable and also conceptually more transparent. For this reason, in this section I focus on the supply side, abstracting from Engel's Law throughout, and will only return to the combination of the supply-side and the demand-side factors in Exercise 20.17.

20.2.1. General Insights. At some level, Baumol's theory of non-balanced growth can be viewed as self-evident—if some sectors have higher rates of technological progress, there must be some non-balanced elements in equilibrium. My first purpose in this section is to show that there are more subtle and compelling reasons for supply-side non-balanced growth than those originally emphasized by Baumol. In particular, most growth models, like the Kongsamut, Rebelo and Xie model presented in the previous section, assume that production functions in different sectors are identical. In practice, however, industries differ considerably in terms of their capital intensity and also in terms of the intensity with which they use other factors (for example, compare the retail sector to durables manufacturing or transport). In short, different industries have different *factor proportions*. The main economic point I would like to emphasize in this section is that factor proportion differences across sectors combined with *capital deepening* will lead to non-balanced economic growth.

I will illustrate this point first using a simple but fairly general environment. This environment consists of two sectors each with a constant returns to scale production function and arbitrary preferences over the goods that are produced in these two sectors. Both sectors employ capital, K , and labor, L . To highlight that the exact nature of the accumulation process is not essential for the results, I take the process of capital and labor supplies, $[K(t), L(t)]_{t=0}^{\infty}$, as given and assume that labor is supplied inelastically.

Preferences are defined over the final output or a consumption aggregator as in (20.3) in the previous section. Whether we use the specification with a consumption aggregator or a

formulation with intermediates used competitively in the production of a final good makes no difference for any of the results. With this in mind, let final output be denoted by Y and assume that it is produced as an aggregate of the output of two sectors, Y_1 and Y_2 ,

$$Y(t) = F(Y_1(t), Y_2(t)).$$

Let us also assume that F satisfies Assumptions 1 and 2, so that, in particular, it exhibits constant returns to scale and is twice differentiable. Sectoral production functions are given by

$$(20.23) \quad Y_1(t) = A_1(t) G_1(K_1(t), L_1(t))$$

and

$$(20.24) \quad Y_2(t) = A_2(t) G_2(K_2(t), L_2(t)),$$

where $L_1(t)$, $L_2(t)$, $K_1(t)$ and $K_2(t)$ denote the amount of labor and capital employed in the two sectors, and the functions G_1 and G_2 are also assumed to satisfy the equivalents of Assumptions 1 and 2. The terms $A_1(t)$ and $A_2(t)$ are Hicks-neutral technology terms.

Market clearing for capital and labor implies that

$$(20.25) \quad \begin{aligned} K_1(t) + K_2(t) &= K(t), \\ L_1(t) + L_2(t) &= L(t), \end{aligned}$$

at each t . Without loss of any generality, I ignore capital depreciation.

Let us take the final good as the numeraire in every period and denote the prices of Y_1 and Y_2 by p_1 and p_2 , and wage and rental rate of capital (interest rate) by w and r . Product and factor markets are competitive, so that product and factor prices satisfy

$$(20.26) \quad \frac{p_1(t)}{p_2(t)} = \frac{\partial F(Y_1(t), Y_2(t)) / \partial Y_1}{\partial F(Y_1(t), Y_2(t)) / \partial Y_2}$$

and

$$(20.27) \quad \begin{aligned} w(t) &= p_1(t) \frac{\partial A_1(t) G_1(K_1(t), L_1(t))}{\partial L_1} = p_2(t) \frac{\partial A_2(t) G_2(K_2(t), L_2(t))}{\partial L_2} \\ r(t) &= p_1(t) \frac{\partial A_1 G_1(K_1(t), L_1(t))}{\partial K_1} = p_2(t) \frac{\partial A_2 G_2(K_2(t), L_2(t))}{\partial K_2}. \end{aligned}$$

An equilibrium, given factor supply processes, $[K(t), L(t)]_{t=0}^{\infty}$, is a process of product and factor prices, $[p_1(t), p_2(t), w(t), r(t)]_{t=0}^{\infty}$ and factor allocations, $[K_1(t), K_2(t), L_1(t), L_2(t)]_{t=0}^{\infty}$, such that (20.25), (20.26) and (20.27) are satisfied.

Let the shares of capital in the two sectors be defined as

$$(20.28) \quad \sigma_1(t) \equiv \frac{r(t) K_1(t)}{p_1(t) Y_1(t)} \text{ and } \sigma_2(t) \equiv \frac{r(t) K_2(t)}{p_2(t) Y_2(t)}.$$

There is *capital deepening* at time t if $\dot{K}(t)/K(t) > \dot{L}(t)/L(t)$. There are *factor proportion differences* at time t if $\sigma_1(t) \neq \sigma_2(t)$. And finally, technological progress is *balanced* at time t if $\dot{A}_1(t)/A_1(t) = \dot{A}_2(t)/A_2(t)$. Notice that factor proportion differences, that is, $\sigma_1(t) \neq \sigma_2(t)$, refers to the equilibrium factor proportions in the two sectors at time t . It does not necessarily mean that these will not be equal at some future date. The following

proposition shows the supply side forces leading to structural change in the simplest possible way:

PROPOSITION 20.5. *Suppose that at time t , there are factor proportion differences between the two sectors, technological progress is balanced, and there is capital deepening, then growth is not balanced, that is, $\dot{Y}_1(t)/Y_1(t) \neq \dot{Y}_2(t)/Y_2(t)$.*

PROOF. First define the capital to labor ratio in the two sectors as

$$k_1(t) \equiv \frac{K_1(t)}{L_1(t)} \text{ and } k_2(t) \equiv \frac{K_2(t)}{L_2(t)},$$

and the “per capita production functions” (without the Hicks-neutral technology terms) as

$$(20.29) \quad g_1(k_1(t)) \equiv \frac{G_1(K_1(t), L_1(t))}{L_1(t)} \text{ and } g_2(k_2(t)) \equiv \frac{G_2(K_2(t), L_2(t))}{L_2(t)}.$$

Since G_1 and G_2 are twice differentiable by assumption, so are g_1 and g_2 and denote their first and second derivatives by g'_1 , g'_2 , g''_1 and g''_2 .

Now, differentiating the production functions for the two sectors,

$$\frac{\dot{Y}_1(t)}{Y_1(t)} = \frac{\dot{A}_1(t)}{A_1(t)} + \sigma_1(t) \frac{\dot{K}_1(t)}{K_1(t)} + (1 - \sigma_1(t)) \frac{\dot{L}_1(t)}{L_1(t)}$$

and

$$\frac{\dot{Y}_2(t)}{Y_2(t)} = \frac{\dot{A}_2(t)}{A_2(t)} + \sigma_2(t) \frac{\dot{K}_2(t)}{K_2(t)} + (1 - \sigma_2(t)) \frac{\dot{L}_2(t)}{L_2(t)}.$$

To simplify the notation, I drop the time arguments for the remainder of this proof.

Suppose, to obtain a contradiction, that $\dot{Y}_1/Y_1 = \dot{Y}_2/Y_2$. Since F exhibits constant returns to scale, $\dot{Y}_1/Y_1 = \dot{Y}_2/Y_2$ together with (20.26) implies

$$(20.30) \quad \frac{\dot{p}_1}{p_1} = \frac{\dot{p}_2}{p_2} = 0.$$

Given the definition in (20.29), eq. (20.27) gives the following conditions characterizing the equilibrium interest rate and wage:

$$(20.31) \quad \begin{aligned} r &= p_1 A_1 g'_1(k_1) \\ &= p_2 A_2 g'_2(k_2), \end{aligned}$$

and

$$(20.32) \quad \begin{aligned} w &= p_1 A_1 (g_1(k_1) - g'_1(k_1) k_1) \\ &= p_2 A_2 (g_2(k_2) - g'_2(k_2) k_2). \end{aligned}$$

Differentiating the interest rate condition, (20.31), with respect to time and using (20.30):

$$\frac{\dot{A}_1}{A_1} + \varepsilon_{g'_1} \frac{\dot{k}_1}{k_1} = \frac{\dot{A}_2}{A_2} + \varepsilon_{g'_2} \frac{\dot{k}_2}{k_2}$$

where

$$\varepsilon_{g'_1} \equiv \frac{g''_1(k_1) k_1}{g'_1(k_1)} \text{ and } \varepsilon_{g'_2} \equiv \frac{g''_2(k_2) k_2}{g'_2(k_2)}.$$

Since $\dot{A}_1/A_1 = \dot{A}_2/A_2$,

$$(20.33) \quad \varepsilon_{g'_1} \frac{\dot{k}_1}{k_1} = \varepsilon_{g'_2} \frac{\dot{k}_2}{k_2}.$$

Differentiating the wage condition, (20.32), with respect to time, using (20.30) and some algebra gives:

$$\frac{\dot{A}_1}{A_1} - \frac{\sigma_1}{1 - \sigma_1} \varepsilon_{g'_1} \frac{\dot{k}_1}{k_1} = \frac{\dot{A}_2}{A_2} - \frac{\sigma_2}{1 - \sigma_2} \varepsilon_{g'_2} \frac{\dot{k}_2}{k_2}.$$

Since $\dot{A}_1/A_1 = \dot{A}_2/A_2$ and $\sigma_1 \neq \sigma_2$, this equation is inconsistent with (20.33), yielding a contradiction and proving the claim. $\square$

The intuition for this result is straightforward. Suppose that there is capital deepening and that, for concreteness, sector 2 is more capital-intensive ($\sigma_1 < \sigma_2$). Now, if both capital and labor were allocated to the two sectors at constant proportions over time, the more capital-intensive sector, sector 2, would grow faster than sector 1. In equilibrium, the faster growth in sector 2 would change equilibrium prices, and the decline in the relative price of sector 2 would cause some of the labor and capital to be reallocated to sector 1. However, this reallocation *could not* entirely offset the greater increase in the output of sector 2, since, if it did, the relative price change that stimulated the reallocation in the first place would not occur. Consequently, equilibrium growth must be non-balanced.

Proposition 20.5 is related to the well-known Rybczynski's Theorem in international trade. Rybczynski's Theorem states that in an open economy within the "cone of diversification" (where factor prices do not depend on factor endowments), changes in factor endowments will be absorbed by changes in the sectoral output mix. Proposition 20.5 can be viewed both as a closed-economy analog and also as a generalization of Rybczynski's Theorem; it shows that changes in factor endowments (capital deepening) will be absorbed by faster growth in one sector than the other, even though relative prices of goods and factors will change in response to the change in factor endowments.

It is also straightforward to generalize Proposition 20.5 to an economy with $N \geq 2$ sectors. In particular, suppose that aggregate output is given by the constant returns to scale production function

$$Y = F(Y_1(t), Y_2(t), \dots, Y_N(t)).$$

Defining $\sigma_j(t)$ as the capital share in sector $j = 1, \dots, N$ as in (20.28):

PROPOSITION 20.6. *Suppose that at time t , there are factor proportion differences among the N sectors in the sense that there exists i and $j \leq N$ such that $\sigma_i(t) \neq \sigma_j(t)$, technological progress is balanced between i and j , that is, $\dot{A}_i(t)/A_i(t) = \dot{A}_j(t)/A_j(t)$, and there is capital deepening, that is, $\dot{K}(t)/K(t) > \dot{L}(t)/L(t)$, then growth is not balanced and $\dot{Y}_i(t)/Y_i(t) \neq \dot{Y}_j(t)/Y_j(t)$.*

PROOF. See Exercise 20.11. $\square$

20.2.2. Balanced Growth and Kuznets Facts. The previous subsection provided general insights about how supply-side factors can lead to non-balanced growth. To obtain a general result on the implications of capital deepening and factor proportion differences across sectors on non-balanced growth, Proposition 20.5 was stated for a given (arbitrary) process of capital and labor supplies, $[K(t), L(t)]_{t=0}^{\infty}$. However, without endogenizing the path of capital accumulation (and specifying the pattern of population growth) we cannot address whether a model relying on supply-side factors can also provide a useful framework for thinking about the Kuznets facts without significantly deviating from the balanced growth patterns exhibited by many relatively developed economies.

For this purpose, I now specialize the environment of the previous subsection by incorporating specific preferences and production functions and then provide a full characterization of a simpler economy. The economy is again in infinite horizon and population grows at the exogenous rate $n > 0$ according to (20.1). Let us also assume that the economy admits a representative household, with standard preferences given by (20.2), who also supplies labor inelastically. Proposition 20.5 emphasized the importance of capital deepening, which will now result from exogenous technological progress.

Instead of a general production function for the final good as in the previous subsection, I now assume that the unique final good is produced with a CES aggregator:

$$(20.34) \quad Y(t) = \left[\gamma Y_1(t)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\gamma) Y_2(t)^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}},$$

where $\varepsilon \in [0, \infty)$ is the elasticity of substitution between the two intermediates and $\gamma \in (0, 1)$ determines the relative importance of the two goods in aggregate production. Let us ignore capital depreciation again and also assume that the final good is distributed between consumption and investment,

$$(20.35) \quad \dot{K}(t) + L(t)c(t) \leq Y(t),$$

where $c(t)$ is consumption per capita.

The two intermediates Y_1 and Y_2 are produced competitively with aggregate production functions

$$(20.36) \quad Y_1(t) = A_1(t) K_1(t)^{\alpha_1} L_1(t)^{1-\alpha_1} \text{ and } Y_2(t) = A_2(t) K_2(t)^{\alpha_2} L_2(t)^{1-\alpha_2}.$$

Throughout, I impose that

$$(20.37) \quad \alpha_1 < \alpha_2,$$

which implies that sector 1 is less capital-intensive than sector 2. This is without loss of any generality, since in the case in which $\alpha_1 = \alpha_2$, there are no supply-side effects and thus the issues I am concerned with in this section do not arise.

In (20.36), A_1 and A_2 correspond to Hicks-neutral technology terms that grow at exogenous and potentially different rates given by

$$(20.38) \quad \frac{\dot{A}_1(t)}{A_1(t)} = a_1 > 0 \text{ and } \frac{\dot{A}_2(t)}{A_2(t)} = a_2 > 0.$$

Labor and capital market clearing again require that at each t ,

$$(20.39) \quad L_1(t) + L_2(t) = L(t),$$

and

$$(20.40) \quad K_1(t) + K_2(t) = K(t).$$

Let us also denote the wage and the interest rate (the rental rate of capital) by $w(t)$ and $r(t)$ and the prices of the two intermediate goods by $p_1(t)$ and $p_2(t)$. We again normalize the price of the final good to 1 at each instant. An equilibrium is defined in the usual manner, as processes of labor and capital allocations and prices, such that $[K_1(t), K_2(t), L_1(t), L_2(t)]_{t=0}^{\infty}$ maximize intermediate sector profits given the prices, $[w(t), r(t), p_1(t), p_2(t)]_{t=0}^{\infty}$, and the aggregate capital and labor supplies, $[K(t), L(t)]_{t=0}^{\infty}$; intermediate and factor markets clear at the prices $[w(t), r(t), p_1(t), p_2(t)]_{t=0}^{\infty}$; $[K(t), c(t)]_{t=0}^{\infty}$ maximize utility of the representative household given the prices $[w(t), r(t), p_1(t), p_2(t)]_{t=0}^{\infty}$; and population evolves according to (20.1).

It is useful to break the characterization of equilibrium into two pieces: *static* and *dynamic*. The static part takes the state variables of the economy, which are the capital stock, the labor supply and the technology, K, L, A_1 and A_2 , as given and determines the allocation of capital and labor across sectors and the equilibrium factor and intermediate prices. The dynamic part of the equilibrium determines the evolution of the endogenous state variable, K (the dynamic behavior of L is given by (20.1) and those of A_1 and A_2 by (20.38)).

The choice of numeraire implies that at each instant

$$1 = \left[\gamma^\varepsilon p_1(t)^{1-\varepsilon} + (1-\gamma)^\varepsilon p_2(t)^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}},$$

and profit maximization of the final good sector implies

$$(20.41) \quad p_1(t) = \gamma \left(\frac{Y_1(t)}{Y(t)} \right)^{-\frac{1}{\varepsilon}} \quad \text{and} \quad p_2(t) = (1-\gamma) \left(\frac{Y_2(t)}{Y(t)} \right)^{-\frac{1}{\varepsilon}}.$$

Given this specification (and the fact that capital does not depreciate), the equilibrium allocation of resources will equate the marginal product of capital and labor into two sectors. The following equations give these equilibrium conditions and also provide expressions for the factor prices (see Exercise 20.12). The equilibrium conditions are

$$(20.42) \quad \gamma(1-\alpha_1) \left(\frac{Y(t)}{Y_1(t)} \right)^{\frac{1}{\varepsilon}} \frac{Y_1(t)}{L_1(t)} = (1-\gamma)(1-\alpha_2) \left(\frac{Y(t)}{Y_2(t)} \right)^{\frac{1}{\varepsilon}} \frac{Y_2(t)}{L_2(t)},$$

and

$$(20.43) \quad \gamma\alpha_1 \left(\frac{Y(t)}{Y_1(t)} \right)^{\frac{1}{\varepsilon}} \frac{Y_1(t)}{K_1(t)} = (1-\gamma)\alpha_2 \left(\frac{Y(t)}{Y_2(t)} \right)^{\frac{1}{\varepsilon}} \frac{Y_2(t)}{K_2(t)},$$

while the factor prices can be expressed as

$$(20.44) \quad w(t) = \gamma(1-\alpha_1) \left(\frac{Y(t)}{Y_1(t)} \right)^{\frac{1}{\varepsilon}} \frac{Y_1(t)}{L_1(t)},$$

and

$$(20.45) \quad r(t) = \gamma \alpha_1 \left(\frac{Y(t)}{Y_1(t)} \right)^{\frac{1}{\varepsilon}} \frac{Y_1(t)}{K_1(t)}.$$

The key to the characterization of the static equilibrium is to determine the fraction of capital and labor employed in the two sectors. Let us define $\kappa(t) \equiv K_1(t)/K(t)$ and $\lambda(t) \equiv L_1(t)/L(t)$. Combining eq.'s (20.39), (20.40), (20.42), and (20.43):

$$(20.46) \quad \kappa(t) = \left[1 + \frac{\alpha_2}{\alpha_1} \left(\frac{1-\gamma}{\gamma} \right) \left(\frac{Y_1(t)}{Y_2(t)} \right)^{\frac{1-\varepsilon}{\varepsilon}} \right]^{-1},$$

and

$$(20.47) \quad \lambda(t) = \left[1 + \frac{\alpha_1}{\alpha_2} \left(\frac{1-\alpha_2}{1-\alpha_1} \right) \left(\frac{1-\kappa(t)}{\kappa(t)} \right) \right]^{-1}.$$

Equation (20.47) makes it clear that the share of labor in sector 1, λ , is monotonically increasing in the share of capital in sector 1, κ . This implies that in equilibrium capital and labor will be reallocated towards the same sector. The structure of the static equilibrium depends on how the allocation of capital and labor depends on the aggregate amount of capital and labor available in the economy. The following proposition answers this question.

PROPOSITION 20.7. *In equilibrium,*

$$(20.48) \quad \frac{d \ln \kappa(t)}{d \ln K(t)} = - \frac{d \ln \kappa(t)}{d \ln L(t)} = \frac{(1-\varepsilon)(\alpha_2 - \alpha_1)(1 - \kappa(t))}{1 + (1-\varepsilon)(\alpha_2 - \alpha_1)(\kappa(t) - \lambda(t))} > 0 \text{ if and only if } (\alpha_2 - \alpha_1)(1 - \varepsilon) > 0.$$

$$(20.49) \quad \frac{d \ln \kappa(t)}{d \ln A_2(t)} = - \frac{d \ln \kappa(t)}{d \ln A_1(t)} = \frac{(1-\varepsilon)(1 - \kappa(t))}{1 + (1-\varepsilon)(\alpha_2 - \alpha_1)(\kappa(t) - \lambda(t))} > 0 \text{ if and only if } \varepsilon < 1.$$

PROOF. See Exercise 20.13. □

Equation (20.48) states that when the elasticity of substitution between sectors, ε , is less than 1, the fraction of capital allocated to the capital-intensive sector declines in the stock of capital (and conversely, when $\varepsilon > 1$, this fraction is increasing in the stock of capital). Intuitively, if K increases and κ remains constant, then the capital-intensive sector, sector 2, will grow by *more* than sector 1. Equilibrium prices given in (20.41) then imply that when $\varepsilon < 1$, the relative price of the capital-intensive sector will fall more than proportionately, inducing a greater fraction of capital to be allocated to the less capital-intensive sector 1. The intuition for the converse result when $\varepsilon > 1$ is similar.

Moreover, eq. (20.49) implies that when the elasticity of substitution, ε , is less than one, an improvement in the technology of a sector causes the share of capital going to that sector to fall. The intuition is again the same: when $\varepsilon < 1$, increased production in a sector causes a more than proportional decline in its relative price, inducing a reallocation of capital away from it towards the other sector (again the converse results and intuition apply when $\varepsilon > 1$).

Combining (20.44) and (20.45), we also obtain relative factor prices as

$$(20.50) \quad \frac{w(t)}{r(t)} = \frac{1 - \alpha_1}{\alpha_1} \left(\frac{\kappa(t) K(t)}{\lambda(t) L(t)} \right),$$

and the capital share in the economy as:

$$(20.51) \quad \sigma_K(t) \equiv \frac{r(t) K(t)}{Y(t)} = \gamma \alpha_1 \left(\frac{Y_1(t)}{Y(t)} \right)^{\frac{\varepsilon-1}{\varepsilon}} \kappa(t)^{-1}.$$

PROPOSITION 20.8. *In equilibrium,*

$$(20.52) \quad \frac{d \ln(w(t)/r(t))}{d \ln K(t)} = - \frac{d \ln(w(t)/r(t))}{d \ln L(t)} = \frac{1}{1 + (1 - \varepsilon)(\alpha_2 - \alpha_1)(\kappa(t) - \lambda(t))} > 0.$$

$$(20.53) \quad \frac{d \ln(w(t)/r(t))}{d \ln A_2(t)} = - \frac{d \ln(w(t)/r(t))}{d \ln A_1(t)} = - \frac{(1 - \varepsilon)(\kappa(t) - \lambda(t))}{1 + (1 - \varepsilon)(\alpha_2 - \alpha_1)(\kappa(t) - \lambda(t))} < 0$$

if and only if $(\alpha_2 - \alpha_1)(1 - \varepsilon) > 0$.

$$(20.54) \quad \frac{d \ln \sigma_K(t)}{d \ln K(t)} < 0 \text{ if and only if } \varepsilon < 1.$$

$$(20.55) \quad \frac{d \ln \sigma_K(t)}{d \ln A_2(t)} = - \frac{d \ln \sigma_K(t)}{d \ln A_1(t)} < 0 \text{ if and only if } (\alpha_2 - \alpha_1)(1 - \varepsilon) > 0.$$

PROOF. The results in (20.52) and (20.53) follow from differentiating eq. (20.50) and Proposition 20.7. To prove the remaining claims, let me suppress time arguments and write:

$$\begin{aligned} \left(\frac{Y_1}{Y} \right)^{\frac{\varepsilon-1}{\varepsilon}} &= \left[\gamma + (1 - \gamma) \left(\frac{Y_1}{Y_2} \right)^{\frac{1-\varepsilon}{\varepsilon}} \right]^{-1} \\ &= \gamma^{-1} \left(1 + \frac{\alpha_1}{\alpha_2} \left(\frac{1}{\kappa} - 1 \right) \right)^{-1} \end{aligned}$$

Then, using the results of Proposition 20.7 and the definition of σ_K from (20.51):

$$(20.56) \quad \frac{d \ln \sigma_K}{d \ln K} = -\Omega \frac{1 - \sigma_K}{\sigma_K} \frac{\alpha_1}{\alpha_2} \frac{(1 - \varepsilon)(\alpha_2 - \alpha_1)(1 - \kappa)/\kappa}{1 + (1 - \varepsilon)(\alpha_2 - \alpha_1)(\kappa - \lambda)}$$

and

$$(20.57) \quad \frac{d \ln \sigma_K}{d \ln A_2} = - \frac{d \ln \sigma_K}{d \ln A_1} = \Omega \frac{1 - \sigma_K}{\sigma_K} \frac{\alpha_1}{\alpha_2} \frac{(1 - \varepsilon)(1 - \kappa)/\kappa}{1 + (1 - \varepsilon)(\alpha_2 - \alpha_1)(\kappa - \lambda)},$$

where

$$\Omega \equiv \left[\left(1 + \frac{\alpha_1}{\alpha_2} \left(\frac{1}{\kappa} - 1 \right) \right)^{-1} - \left(\frac{1 - \alpha_1}{1 - \alpha_2} + \frac{\alpha_1}{\alpha_2} \left(\frac{1}{\kappa} - 1 \right) \right)^{-1} \right].$$

Clearly, $\Omega > 0$ if and only if $\alpha_1 < \alpha_2$, which is satisfied in view of (20.37). Equations (20.56) and (20.57) then imply (20.54) and (20.55). $\square$

The most important result in this proposition is (20.54), which links the equilibrium relationship between the capital share in national income and the capital stock to the elasticity of substitution. Since a negative relationship between the share of capital in national income and the capital stock is equivalent to capital and labor being gross complements in the aggregate, this result also implies that the elasticity of substitution between capital and labor is less than one if and only if ε is less than one. Recall from the discussion in Section 15.6 in Chapter 15 that a variety of different approaches suggest that the elasticity of substitution between capital and labor is less than one.

The intuition for Proposition 20.8 is informative about the workings of the model. Consistent with the discussion of Proposition 20.5 above, when $\varepsilon < 1$, an increase in the capital stock of the economy causes the output of the more capital-intensive sector, sector 2, to increase relative to the output in the less capital-intensive sector (despite the fact that the share of capital allocated to the less-capital intensive sector increases as shown in eq. (20.48)). This then increases the production of the more capital-intensive sector and reduces the relative reward to capital (and the share of capital in national income). The converse result applies when $\varepsilon > 1$.

Recall also from Section 15.2 in Chapter 15 that when $\varepsilon < 1$, (20.55) in Proposition 20.8 implies that an increase in A_1 is “capital biased” and an increase in A_2 is “labor biased”. The intuition for why an increase in the productivity of the sector that is intensive in capital is biased toward labor (and vice versa) is again similar: when the elasticity of substitution between the two sectors, ε , is less than one, an increase in the output of a sector (this time driven by a change in technology) decreases its price more than proportionately, thus reducing the relative compensation of the factor used more intensively in that sector. When $\varepsilon > 1$, the converse pattern applies, and an increase in A_2 is “capital biased,” while an increase in A_1 is “labor biased”

I next turn to the characterization of the dynamic equilibrium path of this economy. The Euler equation for consumers follows from the maximization of (20.2) and takes the familiar form,

$$(20.58) \quad \frac{\dot{c}(t)}{c(t)} = \frac{1}{\theta}(r(t) - \rho).$$

Since the only asset of the representative household in this economy is capital, the transversality condition takes the standard form:

$$(20.59) \quad \lim_{t \rightarrow \infty} K(t) \exp \left(- \int_0^t r(\tau) d\tau \right) = 0,$$

which, together with the Euler equation (20.58) and the resource constraint (20.35), determines the dynamic behavior of consumption per capita and capital stock, c and K . Equations (20.1) and (20.38) give the behavior of L , A_1 and A_2 .

A dynamic equilibrium is given by paths of wages, interest rates, labor and capital allocation decisions, w , r , λ and κ , satisfying (20.44), (20.42), (20.45), (20.43), (20.46) and (20.47),

and of consumption per capita, c , capital stock, K , employment, L , and technology, A_1 and A_2 , satisfying (20.1), (20.35), (20.38), (20.58), and (20.59).

Let us also introduce the following notation for growth rates of the key objects in this economy:

$$\frac{\dot{L}_s(t)}{L_s(t)} \equiv n_s(t), \frac{\dot{K}_s(t)}{K_s(t)} \equiv z_s(t), \frac{\dot{Y}_s(t)}{Y_s(t)} \equiv g_s(t) \text{ for } s = 1, 2 \text{ and } \frac{\dot{K}(t)}{K(t)} \equiv z(t), \frac{\dot{Y}(t)}{Y(t)} \equiv g(t),$$

Whenever they exist, we can also define the corresponding (limiting) asymptotic growth rates as follows:

$$n_s^* = \lim_{t \rightarrow \infty} n_s(t), \quad z_s^* = \lim_{t \rightarrow \infty} z_s(t) \text{ and } g_s^* = \lim_{t \rightarrow \infty} g_s(t),$$

for $s = 1, 2$. Similarly denote the asymptotic capital and labor allocation decisions by “ $*$ ” so that

$$\kappa^* = \lim_{t \rightarrow \infty} \kappa(t) \text{ and } \lambda^* = \lim_{t \rightarrow \infty} \lambda(t).$$

With this terminology, the following useful proposition can be established.

- PROPOSITION 20.9. (1) If $\varepsilon < 1$, then $n_1(t) \gtrless n_2(t) \Leftrightarrow z_1(t) \gtrless z_2(t) \Leftrightarrow g_1(t) \gtrless g_2(t)$.
 (2) If $\varepsilon > 1$, then $n_1(t) \gtrless n_2(t) \Leftrightarrow z_1(t) \gtrless z_2(t) \Leftrightarrow g_1(t) \gtrless g_2(t)$.

PROOF. Omitting time arguments and differentiating (20.42) with respect to time,

$$(20.60) \quad \frac{1}{\varepsilon}g + \frac{\varepsilon - 1}{\varepsilon}g_1 - n_1 = \frac{1}{\varepsilon}g + \frac{\varepsilon - 1}{\varepsilon}g_2 - n_2,$$

which implies that $n_1 - n_2 = (\varepsilon - 1)(g_1 - g_2)/\varepsilon$ and establishes the first part of the proposition. Similarly differentiating (20.43) yields

$$(20.61) \quad \frac{1}{\varepsilon}g + \frac{\varepsilon - 1}{\varepsilon}g_1 - z_1 = \frac{1}{\varepsilon}g + \frac{\varepsilon - 1}{\varepsilon}g_2 - z_2$$

and establishes the second part of the result. $\square$

This proposition establishes the straightforward, but at first counter-intuitive, result that, when the elasticity of substitution between the two sectors is less than one, the equilibrium growth rate of the capital stock and labor force in the sector that is growing faster must be *less* than in the other sector. When the elasticity of substitution is greater than one, the converse result obtains. To see the intuition, note that terms of trade (relative prices) shift in favor of the more slowly growing sector. When the elasticity of substitution is less than one, this change in relative prices is more than proportional with the change in quantities and this encourages more of the factors to be allocated towards the more slowly growing sector.

PROPOSITION 20.10. Suppose the asymptotic growth rates g_1^* and g_2^* exist. If $\varepsilon < 1$, then $g^* = \min\{g_1^*, g_2^*\}$. If $\varepsilon > 1$, then $g^* = \max\{g_1^*, g_2^*\}$.

PROOF. Differentiating the production function for the final good (20.34),

$$(20.62) \quad g(t) = \frac{\gamma Y_1(t)^{\frac{\varepsilon-1}{\varepsilon}} g_1(t) + (1-\gamma) Y_2(t)^{\frac{\varepsilon-1}{\varepsilon}} g_2(t)}{\gamma Y_1(t)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\gamma) Y_2(t)^{\frac{\varepsilon-1}{\varepsilon}}}.$$

This equation, combined with $\varepsilon < 1$, implies that as $t \rightarrow \infty$, $g^* = \min\{g_1^*, g_2^*\}$. Similarly, combined with $\varepsilon > 1$, it implies that as $t \rightarrow \infty$, $g^* = \max\{g_1^*, g_2^*\}$. $\square$

Consequently, when the elasticity of substitution is less than 1, the asymptotic growth rate of aggregate output will be determined by the sector that is growing more slowly, and the converse applies when $\varepsilon > 1$.

As in the previous section, let us focus on a *constant growth path* (CGP), again defined as an equilibrium path where the asymptotic growth rate of consumption per capita exists and is constant so that

$$\lim_{t \rightarrow \infty} \frac{\dot{c}(t)}{c(t)} = g_c^*.$$

Let us also define the growth rate of total consumption as $\dot{C}(t)/C(t) \equiv g_C^* = g_c^* + n$, since it will be slightly more convenient to work with the growth rate of total consumption than the growth rate of consumption per capita. From the Euler equation (20.58), the fact that the growth rate of consumption or consumption per capita are asymptotically constant implies that the interest rate must also be asymptotically constant, that is, $\lim_{t \rightarrow \infty} \dot{r} = 0$.

To establish the existence of a CGP, I impose the following parameter restriction:

$$(20.63) \quad \rho - n \geq (1 - \theta) \max \left\{ \frac{a_1}{1 - \alpha_1}, \frac{a_2}{1 - \alpha_2} \right\},$$

which ensures that the transversality condition (20.59) holds. Terms of the form $a_1/(1 - \alpha_1)$ or $a_2/(1 - \alpha_2)$ appear naturally in equilibrium, since they capture the “augmented” rate of technological progress. In particular, recall that associated with the technological progress, there will also be endogenous capital deepening in each sector. The overall effect on labor productivity (and output growth) will depend on the rate of technological progress augmented with the rate of capital deepening. The terms $a_1/(1 - \alpha_1)$ or $a_2/(1 - \alpha_2)$ capture this, since a lower α_s corresponds to a greater share of capital in sector $s = 1, 2$, and thus to a higher rate of augmented technological progress for a given rate of Hicks-neutral technological change. In this light, condition (20.63) can be understood as implying that the augmented rate of technological progress should be low enough to satisfy the transversality condition (20.59).

The next proposition presents the main result of this subsection and characterizes the relatively simple form of the CGP in the presence of non-balanced growth. However, rather than presenting the general case, it is useful to impose the following assumption

$$(20.64) \quad \text{either (i) } a_1/(1 - \alpha_1) < a_2/(1 - \alpha_2) \text{ and } \varepsilon < 1; \text{ or (ii) } a_1/(1 - \alpha_1) > a_2/(1 - \alpha_2) \text{ and } \varepsilon > 1,$$

which will make it easier to state this result. In particular, this condition ensures that sector 1 is the *asymptotically dominant* sector, either because it has a slower rate of technological progress and $\varepsilon < 1$, or it has more rapid technological progress and $\varepsilon > 1$. Notice also that, for the reasons noted above, the appropriate comparison is not between a_1 and a_2 , but between $a_1/(1 - \alpha_1)$ and $a_2/(1 - \alpha_2)$. Exercise 20.14 generalizes the results in this proposition to the case in which the converse of condition (20.64) holds.

PROPOSITION 20.11. *Suppose that conditions (20.37), (20.63) and (20.64) hold. Then, there exists a unique CGP such that*

$$(20.65) \quad g^* = g_C^* = g_1^* = z_1^* = n + g_c^* = n + \frac{1}{1 - \alpha_1} a_1,$$

$$(20.66) \quad z_2^* = n - (1 - \varepsilon) a_2 + (1 + (1 - \varepsilon)(1 - \alpha_2)) \frac{a_1}{1 - \alpha_1} < g^*,$$

$$(20.67) \quad g_2^* = n + \varepsilon a_2 + (1 - \varepsilon(1 - \alpha_2)) \frac{a_1}{1 - \alpha_1} > g^*,$$

$$(20.68) \quad n_1^* = n \text{ and } n_2^* = n - (1 - \varepsilon)(1 - \alpha_2) \left(\frac{a_2}{1 - \alpha_2} - \frac{a_1}{1 - \alpha_1} \right) < n_1^*.$$

PROOF. Suppose first that $g_2^* \geq g_1^* > 0$ and $\varepsilon > 1$. Then, eq.'s (20.46) and (20.47) imply that $\lambda^* = \kappa^* = 1$. In view of this, Proposition 20.10 implies $g^* = g_1^*$. This condition together with equations, (20.36), (20.60) and (20.61), solves uniquely for n_1^* , n_2^* , z_1^* , z_2^* , g_1^* and g_2^* as given in eq.'s (20.65), (20.66), (20.67) and (20.68). Note that this solution is consistent with $g_2^* > g_1^* > 0$, since conditions (20.37) and (20.63) imply that $g_2^* > g_1^*$ and $g_1^* > 0$. Finally, $C(t) \equiv c(t)L(t) \leq Y(t)$, (20.35) and (20.59) imply that the consumption growth rate, g_C^* , is equal to the growth rate of output, g^* . To see why, suppose that this last claim were not correct, then since $C(t)/Y(t) \rightarrow 0$ as $t \rightarrow \infty$, the resource constraint (20.35) would imply that asymptotically $\dot{K}(t) = Y(t)$. Integrating this, we obtain $K(t) \rightarrow \int_0^t Y(s) ds$, and since Y is growing exponentially, this implies that the capital stock grows more than exponentially, violating the transversality condition (20.59).

Next, we can show that the solution with z_1^* , z_2^* , m_1^* , m_2^* , g_1^* and g_2^* satisfies the transversality condition (20.59). In particular, the transversality condition (20.59) will be satisfied if

$$(20.69) \quad \lim_{t \rightarrow \infty} \frac{\dot{K}(t)}{K(t)} < r^*,$$

where r^* is the constant asymptotic interest rate. Since from the Euler equation (20.58) $r^* = \theta g^* + \rho$, (20.69) will be satisfied when $g^*(1 - \theta) < \rho$. Condition (20.63) ensures that this is the case with $g^* = n + a_1/(1 - \alpha_1)$. The argument for the case in which $g_1^* \geq g_2^* > 0$ and $\varepsilon > 1$ is similar and is left to Exercise 20.14.

To complete the proof, we need to establish that in all CGPs $g_2^* \geq g_1^* > 0$ when $\varepsilon < 1$ ($g_1^* \geq g_2^* > 0$ when $\varepsilon > 1$ is again left to Exercise 20.14). Let us separately derive a contradiction for two configurations, (1) $g_1^* \geq g_2^*$, or (2) $g_2^* \geq g_1^*$ but $g_1^* \leq 0$.

(1) Suppose $g_1^* \geq g_2^*$ and $\varepsilon < 1$. Then, following the same reasoning as above, the unique solution to the equilibrium conditions (20.36), (20.60) and (20.61), when $\varepsilon < 1$ is:

$$(20.70) \quad \begin{aligned} g^* &= g_C^* = g_2^* = z_2^* = n + a_2/(1 - \alpha_2), \\ z_1^* &= n - (1 - \varepsilon) a_1 + (1 + (1 - \varepsilon)(1 - \alpha_1)) \frac{a_1}{1 - \alpha_1}, \\ g_1^* &= n + \varepsilon a_1 + (1 - \varepsilon(1 - \alpha_1)) \frac{a_1}{1 - \alpha_1} \end{aligned}$$

and also similar expressions for n_1^* and n_2^* . Combining these equations implies that $g_1^* < g_2^*$, which contradicts the hypothesis $g_1^* \geq g_2^* > 0$. The argument for $\varepsilon > 1$ is analogous.

- (2) Suppose $g_2^* \geq g_1^*$ and $\varepsilon < 1$, then the same steps as above imply that there is a unique solution to equilibrium conditions (20.36), (20.60) and (20.61), which are given by eq.'s (20.65), (20.66), (20.67) and (20.68). But now (20.65) directly contradicts $g_1^* \leq 0$. Finally suppose $g_2^* \geq g_1^*$ and $\varepsilon > 1$, then the unique solution is given by the equations in subpart 1 above. But in this case, (20.70) directly contradicts the hypothesis that $g_1^* \leq 0$, completing the proof.

□

A number of implications of this proposition are worth emphasizing. First, as long as $a_1/(1 - \alpha_1) \neq a_2/(1 - \alpha_2)$, growth is non-balanced. The intuition for this result is the same as Proposition 20.5 in the previous subsection. Suppose, for concreteness, that $\varepsilon < 1$ and $a_1/(1 - \alpha_1) < a_2/(1 - \alpha_2)$ (which would be the case, for example, if $a_1 \approx a_2$). Then, differential capital intensities in the two sectors combined with capital deepening in the economy (which itself results from technological progress) ensures faster growth in the more capital-intensive sector, sector 2. Intuitively, if capital were allocated proportionately to the two sectors, sector 2 would grow faster. Because of the changes in prices, capital and labor would be reallocated in favor of the less capital-intensive sector, and relative employment in sector 1 would increase. However, crucially, this reallocation would not be enough to fully offset the faster growth of real output in the more capital-intensive sector. This result also highlights that the assumption of balanced technological progress in Proposition 20.5 (which, in this context, corresponds to $a_1 = a_2$) was not necessary for the result there, but we simply needed to rule out the knife-edge case where the relative rates of technological progress between the two sectors were exactly in the right proportion to ensure balanced growth (in this context, $a_1/(1 - \alpha_1) = a_2/(1 - \alpha_2)$).

Second, the CGP growth rates are relatively simple, especially because I restricted attention to the set of parameters that ensure that sector 1 is the asymptotically dominant sector (cf., condition (20.64)). If, in addition, we also have $\varepsilon < 1$, the model leads to the richest set of dynamics, whereby the more slowly growing sector determines the long-run growth rate of the economy, while the more rapidly growing sector continually sheds capital and labor, but does so at exactly the right rate to ensure that it still grows faster than the rest of the economy.

Third, in the limiting equilibrium the share of capital and labor allocated to one of the sector tends to one (e.g., when sector 1 is the asymptotically dominant sector, $\lambda^* = \kappa^* = 1$). Nevertheless, at all points in time both sectors produce positive amounts, so this limit point is never reached. In fact, at all times both sectors grow at rates *greater* than the rate of population growth in the economy. Moreover, when $\varepsilon < 1$, the sector that is shrinking grows *faster* than the rest of the economy at all points in time, even asymptotically. Therefore, the rate at which capital and labor are allocated away from this sector is determined in

equilibrium to be *exactly* such that this sector still grows faster than the rest of the economy. This is the sense in which non-balanced growth is not a trivial outcome in this economy (with one of the sectors shutting down), but results from the positive but differential growth of the two sectors.

Finally, it can be verified that the capital share in national income and the interest rate are constant in the CGP (see Exercise 20.15). For example, when condition (20.64) holds, $\sigma_K^* = \alpha_1$. In contrast, when this condition does not hold $\sigma_K^* = \alpha_2$ —in other words, the asymptotic capital share in national income will reflect the capital share of the dominant sector. Therefore, this model based on supply-side sources of non-balanced growth is also broadly consistent with the Kaldor facts as well as the Kuznets facts (though this model also generates significant deviations from the orderly behavior implied by the Kaldor facts when the economy is away from the CGP). The analysis so far does not establish that the CGP is asymptotically stable. This is done in Exercise 20.16, which also provides an alternative proof of Proposition 20.11. Consequently, a model based on supply-side factors can also give useful insights about structural change. Naturally, to understand the sweeping long-run changes in the composition of output and employment, we need to combine the demand-side and the supply-side factors studied in the last two sections. Exercise 20.17 takes a first step in this direction.

20.3. Agricultural Productivity and Industrialization

Although the models presented in the last two sections have highlighted how demand-side and supply-side factors can lead to structural change (and also how structural change can be consistent with a constant BGP and the Kaldor facts), they did not focus on the process of industrialization. Chapter 1 documented that the industrialization process, beginning at the end of the 18th century in Europe, lies at the root of modern economic growth and cross-country income differences. Thus a natural question is why industrialization started and then progressed rapidly in some countries, while it did not in others. In view of the general patterns presented in Chapter 1, this question might hold important clues about the cross-country differences in income per capita today.

It would therefore be useful to have a number of different approaches to this question and evaluate their pros and cons. While this is part of my objective, I will not present these models all in one place. The first approach, based on the model of Acemoglu and Zilibotti (1997), was already presented as an application of stochastic growth models in Section 17.6 in Chapter 17. Although this theory focused on takeoff in general, the most relevant incident of takeoff in history is related to industrialization. Therefore, the theory in Section 17.6 can be interpreted as offering a potential explanation for the origins of industrialization based on whether the investments in different sectors undertaken by different societies turned out to be successful. In particular, societies that happen to have put a substantial fraction of their resources in sectors that turned out to be unlucky, or were ex post discovered not to be as productive, have been less successful than those that have invested in sectors and

projects that were ex post more successful. The theory showed how success breeds success and how a string of good outcomes can lead to a takeoff, whereby the society is able to diversify its sectoral and project-based risks successfully in a deeper financial markets and allocate its funds more productively towards high-return activities. In the next chapter, we will see another approach to the origins of industrialization based on the idea of *the big push* suggested by Rosenstein-Rodan. The model by Murphy, Shleifer and Vishny (1989) in Section 21.5 in the next chapter will formalize this notion and show how, in the presence of technologies with fixed costs and monopolistic competition, coordination failures might prevent industrialization. The attractive feature of this approach will be its close connection to the baseline endogenous growth models studied in Part 4. A potential shortcoming might be its reliance on multiple equilibria, without an explanation for why some societies manage to coordinate to the good equilibrium whereas others end up in the bad equilibrium.

Before turning to market failures in development, it is useful to consider another approach that can shed light on the factors facilitating, or even spurring, industrialization. A common argument in the economic history literature is that 18th-century England was particularly well-placed for industrialization because of its high agricultural productivity (e.g., Nurske, 1953, Rostow, 1960, Mokyr, 1989, or Overton, 2001). The basic idea is that societies with a high agricultural productivity can afford to shift part of their labor force to industrial activities. Some type of increasing returns coming from technology or demand is then invoked to argue that the ability to shift a critical fraction of the labor force to industry is an important element of the early industrial experience.

In this section, I present a model based on Matsuyama (1992), which formalizes this intuition. It also presents a number of comparative static results that are useful in thinking about the process of industrialization. Matsuyama's model naturally complements the models already studied in this chapter, because it is a model of structural change. It combines Engel's Law with learning-by-doing externalities in the industrial sector. The model is not only a tractable framework for the analysis of the relationship between agricultural productivity and industrialization, but it also enables an insightful analysis of the impact of international trade on industrialization (along the lines of the model in Section 19.7 in the previous chapter).

Consider the following infinite-horizon continuous-time economy with a constant population normalized to 1. The preference side is modeled via a representative household with preferences given by

$$(20.71) \quad U(0) \equiv \int_0^\infty \exp(-\rho t) (c^A(t) - \gamma^A)^\eta c^M(t)^{1-\eta} dt.$$

These preferences are similar to those in (20.2); $c^A(t)$ again denotes the consumption of the agricultural good, $c^M(t)$ is the consumption of the manufacturing good at time t , and the parameter γ^A is the minimum (subsistence) food requirement. In addition ρ is the discount factor and $\eta \in (0, 1)$ designates the importance of agricultural goods versus manufacturing goods in the utility function. The representative household supplies labor inelastically. Let

us also focus on the closed economy in the text, leaving the extension to open economy to Exercise 20.20.

Output in the two sectors is produced with the following production functions

$$(20.72) \quad Y^M(t) = X(t) F(L^M(t))$$

and

$$(20.73) \quad Y^A(t) = B^A G(L^A(t)),$$

where as before Y^M and Y^A denote the total production of the manufacturing and the agricultural goods, and L^M and L^A denote the total labor employed in the two sectors. Both production functions F and G exhibit diminishing returns to labor. More formally, F and G are differentiable and strictly concave. In particular, $F(0) = 0$, $F'(\cdot) > 0$, $F''(\cdot) < 0$, $G(0) = 0$, $G'(\cdot) > 0$, and $G''(\cdot) < 0$, where as usual F' and G' denotes first derivatives of these functions. Diminishing returns to labor might arise because they both use land or some other factor of production as well as labor. Nevertheless, it is simpler to assume diminishing returns rather than introduce another factor of production. The fact that there are diminishing returns implies that when labor is priced competitively, there will be equilibrium profits and these are redistributed to households.

The key feature for this model of industrialization is that there is no technological progress in agriculture, but the production function for the manufacturing good, (20.72), includes the term $X(t)$, which will allow for technological progress in manufacturing. Although there is no technological progress in agriculture, the productivity parameter B^A potentially differs across countries, reflecting either previous technological progress in terms of new agricultural methods or differences in land quality (even though here, for simplicity, I am focusing on a single country). Existing evidence shows that there are very large (perhaps too large) differences in labor productivity and TFP of agricultural activities among countries even today, thus allowing for potential productivity differences in agriculture is reasonable. Current research also shows that the image of the agriculture as a quasi-stagnant sector without technological progress is not accurate, and in fact, this sector experiences both substantial capital-labor substitution and major technological change (including the introduction of new varieties of seeds, mechanization and organizational changes affecting productivity). Nevertheless, the current model provides a good starting point for our purposes.

Labor market clearing requires that

$$L^M(t) + L^A(t) \leq 1,$$

since total the labor supply is normalized to 1. Let $n(t)$ denote the fraction of labor employed in manufacturing as of time t . Since there will be full employment in this economy, $L^M(t) = n(t)$ and $L^A(t) = 1 - n(t)$.

The key assumption is that manufacturing productivity, $X(t)$, evolves over time as a result of learning-by-doing externalities as in Romer's (1986a) model in Chapter 11. In particular, suppose that the growth of the manufacturing technology, $X(t)$, is proportional

to the amount of current production in manufacturing

$$(20.74) \quad \dot{X}(t) = \delta Y^M(t),$$

where $\delta > 0$ measures the extent of these learning-by-doing effects and the initial productivity level is $X(0) > 0$ at time $t = 0$ taken as given. As in the Romer model, learning-by-doing effects are external to individual firms. This type of external learning-by-doing effects are too reduced-form to generate insights about how productivity improvements take place in the industrial sector. Nevertheless, our analysis so far makes it clear that one can endogenize technology choices by introducing monopolistic competition and under the standard assumptions made in Part 4 above, this will generate a market size effect and lead to an equation similar to (20.74). Exercise 20.19 asks you to consider such a model.

In equilibrium, each firm will choose its labor demand in order to equate the value of the marginal product to the wage rate, $w(t)$. Let us choose the agricultural good as the numeraire (so that its price is normalized to 1) and also assume that the equilibrium is interior with both sectors being active. Then, equilibrium labor demand equations in the two sectors will be given by

$$w(t) = B^A G'(1 - n(t)) \text{ and } w(t) = p(t) X(t) F'(n(t))$$

where $p(t)$ is the relative price of the manufactured good (in terms of the numeraire, the agricultural good). Market clearing then implies:

$$(20.75) \quad B^A G'(1 - n(t)) = p(t) X(t) F'(n(t)).$$

The presence of the term $\gamma^A > 0$ implies that as in Section 20.1, preferences are non-homothetic and that the income elasticity of demand for agricultural goods will be less than unity (while that for manufacturing goods will be greater than unity). As we have already seen, this is the simplest way of introducing Engel's Law.

Let us also assume that aggregate productivity is high enough to meet the minimum agricultural consumption requirements of the entire population (which, here, is normalized to 1):

$$(20.76) \quad B^A G(1) > \gamma^A > 0.$$

If this inequality were violated, the economy's agricultural sector would not be productive enough to provide the subsistence level of food to all consumers.

Finally, the budget constraint of the representative household at each date t can be written as

$$c^A(t) + p(t) c^M(t) \leq w(t) + \pi(t)$$

where $\pi(t)$ is the profits per representative household, resulting from the diminishing returns in the production technologies.

An equilibrium in this economy is defined in the standard way as processes of consumption levels in the two sectors and allocations of labor between the two sectors at all dates, such

that households maximize utility and firms maximize profits given prices, and goods and factor prices are such that all markets clear.

Maximization of (20.71) implies that for each household, and thus for the entire economy,

$$(20.77) \quad c^A(t) = \gamma^A + \eta p(t) c^M(t) / (1 - \eta).$$

Since the economy is closed, production must equal consumption and thus

$$c^A(t) = Y^A(t) = B^A G(1 - n(t))$$

and

$$c^M(t) = Y^M(t) = X(t) F(n(t))$$

Now combining these equations with (20.75) and (20.77) yields

$$(20.78) \quad \phi(n(t)) = \frac{\gamma^A}{B^A},$$

where

$$\phi(n) \equiv G(1 - n) - \frac{\eta G'(1 - n) F(n)}{(1 - \eta) F'(n)},$$

is a strictly decreasing function. Moreover, $\phi(0) = G(1)$ and $\phi(1) < 0$. The ϕ function can be interpreted as the aggregate “relative demand” function for manufacturing over agriculture. An equilibrium has to satisfy (20.78). From Assumption (20.76) and the properties of the ϕ function, we can conclude that the equilibrium condition (20.78) has a unique interior solution in which

$$n(t) = n^* \in (0, 1).$$

Notice an important implication of this equation. Even though the current model is one of structural change like those in the previous two sections, it only generates changes in the composition of output—the fraction of the labor force working in agriculture remains constant at $1 - n^*$. This implies that, while the current model is useful for interpreting the onset of industrialization, it will not be sufficient to generate insights about why the composition of employment in different sectors of the economy has been changing over the past 150 or 200 years.

Next, using (20.78), the unique equilibrium allocation of labor between the two sectors satisfies

$$(20.79) \quad n^* = \phi^{-1} \left(\frac{\gamma^A}{B^A} \right).$$

Since ϕ is strictly decreasing, so is its inverse function ϕ^{-1} and thus the fraction of the labor force employed in manufacturing, n^* , is strictly increasing in B^A . This is the most important result of the current model and shows that a greater fraction of the labor force will be allocated to the manufacturing sector when agricultural productivity is higher. The reason for this result is intuitive: the Cobb-Douglas production function combined with homothetic preferences would imply a constant allocation of employment between the two sectors independent of their productivity. However, in the current model, preferences are non-homothetic and a certain amount of food production is necessary first. When agricultural

productivity, B^A , is high, a relatively small fraction of the labor force is sufficient to generate this minimal level of food production, and thus a greater fraction of the labor force can be employed in manufacturing.

This results, combined with learning-by-doing in manufacturing, cf. eq. (20.74), is at the root of the relationship between agricultural productivity and industrialization. In particular, eq. (20.74) implies that output in manufacturing grows at the constant rate, $\delta F(n^*)$, which is also positively related to B^A in view of eq. (20.79). Therefore, the current model generates a very simple representation of the often-hypothesized relationship between agricultural productivity and the origins of industrialization.

It is also useful to note that in the equilibrium of this model, because the shares of employment in manufacturing and agriculture are constant and there is no technological progress in the agricultural sector, agricultural output remains constant. All growth is generated by growth of manufacturing production. However, since manufacturing and agricultural goods are imperfect substitutes, the relative prices change, so expenditure on agricultural goods increases (see Exercise 20.18).

We can summarize these results as follows:

PROPOSITION 20.12. *In the above-described model, the combination of learning-by-doing and Engel's Law generates a unique equilibrium in which the share of employment of manufacturing is constant at $n^* \equiv \phi^{-1}(\gamma^A/B^A)$, and manufacturing output and consumption grow at the rate $\delta F(n^*)$, which is increasing in agricultural productivity B^A .*

We have so far characterized the equilibrium in a closed economy. A major result of this model is that higher agricultural productivity leads to faster industrial growth and thus to faster overall growth. The reason for this is intuitive: higher agricultural productivity enables the economy to allocate a larger fraction of its labor force to the knowledge-producing sector, which is manufacturing (where knowledge-production is introduced in a reduced-form manner as in Romer's (1986a) model). Even though the presumption that most important knowledge-producing activities take place in the manufacturing sector is no longer generally accepted, this model provides a useful framework for the analysis of the origins of industrialization. An important advantage of the current model is its tractability. This enables us to adapt it easily to analyze other related questions, such as the impact of trade opening on industrialization. This is done in Exercise 20.20. The striking result in this case is that the implications of the closed and the open economies are very different. For example, that exercise shows that higher agricultural productivity, in the presence of international trade, can lead to delayed industrialization or even to deindustrialization, rather than being the source of rapid industrialization as in the closed economy. The reason for this is related to the forces highlighted in Section 19.7 of Chapter 19; specialization according to comparative advantage may have negative long-run consequences in the presence of sector-specific externalities. However, as already discussed in that section, the evidence for large externalities of

this sort are not very strong. Consequently, the model in this section and its implications regarding the role of international trade in the process of industrialization should be interpreted with some caution. Nevertheless, this model is an important tool in our arsenal of models of long-run economic development, especially because it illustrates in an elegant and tractable manner how agricultural productivity interacts with the process of industrialization.

20.4. Taking Stock

This chapter took a first step towards the analysis of structural changes involved in the process of economic development. Our first step has been relatively modest. The focus has been on the structural changes associated with the shifts in output and employment away from agriculture to manufacturing and to services and with the changes between sectors of different capital intensities. Section 20.1 focused on demand-side reasons for why growth may be non-balanced. It incorporated Engel's Law into the basic neoclassical growth model so that households spend a smaller fraction of their budget on agricultural goods as they become richer. This framework is ideally suited for the analysis of the structural changes across broad sectors such as agriculture, manufacturing and services. Section 20.2, on the other hand, turned to supply-side reasons for non-balanced growth, which were first highlighted by Baumol's (1967) classic paper. However, instead of assuming exogenously-given different rates of technological progress across sectors, this section emphasized how sectoral differences in capital intensity can lead to non-balanced growth. Capital-intensive sectors tend to grow more rapidly as a result of an equi-proportionate increase in the capital-labor ratio. This feature, combined with capital deepening at the economy level, naturally leads to a pattern of non-balanced growth. This type of non-balanced growth may contribute to structural change across agricultural, manufacturing and service sectors, but becomes even more relevant when we look at sectors differentiated according to their capital intensity. A particular focus of both Sections 20.1 and 20.2 has been to reconcile non-balanced growth at the sectoral level with the pattern of relatively balanced growth at the aggregate. As already noted in Chapter 2, balanced growth need not be taken literally. It is at best an approximation to the growth behavior of advanced economies. Nevertheless, it seems to be a particularly accurate approximation to many features of the growth process, since interest rates and the share of capital income in GDP do appear to have been relatively constant over the past 100 years or so in most advanced economies. It is therefore important to understand how significant reallocation of resources at the sectoral and micro levels can coexist with the more "balanced" behavior at the aggregate. The models in Sections 20.1 and 20.2 suggested some clues about why this may be the case, but the answers provided here should be viewed as preliminary rather than definitive.

I also discussed a simple model of the origins of industrialization. This model showed how agricultural productivity might have an important effect on the timing of industrialization, but also demonstrated that the effect of agricultural productivity might depend on whether the economy is open to international trade. The study of the process of industrialization is

important, in part because, as discussed in Chapter 1, existing evidence suggests that the timing and nature of industrialization may have important implications for cross-country income differences we observe today, and thus the investigation of the economic development problem might necessitate an analysis of why some countries industrialized early, while others were delayed or never started the process of industrialization.

Understanding the sources of the structural changes and how they can be reconciled with the broad patterns of balanced growth in the aggregate sheds light on both the process of economic growth and the process of economic development. In this sense, the models in this section enrich our understanding of economic growth considerably. And yet, this is only a modest step towards the investigation of the sweeping structural changes emphasized by Kuznets because we have not departed from the neoclassical approach to economic growth. In particular, Sections 20.1 and 20.2 used generalized versions of the basic neoclassical growth model of Chapter 8, and Section 20.3 used a variant of the Romer (1986a) model from Chapter 11.

It should be emphasized again that the topics discussed in this chapter, though closely related to the basic neoclassical growth model, are areas of frontier research. We are far from a satisfactory framework for understanding the process of reallocation of capital and labor across sectors, how this changes at different stages of development, and how this remains consistent with relatively balanced aggregate growth and the Kaldor facts. I have therefore not attempted to provide a unified framework that combines the transition from agriculture to industrialization, the demand-side reasons for non-balanced growth and the supply-side forces. The development of such unified models as well as richer models of non-balanced growth are areas for future research.

20.5. References and Literature

The early development literature contains many important works documenting the major structural changes taking place in the process of development. Kuznets (1957, 1973) and Chenery (1960) provide some of the best overviews of the broad evidence and the literature, though similar issues were discussed by even earlier development economists such as Rosenstein-Rodan (1943), Nurske (1953), and Rostow (1960). Figure 20.1, which uses data from *The Historical Statistics of the United States*, gives a summary of these broad changes.

The model of non-balanced growth based on Engel's Law presented in Section 20.1 is based on Kongsamut, Rebelo and Xie (2001). Previous work that have analyzed similar models include Murphy, Shleifer and Vishny (1989), Echevarria (1997), Laitner (2000). More recent work building on Kongsamut, Rebelo and Xie (2001) includes Caselli and Coleman (2001) and Gollin, Parente and Rogerson (2002). Many of these models are considerably richer than the Kongsamut, Rebelo and Xie approach. For example, Murphy, Shleifer and Vishny (1989) incorporate monopolistic competition and analyzes the implications of income inequality for the demand for different types of goods. Echevarria (1997) and Laitner (2000) show how the initial phase of transition from agriculture to manufacturing will be associated

with aggregate non-balanced growth. The distinguishing feature of these models is that land is also a factor of production and is more important for agriculture than for manufacturing. Exercise 20.8 provides an example of such a model. The recent literature also places greater emphasis on sources of agricultural productivity and emphasizes that differences in agricultural productivity across countries are often as large as, or even larger, than productivity differences in other sectors. Gollin, Parente and Rogerson (2002) is one of the first papers in this vein.

The works mentioned in the previous paragraph, like the model I presented in Section 20.1, appeal to Engel's Law and model the resulting non-homothetic preferences by positing Stone-Geary preferences as in eq. (20.3). A more flexible and richer approach is to allow for "hierarchies of needs" in consumption, whereby households consume different goods in a particular sequence (for example, food needs to be consumed before textiles, and textile need to be consumed before electronics, and so on). This approach is used in Stokey (1988), Matsuyama (2002), Foellmi and Zweimuller (2002), and Buera and Kaboski (2006) to generate richer models of structural change. Space restrictions precluded me from presenting these hierarchy of needs models, even though they are both insightful and elegant alternatives to the standard approach of using Stone-Geary preferences.

Section 20.2 builds on Acemoglu and Guerrieri (2006). The precursor to this work is Baumol (1967), which emphasized the importance of differential productivity growth on non-balanced growth. However, Baumol did not derive a pattern of non-balanced growth including reallocation of capital and labor across sectors, and assumed differential rates of productivity growth to be exogenous. Ngai and Pissarides (2006) and Zuleta and Young (2006) provide modern versions of Baumol's hypothesis. Instead, the approach in Section 20.2 emphasizes how the combination of different capital intensities and capital deepening in the aggregate can endogenously lead to this pattern.

The model in Section 20.3 is based on Matsuyama (1992) and is also closely related to the model I presented in Section 19.7 in Chapter 19. Excellent accounts of the role of agriculture in industrialization, especially in the British context, are provided in Mokyr (1993) and Overton (2001).

20.6. Exercises

EXERCISE 20.1. (1) Show that the consumption aggregator in (20.3) leads to Engel's Law.

(2) Suggest alternative consumption aggregators that will generate similar patterns.

EXERCISE 20.2. Prove Proposition 20.1.

EXERCISE 20.3. (1) Set up the optimal control problem for a representative household in the model of Section 20.1.

(2) From the Euler equations and the transversality condition, verify part 1 of Proposition 20.2.

(3) Use eq.'s (20.10)-(20.11) to derive parts 2 and 3 of the proposition.

EXERCISE 20.4. (1) Prove Proposition 20.3.

(2) Show that even though a BGP does not exist, an equilibrium path always exists.

EXERCISE 20.5. (1) Prove Proposition 20.4. In particular, show that if (20.21) is not satisfied, a CGP cannot exist, and that this condition is sufficient for a CGP to exist.

(2) Characterize the CGP effective capital-labor ratio, k^* .

EXERCISE 20.6. In the model of Section 20.1, show that as long as condition (20.21) is satisfied, when the economy starts with an effective capital-labor ratio $K(0)/(X(0)L(0))$ different from k^* , the CGP is globally stable and the effective capital-labor ratio will monotonically converge to k^* as $t \rightarrow \infty$.

EXERCISE 20.7. * Consider a generalization of the model of Section 20.1, where the sectoral production functions are given by the following Cobb-Douglas forms

$$\begin{aligned} Y^A(t) &= K^A(t)^{\alpha^A} (B^A(t) L^A(t))^{1-\alpha^A} \\ Y^M(t) &= K^M(t)^{\alpha^M} (B^M(t) L^M(t))^{1-\alpha^M} \\ Y^S(t) &= K^S(t)^{\alpha^S} (B^S(t) L^S(t))^{1-\alpha^S} \end{aligned}$$

and assume that $B^A(t)$, $B^M(t)$ and $B^S(t)$ grow respectively at the rates g^A , g^M and g^S .

- (1) Derive the equivalents of Propositions 20.1 and 20.2.
- (2) Show that as long as preferences are given by (20.3) and $\gamma^A > 0$ and/or $\gamma^S > 0$, balanced growth is impossible.
- (3) Show that there exists a generalization of condition (20.21) such that this model will have a CGP as defined in Section 20.1. [Hint: the generalization includes two separate conditions that depend on technology growth rates as well as preference parameters].

EXERCISE 20.8. Consider a version of the model in Section 20.1, with only manufacturing and agricultural goods. The consumption aggregator is $c(t) = (c^A(t) - \gamma^A)^{\eta^A} c^M(t)^{\eta^M}$, with $\gamma^A > 0$. Assume that the production functions for agricultural and manufacturing goods take the form $Y^A(t) = X(t) (L^A(t))^\zeta (Z)^{1-\zeta}$ and $Y^M(t) = X(t) L^M(t)$, where Z is land. There are no savings or capital.

- (1) Characterize the competitive equilibrium in this economy.
- (2) Show that this economy also exhibits structural change; in particular, show that the share of manufacturing sector grows over time.
- (3) What happens to land rents along the equilibrium path?

EXERCISE 20.9. * In the model of Section 20.1, suppose that condition (20.21) is not satisfied. Assume that the production function F is Cobb-Douglas. Characterize the asymptotic growth path of the economy (the growth path of the economy as $t \rightarrow \infty$).

EXERCISE 20.10. Consider the model of Section 20.1 but assume that there exist a final good produced according to the technology $Y(t) = (Y^A(t) - \gamma^A)^{\eta^A} Y^M(t)^{\eta^M} (Y^S(t) + \gamma^S)^{\eta^S}$.

- (1) Show that all the results in Section 20.1 hold without any change as long as capital goods are produced out of intermediate Y^M as implied by eq. (20.8).
- (2) Next assume that capital goods are produced out of the final good, so that the resource constraint becomes $\dot{K}(t) + c(t)L(t) = Y(t)$, where $c(t)$ is the per capita consumption of the final good. Show that in this model a CGP (as defined in Section 20.1) does not exist.

EXERCISE 20.11. Prove Proposition 20.6.

EXERCISE 20.12. Derive eq.'s (20.42), (20.43), (20.44), and (20.45).

EXERCISE 20.13. Prove Proposition 20.7.

EXERCISE 20.14. (1) Complete the proof of Proposition 20.11 by considering the case in which $\varepsilon > 1$ and $g_1^* \geq g_2^* > 0$.

- (2) State and prove the equivalent of Proposition 20.11, when the converse of condition (20.64) holds.

EXERCISE 20.15. Show that in the allocation in Proposition 20.11, the asymptotic interest rate is constant and derive a closed-form expression for this interest rate.

EXERCISE 20.16. * In this exercise, you are first asked to provide an alternative proof of Proposition 20.11 and then characterize the local transitional dynamics in the neighborhood of the CGP. Throughout, suppose that either $\varepsilon < 1$ and $a_1/(1 - \alpha_1) < a_2/(1 - \alpha_2)$ or that $\varepsilon > 1$ and $a_1/(1 - \alpha_1) > a_2/(1 - \alpha_2)$.

- (1) Reexpress the equilibrium equations in terms of the following three variables $\varphi(t) \equiv c(t)/L(t) A_1(t)^{1/(1-\alpha_1)}$, $\chi(t) \equiv K(t)/(L(t) A_1(t)^{1/(1-\alpha_1)})$ and $\kappa(t)$. In particular, show that the following three differential equations, together with the appropriate transversality condition and initial values $\chi(0)$ and $\kappa(0)$, characterize the dynamic equilibrium

$$\begin{aligned}
 (20.80) \quad \frac{\dot{\varphi}(t)}{\varphi(t)} &= \frac{1}{\theta} \left[\alpha_1 \gamma \eta(t)^{1/\varepsilon} \lambda(t)^{1-\alpha_1} \kappa(t)^{-(1-\alpha_1)} \chi(t)^{-(1-\alpha_1)} - \rho \right] - n - \frac{a_1}{1 - \alpha_1}, \\
 \frac{\dot{\chi}(t)}{\chi(t)} &= \lambda(t)^{1-\alpha_1} \kappa(t)^{\alpha_1} \chi(t)^{-(1-\alpha_1)} \eta(t) - \chi(t)^{-1} \varphi(t) - n - \frac{a_1}{1 - \alpha_1}, \\
 \frac{\dot{\kappa}(t)}{\kappa(t)} &= \frac{(1 - \kappa(t)) \left[(\alpha_2 - \alpha_1) \frac{\dot{\chi}(t)}{\chi(t)} + a_2 - \frac{1-\alpha_2}{1-\alpha_1} a_1 \right]}{(1 - \varepsilon)^{-1} + (\alpha_2 - \alpha_1) (\kappa(t) - \lambda(t))},
 \end{aligned}$$

where $\kappa(t)$ and $\lambda(t)$ are given by (20.46) and (20.47), and

$$(20.81) \quad \eta(t) \equiv \gamma^{\frac{\varepsilon}{\varepsilon-1}} \left[1 + \frac{\alpha_1}{\alpha_2} \left(\frac{1 - \kappa(t)}{\kappa(t)} \right) \right]^{\frac{\varepsilon}{\varepsilon-1}}.$$

[Hint: use the Euler equation of the representative household and the resource constraint of the economy, rearrange these to express the laws of motion of $\varphi(t)$ and $\chi(t)$ in terms of $\kappa(t)$, $\lambda(t)$ and $\eta(t)$ as defined in (20.81), and then differentiate (20.46).]

- (2) State the appropriate transversality condition.

- (3) Show that if an allocation satisfies the three differential equations in (20.80) and the appropriate transversality condition, then it corresponds to an equilibrium path.
- (4) Show that in a CGP equilibrium $\varphi(t)$ must be constant. Using this, show that the CGP requires that $\kappa(t) \rightarrow 1$ and that $\chi(t)$ must also be constant. From these observations, derive an alternative proof of Proposition 20.11.
- (5) Now linearize these three equations around the CGP of Proposition 20.11 and show that the linearized system has two negative and one positive eigenvalues, and using this fact conclude that the CGP is locally stable. [Hint: as part of this argument, explain why $\kappa(t)$ should be considered a state variable with $\kappa(0)$ taken as an initial value].

EXERCISE 20.17. Consider a model that combines the supply-side and the demand-side features discussed in Sections 20.1 and 20.2. In particular, suppose that the consumption aggregator is given by $c(t) = (c^S(t) + \gamma^S)^{\eta^S} c^M(t)^{\eta^M}$, where c^S consumption of services and c^M denotes the consumption of manufacturing goods. Assume that the economy is closed and both services and manufacturing are produced by Cobb-Douglas technologies with the same Hicks-neutral rate of exogenous technological progress, but manufacturing is more capital-intensive. Assume also investment goods are produced from the manufacturing goods alone as in the model of Section 20.1. Characterize the equilibrium of this economy. Show that the relative price and the employment share of services will be increasing over time. Is it possible for the total consumption of manufacturing goods to increase faster than those of services?

EXERCISE 20.18. Consider the model of Section 20.3.

- (1) Show that aggregate food (agricultural) consumption and production stay constant at

$$B^A G(1 - \phi^{-1}(\gamma^A/B^A)) = \gamma^A + B^A \eta G'(1 - \phi^{-1}(\gamma^A/B^A)) \frac{F(\phi^{-1}(\gamma^A/B^A))}{F'(\phi^{-1}(\gamma^A/B^A))}.$$

- (2) Show that this is increasing in B^A and provide the intuition for this result.
- (3) Show that expenditure on agricultural goods increases at the same rate as aggregate output. [Hint: first characterize how $p(t)$ changes along the equilibrium path].

EXERCISE 20.19. * Consider the model of Section 20.3 and suppose that the production function for the manufacturing sector is given by

$$Y^M(t) = \frac{1}{1-\beta} \left[\int_0^{N(t)} x(\nu, t)^{1-\beta} d\nu \right] L^M(t)^\beta,$$

which is similar to the production functions in Part 4 of the book, with $N(t)$ denoting the range of machines (intermediates) and $x(\nu, t)$ corresponding to the amount of machine of type ν used by the manufacturing sector. Assume as in Part 4 that these machines are supplied by technology monopolists with perpetual patents and can be produced by using the manufacturing good at constant marginal cost of $(1 - \beta)$ units of the manufacturing good. Also assume the lab-equipment specification for creating new machines as in Section 15.7.

Characterize the equilibrium of this economy and show that the qualitative features are the same as the model in the text.

EXERCISE 20.20. Consider an open economy version of the model of Section 20.3. In particular, suppose that the economy trades with the rest of the world taking product prices as given. The rest of the world is characterized by the same technology, except that it has an initial level of productivity in the manufacturing sector equal to $X^F(0)$ and an agricultural productivity given by B^F . Suppose that there are no spillovers in learning-by-doing, so that eq. (20.74) applies to the “home” economy and the law of motion of manufacturing productivity in the rest of the world is given by $\dot{X}^F(t) = \delta Y^{M,F}(t)$, where $Y^{M,F}(t)$ is total foreign manufacturing production at time t .

- (1) Show that comparative advantage in this economy is determined by the comparison of $X(0)/B^A$ to $X^F(0)/B^F$. Interpret this.
- (2) Suppose that $X(0)/B^A < X^F(0)/B^F$, so that the home economy has a comparative advantage in agricultural production. Show that the initial share of employment in manufacturing in the home economy, $n^*(0)$, must satisfy

$$\frac{X(0)F'(n^*(0))}{B^AG'(1-n^*(0))} = \frac{X^F(0)F'(n^{F*}(0))}{B^FG'(1-n^{F*}(0))},$$

where $n^{F*}(0)$ is the share of manufacturing employment in the rest of the world. Show that $n^*(0)$ given by this equation is strictly less than n^* as given by (20.79).

- (3) What happens to manufacturing employment in the home economy starting as in part 2 of this exercise? [Hint: derive an equivalent of the equation in part 2 for any t , differentiate this with respect to time and then use the laws of motion of X and X^F].
- (4) Explain why agricultural productivity, which was conducive to faster industrialization in the closed economy, may lead to delayed industrialization or to deindustrialization in the open economy.
- (5) Consider an economy specializing in agriculture as in the earlier parts of this exercise. Is welfare at time $t = 0$ necessarily lower when this economy is open to trade than when it is closed to trade? Relate your answer to the analysis in Section 19.7 of Chapter 19.

Structural Transformations and Market Failures in Development

Together with the process of economic development and the changes in the structure of production, there is also a transformation of the economy, which both involves major social changes and induces greater (and perhaps more “complex”) coordination of economic activities. Loosely speaking, we can think of a society that is relatively developed as functioning along (or at any rate, nearer) the frontier of its production possibilities set, while a less-developed economy may be in the interior of its “notional” production possibilities set. This may be because certain arrangements necessary for an economy to reach the frontier of its production possibility set require a large amount of capital or some specific technological advances (in which case, even though we may think of the society as functioning in the interior of its production possibility set, this may not be the outcome of market failure, thus the qualifier “notional” in the previous sentence). Alternatively, less developed economies may be in the interior of their production possibility set because these societies are subject to severe market failures. In this chapter, I will discuss these approaches to economic development.

I first focus on structural transformations and how these may be limited by amount of capital or technology available in a society. The main economic issues are most simply and effectively illustrated by a simple model in which economic development is accompanied with financial development, enabling better risk sharing and thus investment in higher productivity activities. This model, presented in Section 21.1, has clear similarities to the model in Section 17.6 in Chapter 17, though it focuses on the sharing of idiosyncratic risks rather than diversification of aggregate risks. Section 21.2 is less explicitly about the economy moving from the interior of its production possibilities set towards the frontier. Nevertheless, it discusses a major aspect of the structural transformation of the economy during the development process—the demographic transition. In particular, this section discusses how population and fertility change over the process of development and emphasizes how these structural transformations may be linked to investments in human capital. This model provides a brief introduction to the rich literature on the demographic transition and its role in economic growth. Section 21.3 discusses another important structural change, the increase in the population living in urban areas via migration from the countryside to the cities. This model illustrates both how development is associated with a process of allocating workers to activities in which their marginal product is higher and also how a dual economy structure

may emerge in equilibrium and slow down this reallocation process. All three of these sections are meant to give a flavor of vast literatures dealing with issues of financial development, the demographic transition, population growth, fertility, migration, urbanization, and other social changes taking place in the course of the development process. Some of these areas are at the forefront of current research in economic development but space restrictions preclude me from spending more than a few sections on these important topics.

In Section 21.4 I present a model that is complementary to those in Sections 21.1-21.3, where the stage of development is captured by the distance of an economy's technology to the world technology frontier. This model shows how the distance to the frontier measure of the stage of development influences the equilibrium and optimal organizational production. More specifically, it focuses on whether entrepreneurs that are unsuccessful will be immediately replaced by new entrepreneurs or whether the equilibrium will feature long-term relationships and the survival of experienced entrepreneurs even when they are not very productive early on. The simple framework presented in this section can be used for modeling a range of decisions related to the structure of production and the internal organization of firms over the process of development, which is another aspect of the sweeping structural changes suggested by Kuznets.

The models presented in Sections 21.1-21.4 focus on structural transformations. Part of the focus of these models is on the structural transformation that is an essential part of the process of economic development and the common theme is an investigation of the factors that help or hinder this structural transformation as well as the impact of this transformation on aggregate productivity. Some of these models feature market failures, though the focus is not on market failures per se.

A central question of economic development (and economic growth) is why so many societies have failed to take advantage of innovations and improve their technologies over the past 200 years. The perspective that emphasizes potential differences in efficiency and in the extent of market failures also suggests that some less-developed economies might be suffering disproportionately from market failures. Some economists interpret the implications of these models as stating that less-developed economies are "stuck" in a potential "development trap," that is, an equilibrium or a steady state where efficiency is low and market failures sustain this low-efficiency equilibrium, though a different type of steady state (or equilibrium) with a higher level of income and/or a higher growth rate is also possible. The rest of the chapter discusses the roles of various different types of market failures and the possibility of poverty traps and economic development. Section 21.5 emphasizes the possibility of multiple equilibria due to aggregate demand externalities. In the model presented in this section, one of the multiple equilibria approximates a situation without industrialization and growth, whereas the other one features industrialization. Section 21.6 investigates the importance of income inequality for economic development and shows how the interaction of imperfect capital markets with income inequality can lead to multiple steady states, again with different levels of efficiency and productivity. I also use the models in this section to emphasize

the difference between multiple equilibria and multiple steady states, and I provide a brief discussion of richer models of income inequality dynamics and their implications for economic development. Finally, Section 21.7 provides a reduced-form model that emphasizes some of the common themes in the approaches covered in this chapter. While each model in this chapter makes quite different assumptions, there are sufficiently many common elements that my hope is that an attempt to bring out the similarities, even if in a highly reduced-form way, will provide additional insights.

The topics covered in this chapter are part of a large and diverse literature. My purpose is not to do justice to this literature, but to emphasize how certain major structural transformations take place as part of the process of economic development and also highlight the potential importance of market failures in this process. Given this objective and the large number of potential models, my choice of models is selective and my treatment will be more informal than the rest of the book. In addition, I will often make reduced-form assumptions in order to keep the exposition brief and simple.

21.1. Financial Development

An important aspect of the structural transformation brought about by economic development is a change in financial relations and deepening of financial markets. Section 17.6 in Chapter 17 already presented a model where economic growth goes hand-in-hand with financial deepening. However, the model in that section only focused on a specific aspect of the role of financial institutions. In general, financial development brings about a number of complementary changes in the economy. First, there is greater depth in the financial market, allowing better diversification of aggregate risks, a feature also emphasized in the model of Section 17.6. Second, one of the key roles of financial markets is to allow risk sharing and consumption smoothing for individuals. In line with this, financial development also allows better diversification of *idiosyncratic risks*. We have seen in Section 17.6 that better diversification of aggregate risks leads to a better allocation of funds across sectors/projects. Similarly, better sharing of idiosyncratic risks will lead to a better allocation of funds across individuals. Third, financial development might also reduce credit constraints on investors and thus also directly enable the transfer of funds to individuals with better investment opportunities. The second and the third channels not only affect the allocation of resources in the society but also the distribution of income, because diversification of idiosyncratic risks and relaxation of credit market constraints might lead to better income and risk sharing. On the other hand, as the possibility of such risk-sharing arrangements reduce consumption risk, individuals might also take riskier actions, potentially affecting the distribution of income. A complete analysis of the issues surrounding financial development and its interactions with economic growth are beyond the scope of this chapter. As already hinted, existing evidence suggests that financial development and economic development go hand-in-hand and many economists interpret this as, at least partly, reflecting the causal effect of financial development on economic growth. A full analysis of issues related to financial development must both

study the relevant theoretical issues and also investigate the empirical relationship between finance and growth.

Here I will instead present a simple model of financial development, focusing on the diversification of idiosyncratic risks and complementing the analysis in Section 17.6. The model is inspired by the work of Townsend (1979) and Greenwood and Jovanovic (1990) and adopts some of the modeling features of the model of Acemoglu and Zilibotti (1997) in Section 17.6. It will illustrate how financial development takes place endogenously and interacts with economic growth, and will also provide some simple insights about the implications of financial development for income distribution. Given the similarity of the model to that in Section 17.6, my treatment here will be relatively quick and informal.

I consider an overlapping generations economy in which each individual lives for two periods and has preference given by

$$(21.1) \quad \mathbb{E}_t U(c(t), c(t+1)) = \log c(t) + \beta \mathbb{E}_t \log c(t+1),$$

where $c(t)$ denotes the consumption of the unique final good of the economy and $\mathbb{E}_t$ denotes the expectation operator given time t information. As in Chapter 9 and also in Section 17.6, these preferences are convenient since they ensure a constant savings rate.

There is no population growth and the total population of each generation is normalized to 1. Let us assume that each individual is born with some labor endowment l . The distribution of endowments across agents is given by the distribution function $G(l)$ over some support $[l, \bar{l}]$. This distribution of labor endowments is constant over time with mean $L = 1$ and labor is supplied inelastically by all individuals in the first period of their lives. In the second period of their lives, individuals cannot supply labor and can only consume their capital income.

The aggregate production function of the economy is given by

$$Y(t) = K(t)^\alpha L(t)^{1-\alpha} = K(t)^\alpha,$$

where $\alpha \in (0, 1)$ and the second equality uses the fact that total labor supply will be equal to 1 at each date. As in Section 17.6, the only risk is in transforming savings into capital, thus the lifecycle of an individual looks identical to that shown in Figure 17.3 in that section. Moreover, suppose that agents can either save all of their labor earnings from the first period of their lives using a safe technology with rate of return q (in terms of capital at the next date) or invest all of their labor income in the risky technology with return $Q + \varepsilon$, where ε is a mean zero independently and identically distributed stochastic shock and

$$Q > q.$$

This implies that the risky technology is more productive. The assumption that individuals have to choose one of these two technologies rather than dividing their savings between the two is made for simplicity only (see Exercise 21.1).

Although the model looks very similar to that in Section 17.6, there is a crucial difference. Because ε is identically and independently distributed *across individuals*, if individuals could

pool their resources, they would get rid of the idiosyncratic risk and enjoy the higher return Q . In particular, if a large number (a continuum) all individuals pooled their resources, they would guarantee an average return of Q . Let us assume that this is not possible because of a standard *informational problem*—the actual return of an individual's saving decision is not observed by others unless some financial monitoring is undertaken. Let us assume that this type of financial monitoring costs $\xi > 0$ for each individual. This implies that by paying the cost of ξ , each individual can join the financial market (or in the language of Townsend, he can become part of a “financial coalition”) and in this case, the actual return of his savings become fully observable. Intuitively, this cost captures the fixed costs that individuals have to pay to be engaged in financial markets as well as the fixed cost associated with monitoring or being monitored. An immediate implication of this specification is that joining the financial markets is more attractive for richer individuals, since the fixed cost is less important for them. This feature is both plausible and also generates predictions consistent with microdata, where we observe richer individuals investing in more complex financial securities.

If the individual does not join the financial markets, then no other agent in the economy can observe the realization of the returns on his savings. In this case, no financial contract for sharing of idiosyncratic risks is possible, since such a contract would involve agents that have a high (realized) value of ε making transfers to those who are unlucky and have low realized values of ε . However, without monitoring, each agent will claim to have a low value of ε , thus receive rather than make ex post payments. The anticipation of this type of opportunistic behavior prevents any risk sharing in the absence of monitoring.

Let us also assume that ε has a distribution that places positive probability on $\varepsilon = -Q$. This implies that if an individual undertakes the risky investments, there is a positive probability that all his savings will be lost. This implies that without some type of risk sharing, individuals would always choose the safe project. This observation significantly simplifies the analysis of the model. Suppose that the economy starts with some initial capital stock of $K(0) > 0$, so an individual with labor endowment l_i will have labor earnings of

$$W_i(0) = w(0)l_i,$$

where

$$(21.2) \quad w(t) = (1 - \alpha)K(t)^\alpha$$

is the competitive wage rate at time t . After labor incomes are realized, individuals first make their savings decisions and then choose which assets to invest in. The preferences in (21.1) imply that individuals will save a constant fraction

$$\frac{\beta}{1 + \beta}$$

of their income regardless of their income level or the rate of return (in particular, independent of whether they are investing in the risky or the safe asset). In view of this, the value to not participating in the financial markets for individual i at time t is

$$V_i^N(W_i(t), R(t+1)) = \log\left(\frac{1}{1+\beta}W_i(t)\right) + \beta \log\left(\frac{\beta R(t+1)q}{1+\beta}W_i(t)\right),$$

which takes into account that the rate of return on capital in the second period of the life of the individual will be $R(t+1)$ and the individual will receive a gross return q on his savings of $\beta W_i(t)/(1+\beta)$. In contrast, when the individual decides to take part in financial markets (presuming that there are sufficiently many other individuals also taking part in financial markets to provide risk diversification, which here means a positive measure of individuals doing so), his value will be

$$V_i^F(W_i(t), R(t+1)) = \log\left(\frac{1}{1+\beta}(W_i(t) - \xi)\right) + \beta \log\left(\frac{\beta R(t+1)Q}{1+\beta}(W_i(t) - \xi)\right),$$

which takes into account that the individual will have to spend the amount ξ out of his labor income on the cost of joining the financial market, leaving him a net income of $W_i(t) - \xi$. He will then save a fraction $\beta/(1+\beta)$ of this, but in return, he will receive the higher gross return Q . The reason why the individual will necessarily receive Q , rather than a risky return, is because, conditional on joining the financial market, each individual is able to fully diversify his idiosyncratic risks and therefore receive the average return Q . The comparison of these two expressions immediately gives the threshold level

$$(21.3) \quad W^* \equiv \frac{\xi}{1 - (q/Q)^{\beta/(1+\beta)}} > 0,$$

such that individuals with first-period earnings greater than W^* will join the financial market and those with less than W^* will not. A notable feature of this threshold W^* is that it is independent of the rate of return on capital in the second period of the lives of the individuals, R . This is an implication of log preferences in (21.1).

Given the behavior of individuals concerning whether they will join the financial market, let us turn to determine the evolution of the economy by studying the evolution of individual earnings. Individual earnings are determined by two factors: individual labor endowments and the capital stock at time t , which gives the wage per unit of labor, $w(t)$, as in (21.2). In particular, suppose that at time t the wage is given by $w(t)$. Then, the fraction of individuals who will join the financial market at time t , $g^F(t)$, is given by the fraction of individuals who have $l_i \geq W^*/w(t)$. Alternatively, using the fact that labor endowments have a distribution given by $G(\cdot)$, the fraction of individuals investing in financial markets is obtained as

$$(21.4) \quad g^F(t) \equiv 1 - G\left(\frac{W^*}{w(t)}\right) = 1 - G\left(\frac{W^*}{(1-\alpha)K(t)^\alpha}\right).$$

In view of this, the capital stock at time $t + 1$ can be written as

$$(21.5) \quad K(t+1) = \frac{\beta}{1+\beta} \left[q \left(\int_l^{\frac{W^*}{(1-\alpha)K(t)^\alpha}} l dG(l) \right) (1-\alpha) K(t)^\alpha + Q \int_{\frac{W^*}{(1-\alpha)K(t)^\alpha}}^{\bar{l}} ((1-\alpha) K(t)^\alpha l - \xi) dG(l) \right],$$

which takes into account that all individuals with labor endowment less than $W^*/q(1-\alpha)K(t)^\alpha$ will choose the safe project and receive the gross return q on their savings, while those above this threshold will spend ξ on monitoring and receive the higher return Q . It can be verified that $K(t+1)$ is increasing in $K(t)$ and there will be growth in the capital stock (and thus output) of the economy provided that $K(t)$ is small enough (in particular, less than the “steady-state” level of capital when this is unique; see Exercise 21.2).

Now inspection of the accumulation eq. (21.5) together with the threshold rule for joining the financial market leads to a number of interesting conclusions.

1. As $K(t)$ increases, that is, as the economy develops, eq. (21.4) implies that more individuals will join the financial market. Consequently, a greater level of capital will lead to more risk taking, but these risks will also be shared better. More importantly, economic development also induces a better composition of investment as a greater fraction of the individuals start using their savings more efficiently. Thus with a mechanism similar to the model in Section 17.6, economic development improves the allocation of funds in the economy and . Consequently, this model, like the one in Section 17.6, implies that economic development and financial development go hand-in-hand.

2. However, there is also a distinct sense in which the economy here allows for a potential causal effect of financial development on economic growth. Imagine that societies differ according to their ξ 's, which can be interpreted as a measure of the institutionally- or technologically-determined costs of monitoring or some other costs associated with financial transactions that may depend on the degree of investor protection. Societies with lower ξ 's will have a greater participation in financial markets and this will endogenously increase their productivity. Thus, while the equilibrium behavior of financial and economic development are jointly determined, differences in financial development driven by exogenous institutional factors related to ξ will have a potential causal effect on economic growth.

3. As noted above, at any given point in time it will be the richer agents—those with greater labor endowment—that will join the financial market. Therefore, initially, the financial market will help those who are already well-off to increase the rate of return on their savings. This can be thought of as the *unequalizing* effect of the financial market.

4. The fact that participation in financial market increases with $K(t)$ also implies that as the economy grows, at least at the early stages of economic development, the unequalizing effect of financial intermediation will become stronger. Therefore, presuming that the economy starts with relatively few rich individuals, the first expansion of the financial market will increase the level of overall inequality in the economy as a greater fraction of the agents in the economy now enjoy the greater returns.

5. As $K(t)$ increases even further, eventually the *equalizing* effect of the financial market will start operating. At this point, the fraction of the population joining the financial market and enjoying the greater returns is steadily increasing. If the steady-state level of capital stock K^* is such that $\underline{l} \geq W^*/(1-\alpha)(K^*)^\alpha$, then eventually all individuals will join the financial market and they will all receive the same rate of return on their savings.

The last two observations are interesting in part because the relationship between growth and inequality is a topic of great interest to development economics (one to which I will return later in this chapter). One of the most important ideas in this context is that of the *Kuznets curve*, based on Simon Kuznets's observations, which claims that growth first increases income inequality in the society and then leads to a decline in inequality. Whether the Kuznets curve is a good description of the relationship between growth and inequality is a topic of current debate. While many European societies seem to have gone through a phase of increasing and then decreasing inequality during the growth process over the 19th century, the evidence for the 20th century is more mixed. Nevertheless, the last two observations show that a model with endogenous financial development based on risk sharing among individuals can generate a pattern consistent with the Kuznets curve. Whether there is indeed a Kuznets curve in general and if so, whether the mechanism highlighted here plays an important role in generating this pattern are areas for future theoretical and empirical work.

21.2. Fertility, Mortality and the Demographic Transition

Chapter 1 highlighted the major questions related to growth of income per capita over time and its dispersion across countries today. Our focus so far has been on these per capita income differences. Equally striking differences exist in the level of population across countries and over time. Figure 21.1 uses data from Maddison (2002) and shows the levels and the evolution of population in different parts of the world over the past 2000 years. The figure is in log scale, so a linear curve indicates a constant rate of population growth. The figure shows that starting about 250 years ago there is a significant increase in the population growth rate in many areas of the world. This accelerated population growth continues in much of the world, but importantly, the rate of population growth slows down in Western Europe sometime in the 19th century (though, thanks to immigration, not so in the Western Offshoots). There is no similar slow down of population growth in less-developed parts of the world. On the contrary, in many less developed nations, the rate of population growth seems to have increased over the past 50 or so years. We have already discussed one of the reasons for this in Chapter 4—the spread of antibiotics, basic sanitation and other health-care measures around the world that reduced the very high mortality rates in many countries. However, equally notable is the *demographic transition* in Western Europe, which is the term coined for the decline in fertility sometime during the 19th century (more precisely at different points during this century for different countries). Understanding why population has grown slowly and then accelerated to reach a breakneck speed of growth over the past 150 years and why population growth rates differ across countries are major questions for economic development

and economic growth. These questions are not only interesting because population levels are among the variables we would like to understand and explain, but also because one might sometimes wish to focus on differences in total income across societies rather than on income per capita differences. In this case, differences in population become a variable to focus on directly.

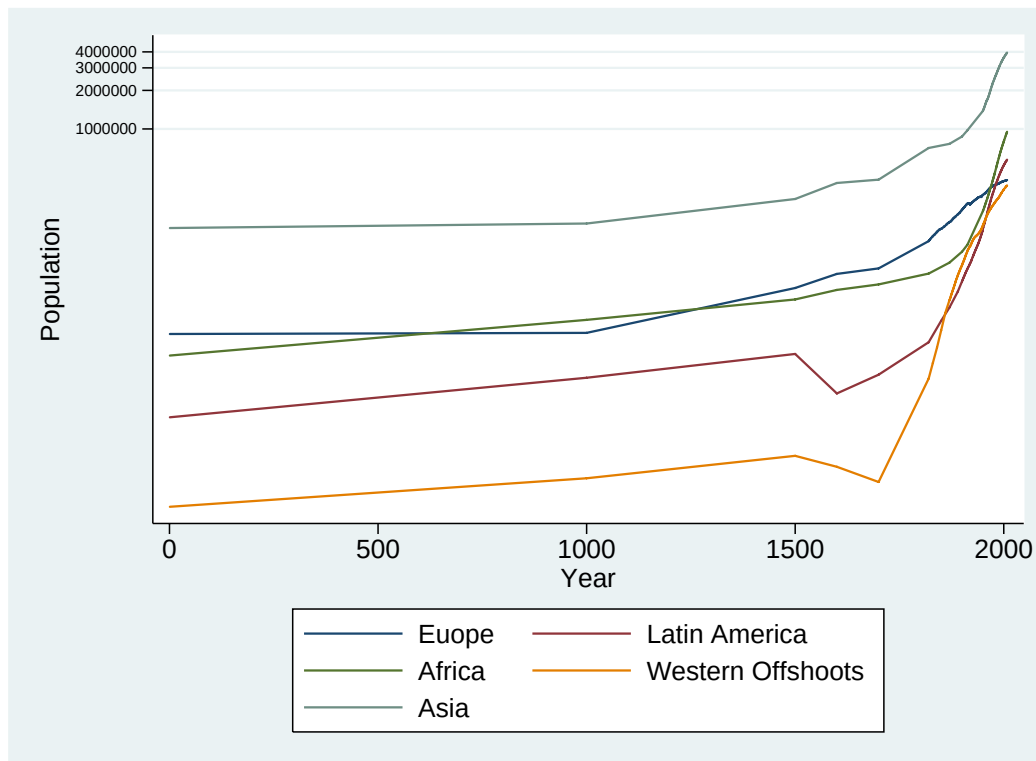


FIGURE 21.1. Total population in different parts of the world over the past 2000 years.

In this section, I present the most basic approaches to population dynamics and fertility. I first discuss a simple version of the famous Malthusian model and then use a variant of this model to investigate potential causes of the demographic transition. Thomas Malthus was one of the most brilliant and influential economists of the 19th century and is responsible for one of the first general equilibrium growth models. The next subsection will present a version of this model. The Malthusian model is responsible for earning the discipline of economics the name “the dismal science” because of its dire prediction that population will adjust up or down (by births or deaths) until all individuals are at the subsistence level of consumption. Nevertheless, this dire prediction is not the most important part of the Malthusian model. Instead, at the heart of this model is the *negative relationship* between population, which is itself endogenously determined, and income per capita. In this sense, it is closely related to the Solow model or the neoclassical growth model, augmented with a behavioral rule that determines the rate of population growth. It is this less extreme version

of the Malthusian model that will be presented in the next subsection. I will then enrich this model by the important and influential idea due to Gary Becker that there is a tradeoff between the quantity and quality of children and that this tradeoff changes over the process of development. I will show how a simple model can capture the process via which parents start valuing the quality (human capital) of their offspring more as the economy becomes richer and values human capital more. This process will eventually lead to the demographic transition, with families bearing in raring fewer, but more skilled children. Since my objective here is to introduce the main ideas rather than give a full account of this active research area, my treatment will be informal.

21.2.1. A Simple Malthusian Model. Consider the following non-overlapping generations model that starts with a population of $L(0) > 0$ at time $t = 0$. Each individual living at time t supplies one unit of labor inelastically and has preferences given by

$$(21.6) \quad c(t)^\beta \left[y(t+1)(n(t+1) - 1) - \frac{1}{2}\eta_0 n(t+1)^2 \right],$$

where $c(t)$ denotes the consumption of the unique final good of the economy by the individual himself, $n(t+1)$ denotes the number of offspring the individual begets and $y(t+1)$ is the income of each offspring, and $\beta > 0$ and $\eta_0 > 0$. The last term in square brackets is the child-rearing costs and are assumed to be convex to reflect the fact that the costs of having more and more children will be higher (for example, because of time constraints of parents, though one can also make arguments for why the costs of child-rearing might exhibit increasing returns to scale over a certain range). Clearly, these preferences introduce a number of simplifying assumptions. First, each individual is allowed to have as many offspring as it likes, which is unrealistic because it does not restrict the number of offspring to a natural number. The technology also does not incorporate possible specialization in child-rearing and market work within the family. Second, these preferences introduce the “warm glow” type altruism we encountered in Chapter 9, so that parents receive utility not from the future utility of their offspring, but from some characteristic of their offspring. Here it is a transform of the total income of all the offspring that features in the utility function of the parent. Third, the costs of child-rearing are in terms of “utils” rather than forgone income, and current consumption multiplies both the benefits and the costs of having additional children. This feature, which is motivated by balanced growth type reasoning, implies that the demand for children will be independent of current income (otherwise, growth will automatically lead to greater demand for children). All three of these assumptions are adopted for simplicity. I have also written the number of offspring that an individual has a time t as $n(t+1)$, since this will determine population at time $t+1$.

Each individual has one unit of labor and there are no savings. The production function for the unique good takes the form

$$(21.7) \quad Y(t) = Z^\alpha L(t)^{1-\alpha},$$

where Z is the total amount of land available for production and $L(t)$ is total labor supply. There is no capital and land is introduced in order to create diminishing returns to labor, which is an important element of the Malthusian model. Without loss of generality, I normalize the total amount of land to $Z = 1$. A key question in models of this sort is what happens to the returns to land. The most satisfactory way of dealing with this problem would be to allocate the property rights to land among the individuals and let them bequeath this to their offspring. This, however, introduces another layer of complication, and since my purpose here is to illustrate the basic ideas, I will follow the unsatisfactory assumption often made in the literature, that land is owned by another set of agents, whose behavior will not be analyzed here.

By definition, population at time $t + 1$ is given as

$$(21.8) \quad L(t+1) = n(t+1)L(t),$$

which takes into account new births as well as the death of the parent.

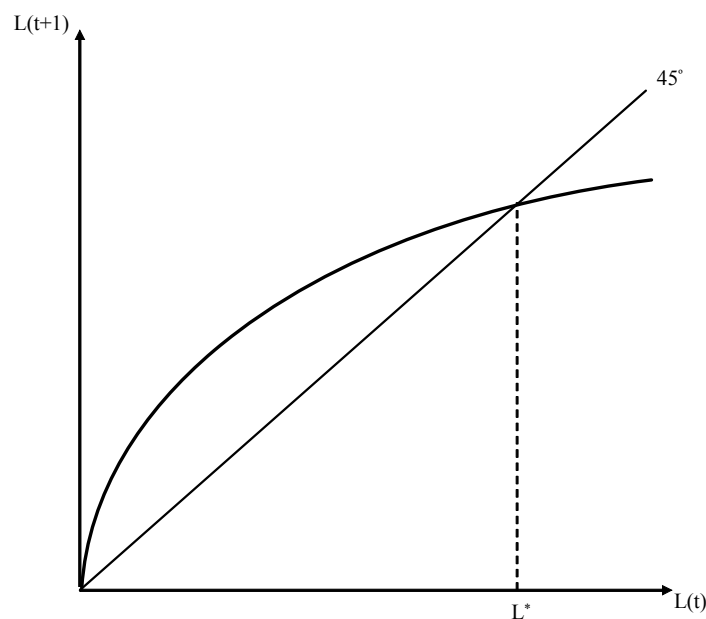


FIGURE 21.2. Population dynamics in this simple Malthusian model.

Labor markets are competitive, so the wage at time $t + 1$ is given as

$$(21.9) \quad w(t+1) = (1 - \alpha)L(t+1)^{-\alpha}.$$

Since there is no other source of income, this is also equal to the income of each individual living at time $t + 1$, $y(t + 1)$. Thus an individual with income $w(t)$ at time t will solve the problem of maximizing (21.6) subject to the constraint that $c(t) \leq w(t)$, together with $y(t + 1) = (1 - \alpha)L(t + 1)^{-\alpha}$, which implies that the individual takes the population level in

the next period as given (since he is infinitesimal). Let us focus on a symmetric equilibrium, which will naturally require the choice of $n(t+1)$ to be consistent with $L(t+1)$ according to eq. (21.8). This maximization problem immediately gives $c(t) = w(t)$ and

$$n(t+1) = (1 - \alpha) \eta_0^{-1} L(t+1)^{-\alpha}.$$

Now substituting for (21.8) and rearranging,

$$(21.10) \quad L(t+1) = (1 - \alpha)^{\frac{1}{1+\alpha}} \eta_0^{-\frac{1}{1+\alpha}} L(t)^{\frac{1}{1+\alpha}}.$$

This equation implies that $L(t+1)$ is an increasing concave function of $L(t)$. In fact, the law of motion for population implied by (21.10) resembles the dynamics of capital-labor ratio in the Solow growth model (or the overlapping generations model) and is plotted in Figure 21.2. The figure makes it clear that starting with any $L(0) > 0$, there exists a unique globally stable state L^* given by

$$(21.11) \quad L^* \equiv (1 - \alpha)^{1/\alpha} \eta_0^{-1/\alpha}.$$

If the economy starts with $L(0) < L^*$, then population will slowly (and monotonically) adjust towards this steady-state level. Moreover, (21.9) shows that as population increases wages will fall. If in contrast, $L(0) > L^*$, then the society will experience a decline in population and rising real wages. It is straightforward to introduce shocks to population and show that in this case the economy will fluctuate around the steady-state population level L^* (with an invariant distribution depending on the distribution of the shocks) and experience cycles reminiscent to the Malthusian cycles, with periods of increasing population and decreasing wages followed by periods of decreasing population and increasing wages (see Exercise 21.3). The main difference of this model from the simplest (or the crudest) version of the Malthusian model is that there is no biologically determined subsistence level of consumption. The level of consumption will tend to a constant given by

$$\begin{aligned} c^* &= (1 - \alpha) (L^*)^{-\alpha} \\ &= \eta_0, \end{aligned}$$

though this is not determined biologically, but by preferences and technology.

21.2.2. The Demographic Transition. I now extend the basic Malthusian model of the previous subsection to study the demographic transition. First, I introduce a quality-quantity tradeoff along the lines of the ideas suggested by Gary Becker. Each parent can choose his offspring to be unskilled or skilled. To make them skilled, the parent has to exert the additional effort for child-rearing denoted by $e(t) \in \{0, 1\}$. If he chooses not to do this, his offspring will be unskilled.

The total population of unskilled individuals at time t is denoted by $U(t)$ and the total population of the skilled are denoted by $S(t)$, clearly with

$$L(t) = U(t) + S(t).$$

The second modification is that there are now two production technologies that can be used for producing the final good. The Malthusian (traditional) technology is still given by (21.7) and any worker can be employed with the Malthusian technology. The modern technology is given by

$$(21.12) \quad Y^M(t) = X(t) S(t).$$

This equation implies that productivity in the modern technology is potentially time varying and also states that only skilled workers can be employed with this technology. It also imposes that all skilled workers will be employed with this technology. Naturally, this need not be true in general (there may be an excess supply of skilled workers). However, this will never be the case in equilibrium, since parents would not choose to exert the additional effort to endow their offspring with skills if they would then work in the traditional sector. In the interest of keeping the exposition brief and simple (and with a slight abuse of notation), eq. (21.12) already incorporates the fact that all skilled workers will be employed in the modern sector.

To model the quality-quantity tradeoff, individual preferences are now modified from eq. (21.6) to

$$(21.13) \quad c(t)^\beta \left[y(t+1)(n(t+1)-1) - \frac{1}{2}(\eta_0(1-e(t)) + \eta_1 X(t+1)e(t))n(t+1)^2 \right].$$

This formulation of the preferences states that if the individual decides to invest in his offspring' skills, instead of the fixed cost η_0 he has to pay a cost that is proportional to the amount of knowledge $X(t+1)$ that the offspring has to absorb to use the modern technology. I assume that $X(0)\eta_1 > \eta_0$, so that even at the initial level of the modern technology rearing a skilled child is more costly than an unskilled child.

Finally, I assume learning-by-doing is external as in Romer (1986a), so that

$$(21.14) \quad X(t+1) = \delta S(t),$$

which implies that the improvement in the technology of the modern sector is a function of the number of skilled workers employed in this sector. This type of reduced-form assumption is clearly unsatisfactory, but as noted above, one could get similar results with an endogenous technology model with the market size effect. Another important feature of this production function is that it does not use land. This assumption is consistent with the fact that most modern production processes make little use of land, instead relying on technology, physical capital and human capital. Equation (21.12) captures this in a simple form, though it does so without introducing physical capital.

The output of the traditional and the modern sectors are perfect substitutes—they both produce the same final good. In view of the observation that all unskilled workers will work in the traditional sector and all skilled workers will work in the modern sector, wages of skilled and unskilled workers at time t are

$$(21.15) \quad w^U(t) = (1-\alpha)U(t)^{-\alpha},$$

and

$$(21.16) \quad w^S(t) = X(t),$$

where (21.15) is identical to (21.9) in the previous subsection, except that it features only the unskilled workers instead of the entire labor force.

Let us next turn to the fertility and quality-quantity decisions of individuals. As before, each individual will consume all his income and his income level has no effect on his fertility and quality-quantity decisions. Thus we do not need to distinguish between high-skill and low-skill parents. Using this observation, let us simply look at the optimal number of offspring that an individual will have when he chooses $e(t) = 0$. This is given by

$$(21.17) \quad \begin{aligned} n^U(t+1) &= w^U(t+1) \eta_0^{-1} \\ &= (1-\alpha) U(t+1)^{-\alpha} \eta_0^{-1}, \end{aligned}$$

where the second line uses (21.15). Instead, if the parent decides to exert effort $e(t) = 1$ and invest in the skills of his offspring, then he will choose the number of offspring equal to

$$(21.18) \quad \begin{aligned} n^S(t+1) &= w^S(t+1) X(t+1)^{-1} \eta_1^{-1} \\ &= \eta_1^{-1}. \end{aligned}$$

The comparison of eq.'s (21.17) and (21.18) suggests that unless unskilled wages are very low, an individual who decides to provide additional skills to his offspring will have fewer offspring. This is because bringing up skilled children is more expensive. Thus the comparison of these two equations captures the quality-quantity tradeoff.

Substituting these equations back into the utility function (21.13), we obtain the utility from the two strategies (normalized by consumption) as

$$V^U(t) = \frac{1}{2} (1-\alpha)^2 U(t+1)^{-2\alpha} \eta_0^{-1}$$

and

$$V^S(t) = \frac{1}{2} X(t+1) \eta_1^{-1}.$$

Inspection of these two expressions shows that we can never have an equilibrium in which all offspring are skilled, since otherwise V^U would become unboundedly large. Therefore, in equilibrium

$$(21.19) \quad V^U(t) \geq V^S(t).$$

This equilibrium condition implies that there are two possible configurations. First, $X(0)$ can be so low that (21.19) will hold as a strict inequality. In this case, all offspring will be unskilled. The condition for this inequality to be strict is

$$X(0) \eta_1^{-1} < (1-\alpha)^2 L(1)^{-2\alpha} \eta_0^{-1},$$

which uses the fact that when there are no skilled workers there is no production in the modern sector and thus $X(1) = X(0)$. If this inequality satisfied, there would be no skilled children at date $t = 0$. However, as long as $L(1)$ is less than L^* as given in (21.11), population will

grow. It is therefore possible that at some point (21.19) holds with equality. The condition for this never to happen is that

$$(21.20) \quad X(0) \eta_1^{-1} < (1 - \alpha)^2 (L^*)^{-2\alpha} \eta_0^{-1}.$$

In this case, the law of motion of population is identical to that in the previous subsection and there is never any investment in skills. We can think of this as a pure Malthusian economy.

If, on the other hand, condition (21.20) is not satisfied, then at least at some point individuals will start investing in the skills of their offspring and the modern sector will have skilled workers to employ. From then on, eq. (21.19) must hold as equality. Let the fraction of parents having unskilled children at time t be denoted by $u(t+1)$. Then, by definition

$$(21.21) \quad \begin{aligned} U(t+1) &= u(t+1) (n^U(t+1) - 1) L(t) \\ &= u(t+1)^{1/(1+\alpha)} (1 - \alpha)^{2/(1+\alpha)} \eta_0^{-1/(1+\alpha)} L(t)^{1/(1+\alpha)} \end{aligned}$$

and

$$(21.22) \quad \begin{aligned} S(t+1) &= (1 - u(t+1)) (n^S(t+1) - 1) L(t) \\ &= (1 - u(t+1)) \eta_1^{-1} L(t). \end{aligned}$$

Moreover, to satisfy (21.19) as equality, we need $(1 - \alpha)^2 U(t+1)^{-2\alpha} \eta_0^{-1} = X(t+1) \eta_1^{-1}$, or

$$(21.23) \quad X(t+1) \eta_1^{-1} = u(t+1)^{-2\alpha/(1+\alpha)} (1 - \alpha)^{2(1-\alpha)/(1+\alpha)} \eta_0^{-(1-\alpha)/(1+\alpha)} L(t)^{-2\alpha/(1+\alpha)}.$$

Equilibrium dynamics are then determined by eq.'s (21.21)-(21.23) together with (21.16). While the details of the behavior of this dynamical system are somewhat involved, the general picture is clear. If an economy starts with both a low level of $X(0)$ and a low level of $L(0)$, but does not satisfy condition (21.20), then the economy will start in the Malthusian regime, only making use of the traditional technology and not investing in skills. As population increases wages fall, and at that point parents start finding it beneficial to invest in the skills of their children and firms start using the modern technology. Those parents that invest in the skills of their children have fewer children than parents rearing unskilled offspring. The rate of population growth and fertility are high at first, but as the modern technology improves and the demand for skills increases, a larger fraction of the parents start investing in the skills of their children and the rate of population growth declines. Ultimately, the rate of population growth approaches η_1^{-1} . As a result, this model gives a very stylized representation of the demographic transition.

In the literature, there are richer models of the demographic transition. For example, there are many ways of introducing quality-quantity tradeoffs in the utility function of the parents, and what spurs a change in the quality-quantity tradeoff may be an increase in capital intensity of production, changes in the wages of workers or even changes in the wages of women differentially affecting the desirability of market and home activities. Nevertheless, the general qualitative features are similar in that the quality-quantity tradeoff is often viewed

as the major reason for the demographic transition. Despite this emphasis on the quality-quantity tradeoff, there is relatively little direct evidence that this tradeoff is important in general or in leading to the demographic transition. Other social scientists have suggested social norms, the large declines in mortality starting in the 19th century, or the reduced need for child labor as potential factors contributing to the demographic transition. As of yet, there is no general consensus on the causes of the demographic transition or on the role of the quality-quantity tradeoff in determining population dynamics. The study of population growth and demographic transition is an exciting and important area, and theoretical and empirical analyses of the factors affecting fertility decisions and how they interact with the reallocation of workers across different tasks (sectors) remain important and interesting questions to be explored.

21.3. Migration, Urbanization and The Dual Economy

Another major structural transformation over the process of development relates to changes in social and living arrangements. For example, as an economy develops, more individuals move from rural areas to cities and also undergo the social changes associated with separation from a small community and becoming part of a larger, more anonymous environment. Other social changes might also be important. For instance, certain social scientists regard the replacement of “collective responsibility systems” by “individual responsibility systems” as an important social transformation. This is clearly related to changes in the living arrangements of individuals (villages versus cities, or extended versus nuclear families). It is also linked to whether different types of contracts are being enforced by social norms and community enforcement, or whether they are enforced by legal institutions. There may also be a similar shift in the importance of the market, as more activities are mediated via prices rather than taking place inside the home or using the resources of an extended family or a broader community. This process of social change is both complex and interesting to study, though a detailed discussion of the literature and possible approaches to these issues falls beyond the scope of the current book. Nevertheless, a brief discussion of some of these social changes are useful to illustrate other, more diverse facets of structural change associated with economic development. I will illustrate the main ideas by focusing on the process of migration from rural areas and urbanization. Another reason to study migration and urbanization is that the reallocation of labor from rural to urban areas is closely related to the popular concept of *the dual economy*, which is an important theme of some of the older literature on development economics. According to this notion, less-developed economies consist of a modern sector and a traditional sector, but the connection between these two sectors is imperfect. The model of industrialization in the previous chapter (Section 20.3) featured a traditional and a modern sector, but these sectors traded their outputs and competed for labor in competitive markets. Dual economy approaches, instead, emphasize situations in which the traditional and the modern sectors function in parallel, but with only limited interactions. Moreover, the traditional sector is often viewed as less efficient than the

modern sector, thus the lack of interaction may also be a way of shielding the traditional economy from its more efficient competitor. A natural implication of this approach will then be to view the process of development as one in which the less efficient traditional sector is replaced by the more efficient modern sector. Lack of development may in turn correspond to an inability to secure such reallocation.

In this section I first present a model of migration that builds on the work by Arthur Lewis (1954). A less-developed economy is modeled as a dual economy, with the traditional sector associated with villages and the modern sector with the cities. The model enables us to study how and whether the reallocation of resources from the traditional sector to the modern sector will take place. I will then present a model inspired by Banerjee and Newman's (1998) article, as well as by Acemoglu and Zilibotti (1999), in which the traditional sector and the rural economy have a comparative advantage in community enforcement, even though in line with the other dual economy approaches, the modern economy (the city) enables the use of more efficient technologies. This model will also illustrate how certain aspects of the traditional sector can shield the less productive firms from more productive competitors and slow down the process of development. Finally, I will show how the import of technologies from more developed economies, along the lines of the models discussed in Section 18.4 of Chapter 18, may also lead to dual economy features as a byproduct of the introduction of more skill-intensive, modern technologies into less-developed economies.

21.3.1. Surplus Labor and the Dual Economy. The main emphasis of Lewis's work was on the idea of *surplus labor*. Lewis argued that less-developed economies typically had surplus labor, that is, unemployed or underemployed labor, often in the villages. The dual economy can then be viewed as the juxtaposition of the modern sector, where workers are gainfully and productively employed, onto the traditional sector where they are underemployed. The general tendency of less-developed economies to have higher levels of unemployment (and lower levels of employment to population ratios) was one of the motivations for Lewis's model. A key feature of Lewis's model is the presence of some barriers preventing, or slowing down, the allocation of workers away from the traditional sector towards urban areas and the modern sector. I now present a reduced-form model that formalizes these notions.

Consider a continuous-time infinite-horizon economy that consists of two sectors or regions, which I will refer to as urban and rural. Total population is normalized to 1. At time $t = 0$, $L^U(0)$ individuals are in the urban area and $L^R(0) = 1 - L^U(0)$ are in the rural area. In the rural area, the only economic activity is agriculture and, for simplicity, suppose that the production function for agriculture is linear, thus total agricultural output is

$$Y^A(t) = B^A L^R(0),$$

where $B^A > 0$. In the urban area, the main economic activity is manufacturing. Manufacturing can only employ workers in the urban area and will employ all of the available workers. The production function therefore takes the form

$$Y^M(t) = F(K(t), L^U(t)),$$

where $K(t)$ is the capital stock, with initial condition $K(0)$. F is a standard neoclassical production function satisfying Assumptions 1 and 2. Let us also assume, for simplicity, that the manufacturing and the agricultural goods are perfect substitutes. Labor markets both in the rural and urban area are competitive. There is no technological change in either sector.

The key assumptions of this model are twofold. First, the marginal product of labor, and thus the wage, in manufacturing will be higher than in agriculture. Second, because of barriers to mobility, there will only be slow migration of workers from rural to urban areas.

In particular, let us capture the dynamics in this model in a reduced-form way whereby capital accumulates only out of the savings of individuals in the urban area, thus

$$(21.24) \quad \dot{K}(t) = sF(K(t), L^U(t)) - \delta K(t),$$

where s is the exogenous saving rate and δ is the depreciation rate of capital. The important feature implied by this specification is that greater output in the modern sector leads to further accumulation of capital for the modern sector. An alternative, adopted in Section 20.3 of the previous chapter that will also be used in the next subsection, is to allow the size of the modern sector to directly influence its productivity growth, for example because of learning-by-doing externalities as in Romer (1986a) or because of endogenous technological change depending on the market size commanded by this sector (e.g., Exercise 20.19). For the purposes of the model in this subsection, which of these alternatives is adopted has no major consequences.

Given competitive labor markets, the wage rates in the urban and rural areas at date t are given by

$$w^U(t) = \frac{\partial F(K(t), L^U(t))}{\partial L} \text{ and } w^R(t) = B^A.$$

Let us assume that

$$(21.25) \quad \frac{\partial F(K(0), 1)}{\partial L} > B^A,$$

so that even if all workers are employed in the manufacturing sector at the initial capital stock, they will have a higher marginal product than working in agriculture.

Migration dynamics are assumed to take the following simple form:

$$(21.26) \quad \dot{L}^R(t) \begin{cases} = -\mu L^R(t) & \text{if } w^U(t) > w^R(t) \\ \in [-\mu L^R(t), 0] & \text{if } w^U(t) = w^R(t) \\ = 0 & \text{if } w^U(t) < w^R(t) \end{cases}$$

This equation implies that as long as wages in the urban sector are greater those in the rural sector, there is a constant rate of migration. The speed of migration does not depend on the wage gap, which is an assumption adopted only to simplify the exposition. We may want to think of μ as small, so that there are barriers to migration and even when there are substantial gains to migrating to the cities, migration will take place slowly. When there is no wage gain to migrating, there will be no migration.

Now (21.25) implies that at date $t = 0$, there will be migration from the rural areas towards the cities. Moreover, assuming that $K(0)/L^U(0)$ is below the steady-state capital-labor ratio, the wage will remain high and will continue to attract further workers. To analyze this process in slightly greater detail, let us define

$$k(0) \equiv \frac{K(0)}{L^U(0)}$$

as the capital-labor ratio in manufacturing (the modern sector). As usual, let us also define the per capita production function in manufacturing as $f(k(t))$. Clearly,

$$w^U(t) = f(k(t)) - k(t)f'(k(t)).$$

Combining (21.24) and (21.26), we obtain that, as long as $f(k(t)) - k(t)f'(k(t)) > B^A$, the dynamics of this capital-labor ratio will be given by

$$(21.27) \quad \dot{k}(t) = sf(k(t)) - (\delta + \mu\nu(t))k(t),$$

where $\nu(t) \equiv L^R(t)/L^U(t)$ is the ratio of the rural to urban population. Notice that when urban wages are greater than rural wages, the rate of migration, μ , times the ratio $\nu(t)$, plays the role of the rate of population growth in the basic Solow model of Chapter 2. In contrast, when $f(k(t)) - k(t)f'(k(t)) \leq B^A$, there is no migration and $\dot{k}(t) = sf(k(t)) - \delta k(t)$. Let us focus on the former case. Define the level of capital-labor ratio $\bar{k}$ such that

$$(21.28) \quad f(\bar{k}) - \bar{k}f'(\bar{k}) = B^A,$$

where urban and rural wages are equalized. Once this level is reached, migration will stop, and so $\nu(t)$ will remain constant. After this level, equilibrium dynamics are given by $\dot{k}(t) = sf(k(t)) - \delta k(t)$. Therefore, the steady state must always involve

$$(21.29) \quad \frac{sf(\hat{k})}{\hat{k}} = \delta.$$

For the analysis of transitional dynamics, which are our primary interest here, there are several cases to study. Let us focus on the one that appears most relevant for the experiences of many less-developed economies (leaving the rest to Exercise 21.4). In particular, suppose that the following conditions hold:

- (1) $k(0) < \hat{k}$, so that the economy starts with lower capital-labor ratio (in the urban sector) than the steady-state level. This assumption also implies that $sf(k(0)) - \delta k(0) > 0$.
- (2) $k(0) > \bar{k}$, which implies that $f(k(0)) - k(0)f'(k(0)) > B^A$, that is, wages are initially higher in the urban sector than in the rural sector.
- (3) $sf(k(0)) - (\delta + \mu\nu(0))k(0) < 0$, so that given the distribution of population between urban and rural areas, the initial migration will lead to a decline in the capital-labor ratio.

In this case, the economy starts with rural to urban migration at date $t = 0$. Since initially $\nu(0)$ is high, this migration reduces the capital-labor ratio in the urban area (which evolves according to the differential equation (21.27)). There are then two possibilities. In

the first, the capital-labor ratio never falls below $\bar{k}$, thus rural to urban migration takes place at the maximum possible rate, μ , forever. Nevertheless, the effect of this migration on the urban capital-labor ratio is reduced over time as $\nu(t)$ declines with migration. Since we know that $sf(k(0)) - \delta k(0) > 0$, at some point the urban capital-labor ratio will start increasing, and it will eventually converge to the unique steady-state level $\hat{k}$. This convergence can take a long time and notably, it is monotone; the capital-labor ratio, and urban wages, first fall and then increase. The second possibility is that the initial surge in rural to urban migration reduces the capital-labor ratio to $\bar{k}$ at some point, say at date t' . When this happens, wages remain constant at B^A in both sectors and the rate of migration $\dot{L}^R(t)/L^R(t)$ adjusts exactly so that capital-labor ratio remains at $\bar{k}$ for a while (recall that when urban and rural wages are equal, (21.26) admits any level of migration between zero and the maximum rate μ). In fact, the urban capital-labor ratio can remain at this level for an extended period of time. Ultimately, however, $\nu(t)$ will again decline sufficiently that the capital-labor ratio in the urban sector must start increasing. Once this happens, migration takes place at the maximal rate μ and the economy again slowly converges to the capital-labor ratio $\hat{k}$ in the urban sector.

Therefore, this discussion illustrates how a simple model of migration can generate rich dynamics of population in rural and urban areas and also wage differences between the modern and the traditional sectors.

The dynamics discussed above, especially in the first case, give the flavor of a *dual economy*. Wages and the marginal product of labor are higher in the urban area than in the rural area. If, in addition, μ is low, the allocation of workers from the rural to the urban areas will be slow, despite the higher wages. Thus the pattern of dual economy may be pronounced and may persist for a long time. It is also notable that rural to urban migration increases total output in the economy, because it enables workers to be allocated to activities in which their marginal product is higher. This process of migration increasing the output level in the economy also happens slowly because of the relatively slow process of migration.

The above discussion implies that, for the parameter configurations on which we have focused, the dual economy structure not only affects the social outlook of the society, which remains rural and agricultural for an extended period of time (especially when μ is small), but also leads to lower output than the economy could have generated by allocating labor more rapidly to the manufacturing sector. One should be cautious in referring to this as a “market failure,” however, since we did not specify the reason why migration is slow. Without providing a micro model for migration, it is difficult to conclude whether the migration decisions are socially optimal or not (in the same sense as without a micro-founded model of savings, we could not talk of whether there was the right amount of savings and capital accumulation in the basic Solow growth model).

The model presented in this subsection therefore gives us a first formalization of a dual economy structure, which many development economists view as a good representation of the workings of less-developed economies. While dual economy features indeed appear to

be important in practice and the model presented here is indeed simple and tractable, there are various reasons for striving for more sophisticated models. First, the migration behavior in the current model is extremely reduced-form. This is important, since the migration behavior is at the heart of the model. The reduced-form formulation implies that we cannot ask questions about whether migration is optimal or suboptimal. Second, the model gives the flavor of too little migration, though in many less-developed economies many urban centers appear to be overpopulated. Thus it is useful to seek more insights on whether there will be too much or too little migration. Finally, the assumption that the manufacturing sector is more productive than agriculture is somewhat crude. While the dual economy structure suggests that one part of the economy will be more productive, it would be more satisfactory and insightful if there are some compensating differentials in the less productive sector. The model presented in the next subsection will rectify some of these shortcomings.

21.3.2. Community Enforcement, Migration and Development. I now present a model inspired by Banerjee and Newman (1998) and Acemoglu and Zilibotti (1999). Banerjee and Newman consider an economy where the traditional sector has low productivity but is less affected by informational asymmetries and thus individuals can engage in borrowing and lending with limited monitoring and incentive costs. In contrast, the modern sector is more productive, but informational asymmetries create more severe credit market problems. Banerjee and Newman discuss how the process of development is associated with the reallocation of economic activity from the traditional to the modern sector and how this reallocation is slowed down by the informational advantage of the traditional sector. Acemoglu and Zilibotti (1999) view the development process as one of “information accumulation,” and greater information enables individuals to write more sophisticated contracts and enter into more complex production relations. This process is then associated with changes in technology, changes in financial relations and social transformations, since greater availability of information and better contracts enable individuals to abandon less efficient and less information-dependent social and productive relationships.

The model in this subsection is simpler than both of these papers, but features a similar economic mechanism. Individuals who live in rural areas are subject to community enforcement. This means that they can enter into economic and social relationships without being unduly affected by moral hazard problems. When individuals move to cities, they can take part in more productive activities, but other enforcement systems are necessary to ensure compliance to social rules, contracts and norms. These systems will typically be associated with certain costs. As in the model of industrialization in the previous section, I will also assume that the modern sector is subject to learning-by-doing externalities. Thus the productivity advantage of the modern sector grows as more individuals migrate to cities and work there. However, the community enforcement advantage of villages slows down this process and may even lead to a development trap. Since the mathematical structure of the model is similar to that in the last section, my treatment will be relatively brief.

Both labor markets are competitive and total population is normalized to 1. There are three differences from the model in the previous subsection. First, migration between the rural and urban areas is costless. Thus at any point in time an individual can switch from one sector to another. Second, instead of capital accumulation, there is an externality, so that output in the modern sector is given by

$$Y^M(t) = X(t) F(L^U(t), Z),$$

where $X(t)$ denotes the productivity of the modern sector, which will be determined endogenously via learning-by-doing externalities. In addition, Z denotes another factor of production in fixed supply (so that there are diminishing returns to labor), and the production function F satisfies our standard assumptions, Assumptions 1 and 2. The returns to factor Z are distributed back to individuals (and how they are distributed has no effect on the results). Moreover, let us assume that the technology in the modern sector evolves according to the differential equation

$$\dot{X}(t) = \eta L^U(t) X(t)^\zeta,$$

where $\zeta \in (0, 1)$. This equation builds in learning-by-doing externalities along the lines of Romer's (1986a) paper and is also similar to the industrialization model of Section 20.3 in the previous chapter. The fact that $\zeta < 1$ implies that these externalities are less than those necessary for sustained growth.

Finally, let us also assume that rural areas have a comparative advantage in *community enforcement*. In particular, individuals engage in many social and economic activities, ranging from financial relations and employment to marriage and social relations. Many of these relationships in cities are anonymous and enforcement is through some type of monitoring by the law and relies on complex institutions. Such institutions often work imperfectly in most societies and particularly in less-developed economies. In contrast, rural areas house a small number of individuals who are typically engaged in long-term relationships. These long-term relationships enable the use of community enforcement in many activities. Thus with long-term relationships, individuals can pledge their reputation to borrow money to smooth consumption, to obtain information about which individual would be most appropriate for a particular job or to ensure cooperation in other work or social relations. I represent these in a reduced-form way by assuming that an individual pays a flow cost of $\xi > 0$ due to imperfect monitoring and lack of community enforcement when he is in the urban area.

All individuals maximize their utility, and since savings do not matter for the key allocation decisions, I do not specify utility functions. The key observation is that all individuals would like to maximize the net present discounted value of their lifetime incomes. But since moving between urban and rural areas is costless, this implies that each individual should work in the sector that has the higher net wage at that time. This implies that in an interior equilibrium (where both the rural and the urban sectors are active), the following wage equalization condition must hold:

$$w^M(t) - \xi = w^A(t).$$

Competitive labor markets then imply that

$$\begin{aligned} w^M(t) &= X(t) \frac{\partial F(L^U(t), Z)}{\partial L} \\ &\equiv X(t) \tilde{\phi}(L^U(t)), \end{aligned}$$

where the second line defines the function $\tilde{\phi}$, which is strictly decreasing in view of Assumption 1 on the production function F . Substituting from the above relationships, labor market clearing implies that

$$X(t) \tilde{\phi}(L^U(t)) = B^A + \xi,$$

or

$$\begin{aligned} L^U(t) &= \tilde{\phi}^{-1}\left(\frac{B^A + \xi}{X(t)}\right) \\ &\equiv \phi\left(\frac{X(t)}{B^A + \xi}\right), \end{aligned}$$

where again the second line defines the function ϕ , which is strictly increasing in view of the fact that $\tilde{\phi}$ (and thus $\tilde{\phi}^{-1}$) is strictly decreasing. Therefore, the evolution of this economy can be represented by the differential equation

$$\dot{X}(t) = \eta \phi\left(\frac{X(t)}{B^A + \xi}\right) X(t)^\zeta.$$

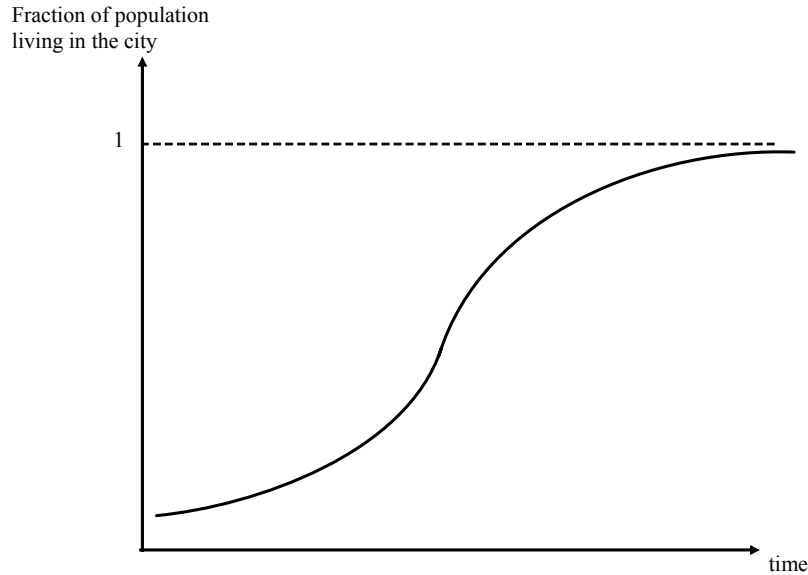


FIGURE 21.3. The dynamic behavior of the population in rural and urban areas.

A number of features about this law of motion are worth noting. First, the typical evolution of $X(t)$ will be given as in Figure 21.3, with an S-shaped pattern. This is because starting with a low initial value of $X(0)$, equilibrium in urban employment, $\phi(X(t) / (B^A + \xi))$,

will also be low during the early stages of development. This implies that there will be limited learning-by-doing and the modern sector technology will progress only slowly. However, as $X(t)$ increases, $\phi(X(t)/(B^A + \xi))$ will also increase, raising the rate of technological change in the modern sector. Ultimately, however, $L^U(t)$ cannot exceed 1, so $\phi(X(t)/(B^A + \xi))$ will tend to a constant, and thus the rate of growth of X will decline. Therefore, this reduced-form model generates an S-shaped pattern of technological change in the modern sector and an associated pattern of migration of workers from rural to urban areas.

Second and more importantly, the process of technological change in the modern sector and migration to the cities is slowed down by the comparative advantage of the rural areas in community enforcement. In particular, the greater is ξ , the slower is technological change and migration into urban areas. Since employment in the urban areas creates positive externalities, the community enforcement system in rural areas slows down the process of economic development in the economy as a whole. We may therefore conjecture that high levels of ξ , corresponding to greater community enforcement advantage of the traditional sector, will generally reduce growth and welfare in the economy. Counteracting this, however, are the static gains created by the better community enforcement system in rural areas. A high level of ξ will increase the initial level of consumption in the economy. Consequently, there is a tradeoff between dynamic and static welfare implications of different levels of ξ and this tradeoff is investigated formally in Exercise 21.5.

Finally, this model also offers a formalization of some of the ideas related to the dual economy. In contrast to the model of subsection 21.3.1, there are no mobility barriers, thus workers in the villages and cities receive the same wage. However, the functioning of the economy and the structure of social relations are different in these two areas. While villages and the rural economy rely on community enforcement, the city uses the modern technology and impersonal institutional checks in order to enforce various economic and social arrangements. Consequently, the dual economy in this model exhibits itself as much in the social dimension as in the economic dimension.

21.3.3. Inappropriate Technologies and the Dual Economy. I now discuss how ideas related to appropriate technology presented in Chapter 18 may provide promising clues about the nature of the dual economy patterns. Recall from Section 18.4 that less-developed economies often import their technologies from more advanced economies and that these technologies are typically designed for different factor proportions than those of the less-developed economy. For example, in Section 18.4, I emphasized the implications of a potential mismatch between the skills of the workforce of a less-developed economy and the skill requirements of modern technologies. However, the model in that section was designed such that the equilibrium (and the best option) for the less-developed economy was always to use the modern technology.

Consider a variant of a model in that section, where each technology is of the Leontief type, so that it requires a certain number of skilled and unskilled workers. For example,

technology A_h will produce a total of $A_h L$ units of the unique final good, where L is the number of unskilled workers, but this technology requires a ratio of skilled to unskilled workers exactly equal to h (for example, the skilled workers will be the managers or the supervisors for the unskilled workers). Suppose A_h is increasing in h , so that more advanced technologies are more productive.

Now consider a less-developed economy that has access to all technologies A_h for $h \in [0, \bar{h}]$ for some $\bar{h} < \infty$. Suppose that the population of this economy consists of H skilled and L unskilled workers, such that $H/L < \bar{h}$. This inequality implies that not all workers can be employed with the most skill-intensive technology. What will be the form of equilibrium be in this economy?

To answer this question, imagine that all markets are competitive, so that the allocation of workers to tasks will simply maximize output (recall the Second Welfare Theorem, Theorem 5.7). Then, the problem can be written as

$$(21.30) \quad \max_{[L(h)]_{h \in [0, \bar{h}]}} \int_0^{\bar{h}} A_h L(h) dh$$

subject to

$$\begin{aligned} \int_0^{\bar{h}} L(h) dh &= L, \text{ and} \\ \int_0^{\bar{h}} h L(h) dh &= H, \end{aligned}$$

where $L(h)$ is the number of unskilled workers assigned to work with technology A_h . The first-order conditions for this maximization problem can be written as

$$(21.31) \quad A_h \leq \lambda_L + h\lambda_H \text{ for all } h \in [0, \bar{h}],$$

where λ_L is the multiplier associated with the first constraint and λ_H is the multiplier associated with the second constraint. The first-order condition is written as an inequality, since not all technologies $h \in [0, \bar{h}]$ will be used, and those that are not active might satisfy this condition with a strict inequality.

Inspection of the first-order conditions implies that if $A_{\bar{h}}$ is sufficiently high and if $A_0 > 0$, the solution to this problem will have a very simple feature. All skilled workers will be employed at technology $\bar{h}$, and together with them, there will be $L(\bar{h}) = H/\bar{h}$ unskilled workers employed with this technology. The remaining $L - L(\bar{h})$ workers will be employed with the technology $h = 0$ (see Exercise 21.6). This equilibrium will then have the feature of a dual economy. Two very different technologies will be used for production, one more advanced (modern), and the other corresponding to the least advanced technology that is feasible. This dual economy structure emerges because of a non-convexity—to maximize output, it is necessary to operate the most advanced technology, but this exhausts all of the available skilled workers, implying that unskilled workers have to be employed in technologies that do not require skilled inputs or supervision. This perspective therefore suggests that a dual economy structure might be the natural outcome of technology transfer, especially

in situations where less-developed economies import their technologies from more advanced nations and these technologies are inappropriate to the needs of less-developed countries.

Models of dual economy based on this type of appropriate technology ideas have not been investigated in detail, though the literature on appropriate technology, which was discussed in Chapter 18, suggests that they may be important in practice. While this model focuses on the dual economy aspect in production, one can easily generalize this framework by assuming that the more advanced technology will be operated in urban areas and with contractual arrangements enforced by modern institutions, while the less advanced technology is operated in villages or rural areas. Thus models based on appropriate (or inappropriate) technology may be able to account for the broad patterns related to dual economy, including rural to urban migration and changes in social arrangements.

21.4. Distance to the Frontier and Changes in the Organization of Production

In this section, I discuss how the structure of production changes over the process of development, and how this might be related both to changes in certain aspects of the internal organization of the firm and to a shift in the “growth strategy” of an economy (meaning whether the engine of growth is innovation or imitation). I will illustrate these ideas using a simple model based on Acemoglu, Aghion and Zilibotti (2006). Because of space restrictions, I only provide a sketch of the model, mainly focusing on the production side.

Consider a less-developed economy that is behind the world technology frontier. There is no need to use country indices, since I focus on a single country, taking the behavior of the world technology frontier as given. Time is discrete and the economy is populated by two-period lived overlapping generations of individuals. Total population is normalized to 1. There is a unique final good, which is also taken as the numeraire. It is produced competitively using a continuum of intermediate inputs according to the standard CES aggregator:

$$(21.32) \quad Y(t) = \int_0^1 A(\nu, t)^\beta x(\nu, t)^{1-\beta} d\nu,$$

where $A(\nu, t)$ is the productivity of the intermediate good in intermediate sector ν at time t , $x(\nu, t)$ is the amount of intermediate good ν used in the production of the final good at time t , and $\beta \in (0, 1)$.

Each intermediate good is produced by a monopolist $\nu \in [0, 1]$ at a unit marginal cost in terms of the unique final good. The monopolist faces a competitive fringe of imitators that can copy its technology and also produce an identical intermediate good with productivity $A(\nu, t)$, but will do so more expensively. In particular, the competitive fringe can produce each intermediate good at the cost of $\chi > 1$ units of final good. The existence of this competitive fringe forces the monopolist to charge a *limit price*:

$$(21.33) \quad p(\nu, t) = \chi > 1.$$

Naturally, this limit price configuration will be an equilibrium when χ is not so high that the monopolist prefers to set a lower unconstrained monopoly price. The condition for this

is simply

$$\chi \leq 1/(1 - \beta),$$

which I impose throughout. Broadly, one can think of the parameter χ as capturing both technological factors and government regulations regarding competitive policy. A higher χ corresponds to a less competitive market. Given the demand implied by the final goods technology in (21.32) and the equilibrium limit price in (21.33), equilibrium monopoly profits are simply:

$$(21.34) \quad \pi(\nu, t) = \delta A(\nu, t),$$

where

$$\delta \equiv (\chi - 1) \chi^{-1/\beta} (1 - \beta)^{1/\beta}$$

is a measure of the extent of monopoly power. In particular it can be verified that δ is increasing in χ for all $\chi \leq 1/(1 - \beta)$.

In this model, the process of economic development will be driven not by capital accumulation—which was the force emphasized in some of the earlier models—but by technological progress, that is, by increases in $A(\nu, t)$. Let us assume that each monopolist $\nu \in [0, 1]$ can increase its $A(\nu, t)$ by two complementary processes: (i) imitation (adoption of existing technologies); and (ii) innovation (discovery of new technologies). The key economic trade-offs in the model arise from the fact that different economic arrangements (both in terms of the organization of firms and in terms of the growth strategy of the economy) will lead to different amounts of imitation and innovation.

To prepare for this point, let us define the average productivity of the economy in question at date t as:

$$A(t) \equiv \int_0^1 A(\nu, t) d\nu.$$

Let $\bar{A}(t)$ denote the productivity at the world technology frontier. The fact that this economy is behind the world technology frontier means that

$$A(t) \leq \bar{A}(t)$$

for all t . The world technology frontier progresses according to the difference equation

$$(21.35) \quad \bar{A}(t) = (1 + g) \bar{A}(t - 1),$$

where the growth rate of the world technology frontier is taken to be

$$(21.36) \quad g \equiv \underline{\eta} + \bar{\gamma} - 1,$$

where $\underline{\eta}$ and $\bar{\gamma}$ will be defined below.

I assume that the process of imitation and innovation leads to the following law of motion of each monopolist's productivity:

$$(21.37) \quad A(\nu, t) = \eta \bar{A}(t - 1) + \gamma A(\nu, t - 1) + \varepsilon(\nu, t),$$

where $\eta > 0$ and $\gamma > 0$, and $\varepsilon(\nu, t)$ is a random variable with zero mean, capturing differences in innovation performance across firms and sectors.

In eq. (21.37), $\eta \bar{A}(t-1)$ stands for advances in productivity coming from *adoption* of technologies from the frontier (and thus depends on the productivity level of the frontier, $\bar{A}(t-1)$), while $\gamma A(t-1)$ stands for the component of productivity growth coming from innovation (building on the existing knowledge stock of the economy in question at time $t-1$, $A(t-1)$). Let us also define

$$a(t) \equiv \frac{A(t)}{\bar{A}(t)}$$

as the (inverse) measure of the country's *distance to the technological frontier* at date t .

Now, we can integrate (21.37) over $\nu \in [0, 1]$, use the fact that $\varepsilon(\nu, t)$ has mean zero, divide both sides by $\bar{A}(t)$ and use (21.35) to obtain a simple linear relationship between a country's distance to frontier $a(t)$ at date t and the distance to frontier $a(t-1)$ at date $t-1$ given by

$$(21.38) \quad a(t) = \frac{1}{1+g}(\eta + \gamma a(t-1)).$$

This equation is similar to the technological catch-up equation in Section 18.2 in Chapter 18. It shows how the dual process of imitation and innovation may lead to a process of convergence. In particular, as long as $\gamma > 1+g$, eq. (21.38) implies that $a(t)$ will eventually converge to 1. Second, the equation also shows that the relative importances of imitation and innovation will depend on the distance to the frontier of the economy in question. In particular when $a(t)$ is large (meaning the country is close to the frontier), γ —thus innovation—matters more for growth. In contrast when $a(t)$ is small (meaning the country is farther from the frontier), η —thus imitation—is relatively more important.

To obtain further insights, let us now endogenize η and γ using a reduced-form approach. Following the analysis in Acemoglu, Aghion and Zilibotti (2006), I will model the parameters η and γ as functions of the investments undertaken by the entrepreneurs and the contractual arrangement between firms and entrepreneurs. The key idea is that there are two types of entrepreneurs: high-skill and low-skill. When an entrepreneur starts a business, his skill level is unknown and is revealed over time through his subsequent performance. This implies that there are two types of “growth strategies” that are possible. The first one emphasizes *selection* of high-skill entrepreneurs and will replace any entrepreneur that is revealed to be low skill. This growth strategy will involve a high degree of churning (creative destruction) and a large number of young entrepreneurs (as older unsuccessful entrepreneurs are replaced by new young entrepreneurs). The second strategy maintains experienced entrepreneurs in place even when they have low skills. This strategy therefore involves an organization of firms relying on “longer-term relationships” (here between entrepreneurs and the credit market), an emphasis on experience and cumulative earnings, and less creative destruction. While low-skill entrepreneurs will be less productive than high-skill entrepreneurs, there are potential reasons for why an experienced low-skill entrepreneur might be preferred to a new young entrepreneur. For example, this may be because entrepreneurial experience increases productivity, so that the low-skill experienced entrepreneur may be better at certain tasks than a high-skill inexperienced entrepreneur. Alternatively, Acemoglu, Aghion and Zilibotti

(2006) show that in the presence of credit market imperfections, the retained earnings of an old entrepreneur may provide him with an advantage in the credit market (because he can leverage his existing earnings to raise more money). I denote the strategy based on selection by $R = 0$, while the strategy that maintains experienced entrepreneurs in place is denoted by $R = 1$.

The key reduced-form assumption here will be that experienced entrepreneurs (either because of the value of experience or because of their retained earnings) are better at increasing the productivity of their company when this involves the imitation of technologies from the world frontier, which can be thought to correspond to relatively “routine” tasks. High-skill entrepreneurs, on the other hand, are more innovative and generate higher growth due to innovation. Thus the tradeoff between $R = 1$ and $R = 0$ and the associated tradeoff between organizational forms boils down to the tradeoff between imitation of technologies from the world technology frontier versus innovation. For this reason, I will refer to the first one as *imitation-based growth strategy* and to the second one as *innovation-based growth strategy*. Motivated by these considerations, let us assume that the equation for the law of motion of the distance to frontier, (21.38), takes the form

$$(21.39) \quad a(t) = \begin{cases} \frac{1}{1+g}(\bar{\eta} + \underline{\gamma}a(t-1)) & \text{if } R(t) = 1 \\ \frac{1}{1+g}(\underline{\eta} + \bar{\gamma}a(t-1)) & \text{if } R(t) = 0 \end{cases}$$

as a function of the contractual/organizational decision at time t , $R(t) \in \{0, 1\}$. In this equation, let us impose

$$(21.40) \quad \bar{\gamma} > \underline{\gamma} < 1 + g \text{ and } \bar{\eta} > \underline{\eta}.$$

The first part of this assumption follows immediately from the notion that high-skill entrepreneurs are better at innovation, while the second part, in particular, that $\bar{\gamma} > \underline{\gamma}$, builds in the feature that experienced entrepreneurs are better at imitation. When the imitation-based growth strategy is pursued, experienced entrepreneurs are not replaced, and consequently, there is greater transfer of technology from the world technology frontier. The final part of this assumption, $\underline{\gamma} < 1 + g$, simply ensures that imitation-based growth will not lead to faster growth than the world technology frontier. Also in terms of (21.39), we can interpret Assumption (21.36) as stating that the world technology frontier advances due to innovation-based growth strategy, which is natural since a country at the world technology frontier cannot imitate from others. This is what eq. (21.40) imposes.

Figure 21.4 draws eq. (21.39), and shows that the economy with long-term contracts ($R = 1$) achieves greater growth (higher level of $a(t)$ for given $a(t-1)$) through the imitation channel, but lower growth through the innovation channel. The figure also shows that which regime maximizes the growth rate of the economy depends on the level of $a(t-1)$, that is, on the distance of the economy to the world technology frontier. In particular, inspection of

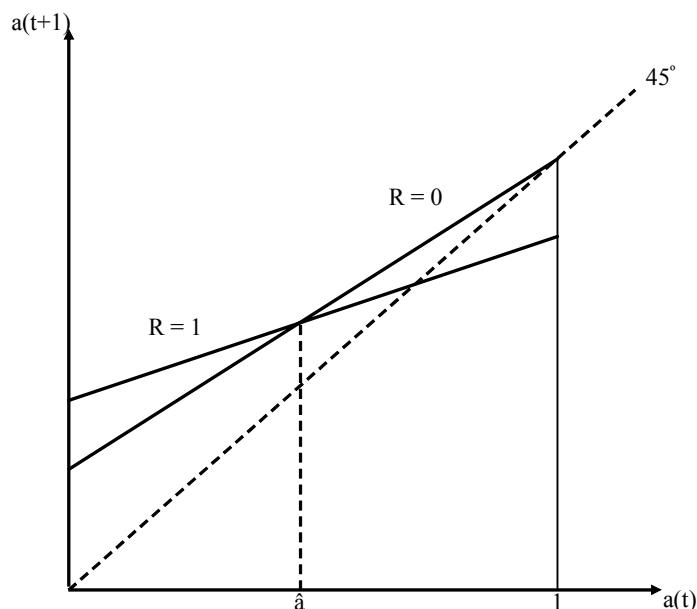


FIGURE 21.4. The growth-maximizing threshold and the dynamics of the distance to frontier in the growth-maximizing equilibrium.

(21.39) is sufficient to establish that there exists a threshold

$$(21.41) \quad \hat{a} \equiv \frac{\bar{\eta} - \underline{\eta}}{\bar{\gamma} - \underline{\gamma}} \in (0, 1)$$

such that when $a(t-1) < \hat{a}$, the imitation-based strategy, $R = 1$ leads to greater growth, and when $a(t-1) > \hat{a}$, the innovation-based strategy, $R = 0$, achieves higher growth. Thus if the economy were to pursue a growth-maximizing sequence of strategies, it would start with $R = 1$ and then switch to an innovation-based strategy, $R = 0$, once it is sufficiently close to the world technology frontier. In the imitation-based regime, incumbent entrepreneurs are sheltered from the competition of younger entrepreneurs and this may enable the economy to make better use of the experience of older entrepreneurs or to finance greater investments out of the retained earnings of incumbent entrepreneurs. In contrast, the innovation-based regime is based on an organizational form relying on greater selection of entrepreneurs and places greater emphasis on maximizing innovation at the expense of experience, imitation and investment.

Figure 21.4 describes the law of motion of technology in an economy as a function of the organization of firms (markets), captured by R . It does not specify what the equilibrium sequence of $\{R(t)\}_{t=0}^{\infty}$ is. To determine this sequence, we need to specify the equilibrium behavior, which involves the selection of entrepreneurs as well as the functioning of credit markets. Space restrictions preclude me from providing a full analysis of the equilibrium in such a model. Instead, I will informally discuss some of the main insights of such an analysis.

Conceptually, one might want to distinguish among four configurations, which may arise as equilibria under different institutional settings.

1. *Growth-maximizing equilibrium*: the first and the most obvious possibility is an equilibrium that is growth maximizing. In particular, if markets and entrepreneurs have growth maximization as their objective and are able to solve the agency problems, have the right decision-making horizon and are able to internalize the pecuniary and non-pecuniary externalities, we would obtain an efficient equilibrium. This equilibrium will take a simple form:

$$R(t) = \begin{cases} 1 & \text{if } a(t-1) < \hat{a} \\ 0 & \text{if } a(t-1) \geq \hat{a} \end{cases}$$

so that the economy achieves the upper envelope of the two lines in Figure 21.4. In this case, there is no possibility of outside intervention to increase the growth rate of the economy.¹ Moreover, an economy starting with $a(0) < 1$ always achieves a growth rate greater than g , and will ultimately converge to the world technology frontier, that is, $a(t) \rightarrow 1$. In this growth-maximizing equilibrium, the economy first starts with a particular set of organizations/institutions, corresponding to $R = 1$. Then, consistent with Kuznets' vision of a structural transformation emphasized above, the economy undergoes a change in its organizational form and growth strategy, and switches from $R = 1$ to $R = 0$. In our simple economy, this structural transformation takes the form of long-term relationships disappearing and being replaced by shorter-term relationships, by greater competition among entrepreneurs and firms and by better selection of entrepreneurs.

2. *Underinvestment equilibrium*: the second potential equilibrium configuration involves the following equilibrium organizational form:

$$R(t) = \begin{cases} 1 & \text{if } a(t-1) < a_r(\delta) \\ 0 & \text{if } a(t-1) \geq a_r(\delta) \end{cases}$$

where $a_r(\delta) < \hat{a}$. Figure 21.5 depicts this visually, with the thick black lines corresponding to the equilibrium law of motion of the distance to the frontier, a . How is $a_r(\delta)$ determined? Acemoglu, Aghion and Zilibotti (2006) show that when investments by young and old entrepreneurs are important for innovation and credit markets are imperfect, then the retained earnings of old (experienced) entrepreneurs enable them to undertake greater investments. However, because of monopolistic competition, there is the standard *appropriability effect*, whereby an entrepreneur that undertakes a greater investment does not capture all the surplus generated by this investment because some of it accrues to consumers in the form of greater consumer surplus. The appropriability effect always discourages investments, and in this context since greater investments are associated with more experienced, older entrepreneurs, it discourages the investment-based strategy. This description also explains why this equilibrium is referred to as the “underinvestment equilibrium”; in the range $a \in (a_r(\delta), \hat{a})$, the

¹However, recall that growth-maximization is not necessarily the same as welfare-maximization. Depending on how preferences and investments are specified, the growth-maximizing allocation may not be welfare-maximizing.

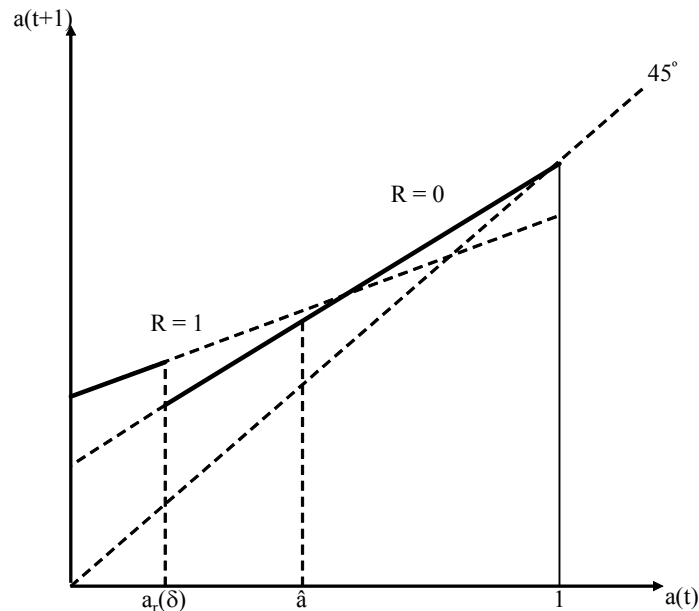


FIGURE 21.5. Dynamics of the distance to frontier in the underinvestment equilibrium.

economy could reach a higher growth rate (as shown in the figure) by choosing $R(t) = 1$, but because the appropriability effect discourages investments, there is a switch to the innovation-based equilibrium and the associated organizational forms earlier than the growth-maximizing threshold.

A notable feature is that although the equilibrium is different from the previous case, it again follows the sequence of $R = 1$ followed by a structural transformation and a switch to greater competition among and selection of entrepreneurs with the innovation-based regime ($R = 0$). Therefore, this equilibrium also exhibits the feature that the process of growth and economic development is associated with structural transformation. Moreover, the economy still ultimately converges to the world technology frontier, that is, $a(t) = 1$ is reached as $t \rightarrow \infty$. The only difference is that the structural transformation from $R = 1$ to $R = 0$ happens too soon, at $a(t-1) = a_r(\delta)$, rather than at the growth-maximizing threshold $\hat{a}$.

Consequently, in this case, a *temporary* government intervention may increase the growth rate of the economy. The temporary aspect is important here, since the best that the government can do is to increase the growth rate while $a \in (a_r(\delta), \hat{a})$. How can the government achieve this? Subsidies to investment would be one possibility. Acemoglu, Aghion and Zilibotti (2006) show that the degree of competition in the product market also has an indirect effect on the equilibrium, as emphasized by the notation $a_r(\delta)$. In particular, a higher level of δ , which corresponds to lower competition in the product market (higher χ), will increase $a_r(\delta)$, and thus may close the gap between $a_r(\delta)$ and $\hat{a}$. Nevertheless, it has to be noted that reducing competition will create other, static distortions (because of higher markups).

Moreover and more importantly, we will see in the next two configurations that reducing competition can have much more detrimental effects on economic growth, so any use of competition policy for this purpose must be subject to serious caveats.

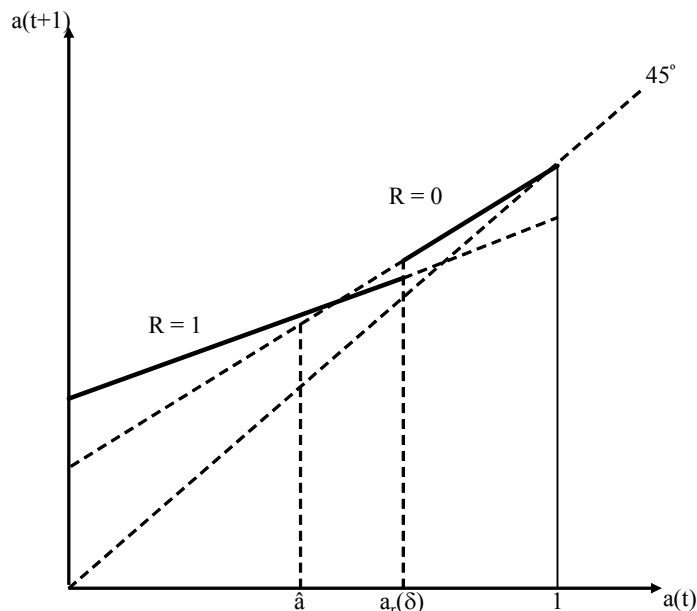


FIGURE 21.6. Dynamics of the distance to frontier in the sclerotic equilibrium.

3. *Sclerotic equilibrium*: the third possibility is a sclerotic equilibrium in which $a_r(\delta) > \hat{a}$, so that incumbent low-skill, low-productivity firms survive even when they are potentially damaging to economic growth. Acemoglu, Aghion and Zilibotti (2006) show that this configuration can also arise in equilibrium because the retained earnings of incumbent entrepreneurs act as a *shield* protecting them against the creative destruction forces brought about by new entrepreneurs. Consequently, in general, the retained earnings or other advantages of experienced entrepreneurs both have (social) benefits and costs, and which of these will dominate will depend on the details of the model and the parameter values. When the benefits dominate, the equilibrium may feature too rapid a switch to the innovation-based strategy, and when the costs dominate, the economy may experience sclerosis with the imitation-based strategy and excessive protection of incumbents.

The resulting pattern in this case is drawn in Figure 21.6. Now the economy fails to achieve the maximum growth rate for a range of values of a such that $a \in (\hat{a}, a_r(\delta))$. In this range, the innovation-based regime would be growth-maximizing, but the economy is stuck with the imitation-based regime because of the retained earnings and the power of the incumbents prevent the transition to the more efficient organizational forms. An interesting feature is that, as Figure 21.6 shows, this economy also follows a pattern in line with Kuznets's vision; it starts with a distinct set organizations, represented by $R = 1$, and then switches

to a different set of arrangements, $R = 0$. Like the previous two types of equilibria, this case also features convergence to the world technology frontier, that is, to $a = 1$.

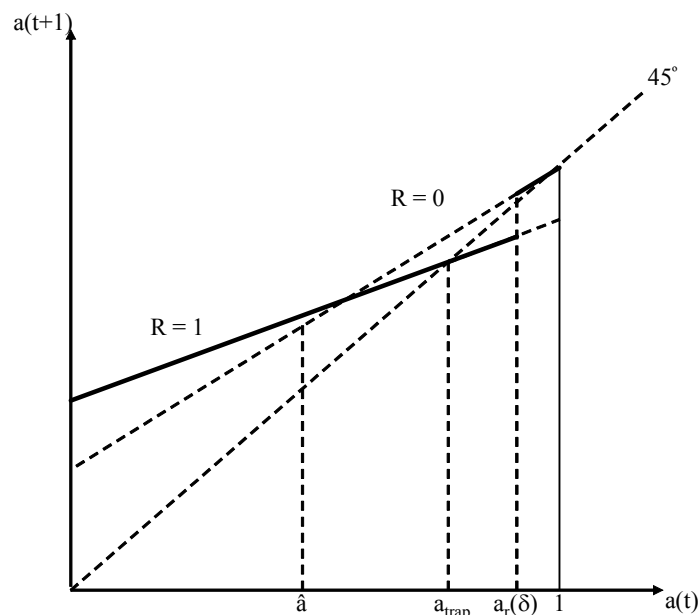


FIGURE 21.7. Dynamics of the distance to frontier in a non-convergence trap. If the economy starts with $a(0) < a_{trap}$, it fails to converge to the world technology frontier and instead converges to a_{trap} .

4. *Non-convergence trap equilibrium*: the fourth possibility is related to the third one and also involves $a_r(\delta) > \hat{a}$. However, now the gap between $a_r(\delta)$ and $\hat{a}$ is even larger as depicted in Figure 21.7, and includes the level of a , a_{trap} , such that

$$a_{trap} \equiv \frac{\bar{\eta}}{1 + g - \underline{\gamma}}.$$

Inspection of (21.39) immediately reveals that if $a(t-1) = a_{trap}$ and $R(t) = 1$, the economy will remain at a_{trap} . Therefore, in this case, the retained earnings or the experience of incumbent firms afford them so much protection that the economy never transitions to the innovation-based equilibrium. This not only retards growth for a temporary interval, but also pushes the economy into a non-convergence trap. In particular, this is the only equilibrium pattern in which the economy fails to converge to the frontier; with the imitation-based regime, $R = 1$, the economy does not grow beyond a_{trap} , and at this distance to frontier, the equilibrium keeps choosing $R = 1$.

This equilibrium therefore illustrates the most dangerous scenario—that of non-convergence. Encouraging imitation-based growth, for example by supporting incumbent firms, may at first appear as a good policy.² But in practice, it may condemn the economy

²The reader may notice that this type of policy to encourage growth has many of the features of “industrial policy” pursued by many less-developed economies. The evidence is that experiments with industrial policy

to non-convergence. This is also the only case in which the Kuznetsian structural transformation does *not* occur because the economy remains trapped. In many ways, this is in line with Kuznets' vision; the resulting economy is an underdeveloped one, unable to realize the structural transformation necessary for the process of economic development.

Taken together the four scenarios suggest that depending on the details of the model, there should be no presumption that the efficient or the growth-maximizing sequence of growth strategies will be pursued. Thus, some degree of government intervention might be useful. However, the third and the fourth cases also emphasize that government intervention can have very negative unintended consequences. Such intervention might improve growth performance during a limited period of time (in the second scenario this will be when $a \in (a_r(\delta), \hat{a})$), but it may subsequently create much more substantial costs by leading to a non-convergence trap as shown in Figure 21.7. Therefore, unless there is very precise information and some way of reversing policies that protect incumbents (a very difficult practice because of political economy reasons, which will be discussed in greater detail in Chapter 22), government interventions to spearhead economic development might backfire.

Even though the implications of these four scenarios for government intervention are mixed, their implications for changes in the structure of organization over the development process are clearer; regardless of which scenario applies, the economy starts with a distinct organization of production, where longer-term contracts, the incumbent producers, experience and imitation are more important, and then, except in the non-convergence trap equilibrium, it ultimately switches to an equilibrium with greater creative destruction, shorter-term relationships, younger entrepreneurs and more innovation. This type of transformation is another facet of the structural transformations emphasized by Kuznets as part of the process of economic development. The framework presented here, though reduced-form, can also be used to study other aspects of the transformation of the production of organization. Exercise 21.7 shows how the ideas in this section can be used to study the changes in other aspects of the internal organization of the firm over the course of the process of development.

21.5. Multiple Equilibria From Aggregate Demand Externalities and the Big Push

I now present a simple model of multiple equilibria arising from aggregate demand externalities, based on Murphy, Shleifer and Vishny's (1989) "big push" model. This model formalizes ideas first proposed by Rosenstein-Rodan (1943), Hirschman (1958) and Nurske (1958), that economic development can (or should) be viewed a move from one (Pareto inefficient) equilibrium to another, more efficient equilibrium. Moreover, these early development economists argued that this type of move requires coordination among different individuals and firms in the economy, thus a *big push*. As already discussed in Chapter 4, multiple equilibria, literally interpreted, are unlikely to be the root cause of persistently low levels

also seem to have backfired, and created powerful incumbents and subsequent sluggish growth in most instances.

of development, since if there is indeed a Pareto improvement—a change that will make *all* individuals better-off—it is unlikely that the necessary coordination cannot be achieved for decades or even centuries. Nevertheless, the forces leading to multiple equilibria highlight important economic mechanisms that can be associated with market failures slowing down, or even preventing, the process of development. Moreover, dynamic versions of models of multiple equilibria can lead to multiple state states, whereby once an economy ends up in a steady state with low economic activity, it may get stuck there (and there is no possibility of a coordination to jump to the other steady state). Models with multiple steady states, which are more useful for thinking about the process of long-run development than models with multiple equilibria, will be discussed in the next section.

Murphy, Shleifer and Vishny consider the following two-period economy, $t = 1$ and 2. The economy admits a representative household with preferences given by

$$U = \frac{C(1)^{1-\theta} - 1}{1-\theta} + \beta \frac{C(2)^{1-\theta} - 1}{1-\theta}$$

where $C(1)$ and $C(2)$ denote consumption at the two dates; β is the discount factor of the households; and θ plays a similar to before; $1/\theta$ is the intertemporal elasticity of substitution and determines how willing individuals are to substitute consumption between date 1 and date 2. The representative household supplies labor inelastically and the total labor supply is denoted by L .

The resource constraint for the economy is

$$\begin{aligned} C(1) + I(1) &\leq Y(1) \\ C(2) &\leq Y(2), \end{aligned}$$

where $I(1)$ denotes investment in the first date, $Y(t)$ is total output at date t , and investment is only possible in the first date.

Households can borrow and lend, so their budget constraint can be represented as

$$C(1) + \frac{C(2)}{R} \leq w(1) + \pi(1) + \frac{w(2) + \pi(2)}{R},$$

where $\pi(t)$ denotes the profits accruing to the representative household, and $w(t)$ is the wage rate at time t . R is the gross interest rate between periods 1 and 2. Although individuals can borrow and lend, in the aggregate the resource constraints have to hold, so R will be determined in equilibrium to ensure this.

As in the endogenous technological progress models in Part 4 and in the model of the previous section, the final good is assumed to be a CES aggregate of a continuum 1 of differentiated intermediate goods, and is thus given as

$$Y(t) = \left[\int_0^1 y(\nu, t)^{\frac{\varepsilon-1}{\varepsilon}} d\nu \right]^{\frac{\varepsilon}{\varepsilon-1}},$$

where $y(\nu, t)$ is the output level of intermediate ν at date t . The fact that many features of the model here are similar to the baseline endogenous technological change model highlights that the aggregate demand externalities that may lead to development traps here are already

present in our workhorse endogenous growth models. As usual ε is the elasticity of substitution between intermediate goods within a given period and is assumed to be strictly greater than one ($\varepsilon > 1$).

The production functions of intermediate goods in the two periods are as follows:

$$y(\nu, 1) = l(\nu, 1)$$

and

$$(21.42) \quad y(\nu, 2) = \begin{cases} l(\nu, 2) & \text{with old technology} \\ \alpha l(\nu, 2) & \text{with new technology} \end{cases}$$

where $\alpha > 1$ and $l(\nu, t)$ denotes labor devoted to the production of intermediate good ν at time t . Labor market clearing, naturally, requires

$$(21.43) \quad \int_0^1 l(\nu, t) d\nu \leq L.$$

At date 1, there is a designated producer for each intermediate, which I will also refer to as a “monopolist”. A competitive fringe of firms can also enter and produce each good as productively as the designated producer. At date 1, the designated producer can also invest in the new technology, which costs F per firm. If this investment is undertaken, this producer’s productivity at date 2 will be higher by a factor $\alpha > 1$ as indicated by eq. (21.42). In contrast, the fringe will not benefit from this technological improvement, thus the designated producer will have some degree of monopoly power. The profits from intermediate producers are naturally allocated to the representative household.

Since this is a two-period economy, we will be looking for a subgame perfect equilibrium. Moreover, to simplify the discussion, let us focus on pure-strategy symmetric Subgame Perfect Equilibria (SSPE). As in the model of Section 18.5 in Chapter 18, a SSPE consists of an allocation of labor across firms, investment decisions for firms, wages for both periods and an interest rate linking consumption between the two periods.

First, since all goods are symmetric, the first period labor market clearing is straightforward and requires

$$l(\nu, 1) = L \text{ for all } \nu \in [0, 1]$$

(recall that the measure of sectors and firms is normalized to 1). This implies that

$$Y(1) = L.$$

At date 2, the equilibrium will depend on how many firms have adopted the new technology. Since the focus is on SSPE, it is sufficient to consider the two extreme allocations, where all firms adopt the new technology and where no firm adopts. In either case, the marginal productivity of all sectors are the same, so labor will be allocated equally so that

$$l(\nu, 2) = L \text{ for all } \nu \in [0, 1].$$

Consequently, when the technology is not adopted,

$$Y(2) = L$$

and when the technology is adopted by all the firms,

$$Y(2) = \alpha L.$$

I now turn to the pricing decisions. In the first date, the designated producers have no monopoly power because of the competitive fringe, thus they charge price equal to marginal cost, which is $w(1)$, and make zero profits. Since total output is equal to $Y(1) = L$, this also implies that the equilibrium wage rate is equal to

$$w(1) = 1.$$

In the second date, if the technology is not adopted, the equilibrium is identical to that at date 1, so

$$w(2) = 1$$

and thus no profits. In this case there is also no investment, so consumption at both dates is equal to L , thus the interest rate that makes individuals happy to consume this amount in both periods is

$$(21.44) \quad \hat{R} = \beta^{-1}.$$

To see this more formally, recall that the standard Euler equation in this case is

$$(21.45) \quad C(1)^{-\theta} = R\beta C(2)^{-\theta},$$

which can only be satisfied with $C(1) = C(2)$, if the gross interest rate is $\hat{R}$ as given in (21.44).

Next consider the situation in which the designated producers have invested in the advanced technology. Now they can produce α units of output with one unit of labor, while the fringe of competitive firms still produces one unit of output with one unit of labor. This implies that the designated producers have some monopoly power. The extent of this monopoly power depends on the comparison of ε and α .

Let us first determine the demand facing each producer, which is given as a solution to the following program of profit maximization for the final good sector:

$$\max_{[y(\nu,2)]_{\nu \in [0,1]}} \left[\int_0^1 y(\nu,2)^{\frac{\varepsilon-1}{\varepsilon}} d\nu \right]^{\frac{\varepsilon}{\varepsilon-1}} - \int_0^1 p(\nu,2) y(\nu,2) d\nu,$$

where $p(\nu,2)$ is the price of intermediate ν at date 2. The first-order condition to this program implies

$$y(\nu,2)^{-1/\varepsilon} Y(2)^{1/\varepsilon} = p(\nu,2),$$

or

$$(21.46) \quad y(\nu,2) = p(\nu,2)^{-\varepsilon} Y(2).$$

This expression is useful in laying the foundations for the aggregate demand externalities; the demand for intermediate ν depends on the total amount of production, $Y(2)$. The familiar feature of the demand curve (21.46) is that it is isoelastic. To make further progress, first imagine the situation in which there is no fringe of competitive producers. In that case, each

designated producer will act as an unconstrained monopolist and maximize its profits given by price minus marginal cost times quantity, that is,

$$\pi(\nu, 2) = \left(p(\nu, 2) - \frac{w(2)}{\alpha} \right) y(\nu, 2).$$

substituting from (21.46), the firm maximization problem is

$$\max_{p(\nu, 2)} \pi(\nu, 2) = \left(p(\nu, 2) - \frac{w(2)}{\alpha} \right) p(\nu, 2)^{-\varepsilon} Y(2),$$

which has a first-order condition,

$$p(\nu, 2)^{-\varepsilon} Y(2) - \varepsilon \left(p(\nu, 2) - \frac{w(2)}{\alpha} \right) p(\nu, 2)^{-\varepsilon-1} Y(2) = 0,$$

which implies

$$p(\nu, 2) = \frac{\varepsilon}{\varepsilon - 1} \frac{w(2)}{\alpha}.$$

This is the standard monopoly price formula with a markup related to demand elasticity over the marginal cost, $w(2)/\alpha$. The markup is constant because the demand elasticity is constant.

However, the monopolist can only charge this price if the competitive fringe cannot enter and make profits stealing the entire market at this price. Since the competitive fringe can produce one unit using one unit of labor, the monopolist can only charge this price if $\varepsilon/((\varepsilon - 1)\alpha) \leq 1$. Otherwise, the price would be too high and the competitive fringe would enter. Let us assume that α is not so high as to make the monopolist unconstrained. In other words,

$$(21.47) \quad \frac{\varepsilon}{\varepsilon - 1} \frac{1}{\alpha} > 1.$$

Under this assumption, the monopolist will be forced to charge a *limit price*. It is straightforward to see that this equilibrium limit price would be

$$p^* = w(2).$$

Consequently, given (21.47), each monopolist would make per unit profits equal to

$$w(2) - \frac{w(2)}{\alpha} = \frac{\alpha - 1}{\alpha} w(2).$$

Total profits are then obtained from (21.46) as

$$(21.48) \quad \pi(2) = \frac{\alpha - 1}{\alpha} w(2)^{1-\varepsilon} Y(2).$$

The wage rate can be determined from income accounting. Total production will be equal to $Y(2) = \alpha L$, and this has to be distributed between profits and wages, thus

$$\frac{\alpha - 1}{\alpha} w(2)^{1-\varepsilon} \alpha L + w(2) L = \alpha L,$$

which has a solution of

$$w(2) = 1,$$

exactly the same as in the case without the technological investments. Intuitively, wages in this economy are determined by the demand from the competitive fringe and thus the increased marginal product does not directly benefit workers. Instead, it increases monopolists' profits. Nevertheless, all of these profits are redistributed to the agents, who are the owners of the firms. Thus $C(2) = \alpha L$. However, because there was investment in the new technology at date 1, $C(1) = L - F$. Again the interest rate has to adjust so that individuals are happy to consume these amounts (in other words, so that they have a steep consumption profile without wanting to borrow). The Euler equation, (21.45), now implies

$$(21.49) \quad (L - F)^{-\theta} = \tilde{R} \beta (\alpha L)^{-\theta},$$

which solves for

$$\tilde{R} = \beta^{-1} \left(\frac{\alpha L}{L - F} \right)^{\theta} > \hat{R}.$$

Consequently, the interest rate in this case is higher than the one in which there is no investment. This is natural, since investment implies that individuals are being asked to forgo date 1 consumption for date 2 consumption. Note also that the greater is θ , the higher is $\tilde{R}$, since with a greater θ , there is less intertemporal substitution. Also a higher F , meaning a greater consumption sacrifice at date 1 implies a higher interest rate.

The question is whether a monopolist will find it profitable to undertake the investment at date 1. The reason for the possibility of multiplicity of equilibria is that the answer to this question will depend on whether other firms are undertaking the investment or not. Let us first consider a situation in which no other firm is undertaking the investment, and consider the incentives of a single monopolist to undertake such an investment.

In this case total output at date 2 is equal to L (since the firm considering investment is infinitesimal), and the market interest rate is given by $\tilde{R}$. Moreover, from (21.48) and the fact that $w(2) = 1$, profits at date 2 are

$$\pi^N(2) = \frac{\alpha - 1}{\alpha} L,$$

where the superscript N denotes that no other firm is undertaking the investment. Therefore, the net discounted profits at date 1 for the firm in question is

$$\begin{aligned} \Delta \pi^N &= -F + \frac{1}{\tilde{R}} \frac{\alpha - 1}{\alpha} L \\ &= -F + \beta \frac{\alpha - 1}{\alpha} L. \end{aligned}$$

Next consider the case in which all other firms are undertaking the investment. In this case, profits at date 2 are

$$\pi^I(2) = (\alpha - 1) L,$$

where the superscript I designates that all other firms are undertaking the investment. Consequently, the profit gain from investing at date 1 is

$$\begin{aligned}\Delta\pi^I &= -F + \frac{1}{R}(\alpha - 1)L \\ &= -F + \beta \left(\frac{\alpha L}{L - F} \right)^{-\theta} (\alpha - 1)L.\end{aligned}$$

As discussed above, the idea of the paper by Murphy, Shleifer and Vishny (1989), similar to the ideas of many economists writing on economic development before them, was to generate multiple equilibria, with one corresponding to backwardness and the other to industrialization. In this context, this means that for the same parameter values, both the allocations with no investment in the new technology and with all monopolists investing in the new technology should be equilibria. This is only possible if

$$(21.50) \quad \Delta\pi^N < 0 \text{ and } \Delta\pi^I > 0,$$

that is, when nobody else invests, investment is not profitable, and when all other firms invest, investment is profitable. This is clearly possible because the *aggregate demand externalities* ensure that $\pi^I > \pi^N$; when other firms invest, they produce more, there is greater aggregate demand, and profits from the new technology are higher. Counteracting this effect is the fact that the interest rate is also higher when all firms invest. Therefore, the existence of multiple equilibria requires the interest rate effect not to be too strong. For example, in the extreme case where preferences are linear ($\theta = 0$), we have

$$\Delta\pi^I = -F + \beta(\alpha - 1)L > \Delta\pi^N = -F + \beta\frac{\alpha - 1}{\alpha}L,$$

so the configuration in (21.50) is certainly possible. More generally, the condition for the existence of multiple equilibria is that:

$$(21.51) \quad \beta \left(\frac{\alpha L}{L - F} \right)^{-\theta} (\alpha - 1)L > F > \beta\frac{\alpha - 1}{\alpha}L.$$

It is also straightforward to see that whenever both equilibria exist, the equilibrium with investment Pareto dominates the one without investment, since condition (21.51) implies that all households are better-off with the upward sloping consumption profile giving them higher consumption at date 2 (see Exercise 21.8). Therefore, this analysis establishes that when condition (21.51) is satisfied, there will exist two pure strategy SSPE. In one of these, all firms undertake the investment at date 1 and consumers are better-off, while in the other one there are no investments in new technology and greater market failures. Intuitively, multiple equilibria emerge in this model because of *aggregate demand externalities*; investing in the new technology at date 1 is profitable only when there is sufficient demand at date 2 and there will be sufficient aggregate demand at date 2 when all firms invest in the new technology. This is at the root of the aggregate demand externalities, since the investment decision of a particular firm creates a positive (pecuniary) externality on other firms by increasing the level of demand facing their products. The reason why pecuniary externalities, which are present in all models, play a more important role here and lead to Pareto-ranked multiple

equilibria is that each firm does not realize the full increase in the social product created by its investment, because the monopoly markup implies that, at the margin, further increases in output create a first-order gain for consumers and for other firms that can sell more and make greater profits. The presence of the markup means that the monopolist does not internalize this first-order gain, thus turning the demand linkages into aggregate demand externalities.

The interpretation for this result suggested by Murphy, Shleifer and Vishny is to consider the equilibrium with no investment in the new technology as representing a “development trap,” where the economy remains in “underdevelopment” because no firm undertakes the investment in new technology and this behavior implies that the demand necessary to make such investments profitable is absent. In contrast, the equilibrium with investment in new technology is interpreted as corresponding to “industrialization”. According to this interpretation, societies that can somehow *coordinate* on the equilibrium with investment (either because private expectations are aligned or because of some type of government action) will industrialize and realize both economic growth and Pareto improvement. As such, this model is argued to provide a formalization of the “big push” type industrialization described by economists such as Nurske or Rosenstein-Rodan. Although the idea of the big push and the aggregate demand externalities are attractive, the model here suffers from a number of obvious shortcomings. First, even though the process of industrialization is a dynamic one, the model here is static. Therefore, it does not allow a literal interpretation of a society being first in the no investment equilibrium and then changing to the investment equilibrium and industrializing. Second, as already discussed in Chapter 4, models with multiple equilibria do not provide a satisfactory model of development, since it is difficult to imagine a society remaining unable to coordinate on a simple range of actions that would make *all* households (and firms) better-off. Instead, it is much more likely that the ideas related to aggregate demand externalities (or other potential forces leading to multiple equilibria) are more important as sources of persistence or as mechanisms generating multiple steady states (while still maintaining a unique equilibrium path). Multiple steady states will be discussed in the next section.

21.6. Inequality, Credit Market Imperfections and Human Capital

The previous section illustrated how aggregate demand externalities can generate development traps. Investment by different firms may require coordination, leading to multiple equilibria. Underdevelopment may be thought to correspond to a situation in which the coordination is on the bad equilibrium, and the development process starts with the “big push,” ensuring coordination to the high-investment equilibrium.

Here I illustrate these issues focusing on how the distribution of income and the organization of financial markets affect human capital investments. The models presented in this section will not only show the possibility of multiple steady states, but will also shed light on more substantive questions related to the role of inequality and credit markets in the process of development. Although in this section I focus on human capital, the interaction between

inequality and credit market problems influences not only human capital investments, but also on business creation, occupational choices and other aspects of the organization of production. Nevertheless, the models focusing on the link between inequality and human capital are both more tractable and also constitute a natural continuation of the models of human capital investments presented in Chapter 10.

21.6.1. A Simple Case With No Borrowing. When credit markets are imperfect, a major determinant of human capital investments will be the distribution of income (as well as the degree of imperfection in the credit markets). I start with a discussion of the simplest case in which there is no borrowing or lending, which introduces an extreme form of credit market problems. I will then enrich this model by introducing credit markets that allow borrowing and lending, but introduce credit market imperfections by making the cost of borrowing greater than the interest rate received by households engaged in saving.

The economy consists of continuum 1 of dynasties. Each individual lives for two periods, childhood and adulthood, and begets an offspring in his adulthood. There is consumption only at the end of adulthood. Preferences are given by

$$(1 - \delta) \log c_i(t) + \delta \log e_i(t + 1)$$

where c is consumption at the end of the individual's life, and e is the educational spending on the offspring of this individual. The budget constraint is

$$c_i(t) + e_i(t + 1) \leq w_i(t),$$

where w is the wage income of the individual. Notice that preferences here have the “warm glow” type altruism which we encountered in Chapter 9 and in Section 21.2 above. In particular, parents do not care about the utility of their offspring, but simply about what they bequeath to them, here education. As usual, this significantly simplifies the analysis. Moreover, preferences are logarithmic and will imply a constant saving rate, here in terms of educational investments.

The labor market is competitive, and wage income of each individual is simply a linear function of his human capital:

$$w_i(t) = Ah_i(t)$$

Human capital of the offspring of individual i of generation t in turn is given by

$$(21.52) \quad h_i(t + 1) = \begin{cases} e_i(t)^\gamma & \text{if } e_i(t) \geq 1 \\ \bar{h} & \text{if } e_i(t) < 1 \end{cases},$$

where $\gamma \in (0, 1)$ and $\bar{h} \in (0, 1)$ is some minimum level of human capital that the individual will attain even without any educational spending. Once spending exceeds a certain level (here set equal to 1), the individual starts benefiting from the additional spending and accumulates further human capital (though with diminishing returns since $\gamma < 1$).

This equation introduces a crucial feature necessary for models of credit market imperfections to generate multiple equilibria or multiple steady states; a *nonconvexity* in the

technology of human capital accumulation. Exercise 21.9 shows that this nonconvexity plays a crucial role in the results of this subsection.

Given this description, the equilibrium is straightforward to characterize. Each individual will choose the spending on education that maximizes his own utility. This immediately implies the following “saving rate” in terms of education

$$(21.53) \quad e_i(t) = \delta w_i(t) = \delta A h_i(t).$$

This rule has one unappealing feature (not crucial for any of the results), which is that because parents derive utility from educational spending on their children, they will spend on education even when $e_i(t) < 1$, in which case educational spendings are in fact wasted (they do not translate into higher human capital of the offspring).

To obtain stark results, let us also assume that

$$(21.54) \quad \delta A > 1 > \delta A \bar{h}.$$

Now, let us look at the dynamics of human capital for a particular dynasty i . If at time 0, $h_i(0) < (\delta A)^{-1}$, then (21.53) implies that $e_i(t) < 1$, so the offspring will have $h_i(1) = \bar{h}$. Given (21.54), $h_i(1) = \bar{h} < (\delta A)^{-1}$, and repeating this argument, $h_i(t) = \bar{h} < (\delta A)^{-1}$ for all t . Therefore, a dynasty that starts with $h_i(0) < (\delta A)^{-1}$ will never reach a human capital level greater than $\bar{h}$.

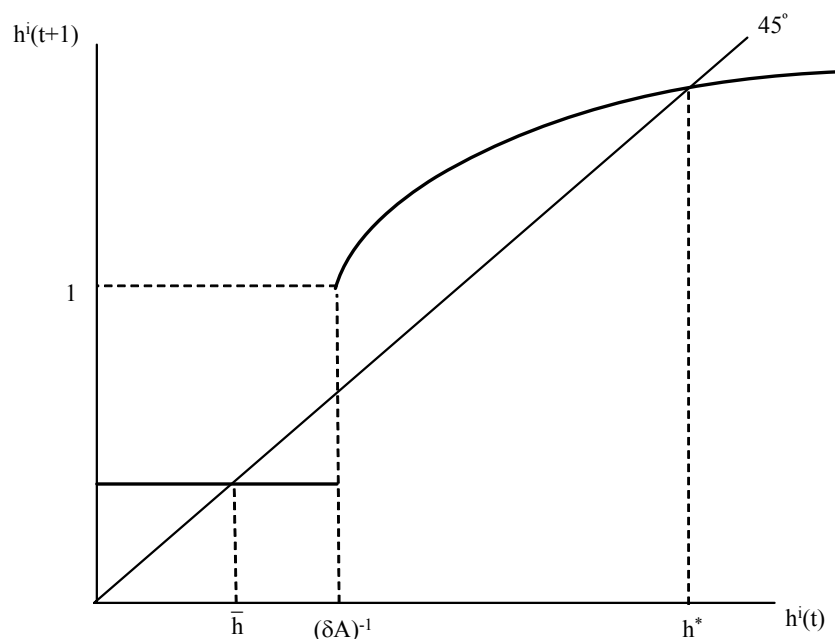


FIGURE 21.8. Dynamics of human capital with nonconvexities and no borrowing.

Next consider a dynasty with $h_i(0) > (\delta A)^{-1}$. Then, from (21.54), $h_i(1) = (\delta A h_i(0))^\gamma > 1$, so this dynasty will gradually accumulate more and more human capital over generations

and ultimately reach the “steady state” given by $h^* = (\delta A h^*)^\gamma$ or

$$h^* = (\delta A)^{\frac{\gamma}{1-\gamma}} > 1.$$

Naturally, this description applies to a dynasty with $h_i(0) \in ((\delta A)^{-1}, h^*)$. If $h_i(0) > h^*$, then the dynasty would have started with too much human capital and would decumulate human capital.

Figure 21.8 illustrates the dynamics of individual human capital decisions. It shows that there are two steady-state levels of human capital for individuals, $\bar{h}$ and $h^* > \bar{h}$. An important question when there are multiple steady states is where the economy (or a particular individual) will converge to given initial conditions. Assume for now that, even though there are multiple steady states, the equilibrium is unique (meaning that given initial conditions there is a unique equilibrium path—this will be the case in all the models discussed in this section). Then, the equilibrium determines a dynamical system, like those we have studied before, with the only difference that there are multiple steady states. Each (locally) asymptotically stable steady state will have a *basin of attraction*, meaning a set of initial conditions, which will ultimately lead to this particular steady state. Both steady states in the model studied here are asymptotically stable and Figure 21.8 plots their basins of attractions. In particular, inspection of this figure shows that dynasties with $h_i(0) < (\delta A)^{-1}$ will tend to the lower steady state level of human capital, $\bar{h}$, while those with $h_i(0) > (\delta A)^{-1}$ will tend to the higher level, h^* .

This figure also reveals why the analysis of the dynamics in this model is so simple; the dynamics of the human capital of a single individual contains all the information relevant for the dynamics of the human capital and income of the entire economy. This is because there are no prices (such as the rate of return to human capital or the interest rate) that are being determined in equilibrium here. For this reason, dynamics in this type of models are sometimes described as *Markovian*—because they are summarized by the Markov process describing the behavior of the human capital of a single individual (without any general equilibrium interactions). Markovian dynamics are much more tractable than dynamics of inequality depending on equilibrium prices. An example of this richer type of model is given in Exercise 21.13.

The most important implication of this analysis is that this simple model features poverty traps due to the nonconvexities created by the credit market problems. This is most clearly illustrated by contrasting two economies subject to the same technology and the same credit market problems, but starting out with different distributions of income. For example, imagine an economy with two groups starting at income levels h_1 and $h_2 > h_1$ such that $(\delta A)^{-1} < h_2$. Now if inequality (poverty) is high so that $h_1 < (\delta A)^{-1}$, a significant fraction of the population will never accumulate much human capital. In contrast, if inequality is limited so that $h_1 > (\delta A)^{-1}$, all agents will accumulate human capital, eventually reaching h^* . This example also illustrates that there are (many) multiple steady states in this economy. Depending on the fraction of dynasties that start with initial human capital below $(\delta A)^{-1}$, any

fraction of the population may end up at the low level of human capital, $\bar{h}$. The greater is this fraction, the poorer is the economy.

At some level, there is a parallel between the multiplicity of steady states here and the multiple equilibria highlighted in the model of the previous section. Nevertheless, the differences are also noteworthy. In the model of the previous section, there are multiple equilibria in a static model. Thus nothing determines which equilibrium the economy will be in. At best, we can appeal to “*expectations*,” arguing that the better equilibrium will emerge when everybody expects the better equilibrium to emerge. One can informally appeal to the role of “*history*,” for example, suggesting that if an economy has been in the low investment equilibrium for a while, it is likely to stay there, but this argument is misleading. First of all, the model is a static one, thus a discussion of an economy “that has been in the low equilibrium for a while” is not quite meaningful. Secondly, even if the model were turned into a dynamic one by repeating it over time, the history of being in one equilibrium for a number of periods will have no effect on the existence of multiple equilibria at the next period. In particular, each static equilibrium would still remain an equilibrium in the “dynamic” environment, and the economy could suddenly jump from one equilibrium to another. This highlights that models with multiple equilibria have a degree of indeterminacy that are both theoretically awkward and empirically difficult to map to reality. Instead, models with multiple steady states avoid these thorny issues. The equilibrium is *unique*, but the initial conditions determine where the dynamical system will end up eventually. Because the equilibrium is unique, there is no issue of indeterminacy or expectations affecting the path of the economy. But also, because multiple steady states are possible, the model can be useful for thinking about potential development traps.

Aside from providing us with a simple example of multiple steady states, this model shows the importance of the distribution of income in an economy with imperfect credit markets (here with no credit markets). In particular, the distribution of income affects which individuals will be unable to invest in human capital accumulation and thus influences the long-run income level of the economy. For this reason, models of this sort (including the one with imperfect capital markets in the next subsection) are sometimes interpreted as implying that an unequal distribution of income will lead to lower output (and growth). In fact, the above example with two classes seems to support this conclusion. However, this is not a general result and it is important to emphasize that this class of models does not make specific predictions about the relationship between inequality and growth. To illustrate this, consider the same economy with two classes, now starting with $h_1 < h_2 < (\delta A)^{-1}$. In this case, neither group will accumulate human capital, but redistributing resources away from group 1 to group 2 (thus increasing inequality), so that we push group 2 to $h_2 > (\delta A)^{-1}$, would increase human capital accumulation. This is a general feature: in models with nonconvexities, there are *no unambiguous* results about whether greater inequality is good or bad for accumulation and economic growth; it depends on whether greater inequality pushes more people below or above the critical thresholds. Somewhat sharper results can be obtained about the effect of

inequality on human capital accumulation and development under additional assumptions. Exercise 21.10 presents a parameterization of inequality in the model here, which delivers the results that greater inequality leads to lower investments in human capital and lower output per capita in relatively rich economies, but to greater investments in human capital in poorer economies.

21.6.2. Human Capital Investments with Imperfect Credit Markets. I now enrich the model of the previous subsection by introducing credit markets. The model I present here is a simplified version of the model by Galor and Zeira (1993). Each individual still lives for two periods. In his youth, he can either work or acquire education. The utility function of each individual is

$$(1 - \delta) \log c_i(t) + \delta \log b_i(t),$$

where again c denotes consumption at the end of the life of the individual. The budget constraint is

$$c_i(t) + b_i(t) \leq y_i(t),$$

where $y_i(t)$ is individual i 's income at time t . Note that preferences still take the “warm glow” form, but the utility of the parent now depends on monetary bequest to the offspring rather than the level of education expenditures. It will now be the individuals themselves who will use the monetary bequests to invest in education. Also, the logarithmic formulation once again ensures a constant saving rate equal to δ .

Education is a binary outcome, and educated (skilled) workers earn wage w_s while uneducated workers earn w_u . The required education expenditure to become skilled is h , and workers acquiring education do not earn the unskilled wage, w_u , during the first period of their lives. The fact that education is binary introduces the aforementioned nonconvexity in human capital investment decisions. As demonstrated in Exercise 21.9, such nonconvexities are important for models with imperfect credit markets to generate multiple steady states.³

Imperfect capital markets are modeled by assuming that there is some amount of monitoring required for loans to be paid back. The cost of monitoring creates a wedge between the borrowing and the lending rates. In particular, assume that there is a linear savings technology open to all agents, which fixes the lending rate at some constant r . However, the borrowing rate is $i > r$, because of costs of monitoring necessary to induce agents to pay back the loans (see Exercise 21.12 for a more micro-founded version of these borrowing costs).

Also assume that

$$(21.55) \quad w_s - (1 + r)h > w_u(2 + r)$$

³An alternative to nonconvexities in human capital investments is presented in Galor and Moav (2004), who show that multiple steady states are possible when there are no nonconvexities, credit markets are imperfect and the marginal propensity to save is higher for richer dynasties. This assumption is motivated by Kaldor's (1957) paper and was discussed in Exercise 2.13 in Chapter 2. Galor and Moav (2004) also show that this “Kaldorian” assumption combined with credit market imperfections can help reconcile the emphasis of the recent literature, which is on the adverse effects of inequality, with the earlier literature (e.g., Lewis, 1954, Kaldor, 1957), which stressed the beneficial effects of inequality via greater savings.

which implies that investment in human capital is profitable when financed at the lending rate r .

Let us now consider an individual with wealth x . If $x \geq h$, assumption (21.55) implies that the individual will invest in education. If $x < h$, then whether it is profitable to invest in education will depend on the wealth of the individual and on the borrowing interest rate, i .

Let us now write the utility of this individual (with $x < h$) in the two scenarios, and also the bequest that he will leave to his offspring. These are

$$\begin{aligned} U_s(x) &= \log(w_s + (1+i)(x-h)) + \log(1-\delta)^{1-\delta} \delta^\delta \\ b_s(x) &= \delta(w_s + (1+i)(x-h)), \end{aligned}$$

when he invests in education. And when he chooses not to invest, the utility and bequest levels are

$$\begin{aligned} U_u(x) &= \log((1+r)(w_u+x) + w_u) + \log(1-\delta)^{1-\delta} \delta^\delta \\ b_u(x) &= \delta((1+r)(w_u+x) + w_u). \end{aligned}$$

Comparing these expressions, it is clear that an individual likes to invest in education if and only if

$$x \geq f \equiv \frac{(2+r)w_u + (1+i)h - w_s}{i-r}$$

The dynamics of the system can then be obtained simply by using the bequests of unconstrained, constrained-investing and constrained-non-investing agents.

More specifically, the equilibrium correspondence describing equilibrium dynamics is

$$(21.56) \quad x(t+1) = \begin{cases} b_u(x(t)) = \delta((1+r)(w_u+x(t)) + w_u) & \text{if } x(t) < f \\ b_s(x(t)) = \delta(w_s + (1+i)(x(t)-h)) & \text{if } h > x(t) \geq f \\ b_n(x(t)) = \delta(w_s + (1+r)(x(t)-h)) & \text{if } x(t) \geq h \end{cases}$$

Equilibrium dynamics can now be analyzed diagrammatically by looking at the graph of (21.56), which is shown in Figure 21.9. As emphasized in the context of the model of the previous subsection, the curve corresponding to equation (21.56) describes both the behavior of the wealth of each individual and the behavior of the wealth distribution in the aggregate economy. This is again a feature of the “Markovian” nature of the current model.

Now define x^* as the intersection of the equilibrium curve (21.56) with the 45 degree line, when the equilibrium correspondence is steeper than the 45 degree line. Such an intersection will exist when the borrowing interest rate, i , is large enough. Suppose this is the case. Then, Figure 21.9 makes it clear that there will be three intersections between (21.56) and the 45 degree line, $\bar{x}_U$, x^* and $\bar{x}_S$. Moreover, inspection of this figure will show that x^* corresponds to an asymptotically unstable steady state, while the other two are locally asymptotically stable.

The basis of attraction of the steady states for $\bar{x}_U$ and $\bar{x}_S$ are also easy to obtained from this figure. In particular, all individuals with $x(t) < x^*$ converge to the wealth level $\bar{x}_U$, while all those with $x(t) > x^*$ converge to the greater wealth level $\bar{x}_S$. Thus the basin

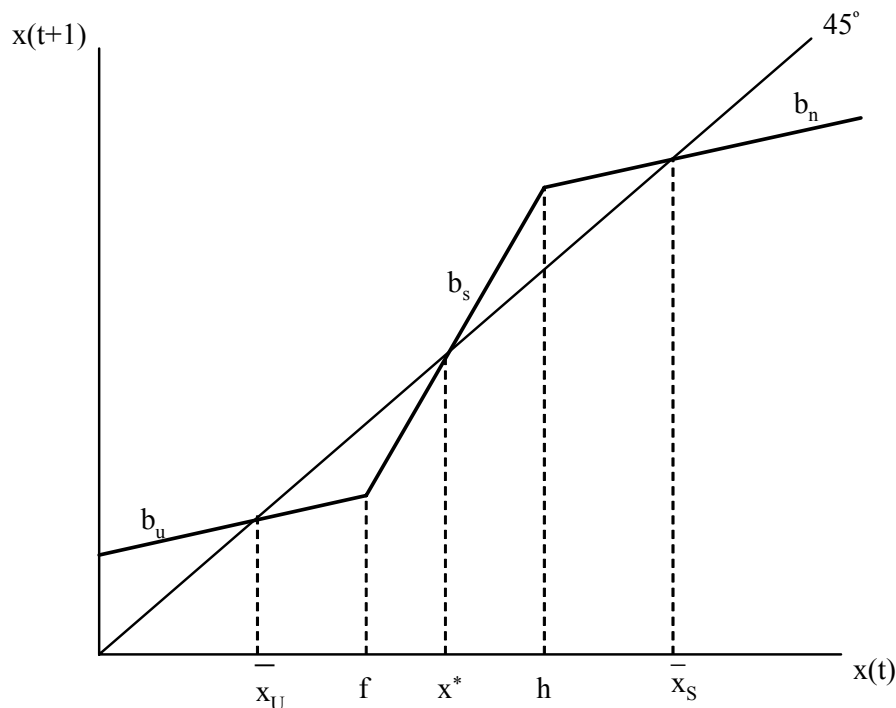


FIGURE 21.9. Multiple steady-state equilibria in the Galor and Zeira model.

of attraction of $\bar{x}_U$ is $[0, x^*]$, and this corresponds to a “poverty trap,” in the sense that individuals (dynasties) with initial wealth will converge to this level of wealth and will be unable to reach the higher level of human capital and income. The initial distribution of income again has a potentially first-order effect on the efficiency and income level of the economy. If the majority of the individuals start with $x(t) < x^*$, the economy will have low productivity, low human capital and low wealth. Therefore, this model extends the insights of the simple model with no borrowing from the previous subsection to a richer environment in which individuals make forward-looking human capital investments. The key is again the interaction between credit market imperfections (which here make the interest rate for borrowing greater than the interest rate for saving) and inequality. As in the model of the previous subsection, it is straightforward to construct examples where an increase inequality can lead to either worse or better outcomes depending whether it pushes more individuals into the basin of attraction of the low steady state.

An important feature of the model of this subsection is that because it allows individuals to borrow and lend in financial markets, it enables an investigation of the implications of financial development for human capital investments. In an economy with better financial institutions, the wedge between the borrowing rate and the lending rate will be smaller, that is, i will be smaller for a given level of r . With a smaller i , more agents will escape the

poverty trap, and in fact the poverty trap may not exist (there may not be an intersection between (21.56) and the 45 degree line where (21.56) is steeper). This shows that financial development not only improves risk sharing as demonstrated in Section 21.1, but in addition, by relaxing credit market constraints, it contributes to human capital accumulation.

Although the model in this section is considerably richer than that in the previous subsection, a number of its shortcomings should also be noted. The most important shortcoming of the model is that, like the one in the previous subsection, it is essentially a partial equilibrium model. Multiple steady states are possible for different individuals as a function of their initial level of human capital (or wealth), but individual dynamics are not affected by general equilibrium prices. Models such as the second model presented in Galor and Zeira (1993), and those in Banerjee and Newman (1994), Aghion and Bolton (1997) and Piketty (1997) consider richer environments in which income dynamics of each dynasty (individual) is affected by general equilibrium prices (such as the interest rate or the wage rate), which are themselves a function of the income inequality at the time. Exercise 21.11 shows that the type of multiple steady states generated by the model presented here may not be robust to the addition of noise in income dynamics—instead of multiple steady states, the long-run equilibrium then corresponds to a stationary distribution of human capital levels, though this stationary distribution will exhibit a large amount of persistence.⁴ In contrast, models in which prices are determined in general equilibrium and affect wealth (income) dynamics can generate more “robust” multiplicity of steady states. The second potential shortcoming of the current model is that it focuses on human capital investments. Some development economists, such as Banerjee and Newman (1994), believe that the effect of income inequality on occupational choices is potentially more important than its effect on human capital investments. Exercise 21.13 presents a simplified version of the Banerjee-Newman model that emphasizes the impact of credit market imperfections on occupational choices.

21.6.3. Heterogeneity, Stratification and the Dynamics of Inequality . The models in the previous two sections investigated the implications of credit market imperfections and income distribution for human capital investments. In this subsection, I consider a slightly more general framework due to Benabou (1996a), which enables a study of the dynamics of inequality and its costs for the efficiency of production resulting from its effects on human capital investments as a function of both the technology of production and how much stratification and segregation there is in the society. In particular, let me use a simplified version of Benabou’s (1996a) model where aggregate output in the economy at time t is given by

$$Y(t) = H(t),$$

⁴The reader will note that this is related to the “Markovian” nature of the model. Markovian models can generate multiple steady states because the Markov chain or the Markov process implied by the model is not ergodic (e.g., poor individuals can never accumulate to become rich). A small amount of noise then ensures that different parts of the distribution “communicate,” making the Markov process ergodic and removing the multiplicity of steady states.

where $H(t)$ is a CES aggregate of the human capital of all the individuals in the society. In particular, normalizing the total population to 1 and denoting the distribution of human capital at time t by $\mu_t(h)$,

$$(21.57) \quad H(t) \equiv \left(\int_0^\infty h^{\frac{\sigma-1}{\sigma}} d\mu_t(h) \right)^{\frac{\sigma}{\sigma-1}},$$

where the parameter σ measures the degree of complementarity or substitutability in the human capital of different individuals. When $\sigma \rightarrow \infty$, the human capital of different individuals become perfect substitutes and $H(t)$ is simply equal to the mean of the distribution of human capital. For any value of $\sigma \in (0, \infty)$, there is some amount of complementarity between the human capital levels of different individuals. For example, each individual may be performing a different task and overall output could be a combination of these tasks. If some individuals have low human capital and are not very successful in the tasks they are supposed to perform, this reduces the productivity of other individuals in the society. The effect of heterogeneity of human capital on aggregate productivity, for a given mean level of human capital in the society, is most severe when the parameter σ is close to 0. Note also that this formulation is also general enough to allow for the case in which greater inequality is productivity-enhancing. In particular, even though this aggregator looks like the CES production function, in contrast to that production function, it is defined for $\sigma < 0$ as well (whereas recall that the CES/Dixit-Stiglitz aggregator is only defined for $\sigma \geq 0$, see Exercise 21.14). When $\sigma < 0$, greater inequality for a given mean level of human capital increases the level of $H(t)$ and thus productivity. For example, in the extreme case where $\sigma \rightarrow -\infty$, we obtain that

$$H(t) = \max_i \{h_i(t)\},$$

that is, it is only the human capital of the highest human capital individual in the society that influences output. Since our interest here is on the potential costs of inequality on human capital investments, let us focus on the case where $\sigma \geq 0$. In this case, a mean preserving spread of the human capital distribution μ will lead to a lower level of $H(t)$ and thus to a lower level of output.

The human capital of an individual from dynasty i at time $t+1$ is given by

$$(21.58) \quad h_i(t+1) = \xi_i(t) B(h_i(t))^\alpha (N_i(t))^\beta (H(t))^\gamma,$$

where B is a positive constant, $h_i(t)$ is the human capital of the individual's parent, $\xi_i(t)$ is a random shock affecting the individual's human capital and $N_i(t)$ is the "average" human capital in the individual's neighborhood. The human capital of the offspring is affected by his parent's human capital either because of natural spillovers within the family or because the parent devotes some of his time to the rearing of his offspring and his time is more valuable in child-rearing because of his higher human capital. In addition, the neighborhood and aggregate human capital levels, $N_i(t)$ and $H(t)$, affect the human capital of the individual through learning spillovers. For example, when the average human capital in the neighborhood is high, this may make it easier for the individual to acquire more human capital. Aggregate human

capital also enters this accumulation equation because the total (or average) level of human capital in the society may affect the type of knowledge that is available for the children to acquire. The presence of this type of aggregate spillover means that a low level of $H(t)$, for example, because of high inequality, will not only reduce income today, but will also slow down further human capital accumulation.

Following Benabou (1996a), let us assume that the neighborhood human capital is also a CES aggregator, this time with an elasticity ε , so that

$$N_i(t) \equiv \left(\int_0^\infty h^{\frac{\varepsilon-1}{\varepsilon}} d\mu_t^i(h) \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

where now $\mu_t^i(h)$ denotes the distribution of human capital in the neighborhood of individual i at time t . The presence of neighborhood human capital in the accumulation equation (21.58) implies that greater heterogeneity in the composition of a neighborhood might also have a negative effect on human capital accumulation. For example, presuming that $\varepsilon \in (0, \infty)$, a mean preserving spread of neighborhood human capital will reduce the human capital of all the offspring. This structure of spillovers may be due to the fact that the presence of some low human capital children slows down learning by those with higher potential (because one “bad apple” will spoil the pack). This type of neighborhood spillovers may then suggest that segregation of high and low human capital parents in different neighborhoods might be beneficial for human capital accumulation. The model here clarifies the conditions under which this might indeed be the case.

The multiplicative structure in (21.58) gives a tractable evolution of the human capital distribution provided that the initial distribution of human capital is log normal and that the random shocks captured by $\xi(t)$ ’s are log normal. In particular, let us assume the following log normal distributions for initial human capital and shocks

$$(21.59) \quad \begin{aligned} \ln h_i(0) &\sim \mathcal{N}(m_0, \Delta_0^2) \\ \ln \xi_i(t) &\sim \mathcal{N}\left(-\frac{\omega^2}{2}, \omega^2\right), \end{aligned}$$

where $\mathcal{N}$ denotes the normal distribution, the draws of $\xi_i(t)$ are independent across time and across individuals, and the distribution of $\ln \xi$ is assumed to have mean $-\omega^2/2$ so that ξ has a mean equal to 1 (that is independent of its variance). It can then be established that the distribution of human capital within every generation will remain log normal, that is,

$$(21.60) \quad \ln h_i(t) \sim \mathcal{N}(m_t, \Delta_t^2),$$

for some endogenous mean m_t and variance Δ_t , which will depend on parameters and the organization of the society, for example, on the extent of segregation and integration of neighborhoods (see Exercise 21.15). Consequently, the analysis of output and inequality dynamics in this economy boils down to characterizing the law of motion of m_t and Δ_t .

Let us now consider two alternative organizations. The first features full segregation so that each parent is in a neighborhood with identical parents. In that case, (21.58) becomes

$$(21.61) \quad h_i(t+1) = \xi_i(t) B(h_i(t))^{\alpha+\beta} (H(t))^\gamma,$$

because the neighborhood human capital is the same as the parent's human capital. The second organization features full mixing, so that each neighborhood is a mirror image of the entire society and thus for all neighborhoods,

$$N^i(t) = N(t) \equiv \left(\int_0^\infty h^{\frac{\varepsilon-1}{\varepsilon}} d\mu_t(h) \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

where notice that μ_t refers to the aggregate distribution. In this case, the accumulation equation becomes

$$(21.62) \quad h_i(t+1) = \xi_i(t) B(h_i(t))^\alpha N(t)^\beta H(t)^\gamma.$$

The intuition above suggests that segregation might be preferable because it will prevent the adverse effects of neighborhood inequality on the human capital accumulation process. We will see, however, that this intuition is not entirely accurate, because whether there is segregation also affects the overall level of inequality in this society, and lack of segregation may reduce long-run inequality leading to a better distribution of income and thus to better economic outcomes as a result.

With full segregation, it is straightforward to see that (see Exercise 21.16):

$$(21.63) \quad \begin{aligned} m_{t+1} &= \ln B - \frac{\omega^2}{2} + (\alpha + \beta + \gamma) m_t + \gamma \left(\frac{\sigma - 1}{\sigma} \right) \frac{\Delta_t^2}{2} \\ \Delta_{t+1}^2 &= (\alpha + \beta)^2 \Delta_t^2 + \omega^2 \end{aligned}$$

whereas with full integration,

$$(21.64) \quad \begin{aligned} \hat{m}_{t+1} &= \ln B - \frac{\omega^2}{2} + (\alpha + \beta + \gamma) \hat{m}_t + \left[\gamma \left(\frac{\sigma - 1}{\sigma} \right) + \beta \left(\frac{\varepsilon - 1}{\varepsilon} \right) \right] \frac{\hat{\Delta}_t^2}{2} \\ \hat{\Delta}_{t+1}^2 &= \alpha^2 \hat{\Delta}_t^2 + \omega^2, \end{aligned}$$

where $\hat{m}_t$ and $\hat{\Delta}_t^2$ refer to the values of the mean and the variance of the distribution under full integration.

A number of features about both of these equations are noteworthy. First, the expression for the mean of the distribution shows that there will be persistence in the distribution of human capital. This is because the human capital of the offspring reflects the human capital of the parents (either through the direct effect of their own parent's human capital, or through neighborhood and spillovers). This explains the autoregressive nature of the behavior of m_t . In addition, the dispersion of the parents' human capital affects the mean of the distribution. In particular, when $\sigma < 1$ or when $\varepsilon < 1$, so that the degree of complementarity in the aggregate or the neighborhood spillovers is high, greater dispersion reduces the mean of the distribution of human capital. More interesting is the behavior of the variance of the distribution. When there is full segregation, the costs of heterogeneity in human capital accumulation resulting from neighborhood spillovers are avoided. But in return, the variance

of log human capital is more persistent under segregation than under full integration. In particular, it is straightforward to verify that when $\varepsilon < 1$, starting with the same m_t and Δ_t , we will have

$$\hat{m}_{t+1} < m_{t+1} \text{ and } \hat{\Delta}_{t+1}^2 < \Delta_{t+1}^2,$$

so that human capital in the next period is higher under segregation. But counteracting this, inequality is also higher and we know from the functional form in (21.57) that inequality has efficiency costs. So whether in the long run segregation or integration will generate greater output and a higher efficiency of production will depend on the dynamics of inequality and the exact structure of spillovers. To answer this question, let us first find the long-run level of inequality under segregation and integration. Equations (21.63) and (21.64) immediately imply that these variances are given by

$$\Delta_\infty^2 = \frac{\omega^2}{1 - (\alpha + \beta)^2} > \hat{\Delta}_\infty^2 = \frac{\omega^2}{1 - \alpha^2},$$

confirming that there will be greater inequality of human capital and income in this society with segregation of neighborhoods. The mean of the two distributions will also be different however. Let us suppose that $\alpha + \beta + \gamma < 1$, so that this steady-state distribution exists under both segregation and integration. Then,

$$m_\infty = \frac{1}{1 - (\alpha + \beta + \gamma)} \left[\ln B - \frac{\omega^2}{2} + \gamma \left(\frac{\sigma - 1}{\sigma} \right) \frac{\omega^2}{2(1 - (\alpha + \beta)^2)} \right],$$

and

$$\hat{m}_\infty = \frac{1}{1 - (\alpha + \beta + \gamma)} \left[\ln B - \frac{s^2}{2} + \left[\gamma \left(\frac{\sigma - 1}{\sigma} \right) + \beta \left(\frac{\varepsilon - 1}{\varepsilon} \right) \right] \frac{s^2}{2(1 - \alpha^2)} \right].$$

The comparison of these two expressions shows that the mean level of human capital in the long run may be higher or lower under segregation. Using the production function above, taking logs on both sides of (21.57) and using log normality,

$$\ln Y(t) = \ln H(t) = m_t + \left(\frac{\sigma - 1}{\sigma} \right) \frac{\Delta_t^2}{2},$$

so that long-run income levels under full segregation and full integration are

$$\begin{aligned} \ln Y(\infty) &= m_\infty + \left(\frac{\sigma - 1}{\sigma} \right) \frac{\Delta_\infty^2}{2} \\ \ln \hat{Y}(\infty) &= \hat{m}_\infty + \left(\frac{\sigma - 1}{\sigma} \right) \frac{\hat{\Delta}_\infty^2}{2}. \end{aligned}$$

Consequently, depending on parameters long-run income levels may be higher or lower under full segregation and full integration (see Exercise 21.17).

This model therefore provides a richer framework for the analysis of the dynamics of income inequality than the models in the previous two subsections and also highlights various different costs arising from income inequality. Counterbalancing this rich structure is that the costs of inequality in this model are introduced in a reduced-form way. While the aggregator in (21.57) is plausible, one may wonder why there could not be segregation in production,

so that high human capital individuals produce with other high human capital individuals, preventing the costs of inequality. One answer to this question is provided in Acemoglu (1997b), where individuals with different levels of human capital are matched with firms via an imperfect matching technology (see Exercise 21.18). Other, more technology-based justifications for (21.57) can also be provided. Another advantage of this framework is that its relative tractability makes it attractive for the study of political economy decisions, such as voting over education budgets, and also for the analysis of issues such as education reform. These topics are addressed in Benabou (1996a,b).

21.7. Towards a Unified Theory of Development and Growth?

There has been a unified theme to the models discussed in this chapter (and even between those discussed in this and the previous chapter). They have either emphasized the transformation of the economy and the society over the process of development or potential reasons for why such a transformation might be halted. This transformation takes the form of the structure of production changing, the process of industrialization getting underway, a greater fraction of the population migrating from rural areas to cities, financial markets becoming more developed, mortality and fertility rates changing via health improvements and the demographic transition, and the extent of inefficiencies and market failures becoming less pronounced over time. In many instances, the driving force for this process is self-reinforced by the structural transformation that it causes. For example, in Section 21.1 and in the model of Section 17.6 in Chapter 17, economic development leads to financial deepening and this in turn enables a better allocation of resources and contributes to further growth and development.

In all of the models, economic development is associated with capital deepening, that is, with greater use of capital instead of human labor (or combined with labor). Thus we can also approximate the growth process with an increase in the capital-labor ratio of the economy, $k(t)$. This does not necessarily mean that capital accumulation is the engine of economic growth. In fact, previous chapters have emphasized how technological change is often at the root of the process of economic growth (and economic development) and thus capital deepening may be the result of technological change. Moreover, Section 21.4 showed how the crucial variable capturing the stage of development might be the distance of an economy's technology to the world technology frontier. Since technological progress appears to play a crucial role in economic growth, we may also wish to take certain aspects of the technological changes taking place during the process of development as endogenous, especially when the links between development and the changes in the extent of market failures are the main focus. Nevertheless, even in these cases, an increase in the capital-labor ratio will take place along the equilibrium path and can thus be used as a proxy for the stage of development (though in this case one must be careful not to confuse increasing the capital-labor ratio with ensuring economic development). With this caveat in mind, in this section I take the

capital-labor ratio as the proxy for the stage of development and for analytical convenience, I use the Solow model to represent the dynamics of the capital-labor ratio.

With the capital-labor ratio as the proxy for development, can we then construct a unified model where a single force drives the process of development and the induced structural transformations contribute to the evolution of this driving force? Developing such a unified theory of development is certainly worthwhile. But I will not offer a unified theory here. This is for two reasons. First, an attempt to pack many different aspects of development into a single model will lead to a framework that is complicated, whereas I believe that relatively abstract representations of reality are more insightful than panoramic approaches. Second, the economic growth and development literatures have not made significant progress towards such unified models. So while I believe there is room for thinking and constructing such unified theories of economic development, I do not think that one (or at least I) can do justice to this challenging task at this point in time and in this limited space.

Instead, I will provide a very reduced-form canonical model of development and structural change. This model is neither meant to be a unified theory of development and growth nor is it meant to be a model that will be informative about the details of the process of development. My purpose is different and more modest. I would like to bring out the common features of the models presented in this chapter, albeit in a very stylized and reduced-form manner.

Consider a continuous-time economy. Suppose that output per capita is given by

$$(21.65) \quad y(t) = f(k(t), x(t)),$$

where $k(t)$ is capital-labor ratio and $x(t)$ is some “social variable,” such as financial development, urbanization, structure of production, the structure of the family and so on. As usual, f is assumed to be differentiable and also increasing and concave in k . Moreover, the social variable x potentially affects the efficiency of the production process and thus is part of the per capita production function in (21.65). As a convention, suppose that an increase in x corresponds to “structural change,” such as a move from the countryside to cities, and thus suppose that f is not only increasing in k , but also in x (so that that is, the partial derivative $f_x \geq 0$). Naturally, not all structural change is beneficial, and certain aspects of the structural changes, such as pollution, may reduce productivity. But here for simplicity’s sake I focus on the case in which f is increasing in x .

Let us assume a highly reduced-form model of social change represented by the differential equation

$$(21.66) \quad \dot{x}(t) = g(k(t), x(t)),$$

where g is also assumed to be twice differentiable. Since x corresponds to structural change associated with development, g should be increasing in k , and in particular, its partial derivative with respect to k is strictly positive, that is, $g_k > 0$. Moreover, standard mean reversion type reasoning suggests that the case in which the derivative g_x is negative is the most reasonable benchmark. If x is above its “natural level,” it should decline and if it is below its natural level, it should increase. Motivated by this, let us also assume that $g_x < 0$.

Capital accumulates according to the Solow growth model is in Chapter 2, so that

$$(21.67) \quad \dot{k}(t) = sf(k(t), x(t)) - \delta k(t),$$

where I have suppressed population growth and there is no technological change for simplicity. For a fixed x , capital naturally accumulates in an identical fashion to that in the basic Solow model. The structure of this economy is slightly more involved because $x(t)$ also changes. Differential equations (21.66) and (21.67) provide a simple reduced-form representation of structural change driven by economic growth (capital accumulation).

To illustrate the types of dynamics and insights implied by this representation, first consider the case in which $f_x(k, x) \equiv 0$ so that the social variable x has no effect on productivity. Dynamics in this case are shown in Figure 21.10. The thick vertical line corresponds to the locus for $\dot{k}(t)/k(t) = 0$ —it represents the zero of the differential equation (21.67). This locus takes the form of a vertical line, since only a single value of $k(t)$, k^* , is consistent with steady state. The upward sloping line, on the other hand, corresponds to (21.66) and shows the locus of the values of k and x such that $\dot{x}(t)/x(t) = 0$. It is upward sloping, since g is increasing in k and decreasing in x . The laws of motion represented by the arrows follow straightforwardly from (21.66) and (21.67). For example, when $k(t) < k^*$, (21.67) implies that $k(t)$ will increase. Similarly, when $x(t)$ is above the $\dot{x}(t)/x(t) = 0$ locus, (21.66) implies that $x(t)$ will decrease. Given the laws of motion implied by the arrows, it is straightforward to see that the dynamical system representing the equilibrium of this model is globally stable and starting with any $k(0) > 0$ and $x(0) > 0$, the economy will travel towards the unique steady state (k^*, x^*) . Now consider the dynamics of a less-developed economy, that is, an economy that starts with a low level of capital-labor ratio, $k(0)$, and a low level of the social variable, $x(0)$. Then, development in this economy will take place with gradual capital deepening and a corresponding increase in $x(t)$ towards x^* , which can be viewed as a reduced-form representation of development-induced structural change.

Next, consider the more interesting case in which $f_x(k, x) > 0$. In this case, the locus for $\dot{k}(t)/k(t) = 0$ will also be upward sloping, since $f_x > 0$ and the right-hand side of (21.67) is decreasing in k by the standard arguments (in particular, because of the fact that by the strict concavity of $f(k, x)$ in k , $f(k, x)/k > f_k(k, x)$ for all k and x , see Exercise 21.19). A steady state is again given by the intersection of the loci for $\dot{k}(t)/k(t) = 0$ and $\dot{x}(t)/x(t) = 0$. Since both of these are now upward sloping, multiple steady states are possible as shown in Figure 21.11. These multiple steady states capture, in a very reduced-form way, the potential multiple equilibria arising from aggregate demand externalities or from the interaction between non-convexities and imperfect credit markets in Sections 21.5 and 21.6. The low steady state (k', x') corresponds to a situation in which the social variable x is low and this depresses productivity, making the economy settle into an equilibrium with a low capital-labor ratio. In contrast in the high steady state (k^*, x^*) , the high level of x supports greater productivity and thus a greater capital-labor ratio consistent with steady state. Moreover, it can be verified that both the low and the high steady states are typically

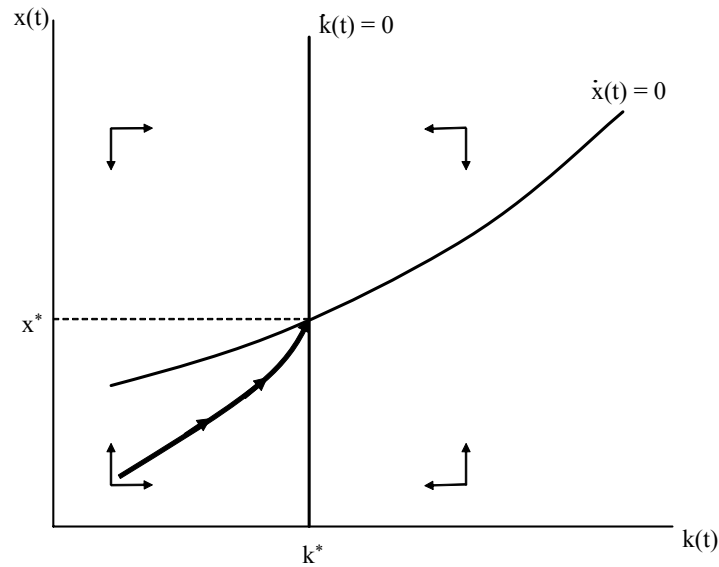


FIGURE 21.10. Capital accumulation and structural transformation without any effect of the “social variable” x on productivity.

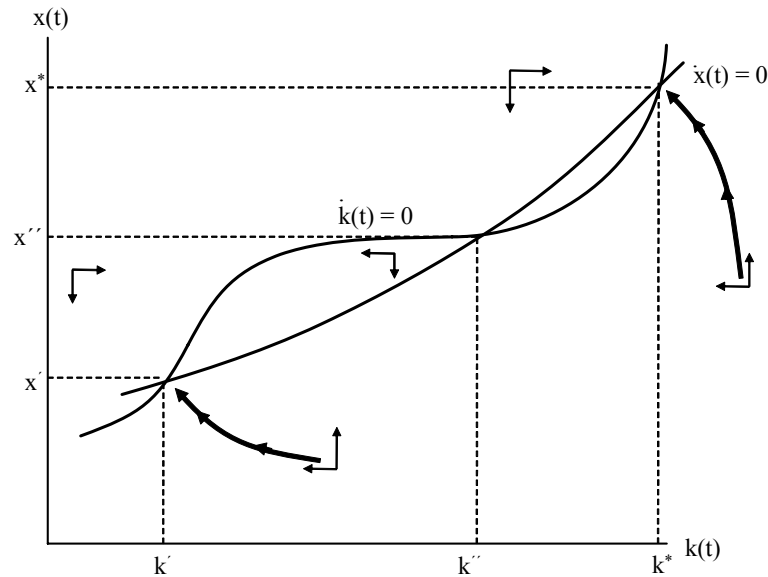


FIGURE 21.11. Capital accumulation and structural transformation with multiple steady states.

locally stable, so that starting from the neighborhood of one, the economy will converge to the nearest steady state and will tend to stay there. This highlights the importance of *historical factors* in the development process. If historical factors or endowments placed the economy in the basin of attraction of the low steady state, the economy will converge to this steady

state corresponding to a “development trap”. Interestingly, this development trap is, at least in part, caused by lack of structural change (that is, by a low value of the social variable x).

Figure 21.11 makes it clear that such multiplicity requires the locus for $\dot{k}(t)/k(t) = 0$ to be relatively flat, at least over some range. Inspection of eq. (21.67) shows that this will be the case when $f_x(k, x)$ is large, at least over some range. Intuitively, multiple steady-state equilibria can only arise when the social variable x has a large effect on productivity, so that the extent of structural change that the economy has undergone should have a large effect on productivity.

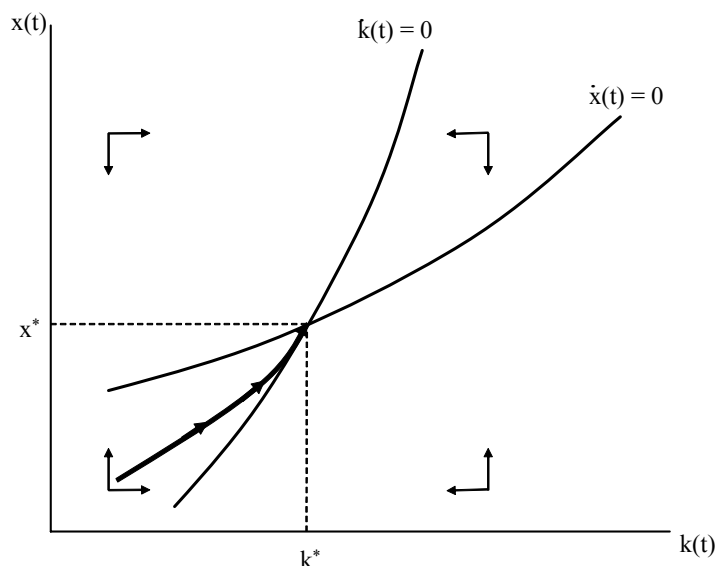


FIGURE 21.12. Capital accumulation and structural transformation when the “social variable” x affects but there exists a unique steady state.

More interesting than multiple steady states is the situation in which the same forces are present, but a unique steady state exists. The same reasoning suggests that this will be the case when $f_x(k, x)$ is relatively small. In this case, the locus for $\dot{k}(t)/k(t) = 0$ will be everywhere steeper than the locus for $\dot{x}(t)/x(t) = 0$. This case is plotted in Figure 21.12 and the unique steady state is given by (k^*, x^*) . The laws of motion represented by the arrows again follow from the inspection of the differential equations (21.66) and (21.67). This figure shows that the unique steady state is globally stable (see Exercise 21.19 for a formal proof). Consider, once again, a less-developed economy starting with a low level of capital-labor ratio, $k(0)$, and a low level of the social variable, $x(0)$. The dynamics in this case are qualitatively similar to those in Figure 21.10. However, the economics is slightly different. Capital accumulation (capital deepening) leads to an increase in $x(t)$ as before, but now this structural change also improves productivity as in the models in Section 17.6 in Chapter 17 and in Sections 21.3 and 21.1 of this chapter. This increase in productivity leads to faster capital accumulation and there is a *self-reinforcing* (“cumulative”) process of

development, with economic growth leading to structural changes facilitating further growth. However, since the effect of x on productivity is limited, this process ultimately takes us towards a unique steady state.

This reduced-form representation of structural change, therefore, captures some of the salient features emphasized in this chapter. It is not meant to be a unified model; on the contrary, rather than combining multiple dimensions of structural change, it presents an abstract representation emphasizing how the process of development, corresponding to capital accumulation, can go hand-in-hand with structural change, which may in turn increase productivity and facilitate further capital accumulation. Results on a truly unified model of economic development and structural change is an interesting area for future work.

21.8. Taking Stock

This chapter provided a large number of models focusing on various aspects of the structural transformation accompanying economic development. As emphasized in the previous section, there is no single framework unifying all these distinct aspects, even though there are many common themes across these models. The previous section was an attempt to bring out these common themes. Instead of repeating these commonalities, I would like to conclude by pointing out that many of the topics covered in this chapter are at the frontier of current research and much still remains to be done. Economic development is intimately linked to economic growth, but it may require different, even specialized, models that do not just focus on balanced growth and the orderly growth behavior captured by the neoclassical and endogenous technology models. These models may also need to take market failures and how these market failures might change over time more seriously. This view stems from the recognition that the essence of economic development is the process of structural transformation, including financial development, the demographic transition, migration, urbanization, organizational change and other social changes. Another important aspect of economic development, again less prominent in the neoclassical growth models, is the possibility that the inefficiencies in the organization of production, credit markets and product markets may culminate in potential development traps. These inefficiencies may stem from lack of coordination in the presence of aggregate demand externalities or from the interaction between imperfect credit markets and human capital investments. These areas not only highlight some of the questions that need to be addressed for understanding the process of economic development, but they also bring a range of issues that are often secondary in the standard growth literature to the forefront of analysis. These include, among other things, the organization of financial markets, the distribution of income and wealth and issues of incentives, such as problems of moral hazard, adverse selection and incomplete contracts both in credit markets and in production relationships. Unfortunately, space restrictions have precluded me from providing a satisfactory discussion of these issues, and instead, I had to incorporate these in simple growth models in reduced-form ways.

The recognition that the analysis of economic development necessitates a special focus on these topics also opens the way for a more constructive interaction between empirical development studies and the theories of economic development surveyed in this chapter. As already noted above, there is now a large literature on empirical development economics, documenting the extent of credit market imperfections, the impact of inequality on human capital investments and occupation choices, the process of social change and various other market failures in less-developed economies. By and large, this literature is about market failures in less-developed economies and sometimes also focuses on how these market failures can be rectified. The standard models of economic growth do not feature these market failures. A fruitful area for future research is then the combination of theoretical models of economic growth and development (that pay attention to market failures) with the rich empirical evidence on the incidence, characterization and costs of these market failures. This combination will have the advantage of being theoretically rigorous and empirically grounded, and perhaps most importantly, it can focus on what I believe to be the essence of development economics—the questions of why some countries are less developed, how they can grow more rapidly and how they can jumpstart the process of structural transformation necessary for economic development.

21.9. References and Literature

By its nature, this chapter has covered a large amount of material. My selection of topics and approaches corresponding to these topics has reflected my own interests and was also motivated by my desire to keep this chapter from becoming even longer than it already is. To obtain an in-depth understanding of the issues in the literature for any one of the topics covered here, the reader would need to study a large literature.

Section 21.1 scratches the surface of a rich literature on financial development and economic growth. On the theoretical side important papers include Townsend (1979), Greenwood and Jovanovic (1990), Bencivenga and Smith (1991), which focus on the interaction between financial development on the one hand and risk sharing and the allocation of funds across different tasks and individuals on the other. Obstfeld (1994), Saint-Paul (1992) and Acemoglu and Zilibotti (1997), which was discussed in Section 17.6, focus on the relationship between financial development and the diversification of risks. There is also a large empirical literature looking at the effect of financial development on economic growth. An excellent survey of this literature is provided in Levine (2005). Some of the most well-known empirical papers include King and Levine (1993), which documents the cross-country correlation between measures of financial development and economic growth, Rajan and Zingales (1998), which shows that lack of financial development has particularly pernicious effects on sectors that have a greater external borrowing needs, and Jayaratne and Strahan (1996), which documents how banking deregulation that increased competition in US financial markets led to more rapid financial and economic growth within the United States. In discussing financial development, I also mentioned the literature on the Kuznets curve. There is no consensus on whether there is a

Kuznets curve. Work that focuses on historical data, such as Lindert and Williamson (1976) or Bourguignon and Morrison (2004), report aggregate patterns consistent with a Kuznets curve, while studies using panels of countries in the postwar era, such as Fields (1994), do not find a consistent pattern resembling the Kuznets curve.

The literature on economic growth and fertility and on the demographic transition is even larger than the literature on financial development. The main trends in world population and cross-country differences in population growth are summarized in Livi-Bacci (1997) and Maddison (2003). The idea that parents face a tradeoff between the numbers and the human capital of their children—the quality and quantity tradeoff—was proposed by Becker (1981). The aggregate patterns in Livi-Bacci (1997) are consistent with this idea, though there is little micro evidence supporting this tradeoff. Recent work on microdata, by Black, Devereux and Salvanes (2005), Angrist, Lavy and Schlosser (2006) and Qian (2007), looks at evidence from Norway, Israel and China, but does not find strong support for the quality-quantity tradeoff. Fertility choices were first introduced into growth models by Becker and Barro (1988) and Barro and Becker (1989). Becker, Murphy and Tamura (1990) provide the first endogenous growth model with fertility choice. More recent work on the demographic transition and the transition from a Malthusian regime to one of sustained growth include Goodfriend and McDermott (1995), Galor and Weil (1996, 1999, 2000), Hansen and Prescott (2002), Tamura (2002), Lagerlof (2003) and Doepke (2004). Kalemli-Ozcan (2002) and Villaverde (2003) focus on the effect of declining mortality on fertility choices in a growth context. A recent series of papers by Galor and Moav (2002, 2004) combine fertility choice, quality-quantity tradeoff and natural selection. Galor (2005) provides an excellent overview of this literature. The first model presented in Section 21.2 is a simplified version of Malthus's classic model in his (1798) book, while the second model is a simplified version of Becker and Barro (1988) and Galor and Weil (1999).

Urbanization is another major aspect of the process of economic development. Bairoch (1988) provides an overview of the history of urbanization and an insightful discussion of some of the economic literature in this area. The first model in Section 21.3 builds on Arthur Lewis's (1954) classic, which argued that early development can be viewed as a situation in which there is surplus labor available to the modern technology, thus growth is constrained by capital and technology but not by labor. A formalization of Lewis's ideas naturally takes us to the realm of the dual economy, since surplus labor for the modern technology can remain only when there is limited interaction between the modern sector (or the cities) and the rural areas. Another well-known model by Harris and Todaro (1970) also emphasize the importance of model of migration, though it features free migration between rural and urban areas and suggests that unemployment in urban areas will be the key equilibrating variable.

The second model, presented in subsection 21.3.2, is inspired by Banerjee and Newman (1998) and Acemoglu and Zilibotti (1999). Banerjee and Newman emphasize the advantage of smaller rural communities in reducing moral hazard problems in credit relations and

show how this interacts with the process of urbanization, which involves individuals migrating to areas where their marginal product is higher. Acemoglu and Zilibotti argue that development—capital accumulation—leads to “information accumulation,” in particular, as more individuals perform similar tasks, more socially useful information is revealed and relative performance of valuations can be used more effectively to filter out common shocks. They show that greater information enables individuals to write more sophisticated contracts and draw the implications of these more complex contractual relations for technology choice, financial development and social transformations associated with economic development. Though inspired by these two papers, the model presented in subsection 21.3.2 was designed to be more reduced-form and simpler so as to communicate the basic ideas in the most economical way. In particular, it did not incorporate credit market relations in villages and urban areas as in Banerjee and Newman or risk-sharing contracts as in Acemoglu and Zilibotti. Instead, it emphasized another important aspect of social and economic relations in less-developed economies, the importance of community enforcement. The sociologist Clifford Geertz (1963), for example, emphasizes the importance of community enforcement mechanisms and how they may sometimes conflict with markets.

Section 21.4 builds on Acemoglu, Aghion and Zilibotti (2004, 2006). Evidence consistent with organizational changes related to the distance to the frontier are provided in Acemoglu, Aghion, Lelarge, Van Reenen and Zilibotti (2007).

Section 21.5 is based on Murphy, Shleifer and Vishny’s famous (1989) paper, which formalized ideas first proposed by Rosenstein-Rodan (1943). Other models that demonstrate the possibility of multiple equilibria in monopolistic competition models featuring nonconvexities include Kiyotaki (1988), who derives a similar result in a model with endogenous labor supply choices as well as investment decisions. Matsuyama (1996) provides an excellent overview of these models, as well as other approaches that can lead to multiple equilibria and multiple steady states. Matsuyama (1996) also provides a very clear discussion of why pecuniary externalities can lead to multiple equilibria in the presence of monopolistic competition.

The distinction between multiple equilibria and multiple steady states is discussed in Krugman (1994) and Matsuyama (1994). Both of these papers highlight that in models with multiple equilibria, expectations determine which equilibrium will be played, while with multiple steady states, there can be (or there often is) a unique equilibrium and initial conditions (history) determine where the economy will end up.

Section 21.6 covers the enormous literature on the role of inequality in human capital investments and occupational choices. The model in subsection 21.6.2 is based on the first model in Galor and Zeira’s well-known (1993) paper. Similar ideas are investigated in Banerjee and Newman (1994) in the context of the effect of inequality on occupational choice, and Aghion and Bolton (1998) and Piketty (1998) in the context of the interaction between inequality and entrepreneurial investments. The model in subsection 21.6.3 is based on Benabou (1996a,b), which investigates the dynamics of inequality, how inequality affects productive efficiency and the implications of different forms of community structures. Other

papers that investigate similar questions include Loury (1981), Tamura (1991, 2001), Durlauf (1996), Fernandez and Rogerson (1996, 1998), Glomm and Ravikumar (1992) and Acemoglu (1997). An excellent survey of this set of papers, together with extensions to analyze the interaction between political economy and inequality and between endogenous technology choices and inequality, is contained in Benabou (2005). There is also a large literature on inequality, human capital and taxation that incorporates political economy features. This literature will be discussed in the next chapter.

21.10. Exercises

EXERCISE 21.1. Analyze the equilibrium of the economy in Section 21.1 relaxing the assumption that each individual has to invest either all or none of his wealth in the risky saving technology. Does this generalization affect the qualitative results derived in the text?

EXERCISE 21.2. Consider the economy in Section 21.1.

- (1) Show that in eq. (21.5), $K(t+1)$ is everywhere increasing in $K(t)$ and that there exists some $\bar{K}$ such that the capital stock will grow over time when $K(t) > \bar{K}$.
- (2) Can there be more than one steady state level of capital stock in this economy? If so, provide an intuition for this type of multiplicity.
- (3) Provide sufficient conditions for the steady state level of capital stock, K^* , to be unique. Show that in this case $K(t+1) > K(t)$ whenever $K(t) > K^*$.

EXERCISE 21.3. In the model of subsection 21.2.1, suppose that the population growth equation takes the form

$$L(t+1) = \varepsilon(t) (n(t+1) - 1) L(t)$$

instead of (21.8), where $\varepsilon(t)$ is a random variable that takes one of two values, $1 - \bar{\varepsilon}$ or $1 + \bar{\varepsilon}$, reflecting random factors affecting population growth. Characterize the stochastic equilibrium. In particular, plot the stochastic correspondence representing the dynamic equilibrium behavior and analyze how shocks affect population growth and income dynamics.

EXERCISE 21.4. Characterize the full dynamics of migration, urban capital-labor ratio and wages in the model of subsection 21.3.1 (that is, consider the cases in which conditions 1, 2 and 3 in that subsection do not all hold together).

EXERCISE 21.5. Consider the model of subsection 21.3.2 and suppose that all individuals have utility given by the standard CRRA preferences,

$$U(0) \equiv \int_0^\infty \exp(-\rho t) \frac{c(t)^{1-\theta} - 1}{1-\theta} dt.$$

Taking the equilibrium path in that subsection as given, find a level of community enforcement advantage ξ that would maximize $U(0)$. What happens if the actual comparative advantage of community enforcement of villages is greater than this level?

EXERCISE 21.6. Consider the maximization problem given in (21.30).

- (1) Explain why this maximization problem characterizes the equilibrium allocation of workers to tasks. What kind of price system will support this allocation?

- (2) Derive the first-order conditions given in (21.31).
- (3) Provide sufficient conditions such that the solution to this problem involves all skilled workers employed at technology $\bar{h}$.
- (4) Provide an example in which no worker will be employed at technology $\bar{h}$ even though $A_{\bar{h}} > A_h$ for all $h \in [0, \bar{h}]$.
- (5) Can there be a solution where more than two technologies are being used in equilibrium? If so, explain the conditions for such an equilibrium to arise.

EXERCISE 21.7. Consider a variant of the model in Section 21.4, where firms have an organizational form decision, in particular, they decide whether or not to vertically integrate. For this purpose, consider a slight modification of eq. (21.37) where

$$A(\nu, t) = \eta \bar{A}(t-1) + \gamma(\nu, t) A(t-1),$$

with

$$\gamma(\nu, t) = \underline{\gamma} + \theta(\nu, t).$$

Suppose that entrepreneurial effort increases $\theta(\nu, t)$, and the internal organization of the firm affects how much effort the entrepreneur devotes to innovation activities. In particular, suppose that $\theta(\nu, t) = 0$ if there is vertical integration, because the entrepreneur is overloaded and has limited time for innovation activities. In contrast, with outsourcing $\theta(\nu, t) = \theta > 0$. However, when there is outsourcing, the entrepreneur has to share a fraction $\beta > 0$ of the profits with the manager (owner) of the firm to which certain tasks have been outsourced (whereas in a vertically integrated structure, he can keep the entire revenue).

- (1) Determine the profit-maximizing outsourcing decision for an entrepreneur as a function of the (inverse) distance to frontier $a(t)$. In particular, show that there exists a threshold $\bar{a}$ such that there will be vertical integration for all $a(t) \leq \bar{a}$ and outsourcing for all $a(t) > \bar{a}$.
- (2) Contrast this equilibrium behavior with the growth-maximizing internal organization of the firm.

EXERCISE 21.8. Show that when multiple equilibria exist in the model of Section 21.5, the equilibrium with investment Pareto dominates the one without.

EXERCISE 21.9. Consider the model of subsection 21.6.1 and remove the nonconvexity in the accumulation equation, (21.52), so that the human capital of the offspring of individual i is given by

$$h_i(t+1) = e_i(t)^\gamma,$$

for any level of $e_i(t)$ and $\gamma \in (0, 1)$. Show that there exists a unique level of human capital to which each dynasty will converge to. Based on this result, explain the role of nonconvexities in generating multiple steady states.

EXERCISE 21.10. Consider the model of subsection 21.6.1 and suppose that initial inequality is given by a uniform distribution with mean human capital of $h(0)$ and support over $[h(0) - \psi, h(0) + \psi]$. Clearly an increase in ψ corresponds to greater inequality.

- (1) Show that when $h(0)$ is sufficiently small, an increase in ψ will increase long-run average human capital and income, whereas when $h(0)$ is sufficiently large, an increase in ψ will reduce long run human capital and income. [Hint: use Figure 21.8 or Figure 21.9].
- (2) What other types of distributions (instead of uniform) would lead to the same result?
- (3) Show that the same result generalizes to the model of 21.6.2.
- (4) On the basis of this result, discuss whether we should expect greater inequality to lead to higher income in poor societies and lower income in rich societies. (If your answer is no, then sketch an environment in which this will not be the case).

EXERCISE 21.11. Consider the model presented in subsection 21.6.2. Make the following two modifications. First, the utility function is now

$$(21.68) \quad (1 - \delta)^{-(1-\delta)} \delta^{-\delta} c^{1-\delta} b^{\delta}$$

and second, unskilled agents receive a wage of $w_u + \varepsilon$ where ε is a mean-zero random shock.

- (1) Suppose that ε is distributed with support $[-\psi, \psi]$, and show that if ψ is sufficiently close to 0, then the multiple steady states characterized in 21.6.2 “survive” in the sense that depending on their initial conditions some dynasties become high skilled and others become low skilled.
- (2) Why was it convenient to change the utility function from the log form used in the text to (21.68)?
- (3) Now suppose that ε is distributed with support $[-\psi, \infty)$, where $\psi \leq w_u$. Show that in this case there is a unique ergodic distribution of wealth and no poverty trap. Explain why the results here are different from those in part 1?
- (4) How would the results be different if, in addition, the skilled wage is equal to $w_s + v$, where v is another mean-zero random shock? [Hint: simply sketch the analysis and the structure of the equilibrium without repeating the full analysis of part 3].

EXERCISE 21.12. Let us now discuss potential microfoundations for the borrowing constraints in the model of subsection 21.6.2.

- (1) Suppose that each individual can run away without paying his debts, and if he does so, he will never be caught. However, a bank that lends to the individual can make sure that the individual is unable to run away by paying a monitoring cost per unit of borrowing equal to m . Suppose that there are many banks competing for lending opportunities, so that Bertrand competition among them will drive them to zero profits. Under these assumptions, show that all bank lending will be accompanied with monitoring, and the lending rate will satisfy $i = r + m$. Show that in this case all of the results in the text apply.
- (2) Next suppose that the bank can prevent individual from running away by paying a fixed monitoring cost of M . Under the same assumptions as in part 1 above show that in this case the interest rate charged to an individual that borrows an amount $x - h$ will be $i(x - h) = r + M/(x - h)$. Given this assumption, characterize the

equilibrium of the model in subsection 21.6.2. How do the conclusions change in this case?

- (3) Next suppose that there is no way of preventing running away by individuals, but if an individual runs away, he will be caught with probability p , and in this case, a fraction $\lambda \in (0, 1)$ of his income will be confiscated. Given this assumption, characterize equilibrium dynamics of the model in subsection 21.6.2. How do the conclusions change in this case?
- (4) Now consider an increase in w_s (for a given level of w_u) so that the skill premium in the economy increases. In which on the three scenarios outlined above will this have the largest effect on human capital investments?

EXERCISE 21.13. In this exercise, you are asked to study Banerjee and Newman's (1994) model of occupational choice, which leads to similar results to the Galor-Zeira model, though with richer dynamics. The utility of each individual again depends on consumption and bequest, with

$$(1 - \delta)^{-(1-\delta)} \delta^{-\delta} c^{1-\delta} b^{\delta} - z$$

where z denotes whether the individual is exerting effort, with cost of effort normalized to 1. Each agent chooses one of four possible occupations. These are (1) subsistence and no work, which leads to no labor income and has a rate of return on assets equal to $\hat{r} < 1/\delta$; (2) work for a wage v ; (3) self-employment, which requires investment I plus the labor of the individual; and (4) entrepreneurship, which requires investment μI plus the employment of μ workers, and the individual himself becomes the boss, monitoring the workers (and does not take part in production). All occupations other than subsistence involve effort. Let us assume that both entrepreneurship and self-employment generate a rate of return greater than subsistence (that is, the mean return for both activities is $\bar{r} > \hat{r}$).

- (1) Derive the indirect utility function associated with the preferences above. Show that no individual will work as a worker for a wage less than 1.
- (2) Assume that $\mu [I(\bar{r} - \hat{r}) - 1] - 1 > I(\bar{r} - \hat{r}) - 1 > 0$. Interpret this assumption. [Hint: it relates the profitabilities of entrepreneurship and self-employment at the minimum possible wage of 1].
- (3) Suppose that only agents that have wealth $w \geq w^*$ can borrow enough to become self-employed and only agents that have wealth $w \geq w^{**} > w^*$ can borrow μI to become an entrepreneur. Explain why this type of borrowing constraint may be present.
- (4) Now compute the expected indirect utility from the four occupations. Show that if

$$v > \bar{v} \equiv \frac{\mu - 1}{\mu} (\bar{r} - \hat{r}) I,$$

than self-employment is preferred to entrepreneurship.

- (5) Suppose the wealth distribution at time t is given by $G_t(w)$. On the basis of the results in part 4, showed that the demand for labor in this economy is given by

$$\begin{aligned} x &= 0 & \text{if } v > \bar{v} \\ x &\in [0, \mu(1 - G_t(w^{**}))] & \text{if } v = \bar{v} \\ x &= \mu(1 - G_t(w^{**})) & \text{if } v < \bar{v}, \end{aligned}$$

- (6) Let $\tilde{v} \equiv (\bar{r} - \hat{r})I > \bar{v}$. Then, show that the supply of labor is given by

$$\begin{aligned} s &= 0 & \text{if } v < 1 \\ s &\in [0, G_t(w^*)] & \text{if } v = 1 \\ s &= G_t(w^*) & \text{if } 1 < v < \tilde{v} \\ s &\in [G_t(w^*), 1] & \text{if } v = \tilde{v}, \\ s &= 1 & \text{if } v > \tilde{v}. \end{aligned}$$

- (7) Show that if $G_t(w^*) > \mu[1 - G_t(w^{**})]$, there will be an excess supply of labor and the equilibrium wage rate will be $v = 1$. Show that if $G_t(w^*) < \mu[1 - G_t(w^{**})]$, there will be an excess demand for labor and the equilibrium wage rate will be $v = \bar{v}$.
- (8) Now derive the individual wealth (bequest) dynamics (for a worker with wealth w) as follows: (1) subsistence and no work: $b(t) = \delta \hat{r}w$; (2) worker: $b(t) = \delta(\hat{r}w + v)$; (3) self-employment: $b(t) = \delta(\bar{r}I + \hat{r}(w - I))$; (4) entrepreneurship: $b(t) = \delta(\bar{r}\mu I + \hat{r}(w - \mu I) - \mu v)$. Explain the intuition for each of these expressions.
- (9) Now using these wealth dynamics show that multiple steady states with different wealth distributions and occupational choices are possible. In particular, show that the steady-state wealth level of a worker when the wage rate is v will be $w_w(v) = \delta v / (1 - \delta \hat{r})$, while the steady-state wealth level of a self-employed individual will be $w_{se} = \delta(\bar{r} - \hat{r})I / (1 - \delta \hat{r})$, and the wealth level of an entrepreneur will be $w_e(v) = \delta(\bar{r}\mu I - \hat{r}\mu I - \mu v) / (1 - \delta \hat{r})$. Now show that when $w_w(v = 1) < w^*$ and $w_e(v = \bar{v}) > w^{**}$, a steady state in which the equilibrium wage rate is equal to $v = 1$ would involve workers not accumulate sufficient wealth to become self-employed, while entrepreneurs accumulate enough wealth to remain entrepreneurs. Explain why this is the case. [Hint: it depends on the equilibrium wage rate].
- (10) Given the result in part 10, show that if we start with a wealth distribution such that $\mu(1 - G(w^{**})) < G(w^*)$, the steady state will involve an equilibrium wage $v = 1$ and no self-employment, on whereas if we start with $\mu(1 - G(w^{**})) > G(w^*)$, the equilibrium wage would be $v = \bar{v}$ and there will be self-employment. Contrast the level of output in these two steady states.
- (11) Is the comparison of the steady states in terms of output in this model plausible? Is it consistent with historical evidence? What are the pros and cons of this model relative to the Galor-Zeira model in subsection 21.6.2?

EXERCISE 21.14. Explain why the aggregator in (21.57) could not be the production function of a final good producer, with each $h(t)$ corresponding to intermediates, but it can be used as an aggregator of the human capital levels of different individuals in the society.

EXERCISE 21.15. Given (21.59), derive (21.60). [Hint: take logs in (21.61) and (21.62)].

EXERCISE 21.16. Derive eq.'s (21.63) and (21.64).

EXERCISE 21.17. In the model of subsection 21.6.3, determine conditions under which the long run income level is higher under full integration than under full segregation.

EXERCISE 21.18. Consider the following non-overlapping generations model, with population normalized 1, where each individual i lives for one period and then begets an offspring. Each individual has preferences given by

$$c_i(t)^{1-\delta} e_i(t+1)^\delta,$$

where $c_i(t)$ is the consumption of the individual and $e_i(t+1)$ is educational investment in the human capital of the offspring. Each individual has some earned income $w_i(t)$ and is subject to the budget constraint $c_i(t) + e_i(t+1) \leq w_i(t)$. The human capital of the offspring, $h_i(t+1)$, is given by eq. (21.52) in subsection 21.6.1 with $\bar{h} = 1$ and $\gamma = 1$. There is also continuum of firms of mass 1, each with the production function

$$y_{i,j} = A(k_j)^\alpha (h_i)^{1-\alpha},$$

where worker i has been matched with firm j . Firms choose the level of physical capital investment at some cost $R > 0$ before matching with workers. Let us also assume that matching is random, so that any worker has the same likelihood of matching with any firm, and in particular, there is no selective process of high human capital workers being allocated to high physical capital firms. If the firm is not happy with the worker that it matches with, it can fire the worker and resample another worker from the remaining (potentially unmatched) distribution of workers. If it does so, it will have lost a fraction $1 - \eta$ of the time devoted for production, so its output will be only a fraction η of the expression given above. Conditional on matching, workers and firms bargain on the wage. Let us suppose that when the distribution of human capital is given by $\mu_t(h)$ and the distribution of physical capital is given by $\nu_t(k)$, the wage of a worker of human capital h_i matched with a firm of physical capital k_j is given by

$$\begin{aligned} w(h_i, k_j, \mu_t, \nu_t) = & \beta A(k_j)^\alpha (h_i)^{1-\alpha} - \beta(1-\beta) A(k_j)^\alpha \int h^{1-\alpha} d\mu_t(h) \\ & + \beta(1-\beta) A(h_i)^{1-\alpha} \int k^\alpha d\nu_t(k). \end{aligned}$$

- (1) Interpret this wage equation. Could you derive it from Nash bargaining? If so, be specific about what assumptions are necessary, particularly concerning what type of worker the worker will match with if it separates from its first partner and vice versa.
- (2) In view of this wage equation, show that there exists some η^* so that for all $\eta < \eta^*$, all firms will choose the same level of physical capital investment at time t given by $k^*[\mu_t, \eta, \beta, R]$. Show that in this case, a mean-preserving spread of the human capital distribution will reduce aggregate output. Provide an intuition for this result.

- (3) Suppose that $\eta < \eta^*$ as determined in part 2 and that the economy starts at time $t = 0$ with two groups of workers, a fraction λ with human capital $h_1(0)$ and a fraction $1 - \lambda$ with human capital $h_2(0) > h_1(0) > \bar{h} = 1$, where $\bar{h} = 1$ is the minimum human capital level defined in (21.52). Let $\phi(t) \equiv h_1(t)/h_2(t)$. Show that the economy will always have two groups of workers and thus its law of motion can be summarized by $\phi(t)$.
- (4) Derive a difference equation for $\phi(t)$ using the optimal capital investment level of firms, $k^*[\mu_t, \eta, \beta, R]$, derived in part 2, and the preferences of individuals regarding investments in their offspring's human capital.
- (5) Prove that there exists some $\bar{\phi} \in (0, 1)$, such that if $\phi(0) > \bar{\phi}$, then the dynamic equilibrium involves $\phi(t) \rightarrow 1$ and the economy achieves a constant growth rate. In contrast, if $\phi(0) < \bar{\phi}$, then $h_1(t) \rightarrow \bar{h}$ and $h_2(t) \rightarrow \tilde{h}$ for some $\tilde{h} > 1$, and the economy converges to no growth. Explain the intuition for this result.
- (6) Compare the model here with the model in subsection 21.6.3. What are its pros and cons? How would you generalize or make this model more realistic?

EXERCISE 21.19. This exercise asks you to analyze the dynamics of the reduced-form model in Section 21.7 more formally.

- (1) Show that when $f_x > 0$, the locus for $\dot{k}/k = 0$ implied by (21.66) is an upward sloping curve.
- (2) Consider the differential equations (21.66) and (21.67), and a steady state (k^*, x^*) . By linearizing the two differential equations around (k^*, x^*) , show that if $f_x(k^*, x^*)$ is sufficiently small, the steady state is locally stable.
- (3) Provide a bound on $f_x(k, x)$ over the entire domain so that there exists a unique steady state. Show that when this bound applies, the unique steady state is globally stable.
- (4) Construct a parameterized example where there are multiple steady states. Interpret the conditions necessary for this example. Do you find them economically likely?

Part 8

Political Economy of Growth

In this part of the book, I turn from the *mechanics* of economic growth to an investigation of potential *causes* of economic growth. Almost all of the models studied so far take economic institutions (such as whether property rights are enforced and what types of contracts can be written), policies (such as tax rates, distortions, subsidies) and often the market structure as given. They then derive implications for economic growth and cross-country income differences. While these models constitute the core of growth theory, they leave some of the central questions raised in Chapters 1 and 4 unanswered: why do some societies choose institutions and policies that discourage growth, while others choose growth-enhancing social arrangements? In this part of the book, I will make a first attempt to provide some answers to these questions based on political economy—that is, on differences in institutions and policies arising from different ways of aggregating individual preferences across societies and on differences in the type and nature of social conflict. In particular, I will emphasize a number of key themes and attempt to provide a tractable and informative formalization of these issues. The main themes are:

1. Different institutions and policies almost always generate winners and losers. In other words, there are a few economic or political reforms that would benefit all members of the society. Consequently, there will be *social conflict* concerning the types of policies and institutions that a society should adopt.

2. Aggregating the preferences of different individuals to arrive to collective choices is nontrivial in the presence of social conflict. Two interrelated factors will be central for the aggregation of these preferences: the form of *political institutions* and the *political power* of different groups. Individuals and groups with significant political power are more likely to be influential and sway policies towards their preferences. Exactly how political power is distributed within the society and how individuals can exercise their political power (resulting from their votes, connections or brute force) will depend on political institutions. For example, a dictatorship that concentrates political power in the hands of a small group will imply a different distribution of political power in the society than a democracy, which corresponds to a society with a greater degree of political equality. We expect that these different political regimes will induce different sets of economic institutions and policies, and thus lead to different economic outcomes. The purpose of the next two chapters is to investigate this process of collective decision-making and the implications of different choices of institutions and policies on economic growth.

3. The technology, the nature of the endowments and the distribution of income and endowments in the economy will influence both the preferences of different individuals and groups towards policies (or specific institutions) and also their political powers. For example, the nature of political conflict and the resulting political economy equilibrium is likely to be different in a society where much of the land and the capital stock is concentrated in the hands of a few individuals and families than one in which there is a more equitable distribution of resources. We would also expect politics to function differently in a society where the major

assets are in the form of human capital vested in individuals than in a society where natural resources, such as diamonds or oil, are the major assets.

The issues raised and addressed in this part of the book are central to the field of political economy. Since this is a book on economic growth not on political economy, I will not try to do justice to the large and growing literature in this area. Instead, I will focus on topics and models that I deem to be most important for the questions posed above. I will also save space and time by focusing, whenever possible, on the neoclassical growth model rather than some of the richer models that have been presented in this book. This might at first appear an odd choice. Why should we focus on the neoclassical growth model, which does not generate growth (other than via exogenous technological change), to study the political economy of growth? My answer is that the neoclassical growth model offers two significant advantages: first, it provides the most tractable framework to analyze the main political economy conflicts. Second, because the competitive equilibrium in the neoclassical growth model is Pareto optimal, it will make the role of political economy distortions more transparent. Naturally, once the basic forces are understood, it is relatively straightforward to incorporate them into endogenous growth models or other richer structures. Some of the exercises will consider these extensions.

I have organized the material on political economy of growth into two chapters. The first chapter takes political institutions as given and focuses on the implications of distributional conflict under different scenarios. In this chapter, I try to highlight why and when distributional conflict can lead to distortionary policies retarding growth. I will also offer various complementary frameworks for the analysis of these questions. Chapter 23 then turns to the implications of different political regimes for economic growth and also includes a brief discussion of how political institutions themselves are determined endogenously.

Before presenting this material, it is useful to start with an abstract discussion of the relationship between economic institutions, political institutions, and economic outcomes, and individual preferences over economic and political institutions. The distinction between economic and political institutions has already been highlighted in Chapter 4 and will be discussed again below. For now I take this distinction as given. Much of political science literature posits that individuals have (direct) preferences over political institutions (and perhaps also over economic institutions). For example, individuals might derive utility from living under a democratic system. While this is plausible, the approach developed so far emphasizes another, potentially equally important, reason for individuals to have preferences over political institutions. Economic institutions and policies have a direct effect on economic outcomes, however (for example, as illustrated by the effects of tax policies, regulation and contracting institutions in previous chapters). Thus a potentially major determinant of individual preferences on economic institutions (and policies) ought to be the allocations that result from these arrangements. Based on this viewpoint, throughout I will focus on these *induced* preferences on economic institutions.

The same applies to political institutions. These determine the *political rules* under which individuals interact. In direct democracy, for example, key decisions are made by majoritarian voting. In representative democracy, majorities choose representatives who then make the policy choices and face the risk of being removed from office if they pursue policies that are not in line with the preferences of the electorate. In contrast, in nondemocratic regimes, such as dictatorships or autocracies, a small clique, an oligarchy of rich individuals or a junta of generals make the key decisions. These differences in political rules imply that different political institutions lead to the different distributions of *de jure political power*, meaning that the institutionally-sanctioned distribution of political power, and thus the decision-making capacity, is distributed differently within the society. As a result, we would expect different policies and economic institutions to emerge in different political systems. For example, democracies are more likely to choose redistributive policies, whereas a society that is dominated by an economic elite or by a small group of individuals is likely to choose policies, that will further the economic interests of this narrow group. This reasoning suggests that since different political institutions will lead to different economic institutions and policies, and via this channel to different economic allocations, individuals will similarly have *induced preferences* over political institutions.

To emphasize this point, let us represent the chain of causation described above by a set of mappings. Let $\mathcal{P}$ denote the set of political regimes or institutions, $\mathcal{R}$ be the set of feasible economic institutions (or policies), and $\mathcal{X}$ denote the set of feasible allocations (which include different levels of consumption of all goods and services by all individuals in the society). Ignoring any stochasticity in outcomes for simplicity, we can think of each political institution in the set $\mathcal{P}$ leading to some specific set of economic institutions in the set $\mathcal{R}$. Let this be represented by the mapping $\pi(\cdot)$. Similarly, each different set of economic institutions will lead to a different allocation (ignoring again stochastic elements and multiple equilibria), and let this be represented by the mapping $\rho(\cdot)$. Schematically, we can write

$$\mathcal{P} \xrightarrow{\pi(\cdot)} \mathcal{R} \xrightarrow{\rho(\cdot)} \mathcal{X}.$$

Now suppose that each individual i has a utility function $u_i : \mathcal{X} \rightarrow \mathbb{R}$, representing his preferences over possible allocations in $\mathcal{X}$. Suppose also that individuals do not care about economic or political institutions beyond these institutions' influences on allocations. In other words, we presume that individuals are purely *consequentialist* (and thus ignore any direct benefits they may obtain from institutions). Then, their preferences over some economic institution $R \in \mathcal{R}$ is simply given by $u_i(\rho(R)) \equiv u_i \circ \rho : \mathcal{R} \rightarrow \mathbb{R}$. This mapping therefore captures their *induced* preferences over economic institutions (as a function of the economic allocations that these institutions will lead to). Preferences over political institutions are also induced in the same manner. The utility that individual i will derive from some political institution $P \in \mathcal{P}$ is simply given by $u_i(\rho(\pi(P)))$, where clearly $u_i \circ \rho \circ \pi : \mathcal{P} \rightarrow \mathbb{R}$. These preferences over institutions are important, since an equilibrium framework must, at least to

some degree, explain the emergence and change of political institutions as a function of the preferences of the members of the society over these objects.

Throughout, the next two chapters, I adopt this consequentialist view and define individual political preferences, over economic institutions or political institutions, purely in terms of these *induced preferences*—that is, preferences according to economic allocations that will ultimately result from these institutions. The interesting part of the analysis is to understand how economic institutions affect economic outcomes, how this shapes individual attitudes towards different economic institutions and policies and which political institutions will lead to what types of economic institutions. We have already seen in Chapter 4 some empirical approaches to how the cluster of economic and political institutions affect economic outcomes (including growth and distribution of resources). Much on my focus in the next two chapters will be to investigate the same linkages theoretically.

This brief introduction has therefore laid two types of foundations for the rest of this part of the book. (1) Our first task will be to understand how different types of economic institutions (and policies) affect economic outcomes, including economic performance and distribution of resources, which, here, is summarized by the mapping $\rho(\cdot)$. Then, based on this, we should analyze the preferences of different groups over these economic institutions (policies) and determine the conditions under which different groups will have a preference for distortionary, non-growth-enhancing economic arrangements. This will be the topic of the next chapter. (2) In order to understand political change and how it interacts with economic decisions and economic growth, we next need to understand induced preferences over political institutions, that is, understand the mapping $\pi(\cdot)$ and combine it with our insights about $\rho(\cdot)$. This will inform us on how political institutions affect economic arrangements, how economic arrangements influence economic allocations, and on the basis of this, how different groups value different sets of political institutions.

Institutions, Political Economy and Growth

This chapter will make a first attempt towards answering the following fundamental question that has been in the background of much of what we have done so far: *why do similar societies choose different institutions and policies, leading to very different economic growth outcomes?*

The analysis so far has highlighted the role of capital accumulation, human capital and technology in economic growth. We have investigated the incentives to accumulate physical capital and human capital, the process via which technology progresses and how different societies transfer and adopt technologies from others (or from the world technology frontier). Throughout I stressed that the level of physical capital, the extent of human capital and even the technology of societies should be thought of as endogenous and respond to incentives. This brings us to the fundamental question: why do different societies end up with different levels of physical capital, human capital and technology (or organize their production differently)? Although I cannot provide a full and comprehensive answer to this question, the recent literature has made considerable progress in the study of why societies differ in their choices. Chapter 4 has argued against the primacy of geographic and cultural factors and has instead suggested that differences in institutions are likely to be the most important fundamental cause of differences in economic performance. The purpose of this and the next chapter is to investigate this claim in greater detail and provide models that can help us understand why institutions might have such an effect and why institutions themselves differ across societies.

This chapter starts with a brief informal discussion of *political economic* analysis of institutions and policies. I will emphasize that different constellations of institutions and policies will typically create different winners and losers, and consequently, social conflict over collective choices (institutions and policies) will be ubiquitous. Political economy concerns the analysis of how societies resolve—or fail to resolve—these conflicts.

The rest of the chapter will present a number of models to shed light on the impact of distributional conflicts. I start with the simplest environments, highlighting why societies choose distortionary policies. I will then enrich these environments both to investigate the robustness of the channels they highlight and also how these mechanisms interact with each other. Special emphasis will be placed on distortionary policies arising because of two complementary reasons: (1) the desire of individuals or social groups to transfer resources to themselves using limited fiscal instruments, and (2) the potential conflict between different social groups in the marketplace or in the political arena. I will try to highlight why these

two sources of distortionary policies are distinct and I will also argue that the second source of distortions is typically more costly for growth.

I will conclude the chapter by pointing out that in addition to economic policies (such as taxes) and economic institutions (such as the security of property rights and the regulation of entry), the provision of public goods by the government is an important topic in the political economy of growth. Sustained economic growth is impossible, or at the very least very difficult, without the provision of the appropriate public goods and some amount of collective investment in infrastructure. Whether a particular society will invest in the right types of public goods is again a political economy question.

22.1. The Impact of Institutions on Long-Run Development

22.1.1. Institutions and Growth. As already emphasized in Chapter 4 “institutions” matter—at least when we look at clusters of economic and political institutions over long horizons. Moreover, most of the models in the book incorporate this feature, since they highlight various different effects of economic institutions and policies on economic allocations. For example, tax and subsidy policies and market structures may affect physical capital accumulation, human capital investments and technological progress, and contracting institutions and the structure of the credit markets will influence technology choices and the efficiency of production. Perhaps even more important, all of the models studied so far assume a relatively orderly working of the market economy. Add to these models some degree of insecurity of property rights or entry barriers preventing activities by the more productive firms, they will imply major inefficiencies. Both theory and casual empiricism suggest that these factors must be important. We are unlikely to explain the differences in economic growth or income per capita between the United States and much of sub-Saharan Africa by small differences in taxes on capital or in subsidies to R&D. Even differences in discount factors or exogenous technology are unlikely to lead to the huge differences in income per capita and growth documented in Chapter 1. Instead, we have to recognize and understand that doing business is very different in the United States than in sub-Saharan Africa. Entrepreneurs and businessmen in the United States (or pretty much everywhere in the OECD) face relatively secure property rights, and individuals or corporations that wish to create new businesses face relatively few barriers. The situation is very different in much of the rest of the world, for example, in sub-Saharan Africa, in the Caribbean and in large parts of Central America and Asia. Similarly, the lives of the majority of the population are radically different across these societies. In much of the OECD, most citizens have access to a wide variety of public goods and the ability to invest in their human capital, while the situation is different in many less-developed economies.

Economists often summarize these variations across societies as “*institutional differences*” (or differences in institutions and policies). The term is slightly unfortunate, but is one that is widely used and accepted in the literature. Institutions mean many different things in

different contexts, and none of these exactly correspond to the meaning intended here. As already emphasized in Chapter 4, by institutional differences, we are referring to differences in a broad cluster of social arrangements including the security of property rights for businesses as well as for regular citizens and the ability of firms and individuals to write contracts to facilitate their economic transactions (contracting institutions), the entry barriers faced by new firms, the socially-imposed costs and barriers facing individual decisions in human capital accumulation, and incentives of politicians and individuals in providing or contributing to the provision of public goods. This definition of institutions is quite encompassing. To make theoretical and empirical progress, one typically needs a narrower definition of institutions. Towards this goal, I have already distinguished between *economic institutions*, which correspond to the security of property rights, contracting institutions, entry barriers and other economic arrangements, and *political institutions*, which correspond to the rules and regulations affecting political decision-making, including checks and balances against presidents, prime ministers or dictators as well as methods of aggregating the different opinions of individuals in the society (for example, electoral laws). In terms of the notation introduced in the introduction to this part, the effect of economic institutions is summarized by the mapping $\rho(\cdot)$, while the implications of political institutions for the types of economic institutions and policies is captured by the mapping $\pi(\cdot)$.

It is also useful to note at this point that the difference between economic institutions and policies is not always clear, so it is often the combination of economic institutions and policies that matter not simply one or the other. For example, we can refer to security of property rights or to the quality of contracting institutions as economic institutions, but we would not typically refer to tax rates as institutions. Yet, entirely insecure property rights and 100% taxation of income have much in common. One difference might be that institutions are more durable than policies. Motivated by this, in Section 22.4, I will make a distinction between economic institutions and policies whereby economic institutions provide a framework in which policies are set. The role played by the durability of political institutions will be further studied in the next chapter. Finally, in Section 22.8, I will discuss another potential reason why taxation and security of property rights might be different. Nevertheless, the contrast of insecure property rights and 100% taxation illustrates the large overlap between economic institutions and policies, and when the distinction between the two is unclear or unimportant, I will typically use the terms “institutions and policies” and “economic institutions,” interchangeably, or whenever there is no risk of confusion with political institutions, I will simply refer to “institutions”.

The evidence presented in Chapter 4 suggests that institutional differences do matter for economic growth. The focus of this section is not to review this evidence, but to build on it and ask the next question: if economic institutions matter so much for economic growth, why do some societies choose institutions that do not encourage growth? In fact, based on available historical evidence we can go further: why do some societies choose institutions and policies that specifically *block* technological and economic progress? The rest of this chapter

and much of the next chapter will try to provide a framework for answering these questions. I start with an informal discussion of the main building blocks towards an answer.

The first important element of the political economy approach is *social conflict*. There are few (if any) economic changes that would benefit all agents in the society. Thus every change in institutions and policies will create winners and losers relative to the status quo. Take the simplest example: removing entry barriers so that a previously monopolized market becomes competitive. Economic theory tells us that this is desirable in the sense that it removes distortions and creates a “potential Pareto improvement”. In the context of growth, we often focus on the implications of changes in institutions and policies on the level of income or the rate of economic growth. In this respect as well, removing entry barriers is likely to be a beneficial reform, since the removal of monopoly power will increase the quantity transacted in the market and raise real incomes. Nevertheless, not all parties in the economy will be winners from the removal of entry barriers. While consumers will benefit because of lower prices, the monopolist, who was previously enjoying a privileged position and high profits, will be a “loser”. The effect on workers depends on the exact market structure. If the labor market is competitive, workers will also benefit, since the demand for labor will increase with the entry of new firms. But if there are labor market imperfections, so that the employees of the monopolist were previously sharing some of the rents accruing to this firm, they will also be potential losers from the reform. Thus if we start with the status quo of a monopoly and consider the reform of liberalizing markets (removing entry barriers), there will not be unanimous support for this proposal. Put differently, there will be social conflict over the policy of “market liberalization”.

The presence of social conflict over institutions and policies is not specific to reform. If, instead of starting with the status quo, we were deciding how markets should be organized without reference to any past arrangements, the same conflicts would be present. Many firms would prefer arrangements in which they are the monopolist protected by entry barriers, while consumers and potential entrants would prefer a more competitive arrangement.

Therefore, because of the different allocations that they will induce, individuals will have different, conflicting preferences over economic institutions. So if there are conflicting preferences over collective choices in general (and over institutions and policies in particular), how do societies make decisions? Political economy is the formal analysis of this process of collective decision-making. If there is social conflict between a monopolist that wishes to retain entry barriers and consumers that wish to dismantle them, it will be the equilibrium of a political process that decides the outcome. This process may be “orderly” as in democracies, or disorderly or even chaotic as in other political regimes as illustrated by the all too frequent civil wars throughout human history. Whether it is a democratic or a non-democratic process that will lead to the equilibrium policy, the *political power* of the parties with conflicting interests will play a central role. Put simply, if two individuals disagree over a particular choice (for example, about how to divide a dollar), how powerful each is will play an important role on the ultimate choice. In the political arena, this corresponds to the

political power of different individuals and groups. For example, in the monopoly example, we may expect the monopolist to have political power because it has already amassed income and wealth and may be able to lobby politicians. In a non-democratic society where the rule of law is tenuous, we can even imagine the monopolist utilizing thugs and paramilitaries to quash the opposition. On the other hand, in a democracy, consumers may have sufficient political power to overcome the interests and wishes of the monopolist through the ballot box or by forming their own lobbying groups. Whatever the outcome, political power will play an important role.

The second key element of the political economy approach is *commitment problems*, which will act both as a source of inefficiency and also augment the distortions created by social conflict. Political decisions at each date are made by the political process at that date (for example, by those holding political power at that point); commitment to future sequences of political and economic decisions are not possible unless they happen to be “equilibrium commitments” arising as part of the equilibrium (here, we will see that whether we use the concept of Subgame Perfect equilibrium or Markov Perfect Equilibrium will play a role in shaping the extent of available commitments).

At this point, it is important to distinguish between *non-growth-enhancing policies* (or distortionary policies) and *Pareto inefficiency*. Many political economy models will not lead to Pareto inefficiency (though some will). This is because in some reduced-form way their equilibrium can be represented as a solution to a weighted social welfare function (see Section 22.6). The resulting allocation, by virtue of maximizing this weighted social welfare function given the set of available instruments, will be a point along the constrained Pareto frontier of the economy. Nevertheless, many such allocations will involve distortionary and non-growth-enhancing policies (think, for example, of an allocation in which a dictator such as Mobutu in Zaire expropriates all the investors in the country; it is possible to change policies to increase investment and growth, but this will typically imply taking resources and power away from Mobutu and making him worse-off). Interestingly, when commitment problems are present, and especially when we focus on Markovian equilibria, the political equilibrium will typically lead to a constrained Pareto inefficient allocation, because there will often exist *future* policies that can make all parties better-off, but those policies will not arise as part of the equilibrium.

Consider a situation in which political power is in the hands of a specific group or an individual—the *political elite*. To simplify the thought experiment, let us ignore for now any constraints on the exercise of this political power (this is essentially where we will begin in the next section). Then, the elite can set policies in order to induce allocations that are most beneficial for themselves, and thus the political equilibrium can be thought of as the solution to the maximization of a social welfare function giving all the weight to the elite. Even though the resulting equilibrium will not necessarily be Pareto inefficient, it will typically involve non-growth-enhancing policies. Why and under what circumstances will the exercise of political power by the elite lead to such distortionary policies?

I will argue that there are two broad reasons for why those with political power will choose distortionary policies. The first is *revenue extraction*, that is, the attempt by the elite to extract resources from other members of the society using a limited menu of fiscal instruments. Central to this source of distortionary policies is two aspects of the society: (1) a decoupling between political power (which, at least in part, is here in the hands of the elite) and economic power (which lies with the entrepreneurs and the workers); (2) a limited set of fiscal instruments. These two aspects combined imply that the elite will use the available fiscal instruments to transfer resources from the rest of the society to themselves, which is the first potential reason for distortionary policies. We will also see that the same type of distortionary policies emerge even when there is no political elite, but decisions are made democratically (Section 22.7). The restriction to a limited set of fiscal instruments, such as linear taxes that discourage investment or work effort, is important here. Had there been non-distortionary taxes, such as lump-sum taxes, the elite could extract resources from the rest of the society without discouraging economic growth. But lump-sum taxes are often not possible, and more generally, most forms of redistribution do create distortions by reducing incentives for work effort or by discouraging investment.

I will argue, however, that the second reason for the use of distortionary policies by the political elite is potentially more damaging to economic growth. The elite will also choose distortionary policies because it will often be *in competition* with other social groups in society. This competition may be economic. For example, the elite may also engage in production and understand that by taxing and creating distortions on other entrepreneurs, they will be able to reduce their demand for factors (for example, labor) and thus increase their profits. I will refer to this as the *factor price manipulation* motive for distortionary policies. The competition between the elite and other social groups may also be political. The elite might foresee that enrichment by other groups will pose a threat to their political power and to their ability to use and benefit from their power in the future. When this is the case, they will use distortionary policies to impoverish their political competitors. I will refer to this as the *political replacement* motive for distortionary policies. The rest of the chapter will illustrate how distortionary policies can be adopted for extracting resources from different social groups and for factor price manipulation and political replacement motives. An interesting implication of the models I present will be that factor price manipulation and political replacement motives will often lead to greater distortions and will be more damaging to the growth potential of a society than the revenue extraction motive.

This basic framework also enables us to illustrate the additional inefficiencies created by commitment problems. In particular because the elite cannot commit to future policies, there will be a *holdup problem*, whereby investments, once undertaken, may be taxed at prohibitively high rates or expropriate. Holdup problems are likely to be important in a wide range of circumstances, for example, when the relevant investments are in long-term projects and assets, so that a range of policies will be decided after these investments are undertaken.

Much of the current chapter is devoted to understanding how the potential conflict over different economic allocations leads to different preferences over economic institutions and policies. The next two sections focus on distributional conflict in a simple society, consisting of different social groups. Throughout this chapter, I take the distribution of political power as given and in the next few sections, political power—and thus the authority to decide policy than economic institutions—will be in the hands of that group of individuals to whom I will refer to as “the elite”. I will investigate how the desire of this group of individuals to influence the allocation of resources in their favor may lead to distortionary policies that reduce investment and output. I will also highlight how these problems can become more severe in the presence of commitment/holdup problems.

Section 22.4 starts the investigation of how inefficiencies in policies might translate into inefficient (economic) institutions. In particular, in this section I will show how the same forces leading to distortionary policies will affect two aspects of economic institutions, whether effective constitutional limits on taxation and expropriation arise endogenously and whether there will be regulation (blocking) of new technologies. Economic institutions preventing future high taxes may emerge if holdup problems are important and the main source of distortionary policies is revenue extraction. In contrast, when factor price manipulation or political replacement motives are important, economic institutions limiting distortionary policies are unlikely to emerge. On the contrary, in this case, economic institutions that explicitly block the adoption of more efficient technologies may emerge. These results underlie the claim above that factor price manipulation and political replacement motives are typically more damaging to economic growth than the revenue extraction motive.

Throughout this chapter, I will focus on *comparative statics* that illustrate which types of societies are more likely to adopt growth-enhancing policies, and which others are likely to try to block economic growth. The major comparative static exercises will look at the effects of the nature of production technology, the distribution of resources within the society, whether the politically powerful compete with economically productive agents in factor markets, the extent of holdup problems, the importance of natural resources and whether or not political power is contested. While the field of the political economy of growth is still in its infancy, it is only by developing such comparative statics that it can contribute to a systematic analysis of why some societies grow and become rich, while others stagnate.

22.2. Distributional Conflict and Economic Growth in a Simple Society

The discussion in the previous section illustrated the complex set of forces that might affect collective choices concerning economic institutions and policies. The rest of this chapter and the next will focus on various dimensions of social conflict that will make societies adopt different economic institutions and policies, leading to different growth trajectories.

While my ultimate purpose is to present a relatively comprehensive framework, it is useful to start with a minimalist setup. For this reason, in this and the next two sections, I will discuss the implications of distributional conflict for economic growth in *a simple society*, in

which individuals are permanently allocated to certain groups (such as producers, landowners, workers) and the main distributional conflict is among groups. The latter feature is ensured by assuming that individuals within each group are ex ante identical and by restricting the set of fiscal instruments such that it is not possible to redistribute resources from one member of the group to another (at least not along the equilibrium path). The former feature, on the other hand, rules out issues of occupational choice and social mobility, which will be discussed in the next chapter. The main advantage of a simple society for our purposes here is that, thanks for the combination of a limited set of fiscal instruments and the symmetry of individuals within social groups, it enables a tractable aggregation of political preferences of individuals. Models in which there is a non-degenerate distribution of endowments (e.g., wealth) across individuals are studied in Section 22.7 below. While these models are significantly richer than the simple society studied here, the economic forces that shape the political economy equilibrium are similar, which motivates my choice of presenting a detailed analysis of political economy equilibrium in a simple society in the next few sections.

Consider a model in which there are three groups of individuals. The first is workers who supply their labor inelastically. The second group consists of entrepreneurs who have access to a production technology and make the investment decisions. The third group is the elite who make the political decisions (and may also engage in entrepreneurial activities). In particular, below I will assume that the political system is an *oligarchy*, dominated by the elite. The assumption that the elite make the political decisions, while the most important economic decisions, the level of investment, are made by entrepreneurs will highlight the impact of distributional conflict (given the set of fiscal instruments) on equilibrium policies and production in the sharpest possible way. The presence of three groups is important for the modeling of the effect of competition between the elite and other producers in the labor market in Sections 22.3 and 22.4. The model will then be enriched in various different ways in this and the next chapter by introducing additional heterogeneity, incorporating occupational choice and also endogenizing the distribution of political power among the various members of the society.

The baseline model is designed to be as similar to the standard neoclassical model in discrete time studied in Chapters 6 and 8 as possible. The focus on the neoclassical growth model was justified above (so as to abstract from imperfections other than those due to the political economy interactions). The focus on discrete time facilitates both the exposition and the analysis of game-theoretic interactions that are inherent in political-economy situations.

22.2.1. The Basic Environment. The economy is populated by a continuum $1 + \theta^e + \theta^m$ of risk-neutral agents, each with a discount factor equal to $\beta \in (0, 1)$. There is a unique non-storable final good denoted by Y . The expected utility of agent i at time 0 is given by:

$$(22.1) \quad \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t C_i(t),$$

where $C_i(t) \in \mathbb{R}$ denotes the consumption of agent i at time t and $\mathbb{E}_t$ is the expectations operator conditional on information available at time t . The most important feature about these preferences is their linearity (risk-neutrality). The gain in simplicity from the linear preferences more than makes up for the loss of generality—linear preferences remove some interesting transitional dynamics, but in return, enable a complete characterization of the political economy equilibrium. The next section will show that concave preferences complicate the analysis even in the most basic environment and often make it impossible to obtain a tight characterization of equilibria.

There is a continuum of workers, with measure normalized to 1, who supply their labor inelastically. The elite, denoted by e , initially hold political power in this society. There is a total of θ^e elites. As a starting assumption, we suppose that the elite do not take part in productive activities. Political economy interactions become considerably richer and more interesting (but also somewhat more involved) when the elite are also competing with other groups in product or factor markets. This issue will be discussed in Section 22.3. Finally, there are θ^m “middle class” agents, denoted by m , who are the entrepreneurs in the economy with access to the production technology. The label of “middle class” for the entrepreneurs is motivated by some historical examples that will be discussed in the next chapter and plays no role in the formal analysis. The sets of elite and middle class producers are denoted by S^e and S^m respectively. With a slight abuse of notation, I will use i to denote either individual or group (though when referring to groups, I will use i as superscript, and when referring to individuals, as subscript). The identity of the agents (their social group membership) does not change over time.

Each entrepreneur (middle-class agent) $i \in S^m$ has access to the following production technology for producing the final good:

$$(22.2) \quad Y_i(t) = F(K_i(t), L_i(t)),$$

where $Y_i(t)$ is final output produced by entrepreneur i , $K_i(t)$ and $L_i(t)$ are the total amount of capital and labor he uses in production. I assume that F satisfies the standard neoclassical assumptions from Chapter 2, Assumptions 1 and 2, which in particular means that F exhibits constant returns to scale. Without further restrictions, a single entrepreneur can employ the entire labor force and the capital stock of the economy. In this section, whether this is the case or not has no bearing on the results. But in some of the models below, it will be important to have a dispersed distribution of entrepreneurial activity. To ensure this, I also assume that there is a maximum scale for each entrepreneur (for example, because each entrepreneur has a limited span of control when it comes to managing his employees). In particular, suppose that $L_i(t) \in [0, \bar{L}]$ for some $\bar{L} > 0$. This implies that at least after a certain level of employment, there will be diminishing returns to additional capital investments by each entrepreneur. Since the total workforce in the economy is equal to 1, labor market clearing at time t in this economy requires

$$(22.3) \quad \int_{S^m} L_i(t) di \leq 1$$

with $L_i(t) \leq \bar{L}$. As in the standard neoclassical model, a fraction δ of capital depreciates.

The equilibrium of this economy without “political economy” is straightforward. Imagine that there are no taxes and labor markets are competitive. Let $k \equiv K/L$ denote the capital-labor ratio as usual and $f(k)$ be the per capita production function again defined as usual (that is, $f(k) \equiv F(K/L, 1)$). This immediately implies that each entrepreneur will choose the capital-labor ratio given by

$$(22.4) \quad k_i(t) = k^* \equiv (f')^{-1}(\beta^{-1} + \delta - 1)$$

for each t , where $(f')^{-1}(\cdot)$ is the inverse of the marginal product of capital (the derivative of the f function). Equation (22.4) is identical to the standard steady-state equilibrium condition from Chapters 6 and 8, which equates the gross marginal product of capital in steady state, $f'(k^*) + 1 - \delta$, with the inverse of the discount factor, β^{-1} (for example, recall eq. (6.49) in Chapter 6). The difference here is that this equation applies at all points in time, not only in steady state. This is a consequence of linear preferences, and implies that there are no transitional dynamics; regardless of initial conditions, each entrepreneur will immediately choose the capital-labor ratio k^* as in (22.4).

Another special feature of this economy is that it may fail to achieve full employment. Recall that the total labor force is equal to 1. However, eq. (22.5) shows that the level of employment of each employer may be strictly less than $1/\theta^m$ because of the maximum size constraint on firms. When this is the case, $1 - \theta^m \bar{L}$ workers will be unemployed and wages will be equal to 0. When there is excess supply of labor, each entrepreneur $i \in S^m$ will employ $\bar{L}$ workers and total employment will fall short of the total supply. When there is no excess supply, the entire labor force will be employed and the allocation of these workers across the entrepreneurs is arbitrary (since all entrepreneurs would be making zero profits). To simplify the exposition, I assume, without loss of any generality, that even in this case, all entrepreneurs will employ the same number workers, so that the equilibrium labor allocation satisfies

$$(22.5) \quad L_i(t) = L^* \equiv \min \left\{ \bar{L}, \frac{1}{\theta^m} \right\}$$

for each $i \in S^m$ at each t .

In addition, in this section I assume that

$$(22.6) \quad \theta^m \bar{L} > 1,$$

which insures that there will be full employment and thus $L^* = 1/\theta^m$. Under this assumption, the equilibrium wage rate at every date will also be given by the usual expression (which follows from Theorem 2.1 in Chapter 2)

$$(22.7) \quad w(t) = w^* \equiv f(k^*) - k^* f'(k^*),$$

where k^* is given in (22.4) above. I refer to the equilibrium without political economy (with capital-labor ratio k^* and wage rate w^*) as the *first-best equilibrium* and contrast it to political economy equilibria.

22.2.2. Policies and Economic Equilibrium. Before we can characterize political equilibria, we need to specify the set of available fiscal instruments (policies), and then define an economic equilibrium for given sequences of policies. Different economic equilibria will involve different levels of welfare for different agents, thus implicitly defining induced preferences over the policies and economic institutions leading to different economic equilibria (this point is further discussed in the next chapter). The political equilibrium then aggregates these preferences over different sequences of policies, taking into account the economic equilibrium that they will induce, to arrive to collective choices. In the current model, this last step is simplified given our focus on sequences of policies that maximize the utility of the elite. The characterization of the economic equilibrium is, in turn, much simplified thanks to linear preferences. Nevertheless, it is useful to go through each of these steps in order.

As for policies, suppose that the society has access to four different policy instruments at each date t :

- a linear tax rate on output $\tau(t) \in [0, \bar{\tau}]$, where $\bar{\tau} \in (0, 1]$ is a maximum tax rate that may be imposed constitutionally or technologically (for example, when the tax rate is above this level, all activity flees into the informal sector). In this and the next two sections, I take $\bar{\tau} = 1$, so that any tax rate is allowed. I will later analyze whether “constitutional” limit on taxes may be desirable and may emerge as part of the equilibrium.
- lump-sum transfers to each of the three groups (workers, middle-class entrepreneurs and the elite), $T^w(t) \geq 0$, $T^m(t) \geq 0$, and $T^e(t) \geq 0$.

Notice that the lump-sum transfers are assumed to be nonnegative. This rules out lump-sum taxes that could raise revenues without creating distortions. Instead revenues can only be raised using the linear tax on output, which, as we will see, will be distortionary. While lump-sum taxes might sometimes be possible, the ability of individuals to move into the informal sector or stop working puts limits on the use of lump-sum taxes. Nevertheless, the restriction to a simple linear tax rate is quite restrictive and there might often exist more efficient ways of raising revenues. In political economy models such restrictions are sometimes made so as to be able to characterize the equilibrium (for example, when using the Median Voter Theorem, see Section 22.6). Here they are imposed to emphasize how the interaction between the decoupling of political and economic power and a limited menu of fiscal instruments can lead to distortionary policies. I return to the question of why, even with efficient means of raising revenues, political economy motives can lead to non-growth-enhancing policies below (see Section 22.4).

Let us next specify the timing of events within each date. The most important aspect here is the timing of taxes relative to investments (and this is the main reason why discrete-time models are slightly more convenient in this context). To start with, let us assume that there is one period commitment to taxes. In other words, the timing of events is such that at each t , we start with a pre-determined tax rate on output $\tau(t)$, as well as the capital stocks $[K_i(t)]_{i \in S^m}$ of the entrepreneurs. Then, entrepreneurs decide how much labor to

hire $[L_i(t)]_{i \in S^m}$ (and in the process the labor market clears). Output is produced and a fraction $\tau(t)$ of the output is collected as tax revenue. After the tax revenue is observed, the political process (for example the politically powerful elite) decides the transfers, $T^w(t) \geq 0$, $T^m(t) \geq 0$, and $T^e(t) \geq 0$ subject to the government budget constraint

$$(22.8) \quad T^w(t) + \theta^m T^m(t) + \theta^e T^e(t) \leq \tau(t) \int_{S^m} F(K_i(t), L_i(t)) di,$$

where the left-hand side denotes total government expenditure in transfers and the right-hand side is the pre-determined tax rate times the output of all entrepreneurs. Next, the political process announces the tax rate $\tau(t+1)$ that will apply at the next date and entrepreneurs, after observing this tax rate, choose their capital stocks for the next date, $[K_i(t+1)]_{i \in S^m}$. The important feature in this timing of events is that entrepreneurs know exactly what tax rate they will face when choosing their capital stock. The alternative, where the capital stock is chosen before the tax rate, will be discussed in Section 22.3. For now it suffices to say that this alternative will lead to greater distortions because of *holdup problems*. It is therefore more natural to start with the timing of events specified here.

Let us also denote the policy or tax sequence starting at time t by $p^t = \{\tau(s), T^w(s), T^m(s), T^e(s)\}_{s=t}^\infty$, which specifies a *feasible* infinite sequence of policies starting at time t . One has to be a little careful about feasibility here, because whether a policy sequence is feasible or not cannot be determined without reference to the actions of the entrepreneurs (for example, any policy sequence with positive transfers cannot be feasible if all entrepreneurs choose zero capital stock). For our purposes, this is not important, since with linear preferences, only the tax rate sequence matters for capital and production decisions, and the transfers can be determined as residuals to satisfy the government budget constraint (22.8). It should nonetheless be noted that each individual, in particular, each entrepreneur, is infinitesimal, thus ignores his impact on total tax revenues and on the government budget constraint.

Let us define an *economic equilibrium* from time t onwards given a pre-determined distribution of capital stocks among the entrepreneurs, $[K_i(t)]_{i \in S^m}$ and a feasible policy sequence p^t . This economic equilibrium corresponds to a sequence of capital stock and labor decisions for each entrepreneur, $\{[K_i(s+1), L_i(s)]_{i \in S^m}\}_{s=t}^\infty$ and wage rates, $\{w(s)\}_{s=t}^\infty$, such that given $[K_i(t)]_{i \in S^m}$, p^t and $w^t \equiv \{w(s)\}_{s=t}^\infty$, $\{[K_i(s+1), L_i(s)]_{i \in S^m}\}_{s=t}^\infty$ maximizes the utility of entrepreneur i for each $i \in S^m$, and such that given $\{[L_i(s)]_{i \in S^m}\}_{s=t}^\infty$, the labor market clears. While an economic equilibrium appears to be a complicated object, the linearity of the preferences leads to a major simplification, enabling us to focus on the main political economy interactions.

Since workers supply labor inelastically, the only nontrivial decisions are by the entrepreneurs. As a first step towards characterizing the equilibrium, note that given *any* feasible policy sequence p^t and equilibrium wages w^t , the utility of an entrepreneur starting with

capital stock $K_i(t)$ at time t as a function of these policies can be written as
(22.9)

$$\begin{aligned} \mathbf{U}_i(\{K_i(s), L_i(s)\}_{s=t}^{\infty} | p^t, w^t) &= \sum_{s=t}^{\infty} \beta^{s-t} [(1 - \tau(s)) F(K_i(s), L_i(s)) \\ &\quad - (K_i(s+1) - (1 - \delta) K_i(s)) - w(s) L_i(s) + T^m(s)]. \end{aligned}$$

This expression makes use of the fact that preferences are linear, thus the value of the entrepreneur can be written simply in terms of the discounted sum of his consumption. His consumption, on the other hand, is simply given by the term in square brackets, since output is taxed at the rate $\tau(t)$ at time t and moreover, a fraction $(1 - \delta)$ of last period's capital stock is left, so an additional investment of $(K_i(t+1) - (1 - \delta) K_i(t))$ is made for next period. Finally, the labor costs at the current wage are subtracted and the lump-sum transfer to middle-class entrepreneurs is added. A special feature of (22.9) is that it is formulated for a given sequence of policies p^t . Loosely speaking, this could be thought of as the case if the sequence of policies were specified and committed to at some date. Although we are interested in political economy equilibria, where there is *no commitment* to future policies, we can think of the sequence of policies p^t as given from the viewpoint of an individual entrepreneur. The drawback is that this way of writing the maximization problem of the entrepreneur does not give information about how he would react if the political process (here the elite) deviated from p^t , since this might also be associated with a change in the remainder of the policy sequence. Nevertheless, linear preferences again ensure that we do not need to worry about this issue, since, as we will see momentarily, entrepreneurial decisions will only depend on current taxes. This issue of off-the-equilibrium path behavior becomes important when preferences are not linear and will be discussed in the next section.

Maximizing (22.9) with respect to the sequences of capital stock and labor choices, we obtain the following simple first-order condition:

$$(22.10) \quad \beta [(1 - \tau(t+1)) f'(k_i(t+1)) + (1 - \delta)] = 1,$$

where $k_i(t+1)$ denotes the capital-labor ratio chosen by entrepreneur i for time $t+1$ given the tax rate $\tau(t+1)$, which has already been announced (and committed to) at the time of the investment decision. Thanks to the Inada conditions in Assumption 2, this first-order condition holds as equality for any $\tau(t+1) \in [0, 1)$ and Exercise 22.1 shows that there will never be 100% taxation. Thus we do not need to spell out complementary slackness conditions.

Equation (22.10) determines the equilibrium capital-labor ratio. Given (22.6), that is, $\theta^m \bar{L} > 1$, there will be full employment of the total mass 1 of workers, and thus the total capital stock is also given by (22.10).

It can be verified easily that if all taxes were equal to zero ($\tau(t) = 0$ for all t), the unique solution to (22.10) would be identical to the steady-state capital-labor ratio k^* in (22.4) given in the previous subsection. Naturally, when there are positive taxes, the level of capital-labor ratio will be less than k^* (this follows immediately since $f(\cdot)$ is strictly

concave; see (22.12) below). It is also worth noting that while in the equilibrium “without political economy” (in this context, this means without taxes and transfers) the capital stock exhibited no transitional dynamics and immediately jumped to its steady-state value, this may no longer be the case in an economic equilibrium given the policy sequence p^t , since the policy sequence may involve time-varying taxes.

The most noteworthy feature of the equilibrium capital-labor ratio given in (22.10) is that, thanks to linear preferences, the choice of the capital-labor ratio by each entrepreneur at time $t+1$ only depends on the tax rate $\tau(t+1)$, and not on future taxes. We can therefore write the equilibrium capital-labor ratio at time t for all entrepreneurs as $\hat{k}(\tau(t))$:

$$(22.11) \quad \hat{k}(\tau(t)) \equiv (f')^{-1} \left(\frac{\beta^{-1} + \delta - 1}{1 - \tau(t)} \right).$$

The fact that the equilibrium capital-labor ratio depends only on one tax rate and is the same for all entrepreneurs will simplify the analysis of political economy considerably. For future reference, note also that since $F(\cdot, \cdot)$, and thus $f(\cdot)$, is twice differentiable, $\hat{k}(\tau)$ is also differentiable, with derivative

$$(22.12) \quad \hat{k}'(\tau) = \frac{f'(\hat{k}(\tau))}{(1 - \tau) f''(\hat{k}(\tau))} < 0,$$

which follows by directly differentiating (22.11) and is negative in view of the fact that $f'(k) > 0$ and $f''(k) < 0$ for all k (from Assumption 1).

Given the expression for the equilibrium capital-labor ratio in (22.11) and full employment as implied by (22.6), the equilibrium wage at time t is given by the usual expression:

$$(22.13) \quad \hat{w}(\tau(t)) = (1 - \tau(t)) \left[f(\hat{k}(\tau(t))) - \hat{k}(\tau(t)) f'(\hat{k}(\tau(t))) \right],$$

which is similar to (22.7) except for the presence of the tax rate in the front.

While (22.11) gives a very simple expression for the capital stock as a function of the tax rate, without knowing more about the sequence of policies we cannot ascertain whether the sequence of equilibrium capital stocks will converge to some steady-state value (for example, this will not be the case if taxes periodically fluctuate between different levels). Nevertheless, our analysis so far has established the following proposition:

PROPOSITION 22.1. *Suppose that (22.6) holds. Then, for any initial distribution of capital stocks among entrepreneurs, $[K_i(0)]_{i \in S^m}$ and for any feasible sequence of policies, $p^t = \{\tau(s), T^w(s), T^m(s), T^e(s)\}_{s=0}^\infty$, there exists a unique equilibrium in which the sequence of capital-labor ratios for each entrepreneur is $\{\hat{k}(\tau(s))\}_{s=0}^\infty$ and the equilibrium wage sequence is $\{\hat{w}(\tau(s))\}_{s=0}^\infty$ where $\hat{k}(\tau(t))$ is given by (22.11) and $\hat{w}(\tau(t))$ is given by (22.13).*

This proposition is convenient not only because the form of the equilibrium is particularly simple, but also because for any given sequence of policies, the aggregate equilibrium allocation is unique. If some policy sequences led to multiple equilibrium allocations, then

expectations concerning which of these equilibrium would be played conditional on these policy choices would complicate the analysis.¹

22.2.3. Political Economy under Elite Control. As noted at the beginning of this section, our task of characterizing the political economy equilibrium here is considerably simplified by the assumption that political power is entirely in the hands of the elite. There is no issue of political power changing hands or the elite choosing policies in order to appease voters or other social groups. Moreover, there are no fiscal instruments that would redistribute income among the elite. Thus the political economy choices here just involve the choice of fiscal policies at each date that would maximize the net present discounted utility of a representative elite agent.

Throughout this section, I focus on the *Markov Perfect Equilibria* (MPE) of the dynamic political game described here. This notion, which was first used in Chapter 14, requires the policy sequence p^t be such that policies dated t only depend on the dated t *payoff-relevant* variables (see the Appendix Chapter C for a formal definition of MPE). Here the only payoff-relevant variables are the capital stocks of the entrepreneurs. Thus most generally, current policies can depend on the current distribution of capital stocks. Linear preferences again simplify the analysis, and imply that we do not need to keep track of a complicated distribution of capital stocks as the relevant state variable. Since the MPE are a subset of the Subgame Perfect Equilibria (SPE) that do not condition on past history except through payoff-relevant variables, they rule out repeated game punishments (such as those relying on trigger strategies). I return to a discussion of situations in which the SPE are different from the MPE below.

For the characterization of the political economy choices of the elite, recall that the elite also care about their level of consumption (discounted with the discount factor β). From this observation, it is straightforward to see that they would never choose to redistribute to workers or to the middle class, thus in what follows we can restrict attention to sequences of policies that involve $T^w(t) = T^m(t) = 0$ for all t . Next, let us combine this fact with the government budget constraint, (22.8), which must hold as equality (since otherwise the elite could increase their consumption and utility by increasing transfers to themselves), to obtain

$$\begin{aligned}
 T^e(t) &= \frac{1}{\theta^e} \tau(t) \int_{S^m} F(K_i(t), L_i(t)) di \\
 (22.14) \qquad &= \frac{1}{\theta^e} \tau(t) f(\hat{k}(\tau(t))),
 \end{aligned}$$

¹Notice the slight abuse of notation in this proposition, which I will make throughout this and the next chapter: the equilibrium would not be “unique” in general, since the allocation of capital and labor across middle-class entrepreneurs is not pinned down. What is typically unique is the aggregate allocation and also the fact that any entrepreneur who is active must have a capital-labor ratio of $\hat{k}(\tau(t))$ at time t . In the present context, “uniqueness” is achieved by the assumption above that, when indifferent, all firms employ the same amount of labor. Throughout this chapter when the issue arises again, rather than explicitly state that the aggregate allocation implied by the equilibrium is unique, I will refer to the equilibrium as “unique”.

where the first line simply uses the government budget constraint (22.8), while the second line uses the equilibrium characterization in Proposition 22.1 together with the fact that with full employment, the total number of workers is equal to 1.

The problem of maximizing the utility of the elite agents can then be written as

$$(22.15) \quad \max_{\{\tau(t), T^e(t)\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t T^e(t)$$

subject to (22.14) at each t . Notice again that although it appears from (22.15) as if the elite were choosing the tax sequences at date $t = 0$, since there is no commitment to future policies, they are in fact only setting taxes for time $t + 1$ at time t . But this way of writing the elite's program will characterize the MPE since middle-class entrepreneurs' capital-labor ratio decisions at time $t + 1$ only depend on the tax rate announced for time $t + 1$, and not on future or past taxes.

To characterize the equilibrium tax sequence, note that $T^e(t)$ only depends on the tax rate at time t and involves the choice of the tax rate that would maximize tax revenue (i.e., that would put the elite at the peak of the "Laffer curve") at each date. Then, the utility-maximizing tax rate for the elite, $\hat{\tau}$, can be obtained as a solution to the following first-order condition:

$$f(\hat{k}(\hat{\tau})) + \hat{\tau} f'(\hat{k}(\hat{\tau})) \hat{k}'(\tau) = 0.$$

Substituting for $\hat{k}'(\tau)$ from (22.12), we obtain the following expression for $\hat{\tau}$:

$$(22.16) \quad f(\hat{k}(\hat{\tau})) + \frac{\hat{\tau}}{1 - \hat{\tau}} \frac{\left(f'(\hat{k}(\hat{\tau}))\right)^2}{f''(\hat{k}(\hat{\tau}))} = 0.$$

Intuitively, the utility-maximizing tax rate for the elite trades off the increase in revenues resulting from a small increase in the tax rate, $f'(\hat{k}(\hat{\tau}))$, against the loss in revenues that will result because the increase in the tax rate will reduce the equilibrium capital-labor ratio, $\hat{\tau} f'(\hat{k}(\hat{\tau})) \hat{k}'(\tau)$. It can be verified that this tax rate $\hat{\tau}$ is always between 0 and 1 (see Exercise 22.1), though the maximization problem of the elite is not necessarily concave and (22.16) may have more than one solution. If this is the case, $\hat{\tau}$ always corresponds to the global maximum for the elite.²

Notice that (22.16) implies a constant tax rate across different dates, and moreover, if we were to consider the maximization problem of the elite, (22.15), after some arbitrary date t' , exactly the same tax sequence would result. This is the reason why we could, without loss of any generality, focus on the maximization problem in (22.15). We will see in the next section that this is not always the case and is important to take into account the sequential nature of the decision-making by the elite and the entrepreneurs.

This analysis so far has thus established the following result:

²Here I ignore the "pathological" case in which there are multiple global maxima.

PROPOSITION 22.2. *Suppose that (22.6) holds. Then, for any initial distribution of capital stocks among entrepreneurs, $[K_i(0)]_{i \in S^m}$, there exists a unique MPE, where at each $t = 0, 1, \dots$, the elite set the tax $\hat{\tau} \in (0, 1)$ as given in (22.16), and all entrepreneurs choose the capital-labor ratio $\hat{k}(\hat{\tau})$ as given by (22.11) and the equilibrium wage rate is $\hat{w}(\hat{\tau})$ as given by (22.13). We have that $\hat{k}(\hat{\tau}) < k^*$, where k^* is given by (22.4) and $\hat{w}(\hat{\tau}) < w^*$, where w^* is given by (22.7).*

This proposition shows that a unique well-defined political equilibrium exists and involves positive taxation of entrepreneurs by the elite. Consequently, the capital-labor ratio and the wage rate are strictly lower than they would be in an economy without taxation. Strictly speaking, the equilibrium distortionary policies do not change “growth,” since we are focusing on a neoclassical economy without technological progress. As explained above, this is the only for convenience and it is straightforward to extend the framework here to incorporate endogenous growth, so that the distortionary policies affect the equilibrium growth rate of the economy (see Exercise 22.2).

Let us now return to the fundamental question raised at the beginning of this chapter: why would a society impose distortionary taxes on businesses/entrepreneurs? The model in this section gives a simple answer: political power is in the hands of the elite, who would like to extract revenues from the entrepreneurs. Given the available tax instruments, here linear taxes on output, the only way they can achieve this is by imposing distortionary taxes. Thus the source of “inefficiencies” in this economy is the combination of *revenue extraction* motive by the politically powerful combined with *a limited menu of fiscal instruments*.

While the analysis so far shows how distortionary policies can emerge and reduce the level of investment and output below the “first-best” level, it is important to emphasize that the equilibrium here is *not* Pareto inefficient. In fact, given the set of fiscal instruments, the equilibrium allocation is the solution to maximizing a social welfare function that puts all the weight on the elite. Pareto inefficiency requires that, *given the set of instruments and informational constraints*, there should exist an alternative feasible allocation that would make each agent either better-off or at least as well-off as they were in the initial allocation. Such an allocation can be found if we allowed lump-sum taxes. But given the restriction to linear taxes, which in the current economy are “technological,” there is no way of improving the utility of the middle-class entrepreneurs and the workers without making the elite worse-off.³ This is an important observation, since it implies that when we explicitly incorporate political economy aspects into the analysis, there are typically no “free lunches”—that is, no way of

³In a slightly modified environment there exist “mechanisms” that would lead to Pareto improvements, but these mechanisms could not be supported as MPE (but could be supported as SPE). For example, suppose that there is a finite number of entrepreneurs, who can make voluntary donations to a pot of money that is then redistributed to the elite. It can be proved that as the discount rate approaches 1, there exists a SPE in which each entrepreneur makes sufficient donations and chooses the first-best capital-labor ratio and the elite refrain from distortionary taxation. This example suggests that the MPE could easily lead to Pareto inefficient equilibria, even though this is not the case in our baseline economy. It also highlights why models with a continuum of agents, where such mechanisms are not possible, may be intuitive.

making all agents better-off. This is the reason why political economy considerations typically involve tradeoffs between losers and winners in the process of various different changes in institutions and policies. Since the allocation in Proposition 22.2 involves distortionary policies and reduces output relative to the first-best allocation, we might want to refer to this outcome as “inefficient” (despite the fact that it is not “Pareto inefficient”). In fact, this label is often used for such allocations in the literature and I will follow this practice. But as already emphasized above, “inefficiencies” do not mean Pareto inefficiencies.

As a preliminary answer to our motivating question, Proposition 22.2 is a useful starting point. However, it leaves a number of important questions unanswered. First, it does not provide useful comparative statics regarding when we should expect higher rates of distortion of the taxes. Second, it takes the distribution of political power as given, and it appears important for the results that political power rests in the hands of the non-productive elite, who are using the fiscal instruments to extract resources from middle-class producers. If political power were in the hands of the middle-class entrepreneurs rather than the non-productive elite, the choice of fiscal instruments would be very different. A first intuition might be that the entrepreneurs would never tax themselves. However, Exercise 22.3 shows that this is not necessarily the case and the middle-class entrepreneurs may prefer to tax themselves as a way of indirectly changing equilibrium wages. This is important because the role of fiscal policies in changing factor prices is often underappreciated and I will argue in Section 22.3 that it is one of the more important sources of political economy distortions in the process of growth. Third, this analysis takes the menu of available fiscal instruments as given. If the elite had access to lump-sum taxes, it could extract revenue from the entrepreneurs without creating distortions. I will extend the current framework to provide answers to these questions in this and the next chapter. Before doing this, let us first consider a more specific version of the economy analyzed so far where the production function is Cobb-Douglas. This Cobb-Douglas economy, by virtue of its tractability, will be a workhorse model for the analysis in Sections 22.3 and 22.4. Finally, Section 22.5 will consider a generalized version of the environment here where individuals have concave preferences.

22.2.4. The Canonical Cobb-Douglas Model of Distributional Conflict. Consider a specialized version of the economy analyzed so far, with two differences. First, the production function of each entrepreneur takes the following Cobb-Douglas form:

$$(22.17) \quad Y_i(t) = \frac{1}{\alpha} (K_i(t))^\alpha (A_i(t) L_i(t))^{1-\alpha},$$

where $A_i(t)$ is a labor-augmenting group-specific or individual-specific productivity term, which will be used later in this chapter. For now, we can set $A_i(t) = A^m$ for all $i \in S^m$. The term $1/\alpha$ in the front is included as a convenient normalization. We will see that this Cobb-Douglas form will enable an explicit-form characterization of the political equilibrium and will also link the elasticity of output with respect to capital to equilibrium taxes. This is the reason why I refer to this model as the “canonical model” of distributional conflict. Second, the analysis so far has shown that with linear preferences, incomplete depreciation of capital

plays no qualitative role, so I will also simplify the notation by assuming full depreciation of capital, that is, $\delta = 1$. This assumption is without any substantive implications.

The Cobb-Douglas production function in (22.17) implies that the per capita production function is given by

$$f(k_i) = \frac{1}{\alpha} (A^m)^{1-\alpha} k_i^\alpha.$$

Combining this production function with the assumption that $\delta = 1$, eq. (22.10) above implies that at date $t + 1$ each entrepreneur will choose a capital-labor ratio $k(t + 1)$ such that

$$(22.18) \quad k(t + 1) = [\beta(1 - \tau(t + 1))]^{1/(1-\alpha)} A^m.$$

The utility-maximizing tax policy of the elite is still given by eq. (22.16), which combined with the Cobb-Douglas form here implies that the utility-maximizing tax for the elite at each date is given by

$$\hat{\tau} = 1 - \alpha.$$

This formula is both simple and economically intuitive. When α is high, the production function is nearly linear in capital. This implies that the demand for capital as a function of its effective price is highly elastic. With such an elastic demand for capital, high taxes would lead to a large decline in the capital stock. Thus by charging high taxes, the elite would be reducing their own revenues. Put differently, with an elastic demand for capital (which in turn follows from production function that is not very concave in capital), the peak of the Laffer curve for the elite is at a low tax rate. On the other hand, if α is low, the production function is highly concave in capital, thus even a significant tax rate will not lead to a large decline in the equilibrium capital-labor ratio choice of the entrepreneurs. In this case, the elite will find it profitable to charge high taxes.

Both the tractability afforded by the Cobb-Douglas production function and the link between the concavity of the production function and equilibrium taxes that it highlights make this a very useful framework, and it will be used in a number of applications below.

22.3. Distributional Conflict and Competition

In this and the next section, I use the canonical framework with Cobb-Douglas production functions and full depreciation of capital ($\delta = 1$) to illustrate two important issues. I first investigate how competition (in the marketplace or in the political arena) between those with political power and the rest can lead to significantly more distortionary policies than the revenue extraction motive discussed so far. In the next section I will use the same framework to derive some preliminary insights on how distributional conflict can provide perspectives on equilibrium economic institutions regulating the formation of policies.

The model and the setup are essentially identical to the canonical Cobb-Douglas model in Section 22.2. In particular, the elite, of size θ^e , are in political power and decide all the policies. The timing of events is identical to that in Section 22.2. There are three differences,

however. First, the elite as well as the middle class can become entrepreneurs. The productivity of each middle-class agent in terms of this production function is A^m ($A_i = A^m$ for all $i \in S^m$) and the productivity of each elite agent is A^e ($A_i = A^e$ for all $i \in S^e$). Productivity of the two groups may differ, for example, because they are engaged in different economic activities (e.g., agriculture versus manufacturing, old versus new industries), or because they have different human capital or talent. Workers do not have access to these production functions and supply their labor inelastically. As in Section 22.2, each entrepreneur can hire at most $\bar{L}$ workers, and assumption (22.6) is no longer imposed. Second, I reintroduce the constitutional maximum on the tax rate, $\bar{\tau}$, so that $\tau(t) \in [0, \bar{\tau}]$ for all t . Finally, I now allow group-specific taxes so that the elite will choose two tax rates, $\tau^e(t)$, applying to the output of elite entrepreneurs, and $\tau^m(t)$, applying to middle-class entrepreneurs. The government budget constraint then takes the form

$$(22.19) \quad T^w(t) + \theta^m T^m(t) + \theta^e T^e(t) \leq \phi \int_{S^m \cup S^e} \tau^i(t) F(K_i(t), L_i(t)) di + R^N$$

where $\phi \in [0, 1]$ is a parameter that captures how much of tax revenue can be redistributed (with the remaining $1 - \phi$ being wasted). This parameter can be thought of as a measure of “state capacity”—with high ϕ , the state has the capacity to raise and redistribute significant revenues. R^N denotes rents from natural resources or from other sources unrelated to the production activities of the elite and the middle class. In Section 22.2, the government budget constraint, (22.8) involved $\phi = 1$ and $R^N = 0$. These parameters will be useful for comparative static exercises below.

Since there are entrepreneurs both from the elite and the middle class, the condition for full employment is different from (22.6). In particular, I assume throughout that $\theta^e \bar{L} < 1$ and $\theta^m \bar{L} < 1$, so that neither of the two groups generates enough labor demand by itself to employ the entire labor force. The following condition then determines whether the elite and the middle class together will generate enough labor demand for the entire labor force:

$$\text{CONDITION 22.1. } (\theta^e + \theta^m) \bar{L} > 1.$$

When this condition holds, there will be full employment. When it does not (by which I mean $(\theta^e + \theta^m) \bar{L} < 1$, excluding the knife-edge case $(\theta^e + \theta^m) \bar{L} = 1$, where there could be multiple equilibrium wage levels), there is a shortage of labor demand and equilibrium wages will be equal to 0. Whether this condition holds or not will affect the nature of the political equilibrium.

The analysis in Section 22.2, in particular, eq. (22.18), implies that the capital-labor ratio choice of each entrepreneur $i \in S^m \cup S^e$ will be given by

$$(22.20) \quad k_i(t+1) = \hat{k}_i(\tau(t+1)) \equiv (\beta(1 - \tau(t+1)))^{1/(1-\alpha)} A_i,$$

where the expression $\hat{k}_i(\tau)$ is implicitly defined by the second equality. This expression is very similar to eq. (22.11), but is adapted to the Cobb-Douglas production function, with labor-augmenting productivity of entrepreneur i equal to A_i . Substituting $\hat{k}_i(\tau)$ into the production function for each entrepreneur and subtracting the cost of investment, we obtain

that the net revenue per worker is $(1 - \alpha) (\beta (1 - \tau(t+1)))^{1/(1-\alpha)} A_i/\alpha$. This implies that the labor demand for each entrepreneur at time t as a function of the wage rate $w(t)$ will take the form

$$(22.21) \quad L_i(t) \begin{cases} = 0 & \text{if } w(t) > (1 - \alpha) (\beta (1 - \tau(t)))^{1/(1-\alpha)} A_i/\alpha \\ \in [0, \bar{L}] & \text{if } w(t) = (1 - \alpha) (\beta (1 - \tau(t)))^{1/(1-\alpha)} A_i/\alpha \\ = \bar{L} & \text{if } w(t) < (1 - \alpha) (\beta (1 - \tau(t)))^{1/(1-\alpha)} A_i/\alpha \end{cases}.$$

This expression states that if the wage exceeds the net marginal product (profitability) of an entrepreneur, given by $(1 - \alpha) (\beta (1 - \tau(t)))^{1/(1-\alpha)} A_i/\alpha$, then he would hire zero labor and shut down the firm. If the wage is strictly less than this net marginal product, then he would like to hire up to the maximum possible amount of labor, $\bar{L}$. The following proposition is then immediate:

PROPOSITION 22.3. *Consider the canonical elite-dominated politics model with Cobb-Douglas technology. Let the taxes on output of the elite and middle-class entrepreneurs at time t be $\tau^e(t)$ and $\tau^m(t)$, then the equilibrium capital-labor ratio of each entrepreneur is uniquely given by (22.20). In addition, if Condition 22.1 holds, then the equilibrium wage at time t is*

$$(22.22) \quad w(t) = \min \left\langle \frac{1 - \alpha}{\alpha} (\beta(1 - \tau^e(t)))^{1/(1-\alpha)} A^e, \frac{1 - \alpha}{\alpha} (\beta(1 - \tau^m(t)))^{1/(1-\alpha)} A^m \right\rangle.$$

If Condition 22.1 does not hold, then $w(t) = 0$.

The only part of this proposition that requires comment is the form of equilibrium wages, (22.22). This equation states that, because $\theta^e \bar{L} < 1$, $\theta^m \bar{L} < 1$ and Condition 22.1, each worker will receive the lower of his net marginal product in employment by the elite or by the middle class. Labor market clearing also implies that whichever group has lower net marginal product will not be able to employ up to its full capacity.

22.3.1. Competition in the Marketplace: The Factor Price Manipulation Effect. The next proposition is the equivalent of Proposition 22.2, except that it now applies when Condition 22.1 fails to hold. The reason for this is that, when this condition holds, there will also be the competition effect, changing the policy preferences of the elite. Propositions 22.4 and 22.5 below focus on the implications of competition in the factor market.

PROPOSITION 22.4. *Consider the canonical elite-dominated politics model with Cobb-Douglas technology. Suppose that Condition 22.1 does not hold and $\phi > 0$, then the unique MPE features*

$$(22.23) \quad \tau^m(t) = \tau^{RE} \equiv \min \{1 - \alpha, \bar{\tau}\} \text{ and } \tau^e(t) = T^m(t) = T^w(t) = 0$$

for all t . $T^e(t)$ is then determined from (22.19) holding as equality.

PROOF. See Exercise 22.4. □

This proposition thus shows that as in the version of the economy with Cobb-Douglas technology discussed in Section 22.2, the elite would like to set a tax rate of $1 - \alpha$ on middle-class entrepreneurs. If this tax is less than the constitutionally allowed maximum $\bar{\tau}$, the political equilibrium will involve $\tau^m = 1 - \alpha$. If on the other hand, $\bar{\tau} < 1 - \alpha$, the utility maximizing tax rate for the elite is $\tau^m = \bar{\tau}$ (this follows because the maximization program of the elite is strictly concave, see Exercise 22.4). Notice, however, that this proposition is stated under the assumption that Condition 22.1 fails to hold—so that the equilibrium wage rate is $w(t) = 0$ for all t . If this were not the case, the elite would also recognize the effect of their taxation policy on equilibrium wages. This would introduce the competition motive in the choice of policies, which is our next focus. An extreme form of this competition effect is shown in the next proposition. The state this proposition, I introduce one more condition:

CONDITION 22.2. *The maximum tax rate $\bar{\tau}$ is such that $A^e > (1 - \bar{\tau})^{1/(1-\alpha)} A^m$.*

The role of this condition will be discussed below.

PROPOSITION 22.5. *Consider the canonical elite-dominated politics model with Cobb-Douglas technology. Suppose that Condition 22.1 and 22.2 hold and $\phi = 0$, then the unique MPE features $\tau^m(t) = \tau^{FPM} \equiv \bar{\tau}$ and $\tau^e(t) = T^m(t) = T^w(t) = 0$ for all t .*

PROOF. See Exercise 22.5. □

In this case, ϕ is set equal to 0, so that there is no revenue extraction motive in taxation. Instead, the only reason why the elite might want to use taxes is in order to affect the equilibrium wage rate as given in (22.22). Clearly, for this we need Condition 22.1 to hold; otherwise, the wage rate would be equal to zero and the elite could derive benefit from manipulating factor prices. Condition 22.2 is necessary, since otherwise even at the maximal tax rate $\bar{\tau}$, the middle class entrepreneurs are more productive than the elite and the elite make zero profits. The noteworthy conclusion of Proposition 22.5 is that the equilibrium tax rate in this case, τ^{FPM} , is greater than the tax rate when the only motive for taxation was revenue extraction (τ^{RE}). This might at first appear paradoxical, but is quite intuitive. With the factor price manipulation mechanism, the objective of the elite is to reduce the profitability of the middle class as much as possible, whereas for revenue extraction, the elite would like the middle class to invest and generate revenues. Consequently, τ^{RE} puts the elite at the top of the Laffer curve, while τ^{FPM} tries to harm middle-class entrepreneurs as much as possible so as to reduce their labor demand (and thus equilibrium wages). It is also worth noting that, differently from the pure revenue extraction case, the tax policy of the elite is indirectly extracting resources from the workers, whose wages are being reduced because of the tax policy.

The role of the assumption that $\phi = 0$ in this context also needs to be emphasized. Taxing the middle class at the highest rate is clearly inefficient. Why is there not a more efficient way of transferring resources to the elite? The answer again relates to the limited fiscal instruments available to the elite. In particular, $\phi = 0$ implies that they cannot use

taxes to extract revenues from the middle class, so they are forced to use inefficient means of increasing their consumption, by directly impoverishing the middle class. The absence of any means of transferring resources from the middle class to the elite is not essential for the factor price manipulation mechanism, however. This will be illustrated next by combining the factor price manipulation motive with revenue extraction (though the absence of non-distortionary lump-sum taxes is naturally important).

The next proposition derives the equilibrium when Condition 22.1 holds and $\phi > 0$, so that both the factor price manipulation and the revenue extraction motives are present. Proposition 22.5 showed that by itself the factor price manipulation motive leads to the extreme result that the tax on the middle class should be as high as possible. Revenue extraction, though typically another motive for imposing taxes on the middle class, will serve to reduce the power of the factor price manipulation effect. The reason is that high taxes also reduce the revenues extracted by the elite (moving the economy *beyond the peak* of the Laffer curve), and are costly to the elite. To derive the political equilibrium in this case, first note that the elite will again not tax themselves, that is, $\tau^e(t) = 0$ for all t . Next the maximization problem of the elite at time $t - 1$ for setting the tax rate $\tau^m(t)$ can be written as:

$$(22.24) \quad \max_{\tau^m(t)} \left[\frac{1-\alpha}{\alpha} \beta^{1/(1-\alpha)} A^e - w(t) \right] L^e(t) + \frac{1}{\theta^e} \left[\frac{\phi}{\alpha} \tau^m(t) (\beta(1-\tau^m(t)))^{\alpha/(1-\alpha)} A^m \theta^m L^m(t) + R^N \right],$$

subject to (22.22) and

$$(22.25) \quad \theta^e L^e(t) + \theta^m L^m(t) = 1, \text{ and}$$

$$(22.26) \quad L^m(t) = \bar{L} \quad \text{if} \quad (1 - \tau^m(t))^{1/(1-\alpha)} A^m \geq A^e,$$

where $L^m(t)$ denotes equilibrium employment by a middle-class entrepreneur and $L^e(t)$ is equilibrium employment by an elite entrepreneur. The first term in (22.24) is the elite's net revenues and the second term is the transfer they receive. Equation (22.25) is the labor market clearing constraint, while (22.26) ensures that middle class producers employ as much labor as they wish provided that their net productivity is greater than those of elite producers.

The solution to this problem can take two different forms depending on whether (22.26) holds at the optimal solution. If it does, then $w = (1 - \alpha) \beta^{1/(1-\alpha)} A^e / \alpha$, and elite producers make zero profits and their only income is derived from transfers. Intuitively, this corresponds to the case where the elite prefer to let the middle class producers undertake all of the profitable activities and maximize tax revenues. In this case, the equilibrium will be clearly identical to that in Proposition 22.4. If, on the other hand, (22.26) does not hold, then the elite generate revenues both from their own production and from taxing the middle class producers. In this case, the equilibrium wage will be $w(t) = (1 - \alpha) (\beta(1 - \tau^m(t)))^{1/(1-\alpha)} A^m / \alpha$. The next proposition focuses on this case:

PROPOSITION 22.6. *Consider the canonical elite-dominated politics model with Cobb-Douglas technology. Suppose that Condition 22.1 holds, $\phi > 0$, and*

$$(22.27) \quad A^e \geq \phi \alpha^{\alpha/(1-\alpha)} A^m \frac{\theta^m}{\theta^e}.$$

Then, the unique MPE features

$$(22.28) \quad \tau^m(t) = \tau^{COM} \equiv \min \left\{ \frac{\kappa(\bar{L}, \theta^e, \alpha, \phi)}{1 + \kappa(\bar{L}, \theta^e, \alpha, \phi)}, \bar{\tau} \right\},$$

for all t , where

$$(22.29) \quad \kappa(\bar{L}, \theta^e, \alpha, \phi) \equiv \frac{1 - \alpha}{\alpha} \left(1 + \frac{\theta^e \bar{L}}{(1 - \theta^e \bar{L}) \phi} \right).$$

PROOF. See Exercise 22.6. □

A number of features about this proposition are worth noting. First, $\kappa(\bar{L}, \theta^e, \alpha, \phi)$ is always less than infinity, so that the most preferred tax rate by the elite is always less than 1. Recall that with the pure factor price manipulation motive, the elite preferred a tax rate of 100% (though their actual tax policy may be constrained by $\bar{\tau}$). Proposition 22.6 therefore shows that the prospect of raising revenues from the middle class reduces the desired tax rate by the elite. On the other hand, $\kappa(\bar{L}, \theta^e, \alpha, \phi)$ is always strictly greater than $(1 - \alpha)/\alpha$, so that τ^{COM} is always greater than $1 - \alpha$, the desired tax rate with pure resource extraction. Therefore, the factor price manipulation motive always increases taxes above the pure revenue maximizing level, and thus beyond the peak of the Laffer curve (though never to as high as 100%). Naturally, if this level of tax is greater than $\bar{\tau}$, the equilibrium tax will be $\bar{\tau}$.

Second, since Proposition 22.6 incorporates both the revenue extraction and the factor price manipulation motives, it contains the main comparative static results of interest for us. First, the equilibrium tax rate is *decreasing* in ϕ , because as ϕ increases, revenue extraction becomes more efficient and this has a moderating effect on the tax preferences of the elites. Loosely speaking, this shows the positive side of “state capacity”; with greater state capacity, the politically powerful can raise revenues through taxation, thus their motives to impoverish competing groups become weaker (we will see a potentially negative or “dark” side of state capacity below). Second, the equilibrium tax rate is increasing in θ^e . The reason for this is again the interplay between the revenue extraction and factor price manipulation mechanisms. When there are more elite producers, reducing factor prices becomes more important relative to raising tax revenues. This comparative static thus reiterates that when the factor price manipulation effect is more important, there will typically be greater distortions. Third, a decline in α raises equilibrium taxes for exactly the same reason as in the pure revenue extraction case; taxes create fewer distortions and this increases the revenue-maximizing tax rate. Finally, for future reference, note that rents from natural resources, R^N , have no effect on equilibrium policies.

22.3.2. Political Competition: The Political Replacement Effect. The previous subsection illustrated how competition in the factor market between the elite and the middle class induces the elite to choose distortionary policies to reduce the labor demand from the middle class. In this section, I will briefly outline the implications of competition in the political arena for equilibrium taxes. The main difference from the models studied so far is that I now allow for endogenous switches of political power. Institutional change and the implications of different political regimes for economic growth will be discussed in greater detail in the next chapter. For now, let us denote the probability that in period t political power permanently shifts from the elite to the middle class by $\eta(t)$. Once they come to power, the middle class will pursue the policies that maximize their own utility. We can easily derive what these policies would be using the same analysis as in the previous subsection. Since the analysis is identical to that above, this is left to Exercise 22.7. Denote the utility of the elite when they are in control of politics and when the middle class are in control of politics by $V^e(E)$ and $V^e(M)$ respectively.

When the probability of the elite losing power to the middle class, η , is exogenous, the analysis in the previous subsection applies without any significant change. New political economy effects arise, however, when the probability that the elite will lose power is endogenous. To save space while communicating the main ideas, I use a reduced-form model and assume that the probability that the elite will lose power to the middle class is a function of the net income level of the middle class, in particular,

$$(22.30) \quad \eta(t) = \eta(\theta^m C^m(t)) \in [0, 1],$$

where $C^m(t)$ is the net income of a representative middle-class entrepreneur, which is also equal to his consumption. I assume that η is differentiable and strictly increasing, with derivative $\eta'(\cdot) > 0$. This assumption implies that when the middle class producers are richer, they are more likely to gain power, which may be because with greater resources, they may be more successful in solving their collective action problems or they may increase their military power.

To simplify the discussion, let us focus on the case in which Condition 22.1 fails to hold, so that equilibrium wage is equal to 0 and there is no factor price manipulation motive. Thus in the absence of the political replacement motive, the only reason for taxation will be revenue extraction (resulting in an equilibrium tax rate of τ^{RE}). Given these assumptions and the definitions of $V^e(E)$ and $V^e(M)$, we can write the maximization problem of the elite when choosing the tax rate $\tau^m(t)$ at $t - 1$ as

$$\begin{aligned} V^e(E) = & \max_{\tau^m} \{ \beta^{\alpha/(1-\alpha)} A^e \bar{L} / \alpha + [\phi \beta^{\alpha/(1-\alpha)} \tau^m (1 - \tau^m)^{\alpha/(1-\alpha)} A^m \theta^m \bar{L} / \alpha + R^N] / \theta^e \\ & + \beta [(1 - \eta[\tau^m]) V^e(E) + \eta[\tau^m] V^e(M)] \}, \end{aligned}$$

where I wrote $\eta[\tau^m]$ to emphasize the dependence of the replacement probability on the tax rate on the middle class (while economizing on notation by not explicitly spelling out the argument of the $\eta(\cdot)$ function).

The first-order condition for an interior solution for the tax rate τ^m is:

$$\frac{\phi \beta^{\alpha/(1-\alpha)} (1 - \tau^m(t))^{\alpha/(1-\alpha)} A^m \theta^m \bar{L}}{\alpha \theta^e} \left(1 - \frac{\alpha}{1-\alpha} \frac{\tau^m(t)}{1 - \tau^m(t)} \right) - \beta \frac{d\eta \left((\beta(1 - \tau^m))^{\alpha/(1-\alpha)} A^m \theta^m \bar{L} / \alpha \right)}{d\tau^m} (V^e(E) - V^e(M)) = 0.$$

The first-term in this first-order condition corresponds to the revenue extraction motive, while the second term relates to the political replacement effect. Inspection of this condition shows that when $\eta'(\cdot) = 0$, we obtain $\tau^m = \tau^{RE} \equiv \min\{1 - \alpha, \bar{\tau}\}$ as above. However, when $\eta'(\cdot) > 0$ and $V^e(E) - V^e(M) > 0$, $\tau^m(t) = \tau^{PC} > \tau^{RE} \equiv \min\{1 - \alpha, \bar{\tau}\}$. The result that $V^e(E) - V^e(M) > 0$ follows from Exercise 22.7.

The important point here is that, as with the factor price manipulation mechanism, the elite tax *beyond the peak* of the Laffer curve. Their objective is not to increase their current revenues, but to consolidate their political power (in fact, taxes beyond the peak of the Laffer curve *reduce* the current income of the elite). However, higher (more distortionary) taxes are still useful for the elite because they reduce the income of the middle class and their political power. Consequently, there is a higher probability that the elite remain in power in the future, enjoying the benefits of controlling fiscal policy.

A number of new comparative static results follow from the possibility that the elite might lose political power. First, as R^N increases, it is straightforward to verify that the gap between $V^e(E)$ and $V^e(M)$ increases (see Exercise 22.7). This immediately translates into a higher equilibrium tax rate on the middle class. Intuitively, the party in power receives the revenues from natural resources, R^N and when these revenues are higher, *political stakes*—defined as the value of controlling political power—are greater. Consequently, the elite are more willing to sacrifice tax revenue (by overtaxing the middle class) in order to increase the probability that they remain in power (because remaining in power has now become more valuable). This contrasts with the results so far where R^N had no effect on taxes. Moreover, in this case a higher state capacity, ϕ , also increases the gap between $V^e(E)$ and $V^e(M)$ (because this enables the group in power to raise more tax revenues, see Exercise 22.7) and thus creates a force towards higher equilibrium taxes (though this effect might be dominated by the tax-reducing effect of ϕ emphasized in the previous subsection). This comparative static result therefore shows the potential “dark side” of greater state capacity; when there is no political competition, greater state capacity, by allowing more efficient forms of transfers, improves the allocation of resources. In contrast, in the presence of political competition, a greater state capacity, increases the political stakes and may induce more distortionary policies.

Finally, when the replacement of the elite by the middle class is very likely (corresponding to $\eta \approx 1$), or when such political replacement is very unlikely ($\eta(\cdot) \approx 0$), we will have that $\eta'(\cdot)$ will be uniformly low. In these cases, there is only limited increase in the tax rate above the revenue maximizing level. It is only when η takes intermediate values and depends on the wealth level of the middle class that $\eta'(\cdot)$ will be high and the political replacement

effect will induce further distortionary taxes. Therefore, we expect the elite to choose more distortionary policies when they have an intermediate level of security (rather than when they are entirely secure in their political power, i.e., $\eta(\cdot) \approx 0$, or when they definitely expect to be replaced, i.e., $\eta(\cdot) \approx 1$). This is the sense in which the political replacement effect here is very similar to the replacement effect pointed out by Arrow in the context of innovation (recall Chapter 12).

22.3.3. Subgame Perfect Versus Markov Perfect Equilibria. I have so far focused on Markov Perfect Equilibria (MPE). In general, such a focus can be restrictive. A natural question is whether the set of Subgame Perfect Equilibria (SPE) is larger than the set of the MPE and whether some of the SPE can lead to more efficient allocation of resources (see Appendix Chapter C for formal definitions of MPE and SPE and differences between the two concepts). I will first show that in the setup analyzed so far the set of SPE and MPE coincide (this is of course not always the case, for example, as suggested by the discussion in footnote 3). I will then turn to potential holdup problems, exacerbating the commitment problems involved in the economy, and see that the SPE can lead to a more efficient allocation of resources than the MPE because it allows for greater “equilibrium commitment” on the part of the elite.

Essentially, the MPE are generally a subset of the SPE, because the latter include equilibria supported by some type of “history-dependent punishment strategies”. If there is no room for such history dependence, SPEs will coincide with the MPEs. In the models analyzed so far, such punishment strategies are not possible even in the SPE. Intuitively, each individual is infinitesimal and makes its economic decisions to maximize profits. Therefore, (22.20) and (22.21) determine the factor demands uniquely in any equilibrium. Given the factor demands, the payoffs from various policy sequences are also uniquely pinned down. This means that the returns to various strategies for the elite are *independent of history*. Consequently, there cannot be any SPEs other than the MPE characterized above. Therefore:

PROPOSITION 22.7. *The MPEs characterized in Propositions 22.4-22.6 are the unique SPEs.*

PROOF. See Exercise 22.9. □

In addition, Exercise 22.10 shows that the MPE in the model of subsection 22.3.2 is also the unique SPE. This last result, however, depends on the assumption that there is only one possible power switch (from the elite to the middle class). If, instead, there were continuous power switches, potential punishment strategies could be constructed and the set of SPEs could include non-Markovian equilibria.

22.3.4. Lack of Commitment—Holdup. The models discussed so far featured full commitment to one-period ahead taxes by the elites. In particular, at the end of period t , the elite can commit to the tax rate on output that will apply at time $t + 1$. Using a term from organizational economics, this corresponds to the situation without any “holdup”. *Holdup*,

on the other hand, corresponds to a situation without commitment to taxes or policies, so that after entrepreneurs have undertaken their investments they can be “held up” by higher rates of taxation or by expropriation. These types of holdup problems are endemic in political economy situations, since commitments to future policies is difficult or impossible. Those who have political power at a certain point in time are likely to make the relevant decisions at that point. Moreover, when the key investments are long-term (so that once an investment is made, it is irreversible), there will be a holdup problem even if there is a one period commitment (since there will be taxes that will affect this investment in its revenues after the investment decisions are sunk).

The problem with holdup is that the elite will be unable to commit to a particular tax rate before middle-class producers undertake their investments (because taxes will be set after investments). This lack of commitment will generally increase the amount of taxation and distortions. Moreover, in contrast to the allocations so far, which featured distortions but were Pareto optimal, the presence of commitment problems will lead to Pareto inefficiency. To illustrate the main issues that arise in the presence of commitment problems in the simplest possible way, I consider the same model as above, but change the timing of events such that taxes on output at time t are decided in period t , that is, after the capital investments for this period have already been made (instead of at $t - 1$, before these capital investments, as assumed so far). The economic equilibrium is essentially unchanged, and in particular, (22.20) and (22.21) still determine factor demands, with the only difference that τ^m and τ^e now refer to “expected” taxes. Naturally, in equilibrium expected and actual taxes coincide.

What is different is the calculus of the elite in setting taxes. Previously, they took into account that higher taxes on output at date t would discourage investment for production at date t . Since, now, taxes are set after investment decisions are sunk, this effect is absent. As a result, in the MPE, the elite will always want to tax at the maximum rate, so in all cases, there is a unique MPE where $\tau^m(t) = \tau^{HP} \equiv \bar{\tau}$ for all t . This establishes:

PROPOSITION 22.8. *With holdup, there is a unique MPE with $\tau^m(t) = \tau^{HP} \equiv \bar{\tau}$ for all t .*

It is clear that this “holdup equilibrium” is more inefficient than the equilibria characterized above. For example, imagine a situation in which $\bar{\tau} = 1$ and Condition 22.1 fails to hold, so that with the original timing of events (without holdup), the equilibrium tax rate is $\tau^m(t) = 1 - \alpha$. But with holdup, the equilibrium tax is $\tau^m(t) = 1$ and the middle class stop producing. This is not only costly for the middle-class entrepreneurs, but also for the elite since they lose all their tax revenues.

In this model, it is no longer true that the unique MPE is the only SPE, since there is room for an implicit agreement between different groups whereby the elite (credibly) promise a different tax rate than $\bar{\tau}$. Relatedly, the MPE in this model, provided in Proposition 22.8 is Pareto inefficient, and a social planner with access to exactly the same fiscal instruments can improve the utility of all agents in the economy.

To illustrate the difference between the MPE and the SPE (and the associated Pareto inefficiency of the MPE), consider the example where Condition 22.1 fails to hold and $\bar{\tau} = 1$. Recall that the history of the game is the complete set of actions taken up to that point. In the MPE, the elite raise no tax revenue from the middle class producers. Instead, consider the following *trigger-strategy* profile: the elite set $\tau^m(t) = 1 - \alpha$ for all t and the middle class producers invest according to (22.20) with $\tau^m(t) = 1 - \alpha$ as long as the history consists of $\tau^m = 1 - \alpha$ and investments have been consistent with (22.20). If there is any other action in the history, the elite set $\tau_m = 1$ and the middle class producers invest zero. Does this strategy constitute a SPE? First, it is clear that the middle class have no profitable deviation, since at each t , they are choosing their *best response* to taxes along the equilibrium path as implied by (22.20). To check whether the elite have a profitable deviation, note that with this strategy profile, they are raising a tax revenue of $\phi(1 - \alpha)\alpha^{\alpha/(1-\alpha)}\beta^{\alpha/(1-\alpha)}A^m\theta^m\bar{L}/\alpha$ in every period, thus receiving transfers worth

$$(22.31) \quad \frac{\phi}{(1 - \beta)}(1 - \alpha)\alpha^{-(1-2\alpha)/(1-\alpha)}\beta^{\alpha/(1-\alpha)}A^m\theta^m\bar{L}.$$

If, in contrast, they deviate at any point, the most profitable deviation for them is to set $\tau^m = 1$, and they will raise a tax revenue of

$$(22.32) \quad \phi\alpha^{-(1-2\alpha)/(1-\alpha)}\beta^{\alpha/(1-\alpha)}A^m\theta^m\bar{L}$$

in that period. Following such a deviation, the continuation equilibrium involves switching to the unique MPE (which is here the worst possible continuation SPE). We have seen above that, with $\bar{\tau} = 1$ the continuation value of the elite in this case is equal to 0. Therefore, the trigger-strategy profile will be an equilibrium as long as (22.31) is greater than or equal to (22.32), which requires $\beta \geq \alpha$. Therefore:

PROPOSITION 22.9. *Consider the holdup game, and suppose that Condition 22.1 holds and that $\bar{\tau} = 1$. Then, for $\beta \geq \alpha$, there exists a SPE where $\tau^m(t) = 1 - \alpha$ for all t .*

PROOF. See Exercise 22.11. □

An important implication of this result is that in societies where there are greater holdup problems, for example, because typical investments involve longer horizons, the MPE leads to a Pareto inefficient equilibrium allocation and there is room for coordinating on a SPE supported by an implicit agreement (trigger strategy profile) between the elite and the rest of the society. The SPE described above can make all the agents in the society better-off relative to the MPE. This analysis also shows that whether we use the MPE or the SPE equilibrium concept has important implications for the structure of the equilibrium and its efficiency properties. While the use of the equilibrium concept is a choice for the modeler, different equilibrium concepts approximate different real-world situations. For example, MPE may be much more appropriate when the institutional structure, the frequency of interactions or the past history make coordination and mutual trust unlikely, while SPE may be useful in modeling equilibria in societies where some degree of mutual trust can be developed among the different parties with conflicting interests.

22.3.5. Technology Adoption. Another source of holdup comes from the technology adoption decisions of entrepreneurs, which may, in practice, be more important than the timing of taxes. Many important technology adoption decisions are made with the long horizon in mind, thus future tax rates matter for these decisions. The analysis earlier in the book highlighted the importance of technology adoption decisions for economic growth, thus the new types of political economy interactions that arise in the presence of such decisions are of practical as well as of theoretical interest.

To illustrate the main issues raised by the presence of technology adoption decisions, let us go back to the original timing where taxes for time $t + 1$ are set and committed to at time t (so that the source of holdup in the previous subsection is now removed). Instead, suppose that at time $t = 0$ before any economic decisions or policy choices are made, middle class agents can invest to increase their productivity. In particular, suppose that there is a cost $\Gamma(A^m)$ of investing in productivity A^m . The function Γ is nonnegative, differentiable and strictly convex. This investment is made once and the resulting productivity A^m applies forever after.

Once investments in technology are made, the game proceeds as before. Since investments in technology are sunk after date $t = 0$, the equilibrium allocations are the same as those presented above. The interesting question is whether the presence of the technology adoption decisions creates additional inefficiencies (including Pareto inefficiencies). One way of answering this question is to ask whether, if they could, the elite would prefer to commit to a tax rate sequence at time $t = 0$ different from the MPE or the SPE tax sequence characterized above. The following proposition answers this question in the case of pure factor price manipulation affect:

PROPOSITION 22.10. *Consider the game with technology adoption and suppose that Condition 22.1 and 22.2 hold and $\phi = 0$. Then, the unique MPE and the unique SPE involve $\tau^m(t) = \tau^{FPM} \equiv \bar{\tau}$ for all t . Moreover, if the elite could commit to a tax sequence at time $t = 0$, then they would still choose $\tau^m(t) = \tau^{FPM} \equiv \bar{\tau}$.*

The result that the allocation described in the proposition is the unique MPE follows immediately from the analysis so far. The fact that it is also the unique SPE follows from Proposition 22.7 and implies that the elite would choose exactly this tax rate even if they could commit to a tax rate sequence at time $t = 0$. The reason is intuitive: in the case of pure factor price manipulation, the only objective of the elite is to reduce the middle class' labor demand, so they have no interest in increasing the productivity of middle class producers.

The situation is quite different, however, when the elite would also like to extract revenues from the middle class. To illustrate this in the starkest possible way, let us next consider the pure revenue extraction case, where Condition 22.1 fails to hold (so that the equilibrium wage is equal to 0 and there is no factor price manipulation). Once again, the MPE is identical to before and involves a tax of τ^{RE} as in (22.23) at each date. As a result, the first-order

condition for an interior solution to the middle class producers' technology choice is:

$$(22.33) \quad \Gamma'(A^m) = \frac{1 - \alpha\beta}{(1 - \beta)\alpha\beta} (\beta(1 - \tau^{RE}))^{1/(1-\alpha)} \bar{L}.$$

Once again, Proposition 22.7 implies that a tax rate of $\tau^m = \tau^{RE} \equiv \min\{1 - \alpha, \bar{\tau}\}$ and technology choice given by (22.33) is also the unique SPE. Intuitively, once again after the middle-class producers have made their technology decisions, there is no history-dependent action left, and it is impossible to create history-dependent punishment strategies to support a tax rate different than the static optimum for the elite. However, in this case this equilibrium allocation is Pareto inefficient and in fact, if the elite could commit to a tax rate sequence at time $t = 0$, they would choose lower taxes. To illustrate this, suppose that the elite can indeed commit to a constant tax rate at $t = 0$ (it is straightforward to show that they will in fact choose a constant tax rate even without this restriction, but this restriction saves on notation). Therefore, the optimization problem of the elite is to maximize tax revenues taking the relationship between taxes and technology as in (22.33) as given. In other words, they will maximize $\phi\tau^m (\beta(1 - \tau^m))^{\alpha/(1-\alpha)} A^m\theta^m \bar{L}/\alpha$ subject to (22.33). The constraint (22.33) incorporates the fact that (expected) taxes affect technology choice.

The first-order condition for an interior solution can be expressed as

$$A^m - \frac{\alpha}{1 - \alpha} \frac{\tau^m}{1 - \tau^m} A^m + \tau^m \frac{dA^m}{d\tau^m} = 0$$

where $dA^m/d\tau^m$ takes into account the effect of future taxes on technology choice at time $t = 0$. This expression can be obtained by differentiating (22.33) with τ^m instead of τ^{RE} as:

$$\frac{dA^m}{d\tau^m} = -\frac{\beta^{1/(1-\alpha)}}{1 - \beta} \frac{1}{\alpha} \frac{(1 - \tau^m)^{\alpha/(1-\alpha)}}{\Gamma''(A^m)} < 0.$$

This immediately implies that the solution to the maximization problem of the elite when they can commit to a tax rate sequence at $t = 0$ has a solution $\tau^m = \tau^{TA} < \tau^{RE} \equiv \min\{1 - \alpha, \bar{\tau}\}$ (provided that $\tau^{TA} < \bar{\tau}$, for example, because $\bar{\tau}$ is sufficiently close to 1). Hence, if they could, the elite would commit to a lower tax rate in the future in order to encourage the middle-class producers to undertake technological improvements. Their inability to commit to such a tax policy leads to more distortionary policies (and in fact in this case to Pareto inefficiency). The next proposition states this result and to simplify the statement, I assume $\bar{\tau} = 1$.

PROPOSITION 22.11. *Consider the game with technology adoption, and suppose that Condition 22.1 fails to hold, that Condition 22.2 holds, that $\phi > 0$ and that 1. Then, the unique MPE and the unique SPE involve $\tau^m(t) = \tau^{RE} \equiv 1 - \alpha$ for all t . If the elite could commit to a tax policy at time $t = 0$, they would prefer to commit to a tax level $\tau^{TA} < \tau^{RE}$ at $t = 0$.*

An important feature is that in contrast to the pure holdup problem where SPE could prevent the additional inefficiency (when $\beta \geq \alpha$, recall Proposition 22.9), with the technology adoption game, the inefficiency survives the SPE. The reason is that, since middle-class producers invest only once at the beginning, there is no possibility of using history-dependent

punishment strategies (whereby following the deviation, middle-class producers have a best response that involves switching to zero or lower investment). This illustrates the limits of implicit agreements to keep tax rates low. Such agreements not only require a high discount factor ($\beta \geq \alpha$), but also frequent investments by the middle class, so that there is a credible threat against the elite if they deviate from the promised policies. When such implicit agreements fail to prevent the most inefficient policies, there is greater need for economic institutions to play the role of placing limits on future policies.

22.4. Inefficient Economic Institutions: A First Pass

I will now use the framework from the previous section to make a first attempt to understand (i) the conditions under which equilibrium economic institutions might put limits on distortionary policies, and (ii) the conditions under which economic institutions might go on to the other extreme, involving the elite using inefficient instruments to reduce output and *block* economic development. To communicate the ideas in the simplest possible way, I consider two prototypical economic institutions that affect the policy choices by the elite: (1) *Security of property rights*; there may be constitutional or other limits on the extent of redistributive taxation and/or other policies that reduce profitability of producers' investments. In terms of the model above, we can think of this as determining the level of $\bar{\tau}$. (2) *Regulation of technology*, which concerns direct or indirect factors affecting the productivity of producers, in particular, of middle-class producers.

The analysis of factor price manipulation in the previous subsection already provides a partial answer to one of the questions raised above: why would the political system use *inefficient instruments*? A full analysis to this question requires a setup with a richer menu of fiscal instruments, such as lump-sum taxes. A glimpse of how such an analysis might go is provided in Exercise 22.15 below. For now, note that the analysis in Propositions 22.5 and 22.6 already provide the beginning of an answer, since they show that the equilibrium tax rate would be strictly above the revenue-maximizing level. Our first task is to derive some implications from these observations about constitutional limits on taxation by the elite.

22.4.1. Emergence of Secure Property Rights. The environment is the same as in the previous section, with the only difference that at time $t = 0$, before any decisions are taken, the elite can change the constitution so as to reduce $\bar{\tau}$, say from $\bar{\tau}^H$ to some Knuth level in the interval $[0, \bar{\tau}^H]$, thus creating an upper bound on taxes and providing greater security of property rights to the middle class. Here, let us suppose that $\bar{\tau}^H$ is technologically imposed (for example, with taxes above $\bar{\tau}^H$, middle-class entrepreneurs could flee to the informal sector). Naturally, a key question is how a constitution that imposes $\bar{\tau} < \bar{\tau}^H$ would be made credible. For now, I do not address this question and take it as given that such a constitutional limit on future taxes can be imposed (though this, to some degree, goes against the presumption that commitment to future policies is not possible; in some sense, we are relaxing this somewhat by assuming that commitment to an upper bound on policies is possible). My objective in this section is not to investigate the credibility of

various constitutional guarantees, but to investigate whether, when they are feasible, the elite would like to make such promises, that is, whether they prefer $\bar{\tau} = \bar{\tau}^H$ or $\bar{\tau} < \bar{\tau}^H$. Also, to start with, I take the natural benchmark in which economic institutions (here constitutional limits on taxation) are decided by the elite, who hold political power at $t = 0$ when these restrictions are introduced.

The next three propositions answer this question in various different versions of the environment studied so far in this section:

PROPOSITION 22.12. *Without holdup and technology adoption, the elite prefer $\bar{\tau} = \bar{\tau}^H$.*

The proof of this result is immediate, since without holdup or technology adoption, putting further restrictions on the taxes can only reduce the elite's utility. This proposition implies that when economic institutions are decided by the elite, who will also hold political power in the future as well, and there are no holdup issues, then the elite derive no benefits from introducing constitutional limits on their future taxes and will not introduce security of property rights to other producers.

The results are different when there are holdup problems, however. To illustrate this, let us go back to the situation with holdup (where taxes for time t are decided after the capital stock for time t is determined). Let us focus on the general case where both the revenue extraction and factor price manipulation motives are present. Moreover, let us for now focus on the MPE.

PROPOSITION 22.13. *Consider the game with holdup and suppose that Conditions 22.1 and 22.2 hold and $\phi > 0$. Then, the unique MPE involves $\tau^m(t) = \bar{\tau}^H$ for all t . If τ^{COM} given by (22.28) is strictly less than $\bar{\tau}^H$, the elite prefer to set $\bar{\tau} = \tau^{COM}$ at $t = 0$.*

PROOF. See Exercise 22.12. □

The intuition for this proposition is simple: in the presence of holdup problems, Proposition 22.8 shows that the unique MPE involves $\tau = \bar{\tau}^H$. However, this is (Pareto) inefficient and in fact, if the elite could commit to a tax rate of $\bar{\tau} = \tau^{COM}$, they would increase their consumption (and also the consumption levels of the middle class and the workers would achieve greater consumption at each date). If the elite could use economic institutions to regulate future taxes, for example by setting constitutional limits on taxes, then they would like to use these to manipulate equilibrium taxes. By manipulating economic institutions, the elite may approach their desired policy (in fact, in this simple economy, they can exactly commit to the tax rate that maximizes their utility).

This result shows that the elite may wish to change economic institutions to provide additional property rights protection to producers in the presence of holdup problems. Note however that the restriction to MPE is important in this proposition. If we allow history-dependent punishment strategies and look at the SPE, then the elite would be able to improve over the MPE allocation in Proposition 22.9, and depending on parameters, they may even be able to implicitly (and credibly) commit to an equilibrium in which the tax rate at each

date is equal to τ^{COM} . If this were the case, there would be less need for changing economic institutions in order to place limits on future taxes. Whether the MPE or the SPE is more relevant in such a situation depends on what the expectations of the different parties are and what degree of coordination can be achieved among the players. It is generally difficult to ascertain whether one or the other equilibrium concept would be more appropriate without specifying other (institutional or historical) details of the situation.

However, when the source of additional inefficiency is technology adoption rather than the holdup problem (resulting from the timing of taxes), there will be a need for a change in economic institutions even if we focus on the SPE. This is shown in the next proposition:

PROPOSITION 22.14. *Consider the game with technology adoption, and suppose that Condition 22.1 holds and $\phi > 0$. Then, the unique MPE and the unique SPE involve $\tau^m(t) = \tau^{COM}$ given by (22.28) for all t . If τ^{COM} is strictly greater than τ^{TA} defined in Proposition 22.11, then the elite would prefer to set $\bar{\tau} = \tau^{TA}$ at $t = 0$.*

This proposition therefore highlights that in environments where long-term investments or technology adoption decisions are important, implicit promises as in Proposition 22.9 are of little use. Instead, explicit (credible) guarantees through economic institutions are necessary to provide incentives and security to middle-class entrepreneurs so that they undertake the appropriate technology investments. Thus, while implicit promises and other informal arrangements could play the role of economic institutions under some circumstances, there will be limits to how well they can perform this role and in many environments, constitutional limits on distortionary policies and expropriation (if feasible) would endogenously emerge in the political equilibrium.

22.4.2. Blocking Economic Development. The focus in the previous subsection was on choosing economic institutions at $t = 0$ to provide more secure property rights and better investment incentives to middle-class entrepreneurs. These types of economic institutions play an important role in practice and variation in the security of property rights for businesses across societies likely explains part of the variation in economic performance. Nevertheless, security of property rights and limits on taxes are only one aspect of the potential effect of institutions on economic activity and economic development. As briefly discussed in Chapter 4, in many societies, rather than encouraging economic activity, the elite actively try to block economic development. Why would the elite in some societies choose specifically inefficient policies in order to reduce the productivity of entrepreneurs and block economic development?

I now discuss this question and try to shed light on the aspects of equilibrium economic institutions related to the regulation of technology. Once again, to provide the basic ideas in the simplest possible way, I will extend the basic framework in this section in one direction: at time $t = 0$, the government (thus the elite controlling political power) chooses a policy affecting the technology choices of producers, denoted by $g \in \{0, 1\}$. This choice can be thought of investment in infrastructure, protection of intellectual property rights, or the provision of law and order (with $g = 0$ corresponding to not making these investments and

$g = 1$ corresponding to creating a better business environment). Alternatively, $g = 0$ may directly correspond to actions taken by the elite to block the technology adoption decisions of the entrepreneurs. To capture these ideas in the simplest possible way, let us assume that $g \in \{0, 1\}$ affects the productivity of middle-class producers in all future periods, and in particular $A^m = A^m(g)$, with $A^m(1) > A^m(0)$. To simplify the discussion, suppose further that g has no effect on the productivity of the elite and also $g = 1$ has no direct cost relative to $g = 0$. The key question is this: will the elite always choose $g = 1$, increasing the middle class producers' productivity, or will they try to block technology adoption by the middle class?

When the only mechanism at work is revenue extraction, the answer is that the elite would like the middle class to have the best technology:

PROPOSITION 22.15. *Suppose that Condition 22.1 fails to hold and $\phi > 0$. Then, the economic equilibrium always involves $w(t) = 0$, and in the unique MPE, the elite choose $g = 1$.*

Therefore, this proposition shows a range of situations in which the elite would not block the technology adoption decisions of middle-class entrepreneurs. This result follows immediately since $g = 1$ increases the tax revenues and has no other effect on the elite's consumption. Consequently, in this case, the elite benefit from the increase in the productivity of the middle-class entrepreneurs and thus would like them to be as productive as possible. Intuitively, there is no competition between the elite and the middle class (either in factor markets or in the political arena), and when the middle class entrepreneurs are more productive, they generate greater tax revenues for the elite.

However, the situation is different when the elite wish to manipulate factor prices:

PROPOSITION 22.16. *Suppose Condition 22.1 holds and Condition 22.2 holds (with $A^m(g = 0)$ replacing A^m), $\phi = 0$, and $\bar{\tau} < 1$. Then, in any MPE or SPE, the elite choose $g = 0$.*

PROOF. See Exercise 22.14. □

Intuitively, with $\bar{\tau} < 1$, labor demand from the middle class is high enough to generate positive equilibrium wages. Since $\phi = 0$, taxes raise no revenues for the elite, and their only objective is to reduce the labor demand from the middle class and wages as much as possible. This makes $g = 0$ the preferred policy for the elite. Consequently, the factor price manipulation mechanism suggests that, when it is within their power, the elite will choose economic institutions so as to reduce the productivity of competing (middle-class) producers. Proposition 22.16 therefore shows how the elite may take actions to directly reduce the productivity of the (other) entrepreneurs in the economy, thus retarding or blocking economic development.

The next proposition shows that a similar effect applies when the political power of the elite is contested.

PROPOSITION 22.17. *Consider the economy with political replacement. Suppose Condition 22.1 fails to hold and $\phi = 0$. Then, in any MPE or SPE, the elite prefer $g = 0$.*

PROOF. See Exercise 22.15. □

In this case, the elite cannot raise any taxes from the middle class since $\phi = 0$. But differently from the previous proposition, there are no labor market interactions, since there is excess labor supply and wages are equal to zero. Nevertheless, the elite would like the profits from middle class producers to be as low as possible so as to consolidate their political power. They achieve this by creating an environment that reduces the productivity of middle class producers.

Overall, this section has demonstrated how the elite's preferences over policies, and in particular their desire to set inefficient policies, translate into preferences over non-growth enhancing (or "inefficient") economic institutions. When there are no holdup problems, introducing economic institutions that limit taxation or put other constraints on policies provides no benefits to the elite. This is intimately related to the fact that in the absence of holdup problems and given the menu of fiscal instruments, the equilibria characterized above corresponded to allocations maximizing a weighted social welfare function (and were thus constrained Pareto efficient). However, when the elite are unable to commit to future taxes (because of holdup problems), the equilibrium is no longer Pareto efficient and equilibrium taxes may be too high even from the viewpoint of the elite. In this case, using economic institutions to manipulate future taxes may be beneficial for the elite who control the political power of the state. Similarly, the analysis reveals that the elite may want to use economic institutions to discourage productivity improvements by the middle class. Interestingly, this never happens when the main mechanism leading to inefficient policies is revenue extraction. Instead, when factor price manipulation and political consolidation effects are present, the elite may want to discourage or block technological improvements by the middle class.

The analysis so far has focused on the basic forces leading to non-growth enhancing policies and economic institutions in the context of a simple society with linear preferences. The rest of this chapter investigates how relaxing these assumptions changes the insights. The next section is concerned with the case in which preferences are concave, while the following two sections introduce a richer structure of heterogeneity among the agents.

22.5. Distributional Conflict and Economic Growth: Concave Preferences*

In this section, I provide a preliminary analysis of an environment similar to the baseline model studied so far, but with concave preferences. My main purpose is to illustrate how to approach the analysis of such an economy and highlight some of the additional conceptual and technical issues that arise in this case. As a byproduct, this analysis will show how the analysis of political economy was simplified by the assumption of linear preferences.

Relative to the framework in Section 22.2, I make two different assumptions. First, preferences are now assumed to take the form

$$(22.34) \quad \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_i(t)),$$

where $U(\cdot)$ is a strictly increasing, strictly concave and continuously differentiable utility function. This specification assumes that all three groups have the same utility function, though this is not important for the analysis and is simply adopted to reduce notation. The second important assumption is that all financial markets are closed. Consequently, entrepreneurs cannot borrow in order to invest in capital, and have to save by reducing their current consumption. Note that in the absence of political economy (and taxes), this would have no effect on the qualitative features of the dynamics of the model, which still closely resemble those of the baseline neoclassical growth model. In particular, without any taxes, there exists a unique globally (saddle-path) stable steady-state equilibrium, which satisfies eq. (6.49) in Chapter 6. All the other assumptions from Section 22.2, especially those regarding the production functions and the timing of policies, continue to apply, however, I also normalize $\theta^m = \theta^e = 1$ to simplify the expressions.

Using exactly the same notation as in Section 22.2, the dynamic optimization of middle-class entrepreneurs for a given sequence of policies and wages, p^t and w^t , can be written as

$$\begin{aligned} \mathbf{U}_i(\{K_i(s), L_i(s)\}_{s=t}^{\infty} \mid p^t, w^t) = \\ \sum_{s=t}^{\infty} \beta^{s-t} U[(1 - \tau(s)) F(K_i(s), L_i(s)) - K_i(s+1) - w(s) L_i(s) + T^m(s)], \end{aligned}$$

where I have set the depreciation rate of capital δ equal to 1 to simplify the notation. This expression is similar to (22.9), except that the utility of consumption replaces the level of consumption as the instantaneous return at each date. Note that $\mathbf{U}_i$ is strictly concave in the sequence $\{K_i(s), L_i(s)\}_{s=t}^{\infty}$ for any tax sequence with $\tau(t) < 1$ for all t . We know from the analysis so far that there will never be 100% taxation, thus we can restrict attention to such tax sequences and the maximization problem of each entrepreneur is indeed a strictly concave problem.

The relevant necessary and sufficient first-order conditions for entrepreneur i can be written as

$$(22.35) \quad U'[C_i(t)] = \beta(1 - \tau(t+1)) f'(k_i(t+1)) U'[C_i(t+1)],$$

for each t , which looks identical to the Euler equation for the representative household given in (6.47) in Chapter 6. This is not surprising, since each entrepreneur solves a similar program to that facing the representative household or the social planner in the basic neoclassical growth model. The only difference is the presence of the taxes, which implies that one unit of consumption foregone today does not earn the full marginal product of capital, but only that left over from taxes. This equation implies that the capital-labor ratio is chosen by

entrepreneur i will now depend on the entire sequence of taxes, since these taxes will influence the current and future consumption levels. Consequently, we no longer have a simple equation such as (22.11) in Section 22.2. Nevertheless, (22.35) does determine a unique sequence of capital-labor ratio choices for each entrepreneur given the sequence of taxes and their initial capital stock, $K_i(0)$.

To simplify the analysis, let us suppose that all entrepreneurs start with the same initial capital stock, that is, $K_i(0) = K(0)$. Given the symmetric initial conditions and the strict concavity of the problem, the equilibrium will be symmetric as well and each entrepreneur will choose exactly the same capital-labor ratio sequence. Now if the sequence of policies p^0 were indeed given, then we could define a single-valued mapping $\Phi: \mathcal{P} \rightarrow \mathcal{K}$, where $\mathcal{P}$ is the set of all feasible policy sequences with less than 100% taxation at each point and $\mathcal{K}$ is the set of equilibrium capital-labor ratios. The political economy problem would then be for the elite to choose some $p^0 \in \mathcal{P}$ to maximize their discounted utility. This would indeed be the solution to the political economy problem if the elite could commit to a sequence of policies at date $t = 0$. But the assumption so far, which is a natural approximation to reality, is that political decisions are made sequentially and commitment to a future sequence of policies is *not* possible. This is exactly where my treatment in the Section 22.2 cut some corners. With linear utility, it did not matter whether the elite chose the sequence of policies at date $t = 0$ or sequentially as specified in the timing of events. To simplify the discussion there, I did not dwell on this distinction. This distinction now becomes crucial.

The right way to approach this problem is to specify the payoff-relevant state variables and then at each date have the elite make their utility-maximizing policy choices (as a function of the payoff-relevant state variables). The major difference from the analysis in this chapter so far is that once the elite undertake a deviation the future sequence of policies should not remain fixed but also change, because the deviation will have affected the evolution of the state variables and the evolution of the state variables will induce a different set of preferred policies for the elite. With linear preferences the deviation had no effect on future equilibrium policies, thus the analysis in the previous sections did not explicitly specify the effect of a deviation on the future sequence of policies. To show how this can be done in general and what its implications will be are the main focus of this section.

Let us now start developing the notation and the language for such an analysis. Most generally, the relevant state variable at time t would be a distribution of capital stocks or capital-labor ratios across all entrepreneurs denoted by $[K_i(t)]_{i \in S^m}$. This would significantly complicate the analysis, since working with entire distributions as the state variable is difficult. Fortunately, we can circumvent this problem. The same type of argument used above for a specific sequence of policies implies that, even taking the potential changes in future policies into account, the maximization problem of each entrepreneur is strictly concave. In addition, each entrepreneur recognizes that he has no effect on aggregates and also all entrepreneurs start with the same initial condition. In view of this, we can restrict attention to a situation in which at all dates all entrepreneurs will choose the same capital-labor ratio, and the state

variable at time t can be represented by the capital-labor ratio of the “representative” entrepreneur, $k(t) \in \mathbb{R}_+$. Moreover, as in Chapter 6, the Inada conditions in Assumption 2 imply that we can restrict attention to state variables in a compact set $k(t) \in [0, \bar{k}]$.⁴

Given this state variable, the policy choice of the elite can be represented by a *policy function* denoted by

$$P : [0, \bar{k}] \rightarrow [0, 1],$$

which determines the utility-maximizing tax rate for the elite at the next date, $\tau(t+1)$, as a function of the current capital-labor ratio $k(t)$. We could extend this function so that it also determines the amount of transfers. But this is not necessary, since, as in the previous subsection, the elite will always choose $T^w(t) = T^m(t) = 0$ for all t and $T^e(t)$ will be given by the government budget constraint, (22.8).

Let us next write the payoff function of a representative entrepreneur recursively. In his optimization problem, each entrepreneur takes five objects as given: (1) its own capital-labor ratio, k_i ; (2) the capital-labor ratio of all other entrepreneurs, k (this will naturally be equal to its own capital-labor ratio in equilibrium, but the entrepreneur does not control this variable himself); (3) tax rate for today, τ ; (4) the tax rate announced for the next date, τ' ; (5), the policy function P of the elite. The fact that the entrepreneur is taking the policy function P as given implies that he is presuming that even if the elite take a deviation today, from tomorrow onwards, they will follow the policy function P that maximizes their discounted utility. This is simply an application of the *one step ahead deviation principle* (see Theorem C.1 in Appendix Chapter C). Let us also introduce the best-response function of the entrepreneurs at this point. From the viewpoint of entrepreneur i , whose behavior we are looking at now, this is the best-response function of all other entrepreneurs, which he takes as given. In equilibrium, his best response function must coincide with this, thus we will be looking for a fixed point. In particular, let

$$\kappa : [0, \bar{k}] \times [0, 1]^2 \rightarrow [0, \bar{k}]$$

be this best response function, where the first argument is today’s capital stock and the next two arguments are today’s and tomorrow’s tax rates, so that the function takes the form $\kappa(k, \tau, \tau')$, with τ denoting the current tax rate and τ' denoting the tax rate announced for next period.

⁴It may appear that we are cutting some corners here as well. All entrepreneurs choose the same capital-labor ratio *along-the-equilibrium path*. What happens if an entrepreneur takes a deviation? It would appear that at that point, the state variable can no longer be represented by a one-dimensional object, and to take care of behavior off-the-equilibrium path properly, we would need to consider state variables of much higher dimension. Fortunately, this is not an issue, thanks to the fact that there is a continuum of entrepreneurs. If a single entrepreneur takes a deviation, this will have no effect on aggregates. Thus both along-the-equilibrium path and for one-step-ahead deviations from the equilibrium (which are the only ones that matter, see Appendix Chapter C), focusing on the one-dimensional state variable is sufficient.

With this preparation, the maximization problem of a representative entrepreneur i can be written as

$$(22.36) \quad V_i(k_i, k, \tau, \tau' | P, \kappa) = \max_{k' \in [0, \bar{k}]} U(C) + \beta V_i(k', \kappa(k, \tau, \tau'), \tau', P(\kappa(k, \tau, \tau')) | P, \kappa)$$

subject to

$$(22.37) \quad C = (1 - \tau) f(k) - k' - w,$$

where I have suppressed expectations, since there will be no uncertainty in this environment (because there are no exogenous shocks and we are focusing on pure strategies). Note that $V_i(k_i, k, \tau, \tau' | P, \kappa)$ denotes the value function of entrepreneur i , when his capital stock (capital-labor ratio) is given by k_i , those of other entrepreneurs is k , today's tax rate is τ , and tomorrow's tax rate has been announced as τ' . In all of this he takes the policy functions of the elite, P , and the best-response function of other entrepreneurs, κ , as given. The continuation value is therefore $\beta V_i(k', \kappa(k, \tau, \tau'), \tau', P(\kappa(k, \tau, \tau')) | P, \kappa)$. Here, k' is his choice of next period's capital stock, so it will be the first element of the state variable entering his value function. The capital stock of other entrepreneurs will be given as a function of announced tax rate τ' and according to their best-response function, $\kappa(k, \tau, \tau')$. Then, the tax rate announced yesterday becomes the current tax rate, so the third element is τ' , and finally, the policy function of the elite implies that they will choose a tax rate for the day after tomorrow as a function of the capital-labor ratio of entrepreneurs at that point, so this policy function is written as $P(\kappa(k, \tau, \tau'))$. The entrepreneur's current level of consumption is then given by (22.37) by standard arguments, with w denoting the equilibrium wage rate. I did not condition on this equilibrium wage rate to reduce the notation which is already quite plentiful.

The maximization by entrepreneur i at each stage simply involves the choice of next date's capital stock, k' . Let us denote the best-response function corresponding to this choice by $k'(k, \tau, \tau')$. The following proposition can be established using the tools from Chapter 6:

PROPOSITION 22.18. *Consider the maximization problem in (22.36). For any $\kappa(\cdot)$ and $P(\cdot)$ functions, the value function V is uniquely defined, continuous in all its arguments, and differentiable in the interior of its domain. The optimal policy $k'(k, \tau, \tau')$ is defined uniquely and is continuous in all of its arguments.*

PROOF. See Exercise 22.16. □

While the analysis in this section is considerably more complicated than in the models presented so far, Proposition 22.18 is a significant step towards characterizing the equilibrium of this more general economy. In particular, once we have the optimal policy of individual entrepreneur i , $k'(k, \tau, \tau')$, it becomes apparent that the optimal policy of this individual entrepreneur is the same as the best response function of all entrepreneurs, so that

$$\kappa(k, \tau, \tau') \equiv k'(k, \tau, \tau').$$

Therefore we have managed to characterize the behavior of the entrepreneurs in a MPE. Our next step is to take this best response function as given and solve the problem of the elite in setting taxes. For this purpose, let us now write the payoff to the elite recursively. Let $W(k, \tau \mid P, \kappa)$ be the value function of the elite when the current capital-labor ratio chosen by the entrepreneurs is k and the current tax rate is τ . Those are the only two states variables relevant for the elite. In addition, we condition this value function on κ , since this determines how entrepreneurs will react to different tax rates. To simplify the analysis, let us assume that the elite do not have access to any saving technology, thus they have to consume their current tax revenue and also normalize $\theta^e = 1$ without loss of generality. Then, their value function can be written as

$$(22.38) \quad W(k, \tau \mid \kappa) = \max_{\tau'} U(\tau f(k)) + \beta W(\kappa(k, \tau, \tau'), \tau' \mid \kappa).$$

Intuitively, in the current period, the elite receive a pre-determined amount of tax revenue given by the tax rate, τ , announced in the previous period times output produced by the capital stock on the economy (which is also predetermined). The capital stock of the economy is equal to the capital-labor ratio of the representative entrepreneur, since the total labor force is equal to 1 (and (22.6) holds, so that there is full employment). Next period's value is then given by the tax rate announced now, τ' , and the capital-labor ratio choice of the entrepreneurs given by $\kappa(k, \tau, \tau')$.

The next proposition is again established using the tools from Chapter 6.

PROPOSITION 22.19. *The value function given in (22.38) is uniquely defined and is continuous in k and τ .*

PROOF. See Exercise 22.17. □

Unfortunately, this proposition does not establish that the value function W is concave or differentiable. This is because it does depend on the function $\kappa(k, \tau, \tau')$, which may be non-convex. Nevertheless, to make progress with the analysis in the simplest possible way, let us suppose that W is indeed differentiable in both k and τ , and also that $\kappa(k, \tau, \tau')$ is differentiable in all three of its arguments. To write the first-order condition for the choice of the tax rate by the elite, let us denote the partial derivatives of the W and κ functions with respect to their j th argument by W_j and κ_j , so that, for example, the derivative of W with respect to τ is denoted by W_2 , and other terms are defined similarly. Then, the first-order condition takes the form

$$W_2(\kappa(k, \tau, \tau'), \tau' \mid \kappa) + W_1(\kappa(k, \tau, \tau'), \tau' \mid \kappa) \kappa_3(k, \tau, \tau') = 0.$$

As in Chapter 6, to make further progress, we need to evaluate the two derivatives of the value function in the first-order condition, and we do this by differentiating the value function with respect to k and τ and using the Envelope Theorem. These derivatives are then obtained as

$$W_1(k, \tau \mid \kappa) = \tau f'(k) U'(\tau f(k)) + \beta W_2(\kappa(k, \tau, \tau'), \tau' \mid \kappa) \kappa_1(k, \tau, \tau'),$$

and

$$W_2(k, \tau | \kappa) = f(k) U'(\tau f(k)) + \beta W_1(\kappa(k, \tau, \tau'), \tau' | \kappa) \kappa_2(k, \tau, \tau').$$

Now taking both expressions to the next period (from t to $t+1$), and substituting for these, we obtain the following condition for the utility-maximizing tax rate choice of the elite:

$$\begin{aligned} & [f(\kappa(k, \tau, \tau') + \tau' f'(\kappa(k, \tau, \tau')))) \kappa_3(k, \tau, \tau')] U'(\tau f(k)) + \\ & \beta W_3(\kappa(k, \tau, \tau'), \tau', \tau'' | \kappa) [\kappa_2(\kappa(k, \tau, \tau'), \tau', \tau'') + \kappa_1(\kappa(k, \tau, \tau'), \tau', \tau'') \kappa_3(k, \tau, \tau')] = 0, \end{aligned}$$

with $\tau'' \equiv P(\kappa(k, \tau, \tau'))$. This first-order condition has some similarity to (22.16) from Section 22.2, but is clearly much more complicated. As with that expression, it trades off the gain from additional taxation, $f(\kappa(k, \tau, \tau'))$, against the loss that additional taxation will induce by reducing the equilibrium capital-labor ratio (the second term in the first bracket in first line). The second line represents the discounted future change in value arising from the fact that a different tax rate changes the capital-labor ratio tomorrow. Notice that these terms depend both on the best response function of the entrepreneurs, κ , and also on the policy function that the elite will use in the continuation game, P . In general, it is not possible to obtain closed-form solutions for the equilibrium tax rate. The presence of the current capital-labor ratio, k , indicates that the utility-maximizing tax rate will not be a constant. Instead, the equilibrium taxes will evolve over time together with the equilibrium capital-labor ratio.

Unfortunately, a further characterization of equilibrium is not possible without imposing further structure. Typically these types of models are solved under a variety of simplifying assumptions (such as quadratic utility) or the equilibrium is characterized numerically. Even though this more general model does not yield an explicit characterization of the MPE, it highlights the new forces that arise once we incorporate the transitional dynamics in individual entrepreneurs' investment decisions, which, in turn, make it optimal for the elite to choose a non-constant path of taxes. However, since there is full depreciation of capital here, some simple cases, notably logarithmic preferences with a Cobb-Douglas production function, still enable explicit solutions. The next example gives a sketch, with the details provided in Exercise 22.18.

EXAMPLE 22.1. (Closed-Form Solutions with Log Preferences) Suppose that $f(k) = Ak^\alpha$ and $U(c) = \log c$. Also assume, as we will do later in Section 22.7, that there is no distinction between the middle class and the citizens, and all non-elite agents have access to the above production function, but there is no labor market, and entrepreneurs can only work themselves with this technology (thus they are “yeoman-entrepreneurs”). Under these assumptions, the entrepreneurs' problem gives a simple solution. In particular, using (22.35) we obtain

$$(22.39) \quad k(t+1) = \alpha\beta(1 - \tau(t)) Ak(t)^\alpha.$$

The details of the derivation of this decision rule are in Exercise 22.18. The interesting feature of (22.39) is that the capital stock chosen for date $t+1$ does not depend on the tax rate on

date $t + 1$ revenues. This is because of the logarithmic preferences, which ensure that income and substitution effects cancel out. Nevertheless, taxes for date $t + 1$ are still distortionary; they affect the capital stock for date $k(t + 2)$ and all future capital stocks thereafter.

Let us now turn to the maximization problem of the elite. The simplest way of expressing the dynamic maximization problem of the elite is to write the value to a representative elite agent at time $t + 1$ as a function of the tax rate $\tau = \tau(t + 1)$. At this point, the capital stock of entrepreneurs at date $t + 1$, $k = k(t + 1)$ is given from (22.39) as $\alpha\beta(1 - \tau(t))Ak(t)^\alpha$ and is independent of $\tau(t + 1)$. On the other hand, the capital stock at date $t + 2$ depends on this tax rate, and in particular, $k' = k(t + 2) = \alpha\beta(1 - \tau)Ak^\alpha$. Then, recalling that the elite also have logarithmic preferences, their value function can be written as

$$(22.40) \quad W(k) = \max_{\tau \in [0,1]} \{ \log[\tau Ak^\alpha] + \beta W(\alpha\beta(1 - \tau)Ak^\alpha) \}.$$

It is straightforward to apply the standard arguments from Chapter 6 to (22.40) and conclude that W is strictly concave and differentiable for $k > 0$ (and denote the derivative by W'). The Euler equation for the elite is

$$\begin{aligned} \frac{1}{\tau} &= \beta^2 \alpha A k^\alpha W'(k') \\ &= \beta \frac{k' W'(k')}{1 - \tau}. \end{aligned}$$

Now conjecturing that $W(k) = \eta + \gamma \log k$ and using the Envelope condition, we obtain that $\gamma = \alpha / (1 - \alpha\beta)$ (see again Exercise 22.18 both for the derivation and the interpretation of this condition). Therefore, the utility-maximizing strategy for the elite is to set

$$(22.41) \quad \tau(t) = 1 - \alpha\beta$$

for all t (regardless of the level of the capital stock at that point). This is again a feature of the logarithmic preferences. The form of the tax rate in (22.41) is intuitive. In particular, it is decreasing in β , because higher taxes increase revenues today at the expense of future revenues, and it is increasing in α , because a higher α increases the elasticity of the response of future capital stocks to higher taxes today. This characterization establishes that with log preferences and full depreciation, there exists a unique MPE such that starting with a common capital stock for entrepreneurs, $k(0) > 0$, $\tau(t) = 1 - \alpha\beta$ and $k(t) = (\alpha\beta)^2 k(t - 1)$ for all $t \geq 1$.

22.6. Heterogeneous Preferences, Social Choice and the Median Voter*

My next objective is to relax the focus on simple societies, which ensured that the social conflict was between the elite and the entrepreneurs. Instead, I wish to illustrate how a richer and more realistic form of heterogeneity among the members of the society will influence policy choices. I will do this in two steps. In this section, I provide a brief overview of how to deal with aggregation of preferences in a society with heterogeneous agents. The celebrated Arrow's Impossibility Theorem, which we will see shortly, states that this is not possible in general. Nevertheless, under some further assumptions on the structure of preferences

(and limits on the menu of available policy options) such aggregation becomes possible. The main tool in this context, which has wide-ranging applications in political economy models, is the *Median Voter Theorem*, and its cousin, the *Downsian Policy Convergence Theorem*. I will show that these two theorems together provide a useful characterization of democratic politics under (limited) heterogeneity among agents. Then, in the next section, using these results I will show that the qualitative results derived in Section 22.2 generalize to a model with heterogeneity among entrepreneurs. The bottom line of the analysis in the next section will be that the source of distortionary (“inefficient”) policies that arise from the desire of the political system to extract revenues from a subset of the population is more general than in the simple society investigated in Section 22.2. But before doing this, we will get a modicum of basic social choice theory. Strictly speaking, only a simple form of the Median Voter Theorem is necessary for next section, and some of the results here are abstract, hence this section has a “*”.

The Median Voter Theorem (MVT) has a long pedigree in economics and has been applied in many different contexts. Given its wide use in political economy models, I will start with a section stating and outlining this theorem. I will also take this opportunity to provide a brief statement and proof of Arrow’s Theorem, because this theorem makes the value of the MVT more transparent. I will then emphasize that the MVT, despite its simplicity and elegance, is of limited use, because it only applies to situations in which the menu of policies can be reduced such that the disagreement among all the individuals and society is over a one-dimensional (or essentially one-dimensional) policy choice. In situations where the society has to make multiple-dimensional decisions, such as simultaneous taxes on capital and labor or nonlinear income taxation, we cannot use the MVT. I will end this section by outlining some alternative ways of aggregating heterogeneous preferences in such cases, which will also illustrate why in many circumstances the determination of political equilibria can be represented as the maximization of a weighted social welfare function.

22.6.1. Basics. This subsection gives an introductory treatment of the large area of social choice theory. Social choice theory is concerned with the fundamental question of political economy already discussed at the beginning of this chapter: how to aggregate the preferences of heterogeneous agents over policies (collective choices). Differently from the most common political economy approaches, however, social choice theory takes an axiomatic approach to this problem. Nevertheless, a quick detour into social choice theory as an introduction to the Median Voter Theorem is useful.

Let us consider an abstract economy consisting of a finite set of individuals $\mathcal{H}$, with the number of individuals denoted by H . Individual $i \in \mathcal{H}$ has a utility function

$$u(x_i, Y(x, p), p \mid \alpha_i).$$

Here x_i is his action, with a set of feasible actions denoted by X_i ; p denotes the vector of political choices (for example, institutions, policies or other collective choices), with the menu of policies denoted by $\mathcal{P}$; and $Y(x, p)$ is a vector of general equilibrium variables, such as

prices or externalities that result from all agents' actions as well as policies, and x is the vector of the x_i 's. Instead of writing a different utility function u_i for each agent, I have parameterized the differences in preferences by the variable α_i . This is without loss of any generality (simply define $u_i(\cdot) \equiv u_i(\cdot | \alpha_i)$) and is convenient for some of the analysis that will follow. Clearly, the general equilibrium variables, such as prices, represented by $Y(x, p)$ here, need not be uniquely defined for a given set of policies p and vector of individual choices x . Since multiple equilibria are not our focus here, I ignore this complication and assume that $Y(x, p)$ is uniquely defined.

I also assume that, given aggregates and policies, individual objective functions are strictly quasi-concave so that each agent has a unique optimal action $x_i(p, Y(x, p), \alpha_i) = \arg \max_{x \in X_i} u(x_i, Y(x, p), p | \alpha_i)$. Substituting this maximizing choice of individual i into his utility function, we obtain his *indirect utility function* defined over policy as $U(p; \alpha_i)$. Next let us define the *preferred policy*, or the (political) bliss point, of voter i , and to simplify notation, suppose that this is uniquely defined and denote it by

$$p(\alpha_i) = \arg \max_{p \in \mathcal{P}} U(p; \alpha_i).$$

In addition, we can think of a more primitive concept of individual preference orderings, which captures the same information as the utility function $U(p; \alpha_i)$. In particular, if individual i weakly prefers p to p' , we write $p \succeq_i p'$ and if he has a strict preference, we write $p \succ_i p'$. Under the usual assumptions on individual preferences (*completeness*, which allows any two choices to be compared; *reflexivity*, so that $z \succeq_i z$; and *transitivity*, so that $z \succeq_i z'$ and $z' \succeq_i z''$ implies $z \succeq_i z''$), we can equivalently represent individual preferences by the ordering $\succeq_i$ or by the utility function $U(p; \alpha_i)$ (see Exercise 22.19). Throughout, I assume that individual preferences are transitive.

In this context, we can also think of a “political system” as a way of aggregating the set of utility functions, $U(p; \alpha_i)$'s, to a social welfare function $U^S(p)$ that ranks policies for the society. Put differently, a political system is a *mapping* from individual preference orderings to a social preference ordering. Arrow's Theorem shows that if this mapping satisfies some relatively weak conditions, then social preferences have to be “dictatorial” in the sense that they will exactly reflect the preferences of one of the agents. I next present this theorem.

22.6.2. Arrow's (Im)Possibility Theorem. Let us simplify the discussion by assuming that the set of feasible policies, $\mathcal{P}$, is finite and is a subset of the Euclidean space, that is, $\mathcal{P} \subset \mathbb{R}^K$ where $K \in \mathbb{N}$. Let $\mathfrak{R}$ be the set of all weak orders on $\mathcal{P}$, that is, $\mathfrak{R}$ contains information of the form $p_1 \succeq_i p_2 \succeq_i p_3$ and so on, and imposes the requirement of transitivity on these individual preferences. An individual ordering R_i is an element of $\mathfrak{R}$, that is, $R_i \in \mathfrak{R}$. This statement reiterates that we are only considering individuals with well-defined transitive preferences.

Since our society consists of H individuals, we define $\rho = (R_1, \dots, R_H) \in \mathfrak{R}^H$ as the society's *preference profile*. That is, ρ gives the preference ordering of each individual $i \in \mathcal{H}$.

Also $\rho_{|\mathcal{P}'} = (R_{1|\mathcal{P}'}, \dots, R_{H|\mathcal{P}'})$ is the society's preference profile when alternatives are restricted to some subset $\mathcal{P}'$ of $\mathcal{P}$.

Let $\mathfrak{S}$ be the set of all reflexive and complete binary relations on $\mathcal{P}$ (but notice *not necessarily* transitive). A social ordering $R^S \in \mathfrak{S}$ is therefore a reflexive and complete binary relation over all the policy choices in $\mathcal{P}$. Thus, a social ordering can be represented as

$$\phi : \mathfrak{R}^H \rightarrow \mathfrak{S}.$$

This mathematical formalism implies that $\phi(\rho)$ gives the social ordering for the preference profiles in ρ . We can alternatively think of ϕ as a political system mapping individual preferences into a social choice. A trivial example of ϕ is the dictatorial ordering making agent 1 the dictator, so that for any preference profile $\rho \in \mathfrak{R}^H$, ϕ induces a social order that entirely coincides with R_1 .

Note that our formulation already imposes the condition of “*unrestricted domain*,” which says that in constructing a social ordering we should consider all possible (transitive) individual orderings. Therefore, we are not limiting ourselves to a special class of individual orderings, such as those with “single-peaked” preferences as we will do later in this section.

We say that a social ordering is *weakly Paretian* if $[p \succ_i p' \text{ for all } i \in \mathcal{H}] \implies p \succ^S p'$, that is, if all individuals in the society prefer p to p' , then the social ordering must also rank p ahead of p' . This is weakly Paretian (rather than strongly), since we require all agents to strictly prefer p to p' .

Next we say that, for a preference profile $\rho \in \mathfrak{R}^H$, a subset $\mathcal{D}$ of $\mathcal{H}$ is *decisive between* $p, p' \in \mathcal{P}$, if $[p \succeq_i p' \text{ for all } i \in \mathcal{D} \text{ and } p \succ_{i'} p' \text{ for some } i' \in \mathcal{D}] \implies p \succ^S p'$ (given ρ). If $\mathcal{D}' \subset \mathcal{H}$ is decisive between $p, p' \in \mathcal{P}$ for *all* preference profiles $\rho \in \mathfrak{R}^H$, then it is *dictatorial between* $p, p' \in \mathcal{P}$. We say that $\mathcal{D} \subset \mathcal{H}$ is *decisive* if it is decisive between any $p, p' \in \mathcal{P}$, and $\mathcal{D}' \subset \mathcal{H}$ is *dictatorial* if it is dictatorial between any $p, p' \in \mathcal{P}$. If $\mathcal{D}' \subset \mathcal{H}$ is dictatorial and a singleton, then its unique element is a *dictator*, meaning that social choices will exactly reflect his preferences regardless of the preferences of the other members of the society. In this case, we say that a social ordering ϕ is *dictatorial*.

Next a social ordering satisfies *independence from irrelevant alternatives*, if for any ρ and $\rho' \in \mathfrak{R}^H$ and any $p, p' \in \mathcal{P}$,

$$\rho|_{\{p, p'\}} = \rho'|_{\{p, p'\}} \implies \phi(\rho)|_{\{p, p'\}} = \phi(\rho')|_{\{p, p'\}}.$$

The axiom of independence from irrelevant alternatives is essential for Arrow's Theorem. It states that if two preference profiles have the same choice over two policy alternatives, the social orderings that derive from these two preference profiles must also have identical choices over these two policy alternatives, regardless of how these two preference profiles differ for “irrelevant” alternatives. While this condition (axiom) at first appears plausible, it is in fact a reasonably strong one. In particular, it rules out any kind of interpersonal “cardinal” comparisons—that is, it excludes information on how strongly an individual prefers one outcome versus another.

The main theorem of the field of social choice theory is the following:

THEOREM 22.1. (*Arrow's (Im)Possibility Theorem*) *If a social ordering, ϕ , is transitive, weakly Paretian and satisfies independence from irrelevant alternatives, then it is dictatorial.*

PROOF. Suppose to obtain a contradiction that there exists a non-dictatorial and weakly Paretian social ordering, ϕ , satisfying independence from irrelevant alternatives. We will derive a contradiction in two steps.

Step 1: Let a set $\mathcal{J} \subset \mathcal{H}$ be *strongly decisive* between $p_1, p_2 \in \mathcal{P}$ if for any preference profile $\rho \in \mathcal{R}^H$ with $p_1 \succ_i p_2$ for all $i \in \mathcal{J}$ and $p_2 \succ_j p_1$ for all $j \in \mathcal{H} \setminus \mathcal{J}$, $p_1 \succ^S p_2$ ($\mathcal{H}$ itself is strongly decisive since ϕ is weakly Paretian). We first prove that if $\mathcal{J}$ is strongly decisive between $p_1, p_2 \in \mathcal{P}$, then $\mathcal{J}$ is dictatorial (and hence decisive for all $p, p' \in \mathcal{P}$ and for all preference profiles $\rho \in \mathcal{R}^H$). To prove this, consider the restriction of an arbitrary preference profile $\rho \in \mathcal{R}^H$ to $\rho|_{\{p_1, p_2, p_3\}}$ and suppose that we also have $p_1 \succ_i p_3$ for all $i \in \mathcal{J}$. Next consider an alternative profile $\rho'|_{\{p_1, p_2, p_3\}}$, such that $p_1 \succ'_i p_2 \succ'_i p_3$ for all $i \in \mathcal{J}$ and $p_2 \succ'_i p_1$ and $p_2 \succ'_i p_3$ for all $i \in \mathcal{H} \setminus \mathcal{J}$. Since $\mathcal{J}$ is strongly decisive between p_1 and p_2 , $p_1 \succ'^S p_2$. Moreover, since ϕ is weakly Paretian, we also have $p_2 \succ'^S p_3$, and thus $p_1 \succ'^S p_2 \succ'^S p_3$. Notice that $\rho'|_{\{p_1, p_2, p_3\}}$ did not specify the preferences of individuals $i \in \mathcal{H} \setminus \mathcal{J}$ between p_1 and p_3 , but we have established $p_1 \succ'^S p_3$ for $\rho'|_{\{p_1, p_2, p_3\}}$. We can then invoke independence from irrelevant alternatives and conclude that the same holds for $\rho|_{\{p_1, p_2, p_3\}}$, i.e., $p_1 \succ^S p_3$. But then, since the preference profiles and p_3 are arbitrary, it must be the case that $\mathcal{J}$ is dictatorial between p_1 and p_3 . Next repeat the same argument for $\rho|_{\{p_1, p_2, p_4\}}$ and $\rho'|_{\{p_1, p_2, p_4\}}$, except that now $p_4 \succ_i p_2$ and $p_4 \succ'_i p_1 \succ'_i p_2$ for $i \in \mathcal{J}$, while $p_2 \succ'_j p_1$ and $p_4 \succ'_j p_1$ for all $j \in \mathcal{H} \setminus \mathcal{J}$. Then, the same chain of reasoning, using the facts that $\mathcal{J}$ is strongly decisive, $p_1 \succ'^S p_2$, ϕ is weakly Paretian and satisfies independence from irrelevant alternatives, implies that $\mathcal{J}$ is dictatorial between p_4 and p_2 (that is, $p_4 \succ^S p_2$ for any preference profile $\rho \in \mathcal{R}^H$). Now once again using independence from irrelevant alternatives and also transitivity, for any preference profile $\rho \in \mathcal{R}^H$, $p_4 \succ_i p_3$ for all $i \in \mathcal{J}$. Since $p_3, p_4 \in \mathcal{P}$ were arbitrary, this completes the proof that $\mathcal{J}$ is dictatorial (i.e., dictatorial for all $p, p' \in \mathcal{P}$).

Step 2: Given the result in Step 1, if we prove that some individual $h \in \mathcal{H}$ is strongly decisive for some $p_1, p_2 \in \mathcal{P}$, we will have established that it is a dictator and thus ϕ is dictatorial. Let $\mathcal{D}_{ab}$ be the strongly decisive set between p_a and p_b . Such a set always exists for any $p_a, p_b \in \mathcal{P}$, since $\mathcal{H}$ itself is a strongly decisive set. Let $\mathcal{D}$ be the minimal strongly decisive set (meaning the strongly decisive set with the fewest members). This is also well-defined, since there is only a finite number of individuals in $\mathcal{H}$. Moreover, without loss of generality, suppose that $\mathcal{D} = \mathcal{D}_{12}$ (i.e., let the strongly decisive set between p_1 and p_2 be the minimal strongly decisive set). If $\mathcal{D}$ a singleton, then Step 1 applies and implies that ϕ is dictatorial, completing the proof. Thus suppose that $\mathcal{D} \neq \{i\}$. Then, by unrestricted domain, the following preference profile (restricted to $\{p_1, p_2, p_3\}$) is feasible

$$\begin{array}{ll} \text{for } i \in \mathcal{D} & p_1 \succ_i p_2 \succ_i p_3 \\ \text{for } j \in \mathcal{D} \setminus \{i\} & p_3 \succeq_j p_1 \succ_j p_2 \\ \text{for } k \notin \mathcal{D} & p_2 \succ_k p_3 \succ_k p_1. \end{array}$$

By hypothesis, $\mathcal{D}$ is strongly decisive between p_1 and p_2 and therefore $p_1 \succ^S p_2$. Next if $p_3 \succ^S p_2$, then given the preference profile here, $\mathcal{D} \setminus \{i\}$ would be strongly decisive between p_2 and p_3 , and this would contradict that $\mathcal{D}$ is the minimal strongly decisive set. Thus $p_2 \not\succ^S p_3$. Combined with $p_1 \succ^S p_2$, this implies $p_1 \succ^S p_3$. But given the preference profile here, this implies that $\{i\}$ is strongly decisive, yielding another contradiction. Therefore, the minimal strongly decisive set must be a singleton $\{h\}$ for some $h \in \mathcal{H}$. Then, from Step 1, $\{h\}$ is a dictator and ϕ is dictatorial, completing the proof. $\square$

An immediate implication of this theorem is that any set of minimal decisive individuals $\mathcal{D}$ within the society $\mathcal{H}$ must either be a singleton, that is, $\mathcal{D} = \{i\}$, so that we have a dictatorial social ordering, or we have to live with intransitivities.

While this theorem is often referred to as Arrow's Possibility Theorem, it is really an "Impossibility Theorem". An alternative way of stating the theorem is that there exists *no* social ordering that is transitive, weakly Paretian, consistent with independence from irrelevant alternatives and non-dictatorial. Viewed in this light, an important implication of this theorem is that there is no way of avoiding the issue of conflict in preferences of individuals by positing a social welfare function. A social welfare function, respecting transitivity, can only replace the actual political economic process of decision making when it is dictatorial. Naturally, who will become the dictator in the society fundamentally brings back the issue of *political power*, which is also essential for any positive political economy analysis of collective decision-making. In addition, from a modeling point of view, Arrow's Theorem means that, if we are interested in non-dictatorial (and transitive) outcomes, we have to look at political systems that either restrict choices or focus on more concrete situations, where we have to be more specific about the distribution of political power and the political institutions regulating the decision-making process. This will be the basis of our analysis for the rest of this chapter and for the next chapter.

Often, economic models restrict the policy space and/or preferences of citizens in order to ensure that Arrow's Impossibility Theorem does not apply. Unfortunately, such restrictions on the policy space have more than technical implications. For example, they often force the modeler to restrict agents to use inefficient methods of redistribution. As a result, some of the inefficiencies that are found in political economy models are not a consequence of the logic of these models, but a consequence of the technical assumptions that the modelers make in restricting the policy space to a single policy. In some circumstances, limits on fiscal instruments might be justified on economic grounds. For example, the assumption that there was only a linear tax on output in Section 22.2 was justified with the argument that lump-sum taxes were technologically infeasible (and this was in turn motivated by our desire to capture the potentially distortionary effects of taxation). In general, it is important to recognize that the limits on the set of fiscal instruments is often responsible for the potential distortions resulting from political economy (as was the case in Section 22.2).

One reaction to Arrow's Theorem might be that the problem of aggregating individual preferences arises because we are not looking at more relevant mechanisms such as voting.

The next subsection shows that the same problems arise when collective choices are made by voting. In fact Arrow's Theorem applies to any possible way of aggregating individual preferences, and if voting were able to solve the problems raised by the theorem, it would be a contradiction to the theorem! Nevertheless, voting can be useful in situations where we put more structure on preferences and on how individuals vote, which will essentially amount to either giving up the "unrestricted domain" assumption on choices or relaxing the independence from irrelevant alternatives.

22.6.3. Voting and the Condorcet Paradox. Let us illustrate how voting also runs into exactly same problems as those highlighted by Arrow's Theorem by using a well-known example, *the Condorcet paradox*. The underlying reason for this paradox is related to Arrow's Theorem and will also illustrate why, to obtain the Median Voter Theorem below, we will have to introduce reasonably strong restrictions.

EXAMPLE 22.2. Imagine a society consisting of three individuals, 1, 2, and 3 and three choices. The individuals' preferences are as follows:

$$\begin{array}{ll} 1 & a \succ c \succ b \\ 2 & b \succ a \succ c \\ 3 & c \succ b \succ a \end{array}$$

Moreover, let us make the political mechanism somewhat more specific, and assume that it satisfies the following three requirements, which together make up the "open agenda direct democracy" system.

A1. *Direct democracy.* The citizens themselves make the policy choices via majoritarian voting.

A2. *Sincere voting.* In every vote, each citizen votes for the alternative that gives him the highest utility according to his policy preferences (indirect function) $U(p; \alpha_i)$. This requirement is adopted for simplicity. In many situations, individuals may vote for the outcome that they do not prefer, anticipating the later repercussions of this choice (we refer to this type of behavior as "*strategic voting*"). Whether they do so or not is important in certain situations, but not for the discussion at the moment.

A3. *Open agenda.* Citizens vote over pairs of policy alternatives, such that the winning policy in one round is posed against a new alternative in the next round and the set of alternatives includes all feasible policies. Later, we will replace the open agenda assumption with parties offering policy alternatives, thus moving away from direct democracy some way towards indirect/representative democracy. For now it is a good starting point.

Now, using the three assumptions, consider a contest between policies a and b . In this contest, agents 2 and 3 will vote for b over a , so b is the majority winner. Next, by the open agenda assumption, the other policy alternative c will run against b . Now agents 1 and 3 prefer c to b , which is the new majority winner. Next, c will run against a , but now agents 1 and 2 prefer a , so a is the majority winner. Therefore, in this case we have "cycling" over the

various alternatives, or put differently there is no “equilibrium” of the voting process that selects a unique policy outcome.

For future reference, let us now define a Condorcet winner as a policy choice that does not lead to such cycling. In particular,

DEFINITION 22.1. *A **Condorcet winner** is a policy p^* that beats any other feasible policy in a pairwise vote.*

In light of this definition, there is no Condorcet winner in the example of the Condorcet paradox.

22.6.4. Single-Peaked Preferences. Suppose now that the policy space is unidimensional, so that p is a real number, that is, $\mathcal{P} \subset \mathbb{R}$. In this case, a simple way to rule out the Condorcet paradox is to assume that preferences are *single peaked* for all voters. We will see below that the restriction that $\mathcal{P}$ is unidimensional is very important and single-peaked preferences are not well defined when there are multiple policy dimensions.

We say that voter i has single-peaked preferences if his preference ordering for alternative policies is dictated by their relative distance from his bliss point, $p(\alpha_i)$: a policy closer to $p(\alpha_i)$ is preferred over more distant alternatives. Specifically:

DEFINITION 22.2. *Consider a finite set of $\mathcal{P} \subset \mathbb{R}$ and let $p(\alpha_i) \in \mathcal{P}$ be individual i 's unique bliss point over $\mathcal{P}$. Then, the policy preferences of citizen i are **single peaked** iff:*

$$\begin{aligned} &\text{For all } p'', p' \in \mathcal{P}, \text{ such that } p'' < p' \leq p(\alpha_i) \text{ or } p'' > p' \geq p(\alpha_i), \\ &\text{we have } U(p''; \alpha_i) < U(p'; \alpha_i). \end{aligned}$$

Note that strict concavity of $U(p'; \alpha_i)$ is sufficient for it to be single peaked, but is not necessary. In fact, single-peakedness is equivalent to strict quasi-concavity. This definition could be weakened so that the bliss point of the individual is not unique (that is, it can be weakened from strict quasi-concavity to quasi-concavity). But this added generality is not important for our purposes.

We can easily verify that in the Condorcet paradox, not all agents possessed single-peaked preferences. For example, taking the ordering to be a, b, c , agent 1 who has preferences $a \succ c \succ b$ does not have single-peaked preferences (if we took a different ordering of the alternatives, then the preferences of one of the other two agents would violate the single-peakedness assumption, see Exercise 22.21).

The next theorem shows that with single-peaked preferences, there always exists a Condorcet winner. Before stating this theorem, let us define the *median voter* of the society. Given the assumption that each individual has a unique bliss point over $\mathcal{P}$, we can rank all individuals according to their bliss points, the $p(\alpha_i)$'s. Also, to remove uninteresting ambiguities, let us imagine that H is an odd number (i.e., $\mathcal{H}$ consists of an odd number of individuals). Then, the median voter is the individual who has exactly $(H - 1)/2$ bliss points to his left and $(H - 1)/2$ bliss points to his right. Put differently, his bliss point is exactly

in the middle of the distribution of bliss points. We denote this individual by α_m , and his bliss point (ideal policy) is denoted by p_m .

THEOREM 22.2. (*The Median Voter Theorem*) Suppose that H is an odd number, that A1 and A2 hold and that all voters have single-peaked policy preferences over a given ordering of policy alternatives, $\mathcal{P}$. Then, a Condorcet winner always exists and coincides with the median-ranked bliss point, p_m . Moreover, p_m is the unique equilibrium policy (stable point) under the open agenda majoritarian rule, that is, under A1-A3.

PROOF. The proof is by a “separation argument”. Order the individuals according to their bliss points $p(\alpha_i)$, and label the median-ranked bliss point by p_m . By the assumption that H is an odd number, p_m is uniquely defined (though α_m may not be uniquely defined). Suppose that there is a vote between p_m and some other policy $p'' < p_m$. By definition of single-peaked preferences, for every individual with $p_m < p(\alpha_i)$, we have $U(p_m; \alpha_i) > U(p''; \alpha_i)$. By A2, these individuals will vote sincerely and thus, in favor of p_m . The coalition voting for supporting p_m thus constitutes a majority. The argument for the case where $p'' > p_m$ is identical. $\square$

The assumption that the society consists of an odd number of individuals was made only to shorten the statement of the theorem and the proof. Exercise 22.23 asks you to generalize the theorem and its proof to the case in which H is an even number.

More important than whether there is an odd or even number of individuals in the society is the assumption of sincere voting. Clearly, rational agents could deviate from truthful reporting of their preferences (and thus from truthful voting) when this is beneficial for them. So an obvious question is whether the MVT generalizes to the case in which individuals do not vote sincerely? The answer is yes. To see this, let us modify the sincere voting assumption to strategic voting:

A2'. Strategic voting. Define a *vote function* of individual i in a pairwise contest between p' and p'' by $v_i(p', p'') \in \{p', p''\}$. Let a voting (counting) rule in a society with H citizens be $V: \{p', p''\}^H \rightarrow \{p', p''\}$ for any $p', p'' \in \mathcal{P}$. (For example, the majoritarian voting rule V^M picks p' over p'' when this policy receives more votes than p''). Let $V(v_i(p', p''), v_{-i}(p', p''))$ be the policy outcome from voting rule V applied to the pairwise contest $\{p', p''\}$, when the remaining individuals cast their votes according to the vector $v_{-i}(p', p'')$, and when individual i votes $v_i(p', p'')$. Strategic voting means that

$$v_i(p', p'') \in \arg \max_{\tilde{v}_i(p', p'')} U(V(\tilde{v}_i(p', p''), v_{-i}(p', p'')); \alpha_i).$$

In other words, strategic voting implies that each individual chooses the voting strategy that maximizes utility given the voting strategies of other agents.

Finally, recall that a *weakly-dominant* strategy for individual i is a strategy that gives weakly higher payoff to individual i than any of his other strategies regardless of the strategy profile of other players

THEOREM 22.3. (*The Median Voter Theorem With Strategic Voting*) Suppose that H is an odd number, that $A1$ and $A2'$ hold and that all voters have single-peaked policy preferences over a given ordering of policy alternatives, $\mathcal{P}$. Then, sincere voting is a weakly-dominant strategy for each player and there exists a unique weakly-dominant equilibrium, which features the median-ranked bliss point, p_m , as the Condorcet winner.

PROOF. The vote counting rule (the political system) in this case is majoritarian, denoted by V^M . Consider two policies $p', p'' \in \mathcal{P}$ and fix an individual $i \in \mathcal{H}$. Assume without loss of any generality that $U(p'; \alpha_i) \geq U(p''; \alpha_i)$. Suppose first that for any $v_i \in \{p', p''\}$, $V^M(v_i, v_{-i}(p', p'')) = p'$ or $V^M(v_i, v_{-i}(p', p'')) = p''$, that is, individual i is *not* pivotal. This implies that $v_i(p', p'') = p'$ is a best response for individual i . Suppose next that individual i is pivotal, that is, $V^M(v_i(p', p''), v_{-i}(p', p'')) = p'$ if $v_i(p', p'') = p'$ and $V^M(v_i(p', p''), v_{-i}(p', p'')) = p''$ otherwise. In this case, the action $v_i(p', p'') = p'$ is clearly a best response for i . Since this argument applies for each $i \in \mathcal{H}$, it establishes that voting sincerely is a weakly-dominant strategy and the conclusion of the theorem follows from Theorem 22.2. $\square$

Notice that the second part of the Theorem 22.2, which applied to open agenda elections, is absent in Theorem 22.3. This is because the open agenda assumption does not lead to a well defined game, so a game-theoretic analysis and thus an analysis of strategic voting is no longer possible.

In fact, there is no guarantee that sincere voting is optimal in dynamic situations even with single-peaked preferences. The following example illustrates this:

EXAMPLE 22.3. Consider three individuals with the following preference orderings.

- 1 $a \succ b \succ c$
- 2 $b \succ c \succ a$
- 3 $c \succ b \succ a$

These preferences are clearly single peaked (order them alphabetically to see this). In a one round vote, b will beat any other policy. But now consider the following dynamic voting set up: first, there is a vote between a and b . Then, the winner goes against c , and the winner of this contest is the social choice. Sincere voting will imply that in the first round players 2 and 3 will vote for b , and in the second round, players 1 and 2 will vote for b , which will become the collective choice.

Is such sincere voting “equilibrium behavior”? Exactly the same argument as above shows that in the second round, sincere voting is a weakly dominant strategy. But not necessarily in round one. Suppose players 1 and 2 are playing sincerely. Now if player 3 deviates and votes for a (even though she prefers b), then a will advance to the second round and would lose to c . Consequently, the social choice will coincide with the bliss point of player 3. Exercise 22.24 asks you to characterize the SPE of this game under strategic voting by all players.

Dynamic voting issues become more interesting, and open the way for agenda setting, when there are no Condorcet winners. The following example illustrates this.

EXAMPLE 22.4. Consider the preference profile in Example 22.2 and the following political mechanism. First, all individuals vote between a and b , and then they vote over the winner of this contest and c . With sincere voting, b will win the first round, and then c wins the second round against b . Now consider agent 2. If he changes his vote in the first round to a (thus does not vote sincerely), the first-round winner will be a , which will also win against c , and player 2 prefers this outcome to the outcome of sincere voting, which was c .

This example can also be used to illustrate the role of “agenda setting”. Suppose that in the above game, agent 1 decides the sequence of alternatives presented for voting. In particular, he has to choose between three options (a vs. b first, a vs. c first, and b vs. c , first). Anticipating strategic voting by player 2, he will choose the first option and will ensure that his most preferred alternative becomes the political choice of the society. In contrast, if agent 3 chose the sequence, he would go for a vs. c first, which would induce agent 1 to vote strategically for c , and lead to c as the ultimate outcome.

22.6.5. Party Competition and the Downsian Policy Convergence Theorem.

The focus so far has been on voting between two alternative policies or on open agenda voting, which can be viewed as an extreme form of “direct democracy”. The MVT becomes potentially more relevant and more powerful when applied in the context of indirect democracy, that is, when combined with a simple model of party competition. I now give a brief overview of this situation and derive the Downsian Policy Convergence Theorem, which is the basis of much applied work in political economy.

Suppose that there is a Condorcet winner, and there are two parties, A and B , competing for political office. Assume that the parties do not have an ideological bias, and would like to come to power. In particular, they both maximize the probability of coming to power, for example, because they receive a rent or utility of $Q > 0$ when they are in power.

Assume also that parties simultaneously announce their policy, and are committed to this policy. This implies that the behavior of the two parties can be represented by the following pair of maximization problems:

$$\begin{aligned}
 (22.42) \quad \text{Party } A & : \max_{p_A} P(p_A, p_B)Q \\
 \text{Party } B & : \max_{p_B} (1 - P(p_A, p_B))Q
 \end{aligned}$$

where Q denotes the rents of being in power and $P(p_A, p_B)$ is the probability that party A comes to power when the two parties’ platforms are p_A and p_B respectively. Let the bliss point of the median voter be p_m . When the median voter theorem applies, we have

$$\begin{aligned}
 (22.43) \quad P(p_A = p_m, p_B) &= 0, \quad P(p_A = p_m, p_B) = 1, \text{ and} \\
 P(p_A = p_m, p_B = p_m) &\in [0, 1].
 \end{aligned}$$

This last statement follows since when both parties offer exactly the same policy, it is a best response for all citizens to vote for either party. However, the literature typically makes the following assumption:

A4. *Randomization:*

$$P(p_A = p_m, p_B = p_m) = 1/2.$$

This assumption can be rationalized by arguing that when indifferent individuals, randomize between the two parties, and since there are many many individuals, by the law of large numbers, each party obtains exactly half of the vote.

We then have the following result:

THEOREM 22.4. (*Downsian Policy Convergence Theorem*) *Suppose that there are two parties that first announce a policy platform and commit to it and a set of voters $\mathcal{H}$ that vote for one of the two parties. Assume that A4 holds and that all voters have single-peaked policy preferences over a given ordering of policy alternatives, and denote the median-ranked bliss point by p_m . Then, both parties will choose p_m as their policy platform.*

PROOF. The proof is by contradiction. Suppose not, then there is a profitable deviation for one of the parties. For example, if $p^A > p^B > p_m$, one of the parties can announce p_m and win the election for sure. When $p^A \neq p_m$ and $p^B = p_m$, party A can also announce p_m and increase its chance of winning to $1/2$. $\square$

Exercise 22.25 asks you to provide a generalization of this theorem without Assumption A4.

This theorem is important because it demonstrates that there will be policy convergence between the two parties and that party competition will implement the Condorcet winner among the voters. Therefore, in situations in which the MVT applies, the democratic process of decision making with competition between two parties will lead to a situation in which both parties will choose their policy platform to coincide with the bliss point of the median voter. Thus the MVT and the Downsian Policy Convergence Theorem together enable us to simplify the process of aggregating the heterogeneous preferences of individuals over policies and assert that, under the appropriate assumptions, democratic decision-making will lead to the most preferred policy of the median voter. The Downsian Policy Convergence Theorem is useful in this context, since it gives a better approximation to “democratic policymaking” in practice than open agenda elections.

There is a sense in which Theorem 22.4 is slightly misleading, however. While the theorem is correct for a society with two parties, it gives the impression of a general tendency towards policy convergence in all democratic societies. Many democratic societies have more than two parties. A natural generalization of this theorem would be to consider three or more parties. Unfortunately, as Exercise 22.26 shows the results of this theorem do not generalize to three parties. Thus some care is necessary in applying the Downsian Policy Convergence Theorem in the context of different political institutions.

Another obvious question is what would happen in the party competition game when there is no Condorcet winner. Theorem 22.4 does not generalize to this case either. In particular, if we take a situation in which there is “cycling,” like the above Condorcet paradox example,

it is straightforward to verify that there is no pure strategy equilibrium in the political competition game. This is further discussed in Exercise 22.27.

22.6.6. Beyond Single-Peaked Preferences. Single-peaked preferences played a very important role in the results of Theorem 22.2 by ensuring the existence of a Condorcet winner. However, single peakedness is a very strong assumption and does not have a natural analog in situations in which voting is over more than one policy choice. When there are multiple policy choices (or when voting is over “functions” such as nonlinear taxation), much more structure needs to be imposed over voting procedures and agenda setting to determine equilibrium policies. Those issues are beyond the scope of my treatment here. Nevertheless, it is possible to relax the assumption of single-peaked preferences and also introduce a set of preferences that are “close” to single-peaked in multidimensional spaces. The latter task would take us too far afield from our focus, so will be left to Exercise 22.28. Here we introduce the useful concept of *single-crossing property*, which will enable us to prove a version of Theorem 22.2 under somewhat weaker assumptions.

DEFINITION 22.3. Consider an ordered policy space $\mathcal{P}$ and also order voters according to their α_i 's. Then, the preferences of voters satisfy the **single-crossing property** over the policy space $\mathcal{P}$ when the following statement is true:

$$\begin{aligned} & \text{if } p > p' \text{ and } \alpha_{i'} > \alpha_i, \text{ or if } p < p' \text{ and } \alpha_{i'} < \alpha_i, \text{ then} \\ & U(p; \alpha_i) > U(p'; \alpha_i) \text{ implies that } U(p; \alpha_{i'}) > U(p'; \alpha_{i'}). \end{aligned}$$

EXAMPLE 22.5. To see why single-crossing property is weaker than single-peaked preferences, consider the following example:

$$\begin{aligned} 1 \quad & a \succ b \succ c \\ 2 \quad & a \succ c \succ b \\ 3 \quad & c \succ b \succ a \end{aligned}$$

It can be verified easily that these preferences are not single peaked. The natural ordering is $a > b > c$, but in this case the preferences of player 2 have two peaks, at a and c . To see why these preferences satisfy single crossing, take the same ordering, and also order players as 1, 2, 3. Now,

$$\begin{aligned} \alpha &= 2: c \succ b \implies \alpha = 3: c \succ b \\ \alpha &= 2: \begin{array}{l} a \succ c \\ a \succ b \end{array} \implies \alpha = 1: \begin{array}{l} a \succ c \\ a \succ b \end{array}. \end{aligned}$$

Notice that while single peakedness is a property of preferences only, the single-crossing property refers to a set of preferences over a given policy space $\mathcal{P}$. It is therefore a joint property of preferences and choices. The following theorem generalizes Theorem 22.2 to a situation with single crossing.

THEOREM 22.5. (*Extended Median Voter Theorem*) Suppose that A1 and A2 hold and that the preferences of voters satisfy the single-crossing property. Then, a Condorcet winner always exists and coincides with the bliss point of the median voter (voter α_m).

PROOF. The proof works with exactly the same separation argument as in the proof of Theorem 22.2. Consider the median voter with α_m , and bliss policy p_m . Consider an alternative policy $p' > p_m$. Naturally, $U(p_m; \alpha_m) > U(p'; \alpha_m)$. Then, by the single crossing property, for all $\alpha_i > \alpha_m$, $U(p_m; \alpha_i) > U(p'; \alpha_i)$. Since α_m is the median, this implies that there is a majority in favor of p_m . The same argument for $p' < p_m$ completes the proof $\square$

Given this theorem, the following result is immediate:

THEOREM 22.6. (*Extended Downsian Policy Convergence*) *Suppose that there are two parties that first announce a policy platform and commit to it and a set of voters that vote for one of the two parties. Assume that A4 holds and that all voters have preferences that satisfy the single-crossing property and denote the median-ranked bliss point by p_m . Then, both parties will choose p_m as their policy.*

PROOF. See Exercise 22.27. $\square$

Despite this generalization, which is quite useful in many applications, and the extension of the MVT presented in Exercise 22.28, the MVT-type results do not apply in many situations with multidimensional policies. Exercise 22.29 gives a simple example, which illustrates how widespread the failure of the MVT will be in practice.

22.6.7. Equilibrium Social Welfare Functions. The MVT and the Downsian Policy Convergence Theorems are powerful for the analysis of many models of political economy. However, as Exercise 22.29 illustrates, the assumptions necessary for these theorems do not apply in many interesting (even simple) models. The political economy literature has thus considered a variety of other plausible ways of aggregating heterogeneous preferences within democratic contexts. Three particularly popular approaches are (1) the “probabilistic voting” models, which essentially add some noise in the voting behavior of individuals (for example, because individuals care about some other non-policy characteristic of the parties that are competing for office); (2) models without policy commitment, such as the citizen-candidate models, in which voters elect a politician, who then decides the policies after election; (3) lobbying models, in which some of the individuals or groups in the society can spend money in order to influence the outcome of democratic politics. A full analysis of these models is beyond the scope of the current book. Nevertheless, one feature of many of these formulations is worth noting, especially in light of the discussion of the issue of Pareto efficiency above. Many simple versions of these models lead to equilibria that are equivalent to maximizing a “reduced-form weighted social welfare function”. The form of this social welfare function is derived from the political economy equilibrium and depends on the specific assumptions made in these models. For our purposes, the noteworthy point is that in various (static) environments, the political economy equilibrium involves maximizing a weighted social welfare function and thus emphasizes that political economy equilibria will be Pareto optimal (given the set of policy instruments). This was generally the case in the models in Sections 22.2–22.4, because, even though these models were dynamic, the extent of dynamic interactions were

limited. I now discuss two standard models that leads to this type of equilibrium (weighted) social welfare functions.

22.6.7.1. *Probabilistic Voting and Swing Voters.* Let the society consist of G distinct groups, with a continuum of voters within each group having the same economic characteristics and preferences. As in the Downsian model, there is electoral competition between two parties, A and B , and let π_P^g be the fraction of voters in group g voting for party P where $P = A, B$, and let λ^g be the share of voters in group g and naturally $\sum_{g=1}^G \lambda^g = 1$. Then, the expected vote share of party P is

$$\pi_P = \sum_{g=1}^G \lambda^g \pi_P^g.$$

In our analysis so far, all voters in group g would have cast their votes identically (unless they were indifferent between the two parties). The idea of probabilistic voting is to smooth out this behavior by introducing other considerations in the voting behavior of individuals. Put differently, probabilistic voting models will add “noise” to equilibrium votes, smoothing the behavior relative to models we analyzed so far. In particular, suppose that individual i in group g has the following preferences:

$$(22.44) \quad \tilde{U}_i^g(p, P) = U^g(p) + \tilde{\sigma}_i^g(P)$$

when party P comes to power, where p is the vector of economic policies chosen by the party in power. Suppose that $p \in \mathcal{P} \subset \mathbb{R}^K$, where K is a natural number, possibly greater than 1. Thus $p \equiv (p^1, \dots, p^K)$ is a potentially multidimensional vector of policies. In addition, $U^g(p)$ is the indirect utility of agents in group g as before (previously denoted by $U(p; \alpha_i)$ for individual i) and captures their economic interests. In addition, the term $\tilde{\sigma}_i^g(P)$ captures the non-policy related benefits that the individual will receive if party P comes to power. The most obvious source of these preferences would be ideological. So this model allows individuals within the same economic group to have different ideological preferences.

Let us normalize $\tilde{\sigma}_i^g(A) = 0$, so that

$$(22.45) \quad \tilde{U}_i^g(p, A) = U^g(p), \text{ and } \tilde{U}_i^g(p, B) = U^g(p) + \tilde{\sigma}_i^g$$

In that case, the voting behavior of individual i can be represented as

$$(22.46) \quad v_i^g(p_A, p_B) = \begin{cases} 1 & \text{if } U^g(p_A) - U^g(p_B) > \tilde{\sigma}_i^g \\ \frac{1}{2} & \text{if } U^g(p_A) - U^g(p_B) = \tilde{\sigma}_i^g \\ 0 & \text{if } U^g(p_A) - U^g(p_B) < \tilde{\sigma}_i^g \end{cases},$$

where $v_i^g(p_A, p_B)$ denotes the probability that the individual will vote for party A , p_A is the platform of party A and p_B is the platform of party B , and if an individual is indifferent between the two parties (inclusive of the ideological benefits), he randomizes his vote.

Let us now assume that the distribution of non-policy related benefits $\tilde{\sigma}_i^g$ for individual i in group g is given by a smooth cumulative distribution function H^g defined over $(-\infty, +\infty)$, with the associated probability density function h^g . The draws of $\tilde{\sigma}_i^g$ across individuals are

independent. Consequently, the vote share of party A among members of group g is

$$\pi_A^g = H^g(U^g(p_A) - U^g(p_B)).$$

Furthermore, to simplify the exposition here, suppose that parties maximize their expected vote share. In this case, party A sets this policy platform p_A to maximize:

$$(22.47) \quad \pi_A = \sum_{g=1}^G \lambda^g H^g(U^g(p_A) - U^g(p_B)).$$

Party B faces a symmetric problem and maximizes π_B , which is defined similarly. In particular, since $\pi_B = 1 - \pi_A$, party B 's problem is exactly the same as minimizing π_A . Equilibrium policies will then be determined as the Nash equilibrium of a (zero-sum) game where both parties make simultaneous policy announcements to maximize their vote share. Let us first look at the first-order condition of party A with respect to its own policy choice, p_A , taking the policy choices of the other party, p_B , as given. This is:

$$\sum_{g=1}^G \lambda^g h^g(U^g(p_A) - U^g(p_B)) DU^g(p_A) = 0,$$

where $DU^g(p_A)$ is the gradient of $U^g(\cdot)$ given by

$$DU^g(p_A) = \left(\frac{\partial U^g(p_A)}{\partial p_A^1}, \dots, \frac{\partial U^g(p_A)}{\partial p_A^K} \right)^T,$$

with p_A^k corresponding to the k th component of the policy vector p_A . Since the problem of party B is symmetric, it is natural to focus on pure strategy symmetric equilibria. In fact, if the maximization problems of both parties are strictly concave, such a symmetric equilibrium will exist (see Exercise 22.30). Clearly in this case, we will have policy convergence with $p_A = p_B = p^*$, and thus $U^g(p_A) = U^g(p_B)$. Consequently, symmetric equilibrium policies, announced by both parties, must be given by

$$(22.48) \quad \sum_{g=1}^G \lambda^g h^g(0) DU^g(p^*) = 0.$$

It is now straightforward to see that eq. (22.48) also corresponds to the solution to the maximization of the following weighted utilitarian social welfare function:

$$(22.49) \quad \sum_{g=1}^G \chi^g \lambda^g U^g(p),$$

where $\chi^g \equiv h^g(0)$ are the weights that different groups receive in the social welfare function. This analysis therefore establishes:

THEOREM 22.7. (Probabilistic Voting Theorem) *Consider a set of policy choices $\mathcal{P}$, let $p \in \mathcal{P} \subset \mathbb{R}^K$ be a policy vector and let preferences be given by (22.45), with the distribution function of $\tilde{\sigma}_i^g$ as H^g . Then, if a pure strategy symmetric equilibrium exists, equilibrium policy is given by p^* that maximizes (22.49).*

The important point to note about this result is its seeming generality: as long as a pure strategy symmetric equilibrium in the party competition game exists, it will correspond to a maximum of some weighted social welfare function. This generality is somewhat exaggerated, however, since such a symmetric equilibrium does not always exist. In fact, conditions to guarantee existence of pure strategy symmetric equilibria are rather restrictive and are discussed in Exercise 22.30.

22.6.7.2. *Lobbying.* Consider next a very different model of policy determination, a lobbying model. In a lobbying model, different groups make campaign contributions or pay money to politicians in order to induce them to adopt a policy that they prefer. With lobbying, political power comes not only from voting, but also from a variety of other sources, including whether various groups are organized, how much resources they have available, and their marginal willingness to pay for changes in different policies. Nevertheless, the most important result for us will be that equilibrium policies under lobbying will also look like the solution to a weighted utilitarian social welfare maximization problem.

To see this, I will quickly review the lobbying model due to Grossman and Helpman (1996). Imagine again that there are G groups of agents, with the same economic preferences. The utility of an agent in group g , when the policy that is implemented is given by the vector $p \in \mathcal{P} \subset \mathbb{R}^K$, is equal to

$$U^g(p) - \gamma^g(p)$$

where $U^g(p)$ is the usual indirect utility function, and $\gamma^g(p)$ is the per-person lobbying contribution from group g . We will allow these contributions to be a function of the policy implemented by the politician, and to emphasize this, it is written with p as an explicit argument.

Following Grossman and Helpman, let us assume that there is a politician in power, and he has a utility function of the form

$$(22.50) \quad V(p) \equiv \sum_{g=1}^G \lambda^g \gamma^g(p) + a \sum_{g=1}^G \lambda^g U^g(p),$$

where as before λ^g is the share of group g in the population. The first term in (22.50) is the monetary receipts of the politician, and the second term is utilitarian aggregate welfare. Therefore, the parameter a determines how much the politician cares about aggregate welfare. When $a = 0$, he only cares about money, and when $a \rightarrow \infty$, he acts as a utilitarian social planner. One reason why politicians might care about aggregate welfare is because of electoral politics (for example, they may receive rents or utility from being in power as in the last subsection and their vote share might depend on the welfare of each group).

Now consider the problem of an individual j in group g . By contributing some money, he might be able to sway the politician to adopt a policy more favorable to his group. But he is one of many members in his group, and there is a natural free-rider problem. He might let others make the contribution, and simply enjoy the benefits. This will typically be an outcome if groups are unorganized (for example, there is no effective organization coordinating their

lobbying activity and excluding non-contributing members from some of the benefits). On the other hand, organized groups might be able to collect contributions from their members in order to maximize group welfare.

Suppose that out of the G groups of agents, $G' < G$ are organized as lobbies, and can collect money among their members in order to further the interests of the group. The remaining $G - G'$ are unorganized, and will make no contributions. Without loss of any generality, let us rank the groups such that groups $g = 1, \dots, G'$ to be the organized ones.

The lobbying game takes the following form: every organized lobby g simultaneously offers a schedule $\gamma^g(p) \geq 0$ which denotes the payments they would make to the politician when policy $p \in \mathcal{P}$ is adopted. After observing the schedules, the politician chooses p . Notice the important assumption here that contributions to politicians (campaign contributions or bribes) can be conditioned on the actual policy that's implemented by the politicians. This assumption may be a good approximation to reality in some situations, but in others, lobbies might simply have to make up-front contributions and hope that these help the parties that are expected to implement policies favorable to them get elected.

This is a potentially complex game, since various different agents (here lobbies) are choosing functions (rather than real numbers or vectors). Nevertheless, the equilibrium of this lobbying game takes a relatively simple form.

THEOREM 22.8. (*Lobbying Equilibrium*) *In the lobbying game described above, contribution functions for groups $g = 1, 2, \dots, J$, $\{\hat{\gamma}^g(\cdot)\}_{g=1,2,\dots,J}$ and policy p^* constitute a subgame perfect Nash equilibrium if:*

- (1) $\hat{\gamma}^g(\cdot)$ is feasible in the sense that $0 \leq \hat{\gamma}^g(p) \leq U^g(p)$.
- (2) The politician chooses the policy that maximizes its welfare, that is,

$$p^* \in \arg \max_p \left(\sum_{g=1}^{G'} \lambda^g \hat{\gamma}^g(p) + a \sum_{g=1}^G \lambda^g U^g(p) \right).$$

- (3) There are no profitable deviations for any lobby, $g = 1, 2, \dots, G'$, that is,

$$p^* \in \arg \max_p \left(\lambda^g (U^g(p) - \hat{\gamma}^g(p)) + \sum_{g'=1}^{G'} \lambda^{g'} \hat{\gamma}^{g'}(p) + a \sum_{g'=1}^G \lambda^{g'} U^{g'}(p) \right) \text{ for all } g = 1, 2, \dots, G'.$$

- (4) There exists a policy p^g for every lobby $g = 1, 2, \dots, G'$ such that

$$p^g \in \arg \max_p \left(\sum_{g'=1}^{G'} \lambda^{g'} \hat{\gamma}^{g'}(p) + a \sum_{g'=1}^G \lambda^{g'} U^{g'}(p) \right)$$

and satisfies $\hat{\gamma}^g(p^g) = 0$. That is, the contribution function of each lobby is such that there exists a policy that makes no contributions to the politician, and gives her the same utility.

PROOF. (Sketch) Conditions 1, 2 and 3 are easy to understand. No group would ever offer a contribution schedule that does not satisfy Condition 1. Condition 2 has to hold, since

the politician chooses the policy. If Condition 3 did not hold, then the lobby could change its contribution schedule slightly and improve its welfare. In particular suppose that this condition does not hold for lobby $g = 1$, and instead of p^* , some $\hat{p}$ maximizes (22.51). Denote the difference in the values of (22.51) evaluated at these two vectors by $\Delta > 0$. Consider the following contribution schedule for lobby $g = 1$:

$$\tilde{\gamma}^1(p) = \lambda_1^{-1} \left[\sum_{g=1}^{G'} \lambda^g \tilde{\gamma}^g(p^*) + a \sum_{g=1}^G \lambda^g U^g(p^*) - \sum_{g=2}^{G'} \lambda^g \tilde{\gamma}^g(p) - a \sum_{g=1}^G \lambda^g U^g(p) + \varepsilon c^1(p) \right]$$

where $c^1(p)$ is an arbitrary function that reaches its maximum at $p = \hat{p}$. Following this contribution offer by lobby 1, the politician would choose $p = \hat{p}$ for any $\varepsilon > 0$. To see this note that by part (1), the politician would choose policy $\tilde{p}$ that maximizes

$$\lambda^1 \tilde{\gamma}^1(p) + \sum_{g=2}^{G'} \lambda^g \tilde{\gamma}^g(p) + a \sum_{g=1}^G \lambda^g U^g(p) = \sum_{g=1}^{G'} \lambda^g \tilde{\gamma}^g(p^*) + a \sum_{g=1}^G \lambda^g U^g(p^*) + \varepsilon c^1(p).$$

Since for any $\varepsilon > 0$ this expression is maximized by $\hat{p}$, the politician would choose $\hat{p}$. The change in the welfare of lobby 1 as a result of changing its strategy is $\Delta - \varepsilon c^1(\hat{p})$. Since $\Delta > 0$, for small enough ε , the lobby gains from this change, showing that the original allocation could not have been an equilibrium.

Finally, condition 4 ensures that the lobby is not making a payment to the politician above the minimum that is required. If this condition were not true, the lobby could reduce its contribution function by a constant, still induce the same behavior, and obtain a higher payoff. $\square$

Next suppose that these contribution functions are differentiable.⁵ Then, it has to be the case that for every policy choice, p^k , within the vector p^* , we must have from the first-order condition of the politician that

$$\sum_{g=1}^{G'} \lambda^g \frac{\partial \tilde{\gamma}^g(p^*)}{\partial p^k} + a \sum_{g=1}^G \lambda^g \frac{\partial U^g(p^*)}{\partial p^k} = 0 \text{ for all } k = 1, 2, \dots, K$$

and from the first-order condition of each lobby that

$$\lambda^g \left(\frac{\partial \tilde{\gamma}^g(p^*)}{\partial p^k} - \frac{\partial U^g(p^*)}{\partial p^k} \right) + \sum_{g'=1}^G \lambda^{g'} \frac{\partial \tilde{\gamma}^{g'}(p^*)}{\partial p^k} + a \sum_{g'=1}^G \lambda^{g'} \frac{\partial U^{g'}(p^*)}{\partial p^k} = 0 \text{ for all } k = 1, 2, \dots, K \text{ and } g = 1, 2, \dots, G'.$$

Combining these two first-order conditions, we obtain

$$(22.52) \quad \frac{\partial \tilde{\gamma}^g(p^*)}{\partial p^k} = \frac{\partial U^g(p^*)}{\partial p^k}$$

⁵But there is nothing in the analysis that implies that these functions have to be differentiable; in fact, equilibria with non-differentiable functions are easy to construct. Nevertheless, it is generally thought that equilibria with non-differentiable functions are more “fragile” and thus less relevant. See the discussion in Grossman and Helpman (1994) and Bernheim and Whinston (1986).

for all $k = 1, 2, \dots, K$ and $g = 1, 2, \dots, G'$. Intuitively, at the margin each lobby is willing to pay for a change in policy exactly as much as this policy will bring them in terms of marginal return.

But then this implies that the equilibrium can be characterized as

$$p^* \in \arg \max_p \left(\sum_{j=1}^{G'} \lambda^j U^j(p) + a \sum_{j=1}^G \lambda^j U^j(p) \right).$$

Consequently, the lobbying equilibrium can also be represented as a solution to the maximization of a weighted social welfare function, with individuals in unorganized groups getting a weight of a and those in organized group receiving a weight of $1 + a$. Intuitively, $1/a$ measures how much money matters in politics, and the more money matters, the more weight groups that can lobby receive. As $a \rightarrow \infty$, we converge to the utilitarian social welfare function.

22.7. Distributional Conflict and Economic Growth: Heterogeneity and the Median Voter

Let us now return to the model of Section 22.2 with linear preferences, but relax the assumption that political power is in the hands of an elite. Instead, we will now introduce heterogeneity among the agents and then apply the tools from the previous section, in particular, the Median Voter and Downsian Policy Convergence Theorems, Theorems 22.2-22.5, to analyze the political economy of this model. Recall that these theorems show that if there is a one-dimensional policy choice and individuals have single-peaked preferences (or preferences over the menu of policies that satisfy the single-crossing property), then the political equilibrium will coincide with the most preferred policy of the median voter.

To focus on the main issues in the simplest possible way, I modify the environment from Section 22.2 slightly. First, there are no longer any elites. Instead, economic decisions will be made by majoritarian voting among all the agents. Second, to abstract from political conflict between entrepreneurs and workers, I also assume that there are no workers (recall Exercise 22.3 for why having only entrepreneurs simplifies the analysis; see Exercise 22.31 for an economy where individuals differ both in terms of their productivity and occupation). Instead, the economy consists of a continuum 1 of *yeoman-entrepreneurs*, each denoted by $i \in [0, 1]$ and with access to a neoclassical production function

$$Y_i(t) = F(K_i(t), A_i L_i(t)),$$

where A_i is a time-invariant labor-augmenting productivity measure and will be the only source of heterogeneity among the entrepreneurs. In particular, F satisfies Assumptions 1 and 2. I assume that A_i has a distribution given by $\mu(A)$ among the entrepreneurs. The yeoman-entrepreneur assumption means that each entrepreneur can only employ himself as the worker, so $L_i(t) = 1$ for all $i \in [0, 1]$ and for all t . This assumption is important, since otherwise the most productive entrepreneur would hire the entire labor force. Since heterogeneity is the main focus, introducing diminishing returns for each entrepreneur is

important, and the yeoman-entrepreneur assumption achieves this in a simple way. I also set the depreciation rate of capital δ equal to 1 to simplify notation.

As noted previously, all agents have linear preferences given by (22.1). Linear preferences again simplify the analysis, by separating the political decisions across different periods. As in Section 22.2, the investment decisions at time $t + 1$ will depend only on the tax rate announced for time $t + 1$. This latter feature is particularly important here, since we know from the previous section that the Median Voter Theorem does not generally apply with multidimensional policy choices. The fact that at each point in time there is only one relevant tax policy will enable us to use the Median Voter Theorem.

More specifically, the timing of events is very similar to that in Section 22.2. At each date t :

- (1) there is voting over a linear tax rate on output $\tau(t + 1) \in [0, 1]$ that will apply to all entrepreneurs in the next period (at $t + 1$). Voting is between two parties with policy commitment, so that Theorems 22.2 and 22.5 (and Theorems 22.4 and 22.6) apply.
- (2) the proceeds of the taxation from time $t + 1$ are redistributed as a lump-sum transfer to all agents, denoted by $T(t + 1) \geq 0$.

Let us focus on the MPE of this game. At each stage voting is over the tax rate that will apply in the next period only (with the lump-sum transfer determined from the budget constraint). Moreover, given the linear preferences, each individual takes future taxes as given (independent of current tax decisions and the current capital stock) and only cares about the current tax rate when making its current decisions. Thus, despite the fact that the economy involves an infinite sequence of taxes, the MVT can be applied to the tax decision at each date, provided the other conditions of the theorem are satisfied. I next show that this is the case.

Let us define $k_i(t) \equiv K_i(t)/A_i$ as the effective capital-labor ratio (the ratio of capital to “effective labor”) of entrepreneur i (bearing in mind that its employment level is equal to 1) and recall that p^t determines the sequence of taxes starting from time t . With this definition, we can write the value of each entrepreneur recursively as

$$(22.53) \quad V_i(k_i(t) | p^t) = \max_{k_i(t+1) \geq 0} \{(1 - \tau(t)) A_i f(k_i(t)) - A_i k_i(t + 1) + T(t) + \beta V_i(k_i(t + 1) | p^{t+1})\},$$

where the fact that total output is equal to $A_i f(k_i(t))$ at time t follows from the constant returns to scale property of F (Assumption 1) and the total amount of capital invested is, by definition, $K_i(t + 1) = A_i k_i(t + 1)$.

Applying the same type of reasoning as in Section 22.2 to the maximization problem in (22.53), we obtain that the capital-labor ratio by entrepreneur i at time t will satisfy the following first-order condition (see Exercise 22.32):

$$(22.54) \quad \beta(1 - \tau(t + 1)) f'(k_i(t + 1)) = 1.$$

The noteworthy feature is that the choice of the effective capital-labor ratio $k_i(t+1)$ is independent of A_i . This intuitive result implies that all entrepreneurs will choose the same effective capital-labor ratio regardless of their exact productivity. This is stated in the next proposition:

PROPOSITION 22.20. *Let the tax rate announced for date $t+1$ be τ . Then, in any MPE, each entrepreneur $i \in [0, 1]$ chooses the effective capital labor ratio $\hat{k}(\tau)$ for date $t+1$ given by*

$$(22.55) \quad \hat{k}(\tau) = (f')^{-1} \left((\beta(1-\tau))^{-1} \right),$$

where $(f')^{-1}(\cdot)$ denotes the inverse of the marginal product of capital.

Now given the result in Proposition 22.20, we can calculate total tax revenues, and thus the lump-sum transfer from the government budget constraint, at time $t+1$ as

$$(22.56) \quad \begin{aligned} T(t+1) &= \int_0^1 \tau(t+1) A_i f(\hat{k}(\tau(t+1))) di \\ &= \tau(t+1) \bar{A} f(\hat{k}(\tau(t+1))), \end{aligned}$$

where $\bar{A} \equiv \int_0^1 A_i di$ is the mean productivity among the entrepreneurs and $\hat{k}(\cdot)$ is given by (22.55). The first line simply uses the definition of total tax revenue (and per capita lump-sum transfer) as the sum (integral) of output over all entrepreneurs and uses the fact that all entrepreneurs will choose the effective capital-labor ratio, $\hat{k}(\tau(t+1))$. The second line takes the terms that do not depend on the identity of the entrepreneur out of the integral and uses the definition of mean productivity $\bar{A}$.

Let us next determine the political bliss point of each entrepreneur, that is, their most preferred tax rate. To do this, let us write their continuation utility from the end of period t . Ignoring all the terms that are bygone by this point and substituting for best responses (that is, for the effective capital-labor ratio from (22.55)), the expected discounted utility of entrepreneur i from (22.53) can be written as

$$(22.57) \quad \tilde{V}_i(\tau' | p^{t+1}) = -A_i \hat{k}(\tau') + \beta \left[(1-\tau') A_i f(\hat{k}(\tau')) + \tau' \bar{A} f(\hat{k}(\tau')) + \tilde{V}_i(p^{t+2}) \right],$$

where τ' denotes the tax rate announced for date $t+1$ and I have used the notation $\tilde{V}_i$ to distinguish this value function defined over the current tax rate from the value function V_i defined in (22.53). In addition, $\tilde{V}_i(p^{t+1})$ is defined as the continuation value from the end of date $t+1$ onwards and I have substituted for the transfer $T(t+1)$ from (22.56).

The most preferred tax rate for entrepreneur i can be obtained from the expression for $\tilde{V}_i(\tau' | p^{t+1})$. However, it can be verified easily that $\tilde{V}_i(\tau' | p^{t+1})$ is not necessarily quasi-concave in τ' , thus preferences are not single peaked (see Exercise 22.33). However:

PROPOSITION 22.21. *Preferences given by $\tilde{V}_i(\tau' | p^{t+1})$ in (22.57) over the policy menu $\tau' \in [0, 1]$ satisfy the single crossing property in Definition 22.3.*

PROOF. See Exercise 22.33. □

In view of Proposition 22.21, we can apply Theorems 22.5 and 22.6, and conclude that at each date, the tax rate most preferred by the entrepreneur with the median productivity will be implemented. Let this median productivity be denoted by A^m . From (22.57), this most preferred tax rate satisfies the following first-order condition:

$$(22.58) \quad (\bar{A} - A^m) f(\hat{k}(\tau')) + \tau' \bar{A} \frac{\left(f'(\hat{k}(\tau'))\right)^2}{(1 - \tau') f''(\hat{k}(\tau'))} \leq 0 \text{ and } \tau' \geq 0,$$

with complementary slackness. In writing this expression, I made use of condition (22.55) to simplify the expression and also differentiated (22.55) to express the derivative of $\hat{k}'(\tau')$, as

$$\hat{k}'(\tau') = \frac{f'(\hat{k}(\tau'))}{(1 - \tau') f''(\hat{k}(\tau'))}.$$

This derivative is strictly negative since $f'' < 0$. Therefore, as in Section 22.2, higher taxes lead to lower capital-labor ratios and lower output (higher distortions). The emphasis on complementary slackness in (22.58) is important here, since the most preferred tax rate of the median voter (entrepreneur) may not satisfy the first-order condition as equality, instead corresponding to a corner solution of $\tau' = 0$. The next proposition shows that this is in fact relevant for a range of distributions of productivity among the entrepreneurs.

PROPOSITION 22.22. *Consider the above-described model. Then, there exists $\tau^m \in [0, 1)$ such that the unique MPE involves $\tau(t) = \tau^m$ for all t . Moreover,:*

- *if the distribution of productivity among the entrepreneurs, $\mu(A)$, is such that $A^m \geq \bar{A}$, then $\tau^m = 0$;*
- *if $A^m < \bar{A}$, then $\tau^m > 0$;*
- *suppose that $A^m < \bar{A}$. Then, for given $\bar{A}$, τ^m is strictly decreasing in A^m .*

PROOF. The argument preceding the proposition combined with Theorems 22.5 and 22.6 shows that the tax rate most preferred by the entrepreneur with the median productivity will be chosen at every period. Moreover, clearly $\tau' = 1$ cannot be preferred by any entrepreneur, since it would lead to zero output by each entrepreneur and thus to zero tax revenues (Exercise 22.1), thus the result that there exists $\tau^m \in [0, 1)$ such that $\tau(t) = \tau^m$ for all t follows, with τ^m a solution to (22.58). Note that this equation might have more than one solution, and if so, τ^m corresponds to the global maximizer of (22.57) with productivity evaluated at A^m .

Next suppose that $A^m = \bar{A}$, then the first expression in (22.58) is equal to zero, and the left-hand side of the equation is unambiguously negative for any $\tau' > 0$ and exactly equal to zero for $\tau' = 0$. This establishes that in this case $\tau^m = 0$. If, on the other hand, $A^m > \bar{A}$, then the first expression is strictly negative and the left-hand side of (22.58) is unambiguously negative and the conclusion that $\tau^m = 0$ follows from the complementary slackness conditions.

Finally, suppose that $A^m < \bar{A}$. In this case, the first expression is strictly positive. Suppose, to obtain a contradiction, that $\tau^m = 0$. Then, the second term is exactly equal to 0

(since τ' is in the numerator). Consequently, the left-hand side of (22.58) is strictly positive and $\tau^m = 0$ cannot be a solution. Hence the unique equilibrium tax rate must be $\tau^m > 0$. To obtain the comparative static result, simply apply the Implicit Function Theorem to (22.58) and use the fact that since τ^m is a global maximum, the derivative of (22.58) with respect to τ^m is negative. $\square$

There are a number of important results in this proposition. First, it shows that linear preferences guarantee the existence of a well-defined MPE even when there is heterogeneity among the individuals in terms of their productivity. The important role played by linear preferences in this result cannot be overstated (recall Section 22.5). As discussed at the end of this chapter, there are a number of results in the literature similar to this proposition, but they do not correspond to well-defined MPE because they do not feature linear preferences (instead, they would be equilibria under the assumption that, even though the game is dynamic, individuals vote once at the beginning of time and have no option to change taxes thereafter).

Second, this proposition shows that if the productivity of the median voter is above the average, there will be no redistributive taxation. This is intuitive. As the first term in (22.58) makes it clear, the benefits of taxation are proportional to the average productivity in the economy, while the cost (to the median voter) is related to his productivity. If the median entrepreneur is more productive than the average, there are two forces making him oppose redistributive taxation; he is effectively redistributing away from himself, and there is also the distortionary effect of taxation captured by the second term in (22.58).

Third and more important, in the case in which the productivity of the median voter is below the average, the political equilibrium will involve positive (distortionary) taxation on all entrepreneurs. To obtain the intuition for this result, recall that tax revenues are equal to zero at $\tau = 0$. A small increase in taxes starting at $\tau = 0$ has a second-order loss for each entrepreneur and when $A^m < \bar{A}$, a first-order redistributive gain for the median voter. This result is important in part because most real-world wealth and income distributions appear to be skewed to the left (with the median lower than the mean), thus this configuration is more likely in practice. Furthermore, this result is most interesting in comparison with those in Section 22.2, which also involved positive distortionary taxation, but in an environment in which a non-productive elite was in power. Proposition 22.22 shows that the same *qualitative* result generalizes to the case in which there is democratic politics and the decisive (median) voter is a productive entrepreneur himself, but is less productive than average. This implies that the essence of the results obtained in our analysis of elite-dominated politics applies more generally.

Finally, Proposition 22.22 gives a new comparative static result. It shows that, holding average productivity constant, a decline in the productivity of the median entrepreneur (voter) leads to an increase in the amount of distortionary taxes. Since as in Section 22.2, higher taxes correspond to lower output and the larger gap between the mean and the median of the productivity distribution can be viewed as a measure of inequality, this result suggests

a political mechanism via which greater inequality may translate into higher distortions and lower output. Nevertheless, some care is necessary in interpreting this last result, since the gap between the mean and the median is *not* a (formally appropriate) measure of inequality. Exercise 22.34 gives an example in which a mean preserving spread of the distribution leads to a smaller gap between the mean and the median. This caveat notwithstanding, the literature interprets this last result as suggesting that greater inequality should lead to lower output and lower growth. Exercise 22.35 presents a version of the model here where taxes affect the equilibrium growth rate.

22.8. The Provision of Public Goods: Weak Versus Strong States

The analysis so far has emphasized the distortionary effects of taxation and expropriation. This paints a picture whereby the major determinants of poor economic performance are high taxes or some type of expropriation, and political economy does (or should) focus on the determination of the incentives for redistributive taxation. While the disincentive effects of taxation cannot be denied, whether taxes are high is only one of the dimensions of policy that might affect economic growth. For example, in many endogenous growth models appropriate R&D or industrial policy might encourage faster growth (even if it involves some taxation of capital and labor). More generally, public good provision, investment in infrastructure and provision of law and order are important roles of a government, and the failure of the government to perform these roles may have significantly negative consequences for economic performance. In fact, existing evidence does not support the view that growth (or high levels of output) are strongly associated with taxation. On the contrary, poor economies typically have lower levels of tax revenues and government spending. This is most stark if we compare the OECD to sub-Saharan Africa. Consequently, the political economy of growth must also pay attention to whether governments perform the roles that they are supposed to. The standard non-political-economy (for example the traditional public finance) approach to this question starts by positing the existence of a benevolent government and looks for policy combinations that would maximize social welfare. Once we incorporate political economy considerations, however, we must recognize that the political elite that control the government may not have an interest in investing in public goods. If public good investments (or investments in infrastructure or law and order) are an important determinant of the growth performance of an economy, it becomes essential to investigate under what circumstances governments will undertake such investments.

In this section, I present the simplest model that can shed light on this topic, which is based on Acemoglu (2007b). The economy consists of a political elite controlling the government and a set of citizens with access to production opportunities. Productivity depends on public good investments by the government. The government, on the other hand, will only undertake these investments, if these are beneficial to the political elite. In this section, I investigate the conditions under which a greater amount of investment in public good will be undertaken in a well-defined political equilibrium. Not surprisingly, the extent of public good

provision will depend on the future returns that the political elite can secure by undertaking such investments. This is related to the issue of weak versus strong states. If the state is very weak, the elite will be unable to raise taxes in the future and reap the benefits of their investments. Anticipating this, they will be unwilling to invest in public goods. On the other hand, if there are no checks on the ability of the elite to impose taxes on the population, then the state is too “strong,” and private investment will be stifled. Thus states that have intermediate levels of strength are most conducive to economic growth.

This model will also enable us to discuss the potential distinction between expropriation and taxes, an issue raised at the beginning of this chapter.

22.8.1. The Model. All agents have again linear preferences given in (22.1). The population consists of a set of yeoman-entrepreneurs (citizens), with mass normalized to 1, and a political elite. The elite do not engage in production but control the government. In particular, they decide the levels of taxation and public good provision. Without loss of any generality, I also normalize the size of the political elite to 1.

Each citizen i has access to the following Cobb-Douglas production technology to produce the unique final good in this economy:

$$(22.59) \quad Y_i(t) = \frac{1}{\alpha} K_i(t)^\alpha (A(t) L_i(t))^{1-\alpha},$$

which only differs from (22.17) because here $A(t)$ is time-varying. $A(t)$ will be determined by the public good investments of the government. Given the assumption that citizens correspond to yeoman-entrepreneurs, $L_i(t) = 1$ for all $i \in [0, 1]$ and for all t .

The timing of events is similar to the baseline model with holdup, in that taxes on output are set at time t , whereas capital investments for time t are decided at $t - 1$. There is again a maximum tax rate $\bar{\tau}$. However, instead of the constitutional limits on taxation, here I suppose that this maximum tax rate arises from the possibility that producers might hide their output (or move to the informal sector) if they face very high taxes. For example, if they do so, they lose a fraction $\bar{\tau}$ of their output, so that with a tax rate above $\bar{\tau}$, all producers would prefer to move to the informal sector and tax revenues would be equal to zero. Consequently, the tax rate at any point in time has to be $\tau(t) \in [0, \bar{\tau}]$. With this interpretation, $\bar{\tau}$ corresponds to the (economic) *strength of the state*. When $\bar{\tau}$ is high, we have a strong state, which can raise high taxes. When it is low, the state is unable to raise high taxes.

Given a tax rate $\tau(t) \in [0, \bar{\tau}]$, tax revenues are

$$(22.60) \quad \text{Tax}(t) = \tau(t) \int_0^1 Y_i(t) di = \tau(t) Y(t),$$

where $Y(t)$ is total output. Naturally, if the tax rate is above $\bar{\tau}$, tax revenues are equal to zero, because all production shifts to the informal economy.

The government (the political elite) at time t decides how much to spend on public goods for the next date, $A(t+1)$. I assume that

$$(22.61) \quad A(t) = \left[\frac{\alpha\phi}{1-\alpha} G(t) \right]^{1/\phi}$$

where $G(t)$ denotes government spending on public goods, and $\phi > 1$, so that there are diminishing returns in the investment technology for public goods (a greater ϕ corresponds to greater decreasing returns). The term $[\alpha\phi/(1-\alpha)]^{1/\phi}$ is included as a convenient normalization. In addition, (22.61) implies full depreciation of $A(t)$, which simplifies the analysis below. The consumption of the elite is given by whatever is left over from tax revenues after expenditure and transfers, thus is equal to $C^E(t) = \text{Tax}(t) - G(t)$.

Let us first characterize the first-best level of public good provision, where there are no distortionary taxes on entrepreneurs and the level of public good provision is chosen to maximize the net present discounted value of a representative entrepreneur. Given the production function and the timing of events here, which corresponds to that of the canonical elite-dominated model with Cobb-Douglas technology, the equilibrium capital-labor ratio of an entrepreneur is the same as in (22.18) above. Consequently, the first-best level of public good investment can be computed as

$$(22.62) \quad A(t) = A^{fb} \equiv \beta^{1/(\phi-1)(1-\alpha)}$$

and the first-best levels of the capital-labor ratio and output are

$$k^{fb} \equiv \beta^{\phi/(\phi-1)(1-\alpha)} \text{ and } Y^{fb} \equiv \frac{1}{\alpha} \beta^{(\phi\alpha+1-\alpha)/(\phi-1)(1-\alpha)}.$$

Let us again focus on the MPE of this game. As usual, a MPE is defined as a set of strategies at each date t , such that these strategies only depend on the current (payoff-relevant) state of the economy, $A(t)$, and on prior actions within the same date according to the timing of events above. Thus, a MPE can be represented by $(\tau(A(t)), [k_i(A(t))]_{i \in [0,1]}, G(A(t)))$, where, by definition of a MPE, the key actions, which consist of the tax rate on output, τ , the capital-labor ratio decision of each entrepreneur $[k_i]_{i \in [0,1]}$, and the government expenditure on public good, G , are conditioned on the current payoff-relevant state variable, $A(t)$. Clearly, since each yeoman-entrepreneur employs only himself, the capital-labor ratio, k_i , and the total capital stock, K_i , of each entrepreneur are identical.

It is clear that in any MPE, the unique equilibrium tax rate for the political elite will be

$$(22.63) \quad \tau(t) = \bar{\tau} \text{ for all } t,$$

since investment decisions are already sunk at the time the elite set the taxes.

Next, the capital-labor ratio of entrepreneurs is again given by (22.18), and thus can be written as

$$(22.64) \quad k_i(t) = (\beta(1-\bar{\tau}))^{1/(1-\alpha)} A(t) \text{ for all } i \in [0,1] \text{ and for all } t.$$

Combining this expression with (22.59) and (22.60), we obtain equilibrium tax revenue as a function of the level of public goods as:

$$(22.65) \quad T(A(t)) = \frac{(\beta(1-\bar{\tau}))^{\alpha/(1-\alpha)} \bar{\tau} A(t)}{\alpha}.$$

Finally, the elite will choose public investment, $G(t)$ to maximize his consumption. To characterize this, let us write the discounted net present value of the elite as

$$(22.66) \quad V^e(A(t)) = \max_{A(t+1)} \left\{ T(A(t)) - \frac{1-\alpha}{\alpha\phi} A(t+1)^\phi + \beta V^e(A(t+1)) \right\},$$

which simply follows from writing the discounted payoff of the elite recursively, after substituting for their consumption, $C^E(t)$, as equal to taxes given by (22.65) minus their spending on public goods from eq. (22.61).

Since, for $\phi > 1$, the instantaneous payoff of the elite is bounded, continuously differentiable and concave in A , so Theorems 6.3, 6.4 and 6.6 in Chapter 6 imply that the value function $V^e(\cdot)$ is concave and differentiable. Hence, the first-order condition of the ruler in choosing $A(t+1)$ can be written as:

$$(22.67) \quad \frac{1-\alpha}{\alpha} A(t+1)^{\phi-1} = \beta (V^e)'(A(t+1)),$$

where $(V^e)'$ denotes the derivative of the value function of the elite. This equation links the marginal cost of greater investment in public goods to the greater value that will follow from this. To make further progress, I use the standard Envelope condition, which is obtained by differentiating (22.66) with respect to $A(t)$:

$$(22.68) \quad (V^e)'(A(t+1)) = T'(A(t)) = \frac{(\beta(1-\bar{\tau}))^{\alpha/(1-\alpha)} \bar{\tau}}{\alpha}.$$

The value of greater public goods for the elite is the additional tax revenue that this will generate, which is given by the expression in (22.68).

Combining these conditions, we obtain the unique MPE choice of the elite as:

$$(22.69) \quad A(t+1) = A[\bar{\tau}] \equiv \left(\beta^{1/(1-\alpha)} (1-\alpha)^{-1} (1-\bar{\tau})^{\alpha/(1-\alpha)} \bar{\tau} \right)^{\frac{1}{\phi-1}},$$

which also defines $A[\bar{\tau}]$ as an expression that will be useful below. Substituting (22.69) into (22.66) yields a simple form of the elite's value function:

$$(22.70) \quad V^e(A(t)) = \frac{(\beta(1-\bar{\tau}))^{\alpha/(1-\alpha)} \bar{\tau} A(t)}{\alpha} + \frac{\beta^{1/(1-\alpha)} (\phi-1) (1-\bar{\tau})^{\alpha/(1-\alpha)} \bar{\tau}}{(1-\beta)\phi\alpha} A[\bar{\tau}].$$

The second term in (22.70) follows since the level of public good spending implied by (22.69) is equal to a fraction $1/\phi$ of tax revenue. The value of the elite naturally depends on the current state of public goods, $A(t)$, inherited from the previous period, and from this point on, the equilibrium involves investment levels given by (22.64) and (22.69).

PROPOSITION 22.23. *In the above-described economy, there exists a unique MPE where $\tau(A) = \bar{\tau}$ for all A , $A(t)$ is given by $A[\bar{\tau}]$ as in (22.69) for all $t > 0$, and, the capital-labor*

ratio of each entrepreneur $i \in [0, 1]$ and for all t is given by (22.64). For all $t > 0$, the equilibrium level of aggregate output is:

$$(22.71) \quad Y(t) = Y[\bar{\tau}] \equiv \frac{1}{\alpha} (\beta(1 - \bar{\tau}))^{\alpha/(1-\alpha)} A[\bar{\tau}].$$

PROOF. See Exercise 22.36. □

Because of linear preferences and full depreciation of public goods, the economy reaches the steady-state level of output in one period.

22.8.2. Weak Versus Strong States. The first result implied by Proposition 22.23 is the importance of the strength of the state as parameterized by $\bar{\tau}$, its ability to raise revenues. When $\bar{\tau}$ is high, the state is “economically powerful”—citizens have little recourse against high rates of taxes. In contrast, when $\bar{\tau}$ is low, the state is “economically weak” (and there is “limited government”), since it is unable to raise taxes. With this interpretation, we can now ask whether greater economic strength of the state leads to worse economic outcomes. The answer is ambiguous, as it can be seen from the fact that when $\bar{\tau} = 0$, that is, when the state is extremely weak, the elite will choose $G(t) = 0$, while with $\bar{\tau} = 1$, the citizens will choose zero investments. In both cases, output will be equal to zero.

It is straightforward to determine the level of $\bar{\tau}$ that maximizes output in the society at all dates after the initial one— $Y(t)$ for $t > 0$. It is the solution to $\max_{\bar{\tau}} Y[\bar{\tau}]$, where $Y[\bar{\tau}]$ is given by (22.71). Exercise 22.36 shows that the solution to this program, denoted $\bar{\tau}^*$, is

$$(22.72) \quad \bar{\tau}^* \equiv \frac{1 - \alpha}{1 - \alpha + \alpha\phi}.$$

If the economic power of the state is greater than $\bar{\tau}^*$, then the state is too powerful, and taxes are too high relative to the output-maximizing benchmark. This corresponds to the standard case on which most political economy models focus. In contrast, if the economic power of the state is less than $\bar{\tau}^*$, then the state is not powerful enough for there to be sufficient rents in the future to entice the elite to invest in public goods. This corresponds to the case, where the state is not too powerful, but too weak. Consequently, the elite do not have the correct incentives to invest in productivity-enhancing public goods. Therefore, we have another example of non-growth-enhancing institutions/policies, but this time resulting from the weakness of the state.

There is an interesting parallel to the theory of the firm here. In the theory of the firm, the optimal structure of ownership and control gives ex post bargaining power to the parties that have more important investments. The same principle applies to the allocation of economic strength as captured by the parameter $\bar{\tau}$; greater power for citizens is beneficial when their investments matter more. When it is the state’s investment that is more important for economic development, a higher $\bar{\tau}$ is required (justified).

The above discussion focused on the output-maximizing value of the parameter $\bar{\tau}$. Equally relevant is the level of $\bar{\tau}$, say $\bar{\tau}^e$, which maximizes the beginning-of-period payoff to the elite and $\bar{\tau}^c$, which maximizes the beginning-of-period payoff to the citizens. One might also define

a tax rate $\bar{\tau}^{wm}$ that maximizes the beginning-of-period payoff to a fictitious social planner weighing the elite and the citizens equally. The next proposition shows how these tax rates compare to the output-maximizing tax rate $\bar{\tau}^*$:

PROPOSITION 22.24. *Let $\bar{\tau}^*$, $\bar{\tau}^{wm}$, $\bar{\tau}^e$ and $\bar{\tau}^c$ be the values of $\bar{\tau}$ that respectively maximize output, social welfare, the elite's utility and citizens' utility for all $t > 0$. Then*

$$0 < \bar{\tau}^c < \bar{\tau}^* < \bar{\tau}^e < 1 \text{ and } 0 < \bar{\tau}^c < \bar{\tau}^{wm} < \bar{\tau}^e < 1.$$

PROOF. See Exercise 22.36. □

The main conclusion from this analysis is that when both the state and the citizens make productive investments, it is no longer true that limiting the rents that accrue to the state is always good for economic performance. Instead, there needs to be a certain degree of *balance of powers* between the state and the citizens. When the political elite controlling the power of the state expect too few rents in the future, they have no incentive to invest in public goods. Consequently, excessively weak states are likely to be as disastrous for economic development as the unchecked power and expropriation by excessively strong states.

A number of shortcomings of the analysis in this section should be noted at this point. The first is that it relied on economic exit options of the citizens in the informal sector as the source of their control over the state, whereas, in practice, political controls may be more important. The second is that it focused on the MPE, without any possibility of an implicit agreement between the state and the citizens, whereby the state could raise sufficient tax revenue both to finance public good investments and to redistribute some of it to the elite (or to politicians). Such an allocation, which could be referred to as a *consensually strong state*, would be an equilibrium if the citizens are willing to tolerate relatively high levels of taxation because of the benefits that they receive from public good provision and the elite prefer not to deviate to higher levels of taxes. In Acemoglu (2005), I generalize the results presented here in these directions. I show that similar results can be obtained when the constraints on the power of the state are not economic, but political. In particular, we can envisage a situation in which citizens can (stochastically) replace the government if taxes are too high. In this case, when citizens are politically powerful, this will limit the extent of taxation, but also the amount of public good provision. In addition, using a model with variable political checks on the state, one can analyze the SPE, where there might be an implicit agreement between the state and the citizens to allow for some amount of taxation and also correspondingly high levels of public good provision. In Acemoglu (2005), I referred to this equilibrium configuration as a “consensually-strong state,” since the citizens allow the economic power of the state to be high (partly because they believe they can control the state and political elites by elections or other means). The configuration with the consensually-strong state might provide an insight into why many OECD countries have higher tax rates and higher levels of public good provision than many less-developed economies.

This perspective also suggests a useful distinction between taxation and expropriation. High taxes appear to have similar effects on investment and economic performance as expropriation. One difference between expropriation and taxes might be uncertainty. It can be argued that producers know exactly at what rate they will be taxed, while expropriation is riskier. In the presence of risk aversion, expropriation could be considerably more costly than taxation. However, the analysis here suggests another useful distinction, which comes not from the revenue side, but from the expenditure side. Expropriation might correspond to the government taking a share of the output of the producers for its own consumption, while in an equilibrium with it consensually strong state, taxation, along-the-equilibrium path, involves some of the revenues being spent for public goods, which are useful for the producers. If this distinction is important, one of the reasons why taxation is viewed as fundamentally different from expropriation may be because taxation is often associated with some of the proceeds being given back to the citizens in the form of public goods.

Perhaps the most important implication of the analysis in this section is to emphasize different aspects of growth-enhancing institutions. Economic growth not only requires secure property rights and low taxes, but also complementary investments, often most efficiently undertaken by the government. Provision of law and order, investment in infrastructure and public goods are obvious examples. Thus growth-promoting institutions need to provide some degree of security of property rights to individuals, but also incentivize the government to undertake the appropriate public good investments. In this light, excessively weak governments might be as very costly to economic performance as the unchecked power of the government.

22.9. Taking Stock

This chapter made a first attempt at investigating why institutions and policies differ across societies. Even though this question may be viewed as part of the study of political economy rather than economic growth per se, in the absence of satisfactory answers to this question, our understanding of the process of economic growth will be limited. The evidence provided in Chapter 4 suggested that societies often choose different institutions and policies, with very different implications for economic growth. Thus to understand why some countries are poor and some are rich, we need to understand why some countries choose growth-enhancing policies while others choose policies that block economic development.

This chapter emphasized a number of key themes in developing answers to these questions. First, the sources of different institutions (and non-growth-enhancing institutions) must be sought in *social conflict* among different individuals and groups in the society. Social conflict implies that there is no guarantee that the society will adopt economic institutions and policies that will encourage economic growth. Such social arrangements will benefit many individuals in the society, but they will also create losers, groups whose rents will be destroyed or eroded by the introduction of new technologies or by the process of economic growth. When individuals in the society have conflicting preferences over institutions and policies, the distribution of *political power* in the society plays an important role in determining which

institutions and policies will be chosen (and whether non-growth-enhancing institutions will be reformed).

I emphasized that non-growth-enhancing policies can emerge even without any significant Pareto inefficiencies. In particular, a range of political economy models lead to equilibrium allocations that would also result from the maximization of a weighted social welfare function. The resulting equilibrium allocation will naturally be constrained Pareto efficient. However, Pareto efficiency does not guarantee high output or growth. I illustrated this first by focusing on a “simple society,” where individuals belong to a social group, the conflict of interest is among social groups, and all political power rests in the hands of a political elite. I showed that this environment, combined with linear preferences, implies that even the restrictive MPE concept leads to constrained Pareto efficient allocations. Despite their Pareto efficiency, equilibrium allocations may involve significant “distortions” (suggesting as a byproduct that Pareto efficiency may not be the right concept to focus on the analysis of the political economy of growth). In particular, there are two distinct sets of reasons for distortionary policies, which will discourage investment and economic growth. The first is *revenue extraction*, while the second results from competition between the elite and other social groups in the marketplace or in the political arena, and can take the form of *factor price manipulation effects* or *political replacement effects*. Revenue extraction involves the use of distortionary taxes by the elite (because those are the only fiscal instruments available to them) in order to extract revenues from entrepreneurs and workers in the society. Factor price manipulation results when the elite use policies in order to reduce the factor of other social groups, so that they face more favorable factor prices. Finally, the political replacement effect emerges when the elite try to impoverish social groups that might politically compete with themselves. The analysis demonstrated that revenue extraction, though distortionary, is typically much less harmful to economic growth than factor price manipulation and political replacement effects, because ultimately the elite can only raise revenues if the groups that are being taxed have the incentives (and the ability) to undertake investments and produce. In contrast, the factor price manipulation and the political replacement effects encourage the elite to pursue policies that harm groups that they perceive as their competitors. This typically leads to higher taxes and also to explicit actions to block technology adoption or other productivity-enhancing investments by competing entrepreneurs. The consequences of these types of policies for economic growth can be disastrous. Remarkably, however, all of this can happen while the equilibrium is still constrained Pareto efficient—it is so, because all weight is given to the elite without any regard to the welfare of the other individuals and groups in the society.

In addition to providing a simple and useful framework for the analysis of policy, the framework with political power vested in the elite also leads to a range of comparative static results that shed light on what types of societies will adopt policies that encourage growth and which societies are likely to pursue non-growth-enhancing policies or even try to block economic development. The following are some of the main comparative static results: (1) taxes are likely to be higher when the demand for capital by entrepreneurs is inelastic, because

in this case the revenue-maximizing tax rate for the elite is higher; (2) taxes are likely to be higher when the factor price manipulation effect is more important relative to the revenue extraction effect; (3) taxes are higher when the political power of the elite is contested and reducing the income level of the competing groups will lead to political consolidation for the elite; (4) in the absence of the political replacement effect, greater state capacity leads to lower taxes; (5) when the political replacement effect is important, both greater state capacity and greater rents from natural resources lead to more distortionary policies because they increase the *political stakes* (the value of holding on to political power).

Pareto efficiency is not a general property of political economy models, however. In particular, once the timing of policies is changed so that taxes by the politically powerful are set after entrepreneurial investments or when long-term investments are important, serious holdup problems emerge. At the simple level, the holdup problem corresponds to a situation in which the politically powerful can not commit to not taxing investments once they have been undertaken. Anticipating these high taxes, entrepreneurs refrain from investment. In such an environment, MPE leads to constrained Pareto inefficient equilibria. SPE then sometimes significantly improve over the MPE. Whether the MPE or the SPE is the appropriate equilibrium concept depends on institutional and historical details that determine whether individuals and groups can coordinate their actions in equilibrium. In any case, we also saw that even with SPE the equilibrium might involve a lack of commitment to a desirable tax sequence by the elite. In this case, if feasible, the elite may introduce economic institutions providing greater security of property rights to entrepreneurs so as to encourage investments (and thus increase tax revenues for themselves). Thus the possibility of commitment problems provides us with one perspective for thinking about the emergence of secure property rights.

To move beyond models in which political power rests with a pre-specified group, here the political elite, we need systematic ways of aggregating heterogeneous political preferences. After reviewing some basic political economy theory, I used the well-known Median Voter Theorem (MVT), which applies in certain economic environments, to investigate the determination of equilibrium policies in an economy with heterogeneous entrepreneurs. One of the most interesting results of this analysis is that the revenue extraction mechanism emphasized in the context of elite-dominated politics is also present in more complex societies with heterogeneity among entrepreneurs. In particular, if the median voter is poorer than the average individual (entrepreneur) in the society, he may want to use distortionary policies to transfer resources to himself. This type of distortionary revenue extraction by the median voter is qualitatively similar to the use of distortionary policies by the elite to extract revenues from middle-class entrepreneurs. Nevertheless, this effect exhibits itself in a more general environment with heterogeneity among the entrepreneurs and also leads to a new comparative static result; when the gap between the mean and the median of the productivity distribution is greater, the incentives to extract revenues are stronger and policies are more likely to be distortionary.

Finally, I emphasized that taxation is not the only relevant policy affecting economic growth. The provision of public goods, in the form of securing law and order, investments in infrastructure or even appropriate subsidies, might also be important for inducing a high rate of economic growth. Will the state provide the appropriate amount and type of public goods? In the context of a political economy model, the answer to this question depends on whether the politically powerful groups controlling the state have the incentives to do so. As already discussed above, the elite may want to block economic development in order to affect the factor prices that it faces or to secure their political position. Beyond this, the elite would only invest in public goods if they expect to reap the benefits of these investments in the future. This raises the issue of weak versus strong states. While an emphasis on taxes suggests that checks on the economic or political power of the state should be conducive to more growth-enhancing policies, weak states will be unwilling to invest in public goods because those controlling the state realize that they will not be able to tax future revenues created by these public good investments. Consequently, an intermediate strength of the state might be most conducive to growth-enhancing policies. The more important point here is that an analysis of the effect of economic institutions and policies on growth should take into account both the incentives for private agents and also the incentives to the government for providing the appropriate amount and type of public goods.

The material in this chapter is no more than an introduction to the exciting and important field of political economy of growth. Many issues have not been addressed. Among those omitted, the following appear most important: first, in addition to taxes, expropriation and public goods, whether the society provides a level playing field to a broad cross-section of society is important. For example, broad-based human capital investments, which may be quite important for modern economic growth, require the provision of appropriate incentives not only to a few businesses, but to the entire population. Similarly, security of property rights for existing businesses have to be balanced against the ease of entry for new firms. Second, the entire analysis here took the distribution of political power in the society, and the political institutions that generated this distribution of political power, as given. It is clear, however, that different distributions of power in the society will lead to different policies and thus to different growth trajectories. Consequently, it seems important to understand how the distribution of political power and equilibrium political institutions might evolve endogenously and whether it might interact with the economic equilibrium. Some of these issues will be discussed in the next chapter.

22.10. References and Literature

The material in this chapter draws on the large political economy literature and also on some of the recent work on the political economy of growth. My purpose has not been to provide a balanced survey of these literatures, but to emphasize the most important features pertaining to the sources of differences in economic institutions and policies across societies with the hope of shedding some light on differential cross-country growth performances. As

noted above, I focused throughout on the neoclassical growth model and its variants in order to isolate the contribution of political economy mechanisms and also to keep the exposition manageable.

Persson and Tabellini (2000) provides an excellent survey of much of the work done in political economy in the 1980s and 1990s, though does not focus on the political economy of growth. Drazen (2001) also provides an excellent introduction to this work, with slightly more emphasis on growth issues. Eggertsson (2005) provides a non-formal discussion of a wider set of political economy questions.

The material in Sections 22.2, 22.3 and 22.4 and the discussion of revenue extraction and factor price manipulation effects draw upon Acemoglu (2007a,b), but the setup has been modified to be more consistent with the neoclassical growth model. Versions of the factor price manipulation effect feature in Acemoglu (2008), which will be discussed in the next chapter, and also in Galor, Moav and Vollrath (2005), who emphasize how the land-owning elite may discourage investment in human capital. The political replacement effect is also discussed in Acemoglu (2007a,b), though it originates in Acemoglu and Robinson (2000b). A detailed analysis of why the political elite may block technological innovations in order to increase the likelihood of their survival is presented in Acemoglu and Robinson (2007a). That paper also shows how both relatively secure elites and elites that are in competitive political environments will not have incentives to block technological change, but those with intermediate levels of security that might be challenged by new technologies are likely to adopt policies that will block economic development. It also provides historical examples of this type of behavior. Models with competitive economic behavior by price-taking agents, but strategic political decisions were first developed by Chari and Kehoe (1990, 1993), though the focus in these papers is on the “time-consistency” of the behavior of a benevolent government.

The material in Section 22.5 builds upon the analysis of MPE in competitive economies with capital accumulation in Krusell and Rios-Rull (1997), Klein, Krusell and Rios-Rull (2004) and Hassler, Krusell, Storesletten and Zilibotti (2004). The first two papers compute the MPE numerically in a related environment, while the last one contains characterization results for an economy with quadratic preferences and linear technology. Acemoglu and Robinson (2000, 2001, 2007a) and Hassler, Rodriguez-Mora, Storesletten and Zilibotti (2003) provide explicit characterizations of MPE in simpler political environments.

The material in Section 22.6 is standard. An excellent introduction to social choice theory, with a thorough discussion of Arrow’s Theorem, is provided in Austen-Smith and Banks (2000). My proof of the theorem here builds on the somewhat longer proof in that book. Arrow (1951) is still the classic for the basis of social choice theory and for Arrow’s Theorem, though similar ideas were also developed in earlier work by Black (1948). The single crossing property is introduced in Roberts (1977) and further developed by Gans and Smart (1996). The notion of intermediate preferences introduced in Exercise 22.28 is due to Grandmont (1978). The Downsian model of political competition is introduced in Downs (1957), and builds heavily on Hotelling’s seminal (1929) paper. Austen-Smith and

Banks (2000) and Persson and Tabellini (2000) discuss the Downsian party competition model in detail. The probabilistic voting model is due to Lindbeck and Weibull (1987) and Coughlin (1992). Persson and Tabellini (2000) provide a detailed treatment of this model. My exposition here was simplified by the assumption that parties care about their vote share, not the probability of coming to power. There are many different lobbying models in the literature. The first one was formulated by Becker (1983). The one presented here builds on Grossman and Helpman (1994), which in turn builds on the menu auctions approach of Bernheim and Winston (1986). Grossman and Helpman (2001) provides a more detailed exposition of various different lobbying models.

Section 22.7 presents one of the most standard models of distributional conflict, which uses the celebrated Median Voter Theorem (which was presented in Section 22.6). The Median Voter Theorem was first applied to an economy with linear redistributive taxes by Roberts (1977) and Romer (1975). Meltzer and Richard (1981) used the Roberts-Romer model to relate inequality to taxes and more importantly, to draw implications about the extent of the voting franchise on the size of the government. Meltzer and Richard's work is a classic as it can be viewed as the beginning of positive political economy—the use of political economy models in order to explain cross-country and over-time differences in public policies. A number of authors have since applied the Roberts-Romer model in growth settings. The most notable examples are Alesina and Rodrik (1994), Persson and Tabellini (1994), Saint-Paul and Verdier (1996) and Benabou (2000). The models in Alesina and Rodrik (1994) and Persson and Tabellini (1994) are very similar to the one I developed in Section 22.7, except that they do not characterize a well-defined MPE. Instead, they assume either that voting is at the beginning of time and over a single tax rate that will apply at all future dates or that agents are myopic and do not take into account future votes (though they do take into account their own future economic decisions). In addition, these papers focus on an economy with endogenous growth, so that differences in taxes lead to differences in equilibrium growth rates (see Exercise 22.35). Both Alesina and Rodrik and Persson and Tabellini emphasize the negative effects of inequality on economic growth, interpreting the gap between the mean and the median as a measure of inequality. They also present cross-country evidence suggesting that inequality is negatively correlated with economic growth. This cross-country growth evidence is difficult to interpret, however, since there are many omitted variables in such growth regressions, and other researchers have found no relationship, and some have even found a positive relationship, between inequality and growth (see, for example, Barro, 2000, Banerjee and Duflo, 2003, and Forbes, 1996). Saint-Paul and Verdier (1996), on the other hand, showed that higher inequality can lead to greater growth, because tax revenues may be invested in human capital accumulation. Benabou's important (2000) paper pushes this idea further, and shows how a negative relationship between inequality and growth is consistent with higher inequality leading to less redistribution in a world in which greater redistribution may be growth-enhancing, again because taxes are partly invested in education. None of these papers characterize the MPE of a dynamic economy, instead assuming that voting is

either myopic or is done once at the beginning of time. The model in Section 22.7 is the only one I am aware that derives these results as a well-defined dynamic equilibrium in a simple neoclassical growth model with linear preferences.

Section 22.8 builds on Acemoglu (2005). The idea that weak states might be an important impediment to economic growth is popular among political scientists and political sociologists, and is most famously articulated in Migdal (1988), Tilly (1990), Wade (1990), Herbst (2000) and Evans (2000). These approaches typically do not incorporate the incentives of the politicians or the government in providing public goods or adopt growth-enhancing strategies. Acemoglu (2005) provides the first formal framework to analyze these issues, and the material in this section embeds the baseline model in that paper into a neoclassical growth model.

22.11. Exercises

EXERCISE 22.1. Prove that $\hat{\tau}$ given by (22.16) satisfies $\hat{\tau} \in (0, 1)$.

EXERCISE 22.2. Consider the model in Section 22.2, with the only difference that the production technology is as in the Romer (1986a) model studied in Chapter 11. In particular, recall that each entrepreneur now has access to the production function $Y_i(t) = F(K_i(t), A(t)L_i(t))$ and $A(t) = B \int_0^1 K_i(t) di = BK(t)$. Characterize the MPE in this case and show that distortionary taxes by the elite reduce the equilibrium growth rate of the economy.

EXERCISE 22.3. Consider the model in Section 22.2 and assume that policies are decided by the middle class. Show that the middle class might prefer positive taxation on themselves (with the proceeds redistributed to themselves as lump-sum transfers). Provide a precise intuition for why such taxation may make political-economic sense for middle-class entrepreneurs. Would the same result apply if the proceeds of taxation were redistributed as a lump-sum transfer to every individual in the society (including workers)?

EXERCISE 22.4. Prove Proposition 22.4. Check in particular that the maximization program of the elite is concave, so that when $\bar{\tau} < 1 - \alpha$, the utility-maximizing tax rate for the elite is $\bar{\tau}$.

EXERCISE 22.5. (1) Prove Proposition 22.5.

(2) Explain why Condition 22.2 is necessary in this proposition.

(3) What happens if Condition 22.2 is not satisfied?

EXERCISE 22.6. Prove Proposition 22.6.

EXERCISE 22.7. Consider a model in Section 22.3 and suppose that the middle class are in political power. Characterize the MPE in this case. Derive the discounted utility of the elite when the middle class are in control of politics, denoted by $V_e(M)$, and compare this to their utility when they are in control, $V_e(E)$. [Hint: see Section 23.2 in the next chapter].

EXERCISE 22.8. In the model with political replacement in subsection 22.3.2, suppose that $\eta'(\cdot) < 0$. Show that in this case the tax rate preferred by the elite is less than $1 - \alpha$ and that when the elite can block technology adoption, they will not choose to do so. Explain

the intuition for this result. What types of institutional structures might lead to $\eta'(\cdot) < 0$ as opposed to $\eta'(\cdot) > 0$.

EXERCISE 22.9. Prove Proposition 22.7.

EXERCISE 22.10. In the model with political replacement in subsection 22.3.2, show that the unique MPE is also the unique SPE.

EXERCISE 22.11. (1) Prove Proposition 22.9.

(2) Explain how Proposition 22.9 needs to be modified if $\bar{\tau} < 1$ and provide an analysis of the best stationary SPE in this case (where only stationary strategies are used).

EXERCISE 22.12. Prove Proposition 22.13.

EXERCISE 22.13. Prove Proposition 22.14.

EXERCISE 22.14. Prove Proposition 22.16.

EXERCISE 22.15. (1) Prove Proposition 22.17.

(2) Now suppose that in this proposition ϕ is not equal to 0. Provide an example in which in the MPE, the elite would still prefer $g = 0$.

(3) Now suppose that the elite can charge lump-sum taxes to middle-class entrepreneurs. Provide an example in which in the MPE, the elite would still prefer $g = 0$.

(4) In light of your answers to 2 and 3 above, explain why the political equilibrium might involve the use of inefficient fiscal instruments, even when more efficient alternatives exist.

EXERCISE 22.16. * Prove Proposition 22.18.

EXERCISE 22.17. * Prove Proposition 22.19.

EXERCISE 22.18. In this exercise, you are asked to provide some of the details omitted from Example 22.1 in Section 22.5.

(1) Derive the optimal strategy of entrepreneurs as given in (22.39). In particular, use the entrepreneurs' Euler equation, (22.35), together with their value function. Conjecture a decision rule of the form $k_i(t+1) = \kappa y_i(t)$ for entrepreneur i (where $y_i(t)$ is his output at time t). Verify that this guess is the unique solution for the entrepreneurs' maximization problem for any given sequence of taxes and that $\kappa = \alpha\beta$.

(2) Show that the main theorems from Chapter 6 can be applied to (22.40) and imply that W is strictly increasing, strictly concave and differentiable for $k > 0$.

(3) Show that the optimal strategy of the elite is given by (22.41). In particular, take the decision rule of entrepreneurs as given, and derive the Euler equation of the elite. Conjecture that $W(k) = \eta + \gamma \log(k)$ and using the Envelope condition, show that $\gamma = \alpha/(1 - \alpha\beta)$. Interpret this coefficient. Combining this with the Euler equation, derive (22.41).

EXERCISE 22.19. * Prove that if individual preferences are reflexive, complete and transitive, then they can be represented by a real-valued utility function.

EXERCISE 22.20. *

- (1) Consider a society with two individuals 1 and 2 and three choices, a, b and c . For the purposes of this exercise, only consider strict individual and social orderings (that is, no indifference allowed). Suppose that the preferences of the first agent are given by abc (short for $a \succ b \succ c$, that is, a strictly preferred to b , strictly preferred to c). Consider the six possible preference orderings of the second individual, i.e., $s_2 \in \{abc, acb, bac, \dots\}$, and so on. Define a social ordering as a mapping from the preferences of the second agent (given the preferences of the first) into a social ranking of the three outcomes, i.e., some function ϕ such that the social ranking is $s = \phi(s_2)$. Illustrate Arrow's Impossibility Theorem using this example [Hint: start as follows: $abc = \phi(abc)$, i.e., when the second agent's ordering is abc , the social ranking must be abc ; next, $\phi(acb) = abc$ or acb (why?); then if $\phi(acb) = abc$, we must also have $\phi(cab) = abc$ (why?); and proceeding this way to show that the social ordering is either dictatorial or it violates one of the axioms].
- (2) Now suppose we have the following aggregation rule: individual 1 will (sincerely) rank the three outcomes, his first choice will get 6 votes, the second 3 votes, the third 1 vote. Individual 2 will do the same, his first choice will get 8 votes, the second 4 votes, and the third 0 vote. The three choices are ranked according to the total number of votes. Which of the axioms of the Arrow's Theorem does this aggregation rule violate?
- (3) With the above voting rule, show that for a certain configuration of preferences, either agent has an incentive to distort his true ranking (that is, he has an incentive not to vote sincerely).
- (4) Now consider a society consisting of three individuals, with preferences given by:

$$\begin{array}{ll} 1 & a \succ b \succ c \\ 2 & c \succ a \succ b \\ 3 & b \succ c \succ a \end{array}$$

Consider a series of pairwise votes between the alternatives. Show that when agents vote sincerely, the resulting social ordering will be "intransitive". Relate this to the Theorem 22.1.

- (5) Show that if the preferences of the second agent are changed to $b \succ a \succ c$, the social ordering is no longer intransitive. Relate this to "single-peaked preferences".
- (6) Explain intuitively why single-peaked preferences are sufficient to ensure that there will not be intransitive social orderings. How does this relate to Theorem 22.1?

EXERCISE 22.21. * In the Condorcet paradox example provided in Section 22.6, show that other orderings of the choices a, b and c will also imply that the preferences of at least one of the three individuals is not single peaked.

EXERCISE 22.22. *

- (1) Consider the example of a three-person three-policy society with preferences

$$\begin{array}{ll} 1 & a \succ b \succ c \\ 2 & b \succ c \succ a \\ 3 & c \succ b \succ a \end{array}$$

Voting is dynamic: first, there is a vote between a and b . Then, the winner goes against c , and the winner of this contest is the social choice. Find the SPE voting strategy profiles in this two-stage game (recall that each player's strategy has to specify how they will vote in the first round, and how they will vote in the second round as a function of the outcome of the first round).

- (2) Suppose a generalization whereby the society $\mathcal{H}$ consists of H individuals and there are finite number of policies, $\mathcal{P} = \{p_1, p_2, \dots, p_M\}$. For simplicity, suppose that H is an odd number. Voting takes $M - 1$ stages. In the first stage, there is a vote between p_1 and p_2 . In the second stage, there is a vote between the winner of the first stage and p_3 , until we have a final vote against p_M . The winner of the final vote is the policy choice of the society. Prove that if preferences of all agents are single peaked (with a unique bliss point for each), then the unique SPE implements the bliss point of the median voter.

EXERCISE 22.23. * Prove Theorem 22.2 when H is even.

EXERCISE 22.24. * Characterize the subgame perfect equilibrium of the game in Example 22.3 under strategic voting by all players.

EXERCISE 22.25. * Modify and prove Theorem 22.4 without Assumption A4.

EXERCISE 22.26. * This exercise reviews Downsian party competition and then shows that Theorem 22.4 does not apply if there are three parties competing. In particular, consider Downsian party competition in a society consisting of a continuum 1 of individuals with single-peaked preferences. The policy space $\mathcal{P}$ is the $[0, 1]$ interval and assume that the bliss points of the individuals are uniformly distributed over this space.

- (1) To start with, suppose that there are two parties, A and B. They both would like to maximize the probability of coming to power. The game involves both parties simultaneously announcing $p^A \in [0, 1]$ and $p^B \in [0, 1]$, and then voters voting for one of the two parties. The platform of the party with most votes gets implemented. Determine the equilibrium of this game. How would the result be different if the parties maximized their vote share rather than the probability of coming to power?
- (2) Now assume that there are three parties, simultaneously announcing their policies $p^A \in [0, 1]$, $p^B \in [0, 1]$, and $p^C \in [0, 1]$, and the platform of the party with most votes is implemented. Assume that parties maximize the probability of coming to power. Characterize all pure strategy equilibria.
- (3) Now assume that the three parties maximize their vote shares. Prove that there exists no pure strategy equilibrium.

- (4) In 3, characterize the mixed strategy equilibrium. [Hint: assume the same symmetric probability distribution for two parties, and make sure that given these distributions, the third party is indifferent over all policies in the support of the distribution].

EXERCISE 22.27. * Prove Theorem 22.6.

EXERCISE 22.28. * This exercise involves generalizing the idea of single-crossing property used in Theorem 22.5 to multidimensional policy spaces. The appropriate notion of preferences of individuals turns out to be “intermediate preferences”. Let $\mathcal{P} \subset \mathbb{R}^K$ (where $K \in \mathbb{N}$) and policies p belong to $\mathcal{P}$. We say that voters have *intermediate preferences*, if their indirect utility function $U(p; \alpha_i)$ can be written as

$$U(p; \alpha_i) = J_1(p) + B(\alpha_i)J_2(p),$$

where $B(\alpha_i)$ is monotonic (monotonically increasing or monotonically decreasing) in α_i , and the functions $J_1(p)$ and $J_2(p)$ are common to all voters. Suppose that A2 holds and voters have intermediate preferences. We define the bliss point (vector) of individual i as in the text, as $p(\alpha_i) \in \mathcal{P}$ that maximizes individual i 's utility. Prove that when preferences are intermediate a Condorcet winner always exists and coincides with bliss point of the voter with the median value of α_i , that is, $p_m = p(\alpha_m)$.

EXERCISE 22.29. * Consider a society consisting of three individuals, 1, 2 and 3 and a resource of size 1. The three individuals vote over how to distribute the resource among themselves and each individual prefers more of the resource for himself and does not care about consumption by the other two. Since all of the resource will be distributed among the three individuals, we can represent the menu of policies as $\{(x_1, x_2) : x_1 \geq 0, x_2 \geq 0 \text{ and } x_1 + x_2 \leq 1\}$, where x_i denotes the share of the resource consumed by individual i . A policy vector (x_1, x_2) is accepted if it receives two votes.

- (1) Show that individual preferences over policy vectors do not satisfy single crossing or the conditions in Exercise 22.28.
- (2) Show that there does not exist a policy vector that it is a Condorcet winner.

EXERCISE 22.30. *

- (1) Show that in Theorem 22.7, a necessary condition for a pure strategy symmetric equilibrium, with $p_A = p_B = p^*$, to exist is that the matrix

$$\begin{aligned} & \sum_{g=1}^G \lambda^g h^g(0) D^2 U^g(p^*) \\ & + \sum_{g=1}^G \lambda^g \left| \frac{\partial h^g(0)}{\partial \sigma} \right| (DU^g(p^*)) \cdot (DU^g(p^*))^T \end{aligned}$$

is negative semidefinite, where $D^2 U^g$ denotes the Jacobian matrix of U^g . Explain why the condition takes the absolute values of $\partial h^g(0)/\partial \sigma$.

- (2) Derive a sufficient condition for such a symmetric equilibrium to exist. [Hint: distinguish between local and global maxima].

- (3) Show that without any assumptions on $U^g(\cdot)$'s (beyond concavity), the sufficient condition for a symmetric equilibrium can only be satisfied if all H^g 's are uniform.

EXERCISE 22.31. Consider the following one-period economy populated by a mass 1 of agents. A fraction λ of these agents are capitalists, each owning capital k . The remainder have only human capital, with human capital distribution $\mu(h)$. Output is produced in competitive markets, with aggregate production function

$$Y = K^{1-\alpha} H^\alpha,$$

where uppercase letters denote total supplies. Assume that factor markets are competitive and denote the market clearing rental price of capital by r and that of human capital by w .

- (1) Suppose that agents vote over a linear income tax, τ . Because of tax distortions, total tax revenue is

$$Tax = (\tau - v(\tau)) \left(\lambda r k + (1 - \lambda) w \int h d\mu(h) \right)$$

where $v(\tau)$ is strictly increasing and convex, with $v(0) = v'(0) = 0$ and $v'(1) = \infty$ (why are these conditions useful?). Tax revenues are redistributed lump sum. Find the ideal tax rate for each agent. Find conditions under which preferences are single peaked, and determine the equilibrium tax rate. How does the equilibrium tax rate change when k increases? How does it change when λ increases? Explain the intuition for these results.

- (2) Suppose now that agents vote over capital and labor income taxes, τ_k and τ_h , with corresponding costs $v(\tau_k)$ and $v(\tau_h)$, so that tax revenues are

$$Tax = (\tau_k - v(\tau_k)) \lambda r k + (\tau_h - v(\tau_h)) (1 - \lambda) w \int h d\mu(h)$$

Determine the most preferred tax rates for each agent. Suppose that $\lambda < 1/2$. Does a voting equilibrium exist? How does it change when λ increases? Explain why the results are different from the case with only one tax instrument?

- (3) In this model with two taxes, now suppose that agents first vote over the capital income tax, and then taking the capital income tax as given, they vote on the labor income tax. Does a voting equilibrium exist? Explain. If an equilibrium exists, how does the equilibrium tax rate change when k increases? How does it change when λ increases?

EXERCISE 22.32. Derive expression (22.54).

EXERCISE 22.33. (1) Show that $\tilde{V}_i(\tau' | p^{t+1})$ defined in (22.57) is not necessarily quasi-concave.

- (2) Show that $\tilde{V}_i(\tau' | p^{t+1})$ satisfies the single-crossing property in Definition 22.3.

EXERCISE 22.34. Consider an economy consisting of three groups, a fraction θ_p poor agents each with income y_p , a fraction θ_m middle-class agents with income $y_r > y_m$, and the remaining fraction $\theta_r = 1 - \theta_p - \theta_m$ rich agents with income $y_r > y_p$. Suppose that both θ_p and θ_r

are less than $1/2$, so that the individual with the median income (the “median voter”) is a middle-class individual.

- (1) Construct a change in incomes that leaves mean income in the society unchanged and increases the gap between the mean and the median but does not constitute a mean preserving spread of the distribution.
- (2) Construct a mean preserving spread of the distribution such that the gap between the mean and the median narrows. [Hint: increase y_m and reduce y_p , holding y_r constant].

EXERCISE 22.35. We now consider the model by Alesina and Rodrik, which is similar to the model studied in Section 22.7. There is a continuum 1 of individuals. All individuals have logarithmic instantaneous utility, so that $U_i = \sum_{t=0}^{\infty} \beta^t \ln C_i(t)$, where i denotes the individual and $C_i(t)$ refers to his consumption at time t . Each individual has one unit of labor, which he supplies inelastically. Final output is produced as

$$Y_i(t) = AK_i(t)^{1-\alpha} G(t)^\alpha L_i(t)^\alpha$$

where K_i and L_i denote capital and labor employed by individual i and G is government investment in infrastructure. The only tax instrument is a linear tax on the capital holdings of all individuals at the rate $\tau(t)$ at time t . All the proceeds of this taxation are spent on government investment in infrastructure, so that

$$(22.73) \quad G(t) = \tau(t) \bar{K}(t)$$

where $\bar{K}(t)$ is the average (total) capital stock in the economy. This specification implies that government’s provision of infrastructure creates a Romer-type externality. Denote the initial capital stock of the economy by $\bar{K}(0)$.

- (1) Characterize the equilibrium with a constant tax rate $\tau > 0$ at each date and show that with A sufficiently large, the economy will achieve a constant and positive growth rate. Show that the growth rate of the economy is independent of the distribution of the initial capital stock among the individuals. [Hint: note that the net interest rate faced by consumers is equal to the marginal product of capital minus the tax rate, τ].
- (2) Let the initial capital stock $\bar{K}(0)$ be distributed among the agents with shares θ_i , so that individual i ’s initial capital holding is $K_i(0) = \theta_i \bar{K}(0)$. Show that in equilibrium $K_i(t) = \theta_i \bar{K}(t)$ for any $t = 1, 2, \dots$
- (3) Suppose that the economy will legislate a constant tax rate τ forever. Determine the most preferred tax rate of individual i as a function of his share of initial capital θ_i at time $t = 0$.
- (4) Show that individuals have single-peaked preferences. On the basis of this, appeal to Theorems 22.2 and 22.4 to argue that the tax rate most preferred by the individual with the median capital holdings, θ_m , will be implemented. Show that as this median

capital holdings falls, the rate of capital taxation increases. What is the effect of this on economic growth?

- (5) Show that the equilibrium characterized in 4 above is *not* a MPE. Explain why not. How would you set up the problem to characterize such an equilibrium? [Hint: just describe how you would set up the problem; no need to solve for the equilibrium].

EXERCISE 22.36. (1) Prove Proposition 22.23.

- (2) Derive the output-maximizing tax rate as in (22.72).
 (3) Characterize the tax rates maximizing the utility of the elite and the citizens and establish the results in Proposition 22.24.

CHAPTER 23

Political Institutions and Economic Growth

The previous chapter investigated why societies often choose “inefficient” economic institutions and policies and consequently fail to take advantage of growth opportunities. It emphasized the importance of social conflict between different groups and lack of commitment to future policies as major sources of *non-growth-enhancing* policies. Much of the discussion was in the context of a given set of *political institutions*, which shaped both the extent and kind of social conflict between different individuals and groups, and what types of policies were possible or could be committed to. A natural conjecture is that political institutions will influence a society’s choices of economic institutions and policies and thus its growth trajectory. This conjecture leads to the following two questions: (1) do certain political institutions mediate social conflict more successfully, thus potentially avoiding non-growth-enhancing policies? (2) why do different societies choose or end up with different political institutions.

This chapter provides some preliminary answers to these two questions. I start with a brief summary of the empirical evidence on the effect of different political regimes and other political factors (such as political instability and civil wars) on economic growth. Section 23.2 then uses the baseline model in Section 22.2 from the previous chapter to illustrate that, once we take the existence of conflicting preferences into account, no political regime is perfect and each will create different types of costs and benefits associated with different losers and winners in the society. Whether a particular set of political institutions leads to growth-enhancing policies then depends on the details of how it will function, on the technology and the factor endowments of the society and on which groups will benefit from these institutions. Section 23.3 then turns to the dynamic tradeoffs between different regimes, emphasizing how democratic regimes might compensate for the short-run distortions that they create by generating long-run benefits, both in terms of avoiding sclerotic outcomes and by creating greater flexibility. This section will also emphasize how different political regimes deal with the process of *creative destruction*, which, as we saw in Chapter 14, is one of the engines of modern economic growth. It will suggest that democracies may be better at taking advantage of the forces of creative destruction.

Although Section 23.3 introduces the dynamics of economic allocations under different political regimes, it only gives us a few clues about how political institutions themselves emerge and change. This is a major area of current research in political economy and largely falls beyond the scope of the current book. Sections 23.4 and 23.5 will therefore give a bird’s eye view of some of the main issues involved in the analysis of equilibrium political institutions

and their implications for economic growth. Section 23.4 starts with a general discussion of what types of models we might want to consider for understanding the dynamics of political institutions and endogenous political change. Section 23.5 illustrates some of these general ideas with a specific example, which shows how we can construct dynamic models featuring changes in political and economic institutions, and how such models can shed light on the empirical patterns discussed in Section 23.1.

23.1. Political Regimes and Economic Growth

The centerpiece of our approach to the political economy of growth are the two mappings introduced at the beginning of Part 8 above, $\rho(\cdot)$ and $\pi(\cdot)$. Recall that these determine how political institutions shape economic institutions and how economic institutions map into allocations. The latter has already been discussed in the previous chapter, though it would not be unfair to say that we do not yet have a full understanding of the consequences of different economic institutions (consider, for example, the debate on the implications of privatization, or what the implications of different contract enforcement practices will be on the exact investment behavior and welfare levels of different individuals in a society). Our understanding of the implications of different political institutions on economic outcomes is even more imperfect. Thus the mapping $\rho \circ \pi : \mathcal{P} \rightarrow \mathcal{X}$, which determines how various combinations of political and economic institutions lead to different economic allocations is something we would like to (and should) learn more about.

In this section, I briefly discuss various different types of distinctions can be made among political institutions. Most scholars would probably start by thinking of the contrast between democratic and nondemocratic regimes. But there are many different types and shades of democracies. Democracy is typically defined by a set of procedural rules, for instance, by whether there are free and fair elections in which most adults can participate and there is free entry of parties into politics. But this definition of democracy is quite encompassing. First, it leaves many distinctive institutional features of democracies unspecified. Democracies can be parliamentary or presidential. They can use different electoral rules, giving different degrees of voice to minorities. Perhaps more importantly, there are different degrees of “free and fair” and “most adults”. In particular, most elections, even those in Europe or the United States, involve some degree of fraud and some restrictions on the entry of parties or candidates. Moreover, many individuals are effectively or sometimes explicitly disenfranchised. Political scientists consider Britain and the United States in the late 19th century to have been democratic, though only males had the right to vote. Few people would consider the United States in the 1960s a nondemocracy, though many blacks were disenfranchised. This creates various different shades of democracy that one might wish to take into account. For example, Colombia has been a democratic country for over half a century according to most political scientists, though in many parts of the country elections are very far from “free and fair” and take place under the threat of explicit violence from paramilitaries. Over the same time period, the entry of a Socialist party into Colombian politics has been blocked by various

legal and non-legal means, violating the “free entry” requirement. This discussion alerts us to the fact that we may want to draw distinctions between the “degree” of democracy and the “type” of democracy. With these caveats in mind, one might be tempted to conclude that the label of “democracy” is so encompassing and nondescript as to become meaningless. This is not my view, however. I maintain that the distinction between democracy and nondemocracy is not meaningless, and in fact, it is a particularly useful starting point for our analysis of the effects of political regimes on economic outcomes and the dynamics of political regimes. Nevertheless, when we wish to understand why different democracies behave differently and also the contrast between democratic and nondemocratic regimes, it will be necessary to delve deeper and make systematic distinctions about the nature and functioning of different types of democracies.

The differences between nondemocratic societies are probably even more pronounced. China under the rule of the Communist Party since 1948 is an undisputed case of a non-democratic regime, but it is very different in nature from the oligarchic regime in place in Britain before the process of democratization started with the First Reform Act of 1832. In Britain before 1832, there were prime ministers and parliaments, though they were elected by a small minority of the population—those with wealth, education and privilege, who made up less than 10% of the adult population. Furthermore, the powers of the state never rivaled those of the Communist Party in China. The Chinese example is also different from military dictatorships such as that of Chile under General Pinochet or South Korea under General Park. Once we consider regimes based on personal rule, such as that of Mobutu in Zaire, and monarchies, such as the rule of the Saud family in Saudi Arabia, the contrast is even more marked.

Nevertheless, there is an important commonality among these nondemocracies and an important contrast between nondemocratic and democratic regimes, making these categories still useful for conceptual and empirical analysis. Despite all of their imperfections and different shades, democratic regimes, at least when they have a certain minimal degree of functionality, provide greater *political equality* than nondemocratic regimes. Free entry of parties and one-person one-vote in democracy are the foundations of this and ensure some amount of voice for each individual. When democracies are particularly well functioning, majorities will have some (often a significant) influence on policies—though they themselves may be constrained by certain constitutional restrictions. In contrast, nondemocracies, rather than representing the wishes of the population at large, represent the preferences of a subgroup of the population. In the previous chapter, I referred to this subgroup as the “elite,” and I will continue to do so here. The identity of the elite differs across nondemocratic societies. In China, it is mainly the wishes of the Communist Party that matters. In Chile under Pinochet, most decisions were taken by a military junta, and it was their preferences, and perhaps the preferences of certain affluent segments of the society supporting the dictatorship, that counted. In Britain before the First Reform Act of 1832, it was the small wealthy minority that was politically influential.

With this cautionary introduction on the distinctions between democracies and non-democracies, what are the major differences between these political regimes? First, one might imagine that democracies and nondemocracies will have different growth performances. The first place to look for such differences is the postwar era for which there are better data on economic growth. Unfortunately, the picture here is not very clear. Przeworski and Limongi (1993) and Barro (1999) document, using cross-country regression evidence, that democracies do not appear to perform better than nondemocracies. Nevertheless, there is no universal consensus on this matter. For example, Minier (2004) reports results showing some positive effects of democratizations on growth and more robust negative effects of transitions to nondemocracy on growth. Persson and Tabellini (2007) argue that once one distinguishes between actual democracy and the probability of democracy surviving, there is a significant effect of democratic survival probabilities on economic growth. All in all, however, the bulk of the available evidence supports the conclusions of Przeworski and Limongi (1993) and Barro (1999) and suggests that, on average, democracies do not grow much faster than nondemocracies (at least, once one controls for other potential determinants of economic growth). This is, at first, a surprising and even perhaps a disturbing finding. One might have expected significantly worse growth performances among nondemocracies, since this group includes highly unsuccessful countries such as Iraq under Saddam Hussein, Zaire under Mobutu or Haiti under the Duvaliers. Despite these nondemocratic “basket cases” of growth, there are plenty of unsuccessful democracies, including India until the 1990s and many newly independent former colonies that started their independence period as electoral democracies (though, often quickly falling prey to coups or personal rule of some strongman). There are also many successful nondemocracies, including Singapore under Lee Kwan Yew or South Korea under General Park, or more recently China. Thus to understand how different political institutions affect economic decisions and economic growth we will need to go beyond the distinction between democracy and nondemocracy.

One idea, which I will argue is useful in thinking about these distinctions, is that of *dysfunctional democracies*, that is, the possibility that some democracies are functioning in a very different, and inefficient, manner than we typically envisage. I will further argue that a particularly important reason why democracies might be dysfunctional is because they are captured by elites despite the fact that on paper they are supposed to provide majoritarian decision-making and political equality. According to this definition, a democracy will be “captured” when its modus operandi—its purpose of creating greater political equality than a typical nondemocratic regime—fails. *Captured democracies* are one example of the more general typology of dysfunctional democracies. Another one might be highly *populist democracies*, where a strongman, such as Juan Peron in Argentina or Hugo Chávez in Venezuela, receives majority support but pursues policies that are detrimental to economic growth. This discussion therefore suggests that a satisfactory understanding of the relationship between democracy and growth, and more generally, that between political regimes and growth, necessitates an analysis of how different political regimes function and why some democracies

might become captured or take the populist route. I will discuss some possible answers to these questions in Section 23.5.

If there are no marked growth differences between democracies and nondemocracies, are there instead other significant policy or distributional differences? On this question, there is even more controversy than on the effects of democracy on growth. Rodrik (1999) documents that democracies have higher labor shares and interprets this as the outcome of greater redistribution in democracies. He shows that the same result holds both in cross-sectional and in panel data regressions. Acemoglu and Robinson (2006a) summarize a range of case studies showing how democracies pursue more redistributive policies and how this has an effect on the distribution of income, for example, on the share of capital in national income or on the overall extent of income inequality. Acemoglu (2008a) also shows that nondemocratic regimes often adopt different regulations, for example, erecting greater barriers against entry of new businesses. Finally, Persson and Tabellini (2004) document major differences in fiscal policy between different types of democracies. In contrast, Gil, Mulligan and Sala-i-Martin (2004) use cross-sectional regressions to show that a range of policies, in particular overall government spending and spending on Social Security, do not differ between democracies and dictatorships. Based on this, they argue that the distinctions between democracy and nondemocracy, or between different shades of democracy, are not useful in understanding policymaking and the extent of redistributive policies that societies adopt. Overall, therefore, there is no consensus in the literature on whether democracies pursue significantly different fiscal policies and whether this has a significant impact on the distribution of resources in the society. Nevertheless, the evidence in Rodrik (1999) and some of the evidence summarized in Acemoglu and Robinson (2006a) do indicate that, at least in some cases, democracies pursue significantly more redistributive policies than nondemocracies, and we can take these differences as our starting point, at least as a working hypothesis. Nevertheless, it is useful to bear in mind that the differences in policy between democracies and nondemocracies, even if present, appear to be much less pronounced than one might have expected on the basis of theory alone. I will argue in Section 23.5 that the same factors that are important in thinking about why democracies do not grow faster than nondemocracies are likely to be important in understanding why policies in democracies and nondemocracies do not differ by much (for example, captured democracies are unlikely to pursue highly redistributive policies). However, before turning to these issues, we need a more systematic analysis of how political institutions influence the economic organization and the economic outcomes of a society. This will be the topic of the next two sections.

It should also be noted at this point that the comparison of democracies to nondemocracies over the postwar era might be overly restrictive. When we look at a longer time horizon, it seems to be the case that democracies have significantly better economic growth performance. Most of the countries that industrialized rapidly during the 19th century were more democratic than those that failed to do so. The comparison of the United States to

the Caribbean or to Peru, or of Britain and France to Russia and Austria-Hungary are particularly informative in this context. For example, the United States, which was one of the most democratic societies at the time, was not any richer than the highly nondemocratic and repressive Caribbean colonies at the end of the 18th century, and a range of evidence in fact indicates that the Caribbean colonies may have been significantly richer than Northeastern United States throughout the 17th and 18th centuries. Even when we compare the United States to Peru, which is another repressive, nondemocratic colony, there is no strong evidence that throughout the 17th and 18th centuries, the United States was much richer, or in fact any richer, than Peru. However, the 19th and 20th centuries witnessed rapid growth and industrialization in the United States and stagnation in the entire Caribbean area and in Peru, as well as in much of the rest of South America. This historical episode therefore suggests that the more democratic societies may have been better at taking advantage of the new investment and growth opportunities that came with the age of industrialization of the 19th century. The contrast of Britain and France to Russia and Austria-Hungary is similar. Even though the former two countries were already richer at the beginning of the 19th century than their Russian and Austria-Hungarian counterparts, the income differences were small. Differences in political institutions were much more marked, however. Britain was already on its way to becoming a parliamentary democracy and France had already undergone the Revolution of 1789 and was becoming a much more representative society. Britain and France adopted pro-growth policies throughout much of the 19th century, even when this was costly to existing landowning elites, whereas Russia and Austria-Hungary explicitly blocked industrialization in order to protect the economic and political interests of their landowning aristocracies.

Long-run regressions, such as those discussed in Chapter 4, are also consistent with this pattern and show a significant effect of a broad cluster of institutions on economic growth. While we cannot confidently say that this represents the effect of political institution on growth, this cluster of institutions comprises both political and economic elements and it is likely that the growth-enhancing cluster of institutions could not exist without the political institutions supporting the economic institutions encouraging investment and free entry.

Finally, even though the effects of democracy and nondemocracy on growth might be less clear-cut than we would have liked, there are certain other regularities that are worth noting. The evidence seems to indicate quite strongly that political order is much more conducive to economic growth than political instability. A range of papers, for example, Alesina and Perotti (1993), Alesina et al. (1996), and Svensson (1998), find a negative and significant effect of political instability, as measured by assassinations or civil unrest, on economic growth. Even more clear are the negative effects of civil wars on economic growth. Many of the big growth disasters in Africa over the past half century have been associated with civil wars and infighting among different warlords, such as in Angola, Mozambique, Rwanda, Ethiopia, Sudan, Sierra Leone and Liberia. Therefore, even if the effect of the exact shade of democracy on growth is still unknown, there is strong evidence that political

factors, at least in their extreme form, have an effect on the economic opportunities available to individuals and thus on economic growth. Exercise 23.1 gives an example of conflict between two groups within a society that creates instability and leads to economic crises and bad economic performance.

I next turn to a theoretical investigation of how we might expect different political institutions to affect economic policies and economic outcomes. I will then enrich this framework to shed light on why the relationship between political regimes and economic growth may be more complex than one might have originally expected.

23.2. Political Institutions and Growth-Enhancing Policies

In this section, I consider the canonical Cobb-Douglas model analyzed in subsection 22.2.4 (and then used again in Sections 22.3 and 22.4 of the previous chapter). In that chapter, this model was analyzed under the assumption that the group of producers which I referred to as the “elite” were in power. I showed how the political equilibrium in this case can lead to various different forms of non-growth-enhancing policies. I will now briefly discuss the equilibrium in the same environment when the middle class or the workers are in power and then contrast the resulting allocations.

23.2.1. The Dictatorship of the Middle Class Versus the Dictatorship of the Elite. First, let us suppose that the middle class hold political power, so that we have the *dictatorship of the middle class* instead of the dictatorship of the elites in the previous chapter. The situation is entirely symmetric to that in the previous chapter with the middle class and the elite having exchanged places. In particular, the analysis leading to Proposition 22.6 immediately yields the following result.

PROPOSITION 23.1. *Consider the environment of subsection 22.2.4 with Cobb-Douglas technology, but the middle class instead of the elite holding political power. Suppose that Condition 22.1 holds, $\phi > 0$, and*

$$(23.1) \quad A^m \geq \phi \alpha^{\alpha/(1-\alpha)} A^e \frac{\theta^e}{\theta^m}.$$

Then, the unique MPE features $\tau^m(t) = 0$ and

$$(23.2) \quad \tau^e(t) = \bar{\tau}^{COM} \equiv \min \left\{ \frac{\kappa(\bar{L}, \theta^m, \alpha, \phi)}{1 + \kappa(\bar{L}, \theta^m, \alpha, \phi)}, \bar{\tau} \right\},$$

for all t , where $\kappa(\bar{L}, \theta^e, \alpha, \phi)$ is defined in (22.29).

PROOF. See Exercise 23.2. □

The notable feature about this equilibrium is the strong parallel to Proposition 22.6. The equilibria under elite control and middle class control are identical, except that the two groups have switched places. We therefore have an example of political institutions having a real effect on both the types of economic policies and economic institutions in place, and on the allocation of resources; in the elite-controlled society, the middle class are

taxed both to create revenues for the elite and to reduce their labor demand. In the middle class dominated society, the competing group of producers that are out of political power are the “elite” (even though the name “elite” has the connotation of political power). So now the elite are taxed to generate tax revenue and to create more favorable labor market conditions for the middle class. The contrast between the elite dominated and the middle-class dominated politics approximates certain well-known historical episodes. For example, in the context of the historical development of European societies, political power was first in the hands of landowners, who exercised it to keep labor tied to land and to reduce the power and the profitability of merchants and early industrialists (capitalists). In many cases, these policies favoring landowners were detrimental to economic growth. Nevertheless, with economic changes and the constitutional revolutions taking place in the late medieval period, power shifted away from landowning aristocracies towards the merchants and industrialists, and it was their turn to adopt policies favorable to their own economic interests and costly for landowners. The repeal of the Corn Law in 1846 illustrates this point, even though the conflict between landowners and capitalists was probably weakest in England because many members of the gentry and the previous landowning class had already transitioned into commercial agriculture and other industrial activities. Nevertheless, there were intense political debates surrounding the Corn Law, with landowners supporting the tariffs imposed by the law, which kept the price of their produce high, and industrialists opposing it so that the implicit tax on their inputs, especially labor, would be removed with the import of cheaper corn from abroad.

So, which one of these two sets of political institutions—the dictatorship of the middle class or the dictatorship of the elite—is better? The answer is that they cannot be compared easily. First, as already emphasized in the previous chapter, the equilibrium considered in Section 22.3 was already Pareto optimal; starting from the allocation there, it is not possible to make any member of the society better-off without making the elite worse-off. In the same way, the current allocation of resources is Pareto optimal, but it has picked a different point along the Pareto frontier—a point that favors middle-class agents instead of the elite. However, in the previous chapter we also emphasized that Pareto optimality may be too weak a concept for a useful analysis of the effect of institutions on economic growth, since two allocations that are Pareto optimal may involve significantly different growth rates. And yet, when we compare the growth rates under these two different political regimes, we also find that there is no straightforward ranking. Either of these two societies may achieve a higher level of income per capita. Which one does so depends on which group has more productive investment opportunities. When the middle class has the more productive investment opportunities, a society in which the elite are in power will create significant distortions. In contrast, if the elite have more profitable and socially beneficial production opportunities, then having political power vested with the elite is more beneficial for economic performance than the dictatorship of the middle class.

The following proposition illustrates a particularly simple case of this result.

PROPOSITION 23.2. *Consider the environment of subsection 22.2.4 with Cobb-Douglas technology. Suppose that Conditions 22.1, (22.27) and (23.1) hold, $\theta^e = \theta^m$, and $\phi > 0$. Then, the dictatorship of the middle class generates higher income per capita when $A^m > A^e$ and the dictatorship of the elite generates higher income per capita when $A^e > A^m$.*

PROOF. See Exercise 23.3. □

This proposition therefore gives a simple example of a situation whereby which political institutions will lead to better economic performance (in terms of income per capita) depends on whether the group that is more productive also holds political power. When political power and economic power are decoupled, there is greater inefficiency. An immediate implication of this result is that it is difficult to think of “efficient political institutions” without considering the self-interested objectives of those who hold and wield political power and without fully analyzing how their productivity and their economic activities compare to those of others. Naturally, one can dream of political institutions that will outperform both the elite dominated politics of the previous chapter and the middle-class dominated politics of this section. For example, we can think of a set of political institutions that constitutionally force all taxes to be equal to zero—so that in the context of the simple model we are focusing on here, there are no distortionary policies. In this environment, this alternative arrangement will outperform both elite dominated and middle class dominated politics. However, such political institutions are not realistic. First, there are numerous reasons why societies need to raise taxes, for example, they need to finance productivity-enhancing public goods as in the model of Section 22.8 in the previous chapter, and they also need to engage in some amount of redistribution to ensure a safety net to their citizens. Once we allow for positive taxes, then the social groups or the politicians that are in power can also misuse these tax revenues and the associated fiscal instruments. Second, constitutional limits on taxes are difficult to enforce. Once a particular group is in power and has the capability to dictate policies, there is no easy way of preventing them to rewrite the constitution as has been the practice in many countries over the last two centuries. This discussion indicates that we can think of “ideal political institutions” that may prevent the distortions of simpler institutions that vest power with a particular group of individuals, but such political institutions are difficult to create, implement and maintain. And this implies that the choice of political institutions in practice will be between arrangements that will create different types of distortions and different winners and losers.

23.2.2. Democracy or Dictatorship of the Workers? The previous subsection contrasted the dictatorship of the middle class to the dictatorship of the elite. A third possibility is to have a more democratic political system in which the majority decides policies. Since in realistic scenarios, the workers will outnumber both the elite and the middle-class entrepreneurs, this means the choice of policies that favor the economic interests of the workers (who have so far been passive in this model, simply supplying their labor at the equilibrium wage rate) will be implemented. While such a system does resemble democracy in some ways, it

can also be viewed as a dictatorship of the workers, since it will now be the workers who will dictate policies, in the same way that the elite or the middle class dictated policies under their own dictatorship.¹ This emphasizes once more that different political institutions will create different winners and losers depending on which group has more political power.

The analysis is again straightforward, though the nature of the political equilibrium does depend even more strongly on whether Condition 22.1 holds (whether there is excess supply or not). The following proposition summarizes the equilibrium when workers monopolize political power.

PROPOSITION 23.3. *Consider the environment of subsection 22.2.4 with Cobb-Douglas technology and suppose that workers hold political power.*

- (1) *Suppose that Condition 22.1 fails to hold (so that there is excess labor supply), then the unique MPE features $\tau^m(t) = \tau^e(t) = \tau^{RE} \equiv \min\{1 - \alpha, \bar{\tau}\}$.*
- (2) *Suppose that Condition 22.1 holds (so that there is no excess labor supply) and that $\theta^e = \theta^m = \theta$. Then, unique MPE is as follows:*
 - (a) *if $A^m > A^e$, then $\tau^e(t) = 0$, and $\tau^m(t) = \tau^{Dm}$ where $(1 - \tau^{Dm})^{1/(1-\alpha)} A^m = A^e$, or $\tau^{Dm} = 1 - \alpha$ and $\alpha^{1/(1-\alpha)} A^m \geq A^e$;*
 - (b) *if $A^m < A^e$, then $\tau^m(t) = 0$, and $\tau^e(t) = \tau^{De}$ where $(1 - \tau^{De})^{1/(1-\alpha)} A^e = A^m$, or $\tau^{De} = 1 - \alpha$ and $\alpha^{1/(1-\alpha)} A^e \geq A^m$.*

The most interesting implication of this proposition comes from the comparison of the cases with and without excess supply. When Condition 22.1 fails to hold, there is excess labor supply and taxes have no effect on wages. Anticipating this, workers favor taxes on both groups of producers to raise revenues to be redistributed to themselves. The dictatorship of the workers (“democracy”) will then generate this outcome as the political equilibrium. Clearly, this is more distortionary than either the dictatorship of the elite or the middle class, because in these political scenarios at least one of the producer groups was not taxed (but the resulting allocation is once again Pareto optimal given the fiscal instruments, for the same reasons as stressed above). The situation is very different when Condition 22.1 holds. In that case, recall that both the dictatorships of the elite and of the middle class generated significant distortions owing to the factor price manipulation effect—in particular, they imposed taxes on competing producers precisely to keep wages low. In contrast, workers now dislike taxes precisely because of their effect on wages. Consequently, in this case, workers have more moderate preferences regarding taxation, and democracy generates lower taxes than both the dictatorship of the elite and the dictatorship of the middle class. This proposition therefore again highlights that which set of political institutions will generate a greater level of income per capita (or higher economic growth) depends on investment opportunities and market structure. When workers (or a subgroup that is influential in democracy) can

¹What the difference between the dictatorship of workers or poor segments of the society and a true “democracy” may be is an important question, of interest to philosophers, political scientists and economists. However, since there is as yet no satisfactory answer to this important question and it falls beyond the scope of my focus here, I will not discuss it further.

tax entrepreneurs without suffering the consequences, democracy will generate high levels of redistributive taxation and can lead to a lower income per capita than elite or middle-class dominated politics. However, when workers recognize the impact of taxes on their own wages, democracy will generate more moderate political outcomes.

The simple analysis in this section therefore already gives us some clues about why there are no clear-cut relationships between political regimes and economic growth. If the forces highlighted here are important, we would expect democracy to generate higher growth under certain circumstances, for example, when the equivalent of Condition 22.1 holds. In contrast, democracy will lead to worse economic performance by pursuing populist policies and imposing high taxes when the equivalent of Condition 22.1 fails to hold. Naturally, the model presented here is very simple in many ways, and Condition 22.1 or its close cousins may not be the right ones for evaluating whether democracy or other regimes are more growth-enhancing. Nevertheless, this analysis emphasizes that democratic regimes, like the dictatorships of the elite and of the middle class, will look after the interests of the groups that have political power and the resulting allocations will often involve different types of distortions. Whether these distortions are more or less severe than those generated by alternative political regimes will depend on technology, factor endowments and the types of policies available to the political system. In light of the analysis so far, this result is not surprising, but its implications are nonetheless important to emphasize. In particular, it highlights that there are no a priori theoretical reasons to expect that there should be a simple empirical relationship between democracy and growth. On balance, we may believe that the distortions created by democracy should be less than those created by dictatorships (nondemocracies), but this will be a conclusion to be reached with more detailed theoretical and empirical analysis. Moreover, in Section 23.5 I will present another set of reasons, which I find more compelling than those implied by the simple models here, for why democracies may not generate more growth than dictatorships.

23.3. Dynamic Tradeoffs

The previous section contrasted economic allocations under different political regimes (in particular, the dictatorship of the elite, the dictatorship of the middle class and democracy, which here amounts to the dictatorship of the workers). Although the underlying economic environment was a simplified version of the infinite-horizon neoclassical growth model, the tradeoffs among the regimes were static. In this section, I will study an environment, which also involves entry into entrepreneurship, social mobility and a simple form of creative destruction. Using this environment, I will contrast the implications of democracy to oligarchy for economic performance. The emphasis will be on the dynamic tradeoffs between the two regimes.

23.3.1. The Baseline Model. The model economy is similar to that analyzed in Section 22.2 and more specifically, to the Cobb-Douglas economy in subsection 22.2.4. The economy is populated by a continuum (of measure) 1 of infinitely-lived agents, each with

preferences given by (22.1) as in Section 22.2. In addition, for reasons that will become clear soon, I assume that each individual dies with a small probability $\varepsilon > 0$ in every period, and a mass ε of new individuals are born (with the convention that after death there is zero utility and $\beta \in (0, 1)$ is the discount factor inclusive of the probability of death). I will consider the limit of this economy with $\varepsilon \rightarrow 0$.

There are two occupations in this economy, production workers and entrepreneurs. The key difference between the models in the previous chapter (and that in the previous section) and the one here is the possibility of social mobility (the fact that individuals may choose their occupations). In particular, each agent can either be employed as a worker or set up a firm to become an entrepreneur. I assume that all agents have the same productivity as workers, but their productivity in entrepreneurship differs. In particular, agent i at time t has entrepreneurial talent/skills $a_i(t) \in \{A^L, A^H\}$ with $A^L < A^H$. To become an entrepreneur, an agent needs to set up a firm, if he does not have an active firm already. Setting up a new firm may be costly because of entry barriers created by existing entrepreneurs.

Each agent therefore starts period t with skill level $a_i(t) \in \{A^H, A^L\}$ and some amount of capital $k_i(t)$ invested from the previous date (recall that capital investments are again made one period in advance), and another state variable denoting whether he already possesses a firm. I will denote this by $e_i(t) \in \{0, 1\}$, with $e_i(t) = 1$ corresponding to the individual having chosen entrepreneurship at date $t - 1$ (for date t). If the individual is already an “incumbent” entrepreneur at t , that is, $e_i(t) = 1$, this may make it cheaper for him to choose $e_i(t + 1) = 1$ and become an entrepreneur at date $t + 1$, because potential entry barriers into entrepreneurship do not apply to incumbents. I refer to an agent with $e_i(t) = 1$ as a member of the “elite” at time t , both because he avoids the entry costs and also because in oligarchy, he will be a member of the political elite making the policy choices.

In summary, at each period t , each agent makes the following decisions: an occupation choice, $e_i(t + 1) \in \{0, 1\}$, and if $e_i(t + 1) = 1$, that is, if he becomes an entrepreneur, he also makes an investment decision for next period $k_i(t + 1) \in \mathbb{R}_+$. In addition, those who are currently entrepreneurs (those with $e_i(t) = 1$) decide how much labor $l_i(t) \in \mathbb{R}_+$ to hire.

Agents also make the policy choices in this society. How the preferences of various agents map into policies differs depending on the political regime, which will be discussed below. There are three policy choices. Two of those are similar to the policies we have seen so far; a tax rate $\tau(t) \in [0, \hat{\tau}]$ on output and a lump-sum transfer distributed to all agents denoted by $T(t) \in [0, \infty)$. Notice that I have already imposed an upper bound on taxes $\hat{\tau} < 1$. This upper bound can be derived from the ability of individuals to hide their output in the informal sector or because of the standard distortionary effects of taxation. It simplifies the analysis to take it as given here. The new policy instrument is a cost $B(t) \in [0, \infty)$ imposed on new entrepreneurs setting up a firm. I assume that the entry barrier $B(t)$ is pure waste, for example, corresponding to the bureaucratic procedures that individuals have to go through to open a new business. This implies that lump-sum transfers are financed only from taxes.

An entrepreneur with skill level $a_i(t)$ and capital level $k_i(t)$ produces

$$(23.3) \quad y_i(t) = \frac{1}{\alpha} k_i(t)^\alpha (a_i(t) l_i(t))^{1-\alpha}$$

units of the final good, when he hires $l_i(t) \in \mathbb{R}_+$ units of labor. Notice that entrepreneurial skill enters the production function as a labor-augmenting productivity term. As in subsection 22.2.4, I assume that there is full depreciation of capital at the end of the period, so $k_i(t)$ is also the level of investment of entrepreneur i at time $t - 1$, which is in terms of the unique final good of the economy.

I will further simplify the analysis by assuming that all firms have to operate at the same size, $\bar{L}$, so $l_i(t) = \bar{L}$ (see Exercise 23.6 for the implications of relaxing this assumption). Finally, I adopt the convention that the entrepreneur himself can work in his firm as one of the workers, which implies that the opportunity cost of becoming an entrepreneur is 0.

The most important assumption here is that each entrepreneur has to run the firm himself, so it is his productivity, $a_i(t)$, that matters for output. An alternative would be to allow costly delegation of managerial positions to other, more productive agents. In this case, low-productivity entrepreneurs may prefer to hire more productive managers. If delegation to managers can be done costlessly, entry barriers would create no distortions. Throughout I assume that delegation is prohibitively costly.

To simplify the expressions below, I define $b(t) \equiv \beta^{-1} B(t) / \bar{L}$, which corresponds to discounted per worker entry cost (and will be the relevant object when we look at the profitability of different occupational choices). Profits (the returns to entrepreneur i gross of the cost of entry barriers) at time t are then equal to $\pi_i(t) = (1 - \tau(t)) y_i(t) - w(t) l_i(t) - \beta^{-1} k_i(t)$, which takes into account that the investment cost $k_i(t)$ was incurred in the previous period, thus the opportunity cost of investment (which is forgone consumption) is multiplied by the inverse of the discount factor. This expression for profits takes into account that the entrepreneur produces output $y_i(t)$, pays a fraction $\tau(t)$ of this in taxes, and also pays a total wage bill of $w(t) l_i(t)$. Given a tax rate $\tau(t)$ and a wage rate $w(t) \geq 0$ and using the fact that $l_i(t) = \bar{L}$, the net profits of an entrepreneur with talent $a_i(t)$ at time t are:

$$(23.4) \quad \pi(k_i(t) \mid a_i(t), w(t), \tau(t)) = (1 - \tau(t)) \alpha^{-1} k_i(t)^\alpha (a_i(t) \bar{L})^{1-\alpha} - w(t) \bar{L} - \beta^{-1} k_i(t),$$

where recall that the tax rate is in the range $0 \leq \tau(t) \leq \hat{\tau}$ and the cost of capital is multiplied by β^{-1} because it is incurred in the previous period. Given this expression, the (instantaneous) gain from entrepreneurship for an agent of talent $z \in \{L, H\}$ at time t as a function of the tax rate, $\tau(t)$, and the wage rate, $w(t)$, is:

$$(23.5) \quad \Pi^z(\tau(t), w(t)) = \max_{k_i(t)} \pi(k_i(t) \mid a_i(t) = A^z, w(t), \tau(t)).$$

Note that this is the *net gain* to entrepreneurship since the agent receives the wage rate $w(t)$ in all cases (either working for another entrepreneur when he is a worker or working for himself—thus having to hire one less worker—when he is an entrepreneur). More importantly, the gain to becoming an entrepreneur for an agent with $e_i(t - 1) = 0$ and ability $a_i(t) = A^z$ is $\Pi^z(\tau(t), w(t)) - \beta^{-1} B(t) = \Pi^z(\tau(t), w(t)) - b(t) \bar{L}$, since this agent will have to pay the

additional cost imposed by the entry barriers, which, like the costs of investment, is incurred in the previous period and is thus multiplied by β^{-1} .

Labor market clearing requires the total demand for labor not to exceed the supply. Since entrepreneurs also work as production workers, the supply is equal to 1, so:

$$(23.6) \quad \int_0^1 e_i(t) l_i(t) di = \int_{i \in S_t^E} \bar{L} di \leq 1,$$

where S_t^E is the set of entrepreneurs at time t .

Finally, I specify the law of motion of entrepreneurial talent, $a_i(t)$, I assume that there is imperfect correlation between the entrepreneurial skill over time with the following Markov chain:

$$(23.7) \quad a_i(t+1) = \begin{cases} A^H & \text{with probability } \sigma^H & \text{if } a_i(t) = A^H \\ A^H & \text{with probability } \sigma^L & \text{if } a_i(t) = A^L \\ A^L & \text{with probability } 1 - \sigma^H & \text{if } a_i(t) = A^H \\ A^L & \text{with probability } 1 - \sigma^L & \text{if } a_i(t) = A^L \end{cases},$$

where $\sigma^H, \sigma^L \in (0, 1)$. Here σ^H is the probability that an agent has high skill in entrepreneurship conditional on being high skill in the previous period, and σ^L is the probability of transitioning from low skill to high skill. It is natural to suppose that $\sigma^H \geq \sigma^L > 0$, so that skills are persistent and low skill is *not* an absorbing state. What is essential for the results is imperfect correlation of entrepreneurial talent over time, which is captured by $\sigma^H < 1$. This implies that the identities of the entrepreneurs necessary to achieve productive efficiency change over time and thus necessitate a type of *creative destruction*, with new entrepreneurs replacing old ones.

The imperfect over-time correlation in $a_i(t)$ can be interpreted in three alternative and complementary ways. First, we can suppose that the productivity of an individual is not constant over time and changes in comparative advantage necessitate changes in the identity of entrepreneurs. Second, we can think of the infinitely-lived agents as representing dynasties and the imperfect over-time correlation in $a_i(t)$ may represent imperfect correlation between the skills of parents and children. Third and perhaps most interestingly, it may be that each individual has a fixed competence across different activities and comparative advantage in entrepreneurship changes as the importance of different activities evolves over time. For example, some individuals may be better in industrial entrepreneurship, while some are better in agriculture, and as industrial activities become more profitable than agriculture, individuals who have a comparative advantage in industry should enter into entrepreneurship and those who have a comparative advantage in agriculture should exit. Both of these stories are parsimoniously captured by the Markov chain for talent given in (23.7).

This Markov chain also implies that the fraction of agents with high skill in the stationary distribution is (see Exercise 23.7):

$$(23.8) \quad M \equiv \frac{\sigma^L}{1 - \sigma^H + \sigma^L} \in (0, 1).$$

Since there is a large number (continuum) of agents, the fraction of agents with high skill at any point is M . I also assume that

$$M\bar{L} > 1,$$

so that, without entry barriers, high-skill entrepreneurs generate more than sufficient demand to employ the entire labor supply. Moreover, I think of M as small and $\bar{L}$ as large; in particular, I assume $\bar{L} > 2$, which ensures that the workers are always in the majority and simplifies the political economy discussion below.

The timing of events within every period can be summarized as follows. At the beginning of time t , $a_i(t)$, $e_i(t)$ and $k_i(t)$ are given for all individuals as a result of their decision from date $t - 1$ and the realization of uncertainty regarding ability. Then, the following sequence of moves takes place.

- (1) Entrepreneurs demand labor and the labor market clearing wage rate, $w(t)$, is determined.
- (2) The tax rate on entrepreneurs, $\tau(t) \in [0, \hat{\tau}]$, is set.
- (3) The skill level of each agent for next period, $a_i(t + 1)$, is realized.
- (4) The entry barrier for new entrepreneurs, $b(t + 1)$, is set.
- (5) All agents make occupational choices, $e_i(t + 1)$, and entrepreneurs make investment decisions, $k_i(t + 1)$, for next period.

Entry barriers and taxes will be set by different agents in different political regimes as will be specified below. Notice that taxes are set after the investment decisions. This raises the holdup problems discussed in the previous chapter and acts as an additional source of inefficiency. The fact that $\tau(t) \leq \hat{\tau} < 1$ puts a limit on these holdup problems. It is also important to note that individuals make their occupational choices and investment decisions knowing their ability level, that is, $a_i(t + 1)$ is realized before the decisions on $e_i(t + 1)$ and $k_i(t + 1)$. Notice also that if an individual does not operate his firm, he loses “the license”, so next time he wants to set up a firm, he needs to incur the entry cost (and the assumption that $l_i(t) = \bar{L}$ rules out the possibility of operating the firm at a much smaller scale). Finally, we need to specify the initial conditions: I assume that the distribution of talent in the society is given by the stationary distribution, nobody starts out as an entrepreneur, so that $e_i(-1) = 0$ for all i , and the initial level of capital holdings is not important, since negative consumption is allowed, thus individuals can always increase their capital holdings by choosing a negative level of consumption.

Let us again focus on MPE, where strategies are only a function of the payoff-relevant states. For individual i the payoff-relevant state at time t includes his own state $(e_i(t), a_i(t), k_i(t), a_i(t + 1))$,² and potentially the fraction of entrepreneurs that are high skill, denoted by $\mu(t)$, and defined as

$$\mu(t) = \Pr(a_i(t) = A^H \mid e_i(t) = 1) = \Pr(a_i(t) = A^H \mid i \in S_t^E).$$

²Here $e_i(t)$, $k_i(t)$ and $a_i(t)$ are part of the individual’s state at time t , because they influence an entrepreneur’s labor demand. In addition, $a_i(t + 1)$ is revealed at time t and influences his occupational choice and investment decisions $e_i(t + 1)$ and $k_i(t + 1)$ for $t + 1$ and is also part of his state.

The equilibrium can be characterized by writing the net present discounted values of different agents recursively and then characterizing the optimal strategies within each time period by backward induction. I start with the *economic equilibrium*, which is the equilibrium of the economy described above given a policy sequence $\{b(t), \tau(t)\}_{t=0,1,\dots}$. Let $x_i(t) = (e_i(t+1), k_i(t+1))$ be the vector of choices of agent i at time t (for entrepreneurship and capital investment at time $t+1$) and let $x(t) = [x_i(t)]_{i \in [0,1]}$ denote the choice profile for all agents, and $p(t) = (\tau(t), b(t+1))$ denote the vector of policies at time t . Moreover, let $p^t = \{p(n)\}_{n=t}^\infty$ denote the infinite sequence of policies from time t onwards, and similarly w^t and x^t denote the sequences of wages and choices from t onwards. Then, $\hat{x}^t$ and a sequence of wage rates $\hat{w}^t$ constitute an economic equilibrium given a policy sequence p^t if, given $\hat{w}^t$ and p^t and his state $(e_i(t-1), a_i(t))$, $x_i(t)$ maximizes the utility of agent i , and $\hat{w}_t$ clears the labor market at time t , that is, eq. (23.6) holds. Each agent's type in the next period, $(e_i(t+1), a_i(t+1))$, is then given by his decision at time t regarding whether to become an entrepreneur and by the law of motion in (23.7).

I now characterize this equilibrium. Since $l_i(t) = \bar{L}$ for all $i \in S_t^E$ (where, recall that, S_t^E is the set of entrepreneurs at time t), profit-maximizing investments are given by:

$$(23.9) \quad k_i(t) = (\beta(1 - \tau(t)))^{1/(1-\alpha)} a_i(t) \bar{L},$$

where $\tau(t)$ is the equilibrium tax rate that entrepreneurs anticipate correctly along the equilibrium path. This equation implies that the level of investment is increasing in the skill level of the entrepreneur, $a_i(t)$, and the level of employment, $\bar{L}$, and decreasing in the tax rate, $\tau(t)$.

Now using (23.9), the net current gain to entrepreneurship for an agent of type $z \in \{L, H\}$ (of skill level A^L or A^H) can be obtained as:

$$(23.10) \quad \Pi^z(\tau(t), w(t)) = (1 - \alpha) \alpha^{-1} (\beta(1 - \tau(t)))^{1/(1-\alpha)} A^z \bar{L} - w(t) \bar{L}.$$

Moreover, the labor market clearing condition (23.6) implies that the total measure of entrepreneurs at any time is $\int_{i \in S_t^E} di = 1/\bar{L}$. Tax revenues at time t and the per capita lump-sum transfers are then given as:

$$(23.11) \quad T(t) = \int_{i \in S_t^E} \tau(t) y_i(t) di = \alpha^{-1} \tau(t) (\beta(1 - \tau(t)))^{\alpha/(1-\alpha)} \bar{L} \int_{i \in S_t^E} a_i(t) di.$$

To economize on notation, let us now denote the sequence of future policies and equilibrium wages by $q^t \equiv (p^t, w^t)$. Then, the time t value of an agent with skill level $z \in \{L, H\}$ if he chooses production work (for time t) is

$$(23.12) \quad W^z(q^t) = w(t) + T(t) + \beta C W^z(q^{t+1}),$$

where it is explicitly conditioned on future policies and wages, q^t , since these influence continuation values, and $C W^z(q^{t+1})$ is the relevant continuation value for a worker of type z from time $t+1$ onwards, given by

$$(23.13) \quad C W^z(q^{t+1}) = \sigma^z \max \{W^H(q^{t+1}); V^H(q^{t+1}) - b(t+1) \bar{L}\} \\ + (1 - \sigma^z) \max \{W^L(q^{t+1}); V^L(q^{t+1}) - b(t+1) \bar{L}\},$$

where $V^z(q^t)$ is defined similarly to $W^z(q^t)$ and is the time t value of an agent of skill z when he is an entrepreneur. The expressions for both (23.12) and (23.13) are intuitive. A worker of type $z \in \{L, H\}$ receives a wage income of $w(t)$ (independent of his skill), a transfer of $T(t)$, and the continuation value $CV^z(q^{t+1})$. This continuation value encodes the major dynamic tradeoffs facing individuals in this model. A worker of type $z \in \{L, H\}$ today—that is, an individual i with $e_i(t) = 0$ —will be high skill in the next period with probability σ^z , and in this case, he can either choose to remain a worker, receiving value W^H , or incur the entry cost $b(t+1)\bar{L}$ and become an entrepreneur ($e_i(t+1) = 1$), receiving the value of a high-skill entrepreneur, V^H . The reason why this individual has to pay the cost $b(t+1)\bar{L}$ when he chooses $e_i(t+1) = 1$ is that he is not currently an entrepreneur (that is, $e_i(t) = 0$), thus he has to pay the costs associated with the entry barriers. The max operator makes sure that the individual chooses whichever option gives higher value. With probability $1 - \sigma^z$, he will be low skill, and receives the corresponding values.

Similarly, the value functions for entrepreneurs are given by:

$$(23.14) \quad V^z(q^t) = w(t) + T(t) + \Pi^z(\tau(t), w(t)) + \beta CV^z(q^{t+1}),$$

where Π^z is given by (23.10) and now crucially depends on the skill level of the agent, and $CV^z(q^{t+1})$ is the continuation value for an entrepreneur of type z :

$$(23.15) \quad CV^z(q^{t+1}) = \sigma^z \max \{W^H(q^{t+1}); V^H(q^{t+1})\} + (1 - \sigma^z) \max \{W^L(q^{t+1}); V^L(q^{t+1})\}.$$

An entrepreneur of ability A^z also receives the wage $w(t)$ (working for his own firm) and the transfer $T(t)$, and in addition makes profits equal to $\Pi^z(\tau(t), w(t))$. The following period, this entrepreneur has high skill with probability σ^z and low skill with probability $1 - \sigma^z$, and conditional on the realization of this event, he decides whether to remain an entrepreneur or become a worker. Two points are noteworthy here. First, in (23.15), in contrast to the expression in (23.13), there is no additional cost of becoming an entrepreneur since this individual already owns a firm. Second, if an entrepreneur decides to become a worker, he obtains the value as given by the expressions in (23.13) so that the next time he wishes to operate a firm, he has to incur the cost of doing so.

Inspection of (23.13) and (23.15) immediately reveals that the occupational choices of individuals for time t will depend on the *net value* of entrepreneurship conditional on their current occupational status, $e_i(t-1) = \mathbf{e}$. Let us write this is

$$NV(q^t | a_i(t) = A^z, e_i(t-1) = \mathbf{e}) = V^z(q^t) - W^z(q^t) - (1 - \mathbf{e})b(t)\bar{L},$$

which is defined as a function of an individual's skill a and current entrepreneurship status, $\mathbf{e}$. The last term is the entry cost incurred by agents with $\mathbf{e} = 0$. The max operators in (23.13) and (23.15) imply that if $NV > 0$ for an agent, then he prefers to become an entrepreneur.

Who will become an entrepreneur in this economy? The answer depends on the NV 's. Standard dynamic programming arguments from Chapter 16, combined with the fact that instantaneous payoffs are strictly monotone, imply that $V^z(q^t)$ is strictly monotonic in $w(t)$, $T(t)$ and $\Pi^z(\tau(t), w(t))$, so that $V^H(q^t) > V^L(q^t)$ (see Exercise 23.5). By the same

arguments, $NV(q^t | a_i(t) = A^z, e_i(t-1) = \mathbf{e})$ is also increasing in $\Pi^z(\tau(t), w(t))$. This in turn implies that for all a and $\mathbf{e}$,

$$\begin{aligned} NV(q^t | a_i(t) = A^H, e_i(t-1) = 1) &\geq NV(q^t | a_i(t) = a, e_i(t-1) = \mathbf{e}) \\ &\geq NV(q^t | a_i(t) = A^L, e_i(t-1) = 0). \end{aligned}$$

In other words, the net value of entrepreneurship is highest for high-skill existing entrepreneurs, and lowest for low-skill workers. However, it is unclear ex ante whether $NV(q^t | a_i(t) = A^H, e_i(t-1) = 0)$ or $NV(q^t | a_i(t) = A^L, e_i(t-1) = 1)$ is greater, that is, whether entrepreneurship is more profitable for incumbents with low skill or for outsiders with high skill, who will have to pay the entry cost.

We can then define two different types of equilibria:

- (1) *Entry equilibrium*, where all entrepreneurs have $a_i(t) = A^H$.
- (2) *Sclerotic equilibrium*, where agents with $e_i(t-1) = 1$ remain entrepreneurs regardless of their productivity.

An entry equilibrium requires the net value of entrepreneurship to be greater for a non-elite high-skill agent than for a low-skill elite. Let us define $w^H(t)$ as the threshold wage rate such that high-skill non-elite agents are indifferent between entering and not entering entrepreneurship. That is, $w^H(t)$ has to be such that $NV(q^t | a_i(t) = A^H, e_i(t-1) = 0) = 0$. Using (23.12) and (23.14), this threshold is obtained as

$$(23.16) \quad w^H(t) \equiv \max \left\{ (1-\alpha)\alpha^{-1}(\beta(1-\tau(t)))^{1/(1-\alpha)}A^H - b(t) + \frac{\beta(CV^H(q^{t+1}) - CW^H(q^{t+1}))}{\bar{L}}; 0 \right\}.$$

Similarly, define $w^L(t)$ as the wage such that low-skill incumbent producers are indifferent between existing entrepreneurship or not. This implies that $w^L(t)$ is such that $NV(q^t | a_i(t) = A^L, e_i(t-1) = 1) = 0$, or

$$(23.17) \quad w^L(t) \equiv \max \left\{ (1-\alpha)\alpha^{-1}(\beta(1-\tau(t)))^{1/(1-\alpha)}A^L + \frac{\beta(CV^L(q^{t+1}) - CW^L(q^{t+1}))}{\bar{L}}; 0 \right\}.$$

Both expressions are intuitive. For example, in (23.16), the term $(1-\alpha)\alpha^{-1}(\beta(1-\tau(t)))^{1/(1-\alpha)}A^H$ is the per worker profits that a high-skill entrepreneur will make before labor costs. $b(t)$ is the per worker entry cost (discounted total costs, $\beta^{-1}B(t)$, divided by $\bar{L}$). Finally, the term $\beta(CV^H(q^{t+1}) - CW^H(q^{t+1}))$ is the indirect (dynamic) benefit, the additional gain from changing status from a worker to a member of the elite for a high-skill agent. Naturally, this benefit will depend on the sequence of policies, for example, it will be larger when there are greater entry barriers in the future. Consequently, if $w^L(t) < w^H(t)$, the total benefit of becoming an entrepreneur for a non-elite high-skill agent exceeds the cost. Equation (23.17) is explained similarly. Evidently, a wage rate lower than both $w^L(t)$ and $w^H(t)$ would lead to excess demand for labor and could not be an equilibrium. Consequently, the condition for an entry equilibrium to exist at time t

can simply be written as a comparison of the two thresholds determined above:

$$(23.18) \quad w^H(t) \geq w^L(t).$$

A sclerotic equilibrium emerges, on the other hand, when the converse of (23.18) holds.

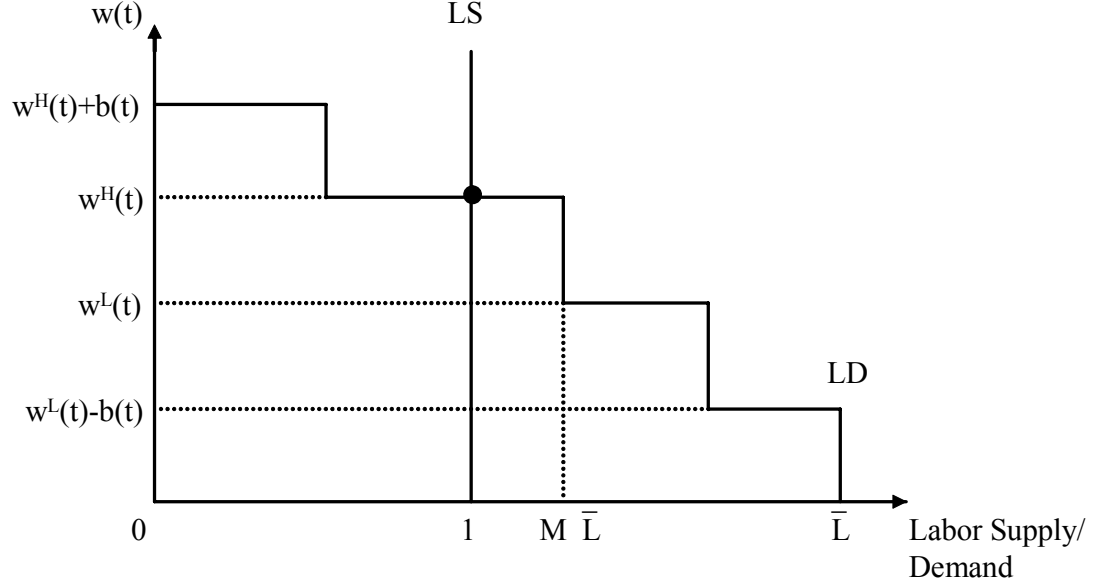


FIGURE 23.1. Labor market equilibrium when (23.18) holds.

Moreover, in an entry equilibrium, that is, an equilibrium where (23.18) holds, $NV(q^t | a_i(t) = A^H, e_i(t-1) = 0) = 0$. If it were strictly positive—in other words, if the wage were less than $w(t)$ —then, all agents with high skill would strictly prefer to become entrepreneurs, which is not possible since, by assumption, $M\bar{L} > 1$. This argument also shows that the total measure of entrepreneurs in the economy will be $1/\bar{L}$. Then, from (23.10), (23.12) and (23.14), the equilibrium wage, which will be denoted $w^E(t)$, is equal to

$$(23.19) \quad w^E(t) = w^H(t).$$

Note also that when (23.18) holds, naturally $NV(q^t | a_i(t) = A^L, e_i(t-1) = 1) \leq 0$, so low-skill incumbents would be worse-off if they remained as entrepreneurs at the equilibrium wage rate $w^E(t)$.

Figure 23.1 illustrates the entry equilibrium diagrammatically by plotting labor demand and supply in this economy. Labor supply is constant at 1, while labor demand is decreasing as a function of the wage rate. This figure is drawn for the case where condition (23.18)

holds, so that there exists an entry equilibrium. The first portion of the curve shows the willingness to pay of high-skill incumbents (those who start with $e_i(t-1) = 1$), but have high entrepreneurial skills $a_i(t) = A^H$. This marginal willingness is $w^H(t) + b(t)$ (since entrepreneurship is as profitable for them as for high-skill potential entrants and they do not have pay the entry cost). The second portion is for high-skill potential entrants—those with $e_i(t-1) = 0$ and $a_i(t) = A^H$ —and is equal to $w^H(t)$. These two groups together demand $M\bar{L} > 1$ workers, ensuring that labor demand intersects labor supply at the wage given in (23.19).

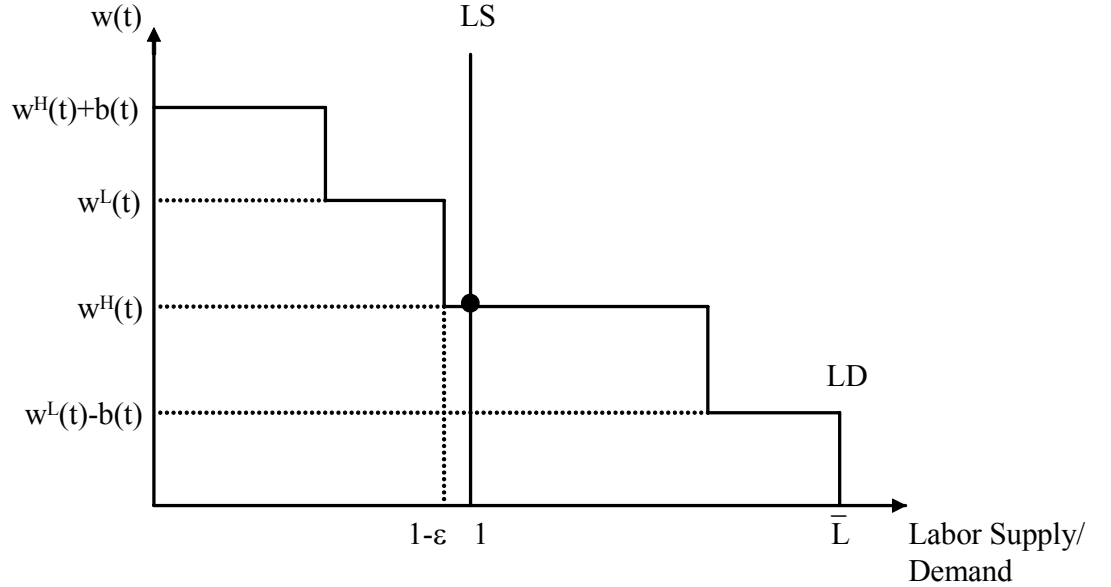


FIGURE 23.2. Labor market equilibrium when (23.18) does not hold.

In a sclerotic equilibrium, on the other hand, $w^H(t) < w^L(t)$, and low-skill incumbents remain in entrepreneurship, that is, $e_i(t) = e_i(t-1)$. If there were no deaths so that $\varepsilon = 0$, the total number of entrepreneurs would be $1/\bar{L}$ and for any $w \in [w^H(t), w^L(t)]$, labor demand would exactly equal labor supply (in other words, $1/\bar{L}$ agents would demand exactly $\bar{L}$ workers each, and a total supply of 1). Hence, there would be multiple equilibrium wages. In contrast, when $\varepsilon > 0$, the total number of entrepreneurs who could pay a wage of $w^L(t)$ will be less than $1/\bar{L}$ for all $t > 0$, thus there would be excess supply of labor at this wage, or at any wage above the lower support of the above range. This implies that the equilibrium wage must be equal to this lower support, $w^H(t)$, which is identical to (23.19). Since at this wage agents with $e_i(t-1) = 0$ and $a_i(t) = A^H$ are indifferent between entrepreneurship and production work, in equilibrium a sufficient number of them enter entrepreneurship, so that total labor demand is equal to 1. In the remainder, I focus on the limiting case of this

economy where $\varepsilon \rightarrow 0$, which picks $w^E(t) = w^H(t)$ as the equilibrium wage even when labor supply coincides with labor demand for a range of wages.³

Figure 23.2 illustrates this case diagrammatically. Because (23.18) does not hold in this case, the second flat portion of the labor demand curve is for low-skill incumbents ($e_i(t-1) = 1$ and $a_i(t) = A^L$) who, given the entry barriers, have a higher marginal product of labor than high-skill potential entrants.

The equilibrium law of motion of the fraction of high-skill entrepreneurs, $\mu(t)$, is:

$$(23.20) \quad \mu(t) = \begin{cases} \sigma^H \mu(t-1) + \sigma^L (1 - \mu(t-1)) & \text{if (23.18) does not hold} \\ 1 & \text{if (23.18) holds} \end{cases},$$

starting with some $\mu(0)$. The exact value of $\mu(0)$ will play an important role below. Recall that $e_i(-1) = 0$ for all i . Under this assumption, any $b(0)$ would apply equally to all potential entrants and as long as it is not so high as to shut down the economy, the equilibrium would involve $\mu(0) = 1$. I consider $\mu(0) = 1$ to be the baseline case.

To obtain a full political equilibrium, we need to determine the policy sequence p^t . I consider two extreme cases: (1) *Democracy*: the policies $b(t)$ and $\tau(t)$ are determined by majoritarian voting, with each agent having one vote. (2) *Oligarchy*: the policies $b(t)$ and $\tau(t)$ are determined by majoritarian voting among the elite—the current entrepreneurs—at time t .

23.3.2. Democracy. A democratic equilibrium is a MPE where $b(t)$ and $\tau(t)$ are determined by majoritarian voting at time t . The timing of events implies that the tax rate at time t , $\tau(t)$, is decided after investment decisions, whereas the entry barriers are decided before. The assumption $\bar{L} > 2$ above ensures that workers (non-elite agents) are always in the majority.

At the time taxes are set, investments are sunk, agents have already made their occupation choices, and workers are in the majority. Therefore, taxes will be chosen to maximize per capita transfers given by

$$\alpha^{-1} \tau(t) k(t)^\alpha \bar{L} \sum_{i \in S_t^E} a_i(t),$$

which takes into account that $k(t)$ is already given from the investment in the previous period. Since this expression is increasing in $\tau(t)$ and $\tau(t) \leq \hat{\tau}$, the optimal tax for a worker is $\tau(t) = \hat{\tau}$ for all t . In view of this, total tax revenues are

$$(23.21) \quad T^E(t) = \alpha^{-1} \hat{\tau} (\beta(1 - \hat{\tau}))^{\alpha/(1-\alpha)} \bar{L} \sum_{i \in S_t^E} a_i(t).$$

The entry barrier, $b(t)$, is then set at the end of period $t-1$ (before occupational choices) to maximize this expression. Low-productivity workers (with $e_i(t-1) = 0$ and $a_i(t) = A^L$)

³In other words, the wage $w^H(t)$ at $\varepsilon = 0$ is the only point in the equilibrium set where the equilibrium correspondence is (lower-hemi) continuous in ε (recall the definition of lower hemi-continuity in Definitions A.31 and A.32 in Appendix Chapter A). In fact, the feature that there will be multiple equilibrium wage levels in dynamic models with entry barriers holds much more generally than the setup here with two types of entrepreneurs. This is demonstrated in Exercise 23.12.

know that they will remain workers, and in MPE, the policy choice at time t has no influence on strategies in the future except through its impact on payoff-relevant state variables. Therefore, given $\tau(t) = \hat{\tau}$, the utility of agent i with $e_i(t-1) = 0$ and $a_i(t) = A^L$ depends on $b(t)$ only through the equilibrium wage, $w^E(t)$, and the transfer, $T^E(t)$. High-productivity workers (those with $e_i(t-1) = 0$ and $a_i(t) = A^H$) may become entrepreneurs, but as the above analysis shows, in this case, $NV(q^t | a_i(t) = A^H, e_i(t-1) = 0) = 0$, $W^H = W^L$, so their utility is also identical to those of low-skill workers. Consequently, all workers prefer a level of $b(t)$ that maximizes $w^E(t) + T^E(t)$. Since the preferences of all workers are the same and they are in the majority, the democratic equilibrium will maximize these preferences.

A democratic equilibrium is therefore given by policy, wage and economic decision sequences, $\hat{p}^t$, $\hat{w}^t$, and $\hat{x}^t$, such that $\hat{w}^t$ and $\hat{x}^t$ constitute an economic equilibrium given $\hat{p}^t$, and $\hat{p}^t = (\hat{\tau}, b(t+1))$ is such that:

$$b(t+1) \in \arg \max_{b(t+1) \geq 0} \{w^E(t+1) + T^E(t+1)\}.$$

Inspection of (23.19) and (23.21) immediately shows that wages and tax revenue are both maximized when $b(t+1) = 0$ for all t , so the democratic equilibrium will not impose any entry barriers. This is intuitive; workers have nothing to gain by protecting incumbents, and a lot to lose, because such protection reduces labor demand and wages. Since there are no entry barriers, only high-skill agents will become entrepreneurs, or in other words $e_i(t) = 1$ only if $a_i(t) = A^H$ at all t . Given this stationary sequence of MPE policies, we can use the value functions (23.12) and (23.14) to obtain

$$(23.22) \quad V^H = W^H = W^L = W = \frac{w^D + T^D}{1 - \beta},$$

where w^D is the equilibrium wage in democracy, and T^D is the level of transfers, given by $\hat{\tau}Y^D$. Since there are no entry barriers now or in the future and $\tau(t) = \hat{\tau}$, eq. (23.16) then implies that $w^D = (1 - \alpha)\alpha^{-1}(\beta(1 - \hat{\tau}))^{\alpha/(1-\alpha)}A^H$. The following proposition therefore follows immediately:

PROPOSITION 23.4. *There exists a unique democratic equilibrium, which features $\tau(t) = \hat{\tau}$ and $b(t) = 0$. Moreover, $e_i(t) = 1$ if and only if $a_i(t) = A^H$, so $\mu(t) = 1$. The equilibrium wage rate is given by*

$$(23.23) \quad w(t) = w^D \equiv (1 - \alpha)\alpha^{-1}(\beta(1 - \hat{\tau}))^{\alpha/(1-\alpha)}A^H,$$

and the aggregate output is

$$(23.24) \quad Y^D(t) = Y^D \equiv \alpha^{-1}(\beta(1 - \hat{\tau}))^{\alpha/(1-\alpha)}A^H.$$

An important feature of the democratic equilibrium is that aggregate output is constant over time. This will contrast with the oligarchic equilibrium, where the skill composition of entrepreneurs and the level of output will change over time. Another noteworthy feature is that there is perfect equality because the excess supply of high-skill entrepreneurs ensures that they receive no rents.

It is useful to note that Y^D corresponds to the level of output inclusive of consumption and investment. “Net output” and consumption can be obtained by subtracting investment costs from Y^D , and in this case, they will be given by $Y_n^D \equiv (\alpha^{-1} - \beta(1 - \hat{\tau}))(\beta(1 - \hat{\tau}))^{\alpha/(1-\alpha)} A^H$. All the results stated for output here also hold for net output. I focus on output only because the expressions are slightly simpler.

23.3.3. Oligarchic Equilibrium. In oligarchy, policies are determined by majoritarian voting among the elite.⁴ At the time of voting over the entry barriers, $b(t)$, the elite consist of those with $e_i(t-1) = 1$, and at the time of voting over the taxes, $\tau(t)$, the elite are those with $e_i(t) = 1$.

Let us start with the taxation decision among those with $e_i(t) = 1$ and also impose the following condition:

CONDITION 23.1.

$$\bar{L} \geq \frac{1}{2} \frac{A^H}{A^L} + \frac{1}{2}.$$

When this condition is satisfied, both high-skill and low-skill entrepreneurs prefer zero taxes, $\tau(t) = 0$. I simplify the analysis here by assuming that this condition holds. Exercise 23.10 discusses the case when this condition is relaxed. Intuitively, Condition 23.1 requires the productivity gap between low and high-skill elites not to be so large that low-skill elites wish to tax profits in order to indirectly transfer resources from high-skill entrepreneurs to themselves.

When Condition 23.1 holds, the oligarchy will always choose $\tau(t) = 0$. Then, at the stage of deciding the entry barriers, high-skill entrepreneurs would like to choose $b(t)$ to maximize V^H , and low-skill entrepreneurs would like to maximize V^L (both groups anticipating that $\tau(t) = 0$). Both of these expressions are maximized by setting a level of the entry barrier that ensures the minimum level of equilibrium wages. Recall from (23.19) that equilibrium wages in this case are still given by $w^E(t) = w^H(t)$, so they will be minimized by ensuring that $w(t) = 0$, or in other words, by choosing any

$$(23.25) \quad b(t) \geq b^E(t) \equiv (1 - \alpha) \alpha^{-1} \beta^{1/(1-\alpha)} A^H + \beta \left(\frac{CV^H(q^{t+1}) - CW^H(q^{t+1})}{\bar{L}} \right).$$

Without loss of any generality, let us assume that they will set the entry barrier as $b(t) = b^E(t)$ in this case.

An oligarchic equilibrium then can be defined as a policy sequence $\hat{p}^t$, wage sequence $\hat{w}^t$, and economic decisions $\hat{x}^t$ such that $\hat{w}^t$ and $\hat{x}^t$ constitute an economic equilibrium given $\hat{p}^t$, and $\hat{p}^t$ involves $\tau(t+n) = 0$ and $b(t+n) = b^E(t+n)$ for all $n \geq 0$. In the oligarchic

⁴Notice that this assumption means political power rests with current entrepreneurs. As discussed in the previous chapter, there may often be a decoupling between economic and political power, so that key decisions are not made by current entrepreneurs, but by those who are politically powerful for historical or other reasons. The analysis in the previous chapter and also in Section 23.2 in this chapter illustrated the distortionary policies that would arise from such decoupling. The model here goes to the other extreme and places all political power in the hands of the current entrepreneurs and highlights a different set of inefficiencies that this will cause.

equilibrium, there is no redistributive taxation and entry barriers are sufficiently high to ensure a sclerotic equilibrium with zero wages.

Imposing $w^E(t+n) = 0$ for all $n \geq 0$, we can solve for the equilibrium values of high- and low-skill entrepreneurs from the value functions (23.14) as

$$(23.26) \quad \tilde{V}^L = \frac{1}{1-\beta} \left[\frac{(1-\beta\sigma^H) A^L + \beta\sigma^L A^H}{(1-\beta(\sigma^H - \sigma^L))} (1-\alpha) \alpha^{-1} \beta^{\alpha/(1-\alpha)} \bar{L} \right],$$

and

$$(23.27) \quad \tilde{V}^H = \frac{1}{1-\beta} \left[\frac{(1-\beta(1-\sigma^L)) A^H + \beta(1-\sigma^H) A^L}{(1-\beta(\sigma^H - \sigma^L))} (1-\alpha) \alpha^{-1} \beta^{\alpha/(1-\alpha)} \bar{L} \right].$$

These expressions are intuitive. First, consider $\tilde{V}^L$ and the case where $\beta \rightarrow 1$; then, starting in the state $e(t-1) = L$, an entrepreneur will spend a fraction $\sigma^L / (1 - \sigma^H + \sigma^L)$ of his future with high skill A^H and a fraction $(1 - \sigma^H) / (1 - \sigma^H + \sigma^L)$ with low skill A^L . The fact that $\beta < 1$ implies discounting and the low-skill states, which occur sooner, are weighed more heavily (since the agent starts out with low skills). The intuition for $\tilde{V}^H$ is identical.

Since there will be zero equilibrium wages and no transfers, it is clear that $W = 0$ for all workers. Therefore, for a high-skill worker, $NV = \tilde{V}^H - b$, implying that

$$(23.28) \quad b(t) = b^E \equiv \frac{1}{1-\beta} \left[\frac{(1-\beta(1-\sigma^L)) A^H + \beta(1-\sigma^H) A^L}{(1-\beta(\sigma^H - \sigma^L))} (1-\alpha) \alpha^{-1} \beta^{\alpha/(1-\alpha)} \bar{L} \right]$$

is sufficient to ensure zero equilibrium wages.

In this oligarchic equilibrium, aggregate output is:

$$(23.29) \quad Y^E(t) = \alpha^{-1} \beta^{\alpha/(1-\alpha)} [\mu(t) A^H + (1 - \mu(t)) A^L],$$

where $\mu(t) = \sigma^H \mu(t-1) + \sigma^L (1 - \mu(t-1))$ as given by (23.20), starting with some $\mu(0)$.

As noted above, if, as in our benchmark assumption, all individuals start with $e_i(-1) = 0$, then the equilibrium will feature $\mu(0) = 1$. In this case, and in fact, for any $\mu(0) > M$, $\mu(t)$ will be a decreasing sequence converging to M and aggregate output $Y^E(t)$ will also be decreasing over time with:

$$(23.30) \quad \lim_{t \rightarrow \infty} Y^E(t) = Y_\infty^E \equiv \alpha^{-1} \beta^{\alpha/(1-\alpha)} [A^L + M(A^H - A^L)].$$

Intuitively, the comparative advantage of the members of the elite in entrepreneurship gradually disappears because of the imperfect correlation between ability over time.

Nevertheless, it is also possible to imagine societies in which $\mu(0) < M$, because there is some other process of selection into the oligarchy in the initial period that is *negatively* correlated with skills in entrepreneurship. In this case, somewhat paradoxically, $\mu(t)$ and thus $Y^E(t)$ would be increasing over time. While interesting in theory, this case appears less relevant in practice, where we would expect at least some positive selection in the initial period, so that high-skill agents are more likely to become entrepreneurs at time $t = 0$ and $\mu(0) > M$.

Another important feature of the oligarchic equilibrium is that there is a high degree of (income) inequality. Wages are equal to 0, while entrepreneurs earn positive profits—in fact, each entrepreneur earns $y_i(t) \bar{L}$ (gross of investment expenses), where $y_i(t)$ depends on the current skill level of the entrepreneur. Since wages are equal to 0, total entrepreneurial earnings are equal to aggregate output. This contrasts with relative equality in democracy.

PROPOSITION 23.5. *Suppose that Condition 23.1 holds. Then, there exists a unique oligarchic equilibrium, with $\tau(t) = 0$ and $b(t) = b^E$ as given by (23.28). The equilibrium is sclerotic, with equilibrium wages $w^E(t) = 0$, and the fraction of high-skill entrepreneurs given by $\mu(t) = \sigma^H \mu(t-1) + \sigma^L(1 - \mu(t-1))$ starting with $\mu(0)$. Aggregate output is given by (23.29) and satisfies $\lim_{t \rightarrow \infty} Y^E(t) = Y_\infty^E$ as in (23.30). Moreover, as long as $\mu(0) > M$, aggregate output is decreasing over time.*

PROOF. See Exercise 23.8. □

23.3.4. Comparison Between Democracy and Oligarchy. Recall that our baseline assumption is that initial selection into entrepreneurship is on the basis of entrepreneurial skills, so $\mu(0) = 1$. Therefore, aggregate output in the initial period of the oligarchic equilibrium, $Y^E(0)$, is greater than the constant level of output in the democratic equilibrium, Y^D . In other words,

$$Y^D = \alpha^{-1} (\beta(1 - \hat{\tau}))^{\alpha/(1-\alpha)} A^H < Y^E(0) = \alpha^{-1} \beta^{\alpha/(1-\alpha)} A^H.$$

Therefore, oligarchy initially generates greater output than democracy, because it is protecting the property rights of entrepreneurs (whereas democracy is imposing distortionary taxes on entrepreneurs). However, the analysis also shows that, in this case, $Y^E(t)$ declines over time, while Y^D is constant. Consequently, the oligarchic economy may subsequently fall behind the democratic society. Whether it does so or not depends on whether Y^D is greater than Y_∞^E as given by (23.30). This will be the case if $(1 - \hat{\tau})^{\alpha/(1-\alpha)} A^H > A^L + M(A^H - A^L)$, or if

CONDITION 23.2.

$$(1 - \hat{\tau})^{\alpha/(1-\alpha)} > \frac{A^L}{A^H} + M \left(1 - \frac{A^L}{A^H} \right).$$

If Condition 23.2 holds, then at some point the democratic society will overtake (“leapfrog”) the oligarchic society.

As noted above, it is possible to imagine societies in which even in the initial period, there are “elites” that are not selected into entrepreneurship on the basis of their skills. In this case, we will typically have $\mu(0) < 1$. In the extreme case where there is negative selection into entrepreneurship in the initial period, $\mu(0) < M$. To analyze these cases, let us define

$$(23.31) \quad \bar{\mu}(0) \equiv \frac{(1 - \hat{\tau})^{\alpha/(1-\alpha)} - A^L/A^H}{1 - A^L/A^H}.$$

It can be verified that as long as $\mu(0) > \bar{\mu}(0)$, oligarchy will generate greater output than democracy in the initial period. Notice also that $\bar{\mu}(0) > M$ if and only if Condition 23.2 holds.

This discussion and inspection of Condition 23.2 establish the following result (proof in the text):

PROPOSITION 23.6. *Suppose that Condition 23.1 holds.*

- (1) *Suppose also that $\mu(0) = 1$. Then, at $t = 0$, aggregate output is higher in an oligarchic society than in a democratic society, that is, $Y^E(0) > Y^D$. If Condition 23.2 does not hold, then aggregate output in oligarchy is always higher than in democracy, that is, $Y^E(t) > Y^D$ for all t . If Condition 23.2 holds, then there exists t' such that for $t \leq t'$, $Y^E(t) \geq Y^D$ and for $t > t'$, $Y^E(t) < Y^D$, so that the democratic society leapfrogs the oligarchic society. Leapfrogging is more likely when $\hat{\tau}$, A^L/A^H and M are low.*
- (2) *Suppose next that $\mu(0) < 1$. If $\mu(0) > \max\{M, \bar{\mu}(0)\}$, then the results from part 1 apply. If Condition 23.2 holds and $\mu(0) < \bar{\mu}(0)$, then aggregate output in oligarchy, $Y^E(t)$, is always lower than that in democracy, Y^D . If Condition 23.2 does not hold and $\mu_0 < M$, then aggregate output in oligarchy, $Y^E(t)$, is always higher than that in democracy, Y^D .*

PROOF. See Exercise 23.9. □

This proposition implies that when $\mu(0)$ is not excessively low (when there is no *negative* correlation between initial entry into entrepreneurship and skills), an oligarchic society will start out as more productive than a democratic society, but will decline over time. It also shows that oligarchy is more likely to be relatively inefficient in the long run:

- (1) when $\hat{\tau}$ is low, meaning that democracy is unable to pursue highly populist policies with a high degree of redistribution away from entrepreneurs/capitalists. The parameter $\hat{\tau}$ may correspond both to certain institutional impediments limiting redistribution, or more interestingly, to the specificity of assets in the economy; with greater specificity, taxes will be limited, and redistributive distortions may be less important.
- (2) when A^H is high relative to A^L , so that the creative destruction process—the selection of high-skill agents for entrepreneurship—is important for the efficient allocation of resources.
- (3) M is low, so that a random selection contains a small fraction of high-skill agents, making oligarchic sclerosis highly distortionary. Alternatively, M is low when σ^H is low, so oligarchies are more likely to lead to low output in the long run when the efficient allocation of resources requires a high degree of “churning” in the ranks of entrepreneurs, which is another measure of the importance of creative destruction.

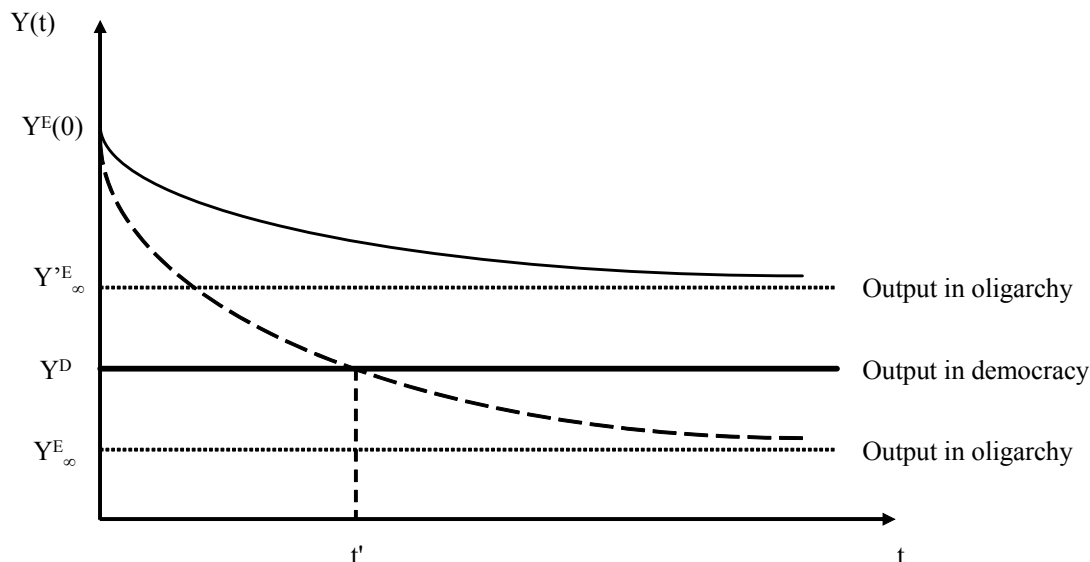


FIGURE 23.3. Dynamic comparison of output in oligarchy and democracy. The dashed line represents output in oligarchy when Condition 23.2 holds, and the solid line represents output in oligarchy when this condition does not hold.

On the other hand, if the extent of taxation in democracy is high and the failure to allocate the right agents to entrepreneurship only has limited costs, then an oligarchic society will generate greater output than a democracy in the long run.

These comparative static results may be useful in interpreting why, as discussed in Section 23.1, the Northeastern United States so conclusively outperformed the Caribbean plantation economies during the 19th century. First, the American democracy was not highly redistributive, corresponding to low $\hat{\tau}$ in terms of the model here. More important, the 19th century was the age of industry and commerce, where the allocation of high-skill agents to entrepreneurship appears to have been probably quite important, and potentially only a small fraction of the population were really talented as inventors and entrepreneurs. This can be thought of as a low value of A^L/A^H and M .

Figure 23.3 illustrates the case with $\mu(0) = 1$ (or $\mu(0) > \max\{M, \bar{\mu}(0)\}$), and depicts both the situation in which Condition 23.2 holds and the converse. The thick flat line shows the level of aggregate output in democracy, Y^D . The other two curves depict the level of output in oligarchy, $Y^E(t)$, as a function of time for the case where Condition 23.2 holds and for the case where it does not. Both of these curves asymptote to some limit, either Y_{∞}^E or Y_{∞}^E , which may lie below or above Y^D . The dashed curve shows the case where Condition 23.2 holds, so after date t' , oligarchy generates less aggregate output than democracy. When

Condition 23.2 does not hold, the solid curve applies, and aggregate output in oligarchy asymptotes to a level higher than Y^D .

The second part of the proposition also highlights the role of selection of individuals into entrepreneurship (and oligarchy) in the initial period. It shows that the results highlighted so far hold even if $\mu(0)$ is less than one, as long as it is not very small. On the other hand, if $\mu(0)$ is very small to start with, oligarchy may always generate less output than democracy, and in fact, if $\mu(0)$ starts out less than M , oligarchy may even have increasing level of output. A very low level of $\mu(0)$ may emerge if the oligarchy is founded by individuals that are talented in non-economic activities (e.g., by elites specialized in fighting in pre-modern times) and these non-economic talents are negatively correlated with entrepreneurial skills. Nevertheless, as noted above, a significant amount of positive selection on the basis of skills even in the initial period seems to be the more reasonable case.

On the basis of this analysis, the current model not only adds to the arguments so far that there is no unambiguous theoretical result on whether democracy or nondemocracy will generate greater growth, but it also highlights a different dimension of the tradeoff between different regimes—that related to the dynamics they imply. While democracy may create short-run distortions, it can lead to better long-run performance because it avoids *political sclerosis*—that is, incumbents becoming politically powerful and erecting entry barriers against new and better entrepreneurs. This model therefore suggests precisely the type of patterns we already discussed in Section 23.1; lack of a clear relationship between democracy and growth over the past 50 years combined with the examples of democracies that have been able to achieve industrialization during critical periods in the 19th century. In fact, a simple extension of the framework here provides additional insights that are useful in thinking about why democracies may be successful in preventing political sclerosis; the forces highlighted here also suggest that democracies are more “*flexible*” than oligarchies. In particular, Exercise 23.11 considers a simple extension of the framework here and demonstrates that democracies will typically be better able to adapt to the arrival of new technologies, because there are no incumbents with rents to protect, who can successfully block or slow down the introduction of new technology. This type of flexibility might be one of the more important advantages of democratic regimes.

Even though the model presented in this section provides a range of ideas and comparative static results that are useful for understanding the comparative development experiences of democratic and nondemocratic regimes, like the model discussed in the previous section, it focuses on the costs of democracy resulting from its more redistributive nature—in particular, it emphasizes that democratic regimes redistribute income away from the rich and the entrepreneurs towards the poorer segments of the society and this leads to distortions reducing income per capita. An alternative source of distortions in democracy, which will complement the mechanisms discussed here, will be the resistance of the elite against democratic redistributive policies, which will often lead to additional inefficiencies. This will be discussed in Section 23.5.

23.4. Understanding Endogenous Political Change

23.4.1. General Insights. The analysis so far has focused on the implications of different political institutions on economic growth and how their economic consequences shape the preferences of different agents over these political institutions. Why do institutions change? Returning to the model of the previous section, one possibility is that oligarchs will voluntarily give up power and institute a democracy. While this might be in their interest under some circumstances, it will generally be costly for them to give up their monopoly of political power and the economic rents that this brings. Not surprisingly, most institutional changes in practice do not happen voluntarily, but result from social conflict.

Consider, for example, the democratization of most Western European nations during the 19th and early 20th centuries or the democratization experience in Latin America during the 20th century. In both cases, democracy was not voluntarily granted by the existing elites, but resulted from the process of social conflict, in which those previously disenfranchised demanded political rights and in some cases were able to secure them. But how does this happen? A nondemocratic regime, by its nature, vests political power with a narrow group. Those who are excluded from this group, the non-elites, do not have the right to vote or nor do they have any voice in collective decisions. So how can they influence, the course of the political equilibrium and induce equilibrium political change? The answer to this question lies in drawing a distinction between *de jure* (formal) and *de facto* political power. *De jure* political power refers to power that originates from the political institutions in society and has been the form of political power on which we have so far focused. One may view it as the more “legitimate” type of political power. Political institutions determine who gets to vote, how representatives make choices and the general rules of collective decision-making in society. In Max Weber’s famous description, they also reserve “the legitimate use of violence” to the state (and to the actors that control the state). These different types of political powers are all of the *de jure* kind. However, there is another, equally important type of political power that features importantly in equilibrium political changes—*de facto* political power. The political power of protesters that marched against the existing regime before the First Reform Act in Britain in 1832 was not of the *de jure* kind. The law of the land did not empower them to influence the political course of actions—in fact, they were quite explicitly disenfranchised. But they had a different kind of power, emanating from their ability to solve the collective action problem and organize protests. This power was also supported by the fact that they were the majority in the society. This type of political power, which lives outside the political institutions, is *de facto* political power.

De facto political power is ever present around us. Civil wars, revolutions and social unrests are manifestations of the use of *de facto* political power by various groups. Military excursions are another example. More interesting for our purposes are the types of *de facto* political power that coexist with *de jure* political power in orderly (or semi-orderly societies). For example, in many Latin American countries governments are elected via democratic means, as it should be according to the political institutions that have specified the

distribution of de jure power in society, but at the same time there is ample fraud, vote buying and use of violence via paramilitaries and other organizations to influence the outcomes of elections. All of these fall within the category of the exercise of de facto political power. Then, there are grey areas. For example, the ability of the rich and well-organized groups to use money for campaign contributions or for lobbying, thus influencing the policy choices and platforms of politicians can be viewed as an example of the exercise of de facto power, though it can also be viewed as part of the regular functioning of political institutions, since in many societies, like the United States, lobbies are legal.

De facto political power is important for political change, since de jure political power itself will act as a source of *persistence*—not of change. For example, consider the model of the previous section. As noted above, the elite will be typically happy to maintain the oligarchic regime. If de jure power is the only source of power, the elite will be the only one with the decision-making powers in the society, and they are unlikely to change the political regime away from oligarchy towards democracy. However, if the non-elites had some source of power—which, by its nature, has to be de facto power—then, political change becomes a possibility. Perhaps in some periods, the non-elites will be able to solve their collective action problem and thus exercise enough pressure on the system to force some changes. In the extreme, they can induce the elite to disband oligarchy and transition to democracy, or they can themselves topple the oligarchic regime.

I will argue that the interaction between de jure and de facto political power is the most promising way to approach the analysis of equilibrium political change. Moreover, this interaction becomes particularly interesting when studied in a dynamic framework. This is for at least two reasons. First, most of the issues we are discussing are dynamic in nature—they refer to political *change*. Second, whether the distribution of de facto political power is permanent or changing stochastically over time has major consequences for the structure of political equilibrium. When a particular (disenfranchised) group has permanent (and unchanging) amount of de facto political power, it can use this at each date to demand concessions from those holding de jure political power. Such a situation may lead to an equilibrium without political change (though the equilibrium will have a very different distribution of resources because of the concessions induced by the de facto power of the disenfranchised group). Next consider a situation in which the de facto political power of the disenfranchised group is highly *transient*—in the sense that, they have been able to solve their collective action problem and can exercise de facto political power today, but it is unlikely that they will have the same type of power tomorrow. Then, the disenfranchised group cannot rely on the use of their de facto political power in the future to receive concessions. If they want concessions and redistribution of resources towards themselves in the future, they have to use their current power in order to secure such a change. This generally involves a change in political institutions as a way of changing the future distribution of de jure power. More explicitly, consider a situation in which a particular group of individuals know that today they have the power to change

institutions and create a playing field favoring themselves in the future, but they also understand that this de facto political power will be gone tomorrow. Thus any limited transfer of resources or other concessions made to them today will be either reversed or will be insufficient relative to the benefits from changing the playing field in their favor. It will therefore be precisely the transient nature of their de facto political power that will encourage them to take actions to change political institutions in order to cement their power more firmly (so that they can change their transient de facto political power into more durable de jure political power). This informal discussion therefore suggests a particular channel via which the interaction between de facto and de jure political power can lead to equilibrium changes in political institutions. I next give a historical example to illustrate this point further.

23.4.2. An Example. As a brief example, consider the development of property rights in Europe during the Middle Ages. There is broad agreement in the literature that lack of property rights for non-elite landowners, merchants and early industrialists was detrimental to economic growth during this epoch. Since political institutions at the time placed political power in the hands of Kings and various types of hereditary monarchies, such rights were largely decided by these monarchs. The monarchs often used their powers to expropriate producers, impose arbitrary taxation, renege on their debts and allocate the productive resources of society to their allies in return for economic benefits or political support. Consequently, economic institutions during the Middle Ages provided little incentive to invest in land, physical or human capital, or technology, and failed to foster economic growth. These economic institutions also ensured that the monarchs and their allies controlled a large fraction of the economic resources in society, solidifying their political power and ensuring the continuation of the political regime.

The 17th century witnessed major changes in the economic and political institutions that paved the way for the development of property rights and limits on monarchs' power, especially in England after the Civil War of 1642 and the Glorious Revolution of 1688, and in the Netherlands after the Dutch Revolt against the Hapsburgs. How did these major institutional changes take place? In England until the 16th century, the King also possessed a substantial amount of de facto political power, and leaving aside civil wars related to royal succession, no other social group could amass sufficient de facto political power to challenge the King. But changes in the English land market and the expansion of Atlantic trade in the 16th and 17th centuries gradually increased the economic fortunes, and consequently the de facto power of landowners and merchants opposed to the absolutist tendencies of the Kings.

By the 17th century, the growing prosperity of the merchants and the gentry, based both on internal and overseas, especially Atlantic, trade, enabled them to field military forces capable of defeating the King. This de facto power overcame the Stuart monarchs in the Civil War and Glorious Revolution, and led to a change in political institutions that stripped the King of much of his previous power over policy. These changes in the distribution of political power led to major changes in economic institutions, strengthening the property rights of both land and capital owners and spurring a process of financial and commercial expansion.

The consequence was rapid economic growth, culminating in the Industrial Revolution, and a very different distribution of economic resources from that in the Middle Ages.

This discussion poses, and also gives clues about the answers to, a crucial question: why do groups with political power want to change political institutions in their favor? In the context of the example above, why did the gentry and merchants use their *de facto* political power to change political institutions rather than simply implement the policies they wanted? The answer lies with the *transient* nature of *de facto* political power and the lack of *commitment* to future policies.

As already discussed in the previous chapter, the commitment problem arises because groups with political power cannot commit to not using their power to change the distribution of resources in their favor. For example, economic institutions that increased the security of property rights for land and capital owners during the Middle Ages would not have been credible as long as the monarch monopolized political power. He could promise to respect others' property rights, but then at some point, he would renege on his promise, as exemplified by the numerous financial defaults by medieval Kings. Credible secure property rights necessitated a reduction in the political power of the monarch. Although these more secure property rights would foster economic growth, they were not appealing to the monarchs who would lose their rents from predation and expropriation as well as various other privileges associated with their monopoly of political power. This is why the institutional changes in England as a result of the Glorious Revolution were not simply conceded by the Stuart Kings. James II had to be deposed for the changes to take place.

The reason why *de facto* political power is often used to change political institutions is closely related to its transient nature and to commitment problems. Individuals care not only about economic outcomes today but also in the future. In the example above, we presume that the gentry and merchants were interested in their profits and therefore in the security of their property rights, not only in the present but also in the future. Therefore, they would have liked to use their (*de facto*) political power to secure benefits in the future as well as the present. However, commitment problems make this difficult. If the gentry and merchants would have been sure to maintain their *de facto* political power, this would not have been a problem. And yet, *de facto* political power is often *transient*, for example because the collective action problems that are solved to amass this power are likely to resurface in the future, or other groups, especially those controlling *de jure* power, can become stronger. The commitment problems, in turn, imply that promises made today cannot be trusted and any change in policies and economic institutions that relies purely on *de facto* political power is likely to be reversed in the future. A plausible story for the nature of political changes in early modern Europe is therefore that when they had the transient political power, the English gentry, merchants and industrialists strove not just to change economic institutions in their favor following their victories against the Stuart monarchy, but they also sought, and managed, to alter the political institutions and the future allocation of *de jure* power in their favor. Using political power to change political institutions then emerges as a useful strategy

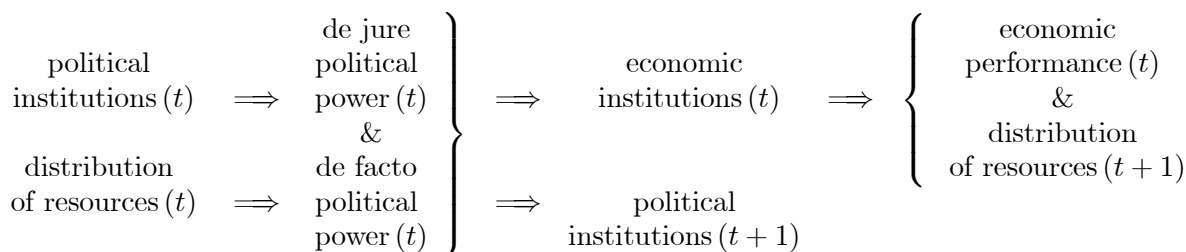
to make gains more durable. Consequently, political institutions are an important way in which future political power can be manipulated, enabling the holders of transient political power to solidify their current power and indirectly shape future, as well as present, economic institutions and outcomes.

23.4.3. Modeling. The discussion so far illustrated how we can use the interaction between *de facto* and *de jure* political power in order to study equilibrium political changes, and their implications for economic growth. While the discussion has given some clues about what the incentives of different parties with and without *de jure* political power will be in a dynamic game, it is so far unclear how one would construct models to analyze these forces and generate useful comparative statics. In this subsection, I suggest a general framework that is useful for thinking about the dynamic interactions between *de facto* and *de jure* political power. In the next subsection and the next section, I will illustrate this framework further.

Imagine a dynamic model in which there are two *state variables*, political institutions and the distribution of resources. For example, $P(t) \in \mathcal{P}$ denotes a specific set of political institutions in place at time t . This can be democracy or nondemocracy, parliamentary versus presidential system, different types of oligarchic institutions, and so on. The set $\mathcal{P}$ denotes the entire set of feasible political institutions relevant for the situation we are studying. Similarly, let $W(t) \in \mathcal{W}$ denote a variable encoding the distribution of resources at time t . For example, in a society consisting of two groups, the rich and the poor, this could be the relative incomes of the two groups. In a society with many individuals, it could be the density function of income or wealth. Again, $\mathcal{W}$ is the set of all possible distributions of resources. It is useful to think of both $P(t)$ and $W(t)$ as state variables for three reasons. First, they are relatively slow-changing, thus corresponding to the loose notion of a state variable. Second, they will typically be the payoff-relevant variables when we set up the problem as a dynamic game (and thus they will naturally be part of the “Markovian states”). Third and perhaps most important, these two variables will determine the two sources of political power essential for understanding equilibrium political change. The variable $P(t)$ will determine the distribution of *de jure* political power, which I denote by $J(t) \in \mathcal{J}$, in particular, it determines who has the right to vote and which politicians are subject to what types of constraints and what decisions they can take. The distribution of resources is not the only variable that affects *de facto* political power, but it is one of its main determinants. In particular, as already hinted in the discussion above, *de facto* political power is typically the result of the ability of certain groups to solve their collective action problem, or it emerges when certain groups have the resources to hire their own armies, paramilitaries and supporters, or simply use the money for lobbying and bribing. Let the distribution of *de facto* political power in the society at time t be $F(t) \in \mathcal{F}$. As in the beginning of Part 8, let us also denote economic institutions by $R(t) \in \mathcal{R}$, and let $Y(t) \in \mathcal{Y}$ be a measure of economic performance, such as income per capita or growth (though it could also include other performance-related variables, such as the level of poverty, health, human capital or other human development indicators).

A dynamic framework that is useful for thinking about political change and its implications for economic growth would consist of a mapping $\varphi : \mathcal{P} \times \mathcal{Z} \rightarrow \mathcal{J}$, which determines the distribution of de jure power at time t as a function of political institutions at time t , $P(t) \in \mathcal{P}$, as well as some potential stochastic elements, captured by $z(t) \in \mathcal{Z}$. It will also consist of a mapping determining the distribution of de facto power in a similar manner, $\phi : \mathcal{W} \times \mathcal{Z} \rightarrow \mathcal{F}$, where the types of stochastic elements influencing whether a particular group has de facto political power will be different than those affecting the distribution of de jure power, but we can summarize both of them with the variable $z(t) \in \mathcal{Z}$. Then, given the realization of $J(t) \in \mathcal{J}$ and $F(t) \in \mathcal{F}$, another mapping $\iota : \mathcal{J} \times \mathcal{F} \rightarrow \mathcal{R} \times \mathcal{P}$ determines both economic institutions today and also one of the future state variables, the political institutions tomorrow, $P(t+1) \in \mathcal{P}$. Put differently, the distribution of de facto and de jure political power regulates what types of economic institutions will be in place today and whether there will be political reform leading to changes in future de jure power (for example, a switch from nondemocracy to democracy so as to increase the future de jure power of the citizens who hold significant de facto power today). Finally, an economic equilibrium mapping $\rho : \mathcal{R} \rightarrow \mathcal{Y} \times \mathcal{W}$ determines both the economic performance variables and the distribution of economic resources. For example, if economic institutions correspond to competitive markets, they may lead to high wages and high output, and if they are repressive labor markets, they will lead to low wages, high profits, but perhaps lower output due to greater distortions they may be creating because of monopsony distortions or the induced misallocation of workers to tasks.

The next chart summarizes this discussion diagrammatically.



This framework and the associated diagram emphasize both the effects of economic institutions on economic performance and the distribution of resources—what we have tried to understand so far. But they have also introduced the dynamics of political power and political institutions. Of course, at this level of generality, such a dynamic framework is somewhat vacuous. It would only be useful and meaningful if we can put more content into the set of political institutions and the distribution of resources that need to be considered, derive the mappings φ , ϕ , ι and ε from economic interactions with sound microfoundations and then conduct useful comparative statics. This is a tall order, and a full dynamic model of the sort, that is able to deliver on all of these counts, does not currently exist. Nevertheless, a number of models focusing on political change and on the interaction between politics and economics exist, and these can be viewed through the lenses of this framework. This suggests that an abstract framework like the one presented here might be useful in emphasizing what the

important frontiers for research are and what types of models we may want to think about for furthering our understanding of political change and the relationship between political institutions and economic growth.

In the next subsection, I provide an informal discussion of one application of this framework, and then in the next section, I present another application in greater detail, because it will not only illustrate the basic ideas highlighted here, but also shed light on the economic tradeoffs between different political regimes.

23.4.4. Another Example: The Emergence of Democracy. The framework presented above is largely inspired by the models of the emergence of democracy developed in Acemoglu and Robinson (2000a, 2006a). James Robinson and I constructed a model of the emergence of democracy based on social conflict between the elite, who originally hold *de jure* political power, and the masses, that are initially without *de jure* political power, but can sometimes solve their collective action problems and gather significant *de facto* political power. A historical case that illustrates the main issues emphasized by in this work is the emergence of democracy in 19th century Europe. Many European nations during the 19th century were run by small elites. Most had elected legislatures, often descendents of medieval parliaments, but the franchise was highly restricted to males with relatively large amounts of assets, incomes or wealth. As the century and the Industrial Revolution progressed, this political monopoly was challenged by the disenfranchised who where able to exercise their *de facto* power (resulting from their sheer numbers) engaged in collective action to force political change.

In response to these developments, the elites responded in three ways. The first response was to use repression in order to prevent social unrest, for example, as was the case in much of Europe during the revolutionary waves of 1848. The second response, used successfully by Bismarck in Germany, was to offer economic concessions to buy off or co-opt part of the opposition. Finally, if neither repression nor concessions were attractive or effective, the third response involved expanding the franchise and giving political power to the previously disenfranchised—they created the precedents of modern democracy.

The first important move towards democracy in Europe came in Britain with the First Reform Act of 1832. This act removed many of the worst inequities under the old electoral system, in particular the “rotten boroughs,” where several members of parliament were elected by very few voters. The 1832 reform also established the right to vote based uniformly on the basis of property and income. The reform was passed in the context of rising popular discontent at the existing political status quo in Britain.

By the 1820s the Industrial Revolution was well under way and the decade prior to 1832 saw continual rioting and popular unrest. Notable were the Luddite Riots from 1811-1816, the Spa Fields Riots of 1816, the Peterloo Massacre in 1819 and the Swing Riots of 1830. Another catalyst for the reforms was the July revolution of 1830 in Paris. Much of this was led and orchestrated by the new middle-class groups who were being created by the spread of industry and the rapid expansion of the British economy. For example, under the pre-1832

system neither Manchester nor Sheffield had any members of the House of Commons, even though they were major centers of industrial growth and new wealth.

There is little this agreement among historians that the motive for the 1832 Reform was to avoid social disturbances. The 1832 Reform Act increased the total electorate from 492,700 to 806,000, which represented about 14.5% of the adult male population. Yet, the majority of British people could still not vote, and the elite still had considerable scope for patronage, since 123 constituencies still contained less than 1,000 voters. There is also evidence of continued corruption and intimidation of voters until the Ballot Act of 1872 and the Corrupt and Illegal Practices Act of 1883. The Reform Act therefore did not create mass democracy; it was instead designed as a strategic concession. As a result, parliamentary reform was still very much on the agenda in the middle of the century, especially with the greater demands of the Chartist movement. Despite its various achievements and fame, the Chartist movement could not secure a major democratic reform, in part, because the *de facto* political power of the disenfranchised groups was not strong enough to force reform. This, however, changed in the latter half of the 19th century, partly because of the sharp business cycle downturn that caused significant economic hardship and the increased threat of violence. Also significant was the founding of the National Reform Union in 1864 and the Reform League in 1865. The Hyde Park Riots of July 1866 provided the most immediate catalyst. Following these events, major electoral reform got underway with the Second Reform Act in 1867, which increased the total electorate from 1.36 million to 2.48 million, and made working class voters the majority in all urban constituencies. The electorate was doubled again by the Third Reform Act of 1884, which extended the same voting regulations that already existed in the boroughs (urban constituencies) to the counties (rural constituencies). The Redistribution Act of 1885 removed many remaining inequalities in the distribution of seats and from this point on Britain only had single member electoral constituencies (previously many constituencies had elected two members—the two candidates who gained the most votes). After 1884, about 60% of adult males were enfranchised. Once again social disorder appears to have been an important factor behind the 1884 act.

The Reform Acts of 1867-1884 were a turning point in the history of the British state. Economic institutions also began to change. In 1871 Gladstone reformed the civil service, opening it to public examination, making it meritocratic. Liberal and Conservative governments introduced a considerable amount of labor market legislation, fundamentally changing the nature of industrial relations in favor of workers. During 1906-1914, the Liberal Party, under the leadership of Asquith and Lloyd George, introduced the modern redistributive state into Britain, including health and unemployment insurance, government financed pensions, minimum wages and a commitment to redistributive taxation. As a result of the fiscal changes, taxes as a proportion of National Product more than doubled in the 30 years following 1870, and then doubled again. In the meantime, the progressivity of the tax system also increased. Finally, there is also a consensus amongst economic historians that inequality in Britain fell after the 1870s. At the same time as the fiscal reforms were taking place, there

were also major educational reforms changing the distribution of resources and distribution of opportunities in the British society in a major way. The Education Act of 1870 committed the government to the systematic provision of universal education for the first time. The school leaving age was set at 11 in 1893. In 1899, it was further increased to 12 and special provisions for the children of needy families were introduced. As a result of these changes, the proportion of 10-year olds enrolled in school that stood at 40 percent in 1870 increased to 100 percent in 1900. Finally, a further act in 1902 led to a large expansion in the resources for schools and introduced the grammar schools, which subsequently became the foundation of secondary education in Britain.

Overall, the picture that emerges from British political history is clear. Beginning in 1832, when Britain was governed by the relatively rich, primarily rural aristocracy, a series of strategic concessions were made over an 86 year period. These concessions were aimed at incorporating the previously disenfranchised into politics since the alternative was seen to be social unrest, chaos and possibly revolution. The concessions were gradual because in 1832, social peace could be purchased by buying off the middle classes. Moreover, the effect of the concessions was diluted by the specific details of political institutions, particularly the continuing unrepresentative nature of the House of Lords. Although challenged during the 1832 reforms, the House of Lords provided an important bulwark for the wealthy against the potential of radical reforms emanating from a democratized House of Commons. Later, as the working classes reorganized through the Chartist movement and through trade unions, further concessions had to be made. World War I and the fallout from it sealed the final offer of full democracy.

Faced with the threat of revolt and social chaos, political elites may also attempt to avoid giving away their political power by making concessions, such as income redistribution or other policies that favor non-elites and the disenfranchised. However, because the promise of concessions is typically non-credible when threats are transient, such promises are typically insufficient to defuse social unrest. Democratization can then be viewed as a credible commitment to the disenfranchised. In particular, democratization is a credible commitment to future redistribution because it redistributes *de jure* political power away from the elites to the masses. In democracy, the poorer segments of the society would be more powerful, and could vote and use their *de jure* political power to implement economic institutions and policies consistent with their interests. Therefore, democratization was a way of transforming the transitory *de facto* power of the disenfranchised poor into more durable *de jure* political power.

The above account of events makes it quite clear that democracy in many Western societies, and particularly in Britain, did not emerge from the voluntary acts of an enlightened elite. Democracy was, in many ways, forced on the elite, because of the threat of revolution. Nevertheless, democratization was not the only potential outcome in the face of pressure from the disenfranchised, or even in the face of the threat of revolution. Many other countries faced the same pressures and political elites decided to repress the disenfranchised rather than make

concessions to them. This happened with regularity in Europe in the 19th century, though by the turn of the 20th century most West European nations had accepted that democracy was inevitable. Repression lasted much longer as the favorite response of elites in Latin America, and it is still the preferred option for current political elites in China or Burma. And yet, repression is costly not only for the repressed, but also for the elites. Therefore, faced with demands for democracy political elites face a tradeoff. If they grant democracy, then they lose power over policy and face the prospect of possibly radical redistribution. On the other hand, repression risks destroying assets and wealth. In the urbanized environment of 19th century Europe (Britain was 70% urbanized at the time of the Second Reform Act), the disenfranchised masses were relatively well organized and therefore difficult to repress. Moreover, industrialization had led to an economy based on physical, and increasingly human, capital. Such assets are easily destroyed by repression and conflict, making repression an increasingly costly option for elites. In contrast, in predominantly agrarian societies like many parts of Latin America earlier in the century or current-day Burma, physical and human capital are relatively unimportant and repression is easier and cheaper. Moreover, not only is repression cheaper in such environments, democracy is potentially much worse for the elites because of the prospect of radical land reform. Since physical capital is much harder to redistribute, elites in Western Europe found the prospect of democracy much less threatening.

So far I have offered a verbal account how one might develop a theoretical model of the democratization process in line with the abstract framework of the previous subsection. Once the main ideas are understood, a formal framework is not difficult to construct. The following is a simplified version of framework we considered in Acemoglu and Robinson (2006a)—see Exercise 23.13. The society consists of two groups, the elite and the masses. Political power is initially in the hands of the elite, but the masses are more numerous. Thus if there is ever democratization, the masses will become politically more powerful and dictate the policies. All individuals are infinitely lived and the elite are richer than the masses. Because the society starts as a nondemocracy, *de jure* power is in the hands of the elite. Let us suppose that the only policy choice is a redistributive tax, τ , the proceeds of which are distributed lump sum. The elite prefer zero taxation, $\tau = 0$, since they are richer and any taxation will redistribute income away from them to the poorer masses.

Let us imagine that while *de jure* power in nondemocracy lies with the elite, the poor may have *de facto* political power. In particular, suppose that with probability q in each period, the masses are able to solve their collective action problem and can threaten to undertake a revolution. A revolution is very costly for the elite, but generates only limited gains for the masses. These limited gains may nonetheless be better than living under elite control and the inequitable distribution of resources that this involves. So when they are able to solve their collective action problem (with probability q), the *revolution constraint* of the masses becomes binding. In this case, the rich need to take some action and make concessions to avoid a revolution.

As in the historical account I provided in the previous subsection, the elite in the theory also have three options to defuse the revolutionary threat. The first is to make concessions through redistributive policies today. This will work if q is high. In the limit, where $q = 1$, the masses can undertake a revolution in each date, thus the rich can credibly commit to making redistribution towards them at each date, because if they fail to do so, the masses can immediately undertake a revolution, which is costly for the elite. However, the same strategy does not work when q is small. Consider the polar case where $q \rightarrow 0$. In this case, the masses essentially expect never to have the same type of de facto political power in the future. Presuming that the amount to redistribution that the elite can give to the masses during a particular period is limited, they will not be satisfied by temporary concessions. If temporary concessions are not sufficient, then the elite may want to use repression. Repression will be successful if the revolutionary threat is not very well-organized and it will be profitable for the rich elite if they have a lot to lose from democratization. Thus repression will be the action of choice for elites that fear major redistribution under democracy (which may be in the form of fiscal taxation or land reform), such as the land-based elites in Central America and Burma. But in a highly urbanized and industrialized society like Britain, where the costs repression will be significant and the elite would have less to fear from democratization, the third option, which is enfranchisement, becomes an attractive choice. This third option involves the elite changing the political system and manufacturing a transition to democracy. As long as this change in institutions is credible, the distribution of de jure power has changed and transferred at least some of the decision-making power to the masses. With their newly-gained decision-making power, the masses know that they can choose policies in the future that will create a more equitable distribution of resources for themselves and will typically be happy to accept democratic institutions instead of a revolution that is costly for themselves as well. This shows how one can build a dynamic model of endogenous changes in political institutions.

Compared to the abstract framework in the previous subsection, the model described here is stripped down (and to save space, I have not even provided the equations to establish the main claims). First, the distribution of resources is no longer a state variable (it is constant and does not affect transitions or the distribution of political power). In addition, de jure political power is simply a nonstochastic outcome of political institutions; in nondemocracy, the elite make the decisions; and in democracy, there is one person one vote, and the masses, thanks to their majority, become the decisive voters. Finally, there are very limited economic decisions. The only relevant decision is one of taxation. Thus, in its current form, this is not a satisfactory framework for analyzing the impact of political institutions on economic institutions or the relationship between political regimes and economic growth. In Acemoglu and Robinson (2006a), we presented extensions of the framework, which go some way towards a framework for the analysis of economic institutions and economic growth. Instead of discussing these extensions, in the next section I will present a related model, in which there is more explicit interaction between economic and political institutions, and it will again be

the sum total of de facto and de jure powers that will determine the evolution of equilibrium institutions.

23.5. Dynamics of Political and Economic Institutions: A First Look

23.5.1. Baseline Model. In this section, I discuss a model based on Acemoglu and Robinson (2007), which will feature both the interaction of de jure and de facto political power and also illustrate how democracy can be captured and can lead to poor economic outcomes for this reason. Interestingly, while the model of democratization described at the end of the previous section emphasized how the de facto political power of the citizens can affect equilibrium dynamics, here the emphasis will be on the de facto political power of the elite in democracy and how they can use this to capture democratic politics.

Consider the following infinite-horizon economy in discrete time, with the unique final good. The society is populated by a finite number L of citizens/workers and M elites. Let me assume to simplify the analysis here that citizens are significantly more numerous than the elites, loosely written as $L \gg M$. What the exact relative sizes of the two groups need to be for the main results to apply is discussed in greater detail in Exercise 23.14 below.

Let us use $h \in \{E, C\}$ to denote whether an individual is from the elite or a citizen, and $\mathcal{E}$ and $\mathcal{C}$ to denote the set of elites and citizens, respectively. All agents have the usual risk-neutral preferences given by

$$(23.32) \quad \sum_{t=0}^{\infty} \beta^t c_i^h(t),$$

where $c_i^h(t)$ denotes consumption of agent i from group $h \in \{E, C\}$ at time t in terms of the unique final good and $\beta \in (0, 1)$ is the common discount factor.

Each citizen owns one unit of labor, which they supply inelastically. Each member of the elite $i \in \mathcal{E}$ has access to a linear production function to produce the unique private good with constant marginal productivity of $A > 0$. Let us consider two different *reduced-form economic institutions*. In the first, labor markets are *competitive* and I index these institutions by the subscript c (indicating “pro-citizen” or “competitive”). Let $I(t) \in \{e, c\}$ denote the institutional choice in period t . Given the production technology each elite will make zero returns and each citizen will receive their marginal product of labor, A . When there are competitive labor markets, $I(t) = c$, the wage rate (and the wage earnings of each citizen) is:

$$(23.33) \quad w_c \equiv A.$$

The return to a member of the elite with competitive markets is similarly

$$(23.34) \quad R_c \equiv 0.$$

The alternative set of economic institutions favor the elite and are *labor repressive* ($I(t) = e$) and allow the elite to use their political power to reduce wages below competitive levels. Let us parameterize the distribution of resources under labor repression as follows: $\lambda < 1$

denotes the share of national income accruing to citizens and $\delta \in [0, 1)$ is the fraction of potential national income, AL , that is lost because of the inefficiency of labor repression. For instance, $\delta > 0$ may result from standard monopsony distortions in the labor market. Note, however, that none of the results presented in this paper depend on the value of δ . The case where $\delta = 0$ would correspond to a situation in which there is no distortion from labor repression and the choice of economic institutions is purely redistributive. Alternatively, one could also consider the case in which $\delta < 0$, so that economic institutions favored by the elite are more “efficient” than those preferred by the citizens. However, given the emphasis on “labor repressive” institutions and the focus on whether democracy may fail to generate greater income per capita even when it is more efficient, the case of $\delta > 0$ is more relevant. A straightforward implication of this assumption is that, when economic institutions are labor repressive, there will be lower income per capita and thus worse economic performance.

Another reason for introducing the parameters δ and λ is that the model will have interesting comparative statics with respect to these parameters. For now, it suffices to start with the levels of factor prices under different economic institutions as functions of δ and λ . In particular, factor prices as functions of economic institutions are

$$(23.35) \quad w_e \equiv \lambda(1 - \delta) A,$$

and

$$(23.36) \quad R_e \equiv (1 - \lambda)(1 - \delta) \frac{AL}{M}.$$

Factor prices can then be written as a function of economic institutions as $w(I(t) = e) = w_e$, $R(I(t) = e) = R_e$, $w(I(t) = c) = w_c$ and $R(I(t) = c) = R_c$. For future reference, let us also define

$$(23.37) \quad \begin{aligned} \Delta R &\equiv R_e - R_c \\ &= (1 - \lambda)(1 - \delta) \frac{AL}{M} > 0, \end{aligned}$$

and

$$(23.38) \quad \begin{aligned} \Delta w &\equiv w_c - w_e \\ &= (1 - \lambda(1 - \delta)) A > 0 \end{aligned}$$

as the gains to the elite and the citizens from their more preferred economic institutions. Since the citizens are significantly more numerous, that is, $L \gg M$, (23.37) and (23.38) imply that $\Delta R \gg \Delta w$.

There are two possible political regimes, democracy and nondemocracy, denoted respectively by D and N . The distribution of de jure political power will vary between these two regimes. At time t , the (payoff-relevant) “state” of this society will be represented by $s(t) \in \{D, N\}$, which designates the political regime that applies at that date. Regardless of the political regime (state), the identities of the elites and the citizens do not change.

Overall, in line with the discussion in the previous section, overall political power is determined by the interaction of de facto and de jure political power. Both groups can invest

to garner further de facto political power. In particular, suppose that elite $i \in \mathcal{E}$ spends an amount $\theta_i(t) \geq 0$ as a contribution to activities increasing their group's de facto power. Then, total elite spending on such activities will be $\sum_{i \in \mathcal{E}} \theta_{i,s}(t)$ when the political state is s , and let us assume that their de facto political power is

$$(23.39) \quad P_s^E(t) = \phi_s^E \sum_{i \in \mathcal{E}} \theta_{i,s}(t),$$

where $\phi_s^E > 0$ and dependence on the state $s \in \{D, N\}$ is made explicit to emphasize that investments in de facto power by the elite may be less effective in democracy. The superscript E distinguishes it from the corresponding parameter for the citizens.

Citizens' power comes from three distinct sources. First, they can also invest in their de facto political power. Second, because citizens are more numerous, they may sometimes solve their collective action problem and exercise additional de facto political power. Let us assume that this second source of de facto political power is stochastic and fluctuates over time. The reasoning underlying this assumption is similar to that given in the previous section for why de facto political power resulting from solving the collective action problem is often transient. These fluctuations will cause equilibrium changes in political institutions. Finally, again because they are more numerous, citizens will have greater power in democracy than in nondemocracy. Overall, the power of the citizens when citizen $i \in \mathcal{C}$ spends an amount $\theta_{i,s}(t) \geq 0$ is

$$(23.40) \quad P_s^C(t) = \phi_s^C \sum_{i \in \mathcal{C}} \theta_{i,s}(t) + \omega(t) + \eta I(s(t) = D),$$

where $\phi_s^C > 0$, $\omega(t)$ is a random variable drawn independently and identically over time from a given distribution $F[\cdot]$, $I(s = D) \in \{0, 1\}$ is an indicator function for $s = D$, and η is a strictly positive parameter measuring citizens' de jure power in democracy. Equation (23.40) implies that in democracy the political power of the citizens shifts to the right in the sense of first-order stochastic dominance. To simplify the discussion, let me impose the following assumptions on F : F is defined over $(\underline{\omega}, \infty)$ for some $\underline{\omega} < 0$, is everywhere strictly increasing and twice differentiable (so that its density f and the derivative of the density, f' , exist everywhere). Moreover, $f[\omega]$ is single peaked (in the sense that there exists ω^* such that $f'[\omega] > 0$ for all $\omega < \omega^*$ and $f'[\omega] < 0$ for all $\omega > \omega^*$) and satisfies $\lim_{\omega \rightarrow \infty} f[\omega] = 0$.

Let us also introduce the variable $\pi(t) \in \{e, c\}$ to denote whether the elite have more (total) political power at time t . In particular, when $P_s^E(t) \geq P_s^C(t)$, $\pi(t) = e$ and the elite have more political power and will make the key decisions. In contrast, whenever $P_s^E(t) < P_s^C(t)$, $\pi(t) = c$ and citizens have more political power, and they will make the key decisions.

Finally, suppose that the group with greater political power will decide both economic institutions at time t , $I(t)$, and the state variable tomorrow (the political regime), $s(t+1)$. Moreover, let us assume that when the elite have more political power, a representative elite agent makes the key decisions, and when citizens have more political power, a representative

citizen does so. Since the political preferences of all elites and all citizens are the same, these representative agents will always make the decisions favored by their group.

Summarizing the timing of events, at each date t , the society starts with the state variable $s(t) \in \{D, N\}$. Then:

- (1) Each elite agent $i \in \mathcal{E}$ and each citizen $i \in \mathcal{C}$ simultaneously chooses how much to spend to acquire de facto political power for their group, $\theta_i(t) \geq 0$, and $P^E(t)$ is determined according to (23.39).
- (2) The random variable $\omega(t)$ is drawn from the distribution F , and $P^C(t)$ is determined according to (23.40).
- (3) If $P^E(t) \geq P^C(t)$ (i.e., $\pi(t) = e$), a representative elite agent chooses $(I(t), s(t+1))$, and if $P^E(t) < P^C(t)$ (i.e., $\pi(t) = c$), a representative citizen chooses $(I(t), s(t+1))$.
- (4) Given $I(t)$, $R(t)$ and $w(t)$ are determined and paid to elites and citizens respectively, and consumption takes place.

Let us first focus on the symmetric MPE of this game (the results with non-symmetric MPE and SPE are discussed in Exercise 23.15). As usual, a MPE imposes the restriction that equilibrium strategies are mappings from payoff-relevant states, which here only include $s \in \{D, N\}$, and since we formulate the model recursively we drop time subscripts from now on. In a MPE strategies are not conditioned on the past history of the game over and above the influence of this past history on the payoff-relevant state s . A MPE consists of contribution functions $\{\theta_{i,s}(t)\}_{i \in \mathcal{E}}$ for each elite agent as a function of the political state, a corresponding vector of functions $\{\theta_{i,s}(t)\}_{i \in \mathcal{C}}$ for the citizens, and decision variables, $I(\pi)$ and $s'(\pi)$ as a function of the state s and $\pi \in \{e, c\}$, and equilibrium factor prices as given by (23.33)-(23.36). Here the function $I(\pi)$ determines the equilibrium decision about labor repression conditional on who has power and the function $s'(\pi) \in \{D, N\}$ determines the political state at the start of the next period. Symmetric MPE in addition imposes the requirement that contribution functions take the form θ_s^E and θ_s^C , and thus they do not depend on the identity of the individual elite or citizen, $i \in \mathcal{E} \cup \mathcal{C}$.

As usual, MPE can be characterized by backward induction within the stage game at some arbitrary date t , given the state $s \in \{D, N\}$, and taking future plays (as functions of future states) as given. Clearly, whenever $\pi = e$ so that the elite have political power, they will choose economic institutions that favor them ($I(e) = e$) and a political system that gives them more power in the future ($s'(e) = N$). In contrast, whenever citizens have political power, $\pi = c$, they will choose $I(c) = c$ and $s'(c) = D$. This implies that choices over economic institutions and political states are straightforward. Moreover the determination of market prices under different economic institutions has already been specified above by eq.'s (23.33)-(23.36). The only remaining decisions are the contributions of each agent to their de facto power, $\theta_{i,s}(t)$ for $i \in \mathcal{E} \cup \mathcal{C}$ and $s \in \{D, N\}$. A symmetric MPE can thus be summarized by two pairs of contribution vectors $\theta^E = (\theta_D^E, \theta_N^E)$ and $\theta^C = (\theta_D^C, \theta_N^C)$. The MPE can be characterized by writing the payoff to agents recursively, and for this reason, I denote the

equilibrium value of an elite agent in state $s \in \{D, N\}$ by V_s^E (V_D^E for democracy and V_N^E for nondemocracy).

Since we are focusing on symmetric MPE, suppose that all other elite agents, except $i \in \mathcal{E}$, have chosen a level of contribution to de facto power equal to θ_s^E and all citizens have chosen a contribution level θ_s^C . Consequently, when agent $i \in \mathcal{E}$ chooses $\theta_i(t)$, the total power of the elite will be

$$P^E(\theta_i, \theta_s^E, \theta_s^C | s) = \phi_s^E((M-1)\theta_s^E + \theta_i).$$

The elite will have political power if

$$(23.41) \quad P^E(\theta_i, \theta_s^E, \theta_s^C | s) \geq \phi_s^C L \theta_s^C + \eta I(s = D) + \omega(t).$$

Expressed differently, the probability that the elite have political power in state $s \in \{N, D\}$ is

$$(23.42) \quad p(\theta_i, \theta_s^E, \theta_s^C | s) = F[\phi_s^E((M-1)\theta_s^E + \theta_i) - \phi_s^C L \theta_s^C - \eta I(s = D)].$$

As noted above, backward induction within the stage game implies that $I(e) = e$, $I(c) = c$, $s'(e) = N$ and $s'(c) = D$. Thus returns to the citizens and the elite will be w_e and R_e as given by (23.35) and (23.36) when $\pi = e$, and w_c and R_c as in (23.33) and (23.34) when $\pi = c$. Incorporating these best responses and using the one-step-ahead deviation principle (see Appendix Chapter C), we can write the payoff of an elite agent i recursively as follows:

$$(23.43) \quad V^E(N | \theta^E, \theta^C) = \max_{\theta_i \geq 0} \{-\theta_i + p(\theta_i, \theta_N^E, \theta_N^C | N)(R_e + \beta V^E(N | \theta^E, \theta^C)) \\ (1 - p(\theta_i, \theta_N^E, \theta_N^C | N))(R_c + \beta V^E(D | \theta^E, \theta^C))\}.$$

This equation incorporates the fact that with probability $p(\theta_i, \theta_N^E, \theta_N^C | N)$ the elite will remain in power and choose $I = e$ and $s' = N$, and with the complementary probability, the citizens will come to power and choose $I = c$ and $s' = D$. Finally, this expression also makes use of the one-step-ahead deviation principle in writing the continuation values as $V^E(N | \theta^E, \theta^C)$ and $V^E(D | \theta^E, \theta^C)$, that is, it restricts attention to symmetric MPE after the current period, where all citizens and elites choose the contribution levels given by the vectors θ^C and θ^E .

Since F is differentiable, $p(\theta_i, \theta_N^E, \theta_N^C | N)$ is also differentiable. Moreover, the continuation values $V^E(D | \theta^E, \theta^C)$ and $V^E(N | \theta^E, \theta^C)$ are taken as given, so the first-order necessary condition for the optimal choice of θ_i by elite agent i can be written as

$$(23.44) \quad \phi_N^E f[\phi_N^E((M-1)\theta_N^E + \theta_i) - \phi_N^C L \theta_N^C] [\Delta R + \beta \Delta V^E] \leq 1,$$

and $\theta_i \geq 0$, with complementary slackness, where recall that ΔR is defined in (23.37), f is the density function of the distribution function F , and

$$\Delta V^E \equiv V^E(N | \theta^E, \theta^C) - V^E(D | \theta^E, \theta^C)$$

is the difference in value between nondemocracy and democracy for an elite agent in the symmetric MPE. Intuitively, (23.44) requires the cost of one more unit of investment in de facto political power to be no less than the benefit. The benefit is given by the increased

probability that the elite will control politics induced by this investment, $\phi^E(N)$, times the density of the F function evaluated at the equilibrium investments, multiplied by the benefit from controlling politics, which is the current benefit ΔR plus the discounted increase in continuation value, $\beta \Delta V^E$. In addition, the second-order sufficient condition is $f'[\phi_N^E((M-1)\theta_N^E + \theta_i) - \phi_N^C L \theta_N^C] < 0$.⁵ For future reference, let us also introduce the notation that $\theta_i \in \Gamma^E[\theta^E, \theta^C | N]$ if θ_i is a solution to (23.44) that satisfies the second-order condition.

Similarly, the value function for a citizen when the initial political state is $s = N$ is

$$(23.45) \quad V^C(N | \theta^E, \theta^C) = \max_{\theta_i \geq 0} \{ -\theta_i + p_0(\theta_i, \theta_N^E, \theta_N^C | N) (w_e + \beta V^C(N | \theta^E, \theta^C)) \\ (1 - p_0(\theta_i, \theta_N^E, \theta_N^C | N)) (w_c + \beta V^C(D | \theta^E, \theta^C)) \},$$

which is very similar to (23.43) except that the labor market rewards are now given by w_e and w_c instead of R_e and R_c , and the probability that $\pi = e$ is now given by the function

$$(23.46) \quad p_0(\theta_i, \theta_s^E, \theta_s^C | s) = F[\phi_s^E M \theta_s^E - \phi_s^C ((L-1)\theta_s^C + \theta_i) - \eta I(s = D)],$$

which is the probability that the elite have more power than the citizens in state $s \in \{D, N\}$, when all elite agents choose investment in de facto power, θ_s^E , all citizens except i choose θ_s^C , and individual i chooses θ_i . The first-order necessary condition is similar to (23.44) and can be written as

$$(23.47) \quad \phi_N^C f[\phi_N^E M \theta_N^E - \phi_N^C ((L-1)\theta_N^C + \theta_i)] [\Delta w + \beta \Delta V^C] \leq 1$$

and $\theta_i \geq 0$ with complementary slackness, and

$$\Delta V^C \equiv V^C(D | \theta^E, \theta^C) - V^C(E | \theta^E, \theta^C).$$

The interpretation of this condition is the same as that of (23.44). The second-order sufficient condition is $f'[\phi_N^E M \theta_N^E - \phi_N^C ((L-1)\theta_N^C + \theta_i)] > 0$. If θ_i is a solution to (23.47), we denote this by $\theta_i \in \Gamma^C[\theta^E, \theta^C | N]$.

By analogy, the value function for the elite in democracy is given by:

$$(23.48) \quad V^E(D | \theta^E, \theta^C) = \max_{\theta_i \geq 0} \{ -\theta_i + p(\theta_i, \theta_D^E, \theta_D^C | D) (R_e + \beta V^E(N | \theta^E, \theta^C)) \\ + (1 - p(\theta_i, \theta_D^E, \theta_D^C | D)) (R_c + \beta V^E(D | \theta^E, \theta^C)) \},$$

where $p(\theta_i, \theta_D^E, \theta_D^C | D)$ is again given by (23.42). The first-order necessary condition for the investment of an elite agent in democracy then becomes:

$$(23.49) \quad \phi_D^E f[\phi_D^E ((M-1)\theta_D^E + \theta_i) - \phi_D^C L \theta_D^C - \eta] [\Delta R + \beta \Delta V^E] \leq 1,$$

and $\theta_i \geq 0$, again with complementary slackness and with the second-order condition $f'[\phi_D^E ((M-1)\theta_D^E + \theta_i) - \phi_D^C L \theta_D^C - \eta] < 0$. We write $\theta_i \in \Gamma^E[\theta^E, \theta^C | D]$ if θ_i solves

⁵This condition with strict inequality is sufficient, while with a weak inequality, it would be necessary but not sufficient. I impose the sufficient condition throughout to simplify the discussion.

(23.49) and satisfies the second-order condition. Finally, for the citizens in democracy,

$$(23.50) \quad V^C(D | \theta^E, \theta^C) = \max_{\theta_i \geq 0} \left\{ -\theta_i + p_0(\theta_i, \theta_D^E, \theta_D^C | D) (w_e + \beta V^C(N | \theta^E, \theta^C)) \right. \\ \left. + (1 - p_0(\theta_i, \theta_D^E, \theta_D^C | D)) (w_c + \beta V^C(D | \theta^E, \theta^C)) \right\},$$

where $p_0(\theta_i, \theta_D^E, \theta_D^C | D)$ is given by (23.46). The first-order necessary condition is now

$$(23.51) \quad \phi_D^C f[\phi_D^E M \theta_D^E - \phi_D^C ((L-1)\theta_D^C + \theta_i) - \eta] [\Delta w + \beta \Delta V^C] \leq 1,$$

and $\theta_i \geq 0$, with complementary slackness and the second-order condition

$$f'[\phi_D^E M \theta_D^E - \phi_D^C ((L-1)\theta_D^C + \theta_i) - \eta] > 0. \text{ I denote solutions to this problem by } \theta_i \in \Gamma^C[\theta^E, \theta^C | D].$$

With these definitions, a symmetric MPE consists of contribution vectors $\theta^E = (\theta_N^E, \theta_D^E)$ and $\theta^C = (\theta_N^C, \theta_D^C)$ such that $\theta_N^E \in \Gamma^E[\theta^E, \theta^C | N]$, $\theta_D^E \in \Gamma^E[\theta^E, \theta^C | D]$, $\theta_N^C \in \Gamma^C[\theta^E, \theta^C | N]$ and $\theta_D^C \in \Gamma^C[\theta^E, \theta^C | D]$. In addition, policy, economic and political decisions $I(\pi)$ and $s'(\pi)$ must be such that, $I(e) = e$, $s'(e) = N$, $I(c) = c$ and $s'(c) = D$, and factor prices must be given by (23.33)-(23.36) as a function of $I \in \{e, c\}$.

The comparison of (23.44) and (23.47) immediately implies that these first-order conditions cannot generally hold as equalities both for the elite and the citizens. The comparison of (23.49) and (23.51) also leads to the same conclusion. In particular, “generically” only one of the two groups will invest to increase their de facto political power. Which group will be the one to invest in their political power? Loosely speaking, the answer is: whichever group has higher gains from doing so. Here the difference in numbers becomes important. In particular, recall that $L \gg M$ implies $\Delta R \gg \Delta w$. Consequently, it will be the elite that have more to gain from controlling politics and that will invest to increase their de facto power. This leads to the following proposition, with the proof left as an exercise, in which the exact threshold for the number of citizens relative to elites necessary for this result to hold has to be determined.

PROPOSITION 23.7. *Suppose that $L \gg M$. Then, any symmetric MPE involves $\theta_D^C = \theta_N^C = 0$.*

PROOF. See Exercise 23.14. □

This proposition simplifies the characterization of equilibrium, which is now reduced to the characterization of two investment levels, θ_N^E and θ_D^E , such that $\theta_N^E \in \Gamma^E[\theta^E, 0 | N]$ and $\theta_D^E \in \Gamma^E[\theta^E, 0 | D]$. Given Proposition 23.7, we can also write the equilibrium probabilities that the elite will have more political power as:

$$(23.52) \quad p_N \equiv F[\phi_N^E M \theta_N^E] \text{ and } p_D \equiv F[\phi_D^E M \theta_D^E - \eta].$$

Next, incorporating symmetry and the fact that $\theta_D^C = \theta_N^C = 0$ into the first-order conditions (23.44) and (23.49), and assuming the existence of an interior solution (with $\theta_N^E > 0$ and $\theta_D^E > 0$), the following two equations characterize interior equilibria:

$$(23.53) \quad \phi_N^E f[\phi_N^E M \theta_N^E] [\Delta R + \beta \Delta V^E] = 1,$$

and

$$(23.54) \quad \phi_D^E f [\phi_D^E M \theta_D^E - \eta] [\Delta R + \beta \Delta V^E] = 1.$$

The question is whether there exists such an interior solution. The following assumption imposes that the additional rents that the elite will gain from labor repressive institutions are sufficiently large and ensures that this is the case.

CONDITION 23.3.

$$\min \{ \phi_N^E f [0] \Delta R, \phi_D^E f [-\eta] \Delta R \} > 1.$$

PROPOSITION 23.8. *Suppose that $L \gg M$ and that Condition 23.3 holds. Suppose also that $\phi_N^E = \phi_D^E$. Then, there exists a unique symmetric MPE. This equilibrium involves $p_D = p_N \in (0, 1)$, so that the probability distribution over economic institutions is non-degenerate and independent of whether the society is democratic or nondemocratic.*

PROOF. Using Proposition 23.7, $\theta_D^C = 0$ and $\theta_N^C = 0$. Then, Condition 23.3 implies that $\theta_D^E = 0$ and $\theta_N^E = 0$ cannot be part of an equilibrium. Since $f[\omega]$ is continuous and $\lim_{\omega \rightarrow \infty} f[\omega] = 0$, both conditions (23.53) and (23.54) must hold as equalities for some interior values of θ_D^E and θ_N^E establishing existence of an equilibrium. The result that $p_D = p_N > 0$ then follows immediately from the comparison of these two equalities, which establishes (23.55). The fact that $p_D = p_N < 1$ follows from the assumption on F , which implies that it is strictly increasing throughout its support, so for any interior θ_D^E and θ_N^E , $F[\phi^E M \theta_D^E - \eta] = F[\phi^E M \theta_N^E] < 1$. In addition, again from the assumption on F (that $f[\omega]$ is single peaked), only a unique pair of θ_D^E and θ_N^E could satisfy (23.53) and (23.54) with $f'[\phi^E M \theta_N^E] < 0$ and $f'[\phi^E M \theta_D^E - \eta] < 0$ for given ΔV^E . The fact that $\Delta V^E = \theta_D^E - \theta_N^E = \eta / (\phi^E M)$ is uniquely determined (from eq. (23.55) below) then establishes the uniqueness of the symmetric MPE. $\square$

This is a quite striking result: when $\phi_N^E = \phi_D^E$, the effects of changes in political institutions are totally offset by changes in investments in de facto power. Consequently, the stochastic distribution for economic institutions is identical starting in democracy or in nondemocracy. The intuition for this result is straightforward and can be obtained by comparing (23.53) and (23.54) in the special case where $\phi_N^E = \phi_D^E = \phi^E$. These two conditions can hold as equality only if

$$(23.55) \quad f[\phi^E M \theta_N^E] = f[\phi^E M \theta_D^E - \eta].$$

The fact that f is single peaked (which has been assumed above) combined with the second-order conditions implies that $\phi^E M \theta_N^E = \phi^E M \theta_D^E - \eta$, or in other words,

$$(23.56) \quad \theta_D^E = \theta_N^E + \frac{\eta}{\phi^E M}.$$

(23.52) then implies that $p_D = p_N$.

Intuitively, in democracy the elite invest sufficiently more to increase their de facto political power so that they entirely offset the democratic (de jure power) advantage of the

citizens. A more technical intuition for this result is that the optimal contribution conditions for the elite both in nondemocracy and democracy equate the marginal cost of contribution, which is always equal to 1, to the marginal benefit. Since the marginal costs are equal, equilibrium benefits in the two regimes also have to be equal. The marginal benefits consist of the immediate gain of economic rents, ΔR , plus the gain in continuation value, which is also independent of current regime. Consequently, marginal costs and benefits can only be equated if $p_D = p_N$. This result illustrates how institutional change and persistence can coexist—while political institutions change frequently, the equilibrium process for economic institutions remains unchanged.

The most important implication of Proposition 23.8 relates to the potential inefficiency of democracy relative to nondemocracy and thus to the discussion in Section 23.1. Recall that the relatively poor performances of democracies in the postwar era is a potential puzzle, especially viewed in light of the presence of some disastrous, kleptocratic nondemocracies. In the current model, an unconstrained democracy would choose competitive labor market institutions, which increase wages and serve the majority of the population, and these institutions would also lead to higher income per capita in the economy, because $\delta > 0$. However, because the elite can invest in de facto political power in democracy to offset the de jure power advantage of the masses, the equilibrium looks very different. In fact, an allocation starting from nondemocracy *weakly Pareto dominates* one that starts in democracy, even though labor repression is socially costly (that is, $\delta > 0$). This is because citizens are equally well off in the two allocations, while starting in democracy the elite receive the same economic payoff but invest more in de facto power and thus are worse-off. This analysis therefore suggests that the high levels of investment in de facto political power by the elite in democracy, which are socially costly, may be one of the reasons why many democratic societies have disappointing economic performances. This source of inefficiency in democracy complements the distortionary effects of democracies emphasized so far (which resulted from distortionary redistributive taxes imposed by democratic regimes). Here democracy is inefficient because the actions by the elites to prevent regular democratic politics turn this regime into a *dysfunctional democracy*.

Proposition 23.8 may be viewed as a special case, because it depends on the assumption that the technology for de facto political power for the elite is the same in democracy and nondemocracy, that is, $\phi_N^E = \phi_D^E$. One may reasonably suspect that the elite may be less effective in using its resources to garner de facto political power in democratic regimes, which may successfully place constraints on their behavior. In this case, we may want to assume that $\phi_N^E > \phi_D^E$ instead of $\phi_N^E = \phi_D^E$. The next proposition presents the relevant results under this assumption.

PROPOSITION 23.9. *Suppose that $L \gg M$ and that Condition 23.3 holds. Then, any symmetric MPE leads to a Markov regime switching structure where the society fluctuates*

between democracy with associated competitive economic institutions ($I = c$) and nondemocracy with associated labor repressive economic institutions ($I = e$), with switching probabilities $1 - p_N \in (0, 1)$ and $1 - p_D \in (0, 1)$. Moreover, provided that $\phi_N^E > \phi_D^E$, $p_D < p_N$.

PROOF. The proof of this proposition builds on the proof of Proposition 23.8. In particular, suppose that $\phi_N^E > \phi_D^E$. Proposition 23.7 and the same argument as in the proof of Proposition 23.8 again imply that $\theta_D^C = 0$ and $\theta_N^C = 0$, and also $\theta_D^E > 0$ and $\theta_N^E > 0$. Again by assumption $f[\omega]$ is continuous with $\lim_{\omega \rightarrow \infty} f[\omega] = 0$, so that both conditions (23.53) and (23.54) must hold as equalities for some interior values of θ_D^E and θ_N^E establishing existence. $p_D > 0$ and $p_N > 0$ follow from the fact that $\theta_D^E > 0$ and $\theta_N^E > 0$, and $p_D < 1$ and $p_N < 1$ follow by the assumptions on F . To complete the proof, we need to establish that when $\phi_N^E > \phi_D^E$, $p_D < p_N$. Suppose, to obtain a contradiction, that $p_D \geq p_N$. Then, the assumption on F combined with the second-order conditions implies that $\phi_D^E M \theta_D^E - \eta \geq \phi_N^E M \theta_N^E$ and $f[\phi_D^E M \theta_D^E - \eta] \leq f[\phi_N^E M \theta_N^E]$. But combined with the hypothesis that $\phi_N^E > \phi_D^E$, this implies that (23.53) and (23.54) cannot both hold, thus leads to a contradiction and establishes that $p_D < p_N$. $\square$

This proposition also has a number additional implications relative to Proposition 23.8. First, the equilibrium now involves endogenous switches between different political regimes. Second, there is “state dependence” or persistence, in the sense that democracy is more likely to follow democracy than it is to follow nondemocracy ($p_D < p_N$). Third, the effects of the changes in the distribution of de jure power induced by political regime change are partially offset by changes in investments in de facto power (though not fully offset as in Proposition 23.8). This offset is due to the elite’s investments in their de facto political power.

Both Propositions 23.8 and 23.9 rely on Condition 23.3, which ensures that investment in de facto power is always profitable for the elite. When this is not the case, democracy can become an absorbing state and changes in political institutions will have more important effects. This is stated in the next proposition.

PROPOSITION 23.10. Suppose that $L \gg M$ and that there exists $\bar{\theta}_N^E > 0$ such that

$$(23.57) \quad \phi_N^E f[\phi_N^E M \bar{\theta}_N^E] \left[\frac{\Delta R - \beta \bar{\theta}_N^E}{1 - \beta F[\phi_N^E M \bar{\theta}_N^E]} \right] = 1;$$

and

$$(23.58) \quad \eta > -\underline{\omega}.$$

Then, there exists a symmetric MPE in which $p_N \in (0, 1)$ and $p_D = 0$.

PROOF. See Exercise 23.16. $\square$

Therefore, if we relax part of Condition 23.3, symmetric MPEs with democracy as an absorbing state may arise. Clearly, Condition (23.58), which leads to this outcome, is more likely to hold when η is high. This implies that if democracy creates a substantial advantage

in favor of the citizens, it may destroy the incentives of the elite to engage in activities that increase their de facto power and undermine democracy. This will then change the future distribution of political regimes and economic institutions.

It is also interesting to note that even when (23.58) holds, the equilibrium with $p_D, p_N > 0$ characterized in Propositions 23.8 and 23.9 may still exist, leading to a symmetric MPE with $p_D = p_N$. Consequently, whether democracy becomes an absorbing state (or equivalently, whether it becomes fully consolidated) may depend on expectations.

The analysis so far has established how the interplay between de facto and de jure political power leads to the coexistence of persistence in economic institutions and change in political regimes. Equally important, however, is how the likelihood of different institutional outcomes are related to the underlying parameters. I now present a number of comparative static results shedding light on this question. To simplify the analysis, let us focus on Proposition 23.8, where $\phi_N^E = \phi_D^E = \phi^E$. How these results generalize to the case where $\phi_N^E > \phi_D^E$ is discussed in Exercise 23.18. When $\phi_N^E = \phi_D^E$, comparative statics are straightforward since eq.'s (23.43), (23.48) and (23.56) immediately imply that

$$(23.59) \quad \Delta V^E = \frac{\eta}{\phi^E M} > 0.$$

This equation is intuitive. Proposition 23.8 implies that from the viewpoint of the elite, there is only one difference between democracy and nondemocracy; in democracy the elite have to spend more in contributions in order to retain the same political power. In particular, the per elite additional spending is equal to $\eta/\phi^E M$, which is increasing in the de jure political power advantage that democracy creates for the citizens (since, in equilibrium, the elite totally offset this advantage).

Using (23.53) and (23.59) and denoting the equilibrium level of θ_N^E by θ_N^* , we obtain:

$$(23.60) \quad \phi^E f[\phi^E M \theta_N^*] \left[\Delta R + \beta \frac{\eta}{\phi^E M} \right] = 1.$$

Similarly, denoting the equilibrium level of θ_D^E by θ_D^* , we also have

$$(23.61) \quad \phi^E f[\phi^E M \theta_D^* - \eta] \left[\Delta R + \beta \frac{\eta}{\phi^E M} \right] = 1.$$

Finally, let us denote the probability that the elite will have political power by $p^* = p_D = p_N$. This probability corresponds both to the probability that the elite will control political power, and also to the probability that the society will be nondemocratic and economic institutions will be labor repressive rather than competitive. Thus this probability summarizes most of the economic implications of the model.

PROPOSITION 23.11. *Suppose that $L \gg M$ and Condition 23.3 holds. Suppose also that $\phi_N^E = \phi_D^E = \phi^E$. Then, θ_N^* , θ_D^* and p^* are strictly increasing in ΔR , β and η , and strictly decreasing in M . Moreover, p^* is strictly increasing in ϕ^E .*

PROOF. See Exercise 23.17. □

Many of the comparative statics in Proposition 23.11 are intuitive and do not require much elaboration. For example, the effect of the number of elite agents, M , on investments in de facto power and the equilibrium probability of nondemocracy and the effect of ϕ^E on the equilibrium probability of nondemocracy are straightforward to understand. Observe that M also has an indirect effect on the equilibrium, which goes in the same direction; greater M reduces ΔR (cf. eq. (23.37)) and further discourages investments in de facto power via this channel.

The fact that an increase in ΔR increases the probability that the elite control political power is also natural, since ΔR is a measure of how much the elite have to gain by controlling political power. But this latter result also has interesting economic implications. Since ΔR will be high when λ or δ are low, we also have $\partial p^*/\partial \lambda < 0$, and $\partial p^*/\partial \delta < 0$, so that political and economic institutions favoring the elite are more likely to arise when the elite will be able to use labor repressive institutions effectively or when the costs of repression are relatively low. A major reason why λ and δ may vary across societies is because of differences in economic structure, economic institutions and factor endowments. For example, we may expect both parameters to be higher in societies where agriculture is more important and physical or human capital-intensive sectors are less important, since labor repression may be more effective in reducing wages and may also create less distortion in such societies than in those with more complex production relations. This interpretation is consistent with the greater prevalence of labor repressive practices in predominantly agricultural societies.

The fact that a higher β also increases the likelihood of labor repressive institutions is somewhat more surprising. In many models, a higher discount factor leads to better allocations. Here, in contrast, a higher discount factor leads to more wasteful activities by the elite and to labor repressive economic institutions. The reason is that the main pivotal agents in this model are the elite, which, by virtue of their smaller numbers, are the ones investing in their de facto political power (recall Proposition 23.7) and thus they take the effect of their contributions on equilibrium allocations into account. Contributing to de facto political power is a form of investment and some of the returns accrue to the elite in the future (when they secure nondemocracy instead of democracy). Therefore a higher level of β encourages them to invest more in their political power and makes nondemocracy and labor repressive economic institutions more likely.

The most surprising and interesting comparative static result concerns the effects of η . Since a higher η corresponds to a greater de jure power advantage for the citizens in democracy, one might have expected a greater η to lead to better outcomes for the citizens. In contrast, we find that higher η makes nondemocracy and labor repressive economic institutions more likely (as long as Condition 23.3 still holds). This is because a higher η makes democracy more costly for the elite, inducing each elite agent to invest more in the group's political power in order to avoid democracy. This effect is strong enough to increase the probability that they will maintain political power. However, the overall impact of η on the likelihood of democracy is non-monotonic: if η increases so much that Condition 23.3 no

longer holds, then Proposition 23.10 applies and democracy becomes fully consolidated (an absorbing state).

Some of the comparative static results in Proposition 23.11 are the outcome of two competing forces. The fact that the cost of investing in de facto political power is linear and the assumption that $\phi_N^E = \phi_D^E$ are important for these results. In particular, in the case where $\phi_N^E > \phi_D^E$, the comparative statics with respect to ΔR , β and M still hold. But those with respect to η become ambiguous; a greater democratic advantage for citizens helps them gain power in democracy, but also induces the elite to invest more in their de facto political power. Which effect dominates cannot be determined without imposing further structure.

23.6. Taking Stock

This chapter provided a brief overview of some of the issues related to the effects of political institutions on economic growth. How societies can choose economic institutions and policies that are not conducive to economic growth was the focus of the previous chapter. A natural conjecture based on the analysis there is to relate differences in economic institutions to political institutions. For example, if political power is in the hands of an elite that is opposed to growth, growth-enhancing policies are less likely to emerge. Our analysis in the previous chapter hinted that such considerations could be important. The empirical evidence in Chapter 4 also provided support for such a view, whereby the cluster of economic institutions that provide secure property rights to a broad cross-section of society together with political institutions that place constraints on elites and politicians appear to be conducive to economic growth. Nevertheless, the relationship between political regimes and growth is more complicated for a number of reasons. First, the empirical evidence is less clear-cut than we may have originally presumed—while there are historical examples of the positive effects of democratic institutions on economic growth, the postwar evidence does not provide strong support for the view that democracies and political institutions that constrain rulers and politicians always generate more economic growth. Second, political institutions themselves are not given, but are endogenous and change dynamically. These two factors imply that we need to understand how political institutions affect economic outcomes more carefully and should also consider the modeling of equilibrium political institutions. Both of these areas are at the forefront of research in political economy and are likely to play a more important role in the research on economic growth in the coming years. Thus there is no established framework and no general consensus on what types of models are most useful in thinking about the issues raised in this chapter (nor any consensus on what the facts are when it comes to the effect of political regimes on economic growth).

In this light, I presented a number of models to highlight the ideas about political equilibria and the relationship between political institutions and economic growth that I view to be most important and most promising for future work. I emphasized how preferences over political institutions need to be derived from the implications of these political institutions for economic allocations. I then highlighted how ideal (or perfect) political institutions are

unlikely to exist, because different political institutions, by creating different sets of winners and losers, also create tradeoffs. Oligarchies, for example, favor the already rich, which brings a range of distortions. Democracies, on the other hand, typically involve higher taxes on the rich to generate income to redistribute to the less well-off and in the process create distortions on investment and other economic choices. In general, it is impossible to unambiguously conclude whether democracies or oligarchies (or yet other political systems favoring other groups) will be more growth-enhancing. However, certain ideas seem both plausible and consistent with the data. One aspect I tried to emphasize is that the dynamic tradeoffs between democracies and other regimes may be different than the static tradeoffs. While democracies may create static distortions because of their greater redistributive tendencies, they are likely to outperform oligarchies in the long run because they avoid political sclerosis, whereby the incumbents are able to dominate the political system and erect entry barriers to protect their businesses, even when efficiency dictates that new individuals should enter and form new businesses to replace theirs. Thus democracy may be more conducive to the process of creative destruction that is part of modern capitalist growth than other political regimes. A related idea, which also receives support from casual empiricism and the model presented in Section 23.3, is that democracies might be more flexible and adaptable to the arrival of new technologies.

Another important idea I tried to emphasize is that democracies may lead to inefficient outcomes because they may sometimes be *dysfunctional*. The main functional characteristic of democracies is that they create political equality, providing voice to the masses, free entry of parties, and free and fair elections. However, as already emphasized, political equality often comes with a tendency to choose redistributive policies that are costly for the elites (especially, for elites that have most of their assets in land or in other easily taxable forms). This creates an incentive for the elite to invest to increase their de facto political power in order to capture democratic politics. Captured democracies are particular example of dysfunctional democracies and may lead to a range of inefficiencies not because the regime is democratic, but precisely because its true democratic nature remains unfulfilled. The model in Section 23.5 provides a stark example of this tendency, whereby democracy may lead to a Pareto dominated allocation, even though without the intervention from the elite, democratic politics would lead to higher income per capita and much less distorted allocations. Whether, in practice, the majority of distortions in democracies are related to their redistributive tendencies or to the fact that they are captured by the elites is an empirical question that has not been addressed yet. Future empirical work might shed light on this important set of questions. Another type of dysfunctional democracy, discussed at the beginning of this chapter, is exemplified by the populist regimes such as those of Perón in Argentina and Chavez in Venezuela. While these regimes engage in some amount of redistribution, they also pursue highly distortionary policies. Why such regimes arise and sometimes even receive support from the electorate is another important part of the puzzle of the relatively disappointing

performances of some democracies, but has not received much attention in the economics literature.

Finally, I also gave a very brief overview of some of the issues that arise when we wish to model the dynamics of political institutions themselves. Section 23.4 provided both a general discussion of the types of models that would be useful for such an analysis and examples of how these models can be developed and applied to various situations. Once again, this is an area of active current research and the material presented here is no more than the tip of the iceberg. It thus meant to entice the reader to think more about these issues and to introduce the bare minimum that is necessary for a more coherent discussion of the relationship between political institutions and economic growth.

23.7. References and Literature

This chapter relates to a large literature in political economy and political science. Because of space constraints, I will not provide a comprehensive literature review. Instead, I will simply refer to the relevant books and papers on which the material I presented draws on. The literature on the relationship between political regimes and economic growth, discussed in Section 23.1, is relatively large. The key references are discussed in that section, so it is not necessary to repeat them here.

Section 23.2 built on the models presented in the previous chapter, which themselves were based on Acemoglu (2006). Section 23.3 is directly based on Acemoglu (2008a). Other models that discuss the functioning of oligarchic societies include Leamer (1998), Bourguignon and Verdier (2000), Robinson and Nugent (2001), Galor, Moav and Vollrath (2005), and Sonin (2003). Section 23.4 provided an abstract discussion of the issues related to the modeling of political change based on Acemoglu and Robinson (2006a) and Acemoglu, Johnson and Robinson (2005a). The distinction between *de jure* and *de facto* political power is introduced in Acemoglu and Robinson (2006a) and is also discussed in Acemoglu, Johnson and Robinson (2005a). There are more details on the historical examples discussed in this section in both of these references. Interesting examples of the use of *de facto* political power by elites in the context of Latin America are provided by Paige (1997) for Central America, by Smith (1979) for Mexico, by Klein (1999) and Mazzuca and Robinson (2004) for Colombia, and by Key (1949), Woodward (1955), Wright (1986) and Ransom and Sutch (2001) for the US South after the Civil War.

Key references on changes in political and economic institutions in medieval Europe include Tawney (1941), Brenner (1976, 1982, 1993), Brewer (1988), Hilton (1981), Ertman (1997) and North and Weingast (1989). The role of Atlantic trade in changing the economic and political landscape of many European nations is emphasized in Davis (1983) and Acemoglu, Johnson and Robinson (2005b). The literature on democratization in Europe and Latin America is summarized in Acemoglu and Robinson (2006a). Important modern reference include Evans (1983), Lee (1994), Lang (1999) and Collier (2000). The fiscal reforms following

democratization are documented and discussed in Lindert (2000, 2004), and the educational reforms are discussed in Ringer (1979) and Mitch (1983).

Engerman (1981), Coatsworth (1993), Eltis (1995), Engerman and Sokoloff (1997) and Acemoglu, Johnson and Robinson (2002) provide information on the prosperity the United States in the 17th and 18th centuries relative to the Caribbean and South America. The contrast of industrialization in Britain and France against the experiences of Russia and Austria-Hungary draws on Acemoglu and Robinson (2006b), which includes references to the original literature. Mosse (1992) and Gross (1973) provide an excellent introduction to the policies of Russian and Austria-Hungarian monarchies concerning industrialization and economic development.

The model sketched at the end of Section 23.4 builds on Acemoglu and Robinson (2000a, 2006a). Finally, the model presented in Section 23.5 is based on Acemoglu and Robinson (2007).

23.8. Exercises

EXERCISE 23.1. Consider the following infinite-horizon economy populated by two groups of equal size, denoted 1 and 2. All agents in both groups maximize the expected present discounted value of income, with discount factor $\beta \in (0, 1)$. In any period one of the groups is in power while the other group is out of power. When either group is in power, it loses power with probability $q < 1/2$ in every period. Income is generated in the following way: group j has an asset stock of $A_j(t)$ at time t . Using these assets, it can produce income $A_j(t) f(I_j(t))$ if it invests $I_j(t)$ and this costs $I_j(t)$ in terms of utility. Investments are made before nature realizes whether the group in power will lose its position. Initially, the asset stock of both groups is the same, $A_1(0) = A_2(0) = A$. Assume that, as long as the assets are not expropriated, income can be hidden in a non-taxable sector that generates a net return of $(1 - \tau) A_j(t) f(I_j(t)) - I_j(t)$. Suppose that the net return to group j is

$$(1 - e_j(t))(1 - \tau_j(t)) A_j(t) f(I_j(t)) + G_j(t) - I_j(t),$$

where $\tau_j(t)$ is a tax rate faced by this group, $e_j(t) \in [0, 1]$ denotes the proportion of group j 's assets that are expropriated in period t , and $G_j(t)$ is a transfer to group j in period t . The law of motion of assets, as a function of expropriation of assets, is given by:

$$\begin{aligned} A_1(t) &= A_1(t-1) - e_1(t) A_1(t-1) + e_2(t) A_2(t-1) \\ A_2(t) &= A_2(t-1) - e_2(t) A_2(t-1) + e_1(t) A_1(t-1). \end{aligned}$$

- (1) First suppose that asset expropriation is not allowed, so $e_j(t) = 0$, and the only decision each group takes is the tax rate it sets when in power. Characterize the pure strategy MPE. Show that the output level is less than first-best and is constant over time.
- (2) Next suppose that the group in power can expropriate the assets of the other group (so the two decisions now are taxes and expropriation). Characterize the MPE, and show that output can actually be higher in this economy than the economy without

asset expropriation. Explain why. Show also that now output is no longer constant, but fluctuates over time.

- (3) Next consider a model endogenizing q . In particular, imagine that the group out of power can choose to take power in any period, but to do so, it must pay a non-pecuniary cost c . This cost c is drawn each period from the cumulative distribution $G(c)$. First consider the case without asset expropriation. Show that there will exist a level of c^* such that when $c \leq c^*$, the group out of power will take power [Hint: write the value functions of the members of the two groups in terms of c^* —or the probability of regime change in the future—and obtain a fixed-point recursion for c^*].
- (4) Next consider the case with asset expropriation (where the group that comes to power can expropriate the assets of the other group). Show that there will exist a level of c^{**} such that when $c \leq c^{**}$, the group out of power will take power, and show that $c^{**} > c^*$. Show also that this economy with endogenous power switches has higher volatility than the corresponding economy with exogenous power switches.
- (5) Discuss whether the two theoretical channels highlighted by this model, linking security of property rights to economic instability, are plausible.

EXERCISE 23.2. Prove Proposition 23.1.

EXERCISE 23.3. (1) Prove Proposition 23.2.

- (2) Generalize the result in Proposition 23.2 to the case where $\theta^e \neq \theta^m$. In particular, derive an inequality that determines when the dictatorship of the elite will generate greater output per capita than the dictatorship of the middle class.

EXERCISE 23.4. Prove Proposition 23.3. [Hint: to prove the second part of this proposition, first note that equilibrium wage will be given by whichever group has lower net (after tax) productivity. Then, write the utility of workers under two scenarios, first, when the elite have lower net productivity, and second when the middle class have lower net productivity. In writing these expressions, recall that the group with the lower productivity will employ $1 - \theta\bar{L}$ workers, since Condition 22.1 holds. Derive the optimal tax policies for workers in these two scenarios and then compare the utility at these optimal policies].

EXERCISE 23.5. In the model of Section 23.3, prove that $V^z(q^t)$ given in (23.14) is strictly monotonic in $w(t)$, $T(t)$ and $\Pi^z(\tau(t), w(t))$, and therefore that $V^H(q^t) > V^L(q^t)$.

EXERCISE 23.6. In the model of Section 23.3, suppose that $l_i(t)$ is unbounded above. What problems would this create? Next suppose that $l_i(t)$ could be arbitrarily small. What problems will this raise for the equilibrium in this section? Could you generalize the results in this section to an environment in which $l_i(t) \in [\underline{L}, \bar{L}]$, where $\underline{L} > 0$ and $\bar{L} < \infty$?

EXERCISE 23.7. Derive eq. (23.8).

EXERCISE 23.8. Prove Proposition 23.5.

EXERCISE 23.9. Prove Proposition 23.6.

EXERCISE 23.10. Suppose that Condition 23.1 does not hold. Generalize the results in Propositions 23.5 and 23.6.

EXERCISE 23.11. Consider the model in Section 23.3, starting with $\mu(0) = 1$ and an oligarchic regime. Suppose that at some time $t' < \infty$ a new technology arises, which is ψ -times as productive as the old technology, where $\psi > 1$. However, entrepreneurial skills with this new technology are uncorrelated with entrepreneurial skills relevant for the old technology. In particular, suppose that entrepreneurial skills for new technology are given by

$$\hat{a}_i(t+1) = \begin{cases} A^H & \text{with probability } \hat{\sigma}^H & \text{if } \hat{a}_i(t) = A^H \\ A^H & \text{with probability } \hat{\sigma}^L & \text{if } \hat{a}_i(t) = A^L \\ A^L & \text{with probability } 1 - \hat{\sigma}^H & \text{if } \hat{a}_i(t) = A^H \\ A^L & \text{with probability } 1 - \hat{\sigma}^L & \text{if } \hat{a}_i(t) = A^L \end{cases}.$$

- (1) Show that there exists $\bar{\psi}$ such that if $\psi > \bar{\psi}$, all existing entrepreneurs will increase entry barriers and switch to the new technology.
- (2) Show that if $\psi < \bar{\psi}$, then again entry barriers will be increased and now only entrepreneurs who have low skills with the old technology will switch to the new technology.
- (3) Now analyze the response of democracy to the arrival of the same technology.
- (4) Compare output per capita in democracy and oligarchy after the arrival of new technology, and explain why democracy is more “flexible” in dealing with the arrival of new technologies.

EXERCISE 23.12. This exercise shows that entry barriers typically lead to multiple equilibrium wages in dynamic models. Consider the following two-period model. The production function is given by (23.3) and the distribution of entrepreneurial talent is given by a continuous cumulative density function $G(a)$. There is an entry cost into entrepreneurship equal to b at each date and each entrepreneur hires one worker (and does not work as a worker himself). Total population is equal to 1.

- (1) First, ignore the second period and characterize the equilibrium wage and determine which individuals will become entrepreneurs. Show that the equilibrium is unique.
- (2) Now consider the second period and suppose that all agents discount the future at the rate β . Show that there are multiple equilibrium wages in the second period and as a result, multiple equilibrium wages in the initial period.
- (3) Now suppose that a fraction ε of all agents die in the second period and are replaced by new agents. New agents have to pay the entry cost into entrepreneurship if they want to become entrepreneurs. Suppose that their talent distribution is also given by $G(a)$. Characterize the equilibrium in this case and show that it is unique.
- (4) Now consider the limiting equilibrium in part 3 with $\varepsilon \rightarrow 0$. Explain why this limit leads to a unique equilibrium while there are multiple equilibria at $\varepsilon = 0$.

EXERCISE 23.13. Consider an economy populated by λ rich agents who initially hold power, and $1 - \lambda$ poor agents who are excluded from power, with $\lambda < 1/2$. All agents are infinitely lived and discount the future at the rate $\beta \in (0, 1)$. Each rich agent has income θ/λ while each poor agent has income $(1 - \theta)/(1 - \lambda)$ where $\theta > \lambda$. The political system determines a linear tax rate, τ , the proceeds of which are redistributed lump-sum. Each agent can hide

their money in an alternative non-taxable production technology, and in the process they lose a fraction ϕ of their income. There are no other costs of taxation. The poor can undertake a revolution, and if they do so, in all future periods, they obtain a fraction $\mu(t)$ of the total income of the society (i.e., an income of $\mu(t)/(1-\lambda)$ per poor agent). The poor cannot revolt against democracy. The rich lose everything and receive zero payoff after a revolution. At the beginning of every period, the rich can also decide to extend the franchise to the poor, and this is irreversible. If the franchise is extended, the poor decide the tax rate in all future periods.

- (1) Define MPE in this game.
- (2) First suppose that $\mu(t) = \mu^l$ at all times. Also assume that $0 < \mu^l < 1 - \theta$. Show that in the MPE, there will be no taxation when the rich are in power, and the tax rate will be $\tau = \phi$ when the poor are in power. Show that in the MPE, there is no extension of the franchise and no taxation.
- (3) Suppose that $\mu^l \in (1 - \theta, (1 - \phi)(1 - \theta) + \phi(1 - \lambda))$. Characterize the MPE in this case. Why is the restriction $\mu^l < (1 - \phi)(1 - \theta) + \phi(1 - \lambda)$ necessary?
- (4) Now consider the SPE of this game when $\mu^l > 1 - \theta$. Construct an equilibrium where there is extension of the franchise along the equilibrium path. [Hint: first, to simplify, take $\beta \rightarrow 1$, and then consider a strategy profile where the rich are always expected to set $\tau = 0$ in the future; show that in this case the poor would undertake a revolution; also explain why the continuation strategy of $\tau = 0$ by the rich in all future periods could be part of a SPE]. Why is there extension of the franchise now? Can you construct a similar non-Markovian equilibrium when $\mu^l < 1 - \theta$?
- (5) Explain why the MPE led to different predictions than the non-Markovian equilibria. Which one is more satisfactory?
- (6) Now suppose that $\mu(t) = \mu^l$ with probability $1 - q$, and $\mu(t) = \mu^h$ with probability q , where $\mu^h > 1 - \theta > \mu^l$. Construct a MPE where the rich extend the franchise, and from there on, a poor agent sets that tax rate. Determine the parameter values that are necessary for such an equilibrium to exist. Explain why extension of the franchise is useful for rich agents?
- (7) Now consider non-Markovian equilibria again. Suppose that the unique MPE has franchise extension. Can you construct a SPE equilibrium, as $\beta \rightarrow 1$, where there is no franchise extension?
- (8) Contrast the role of restricting strategies to be Markovian in the two cases above [Hint: why is this restriction ruling out franchise extension in the first case, while ensuring that franchise extension is the unique equilibrium in the second?].

EXERCISE 23.14. (1) Show that if $L \geq \bar{L}$, then

$$(23.62) \quad \bar{L} \equiv \max \left\{ \frac{\phi_D^C}{\phi_D^E}, \frac{\phi_N^C}{\phi_N^E} \right\} \frac{(1 - \lambda(1 - \delta))}{(1 - \beta)(1 - \lambda)(1 - \delta)} M \in (0, \infty).$$

Then, any MPE involves $\theta_D^C = \theta_N^C = 0$.

- (2) Given the result in part 1, prove Proposition 23.7.

EXERCISE 23.15. *

- (1) Generalize Proposition 23.8 to non-symmetric MPE.
 (2) Show that for β sufficiently high, SPE that are on the Pareto frontier for the elite (meaning that it is impossible to make one elite agent better-off without making another one worse-off) have the same qualitative features as the equilibrium in Proposition 23.8.

EXERCISE 23.16. Prove Proposition 23.10.

EXERCISE 23.17. Prove Proposition 23.11.

EXERCISE 23.18. Generalize the results on the effects of ΔR , β and M on equilibrium objects in Proposition 23.11 to the case where $\phi_N^E > \phi_D^E$.

Epilogue: Mechanics and Causes of Economic Growth

This chapter contains concluding remarks. Instead of summarizing the models and ideas presented so far, I will end with a brief discussion of what we have learned from the models and analyses presented in this book and how they offer a useful perspective on world growth and cross-country income differences. I will then provide a very quick overview of some of the many remaining questions, which are important to emphasize both as a measure of our ignorance and as potential topics for future research.

24.1. What Have We Learned?

It may be useful to first summarize the most important aspects and takeaway lessons of our analysis.

1. *Growth as the source of current income differences*: at an empirical level, the investigation of economic growth is important not only for understanding the growth process, but also because the analysis of the sources of cross-country income differences today requires us to understand why some countries have grown rapidly over the past 200 years while others have not (Chapter 1).

2. *The role of physical capital, human capital, and technology*: cross-country differences in economic performance and growth over time are related to physical capital, human capital and technology. Part of our analysis has focused on theoretical and empirical investigation of the contributions of these factors to production and growth (Chapters 2 and 3), and the remaining, larger part, has focused on understanding physical capital accumulation, human capital accumulation, and technology creation and adoption decisions (Chapters 8-15 and 18-19). One conclusion that has emerged concerns the importance of technology in understanding both cross-country and over-time differences in economic performance. Here, technology refers both to advances in techniques of production, thus to the accumulation of knowledge and blueprints for more efficient machinery. It also refers to the general efficiency of the organization of production, which will be affected by the incentives that a society (and its government) provides to firms and workers, by its contracting institutions, and by the types of market failures that prevent the development of more productive economic relationships (Section 18.5 in Chapter 18 and Chapter 21).

3. *Endogenous investment decisions*: while we can make considerable progress by understanding the role of physical capital and human capital, and using cross-country data on differences in investments in machinery and in education to account for the process of economic growth and development, we also need to endogenize these investment decisions.

Investments in physical and human capital are forward-looking and depend on the rewards that individuals expect from their investments. Understanding differences in these investments is therefore intimately linked to understanding how “reward structures”—that is, the pecuniary and nonpecuniary rewards and incentives for different activities—differ across societies and how individuals respond to differences in reward structures (Chapters 8 and 10).

4. *Endogenous technology*: likewise, technology should be thought of as endogenous, not as *mana* from heaven. There are good empirical and theoretical reasons for thinking that new technologies are created by profit-seeking individuals and firms, via research, development and tinkering. In addition, decisions to adopt new technologies are likely to be highly responsive to profit incentives. Since technology appears to be a prime driver of economic growth over time and a major factor in cross-country differences in economic performance, we must understand how technology responds to factor endowments, market structures, and rewards. Developing a conceptual framework that emphasizes the endogeneity of technology has been one of the major objectives of this book. The modeling of endogenous technology necessitates ideas and tools that are somewhat different from those involved in the modeling of physical and human capital investments. Three factors are particularly important. First, fixed costs of creating new technologies combined with the non-rival nature of technology necessitates the use of models where innovators have *ex post* (after innovation) monopoly power. The same might apply, though perhaps to a lesser degree, to firms that adopt new technologies. The presence of monopoly power changes the welfare properties of decentralized equilibria and creates a range of new interactions and externalities (Chapters 12 and 13 and Section 21.5 in Chapter 21). Second, the process of innovation is implicitly one of competition and creative destruction. The modeling of endogenous technology necessitates more detailed models of the industrial organization of innovation and R&D, and how this impacts competition among firms and how new firms may or may not be able to replace incumbents. These models shed light on the impact of market structure, competition, regulation and intellectual property rights protection on innovation and technology adoption (Chapters 12 and 14). Third, endogenous technology implies that not only the aggregate rate of technological change but also the types of technologies that are developed will be responsive to rewards. Key factors influencing the types of technologies that societies develop are again reward structures and factor endowments. For example, changes in relative supplies of different factors are likely to effect which types of technologies will be developed and adopted (Chapter 15).

5. *Linkages across societies and balanced growth at the world level*: while endogenous technology and endogenous growth are major ingredients in our thinking about the process of economic growth in general and the history of world economic growth in particular, it is also important to recognize that most economies do not invent their own technologies, but adopt them from the world technology frontier or adapt them from existing technologies (Chapter 18). In fact, the process of technology transfer across nations might be one of the reasons why after the initial phase of industrialization, countries that have been part of the global economy have grown at broadly similar rates (Chapter 1). Therefore, the modeling of cross-country

income differences and the process of economic growth for a large part of the world necessitates a detailed analysis of technology adoption, technology diffusion and technology transfer. Two topics deserve special attention in thinking about technological linkages across countries and technology adoption decisions. The first is the contracting institutions supporting contracts between upstream and downstream firms, between firms and workers, and between firms and financial institutions. These institutional arrangements will affect the amount of investment, the selection of entrepreneurs and firms, and also the efficiency with which different tasks are allocated across firms and workers. There are marked differences in contracting institutions across societies and these differences appear to be a major factor influencing technology adoption and diffusion in the world economy. Contracting institutions not only have a direct effect on technology and prosperity, but they also shape the internal organization of firms, which contributes to the efficiency of production and influences how innovative firms will be (Section 18.5 in Chapter 18). The second is international trading relationships. International trade not only generates static gains familiar to economists, but also influences the innovation and growth process. The international division of labor and the product cycle are examples of how international trading relationships help the process of technology diffusion and enable a more productive specialization of production (Chapter 19).

6. *Takeoffs and failures*: the last 200 years of world economic growth stand in stark contrast to the thousands of years before. Despite intermittent growth in certain parts of the world during certain epochs, the world economy was largely stagnant until the end of the 18th century. This stagnation had multiple aspects. These included low productivity, high volatility in aggregate and individual outcomes, a largely rural and agricultural economy and a Malthusian-type configuration where increases in output were often accompanied by increases in population, thus only having a limited effect on per capita income. Another major aspect of stagnation has been the failed growth attempts; many societies grew for certain periods of time and then lapsed back into depressions and stagnation. This changed at the end of the 18th century. We owe our prosperity today to the takeoff in economic activity, closely related to industrialization, that started in Britain and Western Europe, and spread to certain other parts of the world, most notably to West European offshoots, such as the United States and Canada. Therefore, the nations that are rich today are precisely those where this process of takeoff originated or else those that were able to rapidly adopt and build upon the technologies underlying this takeoff (Chapter 1). Understanding current income differences across countries therefore necessitates understanding why some countries failed to take advantage of the new technologies and production opportunities.

7. *Structural changes and transformations*: modern economic growth and development are accompanied by a set of sweeping structural changes and transformations. These include changes in the composition of production and consumption (the shift from agriculture to industry and from industry to services), urbanization, financial development, changes in inequality of income and inequality of opportunity, the transformation of social and living arrangements, changes in the internal organization of firms and the demographic transition.

While the process of economic development is multifaceted, much of its essence lies in the structural transformation of the economy and society at large (Section 17.6 in Chapter 17, Chapters 20 and 21). Many of these transformations are interesting to study for their own sake. Moreover, they are also important ingredients for sustained growth. Lack of structural transformation is not only a symptom of stagnation, but often also one of its causes. Societies may fail to takeoff and benefit from the available technology and investment opportunities, partly because they have not managed to undergo the requisite structural transformations and thus lack the right type of financial relations, the appropriate skills, or the types of firms that would be necessary as conduits of new technologies.

8. *Policy, institutions and political economy*: the reward structures faced by firms and individuals play a central role in shaping whether they undertake the investments in new technology and in human capital necessary for takeoff, industrialization, and economic growth. These reward structures are determined by policies and institutions. Policies and institutions also directly affect whether a society can embark upon modern economic growth for a variety of interrelated reasons (Chapter 4). First, they directly determine the society's reward structure, thus shaping whether investments in physical and human capital and technological innovations are profitable. Second, they determine whether the infrastructure investments and contracting arrangements necessary for modern economic relations are present. For example, modern economic growth would be impossible in the absence of some degree of contract enforcement, the maintenance of law and order, and at least a minimum amount of investment in public infrastructure. Third, they influence and regulate the market structure, thus determining whether the forces of creative destruction will be operational and whether new and more efficient firms can replace less efficient incumbents. Finally, institutions and policies may sometimes (or perhaps often) go in the opposite direction and block the adoption and use of new technologies as a way of protecting politically powerful incumbent producers or to stabilize the established political regime. In this light, to understand why takeoff into sustained growth started 200 years ago and not before, why it started in some countries and not others, and why it was followed by some countries and not others, we need to understand the policy and institutional choices that societies make. This means we need to investigate the political economy of growth, paying special attention to which individuals and groups will be the winners from economic growth and which will be the losers. When losers cannot be compensated and have sufficient political power, we may expect the political economy equilibrium to lead to policies and institutions that are not growth enhancing. Our basic analysis of political economy generates insights about what types of distortionary policies may block growth, when these distortionary policies will be adopted, and how technology, market structure and factor endowments interact with the incentives of the social groups in power to encourage or discourage economic growth (Chapter 22).

9. *Endogenous political institutions*: policies and institutions are central to understanding the growth process over time and cross-country differences in economic performance, but these social choices are in turn determined within society's political institutions. Democracies

and dictatorships are likely to make different policy choices and create different types of reward structures. But political institutions themselves are not exogenous in the long-run equilibrium of a society. They are both determined in equilibrium and change along the equilibrium path as a result of their own dynamics and as a result of stimuli coming from changes in technology, trading opportunities, and factor endowments (Chapter 23). We have already seen some simple models that provide various useful insights, but much remains to be done. Towards a more complete understanding of world economic growth and the income differences today, we therefore need to study: (1) how political institutions affect policies and economic institutions, thus shaping incentives for firms and workers; (2) how political institutions themselves change, especially interacting with economic outcomes and technology; (3) why political institutions and the associated economic institutions did not lead to sustained economic growth throughout history, why they enabled economic takeoff 200 years ago, and why in some countries, they blocked the adoption and use of superior technologies and they derailed the process of economic growth.

In this brief summary, I focused on the ideas most relevant for thinking about the process of world economic growth and cross-country income differences we observe today. The reader will recall that the focus in the book has been not only on ideas, but also on careful mathematical modeling of these ideas, so that coherent and rigorous theoretical approaches to these core issues can be developed. Nevertheless, to avoid repetition I will not mention the theoretical foundations of these ideas, which range from basic consumer, producer and general equilibrium theory to dynamic models of accumulation, models of monopolistic competition, models of world equilibria and dynamic models of political economy. But I wish to emphasize again that a thorough study of the theoretical foundations of these ideas is necessary both to develop a satisfactory understanding of the main issues and also to find the best way of making them empirically operational.

24.2. A Possible Perspective on Growth and Stagnation over the Past 200 Years

The previous section gave a brief summary of the most important ideas highlighted in this book. I now discuss how some of these ideas might be useful in shedding light on the process of world economic growth and cross-country divergence that have motivated our investigation starting in Chapter 1. The central questions are these:

- (1) Why did the world economy not experience sustained growth before 1800?
- (2) Why did economic takeoff start around 1800 and in Western Europe?
- (3) Why did some societies manage to benefit from the new technologies and organizational forms that emerged starting in 1800, while others steadfastly refused or failed to do so?

I will now offer a narrative that provides some tentative answers to these three questions. While certain parts of the mechanisms I propose here have been investigated econometrically and certain other parts receive support from historical evidence, the reader should view these as a first attempt at providing a coherent answers to these central questions. Two

aspects of these answers are noteworthy. First, they build on the theoretical insights that the models in this book generate. Second, in the spirit of the discussion in Chapter 4, they link the proximate causes of economic phenomena to fundamental causes, and in particular to institutions. And here, I take a shortcut. Although I emphasized in Chapter 23 that there are no perfect political institutions and that each set of different political arrangements is likely to favor some groups at the expense of others, I will simplify the discussion here by making a core distinction between two sets of institutional arrangements, one less conducive to growth than the other one. The first, which I will refer to as *authoritarian political systems*, encompasses absolutist monarchies, dictatorships, autocracies and various types of oligarchies that concentrate power in the hands of a small minority and pursue economic policies that are favorable to the interests of this minority. Authoritarian systems often rely on some amount of repression because they seek to maintain an unequal distribution of political power and economic benefits. They also adopt economic institutions and policies that protect incumbents and create rents for those who hold political power. The second set of institutions are *participatory regimes*. These regimes place constraints on rulers and politicians, thus preventing the most absolutist tendencies in political systems, and give voice to new economic interests, so that a strict decoupling between political and economic power is avoided. Such regimes include constitutional monarchies, where broader sections of the society take part in economic and political decision-making, and democracies where political participation is greater than in nondemocratic regimes. The distinguishing feature of participatory regimes is that they provide voice and (economic and political) security to a broader cross-section of society than authoritarian regimes. As a result, they are more open to entry by new businesses and provide a more level playing field and better security of property rights to a relatively broad section of the society. Thus in some ways, the contrast between authoritarian political systems and participatory regimes is related to the contrast between the growth-promoting cluster of institutions and the growth-blocking, extractive institutions emphasized and illustrate in Chapter 4. The reader should note that many different terms could have been used instead of “authoritarian” and “participatory,” and some details of the distinction may be arbitrary. More importantly, it should be borne in mind that even the most participatory regime will involve an unequal distribution of political power and those that have more political power can use the fiscal and political instruments of the state for their own benefits and for the detriment of the society at large (Chapter 23). Why this type of behavior is sometimes a successfully curtailed or limited is a question at the forefront of current research and I will not dwell on it here.

Why Did the World Not Experienced Sustained Growth Before 1800? While sustained growth is a recent phenomenon, growth and improvements in living standards certainly did occur many times in the past. The human history is also full of major technological breakthroughs. Even before the Neolithic Revolution, many technological innovations increased the productivity of hunter gatherers. The transition to farming after about 9000BC

is perhaps the most major technological revolution of all times; it led to increased agricultural productivity and to the development of socially and politically more complex societies. Archaeologists have also documented various instances of economic growth in pre-modern periods. Historians estimate that consumption per-capita doubled during the great flowering of Ancient Greek society from 800BC to 50BC (Morris, 2005). Similar improvements in living standards were experienced by the Roman Republic and Empire after 400BC (Hopkins, 1980), and appear to have been experienced by pre-Columbian civilizations in South America, especially by the Olmec, the Maya and even perhaps the Inca (Mann, 2004, Webster, 2003). Although data on these ancient growth experiences are limited, the available evidence suggests that basic economic models, where growth relies on physical capital accumulation and some technological change, provide a good first description of the developments in these ancient economies (see, for example, Morris, 2005, on capital accumulation and limited technological change in Ancient Greece).

Importantly, however, these growth experiences were qualitatively different from those that the world experienced after its takeoff starting in the late 18th century. Four factors appear to have been particularly important and set these growth episodes apart from modern economic growth. The first is that they were relatively short-lived or took place at relatively slow pace.¹ In most cases, the initial spurt of growth soon crumbled for one reason or another, somehow reminiscent of the failed takeoffs in the model of Section 17.6 in Chapter 17. Secondly and relatedly, growth was never based on continuous technological innovations, thus never resembled the technology-based growth emphasized in Chapters 13-15. Thirdly, in most cases economic institutions that would be necessary to support sustained growth did not develop. Financial relations were generally primitive, contracting institutions remained informal, markets were heavily regulated with various internal tariffs and incomes and savings did not reach levels necessary for the mass market and for simultaneous investments in a range of activities. Put differently, the structural transformations accompanying development discussed in Chapter 21 did not take place. The final factor is arguably more important and is possibly the cause of the first three; all these episodes took place within the context of authoritarian political regimes. They were not broad-based growth experiences. Instead, this was elite-driven growth for the benefit of the elite and exploiting existing comparative advantages. Thus it is not surprising that the improvements in living standards did not affect the entire society, but only a minority.

Why did these growth episodes not turn into a process of takeoff, ultimately leading to sustained growth? My main answer is related to that offered in Section 23.3 in Chapter 23. Growth under authoritarian regimes is possible. Entrepreneurs and workers can become better at what they do, achieve a better division of labor and improve the technologies they work with by tinkering and learning-by-doing. Moreover, those with political power and their

¹For example, in Ancient Greece Morris (2005) estimates that income per capita doubled or at most tripled in the 500 years between 800 B.C. and 300 B.C., and this was largely caused by “catch-up” type growth starting from unusually low levels in 800 B.C.

allies do have the necessary security of property rights to undertake investments. And some technological breakthroughs can always happen by chance. Nevertheless, a distinguishing feature of growth under authoritarian institutions is that it will look after the interests of the current elite. This means that, in the final analysis, growth must always rely on existing techniques and production relationships. It will never unleash the process of creative destruction and the entry of new talent in new businesses necessary to carry a nation to the state of sustained growth. In addition, technological constraints may have also played a role. For example, one may argue that the relatively rapid growth in the 19th century required skilled workers and it would have been prohibitively costly for a critical mass of workers to acquire the necessary human capital before the printing press was invented. Although the progress of technological knowledge is not monotone (and useful production techniques are sometimes forgotten), the technological know-how available to potential entrepreneurs at the end of the 18th century was undoubtedly greater than that available to potential entrepreneurs in Rome or Ancient Greece.

Let me focus on the political economy aspect and provide a few examples that may be useful to illustrate the main limits to growth under authoritarian regimes. The Chinese Empire has been technologically innovative during many distinct phases of its history. Productivity in the Chinese economy, especially in the Yangtze Delta and other fertile lands, has been high enough to support a high density of population. But the Chinese economy never came close to sustained growth. Authoritarian political institutions regulated economic activity tightly for most of Chinese history. The society has typically been hierarchical, with a clear distinction between the elite and the masses. This system did not allow free entry into business by new entrepreneurs that would adopt and exploit new technologies and unleash the powers of creative destruction. When prospects for economic growth conflicted with political stability, the elite always opted for maintaining stability, even if this came at the expense of potential economic growth. Thus China tightly controlled overseas and internal trade, did not develop the property rights and contracting institutions necessary for modern economic growth, and did not allow an autonomous middle-class to emerge as an economic and political force (Elvin, 1973, Mokyr, 1991, Wong, 1997).

The Ancient Greek and Roman civilizations are often viewed as the first democratic societies. One might therefore be tempted to count them as participatory regimes that should have achieved sustained economic growth. But this is not necessarily the case. First, as noted above, participatory regimes do not guarantee sustained economic growth when other preconditions have not been met. But more importantly, these societies were democratic only in comparison to others at the time. Both societies were representative only for a small fraction of the population. Production relied on slavery and coercion. Moreover, despite certain democratic practices, there was a clear distinction between a small elite, which monopolized economic and political power, and the masses, which consisted of both free plebs and slaves. Economic growth in both Greece and Rome did not rely on continuous innovation. Both societies managed to achieve high levels of productivity in agriculture, but

without changing the organization of production in a radical manner. Both societies benefited from their military superiority for a while, and challenges to their military power were also important factors in their decline.

The Ottoman Empire provides another example of a society that was successful for an extended period of time, but without ever transitioning into sustained growth. The Ottoman Empire, especially during the 14th, 15th and 16th centuries, achieved relative prosperity and great military strength. Agricultural productivity was high in many parts of the empire and military tribute contributed to state coffers and generated revenues to be distributed to parts of the population. But the state elite, controlling decision-making within the empire, never encouraged broad-based economic growth. There was no private property in land, trade was encouraged as long as it was consistent with the state's objectives, but always tightly controlled, and any new technology that could destabilize the power of the state was steadfastly blocked. Like China, Greece and Rome, the Ottoman growth first tapered off and then turned into decline (Pamuk, 2007).

The final example I will mention is the Spanish monarchy. By the beginning of the 16th century, the Spanish crown had achieved both political dominance over its own lands under Ferdinand and Isabella and control of the large overseas empire through its colonial enterprises. Many parts of greater Spain, including the South that was recently reconquered from the Moors and the lands of Aragon, were already prosperous in the 15th century. The whole of Spain became much wealthier with the transfer of gold, silver and other resources from the colonies in the 16th century. But this wealth did not translate into sustained growth. The colonial experiment was managed under a highly authoritarian regime set up by Ferdinand and Isabella, and the most lucrative businesses were allocated to the allies of the crown. The greater revenues generated from the colonies only helped to tighten the grip of the crown on the rest of the society and the economy. Instead of abating, absolutism increased. Trade and industry remained highly regulated, and groups not directly allied to the crown were viewed suspiciously and discriminated against. The most extreme example of this, the persecution of Jews that had started under the Inquisition, continued and spilled over to other independent merchants. Spain enjoyed the transfer of wealth from the colonies, but then experienced a very lengthy period of stagnation, with economic and political decline (Elliott, 1963).

It is also remarkable that in none of these cases did complementary economic institutions develop. Financial institutions were always rudimentary. The Roman Republic developed a precursor to the modern corporation and allowed some contracts between free citizens, but by and large, economic prosperity was built on traditional economic activities that did not necessitate complex relationships among producers and between firms and workers. Consequently, the structural transformations that accompany economic growth never took place in these societies. Life was largely rural, social relations were dominated by the state and community enforcement, and financial markets were rudimentary or nonexistent. Perhaps more important, there was little investment in human capital, except for the elite for whom education

was often not a means towards higher productivity. Without broad-based human capital and political rights, creative destruction becomes even more difficult as entrepreneurial activities by a large fraction of the population is essentially ruled out.

I suspect that these patterns are not coincidences. Economic life under authoritarian political regimes often tends to have many of these features. Growth relying on practices that increase the productivity of the elite in traditional activities can secure growth for a while. But it will never engender creative destruction. Growth will go hand-in-hand with the political domination of the elite and thus with entry barriers protecting the status and the power of the elite. Therefore, the answer to the question of “why not before 1800” is twofold. First, no society before 1800 invested in human capital, allowed new firms to bring new technology, and generally unleashed the powers of creative destruction. This might have been partly due to the difficulty of undertaking broad-based human capital investments in societies without the printing press and with only limited communication technologies. But it was also related to the reward structures for workers and firms. An important consequence of this pattern of growth is that no society experienced the sweeping structural transformations that are an essential part of modern economic growth (recall Chapter 21). Second, no society took these crucial steps toward sustained growth because all these societies lived under authoritarian political regimes.

Why did economic takeoff start in 1800 and in Western Europe? Before developing an answer to this question, let me take a slight digression. Faced with the patterns documented so far, one can adopt one of two distinct approaches. Either we can think that stagnation is the usual state of mankind and that something quite unusual, perhaps unlikely, needs to happen in order to break the cycle and lead to economic takeoff (Brenner, 1966). If this perspective is correct, there is no reason to expect that a similar takeoff would happen in other societies unless they were subject to similar, and similarly unusual, shocks or some other process of intervention or change that induced them to undergo similar changes. Alternatively, one might suppose that the impetus for growth is ever present and is kept in check by certain non-growth-enhancing institutions or market failures (Jones, 1988). Once the growth process starts, it is likely to continue and spread. Then, the question is pinpointing what the growth-blocking institutions and market failures are. My perspective is a mixture of these two views.

The division of labor emphasized by Adam Smith and capital accumulation always present growth opportunities to societies. Furthermore, human ingenuity is strong enough to create room for major technological breakthroughs in almost any environment. Thus, consistent with the second perspective, there is always a growth impetus in human societies. Nevertheless, this growth impetus may only be *latent* because it lives within the context of a set of political (economic) institutions. When these institutions are not encouraging growth—when they do not provide the right kind of reward structure, punish rather than reward innovators and investors—we do not expect the growth impetus to lead to sustained growth. Even in such environments, economic growth is possible and this is why China, Greece, Rome and

other empires experienced growth for part of their history. But this prosperity did not exploit the full growth impetus. In fact, such prosperity relied on political regimes that, by their nature, had to control the growth impetus, because the growth impetus would ultimately bring these same regimes down. Therefore, growth under authoritarian political institutions must be growth *despite* the reward structures, not because of them.

West European growth starting in the late 18th century was different because Western Europe underwent three important structural transformations starting in the late Middle Ages. These structural transformations created an environment in which the latent growth impetus could turn into an engine of sustained growth.

The first was the collapse of one of the pillars of the ancient regime, with the decline of feudal relations in Western Europe. Starting in the 13th century and especially after the Black Death during the mid-14th century, the feudal economic relations crumbled in many parts of Western Europe. Serfs were freed from their feudal dues either by default (because the relationship collapsed) or by fleeing to the already expanding city centers (Postan, 1966). This heralded the beginning of an important social transformation; urbanization and changes in social relations. But perhaps more importantly, it created a labor force ready to work at cheap wages in industrial and commercial activities. It also removed one of the greatest sources of conflict between existing elites and new entrepreneurs—competition in the labor market (recall from the analysis in Chapter 22 how competition in factor markets can be the source of a range of distortionary policies). The decline of the feudal order also weakened the power base of the European authoritarian regimes that were largely unchallenged until the end of the Middle Ages (Pirenne, 1937).

The second structural transformation was related. With the decline in population, real incomes increased in much of Europe and many cities created sufficiently large markets for merchants to seek new imports and for industrialists to seek new products (recall the impact of a decline in population on income per capita in the Solow or the neoclassical growth model, Chapters 2 and 8, or in the Malthusian model of Section 21.2 in Chapter 21, and the evidence in Chapter 4; recall also the importance of aggregate demand in jumpstarting industrialization emphasized in Section 21.5 in Chapter 21). During the Middle Ages, a range of important technologies in metallurgy, armaments, agriculture and basic industry, such as textiles, were already perfected (White, 1978, Mokyr, 1991, 2002). Thus the European economy had reached a certain level of technological maturity and perhaps created a platform for entrepreneurial activity in a range of areas and sufficient income levels to generate investment in physical capital and technology necessary for new production relations.

The more important change, however, was the political one. The late Middle Ages also witnessed the start of a political process that inexorably led to the collapse of absolutist monarchies and to the rise of constitutional regimes. The constitutional regimes that emerged in the 16th and 17th centuries in Western Europe were the first examples of participatory regimes, because they shifted political power to a large group of individuals that

were previously “outsiders” to political power, including the gentry, small merchants, proto-industrialists as well as overseas traders and financiers (Section 23.3 in Chapter 23). These regimes then provided secure property rights and growth-enhancing institutions for a broad cross-section of society. These institutional changes created the environment necessary for new investments and technological changes and the beginning of sustained growth, which would culminate in the Commercial Revolution in the Netherlands and Britain during the 17th century and in the British Industrial Revolution at the end of the 18th century. By 19th century, industry and commerce had spread to much of Western Europe (North and Thomas, 1973, Chapter 4).

It is noteworthy to emphasize that constitutional monarchies were not democracies as we understand them today. There was no one-person one-vote and the distinction between the rich and the poor was quite palpable. Nevertheless, these regimes were responses to the demands by the merchants, industrialists, and those with the resources who wished to participate in economic activity. More importantly, these constitutional regimes not only reformed the political institutions of Western Europe, but undertook a series of economic reforms facilitating modern capitalist growth. Internal tariffs and regulations were lifted. Entry into domestic businesses and foreign trade was greatly facilitated. For example, the process of financial development in Britain got underway with the founding of the Bank of England and other financial reforms.

These constitutional regimes that emerged, first in Britain and the Netherlands, then in France and other parts of Western Europe, paved the way for sustained economic growth based on property rights for a broad cross-section of society, investment in contract enforcement, law and order, and infrastructure, and free entry into existing and new business lines. According to the theoretical perspective developed in earlier chapters, these improved conditions should have led to greater investments in physical capital, human capital and technology (Chapters 4 and 22). This is indeed what happened and the process of modern economic growth, unprecedented for the world economy, started. Economic relations now relied on new businesses investing in industry and commerce and on the formation of complex organizational form and production relations. Growth did not immediately accelerate. Economic growth was present but modest during the 17th and 18th centuries (Maddison, 2001). But these institutional changes laid the foundations of the more rapid growth that was soon to come. Financial institutions developed, the urban areas expanded further, new technologies were invented, and markets became the primary arena for transactions and competition (North and Thomas, 1973). By 1800, the process of technological change and investment had progressed enough that historians view this as the beginning of the Industrial Revolution (Ashton, 1968, Mokyr, 1989). The first phase of the Industrial Revolution was followed by the production of yet newer technologies, more complex organizations, greater reliance on skills and human capital in the production process and increasing globalization of the world economy. By the second half of the 19th century, Western Europe had reached growth levels that were unprecedented.

Naturally, a complete answer to the question in the title of this subsection requires an explanation for why the constitutional regimes that were so important for modern economic growth emerged in Western Europe starting in the late 16th and 17th centuries. These institutions had their roots in the late medieval aristocratic parliaments in Europe, but more importantly, they were the outcome of radical reform resulting from the change in the political balance of power in Europe starting in the 16th century (Ertman, 1997). The 16th century witnessed a major economic transformation of Europe following the increase in international trade due to the discovery of the New World and the rounding of the Cape of Good Hope (Davis, 1973, Acemoglu, Johnson and Robinson, 2005a). Together with increased overseas trade came greater commercial activity within Europe. These changes led to a modest increase in living standards and more importantly, to greater economic and political power for a new group of merchants, traders and industrialists. These new men were not the traditional allies of the European monarchies. They therefore demanded, and often were powerful enough to obtain, changes in political institutions that provided them with greater security of property rights and government action to help them in their economic endeavors. By this time, with the collapse of the feudal order, the foundations of the authoritarian regimes that were in place in the Middle Ages were already weak. Nevertheless, the changes leading to the constitutional regimes did not come easy. The Dutch had to fight the Hapsburg monarchy to gain their independence as a republic. Britain had to endure the Civil War and the Glorious Revolution. France had to go through the Revolution of 1789 (Acemoglu, Johnson and Robinson, 2005a). But in all cases, the *ancien régime* gave way to more representative institutions, with greater checks on absolute power and greater participation by merchants, industrialists and entrepreneurs. It was important that the social changes led to a new set of political institutions, not simply to concessions. This is related to the theoretical ideas emphasized in Chapter 23; the nascent groups demanded long-term guarantees for the protection of their property rights and their participation in economic life, and this was most easily delivered by changes in political institutions, not by short-term concessions.

These changes created the set of political institutions that would then enable the creation of the economic institutions mentioned above. The collapse of the authoritarian political regimes and the rise of the first participatory regimes then opened the way for modern economic growth.

Why did some societies manage to benefit from the new technologies while others failed? The economic takeoff started in Western Europe, but quickly spread to certain other parts of the world. The chief importer of economic institutions and economic growth was the United States. The United States, founded by settler colonists, who had just defeated the British crown to gain their independence and set up a smallholder society, already had participatory political institutions. This was a society built by the people who would live in it, and they were particularly keen on creating checks and balances to prevent a strong political or economic elite. This environment turned out to be a perfect conduit for modern economic growth. The lack of a strong political and economic elite meant that a

broad cross-section of society could take part in economic activity, import technologies from Western Europe and then build their own technologies to quickly become the major industrial power in the world (Galenson, 1996, Engerman and Sokoloff, 1997, Keyssar, 2000, Acemoglu, Johnson and Robinson, 2002). In the context of this example, the importance of technology adoption from the world technology frontier is in line with the emphasis in Chapter 18, while the growth-promoting effects of a lack of elite creating entry barriers is consistent with the approach in Section 23.3 in Chapter 23.

Similar processes took place in other West European offshoots, for example, in Canada. Yet in other parts of the world, adoption of new technologies and the process of economic growth came as part of a movement towards defensive modernization. Japan started its economic and political modernization with the Meiji restoration (or perhaps even before) and a central element of this modernization effort was the importation of new technologies.

But these attitudes to new technologies were by no means universal. New technologies were not adopted, they were in fact resisted, in many parts of the world. This included most of Eastern Europe, for example, Russia and Austria-Hungary, where the existing land-based elites saw the technologies as a threat both to their economic interests, because they would lead to the end of the feudal relations that still continued in this part of Europe, and to their political interests, which relied on limiting the power of new merchants and slowing down the process of peasants migrating to cities to become the new working class (see Freudenberger, 1967, and Mosse, 1992, for evidence and Chapter 22 for a theoretical perspective). Similarly, the previously-prosperous plantation economies in the Caribbean had no interest in introducing new technologies and allowing free entry by entrepreneurs. These societies continued to rely on their agricultural staples. Industrialization, competition in free labor markets and workers investing in their human capital were seen as potential threats to the economic and political powers of the elite. The newly independent nations in Latin America were also dominated by a political elite, which continued the tradition of the colonial elites and showed little interest in industrialization. Much of South East Asia, the Indian subcontinent, and almost all of sub-Saharan Africa were still West European colonies, and were run under authoritarian and repressive regimes (often as producers of raw materials for the rapidly industrializing West European nations or as sources of tribute income). Free labor markets, factor mobility, entrepreneurial spirit, creative destruction and new technologies did not feature in the colonial political trajectories of these countries (Chapter 4).

Thus the 19th century, the age of industrialization, was only to see the industrialization of a few select places. Modern economic growth did not start in much of the world until early 20th century. By the 20th century, however, more and more nations had started importing some of the technologies that had been developed and used in Western Europe. The process of technology transfer pulling all of the countries integrated into global economy towards greater income levels (Chapter 19), thus started by the 20th century, but still not for all nations. Many more had to wait for their independence from their colonial masters, and even then, the end of colonialism led to a period of instability and infighting among would-be elites.

Once some degree of political stability was achieved and economic institutions that encourage growth were put in place, growth started. For example, growth in Australia and New Zealand was followed by Hong Kong, then by South Korea, then by the rest of South Asia and finally by India. In each of these cases, as emphasized in Chapters 20 and 21, growth went hand-in-hand with structural transformations. Once the structural transformations were under way, they facilitated further growth. Consistent with the picture in Chapter 19, societies integrated into the global economy started importing technologies and achieved average growth rates in line with the growth of the world technology frontier (and often exceeding those during their initial phase of catch-up). In most cases, this meant growth for the new members of the global economy, but not necessarily the disappearance of the income gap between these new members and the earlier industrializers.

In the meantime, many parts of the world continued to suffer political instability discouraging investment in capital and new technology, or even exhibited overt hostility to new technologies. These included parts of sub-Saharan Africa and until recently much of Central America. Returning to some of the examples discussed in Chapter 1, Nigeria and Guatemala, for example, failed to create incentives for its entrepreneurs or workers both during its colonial period and after independence. Both of these countries also experienced significant political instability and economically disastrous civil wars in the postwar era (recall the discussion of the implications of political instability in Chapter 23). Brazil managed to achieve some degree of growth, but this was mostly based on investment by large, heavily protected corporations and not on a sustained process of technological change and creative destruction (thus more similar to the oligarchic growth in terms of the model of Section 23.3 in Chapter 23). In these cases and others, policies that failed to provide secure property rights to new entrepreneurs and those that blocked the adoption of new technologies, as well as political instability and infighting among the elites, seem to have played an important role in the failure to join the world economy and its growth process. Overall, these areas fell behind the world average in the 19th century and continued to do so for most of the 20th century. Many nations in sub-Saharan Africa, such as Congo, Sudan and Zimbabwe, are still amidst political turmoil and fail to offer even the most basic security of property rights to their entrepreneurs. Consequently, many are still falling further behind the world average.

24.3. Many Remaining Questions

The previous section provided a narrative emphasizing how technological changes transformed the world economy starting in the 18th century and how certain societies took the advantage of these changes while others failed to do so. Parts of the story receive support from the data. The importance of industrialization in the initial takeoff is now well documented. There is broad consensus that economic institutions protecting property rights and allowing for free entry and introduction of new technologies were important in the 19th century and continue to be important today in securing economic growth (see Chapter 4). There is also general consensus that political instability, weak property rights, and lack of infrastructure

are major impediments to growth in sub-Saharan Africa. Nevertheless, the narrative above is speculative. These factors might be important, but they do not need to be the main ones explaining the evolution of the world income distribution over the past 200 years. Moreover, as yet there is no consensus on the role of political institutions in this process.

Thus what I have presented above should be taken for what it is; a *speculative* answer that needs to be investigated more. My purpose in outlining it was not only that I suspect this answer is much truth to it, but also to show how the various models developed in this book can help us better frame answers to fundamental questions of economic growth (and of economics and social sciences in general). I should add that further investigation of the causes of the world's takeoff into sustained growth and the failure of some nations to take advantage of this process is only one of the many remaining questions. The political economy of growth is important because it enables us to ask and answer questions about the fundamental causes of economic growth. But many other aspects of the process of growth require further investigation. In some sense, the field of economic growth is one of the more mature areas in economics, and certainly within macroeconomics it is the area where there is broadest agreement on what types of models are useful for the study of economic dynamics and for empirical analysis. And yet, there is so much that we still do not know.

I will now end by mentioning a few of the areas where the potential for more theoretical and empirical research is clear. First, while in this chapter I have largely focused on factors facilitating or preventing the adoption of technologies in less-developed nations, there is still much to be done to understand the pace at which technological progress happens in the frontier economies. Our models of endogenous technological change give us the basic framework for thinking about how profit incentives shape investments in new technologies. But much needs to be done to understand how market structure affects innovation. Chapter 12 highlighted how different market structures will create different incentives for technological change. We saw in Chapter 14 how competition among firms within an industry can be important for the growth process. But most of our understanding of these issues is at a qualitative level. For example, we lack a framework similar to that used for the analysis of the effects of capital and labor income taxes and indirect taxes in public finance, which we could use to analyze the effects of various regulations, of intellectual property right policies and of anticompetitive laws on innovation and economic growth. Since the pace at which the world technology frontier progresses has a direct effect on the growth of many nations, even small improvements in the environment for innovation in advanced economies could have important dividends for the rest of the world.

In addition to the industrial organization of innovation, the contractual structure of innovation needs further study. We live in a complex society, in which most firms are linked to others as suppliers or downstream customers, and most firms are connected to the rest of the economy indirectly through their relationship with the financial markets. All of these relationships are mediated by various explicit and implicit contracts. Similarly, the employment relationship that underlies the productivity of most firms relies on contractual relations

between employers and employees. We know that moral hazard and holdup problems occur in all these contractual relationships. But how important are they for the process of economic growth? Can improvements in contracting institutions improve innovation and technological upgrading in frontier economies? Can they also facilitate technology transfer? These are basic, but as yet, unanswered questions. The contractual foundations of economic growth are still in their infancy and require much work.

The previous section emphasized how many economies started the growth process by importing technologies and thus integrating into the global economy. Today we live in an increasingly globalized and globalizing economy. But there is still much to understand about how technology is transferred from some firms to others, and from advanced economies to less-developed ones. The models I presented in Chapter 19 emphasized the importance of human capital, barriers to technology adoption, issues of appropriate technology and contracting problems. Nevertheless, most of the models are still at the qualitative level and we lack a framework that can make quantitative predictions about the pace of technology diffusion. We have also not yet incorporated many important notions related to technology transfer into our basic frameworks. These include ideas related to tacit knowledge, the workings of the international division of labor, the role of trade secrecy and the system of international intellectual property rights protection.

The reader will have also noticed that the material presented in Chapter 21 is much less unified and perhaps more speculative than the rest of the book. Although some of this reflects the fact that I had to simplify a variety of models to be able to present them in a limited space, much of it is because we are far from a unified framework for understanding the process of economic development and the structural transformations that it involves. We know that economic growth is accompanied by structural change. Some of the structural change can be viewed as a simple byproduct of economic growth, such as the increase in services relative to agriculture. But other structural transformations, including developments in financial markets, changes in contract enforcement regimes, urbanization and the amount and composition of human capital investments are not simple byproducts of economic growth. These structural transformations are intimately linked to the process of economic development, and they may be facilitators or even preconditions for growth. Thus lack of significant structural transformation might be an important factor in delaying or preventing economic growth. To understand these questions, we require models with stronger theoretical foundations, a unified approach to these related issues, and a greater effort to link the models of economic development to the wealth of empirical evidence that the profession has now accumulated on economic behavior in less-developed economies.

Last but not least, given the narrative in the last section and the discussion in Chapters 4, 22 and 23, the reader will not be surprised that I think many important insights about economic growth lie in political economy. But understanding politics is in many ways harder than understanding economics, because political relations are even more multifaceted. Although I believe that the political economy and growth literatures have made important

advances in this area over the past decade or so, much remains to be done. The political economy of growth is in its infancy and as we further investigate why societies make different choices, we will gain a better understanding of the process of economic growth.

Part 9

Mathematical Appendices

CHAPTER A

Odds and Ends in Real Analysis and Applications to Optimization

This chapter is included as a review of some basic material from real analysis. Its main purpose is to make the book self-contained and also include explicit statements of some of the theorems that are used in the text. The material here is not meant to be a comprehensive treatment of real analysis. Space restrictions preclude me from attempting to do justice to any of the topics here, so my purpose is only a brief review. Accordingly, many results will be stated without proof and other important results will be omitted as long as they are not referred to in the text or do not play an important role in the development (or the proof of some of the other results presented here). I will state some useful results as *Fact* (often leaving their proof as an exercise). These results are typically used or referred to in the text, or are inputs into proving the more important results in this appendix. The more important results will be stated as *Theorem*.

It should be emphasized that the material here is not a substitute for a basic “*Mathematics for Economists*” type review or textbook. An excellent book of this sort is Simon and Blume (1994), and I will presume that the reader is familiar with most of the material in this or a similar book. In particular, I assume that the reader is comfortable with linear algebra, functions, relations, set theoretic language, calculus of multiple variables and basic proof techniques.

To gain a deeper understanding and appreciation of the material here, the reader is encouraged to consult one of many excellent books on real analysis, functional analysis and general topology. Some of the material here is simply a review of introductory real analysis more or less at the level of the classic books by Rudin (1976) or Apostol (1974). Some of the material, particularly those concerning topology and infinite-dimensional analysis, is more advanced and can be found in Conway (1990), Kelley (1955), Kolmogorov and Fomin (1970), Royden (1994) and Aliprantis and Border (1999). Excellent references for applications of these ideas to optimization problems include Luenberger (1969) and Berge (1963). A recent treatment of some of these topics with economic applications is presented in Ok (2007).

A.1. Distances and Metric Spaces

Throughout, X denotes a set and $x \in X$ is a generic element of the set X . A set X can be viewed as a space or as a subset of a larger set (space) Z . I denote a subset Y of X as

$Y \subset X$ (which includes the case where $Y = X$ as well as the empty set, $\emptyset$). For any $Y \subset X$, $X \setminus Y$ stands for the *complement* of Y in X , i.e., $X \setminus Y = \{x : x \in X \text{ and } x \notin Y\}$.¹

Of special importance for our purposes here are two types of spaces: (1) finite-dimensional Euclidean spaces, which I will denote by $X \subset \mathbb{R}^K$ ($K \in \mathbb{N}$); (2) infinite-dimensional spaces, such as spaces of sequences or spaces of functions, which feature in discrete-time and continuous-time dynamic optimization problems. For our purposes, the most useful sets are those that are equipped with a *metric*, so that they can be treated as a *metric space*. Metric spaces play a major role in the analysis of dynamic programming problems in Chapters 6 and 16.

DEFINITION A.1. *Let X be a nonempty set. A function $d : X \times X \rightarrow \mathbb{R}_+$ is a **metric** (distance function) if, for any x, y , and z in X , it satisfies the following three conditions:*

- (1) (**Properness**) $d(x, y) = 0$ if and only if $x = y$.
- (2) (**Symmetry**) $d(x, y) = d(y, x)$.
- (3) (**Triangle Inequality**) $d(x, y) \leq d(x, z) + d(z, y)$.

*A nonempty set X equipped with a metric d constitutes a **metric space** (X, d) .*

In this definition, as in all mathematical definitions that follow, “if” is used instead of “if and only if,” since the context makes it clearer that the notion (e.g., “metric”) is being defined by the mathematical statements following it, thus “if and only if” is implicit. I will adhere to this convention throughout.

The same set can be equipped with different metrics. In many cases, different metrics give equivalent results (in particular, they imply the same topological properties). When this is the case, we say that two metrics are *equivalent*, and the definition for this is given below in Definition A.4 (but I am mentioning this here, since I will refer to equivalent metrics in the next example).

EXAMPLE A.1. The following are examples of metric spaces. In each case, properness and symmetry are easy to verify, but verifying that the triangle inequality holds requires some work (see Exercise A.2).

- (1) For any $X \subset \mathbb{R}^K$, let x_i be the i th component of $x \in X$. Then, the usual *Euclidean distance* $d(x, y) = \left(\sum_{i=1}^K |x_i - y_i|^2 \right)^{1/2}$ is a metric and thus the Euclidean space with its usual distance constitutes a metric space. It is typically referred to as the *K -dimensional Euclidean space*. Moreover, one can construct alternative metrics for Euclidean spaces that are *equivalent*, in the sense that which of these metrics one uses has no bearing on the topological properties or on any of the other issues we focus on here. These metrics include the family $d_p(x, y) = \left(\sum_{i=1}^K |x_i - y_i|^p \right)^{1/p}$ for $1 \leq p < \infty$. An extreme element of this family, which also defines an equivalent metric on finite-dimensional Euclidean spaces, is $d_\infty(x, y) = \sup_i |x_i - y_i|$.

¹Throughout this chapter, I will simplify the notation and use “=” for definitions instead of “ $\equiv$ ”.

- (2) For any nonempty set X , one can construct the *discrete metric* defined as $d(x, y) = 1$ if $x \neq y$ and $d(x, y) = 0$ if $x = y$. In this case (X, d) is a *discrete space*.
- (3) Let $X \subset \mathbb{R}^K$ and consider the set of continuous and bounded (real-valued) functions $f : X \rightarrow \mathbb{R}$ denoted by $\mathbf{C}(X)$. A natural metric for $\mathbf{C}(X)$ is the *sup metric* $d_\infty(f, g) = \sup_{x \in X} |f(x) - g(x)|$. Thus $(\mathbf{C}(X), d_\infty)$ is a metric space. The same metric can be used for the set of bounded (but not necessarily continuous) functions, $\mathbf{B}(X)$, leading to the metric space $(\mathbf{B}(X), d_\infty)$.
- (4) Let $\ell \subset \mathbb{R}^\infty$ be a set consisting of infinite sequences of real numbers. For example, $x = (x_1, x_2, x_3, \dots)$ would be a typical element of ℓ provided that $x_i \in \mathbb{R}$ for each $i = 0, 1, 2, \dots$. A family of metrics for this set is given by $d_p(x, y) = (\sum_{i=1}^\infty |x_i - y_i|^p)^{1/p}$ for $1 \leq p < \infty$ (provided that $(\sum_{i=1}^\infty |x_i|^p)^{1/p} < \infty$ for all $x \in \ell$) or by $d_\infty(x, y) = \sup_i |x_i - y_i|$ (provided that $\sup_i |x_i| < \infty$ for all $x \in \ell$). For any $p \in [1, \infty]$, (ℓ, d_p) is a metric space, sometimes denoted by ℓ_p .

Metric spaces are particularly useful because they enable us to define neighborhoods and open sets, which are the building blocks of mathematical analysis and essential elements for our investigation of optimization problems. Below, I will define notions of neighborhood and openness somewhat more generally, but it is useful to start from the following simpler definition.

DEFINITION A.2. Let (X, d) be a metric space and $\varepsilon > 0$ be a real number. Then, for any $x \in X$,

$$\mathcal{N}_\varepsilon(x) = \{y \in X : d(x, y) < \varepsilon\}$$

is the ε -**neighborhood** of x .

EXAMPLE A.2. In the simplest case where $X \subset \mathbb{R}$ and $x \in X$, $d(x, y) = |x - y|$, $\mathcal{N}_\varepsilon(x) = (x - \varepsilon, x + \varepsilon) \cap X$.

DEFINITION A.3. Let (X, d) be a metric space. Then, $Y \subset X$ is **open** in X if for each $y \in Y$, there exists $\varepsilon > 0$ such that $\mathcal{N}_\varepsilon(y) \subset Y$. $Z \subset X$ is **closed** in X if $X \setminus Z$ is open in X .

The *closure* of a set Y in X is $\overline{Y} = \{y \in X : \text{for each } \varepsilon > 0, \mathcal{N}_\varepsilon(y) \cap Y \neq \emptyset\}$, that is, every neighborhood of each point in $\overline{Y}$ contains at least one point of Y . Clearly, $Y \subset \overline{Y}$. Moreover, if Y is closed, then $Y = \overline{Y}$. The *interior* of a set Y in X can then be defined as $\text{Int}Y = Y \setminus \overline{(X \setminus Y)}$. If Y is an open subset of X , then $\overline{(X \setminus Y)} = X \setminus Y$, and therefore $\text{Int}Y = Y$.

EXAMPLE A.3. Again in the simplest case where $X = [0, 1]$ and $d(x, y) = |x - y|$, for any $x \in (0, 1)$ and $\varepsilon > 0$ sufficiently small, $(x - \varepsilon, x + \varepsilon)$ is open in X , whereas $[0, 1] \setminus (x - \varepsilon, x + \varepsilon) = [0, x - \varepsilon] \cup [x + \varepsilon, 1]$ is closed in X . Also, $\text{Int}(x - \varepsilon, x + \varepsilon) = (x - \varepsilon, x + \varepsilon)$, $\text{Int}([0, 1] \setminus (x - \varepsilon, x + \varepsilon)) = (0, x - \varepsilon) \cup (x + \varepsilon, 1)$, $\overline{(x - \varepsilon, x + \varepsilon)} = [x - \varepsilon, x + \varepsilon]$, and $\overline{[0, 1] \setminus (x - \varepsilon, x + \varepsilon)} = [0, x - \varepsilon] \cup [x + \varepsilon, 1]$.

FACT A.1. Let (X, d) be a metric space. X and $\emptyset$ are both open and closed sets.

The importance of the following theorem will become clear once we turn to the somewhat more abstract topological characterization of closed and open sets. We say that A is a totally (linearly) *ordered set*, if there exists a *transitive* relation “ $\geq$ ” such that for any distinct α, α' in A , we have either $\alpha \geq \alpha'$ or $\alpha' \geq \alpha$. The simplest examples would be $A \subset \mathbb{R}$ or $A = \mathbb{N}$. Then, for an ordered set, $\{X_\alpha\}_{\alpha \in A}$ is a *collection of sets*. If A is countable [finite], then $\{X_\alpha\}_{\alpha \in A}$ is a countable [finite] collection of sets, but it can also be an arbitrary collection of sets. Let us also use X_α^c to denote the complement of X_α in X , i.e., $X_\alpha^c = X \setminus X_\alpha$.

THEOREM A.1. (*Properties of Open and Closed Sets*) Let (X, d) be a metric space and $\{X_\alpha\}_{\alpha \in A}$ be a collection of sets with $X_\alpha \subset X$ for all $\alpha \in A$.

- (1) If each X_α is open in X , then $\bigcup_{\alpha \in A} X_\alpha$ is open.
- (2) If each X_α is open in X and $\{X_\alpha\}_{\alpha \in A}$ is a finite collection of sets (i.e., A is finite), then $\bigcap_{\alpha \in A} X_\alpha$ is open.
- (3) If each X_α is closed in X , then $\bigcap_{\alpha \in A} X_\alpha$ is closed.
- (4) If each X_α is closed in X and $\{X_\alpha\}_{\alpha \in A}$ is a finite collection of sets, then $\bigcup_{\alpha \in A} X_\alpha$ is closed.

PROOF. (Part 1) Let $\{X_\alpha\}_{\alpha \in A}$ be an arbitrary collection of open sets in X . If $\bigcup_{\alpha \in A} X_\alpha$ is empty, then it is open by Fact A.1. If it is nonempty, then for each $x \in \bigcup_{\alpha \in A} X_\alpha$, it must be the case that $x \in X_{\alpha'}$ for some $\alpha' \in A$. Since $X_{\alpha'}$ is open, there exists $\varepsilon > 0$ such that $\mathcal{N}_\varepsilon(x) \subset X_{\alpha'} \subset \bigcup_{\alpha \in A} X_\alpha$, establishing that for each $x \in \bigcup_{\alpha \in A} X_\alpha$ there exists an ε -neighborhood of x in $\bigcup_{\alpha \in A} X_\alpha$ so that $\bigcup_{\alpha \in A} X_\alpha$ is open.

(Part 2) Let $\{X_\alpha\}_{\alpha \in A}$ be a finite collection of open sets in X (enumerated by $\alpha = 1, 2, \dots, N$). Once again, if $\bigcap_{\alpha \in A} X_\alpha$ is empty, it is open by Fact A.1. If it is nonempty, then for each $x \in \bigcap_{\alpha \in A} X_\alpha$, $x \in X_\alpha$ for $\alpha = 1, 2, \dots, N$. Since X_α is open, then by definition there exists $\varepsilon_\alpha > 0$ such that $\mathcal{N}_{\varepsilon_\alpha}(x) \subset X_\alpha$ for each $\alpha = 1, 2, \dots, N$. Let $\varepsilon = \min\{\varepsilon_1, \varepsilon_2, \dots, \varepsilon_N\}$. Clearly, $\varepsilon > 0$. Moreover, by construction, $\mathcal{N}_\varepsilon(x) \subset \mathcal{N}_{\varepsilon_\alpha}(x) \subset X_\alpha$ for each $\alpha = 1, 2, \dots, N$. Therefore, $\mathcal{N}_\varepsilon(x) \subset \bigcap_{\alpha \in A} X_\alpha$, proving the claim.

(Parts 3 and 4) These follow immediately from De Morgan's Law, which states that

$$\left(\bigcup_{\alpha \in A} X_\alpha \right)^c = \bigcap_{\alpha \in A} X_\alpha^c.$$

□

The restriction to finite collections is important in Part 2 of Theorem A.1. Consider the following example.

EXAMPLE A.4. Let $X = \mathbb{R}$ with the Euclidean metric $d(x, y) = |x - y|$. Take the subsets $X_\alpha = (0, 1 + \alpha^{-1})$ of X for $\alpha \in \mathbb{N}$ and consider the infinite intersection $\bigcap_{\alpha \in \mathbb{N}} X_\alpha$. It can

be verified that $\bigcap_{\alpha \in \mathbb{N}} X_\alpha = (0, 1]$, which is not an open set. An even simpler example is the subsets $X_\alpha = (-1/\alpha, 1/\alpha)$, where $\bigcap_{\alpha \in \mathbb{N}} X_\alpha = \{0\}$, which is not open.

DEFINITION A.4. Two metrics d and d' defined on X are **equivalent** if they both generate the same collection of open sets in X . Alternatively, let $\mathcal{N}_\varepsilon$ and $\mathcal{N}'_\varepsilon$ refer to neighborhoods defined by these metrics. Two metrics are equivalent if for each $x \in X$ and $\varepsilon > 0$, there exists $\delta > 0$ and $\delta' > 0$ such that $\mathcal{N}'_\varepsilon(x) \subset \mathcal{N}_\delta(x)$ and $\mathcal{N}_\varepsilon(x) \subset \mathcal{N}'_{\delta'}(x)$.

Exercise A.4 verifies that the two parts of Definition A.4 imply each other.

DEFINITION A.5. Let (X, d) be a metric space. Then, $Y \subset X$ is **bounded** if there exists $x \in X$ and $\delta \in (0, +\infty)$ such that $Y \subset \mathcal{N}_\delta(x)$. If $Y \subset X$ is not bounded, then it is **unbounded**.

EXAMPLE A.5. Let $X = \mathbb{R}$ and $d(x, y) = |x - y|$. The subsets $(0, 1)$ and $[0, 1]$ of $\mathbb{R}$ are bounded, while the subset $\mathbb{R}_+ = [0, \infty)$ of $\mathbb{R}$ is unbounded.

A.2. Mappings, Functions, Sequences, Nets and Continuity

A *mapping* ϕ from X to Y is a subset of $X \times Y$ such that for each $x \in X$, there exists some $y \in Y$ with $(x, y) \in \phi$. As is usual, I denote this by $\phi : X \rightarrow Y$. Throughout $\phi : X \rightarrow Y$ implies that $\phi(x)$ is defined for each $x \in X$. I have also adopted the convention that ϕ assigns a single element of the set Y to $x \in X$ and thus write $\phi(x)$ as an element of Y , that is, $\phi(x) \in Y$ (and in terms of the definition above, if $(x, y) \in \phi$ and $(x, z) \in \phi$, then $y = z$). This is without any loss of generality, since the space Y is not restricted. For example, for a set Z , we could specify $Y = \mathcal{P}(Z)$ (where, recall that, $\mathcal{P}(Z)$ denotes the set of all subsets of Z). In this case, an element of Y would be a subset of Z . Thus, one can alternatively write that for $x \in X$, $\phi(x) \in Y$ or $\phi(x) \subset Z$. I will also use the notation $\phi(X')$ for some $X' \subset X$ to designate the *image* of the set X' , defined as

$$\phi(X') = \{y \in Y : \exists x \in X' \text{ with } \phi(x) = y\}.$$

For a mapping $\phi : X \rightarrow Y$, X is also referred to as the *domain* of ϕ , while Y is its *range*. One might want to reserve the term range to $Y' \subset Y$ such that $Y' = \phi(X)$. For our purposes here, this distinction is not important.

The notation ϕ^{-1} is standard to denote the inverse of the mapping ϕ . Notice that ϕ^{-1} may not be single-valued even if ϕ is single valued, since more than one x in X can have the same image in Y . For $Y' \subset Y$, let

$$\phi^{-1}(Y') = \{x \in X : \exists y \in Y' \text{ with } \phi(x) = y\}.$$

By a *function* f I typically refer to a *real-valued* mapping, i.e., $f : X \rightarrow \mathbb{R}$ for some arbitrary set X . I will use lowercase letters to refer to functions. I will use the term *correspondence* to refer to a set-valued mapping, i.e., $F : X \rightarrow \mathcal{P}(Z)$ for some set Z . This means that the mapping F assigns a subset of Z to each element of x . I will use uppercase

letters to refer to correspondences. Since they will play an important role below, the following common notation will be used for correspondences: $F : X \rightrightarrows Z$. When the range of the correspondence is the real numbers, then we naturally have $F : X \rightrightarrows \mathbb{R}$.

DEFINITION A.6. Let (X, d) be a metric space. A **sequence**, denoted by $\{x_n\}_{n=1}^\infty$ (or simply by $\{x_n\}$) is a mapping ϕ with domain given by the natural numbers, $\mathbb{N}$, and range given by X .

The important point is that in all cases the domain of the mapping ϕ that defines the sequence is $\mathbb{N}$, so that $\{x_n\}_{n=1}^\infty$ is a countable (infinite) sequence. One can easily generalize the notion of a sequence to that of *nets*, which is particularly useful in problems of continuous-time optimization.

DEFINITION A.7. A **net**, denoted by $\{x_\alpha\}_{\alpha \in A}$ for some ordered set A , is a real-valued function with domain given by A .

Whenever sequences and nets have real numbers as elements, the underlying metric space relevant for the convergence is $(\mathbb{R}, d)$, with d referring to the usual Euclidean metric $d(x, y) = |x - y| = \sqrt{|x - y|^2}$.

EXAMPLE A.6. $\{x_n\}_{n=1}^\infty$ such that $x_n = 1/n$ for each $n \in \mathbb{N}$ is a sequence, while $\{x_\alpha\}_{\alpha \in A}$ $x_\alpha = 1/\alpha$ for each $\alpha \in (0, 1]$ is a net.

DEFINITION A.8. Consider the sequence $\{n_k\}_{k=1}^\infty$ of positive increasing natural numbers (such that $n_k > n_{k'}$ whenever $k > k'$). Then, for a given sequence $\{x_n\}_{n=1}^\infty$, $\{x_{n_k}\}$ is a **subsequence** of $\{x_n\}_{n=1}^\infty$.

A subnet can be defined in a similar manner.

DEFINITION A.9. Let (X, d) be a metric space. A sequence $\{x_n\}_{n=1}^\infty$ in X is **convergent** and has **limit point** $x \in X$ if for every $\varepsilon > 0$, there exists $N(\varepsilon) \in \mathbb{N}$ such that $n \geq N(\varepsilon)$ implies $d(x_n, x) < \varepsilon$. We write this as $\lim_{n \rightarrow \infty} x_n = \lim x_n = x$ or simply as $\{x_n\}_{n=1}^\infty \rightarrow x$.

DEFINITION A.10. Let $\{x_\alpha\}_{\alpha \in A}$ be a net in a metric space (X, d) . Then, $\{x_\alpha\}_{\alpha \in A}$ is **convergent** and has **limit point** x if for each $\varepsilon > 0$, there exists $\bar{\alpha}$ such that for all $\alpha \geq \bar{\alpha}$, $x_\alpha \in \mathcal{N}_\varepsilon(x)$.

FACT A.2. If a sequence $\{x_n\}_{n=1}^\infty$ or a net $\{x_\alpha\}_{\alpha \in A}$ in X is convergent, then it has a unique limit point $x \in X$.

PROOF. See Exercise A.6. □

FACT A.3. $\{x_n\}_{n=1}^\infty$ in X is convergent if and only if every subsequence of $\{x_n\}_{n=1}^\infty$ in X is convergent.

PROOF. See Exercise A.7. □

EXAMPLE A.7. Note, however, that convergence of a subsequence (or a subnet) does not guarantee convergence of the original sequence. Consider the sequence $\{x_n\}_{n=1}^{\infty}$ such that $x_n = (-1)^n$. Clearly, this sequence is not convergent. But picking $\{n_k\}_{k=1}^{\infty}$ as the even natural numbers, we construct a convergent subsequence $\{x_{n_k}\}$ with limit point 1.

Let $\bar{\mathbb{R}}$ denote the *extended real numbers*, that is, $\bar{\mathbb{R}} = \mathbb{R} \cup \{-\infty\} \cup \{+\infty\}$. It is straightforward to verify that $(\bar{\mathbb{R}}, \bar{d})$ is a metric space, where $\bar{d}(x, y) = d(x, y) / (1 + d(x, y))^2$, with $d(x, y)$ denoting the standard Euclidean metric now allowed to take infinite values.

FACT A.4. *Let $\{x_n\}$ be a sequence or net in $\bar{\mathbb{R}}$ (equipped with its usual metric). If $\{x_n\}$ is monotone (nondecreasing or nonincreasing), then it is convergent.*

PROOF. See Exercise A.8. □

DEFINITION A.11. *Let $X \subset \mathbb{R}$. Then, the **supremum** of X , denoted by $\sup X$, is the smallest $\bar{x} \in \bar{\mathbb{R}}$ such that $\bar{x} \geq x$ for all $x \in X$. If there does not exist $\bar{x} \in \mathbb{R}$ for which this is true, then clearly $\bar{x} = \infty$. Similarly, the **infimum** of X , denoted by $\inf X$, is the greatest $\underline{x}$ such that $\underline{x} \leq x$ for all $x \in X$, where again $\underline{x} = -\infty$ is allowed. If $\bar{x} = \sup X \in X$, then we refer to $\bar{x}$ as the **maximum** of X and denote it by $\bar{x} = \max X$. Similarly, if $\underline{x} = \inf X \in X$, then $\underline{x}$ is the **minimum** of X and is denoted by $\underline{x} = \min X$.*

Since X here itself can be taken to be a sequence of numbers, supremum and infimum can be defined for sequences. In particular, for $\{x_n\}_{n=1}^{\infty}$ in $\mathbb{R}$ construct the sequences $\{x'_n\}_{n=1}^{\infty}$ and $\{x''_n\}_{n=1}^{\infty}$ such that $x'_n = \sup_{k \geq n} \{x_k\}$ and $x''_n = \inf_{k \geq n} \{x_k\}$. Clearly, $\{x'_n\}_{n=1}^{\infty}$ is monotone (nonincreasing) and $\{x''_n\}_{n=1}^{\infty}$ is monotone (nondecreasing). Therefore, by Fact A.4, $\lim_{n \rightarrow \infty} x'_n$ exists. Let us denote it by $\limsup x_n$, and also $\lim_{n \rightarrow \infty} x''_n$ exists and is denoted by $\liminf x_n$. The same construction works with nets without any modification. The following results on limits (for sequences or nets) will be used in various proofs, especially in Chapter 7.

FACT A.5. *Let $\{x_n\}$ be a sequence or net in $\bar{\mathbb{R}}$. Then:*

- (1) $\inf_n x_n, \liminf x_n, \limsup x_n$ and $\sup_n x_n$ exist and satisfy

$$\inf_n x_n \leq \liminf x_n \leq \limsup x_n \leq \sup_n x_n.$$

- (2) $\{x_n\}$ is convergent if and only if

$$\liminf x_n = \limsup x_n,$$

and in this case, both of these are denoted as $\lim x_n = x$.

- (3) Let $\{y_n\}$ be such that $x_n \leq y_n$ for all n . Then

$$\liminf x_n \leq \liminf y_n \text{ and } \limsup x_n \leq \limsup y_n,$$

and moreover, if the limits exist,

$$\lim x_n \leq \lim y_n.$$

(4) If $\lim x_n y_n = 0$, then

$$\lim x_n |y_n| = \lim |x_n| y_n = \lim |x_n| |y_n| = 0.$$

Moreover, either $\lim x_n = 0$ or $\lim y_n = 0$.

(5) Suppose $\lim x_n = x \in \mathbb{R}$ (i.e., finite). If $\lim [x_n + y_n]$ exists, then

$$\lim [x_n + y_n] = x + \lim y_n.$$

If $\lim x_n y_n$ exists and $x \neq 0$, then

$$\lim x_n y_n = x \lim y_n.$$

In both cases, $\lim y_n$ also exists (though may not be finite).

PROOF. See Exercise A.12. □

Another useful result is the following.

FACT A.6. Let $\{x_n\}$ be a sequence or net in $\bar{\mathbb{R}}$. If all convergent subsequences or subnets $\{x_{n_k}\}$ have the same limit point $x^* \in \bar{\mathbb{R}}$, then $\{x_n\}$ also converge to x^* .

PROOF. If all convergent subsequences or subnets have the same limit point x^* , then $\liminf x_n = \limsup x_n = x^*$, so the result follows from Fact A.5(2). □

DEFINITION A.12. Let (X, d) be a metric space. A sequence $\{x_n\}_{n=1}^{\infty}$ in X is a **Cauchy sequence** if for each $\varepsilon > 0$, there exists $M(\varepsilon) \in \mathbb{N}$ such that for any $n, m \geq M(\varepsilon)$, $d(x_n, x_m) < \varepsilon$.

LEMMA A.1. Let (X, d) be a metric space and $\{x_n\}_{n=1}^{\infty}$ be a convergent sequence in X . Then, it is a Cauchy sequence.

PROOF. Fix $\varepsilon > 0$. Since $\{x_n\}_{n=1}^{\infty}$ is convergent, $\lim_{n \rightarrow \infty} x_n = x$ exists. Then, by the triangle inequality, for any x_n, x_m ,

$$(A.1) \quad d(x_n, x_m) \leq d(x_n, x) + d(x_m, x).$$

Since $\lim_{n \rightarrow \infty} x_n = x$, by Definition A.9 there exists $M(\varepsilon)$ such that for any $n \geq M(\varepsilon)$, $d(x_n, x) < \varepsilon/2$. Combining this with (A.1) implies that $d(x_n, x_m) < \varepsilon$, establishing the desired result. □

The converse of this lemma is not true, as illustrated by the following example.

EXAMPLE A.8. Let $X = (0, 1]$ and $d(x, y) = |x - y|$. Consider the sequence $x_n = 1/n$. This is clearly Cauchy, but does not converge to any point in X , and is thus not convergent.

DEFINITION A.13. A metric space (X, d) is **complete** if every Cauchy sequence in (X, d) is convergent.

Examples of complete spaces include any closed subset of the Euclidean space, as well as the metric space of continuous bounded (real-valued) functions with the sup metric, $(\mathbf{C}(X), d_\infty)$, introduced in Example A.1 (see Exercise A.9). The importance of complete metric spaces is illustrated by the Contraction Mapping Theorem, Theorem 6.7, which was presented in Section 6.4. The following fact is straightforward.

FACT A.7. *Let (X, d) be a complete metric space. A closed subset X' of X is also complete.*

I now briefly discuss continuity of mappings and functions in metric spaces.

DEFINITION A.14. *Let (X, d_X) and (Y, d_Y) be metric spaces and consider a mapping $\phi : X \rightarrow Y$. ϕ is **continuous at** $x \in X$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that whenever $d_X(x, x') < \delta$, then $d_Y(\phi(x), \phi(x')) < \varepsilon$. ϕ is **continuous on** X if it is continuous at each $x \in X$.*

FACT A.8. *Equivalently, ϕ is continuous at x if for all $\{x_n\}_{n=1}^\infty \rightarrow x$, $\{\phi(x_n)\}_{n=1}^\infty \rightarrow \phi(x)$.*

FACT A.9. *Let (X, d_X) , (Y, d_Y) and (Z, d_Z) be metric spaces and consider the mappings $\phi : X \rightarrow Y$ and $\gamma : Y \rightarrow Z$. If ϕ is continuous at x' and γ is continuous at $\phi(x')$, then $\gamma \circ \phi = \gamma(\phi(x))$ is continuous at x' .*

PROOF. See Exercise A.13. □

Similarly, sums and products of continuous functions are continuous and ratios of real-valued continuous functions are continuous as long as the denominator is not equal to zero.

The following is an important theorem in its own right and will also motivate the somewhat more general treatment of continuity in the next section.

THEOREM A.2. (**Open Sets and Continuity I**) *Let (X, d_X) and (Y, d_Y) be metric spaces and consider the mapping $\phi : X \rightarrow Y$. ϕ is continuous if and only if for every $Y' \subset Y$ that is open in Y , $\phi^{-1}(Y')$ is open in X .*

PROOF. ($\implies$) Suppose that ϕ is continuous and Y' is open in Y . Then, take any $x \in \phi^{-1}(Y')$. Since Y' is open, there exists $\varepsilon > 0$ such that $d_Y(\phi(x), y) < \varepsilon$ implies $y \in Y'$. Since ϕ is continuous at x , for the same $\varepsilon > 0$ there exists $\delta > 0$ such that for all x' with $d_X(x, x') < \delta$, $d_Y(\phi(x), \phi(x')) < \varepsilon$. This establishes that $\phi(x') \in Y'$ and thus $\phi^{-1}(Y')$ is open in X .

($\impliedby$) Suppose that $\phi^{-1}(Y')$ is open in X for every open Y' in Y . For given $\varepsilon > 0$ and $x \in X$, let $Y' = \mathcal{N}_\varepsilon(\phi(x))$ (i.e., $Y' = \{y \in Y : d_Y(\phi(x), y) < \varepsilon\}$), which is clearly an open set and thus $\phi^{-1}(Y')$ is open in X . Therefore, there exists $\delta > 0$ such that $x' \in \phi^{-1}(Y')$ whenever $d_X(x, x') < \delta$. Next $x' \in \phi^{-1}(Y')$ implies that $\phi(x') \in Y'$, so that $d_Y(\phi(x), \phi(x')) < \varepsilon$, completing the proof. □

Before moving to a more abstract treatment of continuity, let us turn to a simple but useful theorem.

THEOREM A.3. (*The Intermediate Value Theorem*) Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Suppose that $f(a) \neq f(b)$. Then, for c intermediate between $f(a)$ and $f(b)$ (e.g., $c \in (f(a), f(b))$ if $f(a) < f(b)$), there exists $x^* \in (a, b)$ such that $f(x^*) = c$.

PROOF. The simplest way to prove this result as follows: the image of the interval $[a, b]$ under the function f , $f([a, b])$, must be connected, in the sense that the set $f([a, b])$ cannot be the union of two disjoint open sets W, W' (that is, $f([a, b]) \neq W \cup W'$ for any W, W' open and satisfying $W \cap W' = \emptyset$). Suppose not. Then, there would exist two disjoint open sets V and V' such that $f([a, b]) \subset V \cup V'$. But from Theorem A.2, this implies that $f^{-1}(V)$ and $f^{-1}(V')$ are open in $[a, b]$, and by the fact that $f([a, b]) \subset V \cup V'$, we have $[a, b] \subset f^{-1}(V) \cup f^{-1}(V')$, which implies that $[a, b]$ is not connected, which is clearly incorrect and yields a contradiction. Theorem A.3 then follows immediately since $f([a, b])$ is connected and thus includes any value between $f(a)$ and $f(b)$. $\square$

The Intermediate Value Theorem is, in many ways, the simplest “fixed point theorem” that economists use in some applications (see Theorems A.19 and A.20 below for more general fixed point theorems). Fixed point theorems provide conditions such that given a mapping $\phi : X \rightarrow X$, there exists $x^* \in X$ with $x^* = \phi(x^*)$. The usefulness of this construction stems from the fact that many equilibrium problems can be formulated as fixed point problems. It is also clear that a fixed point is nothing but a “zero” of a slightly different mapping. In particular, define $\tilde{\phi}(x) = \phi(x) - x$. Then, a fixed point of ϕ corresponds to a zero of $\tilde{\phi}$. Perhaps the most useful application of the Intermediate Value Theorem is for the case in which $f(a) < 0$ and $f(b) > 0$ (or $f(a) > 0$ and $f(b) < 0$). In this case, the theorem states that the continuous function f has a “zero” over the interval $[a, b]$, that is, there exists some value $x^* \in (a, b)$ such that $f(x^*) = 0$. This motivates my description of the Intermediate Value Theorem as the “simplest fixed point theorem”.

A.3. A Minimal Amount of Topology: Continuity and Compactness*

Theorem A.2 implies that only the structure of open sets is relevant for thinking about continuity of mappings. This motivates our brief introduction to topology. Topology is the study of open sets and their properties. Our main interest in introducing notions from topology is to be able to talk about *compactness*. While compactness can be discussed just using ideas from metric spaces, for some of the results on infinite-dimensional (dynamic) optimization, a slightly more general treatment of compactness is necessary. I first define a topology.

DEFINITION A.15. Let A be an ordered set. A **topology** $\tau = \{V_\alpha\}_{\alpha \in A}$ on a nonempty set X is a collection of subsets $\{V_\alpha\}_{\alpha \in A}$ of X , such that

- (1) $\emptyset \in \tau$ and $X \in \tau$.
- (2) For any $A' \subset A$, $\bigcup_{\alpha \in A'} V_\alpha$ is in τ .
- (3) For any finite $A' \subset A$, $\bigcap_{\alpha \in A'} V_\alpha$ is in τ .

Given a topology τ on X , V is an **open** set in X if $V \in \tau$ and it is a **closed** set in X if $X \setminus V \in \tau$.

The pair (X, τ) is a **topological space**.

The parallel between this definition and the properties of unions and intersections of open sets given in Theorem A.1 is obvious. Sometimes it is convenient to describe a topology not by all of the open sets, but in a more economical fashion. Two convenient ways of doing this are as follows. First, a topological space can be derived from a metric space. In particular, since a topological space (X, τ) is defined by a collection of open sets and a metric space (X, d) defines the collection of open sets in the space X , it also immediately defines a topological space with the topology *induced* by the metric d . Second, a topological space can be described by a smaller collection of sets (instead of the collection of open sets). This leads us to the concept of a *base* for a topology.

DEFINITION A.16. Given a topological space (X, τ) , $\{W_\alpha\}_{\alpha \in A'}$ is a **base** for (X, τ) if for every $V \in \tau$, there exist $A'' \subset A'$ such that $V = \bigcup_{\alpha \in A''} W_\alpha$.

If $\{W_\alpha\}_{\alpha \in A'}$ is a base for (X, τ) , we can also say that τ is *generated* by $\{W_\alpha\}_{\alpha \in A'}$.

The following are some examples of topological spaces. The parallel to the metric spaces in Example A.1 is clear.

EXAMPLE A.9. (1) For any $X \subset \mathbb{R}^K$, define a collection of open sets (in the sense of

Definition A.3) according to the family of metrics $d_p(x, y) = \left(\sum_{i=1}^K |x_i - y_i|^p\right)^{1/p}$ for $1 \leq p < \infty$ and $d_\infty(x, y) = \max_{i=1, \dots, K} |x_i - y_i|$ denoted by τ_p for $p \in [1, \infty]$. Then, (X, τ_p) is a topological space. (X, τ_2) is sometimes referred to as the *Euclidean topology*, though since the other metrics are also equivalent (Exercise A.11), it would not be wrong to refer to any (X, τ_p) as the Euclidean topology.

(2) For any nonempty set X , the *discrete topology* is defined equivalently either by the discrete metric introduced in Example A.1 or by declaring all subsets of X as open sets.

(3) The *indiscrete topology* τ' on X only has $\emptyset$ and X as open sets.

(4) Consider the $(\mathbf{C}(X), d_\infty)$ metric space of all continuous, bounded real-valued functions with the sup metric. Define the collection of open sets on $\mathbf{C}(X)$ according to d_∞ by τ_∞ , then $(\mathbf{C}(X), \tau_\infty)$ is a topological space.

(5) Consider the set of infinite sequences of real numbers $\ell \subset \mathbb{R}^\infty$ and the family of metrics for this set given by $d_p(x, y) = \left(\sum_{i=1}^\infty |x_i - y_i|^p\right)^{1/p}$ for $1 \leq p < \infty$ and by $d_\infty(x, y) = \sup_i |x_i - y_i|$ (again provided that $\left(\sum_{i=1}^\infty |x_i|^p\right)^{1/p} < \infty$ for all $x \in \ell$ in the first case and $\sup_i |x_i| < \infty$ in the second case). For any $p \in [1, \infty]$, d_p defines a topology τ_p and (ℓ, τ_p) is a topological space, which is sometimes denoted by the same symbol as the corresponding metric space, ℓ_p .

As suggested by this example, many topological spaces of interest are derived from a metric space. In this case, we say that they are *metrizable* and for all practical purposes, we can treat metrizable topological spaces as metric spaces. In particular:

DEFINITION A.17. A topological space (X, τ) is **metrizable** if there exists a metric d on X such that whenever $V \in \tau$, then V is also open in the metric space (X, d) (according to Definition A.3).

FACT A.10. If a topological space (X, τ) is metrizable with some metric d , then it defines the same notions of convergence and continuity as the metric space (X, d) .

PROOF. This follows immediately from the fact that (X, τ) and (X, d) have the same open sets. □

The preceding definition and fact are provided, because metric spaces are easier to work with in practice than topological spaces. Nevertheless, sometimes (as with the product topology introduced in the next section), it may be more convenient to work with more general topological spaces. One disadvantage of general topological spaces is that they do not have all of the nice properties of metric spaces. However, this will not be an issue for the properties of topological spaces that are related to continuity and compactness, which we focus on here. Nevertheless, it is useful to note that a particularly relevant property of general topological spaces is the *Hausdorff* property, which requires that any distinct points x and y of a topological space (X, τ) should be separated, that is, there exist $V_x, V_y \in \tau$ such that $x \in V_x$, $y \in V_y$ and $V_x \cap V_y = \emptyset$. It is clear that every metric space will have the Hausdorff property (see Exercise A.14). For our purposes, the Hausdorff property will not be necessary.

Returning to general topological spaces, the notions of convergence of sequences, subsequences, nets and subnets can be stated for general topological spaces. Here I will only give the definitions for convergence of sequences and nets (those for subsequences and subnets are defined very similarly).

DEFINITION A.18. Let (X, τ) be a topological space. A sequence $\{x_n\}_{n=1}^{\infty}$ [a net $\{x_\alpha\}_{\alpha \in A}$] in X is **convergent** and has **limit point** $x \in X$ if for each $V \in \tau$ with $x \in V$, there exists $N \in \mathbb{N}$ [there exists some $\bar{\alpha} \in A$] such that $x_n \in V$ for all $n \geq N$ [$x_\alpha \in V$ for all $\alpha \geq \bar{\alpha}$]. We write this as $\lim_{n \rightarrow \infty} x_n = \lim x_n = x$ or as $\{x_n\}_{n=1}^{\infty} \rightarrow x$.

Continuity is defined in a similar manner.

DEFINITION A.19. Let (X, τ_X) and (Y, τ_Y) be topological spaces and consider the mapping $\phi : X \rightarrow Y$. ϕ is **continuous at** $x \in X$ if for every $U \in \tau_Y$ with $\phi(x) \in U$, there exists $V \in \tau_X$ with $x \in V$ such that $\phi(V) \subset U$. ϕ is **continuous** on X if it is continuous at each $x \in X$.

The parallel between this definition and the equivalent characterization of continuity in metric spaces in Theorem A.14 is evident. In fact:

THEOREM A.4. (*Open Sets and Continuity II*) Let (X, τ_X) and (Y, τ_Y) be topological spaces and consider the mapping $\phi : X \rightarrow Y$. ϕ is continuous if and only if for every $Y' \subset Y$ that is open in Y , $\phi^{-1}(Y')$ is open in X .

The proof of this theorem is essentially identical to that of Theorem A.2 and is thus omitted.

Unfortunately, in general topological spaces, convergence in terms of sequences is not sufficient to characterize continuity. However, convergence in terms of nets is.

THEOREM A.5. (*Continuity and Convergence of Nets*) Let (X, τ_X) and (Y, τ_Y) be topological spaces. The mapping $\phi : X \rightarrow Y$ is continuous at $x \in X$ if and only if $\{\phi(x_\alpha)\}_{\alpha \in A} \rightarrow \phi(x)$ for any net $\{x_\alpha\}_{\alpha \in A} \rightarrow x$.

PROOF. ($\implies$) Suppose ϕ is continuous at x and consider a net $\{x_\alpha\}_{\alpha \in A} \rightarrow x$. Take $U \in \tau_Y$ with $\phi(x) \in U$, $\phi^{-1}(U) \in \tau_X$ and $x \in \phi^{-1}(U)$. Therefore, for some $\bar{\alpha} \in A$, we have that $\alpha \geq \bar{\alpha}$ implies $x_\alpha \in \phi^{-1}(U)$ and thus $\phi(x_\alpha) \in U$ for all $\alpha \geq \bar{\alpha}$, establishing $\{\phi(x_\alpha)\}_{\alpha \in A} \rightarrow \phi(x)$.

($\impliedby$) Suppose that ϕ is not continuous at x . Then, there exists $U \in \tau_Y$ with $\phi(x) \in U$ such that $\phi^{-1}(U) \notin \tau_X$. Let $V = \mathcal{N}(x)$ denote a neighborhood x in X , that is, $V \in \tau_X$ with $x \in V$. Since there exists $U \in \tau_Y$ such that $\phi(x) \in U$ and $\phi^{-1}(U) \notin \tau_X$, for each $V \in \mathcal{N}(x)$ there exists $x_V \in V$ such that $\phi(x_V) \notin U$. Order V 's in $\mathcal{N}(x)$ by inclusion (i.e., $V' \geq V$ if and only if $V' \subset V$). Then, by construction, $\{x_V\}_{V \in \mathcal{N}(x)}$ is a net converging to x , but $\{\phi(x_V)\}_{V \in \mathcal{N}(x)} \not\rightarrow \phi(x)$, completing the proof. $\square$

FACT A.11. Consider a function $f : X \rightarrow \mathbb{R}$ and suppose that X is endowed with the discrete topology. Then, f is continuous.

PROOF. This immediately follows from the fact that any subset X' of X is open in X according to the discrete topology. $\square$

DEFINITION A.20. Let (X, τ) be a topological space with $\tau = \{V_\alpha\}_{\alpha \in A}$ and $X' \subset X$. A collection of open sets $\{V_\alpha\}_{\alpha \in A'}$ for some $A' \subset A$ is an **open cover** of X' if $X' \subset \bigcup_{\alpha \in A'} V_\alpha$.

FACT A.12. Every $X' \subset X$ has an open cover.

PROOF. By Definition A.15 $X \in \tau$, so that $\{X\}$ is an open cover of X' . $\square$

DEFINITION A.21. A subset X' of a topological space (X, τ) [where $X' = X$ is allowed] is **compact** if every open cover of X' contains a **finite subcover**, i.e., for every open cover $\{V_\alpha\}_{\alpha \in A'}$ of X' , there exists a finite set $A'' \subset A'$ such that $X' \subset \bigcup_{\alpha \in A''} V_\alpha$.

Compactness is a major property, since compact sets have many nice features and some of these will be used below. Compactness has a particularly simple meaning in Euclidean spaces, which is given by the following famous theorem.

THEOREM A.6. (*Heine-Borel Theorem*) Let $X \subset \mathbb{R}^K$ be a Euclidean space (with a Euclidean metric or topology). Then, $X' \subset X$ is compact if and only if X' is closed and bounded in $\mathbb{R}^K$.

A proof of this proposition can be found in any real analysis textbook and I will not repeat it here. Its main implication for us is that any K -dimensional segment $\prod_{i=1}^K [a_i, b_i]$,² with $a_i, b_i \in \mathbb{R}$ and $a_i \leq b_i$, is compact. The assumption that X is a Euclidean space is important for Theorem A.6, as illustrated by the following example.

EXAMPLE A.10. Consider the topological space (ℓ, τ) where ℓ is the space of infinite sequences and τ is the topology induced by the discrete metric. Let $\ell' = \{x \in \{0, 1\}^\infty : \sum_{i=1}^\infty x_i^2 = 1\}$. Clearly ℓ' is closed and bounded subset of ℓ , but not every open cover of ℓ' has a finite subcover. In particular, note that each point in ℓ' has the form $v_1 = (1, 0, 0, 0, \dots)$, $v_2 = (0, 1, 0, 0, 0, \dots)$, $v_3 = (0, 0, 1, 0, 0, \dots)$, and so on. Since τ is the discrete topology, $v_n \in \tau$ for each n and moreover the collection $\bigcup_{n \in \mathbb{N}} v_n$ is an open cover of ℓ' . But clearly, this open cover does not have a finite subcover. Equivalently, the sequence $\{v_n\}_{n=1}^\infty$ does not have a convergent subsequence. Exactly the same construction works as an example of a non-compact, closed and bounded set if we take the following subset of the standard “Hilbert” space ℓ_2 (the set of infinite sequences with metric $d_2(x, y) = \left(\sum_{i=1}^\infty |x_i - y_i|^2\right)^{1/2}$), $\ell'_2 = \{x \in [0, 1]^\infty : \sum_{i=1}^\infty x_i^2 \leq 1\}$. This subset is closed and bounded. But v_1, v_2, v_3 above are elements of ℓ'_2 and the sequence $\{v_n\}_{n=1}^\infty$ does not have a convergent subsequence.

Nevertheless, there are important connections between closed sets and compact sets. For example:

LEMMA A.2. Let (X, τ) be a topological space and suppose that $X' \subset X$ is compact. Then:

- (1) Any $X'' \subset X'$ that is closed is also compact (and hence X' itself is closed).
- (2) For any $X'' \subset X$ that is closed, $X'' \cap X'$ is compact.

PROOF. See Exercise A.15. □

One of the important implications of compactness is the following theorem.

THEOREM A.7. (*The Bolzano-Weierstrass Theorem*) Let (X, d) be a metric space and let $\{x_n\}_{n=1}^\infty$ be a sequence in X . If X is compact, then $\{x_n\}_{n=1}^\infty$ has a convergent subsequence.

PROOF. Suppose to obtain a contradiction that no such convergent subsequence exists. Then, each $x \in X$ must have neighborhood V_x that contains at most one element of the sequence $\{x_n\}_{n=1}^\infty$. Clearly, $\{V_x\}_{x \in X}$ is an open cover of X . But since N_K is an infinite set, $\{V_x\}_{x \in X}$ has no finite subcover, contradicting compactness. □

²The product of subsets can also be denoted by $\times_i [a_i, b_i]$ instead of $\prod_i [a_i, b_i]$.

It is possible to state an equivalent of Theorem A.7 for nets and subnets, but this result is not necessary for our purposes here. The reader may also wonder whether an equivalent of Theorem A.7 applies in a general topological space. Unfortunately, this is not the case (but it is true for topological space that have the Hausdorff property and also have a countable base, see Kelley, 1955).

THEOREM A.8. (*Continuity and Compact Images*) *Let (X, τ_X) and (Y, τ_Y) be topological spaces, and consider the mapping $\phi : X \rightarrow Y$. If ϕ is continuous and $X' \subset X$ is compact, then $\phi(X')$ is compact.*

PROOF. Let $\{V_\alpha\}_{\alpha \in A'}$ be an open cover of $\phi(X')$. Since ϕ is continuous, Theorem A.4 implies that $\phi^{-1}(V_\alpha)$ is open for each $\alpha \in A'$. Since X' is compact, every open cover has a finite subcover and therefore there exists a finite $A'' \subset A'$ such that $X' \subset \bigcup_{\alpha \in A''} \phi^{-1}(V_\alpha)$. Since, by definition, $\phi(\phi^{-1}(Y'')) \subset Y''$ for any $Y'' \subset Y$, this implies that

$$\phi(X') \subset \bigcup_{\alpha \in A''} (V_\alpha),$$

thus $\{V_\alpha\}_{\alpha \in A''}$ is a finite subcover of $\{V_\alpha\}_{\alpha \in A'}$, completing the proof. $\square$

Despite its simplicity Theorem A.8 has many fundamental implications. The most important is the Weierstrass's Theorem.³ Recall that for a real-valued function $f : X \rightarrow \mathbb{R}$, $\max_{x \in X} f(x)$ and $\min_{x \in X} f(x)$ are the maximum and the minimum of the function over the set X . These may not exist. When they do, we also define the following non-empty sets $\arg \max_{x \in X} f(x) = \{x' \in X : f(x') = \max_{x \in X} f(x)\}$ and $\arg \min_{x \in X} f(x) = \{x' \in X : f(x') = \min_{x \in X} f(x)\}$.

THEOREM A.9. (*Weierstrass's Theorem*) *Consider the topological space (X, τ) and a function $f : X \rightarrow \mathbb{R}$. If X' is a compact subset of (X, τ) , then $\max_{x \in X'} f(x)$ and $\min_{x \in X'} f(x)$ exist, and $\arg \max_{x \in X'} f(x)$ and $\arg \min_{x \in X'} f(x)$ are nonempty.*

PROOF. By Theorem A.8, $f(X')$ is compact. A compact subset of $\mathbb{R}$ contains a minimum and a maximum, thus $\max_{x \in X'} f(x)$ and $\min_{x \in X'} f(x)$ exist. The nonemptiness of $\arg \max_{x \in X'} f(x)$ and $\arg \min_{x \in X'} f(x)$ then follows immediately. $\square$

This theorem implies that if we can formulate a maximization problem as one of maximizing a real-valued function subject to a constraint set that is a compact subset of a topological space, then the existence of solutions and nonemptiness of the set of maximizers are guaranteed. An immediate corollary is also useful in many applications. A real-valued function $f : X \rightarrow \mathbb{R}$ is *bounded on X* if there exists $M < \infty$ such that $|f(x)| < M$ for all $x \in X$.

COROLLARY A.1. *Consider a topological space (X, τ) . If $f : X \rightarrow \mathbb{R}$ is continuous on X and X is compact, then f is bounded on X .*

³In fact, there are many theorems that go under the name of "Weierstrass's Theorem," including one on uniform continuity of a family of functions and one on approximation of continuous functions by polynomials. However, since these theorems are not used commonly in economic applications, there should be little confusion in referring to Theorem A.9 as Weierstrass's Theorem.

Finally, a stronger version of continuity for real-valued functions is sometimes useful (e.g., in Theorem 7.15 in Section 7.6 of Chapter 7).

DEFINITION A.22. *Let (X, ρ) be a metric space. Then, $f : X \rightarrow \mathbb{R}$ is **uniformly continuous** on X if, given any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x_1, x_2 \in X$ with $\rho(x_1, x_2) < \delta$, we have $|f(x_1) - f(x_2)| < \varepsilon$.*

Notice the difference between continuity at some point $x \in X$ (for example Definition A.14) and uniform continuity. In the former, δ can vary depending on x , whereas with uniform continuity the same δ must be used for all $x \in X$.

THEOREM A.10. (*Uniform Continuity over Compact Sets*) *Let (X, d_X) and (Y, d_Y) be metric spaces. If (X, d_X) is compact and f is continuous on X , then it is uniformly continuous on X .*

PROOF. Suppose, to obtain a contradiction, that f is continuous on the compact metric space (X, d_X) , but not uniformly so. Then, for some $\varepsilon > 0$ and every $n = 1, 2, \dots$, there exists $x_n, x'_n \in X$ such that

$$(A.2) \quad d_X(x_n, x'_n) < \frac{1}{n},$$

but

$$(A.3) \quad d_Y(f(x_n), f(x'_n)) \geq \varepsilon.$$

Now consider the sequence $\{x_n\}_{n=1}^\infty$ in X . Since X is compact, Theorem A.7 implies that there exists a subsequence $\{x_{n_k}\}$ converging to $x \in X$. Then, (A.2) implies that the corresponding subsequence of $\{x'_n\}_{n=1}^\infty$, $\{x'_{n_k}\}$, also converges to the same x . From (A.3), for each n_k ,

$$\varepsilon \leq d_Y(f(x_{n_k}), f(x'_{n_k})) \leq d_Y(f(x), f(x'_{n_k})) + d_Y(f(x), f(x_{n_k})),$$

where the second inequality uses the triangle inequality. This implies that either $d_Y(f(x), f(x_{n_k})) \geq \varepsilon/2$ or $d_Y(f(x), f(x'_{n_k})) \geq \varepsilon/2$ or both. But this contradicts the continuity of f on X . This contradiction establishes the uniform continuity of f on X . $\square$

The converse of this theorem is obvious, since every uniform the continuous function is continuous.

A.4. The Product Topology*

One of the main reasons for introducing topological spaces rather than simply working with metric spaces is to introduce the *product topology*. The product topology is particularly useful when dealing with infinite-dimensional optimization problems, since we can represent the space of sequences, ℓ , as the infinite product of $\mathbb{R}$, i.e., as $\mathbb{R}^\infty$. What are the topological properties of such product spaces? The answer is provided by the famous Tychonoff Theorem, Theorem A.13 below. Before presenting this theorem, it is necessary to introduce a few more concepts. First, we need to rank topologies according to how “weak” or “strong” they are.

DEFINITION A.23. Let τ and τ' be topologies defined on some set X . Then, τ is **weaker** than τ' (τ' is **stronger** than τ), if whenever V_α is open in τ , it is also open in τ' .

Now using this notion we can define the product topology as follows.

DEFINITION A.24. Let $A \subset \mathbb{R}$ and $\{(X_\alpha, \tau_\alpha)\}_{\alpha \in A}$ be a collection of topological spaces. Then, the **product topology** $\tau = \prod_{\alpha \in A} \tau_\alpha$ is the strongest topology such that all sets of the form $\bigcup_{j \in J} V^j$ are open, where $V^j = \prod_{\alpha \in A} V_\alpha^j$ with $V_\alpha^j \in \tau_\alpha$ and $V_\alpha^j = X_\alpha$ for all but finitely many α 's.

A different way of stating this definition is that sets of the form $V^j = \prod_{\alpha \in A} V_\alpha^j$ with $V_\alpha^j \in \tau_\alpha$ and $V_\alpha^j = X_\alpha$ for all but finitely many α s form a *base* for the product topology (recall Definition A.16 above).

A major reason for the usefulness of the product topology is related to the fact that it ensures continuity of the projection maps (which seems like a minimal requirement for any reasonable topology), without introducing too many open sets.

DEFINITION A.25. Let $X = \prod_{\alpha \in A} X_\alpha$ and for each $\alpha \in A$. The **projection map** is defined as $P_\alpha : X \rightarrow X_\alpha$ such that $P_\alpha(x) = x_\alpha$.

THEOREM A.11. (**Projection Maps and the Product Topology**) The product topology is the weakest topology that makes each projection map P_α continuous.

PROOF. Let τ be the product topology and τ' any other topology in which each projection map is continuous. This implies that for each $\alpha \in A$, whenever $V_\alpha \in \tau_\alpha$ is open in X_α , then $P_\alpha^{-1}(V_\alpha)$ is open according to τ' , i.e., $P_\alpha^{-1}(V_\alpha) \in \tau'$. But this implies that finite intersections of all sets of the form $P_\alpha^{-1}(V_\alpha)$ are members of τ' , and therefore all open sets in the product topology τ belong to τ' . Thus τ' must be finer than τ and establishes that the product topology is the weakest topology in which each projection map is continuous. $\square$

The product topology is also referred to as *topology of pointwise convergence* because of the following:

FACT A.13. A sequence $\{x_n\}_{n=1}^\infty$ or a net $\{x_j\}_{j \in J}$ in $X = \prod_{\alpha \in A} X_\alpha$ converges to some $\bar{x}$ if and only if the projections $P_\alpha(x_n)$ or $P_\alpha(x_j)$ converge to $P_\alpha(\bar{x})$ for any $\alpha \in A$.

This implies that the product topology will be the right tool for analyzing convergence of infinite sequences. An alternative to the product topology would be the *box topology*, which is defined similarly, except that it does not have the last qualifier " $V_\alpha^j = X_\alpha$ for all but finitely many α s". This implies that the box topology has an abundance of open sets and thus is stronger than the product topology. Consequently, compactness is difficult to achieve in the box topology. Exercise A.16 investigates this issue further.

An implication of Theorem A.11 is that a mapping $\phi : Y \rightarrow \prod_{\alpha \in A} X_\alpha$ is continuous according to the product topology if $P_\alpha \circ \phi : Y \rightarrow X_\alpha$ is continuous for each $\alpha \in A$. The product topology is particularly useful in dynamic optimization problems because of the following result.

THEOREM A.12. (*Continuity in the Product Topology*) Suppose that $f_n : X_n \rightarrow \mathbb{R}$ is continuous, X_n is a compact metric space for every $n \in \mathbb{N}$, the collection of functions $\{f_n\}_{n \in \mathbb{N}}$ is uniformly bounded in the sense that there exists $M \in \mathbb{R}$ such that $|f_n(x_n)| \leq M$ for all $x_n \in X_n$ and $n \in \mathbb{N}$, and $\beta < 1$. Then, $f = \sum_{n=1}^{\infty} \beta^n f_n : \prod_{n \in \mathbb{N}} X_n \rightarrow \mathbb{R}$ is continuous in the product topology.

PROOF. First note that uniform boundedness of the functions $\{f_n\}_{n \in \mathbb{N}}$ ensures that f is well-defined for all $x \in \prod_{n \in \mathbb{N}} X_n$. From Theorem A.5, f is continuous in the product topology if and only if for any $x^\infty \in \prod_{n \in \mathbb{N}} X_n$, $\{f(x_j)\}_{j \in J} \rightarrow f(x^\infty)$ for any net $\{x_j\}_{j \in J} \in \prod_{n \in \mathbb{N}} X_n$ with $\{x_\alpha\}_{\alpha \in A} \rightarrow x^\infty$ in the product topology. Now take a net $\{x_j\}_{j \in J} \rightarrow x^\infty$. By Fact A.13, $\{x^j\}_{j \in J} \rightarrow x^\infty$ in the product topology if and only if $\{x_n^j\}_{j \in J} \rightarrow x_n^\infty$ for each $n \in \mathbb{N}$. Then, by continuity of each f_n , $\{f_n(x_n^j)\}_{j \in J} \rightarrow f_n(x_n^\infty)$. Fix $\varepsilon > 0$, and let $\bar{n}$ be such that $\frac{\beta^{\bar{n}}}{1-\beta} 2M < \varepsilon/2$. Since $\{f_n(x_n^j)\}_{j \in J} \rightarrow f_n(x_n^\infty)$ for each $n < \bar{n}$, there exists $\bar{j} \in J$ such that $|f_n(x_n^j) - f_n(x_n^\infty)| \leq \varepsilon(1-\beta)/2$ for each $n < \bar{n}$ and $j \geq \bar{j}$. Therefore, for all $j \in J$ such that $j \geq \bar{j}$,

$$\begin{aligned} \left| \sum_{n=1}^{\infty} \beta^n f_n(x_n^j) - \sum_{n=1}^{\infty} \beta^n f_n(x_n^\infty) \right| &\leq \sum_{n=1}^{\bar{n}-1} \beta^n |f_n(x_n^j) - f_n(x_n^\infty)| + \sum_{n=\bar{n}}^{\infty} \beta^n 2M \\ &\leq \sum_{n=1}^{\bar{n}-1} \beta^n \frac{\varepsilon(1-\beta)}{2} + \frac{\varepsilon}{2} < \varepsilon, \end{aligned}$$

where the first line uses the triangle inequality and the fact that $\{f_n\}_{n \in \mathbb{N}}$ is uniformly bounded, and the second line uses the definition of $\bar{j}$. This inequality shows that $\{f(x^j)\}_{j \in J} \rightarrow f(x^\infty)$ and establishes the continuity of f . $\square$

Discounting is important in the previous result. The following example shows why.

EXAMPLE A.11. Suppose that $f_n : X \rightarrow \mathbb{R}$ is continuous and X is a compact metric space, and let $f = \sum_{n=1}^{\infty} f_n : X^\infty \rightarrow \mathbb{R}$. It can be verified that f is not continuous and tends to infinity for any $\{x^j\}_{j=1}^{\infty} \rightarrow x^*$ such that $f_n(x^n) > \varepsilon$ for all n for some $\varepsilon > 0$.

THEOREM A.13. (*Tychonoff's Theorem*) Let $A \subset \mathbb{R}$ and consider the family of topological spaces $\{(X_\alpha, \tau_\alpha)\}_{\alpha \in A}$. If each X_α is compact, then $X = \prod_{\alpha \in A} X_\alpha$ is compact in the product topology, i.e., (X, τ) is compact, where $\tau = \prod_{\alpha \in A} \tau_\alpha$.

The proof of this theorem is somewhat involved, and can be found in Kelley (1955) or Royden (1994).

Combined with Theorem A.12, this theorem implies that problems involving the maximization of discounted utility in standard dynamic economic environments has a continuous objective function in the product topology. We can then appeal to Tychonoff's Theorem to make sure that the relevant constraint set is compact (again in the product topology). This combination then enables us to apply Weierstrass's Theorem, Theorem A.9, to show the existence of solutions (see, for example, Chapters 6 and 16).

A.5. Absolute Continuity and Equicontinuity*

In this section, I provide a number of more advanced results that are useful in establishing existence of solutions in optimal control problems, in particular, in Section 7.6 of Chapter 7. Some of the results presented in this section are typically developed in the context of measure-theoretic analysis. Nevertheless, since I have avoided the use of concepts from measure theory throughout the book, I will continue to do so here.

DEFINITION A.26. *Let $X \subset \mathbb{R}$. Then, $f : X \rightarrow \mathbb{R}$ is **absolutely continuous** if for any $\varepsilon > 0$, there exists $\delta > 0$ such that*

$$\sum_{k=1}^n |f(b_k) - f(a_k)| < \varepsilon$$

for any collection of pairwise disjoint intervals (a_k, b_k) with $\sum_{k=1}^n (b_k - a_k) < \delta$.

In this definition, $X = \mathbb{R}$ is allowed, that is, a function can be absolutely continuous on the entire real line. Absolute continuity arises naturally in the context of (Lebesgue) integration. In particular, the following facts are straightforward and illustrate the context in which absolute continuity plays a useful role.

FACT A.14. *Let $f(x) = \int_0^x g(s) ds$ for all $x \in X$. If $g(s)$ is piecewise continuous on X , then f is absolutely continuous on X .*

Here the integral can be interpreted as the standard Riemann integral (see Appendix Chapter B). However, the same result holds when the integral is the Lebesgue integral and $g(s)$ is simply measurable (rather than piecewise continuous).

FACT A.15. *If $f : X \rightarrow \mathbb{R}$ is absolutely continuous on X , then it is uniformly continuous (and thus continuous) on X .*

FACT A.16. *If $f : X \rightarrow \mathbb{R}$ is differentiable on X , then it is absolutely continuous on X .*

I next introduce a number of concepts that will be useful in establishing compactness of a subset of $\mathbf{C}(X)$. Recall that $\mathbf{C}(X)$ is the set of continuous and bound the real-valued functions defined on X , and in what follows, I will take X to be a compact Euclidean space (a compact subset of the Euclidean space).

DEFINITION A.27. *Let (X, d) be a metric space. For $\varepsilon > 0$, $A \subset X$ is an ε -**net** for X' if for every $x \in X'$, there exists $a \in A$ such that $d(a, x) \leq \varepsilon$.*

DEFINITION A.28. Let (X, d) be a metric space. A subset X' of X is **totally bounded** if for every $\varepsilon > 0$, there exists a finite set $A_\varepsilon \subset X$ that is an ε -net (a finite ε -net) of X' .

In this definition, we can, without loss of any generality, set $A_\varepsilon \subset X'$. Also, X , as a subset of itself, can be totally bounded with the same definition.

The following theorem, which I state without a proof, is an alternative characterization of compactness (for a proof, see, for example, Kolmogorov and Fomin, 1970, pp. 100-102, Theorems 2 and 3).

THEOREM A.14. (**Totally Bounded and Compact Spaces**) A metric space (X, d) is compact if and only if it is totally bounded and complete.

DEFINITION A.29. Let X be a compact Euclidean space. A subset $\mathcal{F}$ of $\mathbf{C}(X)$ is **uniformly bounded** if there exists $K > 0$ such that

$$|f(x)| < K$$

for all $x \in X$ and all $f \in \mathcal{F}$.

DEFINITION A.30. Let X be a compact Euclidean space. A subset $\mathcal{F}$ of $\mathbf{C}(X)$ functions is **equicontinuous** if for any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x_1, x_2 \in X$ with $|x_1 - x_2| < \delta$ and any $f \in \mathcal{F}$,

$$|f(x_1) - f(x_2)| < \varepsilon.$$

THEOREM A.15. (**Arzela-Ascoli Theorem**) Let X be a compact Euclidean space and let $\mathcal{F}$ be a subset of $\mathbf{C}(X)$. The closure of $\mathcal{F}$, $\overline{\mathcal{F}}$, is compact in $\mathbf{C}(X)$ if and only if $\mathcal{F}$ is uniformly bounded and equicontinuous. That is, if $\mathcal{F}$ is uniformly bounded and equicontinuous, whenever $f^n \in \mathcal{F}$ for $n = 1, 2, \dots$, there exists a subsequence $\{f^{n_k}\}$ of $\{f^n\}_{n=1}^\infty$ such that $\{f^{n_k}\} \rightarrow f \in \overline{\mathcal{F}}$.

PROOF. ($\Leftarrow$) Suppose $\overline{\mathcal{F}}$ is compact in $\mathbf{C}(X)$. Then, by Theorem A.14 and Definition A.28, for every $\varepsilon > 0$, there exists a finite $(\varepsilon/3)$ -net $\{f_1, \dots, f_n\}$ in $\mathcal{F}$. This implies that for any $f \in \mathcal{F}$, there exists $i \in \{1, \dots, n\}$ such that

$$(A.4) \quad \sup_{x \in X} |f(x) - f_i(x)| \leq \frac{\varepsilon}{3}.$$

Moreover, each f_i is bounded, since it is continuous on compact X (Corollary A.1) thus for each $i = 1, \dots, n$, there exists $K_i < \infty$ such that

$$|f_i(x)| \leq K_i \text{ for all } x \in X.$$

Set $K = \max\{K_1, \dots, K_n\} + \varepsilon/3$ and using the triangle inequality, rewrite (A.4) as

$$|f(x)| \leq |f_i(x)| + \frac{\varepsilon}{3} \leq K_i + \frac{\varepsilon}{3} \leq K,$$

for all $x \in X$, which establishes that $\mathcal{F}$ is uniformly bounded. Moreover, each one of $f_1, \dots, f_n$ is uniformly continuous (this follows from Theorem A.10, since each f_i is continuous and X is compact). Therefore, for each $i = 1, \dots, n$ and for any $\varepsilon > 0$, there exists $\delta_i > 0$ such that whenever $|x - x'| < \delta_i$ for $x, x' \in X$,

$$|f_i(x) - f_i(x')| < \frac{\varepsilon}{3}.$$

Set $\delta = \max \{\delta_1, \dots, \delta_n\}$. Then, for any $f \in \mathcal{F}$, choose $i \in \{1, \dots, n\}$ such that (A.4) holds, and again using the triangle inequality,

$$\begin{aligned} |f(x) - f(x')| &\leq |f(x) - f_i(x)| + |f_i(x) - f_i(x')| + |f_i(x') - f(x')| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon, \end{aligned}$$

for all $x, x' \in X$ with $|x - x'| < \delta$, implying that $\mathcal{F}$ is equicontinuous, and completing the proof of necessity.

($\Rightarrow$) Take $\varepsilon > 0$. Since $\mathcal{F}$ is equicontinuous, for each $x \in X$, there exists $\delta > 0$ such that

$$|f(x) - f(x')| < \frac{\varepsilon}{4}$$

for all $f \in \mathcal{F}$, whenever $x' \in \mathcal{N}_\delta(x)$ (open δ -neighborhood of x). Since X is compact, it has a finite subcover (Definition A.21) and thus we can choose $X' = \{x_1, \dots, x_n\}$ such that $X = \bigcup_{x_i \in X'} \mathcal{N}_\delta(x_i)$. Moreover, since each $f \in \mathcal{F}$ is bounded on X (Corollary A.1), so the set $\{f(x_i) : f \in \mathcal{F} \text{ and } i = 1, \dots, n\}$ is a totally bounded subset of $\mathbf{C}(X)$. From Definition A.28, there exists a finite $(\varepsilon/4)$ -net $A = \{g_1, \dots, g_m\} \subset \mathcal{F}$ for $X'' = \{f(x_1), \dots, f(x_n)\}$. Now consider the set of functions F_A such that $F_A = \{X' \rightarrow A\}$. Since both X' and A are finite, the set F_A is also finite. Let

$$\mathcal{F}_\phi = \left\{ f \in \mathcal{F} : |f(x_i) - \phi(x_i)| < \frac{\varepsilon}{4} \text{ for } i = 1, \dots, n \right\}.$$

Since A is a $(\varepsilon/4)$ -net for X'' ,

$$\bigcup_{\phi \in F_A} \mathcal{F}_\phi = \mathcal{F}.$$

Moreover, take $f, g \in \mathcal{F}_\phi$ for some $\phi \in F_A$ and observe that, from the triangle inequality,

$$\begin{aligned} |f(x_i) - g(x_i)| &\leq |f(x_i) - \phi(x_i)| + |\phi(x_i) - g(x_i)| \\ &\leq \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2} \end{aligned}$$

for $x_i \in X'$. By the fact that $X = \bigcup_{x_i \in X'} \mathcal{N}_\delta(x_i)$, any $x \in X$ is in $\mathcal{N}_\delta(x_i)$ for some $i \in \{1, \dots, n\}$, and therefore, for any $x \in X$,

$$\begin{aligned} |f(x) - g(x)| &\leq |f(x) - f(x_i)| + |f(x_i) - g(x_i)| + |g(x_i) - g(x)| \\ &< \frac{\varepsilon}{4} + \frac{\varepsilon}{2} + \frac{\varepsilon}{4} = \varepsilon. \end{aligned}$$

This implies that $\mathcal{F}$ is totally bounded. Moreover, since $\mathbf{C}(X)$ is complete, $\overline{\mathcal{F}}$ is also complete (Fact A.7), thus $\overline{\mathcal{F}}$ is compact in $\mathbf{C}(X)$, proving this sufficiency part of the theorem.

Finally, the sufficiency result also implies that if $f^n \in \mathcal{F}$ for $n = 1, 2, \dots$, then there exists a subsequence $\{f^{n_k}\}$ of $\{f^n\}$ with $\{f^{n_k}\} \rightarrow f \in \overline{\mathcal{F}}$, completing the proof of the theorem. $\square$

The following corollary can be proved using an identical argument to that of Theorem A.15.

COROLLARY A.2. *Let X be a compact Euclidean space and let $\mathcal{F}$ be the subset of $\mathbf{C}(X)$ consisting of the family of absolutely continuous functions defined on X . Suppose that $\mathcal{F}$ is*

is uniformly bounded and equicontinuous, then whenever $f^n \in \mathcal{F}$ for $n = 1, 2, \dots$, there exists a subsequence $\{f^{n_k}\}$ of $\{f^n\}_{n=1}^\infty$ such that $\{f^{n_k}\} \rightarrow f \in \overline{\mathcal{F}}$.

A.6. Correspondences and Berge's Maximum Theorem

In this section, I state one of the most important theorems in economic analysis, Berge's Maximum Theorem. This theorem is not only essential for dynamic optimization, but it plays a major role in general equilibrium theory, game theory, political economy, public finance and industrial organization. In fact, it is hard to imagine any area of economics where it does not play a major role. Despite its enormous importance, this theorem is left out of most basic "Mathematics for Economists" courses and textbooks. This motivates my somewhat detailed treatment of it here. The first step for this theorem is to have a brief review of correspondences, which were already mentioned above.

In this and the next three sections, I focus on metric spaces. Recall that F is a correspondence from a metric space (X, d_X) into (Y, d_Y) if to each $x \in X$ it assigns a subset of Y . We write this as

$$F : X \rightrightarrows Y \text{ or } F : X \rightarrow \mathcal{P}(Y) \setminus \emptyset,$$

where $\mathcal{P}(Y)$ is the power set of Y and the empty set $\emptyset$ is explicitly subtracted so that the correspondence is not empty valued. We are interested in correspondences for three fundamental reasons. First, even when a mapping into real numbers is a well behaved function, $f : X \rightarrow \mathbb{R}$, its inverse f^{-1} will typically be set-valued, thus a correspondence. Second, our main interest in most economic problems is with the "arg max" sets defined above, which are the subsets of values in some set X that maximize a function. These will correspond to utility-maximizing consumption, investment or price levels in simple economic problems. Finally, correspondences are also useful in expressing the properties of maximizers in Berge's Maximum Theorem, Theorem A.16 below.

As with functions, for a correspondence $F : X \rightrightarrows Y$, I will use the notation $F(X')$ to denote the image of the set X' under the correspondence F :

$$F(X') = \{y \in Y : \exists x \in X' \text{ with } y \in F(x)\}.$$

DEFINITION A.31. Let (X, d_X) and (Y, d_Y) be metric spaces and consider the correspondence $F : X \rightrightarrows Y$. Let $\mathcal{N}_\varepsilon(x)$ refer to neighborhoods in (X, d_X) . Then

- (1) F is **upper hemi-continuous** at $x \in X$ if for every open subset Y' of Y with $F(x) \subset Y'$, there exists $\varepsilon > 0$ such that $F(\mathcal{N}_\varepsilon(x)) \subset Y'$.
- (2) F is **upper hemi-continuous** on the set X if it is upper hemi-continuous at each $x \in X$.
- (3) F is **lower hemi-continuous** at $x \in X$ if for every open subset Y' of Y with $F(x) \cap Y' \neq \emptyset$, there exists $\varepsilon > 0$ such that $F(x') \cap Y' \neq \emptyset$ for all $x' \in \mathcal{N}_\varepsilon(x)$.
- (4) F is **lower hemi-continuous** on the set X if it is lower hemi-continuous at each $x \in X$.

- (5) F is **continuous** at $x \in X$ if and only if it is both upper- and lower hemi-continuous at $x \in X$.
- (6) F is **continuous** on the set X if and only if it is both upper- and lower hemi-continuous on the set X .

These notions are slightly easier to understand if we specialize them to Euclidean spaces. First, we say that a correspondence $F : X \rightrightarrows Y$ is *closed-valued* [*compact-valued*] if $F(x)$ is closed [compact] in Y for each x . For Euclidean spaces, the following definition (see Exercise A.18) is equivalent to Definition A.31 and in general, as Fact A.17 shows, it implies Definition A.31.

DEFINITION A.32. Let $X \subset \mathbb{R}^{K_X}$ and $Y \subset \mathbb{R}^{K_Y}$ where $K_X, K_Y \in \mathbb{N}$ and consider a compact-valued correspondence $F : X \rightrightarrows Y$.

- (1) F is **upper hemi-continuous** at $x \in X$ if for every sequence $\{x_n\}_{n=1}^\infty \rightarrow x$ and every sequence $\{y_n\}_{n=1}^\infty$ with $y_n \in F(x_n)$ for each n , there exists a convergent subsequence $\{y_{n_k}\}$ of $\{y_n\}_{n=1}^\infty$ such that $\{y_{n_k}\} \rightarrow y \in F(x)$.
- (2) F is **lower hemi-continuous** at $x \in X$ if $F(x)$ is nonempty and for every $y \in F(x)$ and every sequence $\{x_n\}_{n=1}^\infty \rightarrow x$, there exists some $N \in \mathbb{N}$ and a sequence $\{y_n\}_{n=1}^\infty$ with $y_n \in F(x_n)$ for all $n \geq N$, and $\{y_n\}_{n=1}^\infty \rightarrow y$.

Upper hemicontinuity and lower hemicontinuity according to Definition A.32 imply the corresponding concepts in Definition A.31 for general metric spaces.

FACT A.17. Let (X, d_X) and (Y, d_Y) be metric spaces and consider the correspondence $F : X \rightrightarrows Y$. If F is upper hemi-continuous [lower hemi-continuous] at $x \in X$ according to Definition A.32, then it is upper hemi-continuous [lower hemi-continuous] at $x \in X$ according to Definition A.31.

PROOF. Suppose, to obtain a contradiction, that part 1 of Definition A.32 holds at x , but F is not upper hemi-continuous at x . Then, there exists an open set $Y' \subset Y$ such that $F(x) \subset Y'$ but for any $\varepsilon > 0$, $F(\mathcal{N}_\varepsilon(x))$ is not a subset of Y' . Then, for any $\varepsilon > 0$, there exists $x_\varepsilon \in \mathcal{N}_\varepsilon(x)$ and $y_\varepsilon \in F(x_\varepsilon)$ such that $y_\varepsilon \notin Y'$. Construct the sequence $\{(x_n, y_n)\}_{n=1}^\infty$ such that each (x_n, y_n) satisfies this property for $\varepsilon = 1/n$. Clearly, $\{x_n\}_{n=1}^\infty \rightarrow x$. Therefore, by hypothesis, there exists a convergent subsequence $\{y_{n_k}\} \rightarrow y \in F(x)$. Since Y' is open, $Y \setminus Y'$ is closed, and since $y_{n_k} \in Y \setminus Y'$ for each n_k , the limit point y must also be in the closed set $Y \setminus Y'$. But $y \in Y \setminus Y'$ together with $y \in F(x)$ yields a contradiction in view of the fact that $F(x) \subset Y'$, proving the first part of the Fact.

Suppose, to obtain a contradiction, that part 2 of Definition A.32 holds at x , but F is not lower hemi-continuous at x . Then, there exists an open set $Y' \subset Y$ such that $F(x) \cap Y' \neq \emptyset$, but for any $\varepsilon > 0$, there exists $x_\varepsilon \in \mathcal{N}_\varepsilon(x)$ such that $F(x_\varepsilon) \cap Y' = \emptyset$. Consider the sequence $\{x_n\}_{n=1}^\infty$ with $x_n \rightarrow x$, let $\varepsilon = 1/n$, and suppose that this sequence satisfies the property just stated, i.e., for any $\varepsilon > 0$ there exists $x_\varepsilon \in \mathcal{N}_\varepsilon(x)$ such that $F(x_\varepsilon) \cap Y' = \emptyset$. Also let $y \in F(x) \cap Y'$. By part 2 of Definition A.32, there exists a sequence $\{y_n\}_{n=1}^\infty$ and

some $N \geq 1$ such that $y_n \in F(x_n)$ for all $n \geq N$ and $\{y_n\}_{n=1}^\infty \rightarrow y$. However, by the construction of the sequence $\{x_n\}_{n=1}^\infty$, $y_n \notin Y'$. Once again, since $Y \setminus Y'$ is closed, it must be the case that the limit point y also lies in the closed set $Y \setminus Y'$. This contradicts $y \in F(x) \cap Y'$ and establishes the second part of the Fact. $\square$

DEFINITION A.33. Let (X, d_X) and (Y, d_Y) be metric spaces and consider the correspondence $F : X \rightrightarrows Y$. Then, F has a **closed graph** (is closed) at $x \in X$ if for every sequence $\{(x_n, y_n)\}_{n=1}^\infty \rightarrow (x, y)$ such that $y_n \in F(x_n)$ for each n , we also have $y \in F(x)$. F has a closed graph on the set X if it is closed at each $x \in X$.

The following fact is a simple consequence of Definition A.32.

FACT A.18. Let $X \subset \mathbb{R}^{K_X}$ and $Y \subset \mathbb{R}^{K_Y}$ where $K_X, K_Y \in \mathbb{N}$ and consider the correspondence $F : X \rightrightarrows Y$ that is upper hemi-continuous. If $F(x)$ is a closed set in Y (i.e., if F is closed-valued) for each $x \in X$, then F has a closed graph.

PROOF. See Exercise A.20. $\square$

For finite-dimensional spaces, correspondences with closed graph are also upper hemi-continuous, provided that they satisfy a simple boundedness hypothesis.

FACT A.19. Let $X \subset \mathbb{R}^{K_X}$ and $Y \subset \mathbb{R}^{K_Y}$ where $K_X, K_Y \in \mathbb{N}$ and consider a correspondence $F : X \rightrightarrows Y$. Suppose that F has closed graph at $x \in X$ and that there exists a neighborhood V_x of x such that $F(V_x)$ is bounded. Then, F is upper hemi-continuous at x .

PROOF. Consider sequences $\{x_n\}_{n=1}^\infty$ and $\{y_n\}_{n=1}^\infty$ such that $y_n \in F(x_n)$ for each n . Suppose that $\{x_n\}_{n=1}^\infty \rightarrow x$. Then, by definition, there exists $N \in \mathbb{N}$ such that $x_n \in V_x$ for all $n \geq N$, where V_x is the neighborhood specified in the statement of the claim, satisfying the property that $F(V_x)$ is bounded. Since Y is a Euclidean space, this implies that the closure of $V_x, \overline{F(V_x)}$, is compact. Then, by Theorem A.7, $\{y_n\}_{n=1}^\infty$ has a subsequence $\{y_{n_k}\}$ converging to some $y \in Y$. This implies that the (sub)sequence $\{(x_{n_k}, y_{n_k})\} \rightarrow (x, y)$. Moreover, since F has closed graph at x , $y \in F(x)$, which establishes that F is upper hemi-continuous at $x \in X$ according to Definition A.32. Then, from Fact A.17, it is upper hemi-continuous at $x \in X$ according to Definition A.31. $\square$

The hypothesis that there exists a neighborhood V_x with $F(V_x)$ bounded can not be dispensed with in this result. This is shown by the following example:

EXAMPLE A.12. Consider the correspondence $F : [0, 1] \rightarrow \mathbb{R}$ given by $F(x) = \{0\}$ if $x = 0$ and $F(x) = \{\log x, 0\}$ if $x \in (0, 1]$. F has closed graph, but is not upper hemi-continuous at $x = 0$. It can be verified easily that F does not satisfy the hypothesis that there exists a neighborhood V_x with $F(V_x)$ bounded at $x = 0$.

The following fact is useful for using continuous correspondences in optimization problems.

FACT A.20. Let (X, d_X) be a metric space and consider the continuous concave function $g : Y \rightarrow \mathbb{R}$. Then, the set-valued mapping

$$G(x) = \{y \in Y : y \leq g(x)\}$$

defines a continuous correspondence $G : X \rightrightarrows Y$.

PROOF. See Exercise A.21. □

THEOREM A.16. (**Berge's Maximum Theorem**) Let (X, d_X) and (Y, d_Y) be metric spaces. Consider the maximization problem

$$\begin{aligned} & \sup_{y \in Y} f(x, y) \\ & \text{subject to} \\ & y \in G(x), \end{aligned}$$

where $G : X \rightrightarrows Y$ and $f : X \times Y \rightarrow \mathbb{R}$. Suppose that f is continuous and G is compact-valued and continuous at x . Then

- (1) $M(x) = \max_{y \in Y} \{f(x, y) : y \in G(x)\}$ exists and is continuous at x .
- (2) $\Pi(x) = \arg \max_{y \in Y} \{f(x, y) : y \in G(x)\}$ is nonempty, compact-valued, upper hemi-continuous and has closed graph at x .

PROOF. In view of Fact A.17, I will work with Definition A.32.

The fact that $M(x)$ exists and thus $\Pi(x)$ is nonempty for all $x \in X$ follows from Theorem A.9. Consider a sequence $\{y_n\}_{n=1}^\infty \rightarrow y$ such that $y_n \in \Pi(x)$ for each n . Since $G(x)$ is closed, $y \in G(x)$. Moreover, by definition, $f(x, y_n) = M(x)$ for each n . Since f is continuous, $f(x, y) = M(x)$ follows. Therefore, $y \in \Pi(x)$ and thus $\Pi(x)$ is closed. Since $\Pi(x)$ is a closed subset of the compact set $G(x)$, we can invoke Lemma A.2 to conclude that $\Pi(x)$ is compact-valued.

Now again take $\{x_n\}_{n=1}^\infty \rightarrow x$, $\{y_n\}_{n=1}^\infty$ with $y_n \in G(x_n)$ for all n , with a convergent subsequence $\{y_{n_k}\} \rightarrow y$. Since $G(x)$ is upper hemi-continuous, $y \in G(x)$. Take any $z \in G(x)$. Since $G(x)$ is continuous and thus lower-hemi-continuous, there exists $\{z_{n_k}\} \rightarrow z$ with $z_{n_k} \in G(x_{n_k})$ for all n_k . Since $y_{n_k} \in \Pi(x_{n_k})$, $M(x_{n_k}) = f(x_{n_k}, y_{n_k}) \geq f(x_{n_k}, z_{n_k})$. Moreover, since f is continuous, by Fact A.5,

$$M(x) = f(x, y) \geq f(x, z).$$

Since this holds for all $z \in G(x)$, $y \in \Pi(x)$ and therefore $\Pi(x)$ is upper hemi-continuous. Applying Fact A.18 once more, we conclude that $\Pi(x)$ also has a closed graph.

To complete the proof, we need to show that $M(x)$ is continuous at x . This follows from the fact that $\Pi(x)$ is upper hemi-continuous. Take $\{x_n\}_{n=1}^\infty \rightarrow x$, and consider, $\{y_n\}_{n=1}^\infty$ such that $y_n \in \Pi(x_n)$ for each n . Since, Π is upper hemi-continuous, Definition A.32 implies that there exists a subsequence $\{y_{n_k}\}$ converging to $y \in F(x)$. The continuity of f implies that $M(x_{n_k}) = f(x_{n_k}, y_{n_k}) \rightarrow f(x, y) = M(x)$ and establishes that $M(x)$ is continuous at x . □

Note that I wrote the maximization problem as $\sup_{y \in Y}$ instead of $\max_{y \in Y}$. There would have been no loss of generality in using the latter notation, since the theorem establishes that the maximum is attained. Nevertheless, the former might be slightly more appropriate, since when we first consider the problem we do not know whether the maximum is attained or not. Throughout the appendix, I used the “sup” notation, while in the text I typically use the simpler “max” notation.

One difficulty in using Theorem A.16 is that constraint sets do not always define continuous correspondences. This is illustrated in Exercise A.19. However, Fact A.20 shows that in some important cases they do in fact define continuous correspondences.

A.7. Convexity, Concavity, Quasi-Concavity and Fixed Points

Theorem A.16 shows how we can ensure certain desirable properties of the set of maximizers in a variety of problems arising in economic analysis. However, it is not strong enough to assert uniqueness of maximizers or continuity of the set of maximizers (instead, we have upper hemi-continuity, which is weaker than continuity). In this section, I will show how these results can be strengthened when we focus on problems with concave objective functions and convex constraint sets, and then I will provide a brief illustration of how these stronger results can be used. Throughout the rest of this appendix, let V be a *vector space* (or a linear space) so that if $x, y \in V$ and λ is a real number, then $x + y \in V$ and $\lambda x \in V$. Let X and Y be subsets of V . Properties of vector spaces will be discussed further in Section 10.1 below.

DEFINITION A.34. A set X is **convex**, if for any $\lambda \in [0, 1]$, and any $x, y \in X$, $\lambda x + (1 - \lambda)y \in X$.

DEFINITION A.35. A correspondence $G : X \rightrightarrows Y$ is **convex-valued** at x if $G(x)$ is a convex set (in Y).

DEFINITION A.36. Let X be convex, $f : X \rightarrow \mathbb{R}$ be a real-valued function and $\lambda \in (0, 1)$. Suppose that $f(x)$, $f(y)$ and $f(\lambda x + (1 - \lambda)y)$ are well defined. Then:

- (1) f is **concave** if $f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$ for all $\lambda \in (0, 1)$ and all $x, y \in X$.
- (2) f is **strictly concave** if $f(\lambda x + (1 - \lambda)y) > \lambda f(x) + (1 - \lambda)f(y)$ for all $\lambda \in (0, 1)$ and all $x, y \in X$ with $x \neq y$.
- (3) f is **convex** if $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all $\lambda \in (0, 1)$ and all $x, y \in X$.
- (4) f is **strictly convex** if $f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y)$ for all $\lambda \in (0, 1)$ and all $x, y \in X$ with $x \neq y$.
- (5) f is **quasi-concave** if $f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}$ for all $\lambda \in (0, 1)$ and all $x, y \in X$.
- (6) f is **strictly quasi-concave** if $f(\lambda x + (1 - \lambda)y) > \min\{f(x), f(y)\}$ for all $\lambda \in (0, 1)$ and all $x, y \in X$ with $x \neq y$.

- (7) f is **quasi-convex** if $f(\lambda x + (1 - \lambda)y) \leq \max\{f(x), f(y)\}$ for all $\lambda \in (0, 1)$ and all $x, y \in X$.
- (8) f is **strictly quasi-convex** if $f(\lambda x + (1 - \lambda)y) < \max\{f(x), f(y)\}$ for all $\lambda \in (0, 1)$ and all $x, y \in X$ with $x \neq y$.

Naturally, one can define all of these concepts for a subset X' of the domain X of the function f , since a function could be concave over a certain range, but not everywhere.

The following result strengthens the conclusions of Theorem A.16 under additional assumptions.

THEOREM A.17. (Properties of Maximizers) *Consider the maximization problem*

$$\begin{aligned} & \sup_{y \in Y} f(x, y) \\ & \text{subject to} \\ & y \in G(x), \end{aligned}$$

where $G : X \rightrightarrows Y$ and $f : X \times Y \rightarrow \mathbb{R}$. Suppose that f is continuous and G is convex-valued, compact-valued and continuous at x . Then

- (1) If f is quasi-concave, then $\Pi(x) = \arg \max_{y \in Y} \{f(x, y) : y \in G(x)\}$ is nonempty, compact-valued, upper hemi-continuous, has closed graph and is convex-valued at x .
- (2) If f is strictly quasi-concave in a neighborhood of x , then $\Pi(x)$ is a singleton.
- (3) If f satisfies the conditions in part 2 everywhere in X , then $\Pi(x)$ is a continuous (single-valued) function in X .

PROOF. (Part 1) Most of the statements here follow from Theorem A.16. We only need to prove that $\Pi(x)$ is convex-valued. Suppose, to obtain a contradiction, that this is not the case. This implies that there exist y and $y' \neq y$ in $\Pi(x)$ such that for some $\lambda \in (0, 1)$, $y'' = \lambda y + (1 - \lambda)y' \notin \Pi(x)$. But since $G(x)$ is convex-valued, $y'' \in G(x)$. Then, by quasi-concavity, $f(\lambda y + (1 - \lambda)y') \geq \min\{f(y), f(y')\}$. But since $y, y' \in \Pi(x)$, $f(y) = f(y')$, and thus $f(\lambda y + (1 - \lambda)y') \geq f(y) = f(y')$, implying that $y'' = \lambda y + (1 - \lambda)y' \in \Pi(x)$. This yields a contradiction and establishes that $\Pi(x)$ is convex-valued.

(Part 2) Suppose, to obtain a contradiction, that there exist y and $y' \neq y$ in $\Pi(x)$. Since $G(x)$ is convex-valued, $y'' = \lambda y + (1 - \lambda)y' \in G(x)$ for any $\lambda \in (0, 1)$ and moreover, by strict quasi-concavity of f , $f(\lambda y + (1 - \lambda)y') > \lambda f(y) + (1 - \lambda)f(y')$. Again since $y, y' \in \Pi(x)$, $f(y) = f(y')$, and thus $f(\lambda y + (1 - \lambda)y') > f(y) = f(y')$, contradicting that $y, y' \in \Pi(x)$ and establishing the result.

(Part 3) Part 2 implies that $\Pi(x)$ is single-valued everywhere and Part 1 implies that it is upper hemi-continuous. From Definition A.31, this implies that for every sequence $\{x_n\}_{n=1}^{\infty} \rightarrow x$ and every sequence $\{y_n\}_{n=1}^{\infty}$ with $y_n = \Pi(x_n)$ for each n , there exists a convergent subsequence $\{y_{n_k}\}$ of $\{y_n\}_{n=1}^{\infty}$ such that $\{y_{n_k}\} \rightarrow y = \Pi(x)$. When $\Pi(x)$ is single valued, this implies continuity at x (recall Fact A.8). $\square$

Clearly, all of these results can be generalized to minimization problems. In particular, we have the following theorem, which follows by applying Theorems A.16 and A.17 to $-f$.

THEOREM A.18. (*Properties of Minimizers*) Consider the minimization problem

$$\begin{aligned} & \inf_{y \in Y} f(x, y) \\ & \text{subject to} \\ & y \in G(x), \end{aligned}$$

where $G : X \rightrightarrows Y$ and $f : X \times Y \rightarrow \mathbb{R}$. Suppose that f is continuous and G is convex-valued, compact-valued and continuous at x . Then

- (1) If f is quasi-convex, then $\Pi(x) = \arg \min_{y \in Y} \{f(x, y) : y \in G(x)\}$ is nonempty, compact-valued, upper hemi-continuous, has closed graph and is convex-valued at x .
- (2) If f is strictly quasi-convex, then in addition, $\Pi(x)$ is a singleton.
- (3) If f satisfies the conditions in part 2 everywhere, then $\Pi(x)$ is a continuous single-valued mapping.

The following well-known and important theorem shows why convex-valuedness is important.

THEOREM A.19. (*Kakutani's Fixed Point Theorem*) Suppose $X \subset \mathbb{R}^K$ (where $K \in \mathbb{N}$) is a nonempty, compact, convex set and let

$$F : X \rightrightarrows X$$

be a nonempty, convex-valued, and upper hemi-continuous correspondence. Then, F has a **fixed point** in X , that is, there exists $x^* \in X$ such that $x^* \in F(x^*)$.

The proof of this theorem is nontrivial and can be found in Berge (1963), Aliprantis and Border (1999), or Ok (2006). Exercise A.22 shows why convex-valuedness is important and Exercise A.23 presents an application of the results in this section and of Theorem A.19 to the existence of pure strategy Nash equilibria in normal-form games.

While some of the proofs of Kakutani's Fixed Point Theorem start from the slightly simpler Brouwer's Fixed Point Theorem, now that I presented Theorem A.19, Brouwer's Fixed Point Theorem can be obtained as a corollary.

THEOREM A.20. (*Brouwer's Fixed Point Theorem*) Suppose $X \subset \mathbb{R}^K$ (where $K \in \mathbb{N}$) is a nonempty, compact, convex set and let

$$\phi : X \rightarrow X$$

be a continuous map. Then, ϕ has a **fixed point** in X , that is, there exists $x^* \in X$ such that $x^* = \phi(x^*)$.

PROOF. The result follows immediately from Theorem A.19, using Part 3 of Theorem A.17, which shows that a continuous map is a nonempty, convex-valued and upper hemi-continuous correspondence. $\square$

A.8. Differentiation, Taylor Series and the Mean Value Theorem

In this and the next section, I briefly discuss differentiation and some important results related to differentiation that are useful for the analysis in the text. The material in this section should be more familiar, thus I will be somewhat more brief in my treatment. In this section, the focus is on a real-valued function of one variable $f : \mathbb{R} \rightarrow \mathbb{R}$. Functions of several variables and vector-valued functions are discussed in the next section.

The reader will recall that the derivative (function) for $f : \mathbb{R} \rightarrow \mathbb{R}$ has a simple definition. Take a point x in an open set X' on which the function f is defined. Then, when the limit exists (and is finite), the derivative of f at x is defined as

$$(A.5) \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Clearly, the term $f(x+h)$ is well defined for h sufficiently small since x is in the open set X' . Moreover, this limit will exist at point x only if f is continuous at $x \in X$. This is a more general property; differentiability implies continuity (see Fact A.21). Using the elementary properties of limits, expression in (A.5) can be rearranged as

$$(A.6) \quad \lim_{h \rightarrow 0} \frac{f(x+h) - f(x) - L(x)h}{h} = 0,$$

where $L(x) = f'(x)$. This expression emphasizes that we can think of the derivative of the function $f(x)$, $f'(x)$, as a *linear operator*. In fact, one might want to define $f'(x)$ precisely as the *linear operator* $L(x)$ that satisfies eq. (A.6). Note that $f'(x)$ is linear in h not in x . It is generally a nonlinear function of x , but it defines a linear function from X' (the open subset of X where f is defined) to $\mathbb{R}$ that assigns the value $f'(x)h$ to each h such that $x+h \in X'$. This perspective will be particularly useful in the next section.

DEFINITION A.37. When $f'(x)$ exists at x , f is ***differentiable*** at x . If $f'(x)$ exist at all x in some subset $X'' \subset X$, then f is ***differentiable*** on the entire X'' . If, in addition, f' is a continuous function of x on X'' , f is ***continuously differentiable*** and is denoted as a C^1 function.

When X' is a closed set, then f being differentiable or continuously differentiable on X' is equivalent to f being differentiable or continuously differentiable in the interior of X' and then also having an *extension* (or a continuous extension) of its derivative to the boundary of X' . A slightly stronger requirement, which will also guarantee (continuous) differentiability on X' , is that there exists an open set $X'' \supset X'$ such that f is (continuously) differentiable on X'' .

Differentiability is a stronger requirement than continuity. In fact:

FACT A.21. Let $X \subset \mathbb{R}$ and $f : X \rightarrow \mathbb{R}$ be a real-valued function. If f is differentiable at $x \in X$, then it is also continuous at x .

PROOF. See Exercise A.24. □

It is also useful to note that differentiability over some set X' does not imply continuous differentiability. The following example illustrates this point.

EXAMPLE A.13. Consider the function f such that $f(x) = x^2 \sin(1/x)$ for all $x \neq 0$ and $f(0) = 0$. It can be verified that f is continuous and differentiable, with derivative $f'(x) = 2x \sin(1/x) - \cos(1/x)$ and $f'(0) = 0$. But clearly, $\lim_{x \downarrow 0} f'(x) \neq 0$.

Higher order derivatives are defined in a similar manner. Again starting with a real-valued function f , suppose that this function has a continuous derivative $f'(x)$. Then, again taking x in some open set X' where $f'(X')$ is well-defined, the second derivative of f , denoted $f''(x)$, is

$$f''(x) = \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h}.$$

Higher than the second-order derivatives are defined similarly. If a real-valued function f has continuous derivatives up to order n on some set X' , then it is said to be $\mathcal{C}^n$. If f is a $\mathcal{C}^2$ function, we also say that it is *twice continuously differentiable*. A $\mathcal{C}^\infty$ has continuous derivatives of any order (which may be constant after some level, for example, as is the case with polynomials).

The following simple fact shows how first- and second-order derivatives relate to concavity (equivalent results naturally hold for convexity).

FACT A.22. Suppose that $X \subset \mathbb{R}$ and that $f : X \rightarrow \mathbb{R}$ is differentiable. Then:

(1) f is concave on X if and only if

$$(A.7) \quad f(y) - f(x) \leq f'(x)(y - x)$$

for all $x, y \in X$.

(2) f is concave on X if and only if $f'(x)$ is nonincreasing in x for all $x \in X$.

(3) If, in addition, f is twice differentiable, then f is concave on X if and only if $f''(x) \leq 0$ for all $x \in X$.

PROOF. (**Part 1**) Suppose first that f is concave, and take, without loss of any generality, $y > x$. Then, $f(\lambda y + (1 - \lambda)x) \geq \lambda f(y) + (1 - \lambda)f(x)$ for all $\lambda \in (0, 1)$. Rearranging this

$$f(y) - f(x) \leq \frac{f(x + \lambda(y - x)) - f(x)}{\lambda(y - x)}(y - x).$$

Let $\varepsilon = \lambda(y - x)$ and note that this inequality is true for any $\lambda \in (0, 1)$ and thus for any $\varepsilon \geq 0$ in the neighborhood of 0. Therefore,

$$\begin{aligned} f(y) - f(x) &\leq \frac{f(x + \varepsilon) - f(x)}{\varepsilon}(y - x) \\ &\leq f'(x)(y - x), \end{aligned}$$

where the second line follows by taking the limit $\varepsilon \downarrow 0$ and using the fact that, by the differentiability of f , this limit uniquely defines $f'(x)$.

Conversely, suppose that (A.7) holds, then for any $\lambda \in (0, 1)$,

$$f(y) - f(\lambda y + (1 - \lambda)x) \leq (1 - \lambda) f'(\lambda y + (1 - \lambda)x)(y - x),$$

and

$$f(x) - f(\lambda y + (1 - \lambda)x) \leq -\lambda f'(\lambda y + (1 - \lambda)x)(y - x).$$

Multiplying the first inequality with λ and the second with $(1 - \lambda)$, and summing the two, we obtain

$$f(\lambda y + (1 - \lambda)x) \geq \lambda f(y) + (1 - \lambda)f(x)$$

for all $\lambda \in (0, 1)$.

(Part 2) Suppose f is concave or equivalently (A.7) holds. Then, for $y > x$,

$$\begin{aligned} f'(x) &\geq \frac{f(y) - f(x)}{y - x} \\ &= \frac{f(x) - f(y)}{x - y} \\ &\geq f'(y), \end{aligned}$$

where the last inequality uses the fact that $x - y < 0$.

Conversely, if $y > x$ and $f'(x) < f'(y)$, then the previous string of inequalities would imply either $f'(x)(y - x) < f(y) - f(x)$ or $f'(y)(x - y) > f(x) - f(y)$, thus violating (A.7).

(Part 3) This follows immediately from Part 2 when f is twice differentiable. □

Before moving to more general mappings, I present three results that are often very useful in applications. The first one is a generalization of the Intermediate Value Theorem, Theorem A.3, to derivatives.

THEOREM A.21. (Mean Value Theorems) Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is continuously differentiable on $[a, b]$. Then:

- (1) Suppose that $f'(a) \neq f'(b)$, then for any c intermediate between $f'(a)$ and $f'(b)$, there exists $x^* \in (a, b)$ such that $f'(x^*) = c$.
- (2) There exists $x^* \in [a, b]$ such that

$$f'(x^*) = \frac{f(b) - f(a)}{b - a}.$$

PROOF. See Exercise A.25. □

A particular difficulty often encountered in evaluating limits of the form $\lim_{x \rightarrow x^*} f(x)/g(x)$ (where f and g are continuous real-valued functions) is that we may have both $f(x^*) = 0$ and $g(x^*) = 0$. The following result, known as *L'Hospital's Rule* (or *L'Hospital's Theorem*) provides one way of evaluating these types of limits.

THEOREM A.22. (*L'Hospital's Rule*) Suppose that $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ are differentiable functions on $[a, b]$, and suppose that $g'(x) \neq 0$ for $x \in (a, b)$ and let $c \in [a, b]$. If

$$\lim_{x \uparrow c} \frac{f'(x)}{g'(x)}$$

exists and

$$\text{either } \lim_{x \uparrow c} f(x) = \lim_{x \uparrow c} g(x) = 0 \text{ or } \lim_{x \uparrow c} f(x) = \lim_{x \uparrow c} g(x) = \infty,$$

then,

$$\lim_{x \uparrow c} \frac{f(x)}{g(x)} = \lim_{x \uparrow c} \frac{f'(x)}{g'(x)}.$$

The same conclusions also hold for $\lim_{x \downarrow c}$.

PROOF. See Exercise A.26. □

The final result in this section are the Taylor Theorem and the resulting Taylor Series approximation to differentiable real-valued functions. For this theorem, let the n th derivative of a real-valued function f be denoted by $f^{(n)}$ (e.g., $f' = f^{(1)}$, and so on).

THEOREM A.23. (*Taylor's Theorem I*) Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is a $\mathcal{C}^{n-1}$ function and moreover its n th derivative, $f^{(n)}(x)$ exists for all $x \in (a, b)$. Then, for any x and $y \neq x$ in $[a, b]$, there exists z between x and y such that

$$f(y) = f(x) + \sum_{k=1}^{n-1} \frac{f^{(k)}(x)}{k!} (y-x)^k + \frac{f^{(n)}(z)}{n!} (y-x)^n.$$

PROOF. Suppose $y > x$. The proof requires that we show the existence of $z \in (x, y)$ such that

$$f^{(n)}(z) = n!(y-x)^{-n} \left[f(y) - f(x) - \sum_{k=1}^{n-1} \frac{f^{(k)}(x)}{k!} (y-x)^k \right].$$

Let

$$g(t) = f(t) - f(x) - \sum_{k=1}^{n-1} \frac{f^{(k)}(x)}{k!} (t-x)^k - \frac{(t-x)^n}{(y-x)^n} \left(f(y) - f(x) - \sum_{k=1}^{n-1} \frac{f^{(k)}(x)}{k!} (y-x)^k \right).$$

Clearly g is n times differentiable. Thus the proof is equivalent to showing that there exists $z \in (x, y)$ such that $g^{(n)}(z) = 0$. It is straightforward to verify that $g^{(k)}(x) = 0$ for $k = 0, 1, \dots, n-1$ and also $g(x) = g(y) = 0$. The Mean Value Theorem, Theorem A.21, then implies that $g^{(1)}(z_1) = 0$ for some $z_1 \in (x, y)$. Next since $g^{(1)}(x) = g^{(1)}(z_1) = 0$, again from Theorem A.21, we have that there exists $z_2 \in (x, z_1)$ such that $g^{(2)}(z_2) = 0$. Continuing inductively for $n-2$ more steps, establishes the existence of $z \in (x, y)$ such that $g^{(n)}(z) = 0$. □

The following corollary provides both an equivalent form of Theorem A.23 and also an implication of this theorem.

COROLLARY A.3. (1) Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is a $\mathcal{C}^n$ function. Then,

$$f(y) = f(x) + \sum_{k=0}^n \frac{f^{(k)}(x)}{k!} (y-x)^k + o(|y-x|^n),$$

where recall that $o(k)/k \rightarrow 0$ as $k \rightarrow 0$.

(2) Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is a $\mathcal{C}^\infty$ and that $\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{f^{(k)}(x)}{k!} (y-x)^k$ exists. Then

$$f(y) = f(x) + \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{f^{(k)}(x)}{k!} (y-x)^k.$$

PROOF. See Exercise A.27. □

A somewhat more useful corollary, which was used in the text, is

COROLLARY A.4. Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is twice continuously differentiable and concave. Then, for any $x, y \in [a, b]$,

$$f(y) \leq f(x) + f'(x)(y-x).$$

PROOF. By Theorem A.23, $f(y) = f(x) + f'(x)(y-x) + f''(z)(y-x)^2/2$ for some z between x and y . From Fact A.22 $f''(z) \leq 0$ for a concave function and thus the conclusion follows. □

A.9. Functions of Several Variables and the Inverse and Implicit Function Theorems

Throughout this section, I limit myself to differentiation in Euclidean spaces, that is, our interests will be with mappings

$$\phi : X \rightarrow Y,$$

where

$$X \subset \mathbb{R}^{K_X} \text{ and } Y \subset \mathbb{R}^{K_Y},$$

with $K_X, K_Y \in \mathbb{N}$. In the text, when mappings of this form arise and emphasis is needed, I will refer to ϕ as a *vector* function or *vector-valued* function, since $\phi(x) \in \mathbb{R}^{K_Y}$ (for $x \in X$).

The theory of differentiation and the types of results that I will present below can be developed in more general spaces than Euclidean spaces. For example, Luenberger's (1969) classic treatment of general optimization problems considers X and Y to be Banach spaces (complete normed vector spaces, which allow for a convenient definition of *linear operators*, see Section 10.1 below). Nevertheless, for the results presented here, restricting attention to Euclidean spaces is without loss of any generality and enables me to reduce notation and avoid unnecessary complexities.

The case $K_x = K_Y = 1$ was treated in the previous section. Building on the results and the intuitions of that section, let us now move to more general mappings. For $\phi : X \rightarrow Y$ (where $X \subset \mathbb{R}^{K_X}$ and $Y \subset \mathbb{R}^{K_Y}$), the equivalent of the derivative is the linear operator $J(x) : X \rightarrow Y$. In particular, with analogy to (A.6), we have the following definition of

differentiability.⁴ Let $h \in X$ be a vector and let $\|h\|$ denote its Euclidean norm. Then, for $x \in X'$, where X' is an open set with $\phi(X') \subset Y$ well-defined, ϕ is differentiable if the limit

$$(A.8) \quad \lim_{h \rightarrow 0} \frac{\|\phi(x+h) - \phi(x) - J(x)h\|}{\|h\|} = 0$$

at x exists and defines a unique linear operator $J(x)$ (mapping from $\mathbb{R}^{K_X}$ onto $\mathbb{R}^{K_Y}$). In this case, the derivative of $\phi(x)$ is denoted by $J(x)$. The derivative is again a linear operator because, though it depends on x , it assigns the value $J(x)h$ to any vector h such that $x+h \in X'$.

We will refer to $J(x)$ as the *Jacobian matrix* (or as simply the Jacobian) of ϕ at x and often denote it by $D\phi(x)$. The latter is a more convenient notation than $J(x)$, since it indicates which function we are referring to. We will see below that the Jacobian, when it exists, is also the matrix of partial derivatives of ϕ . We can also denote the matrix of *partial derivatives* by $D_{x_1}\phi(x_1, x_2)$ for $x_1 \in \mathbb{R}^{K_1}$, $x_2 \in \mathbb{R}^{K_2}$, and $K_1, K_2 \in \mathbb{N}$.

The following Fact generalizes Fact A.21:

FACT A.23. *Let $X \subset \mathbb{R}^{K_X}$, $Y \subset \mathbb{R}^{K_Y}$ (where $K_X, K_Y \in \mathbb{N}$) and $\phi : X \rightarrow Y$. If ϕ is differentiable at $x \in X$, then it is also continuous at x .*

Let us next take $X \subset \mathbb{R}^{K_X}$, and consider the mapping $\phi : X \rightarrow \mathbb{R}$, also referred to as a *function of several variables*. Its partial derivatives with respect to each component of X are defined identically to the derivative of a real-valued function of one variable (holding all the other variables constant). Let $x = (x_1, \dots, x_{K_X})$ and assume that ϕ is differentiable with respect to its k th component. Then, the k th partial derivative of ϕ is

$$\frac{\partial \phi(x_1, \dots, x_{K_X})}{\partial x_k} = \phi_k(x),$$

where

$$\phi_k(x) = \lim_{h \rightarrow 0} \frac{\phi(x_1, \dots, x_{k-1}, x_k + h, x_{k+1}, \dots, x_{K_X}) - \phi(x_1, \dots, x_{k-1}, x_k, x_{k+1}, \dots, x_{K_X})}{h}.$$

Now assuming that ϕ has partial derivatives with respect to each x_k for $k = 1, \dots, K_X$, the Jacobian in this case is simply a *row vector*,

$$J(x) = \left(\phi_1(x) \quad \cdot \quad \cdot \quad \cdot \quad \phi_{K_X}(x) \right).$$

A general mapping $\phi : X \rightarrow Y$, where Y is a subset of $\mathbb{R}^{K_Y}$ can then be thought of as consisting of K_Y real-valued functions of several variables, $\phi^1(x), \dots, \phi^{K_Y}(x)$. We can define the partial derivatives of each of these functions in a similar fashion and denote them by

⁴More precisely, this is the definition of *Frechet differentiability*. The alternative, weaker notion of *Gateaux differentiability* is also useful in many instances (see, e.g., Luenberger, 1969). For our purposes, there is no need to distinguish between these two notions, since in finite-dimensional spaces they are equivalent. The even weaker notion of directional derivative, discussed in Example A.15, will be used in some proofs.

$\phi_k^j(x)$. The Jacobian can then be written as

$$J(x) = \begin{pmatrix} \phi_1^1(x) & \cdot & \cdot & \cdot & \phi_{K_X}^1(x) \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \phi_1^{K_Y}(x) & \cdot & \cdot & \cdot & \phi_{K_X}^{K_Y}(x) \end{pmatrix}.$$

Higher-order derivatives can be defined in a similar fashion, though they will now correspond to higher-dimensional objects. When $\phi : X \rightarrow X$, $J(x)$ is a $K_X \times K_X$ matrix, and in this case, we can investigate whether it is invertible or not (i.e., whether the inverse $J^{-1}(x)$ at x exists). This will play an important role in the Inverse Function and Implicit Function Theorems below.

One potential source of complication and confusion is that the matrix of partial derivatives may exist, and we may naturally refer to it as the Jacobian, but the mapping in question *may fail to be differentiable* (and this would be true even with weaker notions of differentiability). The following example illustrates the problem.

EXAMPLE A.14. Consider the function of several variables $\phi(x_1, x_2)$ over the entire $\mathbb{R}^2$ such that $\phi(x_1, x_2) = 0$ if $x_1 = x_2 = 0$, and

$$\phi(x_1, x_2) = \frac{x_1^2 x_2^2}{x_1 + x_2} \text{ otherwise.}$$

The partial derivatives of this function are

$$\begin{aligned} \frac{\partial \phi(x_1, x_2)}{\partial x_1} &= \frac{x_1^2 x_2^2 + 2x_1 x_2^3}{(x_1 + x_2)^2} \\ \frac{\partial \phi(x_1, x_2)}{\partial x_2} &= \frac{x_1^2 x_2^2 + 2x_1^3 x_2}{(x_1 + x_2)^2}. \end{aligned}$$

It can be verified that these partial derivatives exist everywhere in $\mathbb{R}^2$, and in particular, $\partial \phi(0, 0) / \partial x_1 = \partial \phi(0, 0) / \partial x_2 = 0$. However, it is also clear that ϕ is not continuous at $x_1 = x_2 = 0$ (consider the limit $x \rightarrow 0$ with $x = x_1 = x_2$, which using L'Hospital's Rule, Theorem A.22, is obtained as $\lim_{x \rightarrow 0} \phi(x, x) = 2$), and thus in view of Fact A.23, ϕ is not differentiable. The fact that ϕ is not differentiable can also be established using directly the definition of differentiability provided above.

The situation illustrated in Example A.14 is important to bear in mind and it implies that a well-defined matrix of partial derivatives does not guarantee differentiability. In view of this, one may wish to distinguish between the linear operator $J(x)$ defined above and the Jacobian, consisting of the partial derivatives, $D\phi(x)$. Throughout this book, the term $D\phi(x)$ refers to the Jacobian or the matrix of partial derivatives. In addition, the problem highlighted in Example A.14 is not central for our focus here, since this type of problem will not arise in the context of the models we study in the current book.

It is also useful to briefly introduce the notion of *directional* derivatives, which refer to derivatives when a specific limit is taken. For example, in the one-dimensional case, the left derivative and the right derivative, $\lim_{h \uparrow 0} [f(x+h) - f(x)]/h$ and $\lim_{h \downarrow 0} [f(x+h) - f(x)]/h$, would be the simplest examples. Directional derivatives are used in the second version of the proof of Theorem 6.6 in Chapter 6. The next example illustrates how simple functions may have directional derivatives but fail to be defensible.

EXAMPLE A.15. An immediate example of a function that has directional derivatives but is not differentiable according to the stronger notion here would be f defined by $f(x) = x$ for $x \geq 0$ and $f(x) = -x$ for $x < 0$, which has derivatives from the left and the right at 0, but is not differentiable according to (A.5) or (A.8), since a unique $f'(x)$ does not exist.

An even stronger notion than differentiability (or Frechet differentiability) is continuous differentiability. I provide a definition of continuous differentiability for vector-valued mappings.

DEFINITION A.38. A mapping ϕ is of class $\mathcal{C}^n$ (n -times continuously differentiable) on some set X' if it has continuous derivatives up to the n th order.

FACT A.24. A mapping $\phi: X \rightarrow Y$, with $X \subset \mathbb{R}^{K_X}$, $Y \subset \mathbb{R}^{K_Y}$ (where $K_X, K_Y \in \mathbb{N}$) and X open, is of class $\mathcal{C}^1$ on X if its partial derivatives $\phi_k^j(x)$ for $k = 1, \dots, K_X$ and $j = 1, \dots, K_Y$ exist and are continuous functions of x for each $x \in X$.

PROOF. See Exercise A.28. □

When there is no need for further generality, I impose that the relevant utility or production functions are continuously differentiable (of class $\mathcal{C}^1$).

Taylor's Theorem and its corollaries can be generalized to mappings discussed here. Let me simply state the result for $\phi: X \rightarrow \mathbb{R}$ with $X \subset \mathbb{R}^{K_X}$. Moreover, let $D\phi$ and $D^2\phi$ denote the vector of first derivatives and the Jacobian of ϕ , let $\|y - x\|$ be the Euclidean norm of the K_X dimensional vector $y - x$, and let z^T be the transpose of vector z . The following is a simpler version of the equivalent form of Taylor's Theorem in Corollary A.3. Its proof is similar to that of Theorem A.23, but is longer and requires more notation, and is thus omitted.

THEOREM A.24. (**Taylor's Theorem II**) Suppose that $\phi: X \rightarrow \mathbb{R}$ is a $\mathcal{C}^1$ function and moreover its second derivative $D^2\phi(x)$ exists for all $x \in X$. Then, for any x and $y \neq x$ in X ,

$$\phi(y) = \phi(x) + D\phi(x)^T(y - x) + o(\|y - x\|^2).$$

If in addition $\phi: X \rightarrow \mathbb{R}$ is a $\mathcal{C}^2$ function with third derivative $D^3\phi(x)$ for all $x \in X$, then for any x and $y \neq x$ in X

$$\phi(y) = \phi(x) + D\phi(x)^T(y - x) + (y - x)^T D^2\phi(x)(y - x) + o(\|y - x\|^3).$$

The following two theorems are the basis of much of the comparative static results in economic models. They are therefore among the most important mathematical results for economic analysis. Consider a mapping $\phi : X \rightarrow X$ for $X \subset \mathbb{R}^{K_X}$. One obvious question is whether this mapping will have an inverse $\phi^{-1} : X \rightarrow X$. If for some subset X' of X , ϕ is single-valued, has an inverse ϕ^{-1} , which is also a single valued, then we say that it is *one-to-one*.

THEOREM A.25. (*The Inverse Function Theorem*) Consider a $\mathcal{C}^1$ mapping $\phi : X \rightarrow X$ for $X \subset \mathbb{R}^{K_X}$. Suppose that the Jacobian of ϕ , $J(x)$ evaluated at some interior point x^* of X is invertible. Then, there exist open sets X' and X'' in X such that $x^* \in X'$, $\phi(x^*) \in X''$, and ϕ is one-to-one on X' with $\phi(X') = X''$. Moreover, $\phi^{-1}(\phi(x)) = x$ for all x in X' and ϕ^{-1} is also a $\mathcal{C}^1$ mapping.

The proof of this theorem is not difficult, but somewhat long and can be found in any real analysis book, so I will not provide it here. The following theorem is directly used in most comparative static exercises in economics.

THEOREM A.26. (*The Implicit Function Theorem*) Consider a $\mathcal{C}^1$ mapping $\phi : X \times Y \rightarrow Y$ with $X \subset \mathbb{R}^{K_X}$ and $Y \subset \mathbb{R}^{K_Y}$. Suppose that $(x^*, y^*) \in X \times Y$, $\phi(x^*, y^*) = 0$, all the entries of the Jacobian of ϕ with respect to (x, y) , $D_{(x,y)}\phi(x^*, y^*)$, are finite and $D_y\phi(x^*, y^*)$ is invertible. Then, there exists an open set $X' \ni x^*$ and a unique $\mathcal{C}^1$ mapping $\gamma : X' \rightarrow Y$ such that $\gamma(x^*) = y^*$ and

$$(A.9) \quad \phi(x, \gamma(x)) = 0$$

for all $x \in X'$.

This theorem is called the Implicit Function Theorem because the mapping γ is defined implicitly. Exercise 6.5 in Chapter 6 provided the proof of a special case of this theorem. The more general case can also be proved with exactly the same methods as in that exercise. An alternative proof uses the Inverse Function Theorem. Since the former proof has already been discussed and the latter one is contained in most real analysis books, I will not provide the proof of Theorem A.26 here.

The main utility of this theorem comes from the fact that since ϕ and γ are $\mathcal{C}^1$ and (A.9) holds for an open set around x^* , (A.9) can be differentiated with respect to x , to obtain an expression for how the solution y to the set of equations $\phi(x, y) = 0$ behaves as a function of x . If we think of x as representing a set of parameters and y as the endogenous variables determined by some economic relationship summarized by (A.9), then this procedure can tell us how the endogenous variables change in response to the changes in the environment captured by the parameter x . I made repeated use of this approach throughout the book.

A.10. Separation Theorems*

In this section, I will briefly discuss the separation of convex disjoint sets using linear functionals (or hyperplanes). These results form the basis of the Second Welfare Theorem,

provided in Theorem 5.7 in Chapter 5. They also provide the basis of many important results in constrained optimization (see Section A.11).

For this section, we take X be a *vector space* (linear space). Recall from Section A.7) that this implies: if $x, y \in X$ and λ is a real number, then $x + y \in X$ and $\lambda x \in X$. The element of X with the property that $x = \lambda x$ for all $\lambda \in \mathbb{R}$ is denoted by θ . Moreover:

DEFINITION A.39. *The real-valued nonnegative function $\|\cdot\| : X \rightarrow \mathbb{R}_+$ is taken to be a **norm** on X , which implies that for any $x, y \in X$ and any $\lambda \in \mathbb{R}$,*

- (1) (**Properness**) $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = \theta$.
- (2) (**Linearity**) $\|\lambda x\| = |\lambda| \|x\|$.
- (3) (**Triangle Inequality**) $\|x + y\| \leq \|x\| + \|y\|$.

A vector space equipped with a norm is a **normed vector space**. A complete normed vector space is a **Banach space**.

If a function $p : X \rightarrow \mathbb{R}_+$ satisfies properness and triangle inequality, but not necessarily the linearity condition, then it is referred to as a *semi-norm*.

Many of the metric spaces given in Example A.1 are also normed vector spaces with the appropriate norm. In fact, a simple way of obtaining the norm in many cases is to take the distance function d and try the norm $\|x\| = d(x, \theta)$. Notice, however, that this will not always work, since metrics do not need to satisfy the *linearity* condition in Definition A.39.

EXAMPLE A.16. The first four spaces are normed vector spaces, while the fifth one is not.

- (1) For any $X \subset \mathbb{R}^K$, let x_i be the i th component of $x \in X$. Then, the K -dimensional *Euclidean space* is a normed vector space with norm given by $\|x\| = \left(\sum_{i=1}^K |x_i|^2\right)^{1/2}$.
- (2) Let $X \subset \mathbb{R}^K$ and consider the set of continuous, bounded real-valued functions $f : X \rightarrow \mathbb{R}$ denoted by $\mathbf{C}(X)$. $\mathbf{C}(X)$ is a normed vector space with $\|f\| = \sup_{x \in X} |f(x)|$.
- (3) $\ell \subset \mathbb{R}^\infty$, the set consisting of infinite sequences of real numbers, is a normed vector space with norm $\|x\|_p = (\sum_{i=1}^\infty |x_i|^p)^{1/p}$ for $1 \leq p < \infty$ or by $\|x\|_\infty = \sup_i |x_i|$. For any $p \in [1, \infty]$, the corresponding normed vector space is denoted by ℓ_p . The one of greatest interest for us is ℓ_∞ .
- (4) Let $\mathbf{c} \subset \mathbb{R}^\infty$, be the set consisting of infinite sequences of real numbers that are equal to zero after some point (e.g., $(x_1, \dots, x_M, 0, 0, \dots)$, where $M \in \mathbb{N}$). Let the sup norm on $\mathbf{c}$ be defined as usual $\|x\|_\infty = \sup_i |x_i|$. Then, $\mathbf{c}$ with the sup norm is a normed vector space.
- (5) For X nonempty, consider the discrete metric $d(x, y) = 1$ if $x \neq y$ and $d(x, y) = 0$ if $x = y$. The metric space (X, d) is *not* a normed vector space.

When the norm is understood implicitly, we refer to X as a normed vector space.

DEFINITION A.40. Let X be a normed vector space. Then, $\phi : X \rightarrow \mathbb{R}$ is a **linear functional** on X if for any $x, y \in X$ and any real numbers λ and μ ,

$$\phi(\lambda x + \mu y) = \lambda \phi(x) + \mu \phi(y).$$

Linear functionals on normed vector spaces have many nice properties. For example, if $X \subset \mathbb{R}^K$, then any linear functional on X can be expressed as an *inner product* of x with another K -dimensional vector η , i.e., $\phi(x) = \eta \cdot x = \sum_{i=1}^K \eta_i x_i$, where $\eta = (\eta_1, \dots, \eta_K) \in \mathbb{R}^K$. Therefore, on Euclidean spaces, linear functionals correspond to inner products. In many other spaces, linear functionals will have properties similar to inner products. Some other nice properties of linear functionals are provided in the following result.

THEOREM A.27. (**Continuity of Linear Functionals**) Let X be a normed vector space. Then:

- (1) The linear functional $\phi : X \rightarrow \mathbb{R}$ is continuous on X if and only if it is continuous at θ .
- (2) The linear functional $\phi : X \rightarrow \mathbb{R}$ is continuous on X if and only if it is bounded in the sense that there exists $M \in \mathbb{R}$ such that $|\phi(x)| \leq M \|x\|$ for all $x \in X$.

PROOF. (**Part 1**) We only need to prove that ϕ is continuous on X if it is continuous at θ . Suppose that it is continuous at θ . Fix an arbitrary $x \in X$ and consider a sequence in X , $\{x_n\}$, converging to x . By the linearity of ϕ ,

$$|\phi(x_n) - \phi(x)| = |\phi(x_n - x + \theta) - \phi(\theta)|.$$

Since $x_n \rightarrow x$, $x_n - x + \theta \rightarrow \theta$ and since ϕ is continuous at θ , $\phi(x_n - x + \theta) \rightarrow \phi(\theta)$. Therefore $|\phi(x_n) - \phi(x)| \rightarrow 0$, proving that ϕ is continuous at x .

(**Part 2**) To prove the “if” part, suppose that ϕ is bounded. Consider a sequence $\{x_n\}$ converging to θ . This implies

$$|\phi(x_n)| \leq M \|x_n\|$$

and since $x_n \rightarrow \theta$, $\|x_n\| \rightarrow 0$, proving that ϕ is continuous at θ . Then, by Part 1, ϕ is continuous on X .

To prove the “only if” part, suppose that ϕ is continuous at θ . Fix $\varepsilon > 0$. Then, there exists $\delta > 0$ such that for $\|x\| \leq \delta$, $|\phi(x)| \leq \varepsilon$. Note that for $x \neq \theta$, the vector $\delta x / \|x\|$ has norm equal to δ . Therefore,

$$\begin{aligned} |\phi(x)| &= \left| \phi\left(\frac{\delta x}{\|x\|}\right) \right| \cdot \frac{\|x\|}{\delta} \\ &= \varepsilon \cdot \frac{\|x\|}{\delta} \\ &= M \|x\|, \end{aligned}$$

with $M = \varepsilon/\delta$, completing the proof. $\square$

The smallest M that satisfies $|\phi(x)| \leq M \|x\|$ for all $x \in X$ is defined as the norm of the linear functional ϕ and is sometimes denoted by $\|\phi\|$. Theorem A.27 therefore implies that a continuous linear functional has a finite norm.

DEFINITION A.41. *Let X be a normed vector space. The space of all continuous linear functionals on X is the **normed dual** of X and is denoted by X^* .*

Dual spaces have many nice features. For example:

FACT A.25. *If X is a normed vector space, then its dual X^* is a Banach space.*

The following example gives the duals of some common spaces (see Exercise A.30).

EXAMPLE A.17. (1) For any $K \in \mathbb{N}$, the dual of $\mathbb{R}^K$ is $\mathbb{R}^K$.

(2) For any $p \in (1, \infty)$, the dual of ℓ_p is ℓ_q where $p^{-1} + q^{-1} = 1$.

A nonobvious fact is the following. Let $\mathbf{c} = \{x = (x_1, x_2, \dots) \in \ell_\infty : \lim_{n \rightarrow \infty} x_n = 0\}$.

FACT A.26. (1) *The dual of ℓ_∞ is not ℓ_1 (it contains ℓ_1).*

(2) *The dual of $\mathbf{c}$ is ℓ_1 .*

Dual spaces are particularly useful in economics, since if X is a commodity space, then its dual, X^* , corresponds to the space of “price functionals” for X . For example, the dual of $X \subset \mathbb{R}^K$ is $X^* \subset \mathbb{R}^K$ and indeed consists of functionals of the form $\phi(x) = \sum_{i=1}^K \eta_i x_i$ as noted above. Loosely speaking, we can interpret the η_i ’s as “prices” corresponding to the commodity vector x , so that $\phi(x)$ is the “cost” of x at the price vector η . The particular usefulness of this construction for economics stems from the following famous theorem. We refer to a linear functional ϕ defined on X as *nonzero* if it is not identically equal to zero for all $x \in X$.

THEOREM A.28. (**Geometric Hahn-Banach Theorem**) *Let X be a normed vector space and let $X^1, X^2 \subset X$. Suppose that X^1 and X^2 are convex, $\text{Int}X^1 \neq \emptyset$ and $X^2 \cap \text{Int}X^1 = \emptyset$, then there exists a nonzero continuous linear functional ϕ on X such that*

$$\phi(x^1) \leq c \leq \phi(x^2) \text{ for all } x^1 \in X^1, x^2 \in X^2 \text{ and some } c \in \mathbb{R}.$$

This theorem is obtained from the Hahn-Banach Theorem. The Hahn-Banach Theorem states that if ϕ is a continuous linear functional on a subspace M of X and is dominated by a semi-norm $p(x)$, i.e., $\phi(x) \leq p(x)$ for all $x \in M$, then there is an extension Φ of ϕ to the entire X such that Φ is a continuous linear functional on X , $\Phi(x) = \phi(x)$ for all $x \in M$ and $\Phi(x) \leq p(x)$ for all $x \in X$. This theorem therefore establishes that normed vector spaces are “abundant” in linear functionals. More important for our purposes, it also implies Theorem A.28. Since its proof is not particularly useful for our purposes here, it is omitted. A proof of this theorem together with further separation theorems can be found in Conway (2000), Kolmogorov and Fomin (1970), and Luenberger (1969).

Notice the non-intuitive requirement that $\text{Int}X^1 \neq \emptyset$, which implies that X^1 should contain an interior point. This is not a stringent requirement when X is a subset of the Euclidean space (and in fact, this condition is not even necessary in that case). However, some common infinite-dimensional normed vector spaces, such as ℓ_p for $p < \infty$ do not contain interior points when we restrict attention to their economically relevant subspaces, that is, ℓ_p^+ ,

which requires all sequences to consist of nonnegative numbers (this is rather nonobvious, but Exercise A.31 illustrates why). This might be a problem if we wished to model the allocations (for example, the sequence of consumption levels or capital stocks) in an infinite-horizon economy as elements of ℓ_p^+ . Nevertheless, this is not an issue when we focus on the economically more natural space of sequences of allocations ℓ_∞ , because ℓ_∞^+ does contain interior points (see Exercise A.32). The only complication that arises from the use of ℓ_∞ is that not all linear functionals on ℓ_∞ have an inner product representation and thus may not correspond to economically meaningful price systems (recall Fact A.26). This problem can be handled, however, by making somewhat stronger assumptions on preferences and technology to ensure that the relevant linear functionals on ℓ_∞ have the desired inner product representation. This is the reason why the Second Welfare Theorem, Theorem 5.7, imposes additional conditions on preferences and technology.

It is also useful to note the following immediate corollary of Theorem A.28.

THEOREM A.29. (*Separating Hyperplane Theorem*) *Let $X \subset \mathbb{R}^K$ and $X^1, X^2 \subset X$. Suppose that X^1 and X^2 are convex and $X^2 \cap \text{Int}X^1 = \emptyset$, then there exists a hyperplane $H = \left\{x \in X: \sum_{i=1}^K \eta_i x_i = c \text{ for } \eta \in \mathbb{R} \text{ and } \eta \neq 0\right\}$ such the H separates X^1 and X^2 , or in other words,*

$$\eta \cdot x^1 \leq c \leq \eta \cdot x^2 \text{ for all } x^1 \in X^1, x^2 \in X^2,$$

where recall that $\eta \cdot x = \sum_{i=1}^K \eta_i x_i$.

Note that the statement of this theorem disposes of the hypothesis that $\text{Int}X^1 \neq \emptyset$, which is not necessary when the two sets are subsets of Euclidean spaces. Moreover, the theorem does not add the qualification that the hyperplane H is “nonzero” (in the same way as Theorem A.28 did for linear functionals), since the definition of the hyperplane already incorporates this requirement.

A.11. Constrained Optimization

Many of the problems we encountered in this book are formulated as constrained optimization problems. Chapters 6, 7, and 16 dealt with dynamic (infinite-dimensional) constrained optimization problems. Complementary insights about these problems can be gained by using the separation theorems of the previous section. Let me illustrate this here by focusing on finite-dimensional optimization problems. Consider the maximization problem

$$\begin{aligned} \text{(A.10)} \quad & \sup_{x \in X} f(x) \\ & \text{subject to} \\ & g(x) \leq 0, \end{aligned}$$

where X is an open subset of $\mathbb{R}^K$, $f: X \rightarrow \mathbb{R}$, $g: X \rightarrow \mathbb{R}^N$, and $N, K \in \mathbb{N}$.

The constrained maximization problem (A.10) satisfies the *Slater* condition if there exists $x' \in X$ such that $g(x') < 0$ (meaning that each component of the mapping g takes a negative value). This is equivalent to the set $G = \{x: g(x) \leq 0\}$ having an interior point. We say that

g is convex, if each component function of g is convex. This implies that the set G is also convex (but the converse is not necessarily true (see Exercise A.33)). As usual, we define the Lagrangian function as

$$\mathcal{L}(x, \lambda) \equiv f(x) - \lambda \cdot g(x)$$

for $\lambda \in \mathbb{R}_+^N$. The vector λ is referred to as the *Lagrange multiplier* and $\lambda \cdot g(x)$ denotes the inner product between two vectors (here λ and the vector-valued function $g(\cdot)$ evaluated at x), thus it is equal to a real number. A central theorem in constrained maximization is the following.

THEOREM A.30. (*The Saddle Point Theorem*) Suppose that in (A.10) f is a quasi-concave function, g is convex and the **Slater condition** is satisfied. Then:

(1) If x^* is a solution to (A.10), then there exists $\lambda^* \in \mathbb{R}_+^N$ such that

$$(A.11) \quad \mathcal{L}(x, \lambda^*) \leq \mathcal{L}(x^*, \lambda^*) \leq \mathcal{L}(x^*, \lambda) \text{ for all } x \in X \text{ and } \lambda \in \mathbb{R}_+^N.$$

In this case, (x^*, λ^*) satisfies the **complementary slackness condition**

$$(A.12) \quad \lambda^* \cdot g(x^*) = 0.$$

(2) If $(x^*, \lambda^*) \in X \times \mathbb{R}_+^N$ satisfies $g(x^*) \leq 0$ and (A.11), then x^* is a solution to (A.10).

PROOF. The proof follows from Theorem A.29.

(Part 1) Consider the space $Y = \mathbb{R}^{N+1}$, with subsets

$$Y^1 = \{(a, b) \in Y : a > f(x^*) \text{ and } b < 0\},$$

and

$$Y^2 = \{(a, b) \in Y : \exists x \in X \text{ with } a \leq f(x) \text{ and } b \geq g(x)\},$$

where $a \in \mathbb{R}$, $b \in \mathbb{R}^N$ and $b < 0$ means that each element of the N -dimensional vector b is negative. Y^1 is clearly convex. Moreover, the quasi-concavity of f and the convexity of g ensure that Y^2 is also convex.

By the hypothesis that x^* is a solution to (A.10), the two sets are disjoint. Then, Theorem A.29 implies that there exists a hyperplane separating these two sets. In other words, there exists a nonzero vector $\eta \in \mathbb{R}^{N+1}$ such that

$$\eta \cdot y^1 \leq c \leq \eta \cdot y^2 \text{ for all } y^1 \in Y^1, y^2 \in Y^2.$$

Moreover, the same conclusion holds for all $y^1 \in \overline{Y^1}$ and $y^2 \in \overline{Y^2}$. Therefore, let $\eta = (\rho, \lambda)$ with $\rho \in \mathbb{R}$ and $\lambda \in \mathbb{R}^N$ so that

$$(A.13) \quad \rho a^1 + \lambda \cdot b^1 \leq \rho a^2 + \lambda \cdot b^2 \text{ for all } (a^1, b^1) \in Y^1, (a^2, b^2) \in \overline{Y^2}.$$

For $(f(x^*), 0) \in \overline{Y^2}$,

$$(A.14) \quad \rho a^1 + \lambda \cdot b^1 \leq \rho f(x^*)$$

for all $(a^1, b^1) \in \overline{Y^1}$. Now taking $a^1 = f(x^*)$ and $b^1 < 0$ implies $\lambda \geq 0$ (suppose instead that one component of the vector λ is negative; then take b^1 to have zeros everywhere except for that component, yielding a contradiction to (A.14)). Similarly, setting $b^1 = 0$ and $a^1 > f(x^*)$,

we obtain $\rho \leq 0$. Moreover, by the definition of a hyperplane, either ρ is negative or a component of λ must be strictly positive.

Next the optimality of x^* implies that for any $x \in X$, $(f(x), g(x)) \in \overline{Y^2}$. Since $(f(x^*), 0) \in \overline{Y^1}$, (A.13) implies

$$(A.15) \quad \rho f(x^*) \leq \rho f(x) + \lambda \cdot g(x) \text{ for all } x \in X$$

Now to obtain a contradiction suppose that $\rho = 0$. Then, by the Slater condition, there exists $x' \in X'$ such that $g(x') < 0$, so that $\lambda \cdot g(x') < 0$ for any nonzero vector λ , violating (A.15). Therefore, $\lambda = 0$. However, this in turn contradicts the fact that the separating hyperplane is nonzero (so that we cannot have both $\rho = 0$ and $\lambda = 0$). Therefore, $\rho < 0$. In view of this, define

$$\lambda^* = -\frac{\lambda}{\rho} \geq 0.$$

The complementary slackness condition now follows immediately from (A.15). In particular, evaluate the right-hand side at $x^* \in X$, which implies $\lambda \cdot g(x^*) \geq 0$. Since $\lambda \geq 0$ and $g(x^*) \leq 0$, we must have

$$\lambda \cdot g(x^*) = -\rho(\lambda^* \cdot g(x^*)) = 0.$$

Now using the complementary slackness condition and (A.15) together with the fact that $\rho < 0$,

$$\mathcal{L}(x, \lambda^*) = f(x) - \lambda^* \cdot g(x) \leq f(x^*) = \mathcal{L}(x^*, \lambda^*) \text{ for all } x \in X,$$

which establishes the first inequality in (A.11). To establish the second inequality, again use the complementary slackness condition and the fact that $g(x^*) \leq 0$ to obtain

$$\mathcal{L}(x^*, \lambda^*) = f(x^*) \leq f(x^*) - \lambda \cdot g(x^*) = \mathcal{L}(x^*, \lambda) \text{ for all } \lambda \in \mathbb{R}_+^N,$$

which completes the proof of the first part.

(Part 2) Suppose to obtain a contradiction that (A.11) holds, but x^* is not a solution to (A.10). This implies that there exists $x' \in X$ with $g(x') \leq 0$ and $f(x') > f(x^*)$. Then,

$$f(x') - \lambda^* \cdot g(x') > f(x^*) - \lambda^* \cdot g(x^*),$$

which exploits the fact that $\lambda^* \cdot g(x^*) = 0$ and $\lambda^* \cdot g(x') \leq 0$ (since $\lambda^* \geq 0$ and $g(x') \leq 0$). But this contradicts (A.11) and establishes the desired result. $\square$

Exercise A.34 shows that the Slater condition cannot be dispensed with in this theorem. Despite their importance, constraint qualification conditions, such as the Slater condition or the linear independence condition in the next theorem, are often not stated explicitly in economic applications, because in most problems they are naturally satisfied. Nevertheless, it is important to be aware that these conditions are necessary and that ignoring them can sometimes lead to misleading results.

An immediate corollary of the first inequality in (A.11) is that if $x^* \in \text{Int}X$ and if f and g are differentiable, then

$$(A.16) \quad D_x f(x^*) = \lambda^* \cdot D_x g(x^*),$$

where, as usual, $D_x f$ and $D_x g$ denote the Jacobians of f and g . (A.16) is the usual first-order necessary condition for interior constrained maximum. In this case, (A.16), together with $g(x^*) \leq 0$, is also sufficient for a maximum.

The next result is the famous Kuhn-Tucker Theorem, which shows that (A.16) is necessary for an interior maximum (provided that f and g are differentiable) even when the quasi-concavity and the convexity assumptions do not hold.

THEOREM A.31. (Kuhn-Tucker Theorem) *Consider the constrained maximization problem*

$$\begin{aligned} & \sup_{x \in \mathbb{R}^K} f(x) \\ & \text{subject to} \\ & g(x) \leq 0 \\ & h(x) = 0 \end{aligned}$$

where $f : x \in X \rightarrow \mathbb{R}$, $g : x \in X \rightarrow \mathbb{R}^N$ and $h : x \in X \rightarrow \mathbb{R}^M$ (for some $K, N, M \in \mathbb{N}$). Let $x^* \in \text{Int}X$ be a solution to this maximization problem and suppose that $N_1 \leq N$ of the inequality constraints are **active**, in the sense that they hold as equality at x^* . Define $\bar{h} : X \rightarrow \mathbb{R}^{M+N_1}$ to be the mapping of these N_1 active constraints stacked with $h(x)$ (so that $\bar{h}(x^*) = 0$). Suppose that the following **constraint qualification condition** is satisfied: the Jacobian matrix $D_x(\bar{h}(x^*))$ has rank $N_1 + M$. Then, the following Kuhn-Tucker condition is satisfied: there exist Lagrange multipliers $\lambda^* \in \mathbb{R}_+^N$ and $\mu^* \in \mathbb{R}^M$ such that

$$(A.17) \quad D_x f(x^*) - \lambda^* \cdot D_x g(x^*) - \mu^* \cdot D_x h(x^*) = 0,$$

and the **complementary slackness condition**

$$\lambda^* \cdot g(x^*) = 0$$

holds.

PROOF. (Sketch) The constraint qualification condition ensures that there exists a $N_1 + M$ -dimensional manifold at x^* , defined by the equality and active inequality constraints. Since g and h are differentiable, this manifold is differentiable at x^* . Let $v_\varepsilon(x)$ denote a feasible direction along this manifold for small $\varepsilon \in \mathbb{R}^K$, in particular, such that $x^* \pm \varepsilon v_\varepsilon(x^* + \varepsilon)$ remains along this manifold and thus satisfies $D_x \bar{h}(x^*) \cdot \varepsilon v_\varepsilon(x^* + \varepsilon) = 0$. For ε sufficiently small, the $N - N_1$ nonactive constraints are still satisfied, thus $x^* \pm \varepsilon v_\varepsilon(x^* + \varepsilon)$ is feasible. If $D_x f(x^*) \cdot \varepsilon v_\varepsilon(x^* + \varepsilon) \neq 0$, then $f(x^* + \varepsilon v_\varepsilon(x^* + \varepsilon)) > f(x^*)$ or $f(x^* + \varepsilon v_\varepsilon(x^* + \varepsilon)) < f(x^*)$, implying that x^* cannot be a local (and thus global) maximum. Now consider the $M + N_1 + 1 \times K$ dimensional matrix A , where the first row is $D_x f(x^*)^T$ and the rest is given by $D_x(\bar{h}(x^*))$. The preceding argument implies that for all nonzero $\varepsilon \in \mathbb{R}^K$ such that $D_x \bar{h}(x^*) \cdot \varepsilon v_\varepsilon(x^* + \varepsilon) = 0$, we also have $A \cdot (\varepsilon + v_\varepsilon(x^* + \varepsilon)) = 0$. Therefore, both $D_x \bar{h}(x^*)$ and A have the same rank, which by the constraint qualification condition is equal to $M + N_1$. Since A has $M + N_1 + 1$ rows, this implies that the first row of A must be a linear combination of its remaining $M + N_1$ rows, which equivalently implies that there exists an $M + N_1$ vector

$\bar{\mu}$ such that $D_x f(x^*) = \bar{\mu} D_x \bar{h}(x^*)$. Assigning zero multipliers to all nonactive constraints, this is equivalent to (A.17). The complementary slackness condition then follows immediately since we have zero multipliers for the nonactive constraints and $g_j(x^*) = 0$ for the active constraints. $\square$

The constraint qualification condition, which required that the active constraints should be linearly independent, plays a similar role to the Slater condition in Theorem A.30. Exercise A.35 shows that this constraint qualification condition cannot be dispensed with (though somewhat weaker conditions can be used instead of the full rank condition used in Theorem A.31).

Let us end this Appendix with the famous and eminently useful Envelope Theorem.

THEOREM A.32. (*The Envelope Theorem*) *Consider the constrained maximization problem*

$$\begin{aligned} v(p) &= \max_{x \in X} f(x, p) \\ \text{subject to} \\ g(x, p) &\leq 0 \\ h(x, p) &= 0 \end{aligned}$$

where $X \in \mathbb{R}^K$, $p \in \mathbb{R}$, and $f : X \times \mathbb{R} \rightarrow \mathbb{R}$, $g : X \times \mathbb{R} \rightarrow \mathbb{R}^N$ and $h : X \times \mathbb{R} \rightarrow \mathbb{R}^M$ are differentiable ($K, N, M \in \mathbb{N}$). Let $x^*(p) \in \text{Int}X$ be a solution to this maximization problem. Denote the Lagrange multipliers associated with the inequality and equality constraints by $\lambda^* \in \mathbb{R}_+^N$ and $\mu^* \in \mathbb{R}^M$. Suppose also that $v(\cdot)$ is differentiable at $\bar{p}$. Then

$$(A.18) \quad \frac{\partial v(\bar{p})}{\partial p} = \frac{\partial f(x^*(\bar{p}), \bar{p})}{\partial p} - \lambda^* \cdot D_p g(x^*(\bar{p}), \bar{p}) - \mu^* \cdot D_p h(x^*(\bar{p}), \bar{p}).$$

PROOF. Since $x^*(p)$ is the solution to the maximization problem,

$$(A.19) \quad v(\bar{p}) = f(x^*(\bar{p}), \bar{p}).$$

By hypothesis, $v(\cdot)$ is differentiable at $\bar{p}$, so $\partial v(\bar{p}) / \partial p$ exists. Moreover, applying the Implicit Function Theorem to the necessary conditions for a maximum given in Theorem A.31, $x^*(\cdot)$ is also differentiable at $\bar{p}$. Therefore, from (A.19) we can write

$$(A.20) \quad \frac{\partial v(\bar{p})}{\partial p} = \frac{\partial f(x^*(\bar{p}), \bar{p})}{\partial p} + D_x f(x^*(\bar{p}), \bar{p}) \cdot D_p x^*(\bar{p}),$$

where, once again, $D_x f(x^*(\bar{p}), \bar{p}) \cdot D_p x^*(\bar{p})$ is the inner product and thus is a real number. Let $\tilde{g} : X \times \mathbb{R} \rightarrow \mathbb{R}^{N_1}$ denote the $N_1 \leq N$ active inequality constraints. Differentiating the active inequality constraints and the equality constraints with respect to p , we also have

$$\begin{aligned} -D_p \tilde{g}(x^*(\bar{p}), \bar{p}) &= D_x \tilde{g}(x^*(\bar{p}), \bar{p}) \cdot D_p x^*(\bar{p}) \\ -D_p h(x^*(\bar{p}), \bar{p}) &= D_x h(x^*(\bar{p}), \bar{p}) \cdot D_p x^*(\bar{p}). \end{aligned}$$

The equivalent of (A.17) for this problem (recall Theorem A.31) implies

$$D_x f(x^*(\bar{p}), \bar{p}) - \lambda^* \cdot D_x g(x^*(\bar{p}), \bar{p}) - \mu^* \cdot D_x h(x^*(\bar{p}), \bar{p}) = 0.$$

Combining this with the previous two equations and noting that the Lagrange multipliers for the inactive constraints are equal to zero,

$$D_x f(x^*(\bar{p}), \bar{p}) \cdot D_p x^*(\bar{p}) = -\lambda^* \cdot D_p g(x^*(\bar{p}), \bar{p}) - \mu^* \cdot D_x(x^*(\bar{p}), \bar{p}).$$

Substituting into (A.20) gives (A.18). $\square$

A special case of this result applies when the problem is one of unconstrained maximization and in that case we simply have

$$\frac{\partial v(\bar{p})}{\partial p} = \frac{\partial f(x^*(\bar{p}), \bar{p})}{\partial p}.$$

A.12. Exercises

EXERCISE A.1. *

- (1) Prove the *Minkowski's inequality* that for any $x = (x_1, x_2, \dots, x_K) \in \mathbb{R}^K$, $y = (y_1, y_2, \dots, y_K) \in \mathbb{R}^K$ with $K \in \mathbb{N}$ and any $p \in [1, \infty)$,

$$\left(\sum_{k=1}^K |x_k + y_k|^p \right)^{1/p} \leq \left(\sum_{k=1}^K |x_k|^p \right)^{1/p} + \left(\sum_{k=1}^K |y_k|^p \right)^{1/p}.$$

- (2) Formulate and prove the generalization of this inequality for $K = \infty$.

EXERCISE A.2. Using Minkowski's inequality show that the metric spaces in Example A.1 part 1 satisfy the triangle inequality.

EXERCISE A.3. Show that the *sup metric* $d_\infty(f, g) = \sup_{x \in X} |f(x) - g(x)|$ on $\mathbf{C}(X)$ in Example A.1 satisfies the triangle inequality.

EXERCISE A.4. Using the definition of equivalent metrics in Definition A.4, show that if d and d' are equivalent metrics on X , and a subset X' of X is open according to the collection of neighborhoods generated by metric d , then it is open according to the collection of neighborhoods generated by metric d' .

EXERCISE A.5. Prove that $X' \subset X$ is closed if and only if every convergent sequence $\{x_n\}_{n=1}^\infty$ in X' has a limit point $x \in X'$.

EXERCISE A.6. Prove Fact A.2.

EXERCISE A.7. Prove Fact A.3.

EXERCISE A.8. Prove Fact A.4.

EXERCISE A.9. Prove that the metric space $(\mathbf{C}(X), d_\infty)$ introduced in Example A.1 is complete.

EXERCISE A.10. Using an argument similar to that in the proof of Theorem A.3 show that if (X, d) is a metric space and $\phi : X \rightarrow Y$ a continuous mapping, then $\phi(X')$ is a connected subset of Y for every connected subset X' of X .

EXERCISE A.11. Prove that all metrics of the family d_p defined on the Euclidean space in Example A.1 are equivalent according to Definition A.4.

EXERCISE A.12. Prove Fact A.5.

EXERCISE A.13. Prove Fact A.9.

EXERCISE A.14. Show that every metric space will have the Hausdorff property.

EXERCISE A.15. Prove Lemma A.2.

EXERCISE A.16. (1) Show that if $X = \prod_{\alpha=1}^K X_{\alpha}$, i.e., if X is a finite-dimensional product, then the box and the product topologies are equivalent in the sense that they define the same open sets (recall Definition A.4 for equivalence of metrics, which applies to the equivalence of topologies as well).

(2) Show that if X is not finite-dimensional, then the box and the product topologies are not equivalent.

(3) Show that projection maps are always continuous in the box topology.

EXERCISE A.17. * Suppose that X_{α} is a metric space for each $\alpha \in A$. Show that the space $X = \prod_{\alpha \in A} X_{\alpha}$ endowed with the product topology satisfies the Hausdorff property.

EXERCISE A.18. Prove that the properties of upper and lower hemi-continuity in Definition A.31 imply the properties in Definition A.32 when X and Y are Euclidean.

EXERCISE A.19. (1) Show that $G(x, y) = \{(x, y) \in \mathbb{R}^2 : xy \leq 0\}$ is not a continuous correspondence.

(2) Show that if $G_1(x)$ and $G_2(x)$ are continuous, their nonempty intersection $G_1(x) \cap G_2(x)$ may fail to be continuous. [Hint: consider $G_1(x) = (-\infty, x]$ and $G_2(x) = \{a, b\}$ for some $a \neq b$].

EXERCISE A.20. Prove Fact A.18.

EXERCISE A.21. Prove Fact A.20.

EXERCISE A.22. Give an example of an upper hemi-continuous correspondence from $[0, 1]$ into $[0, 1]$ that is not convex-valued and does not have a fixed point.

EXERCISE A.23. Consider a N -person normal-form game. Player i 's strategy is denoted by $a_i \in A_i$ and his payoff function is given by the real-valued function $u_i(a_1, \dots, a_N)$.

(1) Using Theorems A.16, A.17, and A.19, prove that if each A_i is nonempty, compact and convex and each u_i is continuous in a_j for $j \neq i$ and continuous and quasi-concave in a_i , there exists a strategy profile $(a_1^*, \dots, a_N^*)$ that constitutes a pure strategy Nash equilibrium.

(2) Give counterexamples showing why each of the assumptions of (1) compactness of A_i , (2) convexity of A_i , (3) continuity of u_i , and (4) quasi concavity of u_i in own strategy cannot be dispensed with.

EXERCISE A.24. Prove Fact A.21.

EXERCISE A.25. Prove Theorem A.21.

EXERCISE A.26. Prove Theorem A.22. [Hint: use Theorem A.21].

EXERCISE A.27. Prove Corollary A.3.

EXERCISE A.28. Prove Fact A.24.

EXERCISE A.29. Show that the first four spaces given in Example A.16 are normed vector spaces, while the fifth one is not. [Hint: in each case, verify the triangle inequality and the linearity conditions].

EXERCISE A.30. Prove the claims in Example A.17.

EXERCISE A.31. Consider the subspace of ℓ_p , ℓ_p^+ , where all elements of the sequence are nonnegative. Suppose $1 \leq p < \infty$. Now consider $x \in \ell_p^+$ and the ε -neighborhood of x , $\mathcal{N}_\varepsilon(x)$. Show that for any $x \in \ell_p^+$ and any $\varepsilon > 0$, $\mathcal{N}_\varepsilon(x) \not\subset \ell_p^+$. [Hint: fix $\varepsilon > 0$ and $x = (x_1, x_2, \dots) \in \ell_p^+$. Since $x \in \ell_p$, for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n| < \varepsilon/2$. Then, define z such that $z_n = x_n$ for all $n \neq N$ and $z_N = x_N - \varepsilon/2$. Show that $z \in \mathcal{N}_\varepsilon(x)$ but $z \notin \ell_p^+$].

EXERCISE A.32. Show that $x = (1, 1, 1, \dots)$ is an interior point of ℓ_∞^+ . [Hint: consider $z_\varepsilon = (1 + \varepsilon, 1 + \varepsilon, \dots)$, and show that $z \in \mathcal{N}_\varepsilon(x) \subset \ell_\infty^+$].

EXERCISE A.33. For the mapping $g : X \rightarrow \mathbb{R}^N$ for some $X \subset \mathbb{R}^K$, construct the set $G = \{x : g(x) \leq 0\}$. Show that even when each component of g is not a convex function, the set G can be convex.

EXERCISE A.34. Consider the problem of maximizing x subject to the constraint that $x^2 \leq 0$. Show that there exists a unique solution to this program, but there exists no Lagrange multiplier contrary to the claim in Theorem A.30. Show that this is because the Slater condition is not satisfied.

EXERCISE A.35. Consider the constrained maximization problem $\max_{x_1, x_2} -x_1$ subject to $x_1^2 \leq x_2$ and $x_2 = 0$. Show that there exists a unique solution, which is $(x_1, x_2) = (0, 0)$. Show that there does not exist a Lagrange multiplier vector (λ, μ) at which $(0, 0)$ satisfies (A.17). Explain how this is related to the failure of the constraint qualification condition.

CHAPTER B

Review of Ordinary Differential Equations

In this chapter, I give a very brief overview of some basic results on differential equations and also include a few results on difference equations. I limit myself to results that are useful for the material covered in the body of the text. In particular, I provide the background for the major theorems on stability, Theorems 2.2, 2.3, 2.4, 2.5, 7.18, and 7.19, which were presented and then extensively used in the text. I will also provide some basic theorems on existence, uniqueness and continuity of solutions to differential equations. Most of the material here can be found in basic differential equation textbooks, such as Boyce and DiPrima (1977). Luenberger (1979) is an excellent reference, since it gives a symmetric treatment of differential and difference equations. The results on existence, uniqueness and continuity of solutions can be found in more advanced books, such as Walter (1991). Before presenting the results on differential equations, I also provide a brief overview of eigenvalues and eigenvectors, and some basic results on integrals. Throughout, I continue to assume basic familiarity with matrix algebra and calculus.

B.1. Review of Eigenvalues and Eigenvectors

Let $\mathbf{A}$ be a $n \times n$ (square) real matrix—meaning that all of its entries are real numbers. The square ($n \times n$) matrix $\mathbf{D}$ is *diagonal* if all of its non-diagonal elements are equal to zero, i.e.,

$$\mathbf{D} = \begin{pmatrix} d_1 & 0 & \cdot & 0 \\ 0 & d_2 & \cdot & \cdot \\ \cdot & \cdot & \cdot & 0 \\ 0 & \cdot & 0 & d_n \end{pmatrix}.$$

The $n \times n$ *identity matrix*, $\mathbf{I}$, is the diagonal matrix with 1's on the diagonal, that is,

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & \cdot & 0 \\ 0 & 1 & \cdot & \cdot \\ \cdot & \cdot & 1 & 0 \\ 0 & \cdot & 0 & 1 \end{pmatrix}.$$

Throughout, this chapter I denote matrices and vectors by boldface letters, so $\mathbf{0}$ is the vector or matrix of zeros, whereas 0 is simply the number zero.

Let the real number $\det \mathbf{A}$ denote the determinant of a square matrix $\mathbf{A}$. A matrix $\mathbf{A}$ is *nonsingular* or *invertible* if $\det \mathbf{A} \neq 0$ or alternatively if the only $n \times 1$ column vector $\mathbf{v}$ that is a solution to the equation

$$\mathbf{A}\mathbf{v} = \mathbf{0}$$

is the zero vector $\mathbf{v} = (0, \dots, 0)^T$. If $\mathbf{A}$ is invertible, then there exists $\mathbf{A}^{-1}$ such that

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}.$$

Conversely, if there exists a nonzero solution $\mathbf{v}$ or if $\det \mathbf{A} = 0$, then $\mathbf{A}$ is singular and does not have an inverse.

Let $a, b \in \mathbb{R}$ and define the imaginary number i such that $i^2 = -1$, so that $i = \pm\sqrt{-1}$. Throughout this appendix, with no loss of generality, I will take $i = \sqrt{-1}$. Then, $\chi = a + bi$ is a *complex number*. A complex number ξ is an *eigenvalue* of $\mathbf{A}$ if

$$\det(\mathbf{A} - \xi\mathbf{I}) = 0,$$

or in other words, if by subtracting ξ times the identity matrix from $\mathbf{A}$, we create a singular matrix. Clearly, if $\mathbf{A}$ is invertible, then none of its eigenvalues are equal to zero. Given an eigenvalue ξ of $\mathbf{A}$, the $n \times 1$ nonzero column vector $\mathbf{v}_\xi$ is an eigenvector of $\mathbf{A}$ if

$$(\mathbf{A} - \xi\mathbf{I})\mathbf{v}_\xi = \mathbf{0}.$$

Clearly, if $\mathbf{v}_\xi$ satisfies this equation, so does $\lambda\mathbf{v}_\xi$ for any $\lambda \in \mathbb{R}$. The linear space $\mathbf{V} = \{\mathbf{v} : (\mathbf{A} - \xi\mathbf{I})\mathbf{v} = \mathbf{0}\}$ is sometimes referred to as the *eigenspace* of $\mathbf{A}$. One of the major uses of eigenvalues and eigenvectors is in “diagonalizing” a non-diagonal square matrix $\mathbf{A}$. In particular, suppose that the $n \times n$ matrix $\mathbf{A}$ has n *distinct* real eigenvalues, then a standard result in matrix algebra implies that

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D},$$

where $\mathbf{D}$ is the diagonal matrix with the eigenvalues $\xi_1, \dots, \xi_n$ on the diagonal and $\mathbf{P} = (\mathbf{v}_{\xi_1}, \dots, \mathbf{v}_{\xi_n})$ is a matrix of the eigenvectors corresponding to the eigenvalues. This result will be used in the proof of Theorem B.5 below and is more generally useful in deriving explicit solutions to systems of linear differential and difference equations.

Note that the eigenvalues of the matrix $\mathbf{A}$ with real entries can be complex numbers (corresponding to complex roots to the $\det(\mathbf{A} - \xi\mathbf{I}) = 0$). In addition, the polynomial $\det(\mathbf{A} - \xi\mathbf{I}) = 0$ may have repeated roots, so that the $n \times n$ matrix $\mathbf{A}$ might have repeated (rather than distinct) eigenvalues. Both of these possibilities create a range of difficulties in diagonalizing matrix $\mathbf{A}$. These difficulties are discussed in most linear algebra, matrix algebra, and differential equations textbooks, and will not be discussed in detail here.

B.2. Some Basic Results on Integrals

Before proceeding to differential equations, it is useful to review some basic results on integrals. Throughout this section, I will focus on *Riemann integrals*. In particular, for real numbers $b > a$, let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Then, the Riemann integral of f between a and b , denoted by $\int_a^b f(x) dx$, is defined as follows. First create a partition of interval $[a, b]$, that is, divide $[a, b]$ into N subintervals of the form $[a, x_1]$, $[x_1, x_2]$, ..., $[x_{N-1}, b]$, with the convention that $a = x_0$ and $b = x_N$. Moreover, take N numbers, ξ_1 ,

$\xi_2, \dots, \xi_N$, respectively from one of each subinterval. Denote the vector of numbers $X_N = (x_1, x_2, \dots, x_{N-1}, \xi_1, \xi_2, \dots, \xi_N)$. Define the *Riemann sum* given X_N as

$$\mathbf{R}(X_N) = \sum_{j=0}^{N-1} f(\xi_j) (x_{j+1} - x_j).$$

Now consider the limit $\lim_{N \rightarrow \infty} \mathbf{R}(X_N)$ corresponding the value of the preceding expression as we take finer and finer partitions of $[a, b]$, that is, as we increase N . If this limit exists and is independent of the partition X_N , it defines the *Riemann integral*, denoted by

$$(B.1) \quad \int_a^b f(x) dx.$$

For example, when it exists the Riemann integral is equal to the Riemann sum resulting from the equal partition of the interval $[a, b]$, that is,

$$\int_a^b f(x) dx = \lim_{N \rightarrow \infty} \frac{b-a}{N} \sum_{j=0}^{N-1} f\left(a + j \frac{b-a}{N}\right).$$

The assumption that f is continuous is not necessary for the Riemann integral to be well-defined (for example, the Riemann integral can be defined for monotone discontinuous functions). But for many functions the Riemann integral is not well-defined. For this reason, it is often more convenient to work with more general integrals, such as the Lebesgue integral. Although I made some references to Lebesgue integrals in the text, here I focus exclusively on Riemann integrals to simplify the discussion. When they both exists, the Riemann and Lebesgue integrals are equivalent. When a function f has a well-defined Riemann integral over the interval $[a, b]$, it is said to be *Riemann integrable* over $[a, b]$.

The following four basic results are useful for our analysis. The proofs can be found in standard real analysis or calculus textbooks, and are not repeated here.

THEOREM B.1. (*Fundamental Theorem of Calculus I*) Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$. For any $x \in [a, b]$, define

$$F(x) = \int_a^x f(t) dt.$$

Then, $F : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$. If f is continuous at some $x_0 \in [a, b]$, then $F(x)$ is differentiable at x_0 with derivative

$$F'(x_0) = f(x_0).$$

THEOREM B.2. (*Fundamental Theorem of Calculus II*) Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$, then there exists a differentiable function $F : [a, b] \rightarrow \mathbb{R}$ on $[a, b]$ (or only with the right derivative at a and the left derivative at b) such that $F'(x) = f(x)$ for all $x \in [a, b]$ and for any such function

$$\int_a^b f(x) dx = F(b) - F(a).$$

THEOREM B.3. (*Integration by Parts*) Let $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be continuous functions and let $F : [a, b] \rightarrow \mathbb{R}$ and $G : [a, b] \rightarrow \mathbb{R}$ be differentiable functions such that $F'(x) = f(x)$ and $G'(x) = g(x)$ for all $x \in [a, b]$. Then, the (product) functions Fg and Gf are integrable, and

$$\int_a^b F(x) g(x) dx = F(b) G(b) - F(a) G(a) - \int_a^b G(x) f(x) dx.$$

THEOREM B.4. (*Leibniz's Rule*) Let $f(x, y)$ be continuous in x on $[a, b]$ and differentiable in y at y_0 and suppose that the functions $a(y)$ and $b(y)$ are differentiable at y_0 with derivatives denoted by a' and b' . Then

$$\left. \frac{d}{dy} \int_{a(y)}^{b(y)} f(x, y) dx \right|_{y=y_0} = \int_{a(y_0)}^{b(y_0)} \frac{\partial f(x, y_0)}{\partial y} dx + b'(y_0) f(b(y_0), y_0) - a'(y_0) f(a(y_0), y_0).$$

Riemann integrals as in (B.1), which specify lower and upper limits, are referred to as *definite integrals*. One can also define *indefinite integrals*, $\int f(x) dx$, which simply refer to the set of functions $F(x)$ with the property that $F'(x) = f(x)$ (this is a set, since if $F(x)$ satisfies this property, so does $F(x) + c$, where c is a constant). For this reason, the indefinite integral $\int f(x) dx$ is also sometimes referred to as an *anti-derivative*. Definite integrals can also be defined for the cases in which $a = -\infty$ and/or $b = \infty$, provided that the limit is finite.

B.3. Linear Differential Equations

Recall the motivation for considering differential equations in dynamic economic models discussed in Chapter 2. In particular, consider a function $x : T \rightarrow \mathbb{R}$, where $T \subset \mathbb{R}$. Suppose that given the real number Δt ,

$$x(t + \Delta t) - x(t) = G(x(t), t, \Delta t),$$

where $G(x(t), t, \Delta t)$ is a real-valued function. Now divide both sides of this equation by Δt and consider the limit as $\Delta t \rightarrow 0$. Suppose that $\lim_{\Delta t \rightarrow 0} G(x(t), t, \Delta t) / \Delta t$ exists and let

$$g(x(t), t) \equiv \lim_{\Delta t \rightarrow 0} \frac{G(x(t), t, \Delta t)}{\Delta t}.$$

Using this, we obtain the following simple *differential equation*

$$(B.2) \quad \frac{dx(t)}{dt} \equiv \dot{x}(t) = g(x(t), t).$$

This is an *explicit* first-order differential equation. The term explicit refers to the fact that $\dot{x}(t)$ is separated from the rest of the terms. This contrasts with *implicit* first-order differential equations of the form

$$H(\dot{x}(t), x(t), t) = 0.$$

For our purposes, it is sufficient to deal with explicit equations.

A differential equation is *autonomous*, if it can be written in the form

$$\dot{x}(t) = g(x(t)),$$

or simply as

$$\dot{x} = g(x),$$

meaning that time is not a separate argument. Alternatively, if it cannot be written this way, it is a *nonautonomous* equation. In addition to first-order differential equations, we can consider, second order or n th order equations, for example,

$$\frac{d^2 x(t)}{dt^2} = g\left(\frac{dx(t)}{dt}, x(t), t\right),$$

or

$$(B.3) \quad \frac{d^n x(t)}{dt^n} = g\left(\frac{d^{n-1} x(t)}{dt^{n-1}}, \dots, \frac{dx(t)}{dt}, x(t), t\right).$$

I will focus on first-order equations, since higher-order equations can always be transformed into a system of first-order equations (see Exercise B.3).

The most common form of differential equation is the so-called *initial value* problem. In this case, a differential equation as in (B.2) is specified together with an initial condition $x(0) = x_0$. We saw many examples of such initial value problems in the text. However, many important problems in economics are not initial value problems, since the boundary conditions are specified by *transversality conditions*, that is, by what the terminal value of the solution $x(t)$ should be at some time $T < \infty$ or at $T = \infty$.

Suppose that a first-order differential equation (B.2) is defined for all $t \in \mathcal{D}$, where $\mathcal{D}$ is an interval in $\mathbb{R}$ and an initial value $x(0) = x_0$ has been specified. A *solution* to this initial value problem is given by a function $x : \mathcal{D} \rightarrow \mathbb{R}$ that satisfies (B.2) for all $t \in \mathcal{D}$ with $x(0) = x_0$. Sometimes, a family of functions $\mathcal{X} = \{x : \mathcal{D} \rightarrow \mathbb{R} \text{ such that } x \text{ satisfies (B.2) for all } t \in \mathcal{D}\}$ is referred to as the *general solution*, while an element of $\mathcal{X}$ that satisfies the boundary condition is called a *particular solution*.¹

B.4. Solutions to Linear First-Order Differential Equations

Let us now first look at linear first-order differential equations. This is a good starting point both because such equations are commonly encountered in economics and they have simple solutions. A *linear first-order differential equation* takes a general form

$$(B.4) \quad \dot{x}(t) = a(t)x(t) + b(t).$$

In addition, if $b(t) = 0$, this is referred to as a *homogeneous equation* and if $a(t) = a$ and $b(t) = b$, then it is an equation with *constant coefficients*.

Let us start with the simplest case, which is a homogeneous linear equation with constant coefficients, i.e.,

$$(B.5) \quad \dot{x}(t) = ax(t).$$

¹This is a somewhat confusing terminology, since general and particular solutions are also used with different meanings in other contexts. Since in this book these other notions are not introduced, there will be no cause for confusion.

A solution to this equation is straightforward to obtain. One can simply guess the solution and then verify that the solution satisfies the differential equation (B.5). Or one can divide both sides by $\dot{x}(t)$, integrate with respect to t and recall that for $x(t) \neq 0$,

$$\int \frac{\dot{x}(t)}{x(t)} dt = \ln |x(t)| + c_0$$

and

$$\int a dt = at + c_1,$$

where c_0 and c_1 are constants of integration. Now taking exponents on both sides, the general solution to (B.5) is obtained as

$$(B.6) \quad x(t) = c \exp(at),$$

where c is a constant of integration combining c_0 and c_1 (in fact, $c = \pm \exp(c_1 - c_0)$). Differentiating (B.6) one can easily obtain (B.5) and verify that (B.6) is indeed a general solution to (B.5). If (B.5) is specified as an initial value problem, then we also have a boundary condition, which, without loss of any generality, can be specified at $t = 0$ as $x(0) = x_0$. This boundary condition pins down the unique value of the constant of integration. In particular, since $\exp(a \times 0) = 1$, $c = x_0$. Therefore, the particular solution with this initial value is

$$x(t) = x_0 \exp(at).$$

Next consider a slightly more general equation that is homogeneous, but not with constant coefficients, that is,

$$(B.7) \quad \dot{x}(t) = a(t)x(t),$$

defined over $t \geq 0$ with an initial condition $x(0) = x_0$. Once again, dividing both sides by $x(t)$, integrating and then finally taking exponents, we obtain

$$(B.8) \quad x(t) = c \exp\left(\int_0^t a(s) ds\right).$$

This follows since the integral of the right-hand side for a bounded function $a(t)$ is simply $\int_0^t a(s) ds + c_1$. Since $\lim_{t \rightarrow 0} \int_0^t a(s) ds = \int_0^0 a(s) ds = 0$, the constant of integration is again pinned down by the initial condition, that is, $c = x_0$. That (B.8) is a solution to (B.7) can be verified by differentiating (B.8) using the Fundamental Theorem of Calculus or Leibniz's Rule from the previous section (Theorems B.1 or B.4).

Next consider an autonomous but nonhomogeneous first-order linear differential equation,

$$(B.9) \quad \dot{x}(t) = ax(t) + b.$$

A similar analysis gives the general solution as

$$(B.10) \quad x(t) = -\frac{b}{a} + c \exp(at).$$

Derivation of (B.10): To derive this solution, we use the following simple change of variables. Let

$$y(t) = x(t) + b/a.$$

It is clear that $\dot{y}(t) = \dot{x}(t)$ (simply differentiate both sides with respect to t). Then, rewrite (B.9) in terms of $y(t)$, which gives

$$\dot{y}(t) = ay(t).$$

Now using the general solution to (B.5) derived above,

$$y(t) = c \exp(at),$$

where c is the appropriate constant of integration. Now transforming this back into $x(t)$, we obtain (B.10) as the general solution to (B.9). ■

Finally, to obtain the particular solution note that the constant of integration must be $c = x_0 + b/a$ in order to ensure that $x(0) = x_0$. Therefore, the particular solution that satisfies the boundary condition is obtained as

$$(B.11) \quad x(t) = -\frac{b}{a} + \left(x_0 + \frac{b}{a}\right) \exp(at).$$

This equation also enables us to have a simple discussion of stability. Recall that, as in the text, a *steady state* of (B.9) refers to a situation which $\dot{x}(t) = 0$ for all t . Clearly in this case,

$$x(t) = x^* \equiv -\frac{b}{a},$$

is the unique steady state. Inspection of (B.11) immediately shows that $x(t)$ will approach the steady-state value x^* as t increases if $a < 0$ and it will diverge away from it if $a > 0$. This is naturally what we would expect from Theorem 2.4, which states that the steady state is asymptotically stable if $a < 0$.

Finally, let us consider the most general case of the first-order linear differential equation, that given in (B.4). The general solution to (B.4) is

$$(B.12) \quad x(t) = \left[c + \int_0^t b(s) \left(\exp \int_0^s a(v) dv \right)^{-1} ds \right] \exp \left(\int_0^t a(s) ds \right).$$

Differentiation using Leibniz's Rule verifies that (B.12) provides the solution to (B.4) (see Exercise B.4). A similar analysis to that above allows us to derive the constant of integration from the initial value $x(0) = x_0$ as $c = x_0$. Notice, however, that in this case there may not exist a steady-state value of x^* for which $\dot{x}(t) = 0$ in (B.4), since $\dot{x}(t) = 0$ implies $x(t) = -b(t)/a(t)$, which is generally not a constant.

Derivation of (B.12): The derivation of (B.12) as the solution to (B.4) requires a somewhat different argument than those used for the special cases above. Rewrite (B.4) as $\dot{x}(t) - a(t)x(t) = b(t)$ and multiply both sides by *the integrating factor* $\exp\left(-\int_0^t a(s) ds\right)$ to obtain

$$\dot{x}(t) \exp\left(-\int_0^t a(s) ds\right) - a(t)x(t) \exp\left(-\int_0^t a(s) ds\right) = b(t) \exp\left(-\int_0^t a(s) ds\right).$$

It can be verified that the left-hand side is exactly the derivative of $x(t) \exp\left(-\int_0^t a(s) ds\right)$. Therefore, integrating both sides of this expression, we obtain

$$x(t) \exp\left(-\int_0^t a(s) ds\right) = \int_0^t b(s) \exp\left(-\int_0^s a(v) dv\right) ds + c,$$

where c is the constant of integration. Dividing both sides by $\exp\left(-\int_0^t a(s) ds\right)$, we obtain (B.12). ■

A byproduct of this brief discussion is that we have also established the existence of unique solutions to linear differential equations (since I provided explicit solutions). Thus linear differential equations (formulated as initial value problems) always have a unique solution. Moreover, the solution is unique. This is a special case of Theorem B.8 below.

B.5. Systems of Linear Differential Equations

The results on the existence of solutions and explicit characterization of solutions for linear first-order differential equations can be extended to systems of differential equations. The general result here is provided in Theorem B.6. However, before presenting this more general result, it is useful to consider the following simpler system of first-order differential equations with *constant coefficients*:

$$(B.13) \quad \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t),$$

where $\mathbf{x}(t) \in \mathbb{R}^n$, $n \in \mathbb{N}$ and $\mathbf{A}$ is a $n \times n$ matrix. The boundary condition again takes the form of an initial value $\mathbf{x}(0) = \mathbf{x}_0 \in \mathbb{R}^n$. This system of equation does not include a constant, so that the steady state is $\mathbf{x}^* = \mathbf{0}$. This is simply a normalization, since as we just saw, a nonhomogeneous differential equation with constant coefficients can be transformed into a homogeneous one by a simple change of variables. This system of differential equations always has a unique solution (this follows from Theorem B.6 for from Theorem B.10, see Exercise B.5). However, when $\mathbf{A}$ has distinct real eigenvalues, the solution to (B.13) takes a particularly simple form. This case is presented in the next result.

THEOREM B.5. (*Solution to Systems of Linear Differential Equations with Constant Coefficients*) Suppose that $\mathbf{A}$ has n distinct real eigenvalues $\xi_1, \dots, \xi_n$, then the unique solution to the system of linear differential equations (B.13) with the initial value $\mathbf{x}(0) = \mathbf{x}_0$ takes the form

$$\mathbf{x}(t) = \sum_{j=1}^n c_j \exp(\xi_j t) \mathbf{v}_{\xi_j},$$

where $\mathbf{v}_{\xi_1}, \dots, \mathbf{v}_{\xi_n}$ denote the eigenvectors corresponding to the eigenvalues $\xi_1, \dots, \xi_n$ and $c_1, \dots, c_n$ denote the constants of integration.

PROOF. The proof follows by diagonalizing the matrix $\mathbf{A}$. In particular, since $\mathbf{A}$ has n distinct real eigenvalues, recall that

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D},$$

where $\mathbf{D}$ is a diagonal matrix with the eigenvalues $\xi_1, \dots, \xi_n$ on the diagonal and $\mathbf{P} = (\mathbf{v}_{\xi_1}, \dots, \mathbf{v}_{\xi_n})$ is the matrix of the eigenvectors corresponding to the eigenvalues. Let $\mathbf{z}(t) \equiv \mathbf{P}^{-1}\mathbf{x}(t)$, which also implies

$$\begin{aligned} \dot{\mathbf{z}}(t) &= \mathbf{P}^{-1}\dot{\mathbf{x}}(t) \\ &= \mathbf{P}^{-1}\mathbf{A}\mathbf{x}(t) \\ &= \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{z}(t) \\ (B.14) \quad &= \mathbf{D}\mathbf{z}(t). \end{aligned}$$

Since $\mathbf{D}$ is a diagonal matrix, writing $\mathbf{z}(t) = (z_1(t), \dots, z_n(t))$, (B.14) implies that $z_1(t) = c_1 \exp(\xi_1 t), \dots, z_n(t) = c_n \exp(\xi_n t)$, where $c_1, \dots, c_n$ are the constants of integration. Now since $\mathbf{x}(t) = \mathbf{P}\mathbf{z}(t)$, the result follows by multiplying the matrix $(\mathbf{v}_{\xi_1}, \dots, \mathbf{v}_{\xi_n})$ with the vector of solutions, $\mathbf{z}(t)$. $\square$

When the matrix $\mathbf{A}$ has repeated or complex eigenvalues, explicit solutions can still be derived but are somewhat more complicated. Therefore, I will instead focus on the more general results in Theorem B.6 below. One important set of implications of Theorem B.5 are Theorems 2.4 and 7.18 in the text. In particular, Theorem B.5 implies that the steady-state value, here $\mathbf{x}^* = \mathbf{0}$, will be stable only when all eigenvalues are negative. If, instead, $m < n$ of the eigenvalues are negative, then there will exist a m -dimensional subspace, such that the solution will tend to the steady state only starting with an initial value on this subspace.

Now consider the most general form of a system of linear differential equations:

$$(B.15) \quad \dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t),$$

where $\mathbf{x}(t) \in \mathbb{R}^n$, $n \in \mathbb{N}$ and $\mathbf{A}(t)$ and $\mathbf{B}(t)$ are $n \times n$ matrices for each t . To simplify the discussion, let us assume that each element of $\mathbf{A}(t)$ and $\mathbf{B}(t)$ is continuous or monotone functions of time, so that they are integrable.

I now characterize the solution to (B.15). I proceed in two steps. First, I introduce the *state-transition matrix* $\Phi(t, s)$ corresponding to $\mathbf{A}(t)$ as the $n \times n$ matrix function that is differentiable in its first argument and is uniquely defined by

$$(B.16) \quad \frac{d}{dt}\Phi(t, s) = \mathbf{A}(t)\Phi(t, s) \text{ and } \Phi(t, t) = \mathbf{I} \text{ for all } t \text{ and } s.$$

The state-transition matrix is useful because it enables us to express the solutions to homogeneous systems and then derive the solutions to (B.15) from the solutions to the corresponding homogeneous systems. In particular, if $\hat{\mathbf{x}}(t)$ is a solution to the homogeneous system

$$(B.17) \quad \dot{\hat{\mathbf{x}}}(t) = \mathbf{A}(t)\hat{\mathbf{x}}(t),$$

then it is straightforward to verify that (see Exercise B.6):

$$(B.18) \quad \hat{\mathbf{x}}(t) = \Phi(t, s)\hat{\mathbf{x}}(s) \text{ for any } t \text{ and } s.$$

Let us next define the *fundamental set of solutions* to (B.17). The $n \times n$ matrix $\mathbf{X}(t)$ is a fundamental set of solutions to (B.17) if its columns consist of vector-valued functions

$\mathbf{x}^1(t), \mathbf{x}^2(t), \dots, \mathbf{x}^n(t)$ that are linearly independent from each other and are solutions to (B.17). In this case, clearly,

$$\dot{\mathbf{X}}(t) = \mathbf{A}(t) \mathbf{X}(t).$$

Then, it can be verified that (see Exercise B.7):

$$(B.19) \quad \Phi(t, s) = \mathbf{X}(t) \mathbf{X}(s)^{-1}.$$

We are now in a position to state the form of the unique solution to the general system of linear equations in (B.15).

THEOREM B.6. (*General Solutions to Systems of Linear Differential Equations*)
The solution to the system of differential equations in (B.15) with initial condition $\mathbf{x}(0) = \mathbf{x}_0$ is given by

$$(B.20) \quad \hat{\mathbf{x}}(t) = \Phi(t, 0) \mathbf{x}_0 + \int_0^t \Phi(t, s) \mathbf{B}(s) ds,$$

where $\Phi(t, s)$ is the state transition matrix corresponding to $\mathbf{A}(t)$.

PROOF. We only need to verify that $\hat{\mathbf{x}}(t)$ given in (B.20) is a solution to (B.15). Let us simplify notation of time derivatives by writing this as $\mathbf{x}(t)$. Differentiating (B.20) with respect to time and using Leibniz's Rule (Theorem B.4), we obtain

$$\frac{d}{dt} \mathbf{x}(t) = \frac{d}{dt} \Phi(t, 0) \mathbf{x}_0 + \int_0^t \frac{d}{dt} \Phi(t, s) \mathbf{B}(s) ds + \Phi(t, t) \mathbf{B}(t).$$

By the definition of the state transition matrix, (B.16), $\Phi(t, t) = \mathbf{I}$ and

$$\frac{d}{dt} \Phi(t, s) = \mathbf{A}(t) \Phi(t, s).$$

Therefore,

$$\begin{aligned} \dot{\mathbf{x}}(t) &\equiv \frac{d}{dt} \mathbf{x}(t) \\ &= \mathbf{A}(t) \Phi(t, 0) \mathbf{x}_0 + \mathbf{A}(t) \int_0^t \Phi(t, s) \mathbf{B}(s) ds + \mathbf{B}(t) \\ &= \mathbf{A}(t) \mathbf{x}(t) + \mathbf{B}(t), \end{aligned}$$

completing the verification that (B.20) satisfies (B.15) with initial condition $\mathbf{x}(0) = \mathbf{x}_0$. $\square$

B.6. Stability for Nonlinear Differential Equations

Systems of nonlinear differential equations can be analyzed in the neighborhood of the steady state by using Taylor's Theorem (Theorem A.24). In particular, consider the system of nonlinear autonomous differential equations

$$(B.21) \quad \dot{\mathbf{x}}(t) = \mathbf{G}(\mathbf{x}(t)),$$

where again $\mathbf{x}(t) \in \mathbb{R}^n$, $n \in \mathbb{N}$ and now $\mathbf{G} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a continuously differentiable mapping. Suppose that this system of differential equations has a steady state $\mathbf{x}^* \in \mathbb{R}^n$ and consider $\mathbf{x}(t)$ in the neighborhood of $\mathbf{x}^*$. Then, from Taylor's Theorem

$$\dot{\mathbf{x}}(t) = D\mathbf{G}(\mathbf{x}^*)(\mathbf{x}(t) - \mathbf{x}^*) + o(\|\mathbf{x}(t) - \mathbf{x}^*\|^2),$$

where recall that $o\left(\|\mathbf{x}(t) - \mathbf{x}^*\|^2\right) / \|\mathbf{x}(t) - \mathbf{x}^*\|^2 \rightarrow \mathbf{0}$ as $\|\mathbf{x}(t) - \mathbf{x}^*\| \rightarrow 0$ and thus as $\mathbf{x}(t) \rightarrow \mathbf{x}^*$. We say that $\mathbf{x}^*$ is a *hyperbolic* steady state if the matrix $D\mathbf{G}(\mathbf{x}^*)$ does not have zero eigenvalues (or complex eigenvalues with zero real parts). Then, as long as $\mathbf{x}^*$ is a hyperbolic steady state, the behavior of $\mathbf{x}(t)$ in the neighborhood of the steady state $\mathbf{x}^*$ can be approximated by the system of linear differential equations

$$\dot{\mathbf{x}}(t) = D\mathbf{G}(\mathbf{x}^*)(\mathbf{x}(t) - \mathbf{x}^*).$$

This is the basis of Theorems 2.5 and 7.19 in the text. In fact, the following theorem, which is a slightly stronger version of those results, can be proved by using linearization arguments.

THEOREM B.7. (*Grobman-Hartman Theorem*) *Let $\mathbf{x}^*$ be a steady state of (B.21) and suppose that $\mathbf{G} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a continuously differentiable mapping. If $\mathbf{x}^*$ is hyperbolic, then there exists an open set of trajectories U of (B.21) around $\mathbf{x}^*$ and an open set of trajectories V of the linear system $\dot{\mathbf{x}}(t) = D\mathbf{G}(\mathbf{x}^*)(\mathbf{x}(t) - \mathbf{x}^*)$ around $\mathbf{x}^*$ such that there exists a one-to-one continuous function $h : U \rightarrow V$ that preserves the direction of trajectories in U and V .*

PROOF. See Walter (1991, Chapter 7.29). □

B.7. Separable and Exact Differential Equations

We cannot obtain explicit solutions to nonlinear differential equations in general (though existence of solutions can be guaranteed under some mild conditions as shown in the next section). Nevertheless, two important special classes of differential equations, *separable* and *exact differential equations*, often enable us to derive explicit solutions. I start with separable differential equations. A differential equation

$$(B.22) \quad \dot{x}(t) = g(x(t), t)$$

is *separable* if g can be written as

$$g(x, t) \equiv f(x) h(t).$$

In that case, the differential equation (B.22) can be expressed as

$$\begin{aligned} \frac{\dot{x}(t)}{f(x(t))} &= h(t). \\ \frac{dx(t)}{f(x(t))} &= h(t) dt. \end{aligned}$$

Integrating both sides, we obtain

$$\int \frac{dx}{f(x)} = \int h(t) dt.$$

This equation typically allows us to obtain an explicit solution. The following example illustrates a particular application.

EXAMPLE B.1. The differential equation

$$\dot{x}(t) = \frac{4t^3 + 3t^2 + 2t + 1}{2x(t)}$$

with initial value $x(0) = 1$ at first looks difficult to solve. However, once we note that it is separable, we can write it as

$$2x \cdot dx = (4t^3 + 3t^2 + 2t + 1) \cdot dt,$$

and integrate

$$2x \cdot dx = (4t^3 + 3t^2 + 2t + 1) \cdot dt,$$

to obtain

$$x^2(t) = t^4 + t^3 + t^2 + t + c,$$

where c is a combination of the two constants of integration. To satisfy the initial value, we need $c = 1$. Therefore, the solution to this initial value problem is given by

$$x(t) = \sqrt{t^4 + t^3 + t^2 + t + 1},$$

where the negative root to the quadratic is eliminated because it does not satisfy the initial value $x(0) = 1$. Another example, which is more relevant for economic applications, is given in Exercise B.9.

Next, consider a differential equation of the form (B.22) again, and suppose that the function g can be written as

$$g(x(t), t) \equiv \frac{G_1(x(t), t)}{G_2(x(t), t)},$$

where

$$G_1(x(t), t) = \frac{\partial F(x(t), t)}{\partial t} \text{ and } G_2(x(t), t) = -\frac{\partial F(x(t), t)}{\partial x},$$

then (B.22) defines an *exact differential equation*. In particular, in this case, we can write

$$\begin{aligned} \dot{x}(t) &= \frac{G_1(x(t), t)}{G_2(x(t), t)} \\ &= -\frac{\partial F(x(t), t) / \partial t}{\partial F(x(t), t) / \partial x}, \end{aligned}$$

or

$$\dot{x}(t) \frac{\partial F(x(t), t)}{\partial x} + \frac{\partial F(x(t), t)}{\partial t} = 0.$$

Let $\hat{x}(t)$ be a solution to this differential equation. Then, we equivalently have that

$$(B.23) \quad \frac{d}{dt} F(\hat{x}(t), t) = 0,$$

where d/dt denotes the total derivative of the function F . (B.23) then implies that

$$(B.24) \quad F(\hat{x}(t), t) = c,$$

where c is the constant of integration. Equation (B.24) implicitly defines the solution $\hat{x}(t)$.

Exact differential equations are straightforward to solve once they have been identified. The following provides a simple example.

EXAMPLE B.2. Consider the differential equation

$$\dot{x}(t) = -\frac{2x(t) \ln x(t)}{t},$$

with initial value $x(1) = \exp(1)$. While this differential equation looks difficult to solve at first, once we recognize that it can be written as

$$\begin{aligned} \dot{x}(t) &= -\frac{2t \ln x(t)}{t^2/x(t)} \\ &= -\frac{\partial(t^2 \ln(x(t))) / \partial t}{\partial(t^2 \ln(x(t))) / \partial x}, \end{aligned}$$

it can be seen to be an exact differential equation. Therefore, its solution $\hat{x}(t)$ is given by

$$t^2 \ln(\hat{x}(t)) = c,$$

which implies

$$\hat{x}(t) = \exp(ct^{-2}),$$

as the general solution, and the initial condition pins down the constant of integration as $c = 1$.

B.8. Existence and Uniqueness of Solutions

Initial value problems generally enable us to establish the existence and uniqueness of solutions under relatively weak conditions. In fact, there are many related existence theorems. I will state the most basic existence theorem here, which extends the original theorem by Picard. Consider a first-order differential equation

$$(B.25) \quad \dot{x}(t) = g(x(t), t)$$

defined on some interval $\mathcal{D} \subset \mathbb{R}$, i.e., defined for all $t \in \mathcal{D}$. Throughout, I assume that 0 is in the interior of $\mathcal{D}$. Let us introduce the following Lipschitz condition:

DEFINITION B.1. *The first-order differential equation (B.25) satisfies the **Lipschitz condition** on the strip $S = \mathcal{X} \times \mathcal{D}$ if there exists a real number $L < \infty$ such that*

$$|g(x, t) - g(x', t)| \leq L |x - x'|$$

for all $x, x' \in \mathcal{X}$ and for all $t \in \mathcal{D}$.

It can be verified that if $g : S \rightarrow \mathbb{R}$ satisfies the Lipschitz condition, then it must be continuous (see Exercise B.8).

THEOREM B.8. (**Picard's Theorem I**) *Suppose that $g : \mathcal{X} \times \mathcal{D} \rightarrow \mathbb{R}$ is continuous in both of its arguments and satisfies the Lipschitz condition in Definition B.1. Then, there exists $\delta > 0$ such that the initial value problem defined by (B.25) with $x(0) = x_0 \in \mathcal{X}$ has a unique solution $x(t)$ over the interval $[-\delta, \delta] \subset \mathcal{D}$.*

This theorem guarantees only the existence of unique solution in the neighborhood of the initial value x_0 . A stronger version of this theorem holds when $\mathcal{D}$ is compact.

THEOREM B.9. (*Existence and Uniqueness on Compact Sets I*) Suppose that g is continuous in both of its arguments and satisfies the Lipschitz condition in Definition B.1, and that $\mathcal{D}$ is compact. Then, the initial value problem defined by (B.25) with $x(0) = x_0$ has a unique solution $x(t)$ over the entire interval $\mathcal{D}$.

There are various different proofs of these theorems. Example 6.3 and Exercise 6.4 in Chapter 6 provide proofs using the Contraction Mapping Theorem, Theorem 6.7.

This theorem can be easily extended to systems of first-order differential equations. Suppose that $\mathbf{x}(t) \in \mathcal{X} \subset \mathbb{R}^n$, where $n \in \mathbb{N}$. Consider the following system of first-order differential equations:

$$(B.26) \quad \dot{\mathbf{x}}(t) = \mathbf{G}(\mathbf{x}(t), t),$$

where

$$\mathbf{G} : \mathcal{X} \times \mathcal{D} \rightarrow \mathcal{X}$$

and $t \in \mathcal{D} \subset \mathbb{R}$.

DEFINITION B.2. The system of first-order differential equation (B.25) satisfies the **Lipschitz condition** over the strip $S = \mathcal{X} \times \mathcal{D}$ if there exists a real number $L < \infty$ such that

$$\|\mathbf{G}(\mathbf{x}, t) - \mathbf{G}(\mathbf{x}', t)\| \leq L \|\mathbf{x} - \mathbf{x}'\|$$

for all $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$ and for all $t \in \mathcal{D}$.

THEOREM B.10. (*Picard's Theorem II*) Suppose that $\mathbf{G}$ is continuous in all of its arguments and satisfies the Lipschitz condition in Definition B.2. Then, there exists $\delta > 0$ such that the initial value problem defined by the system of differential equations in (B.26) with $\mathbf{x}(0) = \mathbf{x}_0$ has a unique solution $\mathbf{x}(t)$ over the interval $[-\delta, \delta] \subset \mathcal{D}$.

THEOREM B.11. (*Existence and Uniqueness on Compact Sets II*) Suppose that $\mathbf{G}$ is continuous in all of its arguments and satisfies the Lipschitz condition in Definition B.2 and that $\mathcal{D}$ is compact. Then, the initial value problem defined by the system of differential equations in (B.26) with $\mathbf{x}(0) = \mathbf{x}_0$ has a unique solution $\mathbf{x}(t)$ over the entire interval $\mathcal{D}$.

The proof of these theorems can be found in Walter (1991, Chapter 3.10).

THEOREM B.12. (*Peano's Theorem*) Suppose that $g : \mathcal{X} \times \mathcal{D} \rightarrow \mathbb{R}$ is continuous in both of its arguments. Then, for each $(x', t') \in \mathcal{X} \times \mathcal{D}$ there exists at least one solution to the differential equation (B.25) that goes through (x', t') .

PROOF. See Walter (1991, Chapter 3.10). □

This theorem can also be extended to systems of differential equations.

B.9. Continuity and Differentiability of Solutions

It is often of interest to know that when some parameter or an initial condition of a differential equation is changed by a small amount, the solution will also change by a small amount. The following theorem provides conditions for this to be true

THEOREM B.13. (*Continuity of Solutions to Differential Equations*) Suppose that $g : \mathcal{X} \times \mathcal{D} \rightarrow \mathbb{R}$ is continuous in both of its arguments and that $\mathcal{D}$ is compact. Let $x(t)$ be a solution to (B.26) with initial condition $x(0) = x_0$. For every $\varepsilon > 0$, $x_0 \in \mathcal{X}$, and continuous function $\tilde{g} : \mathcal{X} \times \mathcal{D} \rightarrow \mathbb{R}$, there exists $\delta > 0$ such that if

$$|\tilde{g}(x, t) - g(x, t)| < \delta \text{ and } |\tilde{x}_0 - x_0| < \delta \text{ for all } (x, t) \in \mathcal{X} \times \mathcal{D},$$

then every solution $\tilde{x}(t)$ to the perturbed initial value problem

$$\dot{x}(t) = \tilde{g}(x(t), t) \text{ with } x(0) = \tilde{x}_0$$

satisfies

$$|\tilde{x}(t) - x(t)| < \varepsilon \text{ for all } t \in \mathcal{D}.$$

PROOF. See Walter (1991, Chapter 3.12). □

This theorem can also be extended to systems of differential equations. Finally, under slightly stronger assumptions, the solution depends on initial values and parameters smoothly.

THEOREM B.14. (*Differentiability of Solutions to Differential Equations*) Suppose that $g : \mathcal{X} \times \mathcal{D} \rightarrow \mathbb{R}$ is differentiable in both of its arguments and that $\mathcal{D}$ is compact. Let $x(t)$ be a solution to (B.26) with initial condition $x(0) = x_0$. For every $\varepsilon > 0$, $x_0 \in \mathcal{X}$, there exists $\delta > 0$ such that if

$$|x'_0 - x_0| < \delta \text{ for all } (x, t) \in \mathcal{X} \times \mathcal{D},$$

then every solution $x'(t)$ to the perturbed initial value problem

$$\dot{x}(t) = g(x(t), t) \text{ with } x(0) = x'_0$$

satisfies

$$|x'(t) - \dot{x}(t)| < \varepsilon \text{ for all } t \in \mathcal{D}.$$

PROOF. See Walter (1991, Chapter 3.13). □

B.10. Difference Equations

Similar to first-order differential equations, a first-order difference equation is defined as

$$x(t+1) = g(x(t), t),$$

where $g : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$. Higher-order difference equations and systems of difference equations are defined similarly.

Solutions to difference equations have many features that are common with the solutions to differential equations. For example, the simple first-order difference equation

$$x(t+1) = ax(t) + b,$$

has a solution similar to the first-order linear differential equation with constant coefficients. In particular, if we specify the initial condition $x(0) = x_0$, then successive substitutions yield

$$\begin{aligned} x(1) &= ax_0 + b \\ x(2) &= a^2x_0 + ab + b, \end{aligned}$$

and so on. By induction, the general solution to this equation is

$$x(t) = \begin{cases} x_0 + bt & \text{if } a = 1 \\ a^t \left(x_0 - \frac{b}{1-a} \right) + \frac{b}{1-a} & \text{otherwise} \end{cases}.$$

The reader will recognize $x^* \equiv b/(1-a)$ as the steady-state value (when $a \neq 1$), and the solution makes it clear that if $|a| < 1$, then the first-term will tend to zero and $x(t) \rightarrow x^*$ (as $t \rightarrow \infty$), which is the essence of the stability results presented in the text. In contrast, when $|a| > 1$, the solution will diverge away from x^* .

Next consider the system of first-order linear difference equations

$$(B.27) \quad \mathbf{x}(t+1) = \mathbf{A}\mathbf{x}(t),$$

where $\mathbf{x}(t) \in \mathbb{R}^n$, $n \in \mathbb{N}$ and $\mathbf{A}$ is a $n \times n$ real matrix. When $\mathbf{A}$ has n distinct real eigenvalues, the solution to the system of equations is very similar to that given in Theorem B.5 above. In particular:

THEOREM B.15. (*Solution to Systems of Linear Difference Equations with Constant Coefficients*) Suppose that $\mathbf{A}$ has n distinct real eigenvalues $\xi_1, \dots, \xi_n$, then the unique solution to the system of linear difference equations (B.27) with the initial value $\mathbf{x}(0) = \mathbf{x}_0$ takes the form

$$\mathbf{x}(t) = \sum_{j=1}^n c_j \xi_j^t \mathbf{v}_{\xi_j},$$

where $\mathbf{v}_{\xi_1}, \dots, \mathbf{v}_{\xi_n}$ denote eigenvectors corresponding to the eigenvalues $\xi_1, \dots, \xi_n$ and $c_1, \dots, c_n$ denote constants determined by the initial conditions.

PROOF. The proof again follows by diagonalizing the matrix $\mathbf{A}$. Recall that since $\mathbf{A}$ has n distinct real eigenvalues, we have $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$, where $\mathbf{D}$ is a diagonal matrix with the eigenvalues $\xi_1, \dots, \xi_n$ on the diagonal and $\mathbf{P} = (\mathbf{v}_{\xi_1}, \dots, \mathbf{v}_{\xi_n})$ is the matrix of the eigenvectors corresponding to the eigenvalues. Let $\mathbf{z}(t) \equiv \mathbf{P}^{-1}\mathbf{x}(t)$ and note that

$$\begin{aligned} \mathbf{z}(t+1) &= \mathbf{P}^{-1}\mathbf{x}(t+1) \\ &= \mathbf{P}^{-1}\mathbf{A}\mathbf{x}(t) \\ &= \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{z}(t) \\ (B.28) \quad &= \mathbf{D}\mathbf{z}(t). \end{aligned}$$

Since $\mathbf{D}$ is a diagonal matrix, writing $\mathbf{z}(t) = (z_1(t), \dots, z_n(t))$, (B.28) implies that $z_1(t) = c_1 \xi_1^t, \dots, z_n(t) = c_n \xi_n^t$. Now since $\mathbf{x}(t) = \mathbf{P}\mathbf{z}(t)$, the result follows by multiplying the matrix $(\mathbf{v}_{\xi_1}, \dots, \mathbf{v}_{\xi_n})$ with the vector of solutions, $\mathbf{z}(t)$. $\square$

Solutions in the more general case where the system of difference equations do not have constant coefficients or eigenvalues may be complex or repeated are given by the analog of Theorem B.6. The matrix of fundamental solutions, $\mathbf{X}(t)$, is defined similarly to the matrix of fundamental solutions to differential equations. In addition, the state transition matrix again satisfies

$$\Phi(t, s) = \mathbf{X}(t) \mathbf{X}(s)^{-1}.$$

Moreover, as before $\Phi(t, t) = \mathbf{I}$. Now consider the general system of first-order difference equations given by

$$(B.29) \quad \mathbf{x}(t+1) = \mathbf{A}(t) \mathbf{x}(t) + \mathbf{B}(t).$$

The solution to this set of difference equations is characterized by the next theorem.

THEOREM B.16. (*General Solutions to Systems of Linear Differential Equations*) *The solution to the system of difference equations in (B.29) with initial condition $\mathbf{x}(0) = \mathbf{x}_0$ is given by*

$$\mathbf{x}(t) = \Phi(t, 0) \mathbf{x}_0 + \sum_{s=0}^{t-1} \Phi(t, s+1) \mathbf{B}(s).$$

PROOF. See Exercise B.10. $\square$

Linearizing systems of nonlinear difference equations then leads to an analog of Theorem B.7. Finally, existence and uniqueness of solutions is somewhat more straightforward for difference equations.

THEOREM B.17. (*Existence and Uniqueness of Solutions to Difference Equations*) *Consider the system of first-order nonlinear difference equations*

$$(B.30) \quad \mathbf{x}(t+1) = \mathbf{G}(\mathbf{x}(t)),$$

where $\mathbf{x}(t) \in \mathbb{R}^n$, $n \in \mathbb{N}$, and $\mathbf{G} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an arbitrary mapping. Suppose that the initial condition is specified as $\mathbf{x}(0) = \mathbf{x}_0$. Then, (B.30) has a unique solution for all $t \in \mathbb{N}$.

PROOF. Given $\mathbf{x}_0$, $\mathbf{x}(1)$ is uniquely defined as $\mathbf{G}(\mathbf{x}_0)$. Proceeding iteratively, we determine a unique $\mathbf{x}(t)$ that satisfies (B.30) for any $t \in \mathbb{N}$. $\square$

Using a similar method to that used for turning higher-order differential equations into a system of first-order differential equations (see Exercise B.3), this theorem also guarantees existence and uniqueness of solutions to higher-order difference equations when the appropriate initial values are specified (see Exercise B.11).

B.11. Exercises

EXERCISE B.1. Use integration by parts as in Theorem B.3 to evaluate $\int_a^b \ln x dx$.

EXERCISE B.2. Consider a household with preferences given by $U(0) = \int_0^\infty \exp(-\rho t) \log C(t) dt$. Suppose that $C(0) = C_0$ and that $C(t)$ grows at a constant proportional rate g (i.e., $C(t) = \exp(gt) C(0)$). Derive the expression for $U(0)$.

EXERCISE B.3. Show that a n th order differential equation as in (B.3) can be written as a system of n first-order equations. [Hint: let $z_j(t) = d^j x(t)/dt^j$ for $j = 1, \dots, n$].

EXERCISE B.4. Show that (B.12) is the general solution to the first-order differential equation (B.4).

EXERCISE B.5. Verify that the system of linear differential equations in (B.13) satisfies the conditions of Theorem B.10.

EXERCISE B.6. Verify (B.18).

EXERCISE B.7. Prove (B.19).

EXERCISE B.8. Show that if $g : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ satisfies the Lipschitz condition in Definition B.1, then $g(x, t)$ is continuous in x .

EXERCISE B.9. This exercise asks you to use the techniques to solve separable differential equations to characterize the family of utility functions with a constant coefficient of relative risk aversion. In particular, recall that the (Arrow-Pratt) measure of relative risk aversion of a twice differentiable utility function u is given by

$$\mathcal{R}_u(c) = -\frac{u''(c)c}{u'(c)}.$$

Suppose that $\mathcal{R}_u(c) = r > 0$ and let $v(c) = u'(c)$, then we obtain

$$\frac{v'(c)}{v(c)} = -\frac{r}{c}.$$

Using this equation, characterize the family of utility functions that have a constant coefficient of relative risk aversion.

EXERCISE B.10. Prove Theorem B.16.

EXERCISE B.11. Consider the n th order difference equation

$$x(t+n) = H(x(t+n-1), \dots, x(t), t),$$

where $H : \mathbb{R}^n \rightarrow \mathbb{R}$. Prove that if the initial values $x(0), x(1), \dots, x(n-1)$ are specified, this equation has a unique solution for any t .

CHAPTER C

Brief Review of Dynamic Games

This chapter provides a very brief overview of some basic definitions, results and notation for infinite-horizon dynamic games. The reader is already assumed to be familiar with basic game theory, and the notions of Nash Equilibrium and Subgame Perfect (Natch) Equilibrium in finite games. A review of these notions as well as much of the material covered here can be found in standard graduate game theory textbooks such as Fudenberg and Tirole (1994), Myerson (1991), Osborne and Rubinstein (1994) as well as Part 2 of MasColell, Whinston and Green (1995). My focus throughout is on games of complete information (or the so-called games of *perfect monitoring*). These types of games were used in Section 14.4 in Chapter 14, as well as in Chapters 22 and 23. The reader is referred to Fudenberg and Tirole (1994) for further details.

C.1. Basic Definitions

I consider the following class of *dynamic infinite-horizon games*. As the name suggests, these games are not finite and they are also somewhat more general than repeated games, since the stage game played at each date is a function of actions taken in the past.

More formally, there is a set of players denoted by $\mathcal{N}$. This set will be either finite, or when it is infinite (especially uncountable), there will be more structure to make the game tractable and thus variants of the theorems presented below will still be applicable. In particular, in many of the applications, especially in those considered in Chapters 22 and 23, there will be a continuum of players, but these will be in distinct finite groups and the game can be viewed as one between those distinct groups. With this motivation, in this Appendix I focus on the case in which $\mathcal{N}$ is finite, consisting of N players. Each player $i \in \mathcal{N}$ has a strategy set $A_i(k) \subset \mathbb{R}^{n_i}$ (with $n_i \in \mathbb{N}$) at every date and in addition, $k \in K \subset \mathbb{R}^n$ is the state vector (with $n \in \mathbb{N}$), with value at time t denoted by $k(t)$. A generic element of $A_i(k)$ at time t is denoted by $a_i(t)$, and $a(t) = (a_1(t), \dots, a_N(t))$ denotes the *vector of actions* (or the “action profile”) at time t , i.e.,

$$a(t) \in A(k(t)) = \prod_{i=1}^N A_i(k(t)).$$

I use the standard notation $a_{-i}(t) = (a_1(t), \dots, a_{i-1}(t), a_{i+1}(t), \dots, a_N(t))$ to denote the vector of actions without i 's action, thus, with a slight abuse of notation, we can also write $a_i(t) = (a_i(t), a_{-i}(t))$. Notice that, consistent with the types of models analyzed in the text, the

action set of each player $A_i(k)$ is only a function of the state variable k and not of calendar time.

Each player has an instantaneous utility function $u_i(k(t), a(t))$ where

$$u_i : K \times A \rightarrow \mathbb{R}$$

is assumed to be continuous and bounded. This notation emphasizes that each player's payoff depends on the entire action profile in that period (and not on past actions) and also on a common vector of state variables, denoted by $k(t)$. Past actions will only have an effect on current payoffs through this vector of state variables.

As usual, each player's objective at time t is to maximize his discounted payoff

$$(C.1) \quad U_i[t] = \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s u_i(k(t+s), a(t+s)),$$

where $\beta \in (0, 1)$ is the discount factor and $\mathbb{E}_t$ is the expectations operator conditional on information available at time t . The games I focus on here contain potential uncertainty about the evolution of the state variable in the future and also strategic uncertainty resulting because of mixed strategies. However, they will not feature *asymmetric information*, since I did not use incomplete information or asymmetric information dynamic games in this book. Consequently, the expectations operator $\mathbb{E}_t$ is not indexed by i .

The law of motion of the state vector $k(t)$ is given by the following Markovian transition function

$$(C.2) \quad q(k(t+1) | k(t), a(t)),$$

which denotes the probability density that next period's state vector is equal to $k(t+1)$ when the time t action profile of all the agents is $a(t) \in A(k(t))$ and the state vector is $k(t) \in K$. I refer to this transition function Markovian, since it only depends on the current profile of actions and the current state. Naturally, the probability of all possible states tomorrow integrate to 1:

$$\int_{-\infty}^{\infty} q(k | k(t), a(t)) dk = 1 \text{ for all } k(t) \in K \text{ and } a(t) \in A(k(t)).$$

Next, we need to specify the information structure of the players. We focus on games with perfect observability or perfect monitoring, so that individuals observe realizations of all past actions. Then, the public history at time t , observed by all agents up to time t , is

$$h^t = (a(0), k(0), \dots, a(t), k(t)).$$

With mixed strategies, the history naturally only includes the realizations of mixed strategies not the actual strategy. Let the set of all potential histories at time t be denoted by H^t . It should be clear that any element $h^t \in H^t$ for any t corresponds to a subgame of this game.¹

¹Sometimes, it may be useful to distinguish calendar time from the nodes within a stage game. In this case, one might want to use the notation h^t to denote the history up to the beginning of time t and then some other variable, say, $j^t \in J^t$ to summarize actions within the stage game at time t . In that case, the proper history at time t would be given by an element of the set $H^t \times J^t$. For our purposes here, this distinction is not necessary.

Let a (pure) strategy for player i at time t be

$$\sigma_i(t) : H^{t-1} \times K \rightarrow A_i,$$

that is, a mapping that determines what to play given the entire past history h^{t-1} and the current-value of the state variable $k(t) \in K$. This is the natural specification of a strategy for time t given that h^{t-1} and $k(t)$ entirely determine which subgame we are in.

A mixed strategy for player i at time t is

$$\sigma_i(t) : H^{t-1} \times K \rightarrow \Delta(A_i),$$

where $\Delta(A_i)$ is the set of all probability distributions over A_i . We are using the same symbol, σ , for pure and mixed strategies to economize on notation. Let $\sigma = (\sigma_1, \dots, \sigma_N)$ be the strategy profile in the infinite game, and let $\sigma_i(t) = (\sigma_1(t), \dots, \sigma_N(t))$ be the *continuation strategy profile* after time t induced by the strategy profile of the infinite game, σ . Let S_i be the set of all feasible strategies for player i in the infinite game, and $S = \prod_{i=1}^N S_i$ be the set of all feasible strategy profiles. Let me also use the notation $S_i(t)$ for the set of continuation strategies for player i starting at time t . Naturally, $S(t) = \prod_{i=1}^N S_i(t)$ and $S_{-i}(t) = \prod_{j \neq i} S_j(t)$ are defined in the usual manner.

As is standard, define the best response correspondence as

$$BR(\sigma_{-i}(t) \mid h^{t-1}, k(t)) = \{\sigma_i(t) \in S_i(t) : \sigma_i(t) \text{ maximizes (C.1) given } \sigma_{-i}(t) \in S_{-i}(t)\}$$

DEFINITION C.1. A **Subgame Perfect Equilibrium (SPE)** is a strategy profile $\sigma^* = (\sigma_1^*, \dots, \sigma_N^*) \in S$ such that $\sigma_i^*(t) \in BR(\sigma_{-i}^*(t) \mid h^{t-1}, k(t))$ for all $(h^{t-1}, k(t)) \in H^{t-1} \times K$, for all $i \in \mathcal{N}$, and for $t = 0, 1, \dots$

Therefore, a SPE requires strategies to be best responses to each other given all possible histories, which is a minimal requirement. What is “strong” (or “weak,” depending on the perspective) about the SPE concept is that strategies are mappings from the entire history. Consequently, in many infinitely repeated games, there will be numerous SPEs. This type of multiplicity sometimes makes it attractive to focus on subsets of equilibria. One possibility would be to look for “stationary” SPEs, motivated by the fact that the underlying game itself is stationary, i.e., payoffs do not depend on calendar time. Another possibility would be to look at the “best SPEs,” i.e., those that are on the Pareto frontier, and maximize the utility of one player subject to the utility of the remaining players not being below certain levels.

Perhaps the most popular alternative concept often used in dynamic games is that of Markov Perfect Equilibrium (MPE). The MPE differs from the SPE in that it only conditions on the *payoff-relevant* “state” of the game. The motivation comes from dynamic programming where, as we have seen, an optimal plan is a mapping from the state vector to the control vector. MPE can be thought of as an extension of this reasoning to game-theoretic situations. The advantage of the MPE relative to the SPE is that many infinite games will have many fewer MPEs than SPEs in general. The disadvantage, naturally, is that some economically interesting SPEs will be ignored when we focus on MPEs.

We could define payoff-relevant history at time t as the smallest (coarsest) partition $\mathcal{P}^t$ of H^t such that any two distinct elements of $\mathcal{P}^t$ necessarily lead to different payoffs or strategy sets for at least one of the players, holding the action profile of all other players constant.

In this case, it is clear that given the Markovian transition function above, the payoff-relevant state is simply $k(t) \in K$. Then, we define a pure Markovian strategy as

$$\hat{\sigma}_i : K \rightarrow A_i,$$

and a mixed Markovian strategy as

$$\hat{\sigma}_i : K \rightarrow \Delta(A_i).$$

Define the set of Markovian strategies for player i by $\hat{S}_i$ and naturally, $\hat{S} = \prod_{i=1}^N \hat{S}_i$.

Notice that I have dropped the dependence on t , since time is not part of the payoff-relevant state. This is a feature of the infinite-horizon nature of the game. With finite horizons, time would necessarily be part of the payoff-relevant state. It is also possible to imagine more general infinite-horizon games where the payoff function is $u_i(t, k(t), a_i(t))$, with calendar time being part of the payoff-relevant state.

Note also that $\hat{\sigma}_i$ has a different dimension than σ_i above. In particular, $\hat{\sigma}_i$ assigns an action (or a probability distribution over actions) to each state $k \in K$, while σ_i does so for each subgame, i.e., for all $(h^{t-1}, k(t)) \in H^{t-1} \times K$ and all t . To compare Markovian and non-Markovian strategies (and to make sure below that we can compare Markovian strategies to deviations that are non-Markovian), it is useful to consider an *extension* of Markovian strategies to the same dimension as σ_i . In particular, let $\hat{\sigma}'_i$ be an extension of $\hat{\sigma}_i$ such that

$$\hat{\sigma}'_i : K \times H^{t-1} \rightarrow \Delta(A_i)$$

with $\hat{\sigma}'_i(k, h^{t-1}) = \hat{\sigma}_i(k)$ for all $h^{t-1} \in H^{t-1}$ and $k(t) \in K$. Define the set of extended Markovian strategies for player i by $\hat{S}'_i$ and $\hat{S}' = \prod_{i=1}^N \hat{S}'_i$. Moreover, as before, let $\hat{\sigma}'_i(t)$ be the continuation strategy of player i induced by $\hat{\sigma}'_i$ after time t , and $\hat{\sigma}'_{-i}(t)$ be the continuation strategy profile of all players other than i induced by their Markovian strategies $\hat{\sigma}_{-i}$. I will refer both to $\hat{\sigma}_i$ and to its extension $\hat{\sigma}'_i$ as “Markovian strategies”.

DEFINITION C.2. A **Markov Perfect Equilibrium (MPE)** is a profile of Markovian strategies $\hat{\sigma}^* = (\hat{\sigma}_1^*, \dots, \hat{\sigma}_N^*) \in \hat{S}$ such that the extension of these strategies satisfy $\hat{\sigma}_i^{I*}(t) \in BR(\hat{\sigma}_{-i}^{I*}(t) \mid h^{t-1}, k(t))$ for all $(h^{t-1}, k(t)) \in H^{t-1} \times K$, for all $i \in \mathcal{N}$ and for all $t = 0, 1, \dots$

Therefore, the only difference between MPE and SPE is that in the former attention is restricted to Markovian strategies. It is important to note that *deviations are not restricted to be Markovian*. This is implicitly emphasized by the extension of the Markovian strategies to $\hat{\sigma}_i^{I*} \in \hat{S}'_i$ and also by the requirement that $\hat{\sigma}_i^{I*}(t) \in BR(\hat{\sigma}_{-i}^{I*}(t) \mid h^{t-1}, k(t))$ (which conditions on history h^t). In particular, for a MPE, a Markovian strategy $\hat{\sigma}_i^*$ must be a best response to $\hat{\sigma}_{-i}^*$ among all strategies $\sigma_i(t) : H^{t-1} \times K \rightarrow \Delta(A_i)$ available at time t .

It should also be clear that a MPE is a SPE, since the extended Markovian strategy satisfies $\hat{\sigma}_i^{I*}(t) \in BR(\hat{\sigma}_{-i}^{I*}(t) \mid h^{t-1}, k(t))$, ensuring that $\hat{\sigma}_i^{I*}$ is a best response to $\hat{\sigma}_{-i}^{I*}$ in all subgames, that is, for all $(h^{t-1}, k(t)) \in H^{t-1} \times K$ and for all t .

C.2. Some Basic Results

The following are some standard results and theorems that are useful to bear in mind in the application of dynamic games. First, we start with the eminently useful *one-stage deviation principle*. Recall that $\sigma(t) = (\sigma_1(t), \dots, \sigma_N(t))$ denotes the continuation play for player i after date t , and therefore $\sigma_i(t) = (a_i(t), \sigma_i^*(t+1))$ designates the strategy involving action $a_i(t)$ at date t and then the continuation play given by strategy $\sigma_i^*(t+1)$.

THEOREM C.1. (*One-Stage Deviation Principle*) *Suppose that the instantaneous payoff function of each player is uniformly bounded, that is, there exists $M < \infty$ such that $\sup_{k \in K, a \in A(k)} u_i(a, k) < M$ for all $i \in \mathcal{N}$. Then, a strategy profile $\sigma^* = (\sigma_1^*, \dots, \sigma_N^*) \in S$ is a SPE [respectively $\hat{\sigma}^* = (\hat{\sigma}_1^*, \dots, \hat{\sigma}_N^*) \in \hat{S}$ is a MPE] if and only if for all $i \in \mathcal{N}$, $(h^{t-1}, k(t)) \in H^{t-1} \times K$ and time t and for all $a_i(t) \in A(k(t))$, $\sigma_i^*(t) = (a_i(t), \sigma_i^*(t+1))$ [resp. $\hat{\sigma}_i^{I*}(t) = (a_{it}, \hat{\sigma}_i^{I*}(t+1))$] yields no higher payoff to player i than $\sigma_i^*(t)$ [resp. $\hat{\sigma}_i^{I*}(t)$].*

PROOF. (Sketch) Fix the strategy profile of other players. Then, the problem of individual i is equivalent to a dynamic optimization problem. Then, since $\lim_{T \rightarrow \infty} \sum_{s=T}^{\infty} \beta^s u_i(k(t+s), a(t+s)) = 0$ for all $\{k(t+s), a(t+s)\}_{s=0}^T$ and all t given the uniform boundedness of instantaneous payoffs and $\beta < 1$, we can apply a slight variant of the principle of optimality from dynamic programming, Theorem 16.2 from Chapter 16. In particular, given the uniform boundedness assumption, the same argument as in the proof of this theorem implies that an optimal plan for an individual, for a fixed profile of strategies of all other players, must be optimal for the next stage given his optimal continuation from then on, and moreover, that any non-optimal plan must violate the principle of optimality at some point. $\square$

This theorem basically implies that in dynamic games, we can check whether a strategy is a best response to other players' strategy profile by looking at one-stage deviations, keeping the rest of the strategy of the deviating player as given. The uniform boundedness assumption can be weakened to require "continuity at infinity," which essentially means that discounted payoffs converge to zero along any history (and this assumption can be relaxed further).

LEMMA C.1. *Suppose that $\hat{\sigma}_{-i}^{I*}$ is Markovian (i.e., it is an extension of a Markovian strategy $\hat{\sigma}_{-i}^*$) and that for $h^{t-1} \in H^{t-1}$ and $k(t) \in K$, $BR(\hat{\sigma}_{-i}^{I*} \mid k(t), h^{t-1}) \neq \emptyset$. Then, there exists $\hat{\sigma}_i^{I*} \in BR(\hat{\sigma}_{-i}^{I*} \mid k(t), h^{t-1})$ that is Markovian.*

PROOF. Suppose $\hat{\sigma}_{-i}^{I*}$ is Markovian. Suppose, to obtain a contradiction, that there exists a non-Markovian strategy σ_i^* that performs strictly better against $\hat{\sigma}_{-i}^{I*}$ than all Markovian strategies. Then, by Theorem C.1, there exists $t, \tilde{t} > t$, $k \in K$, $h^{t-1} \in H^{t-1}$ and $\tilde{h}^{\tilde{t}-1} \in H^{\tilde{t}-1}$ such that the continuation play following these two histories given $k \in K$ are

not the same, i.e., $\sigma_i^*[t](k, h^{t-1}) \in BR(\hat{\sigma}_{-i}^* | k, h^{t-1})$, $\sigma_i^*[\tilde{t}](k, \tilde{h}^{\tilde{t}-1}) \in BR(\hat{\sigma}_{-i}^* | k, \tilde{h}^{\tilde{t}-1})$ and $\sigma_i^*[t](k, h^{t-1}) \neq \sigma_i^*[\tilde{t}](k, \tilde{h}^{\tilde{t}-1})$, where $\sigma_i^*[t](k, h^{t-1})$ denotes a continuation strategy for player i starting from time t with state vector k and history h^{t-1} . Now, construct the continuation strategy $\hat{\sigma}_i'^*[t]$ such that $\hat{\sigma}_i'^*[t](k, \tilde{h}^{\tilde{t}-1}) = \sigma_i^*[t](k, h^{t-1})$. Since $\hat{\sigma}_{-i}^*$ is Markovian, $\hat{\sigma}_{-i}^*[t]$ is independent of $h^{t-1}, \tilde{h}^{\tilde{t}-1}$, and therefore $\hat{\sigma}_i'^*[t](k, h^{t-1}) = \hat{\sigma}_i'^*[t](k, \tilde{h}^{\tilde{t}-1}) \in BR(\hat{\sigma}_{-i}^* | k, h^{t-1}) \cap BR(\hat{\sigma}_{-i}^* | k, \tilde{h}^{\tilde{t}-1})$. Repeating this argument for all instances in which σ_i^* is not Markovian establishes that a Markovian strategy $\hat{\sigma}_i'^*$ is also best response to $\hat{\sigma}_{-i}^*$. $\square$

This lemma states that when all other players are playing Markovian strategies, there exists a best response that is Markovian for each player. This does not mean that there are no other best responses, but since there is a Markovian best response, this gives us hope in constructing MPE. Consequently:

THEOREM C.2. (*Existence of Markov Perfect Equilibria*) Let K and $A_i(k)$ for all $k \in K$ be finite sets, then there exists a MPE $\hat{\sigma}^* = (\hat{\sigma}_1^*, \dots, \hat{\sigma}_N^*)$.

PROOF. (Sketch) Consider an extended game in which the set of players is an element (i, k) of $\mathcal{N} \times K$, with payoff function given by the original payoff functions for player i starting in state k as in (C.1) and strategy set $A_i(k)$. The set $\mathcal{N} \times K$ is finite, and since $A_i(k)$ is also finite, the set of mixed strategies $\Delta(A_i(k))$ for player (i, k) is the simplex over $A_i(k)$. Therefore, the standard proof of existence of Nash equilibrium based on Kakutani's Fixed Point Theorem (Theorem A.19 from Chapter A) applies and leads to the existence of an equilibrium $\left(\hat{\sigma}_{(i,k)}^*\right)_{(i,k) \in \mathcal{N} \times K}$ in this extended game.

Now going back to the original game, construct the strategy $\hat{\sigma}_i^*$ for each player $i \in \mathcal{N}$ such that $\hat{\sigma}_i^*(k) = \hat{\sigma}_{(i,k)}^*$, i.e., $\hat{\sigma}_i^* : K \rightarrow \Delta(A_i)$. This strategy profile $\hat{\sigma}^*$ is Markovian. Consider the extension of $\hat{\sigma}^*$ to $\hat{\sigma}'^*$ as above, i.e., $\hat{\sigma}_i'^*(k, h^{t-1}) = \hat{\sigma}_i^*(k)$ for all $h^{t-1} \in H^{t-1}$, $k(t) \in K$, $i \in \mathcal{N}$ and t . Then, by construction, given $\hat{\sigma}_{-i}^*$, it is impossible to improve over $\hat{\sigma}_i'^*$ with a deviation at any $k \in K$. Theorem C.1 implies that $\hat{\sigma}_i'^*$ is a best response to $\hat{\sigma}_{-i}^*$ for all $i \in \mathcal{N}$ among all Markovian strategies. Lemma C.1 then implies there exists no non-Markovian strategy that can perform strictly better than $\hat{\sigma}_i'^*$ against $\hat{\sigma}_{-i}^*$ for all $i \in \mathcal{N}$. This establishes that $\hat{\sigma}'^*$ is a MPE strategy profile. $\square$

Similar existence results can be proved for countably infinite sets K and $A_i(k)$, and also for uncountable sets, but in this latter instance, some additional requirements are necessary, and these are rather technical in nature. Since they will play no role in what follows, we do not need to elaborate on these.

For the next result, let $\hat{\Sigma} = \{\hat{\sigma}^* \in \hat{S} : \hat{\sigma}^* \text{ is a MPE}\}$ be the set of MPE strategies and $\Sigma^* = \{\sigma \in S : \sigma^* \text{ is a SPE}\}$ be the set of SPE strategies. Let $\hat{\Sigma}'$ be the extension of $\hat{\Sigma}$ to include conditioning on histories. In particular, as defined before, recall that $\hat{\sigma}_i' : K \times H^{t-1} \rightarrow$

$\Delta(A_i)$ is such that $\hat{\sigma}'_i(k, h^{t-1}) = \hat{\sigma}_i(k)$ for all $h^{t-1} \in H^{t-1}$ and $k(t) \in K$, and let

$$\hat{\Sigma}' = \left\{ \hat{\sigma}' \in S: \begin{array}{l} \hat{\sigma}'_i(k, h^{t-1}) = \hat{\sigma}_i(k) \text{ for all } h^{t-1} \in H^{t-1}, k(t) \in K \\ \text{and } i \in \mathcal{N} \text{ and } \hat{\sigma} \text{ is a MPE} \end{array} \right\}.$$

THEOREM C.3. (*Markov Versus Subgame Perfect Equilibria*) $\hat{\Sigma}' \subset \Sigma^*$.

PROOF. This theorem follows immediately by noting that since $\hat{\sigma}^*$ is a MPE strategy profile, the extended strategy profile, $\hat{\sigma}^{/*}$, is such that $\hat{\sigma}_i^{/*}$ is a best response to $\hat{\sigma}_{-i}^*$ for all $h^{t-1} \in H^{t-1}$, $k(t) \in K$ and for all $i \in \mathcal{N}$, thus is subgame perfect. $\square$

This theorem implies that every MPE strategy profile corresponds to a SPE strategy profile and any equilibrium-path play supported by a MPE can be supported by a SPE.

THEOREM C.4. (*Existence of Subgame Perfect Equilibria*) Let K and $A_i(k)$ for all $k \in K$ be finite sets, then there exists a SPE $\sigma^* = (\sigma_1^*, \dots, \sigma_N^*)$.

PROOF. Theorem C.2 shows that a MPE exists and since a MPE is a SPE (Theorem C.3), the existence of a SPE follows. $\square$

When K and $A_i(k)$ are uncountable sets, existence of pure strategy SPEs can be guaranteed by imposing compactness and convexity of K and $A_i(k)$ and quasi-concavity of $U_i[t]$ in $\sigma_i[t]$ for all $i \in \mathcal{N}$ (in addition to the continuity assumptions above). In the absence of convexity of K and $A_i(k)$ or quasi-concavity of $U_i[t]$, mixed strategy equilibria can still be guaranteed to exist under some very mild additional assumptions.

Finally, a well-known theorem for SPE from repeated games also generalizes to dynamic games. Let $p(a | \sigma)$ be the probability distribution over the equilibrium-path actions induced by the strategy profile σ , with the usual understanding that $\int_{a \in A} p(a | \sigma) da = 1$ for all $\sigma \in S$, where A is a set of admissible action profiles. With a slight abuse of terminology, I will refer to $p(a | \sigma)$ as the equilibrium-path action induced by strategy σ . Then, let

$$U_i^M(k) = \min_{\sigma_{-i} \in \Sigma_{-i}} \max_{\sigma_i \in \Sigma_i} \mathbb{E} \sum_{s=0}^{\infty} \beta^s u_i(k(t+s), a(t+s)),$$

starting with $k(t) = k$ and with $k(t+s)$ given by (C.2) be the minmax payoff of player i starting with state k . Moreover, let

$$(C.3) \quad U_i^N(k) = \min_{\sigma \in \Sigma} \mathbb{E} \sum_{s=0}^{\infty} \beta^s u_i(k(t+s), a(t+s)),$$

be the minimum SPE payoff of player i starting in state $k \in K$. In other words, this is player i 's payoff in the equilibrium chosen to minimize this payoff (starting in state k). Then:

THEOREM C.5. (*Punishment with the Worst Equilibrium*) Suppose $\sigma^* \in S$ is a pure strategy SPE with the distribution of equilibrium-path actions given by $p(a | \sigma^*)$. Then, there exists a SPE $\sigma^{**} \in S$ (possibly equal to σ^*) such that $p(a | \sigma^*) = p(a | \sigma^{**})$ and σ^{**} involves a continuation payoff of $U_i^N(k)$ to player i , if i is the first to deviate from σ^{**} at date t after some history $h^{t-1} \in H^{t-1}$ and when the resulting state in the next period is $k(t+1) = k$.

PROOF. **(Sketch)** If σ^* is a SPE, then no player wishes to deviate from it. Suppose that i were to deviate from σ^* at date t after history $h^{t-1} \in H^{t-1}$ and when $k(t) = k$. Denote his continuation payoff starting at time t , with $k(t)$, and h^{t-1} by $U_i^d[t](k(t), h^{t-1} | \sigma^*)$ and denote his equilibrium payoff under σ^* by $U_i^c[t](k(t), h^{t-1} | \sigma^*)$. σ^* can be a SPE only if

$$U_i^c[t](k(t), h^{t-1} | \sigma^*) \geq \max_{a_i(t) \in A_i(k)} \mathbb{E} \left\{ u_i(a_i(t), a_{-i}(\sigma_{-i}^*) | k(t), h^{t-1}) + \beta U_i^d[t+1](k(t+1), h^t | \sigma^*) \right\},$$

where $u_i(a_i(t), a_{-i}(\sigma_{-i}^*) | k(t), h^{t-1}, \sigma^*)$ is the instantaneous payoff of individual i when he chooses action $a_i(t)$ in state $k(t)$ following history h^{t-1} and other players are playing the (potentially mixed) action profiles induced by σ_{-i}^* , denoted by $a_{-i}(\sigma_{-i}^*)$. $U_i^d[t+1](k(t+1), h^t | \sigma^*)$ is the continuation payoff following this deviation, with $k(t+1)$ following from the transition function $q(k(t+1) | k(t), a_i(t), a_{-i}(\sigma_{-i}^*))$ and h^t incorporating the actions $a_i(t), a_{-i}(\sigma_{-i}^*)$. Note that by construction, the continuation play, following the deviation, will correspond to a SPE, since σ_{-i}^* specifies a SPE action for all players other than i in all subgames, and in response, the best that player i can do is to play an equilibrium strategy.

By definition of a SPE and the minimum equilibrium payoff of player i defined in (C.3),

$$U_i^d[t+1](k(t+1), h^t | \sigma^*) \geq U_i^N(k(t+1)).$$

The preceding two inequalities imply

$$U_i^c[t](k(t), h^{t-1} | \sigma^*) \geq \max_{a_i(t) \in A_i(k)} \mathbb{E} \left\{ u_i(a_i(t), a_{-i}(\sigma_{-i}^*) | k(t), h^{t-1}) + \beta U_i^N(k(t)) \right\}.$$

Therefore, we can construct σ^{**} , which is identical to σ^* except replacing $U_i^d[t+1](k(t+1), h^t | \sigma^*)$ with $U_i^N(k(t+1))$ following the deviation by player i from σ^* at date t after some history $h^{t-1} \in H^{t-1}$ and when in the next period, we have $k(t+1) = k$. Since $U_i^N(k(t+1))$ is a SPE payoff, σ^{**} will also be a SPE. $\square$

This theorem therefore states that in characterizing the set of sustainable payoffs in SPEs, we can limit attention to SPE strategy profiles involving the most severe equilibrium punishments. A stronger version of this theorem is the following:

THEOREM C.6. (*Punishment with Minmax Payoffs*) Suppose $\sigma^* \in S$ is a pure strategy SPE with the distribution of equilibrium-path actions given by $p(a | \sigma^*)$. Then, there exists $\bar{\beta} \in (0, 1)$ such that for all $\beta \geq \bar{\beta}$, there exists a SPE $\sigma^{**} \in S$ (possibly equal to σ^*) with $p(a | \sigma^*) = p(a | \sigma^{**})$ and σ^{**} involves a continuation payoff of $U_i^M(k)$ to player i , if i is the first to deviate from σ^{**} at date t after some history $h^{t-1} \in H^{t-1}$ and when $k(t) = k$.

PROOF. The proof is identical to that of Theorem C.5, except that it uses $U_i^M(k)$ instead of $U_i^N(k)$. When β is high enough, the minmax payoff for player i , $U_i^M(k)$, can be supported as part of a SPE. The details of this proof can be found in Abreu (1988) and a further discussion is contained in Fudenberg and Tirole (1994). $\square$

C.3. Application: Repeated Games With Perfect Observability

For repeated games with perfect observability, both SPE and MPE are easy to characterize. Suppose that the same stage game is played an infinite number of times, so that payoffs are given by

$$(C.4) \quad U_i[t] = \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s u_i(a(t+s)),$$

which is only different from (8.3) because there is no conditioning on the state variable $k(t)$. Let us refer to the game $\{u_i(a), a \in A\}$ as the stage game. Define

$$m_i = \min_{a_{-i}} \max_{a_i} u_i(a),$$

as the minmax payoff in this stage game. Let $V \in \mathbb{R}^N$ be the set of feasible per period payoffs for the N players, with v_i corresponding to the payoff to player i (so that discounted payoffs correspond to $v_i/(1-\beta)$). Then:

THEOREM C.7. (*The Folk Theorem for Repeated Games*) Suppose that $\{A_i\}_{i \in \mathcal{N}}$ are compact. Then, for any $v \in V$ such that $v_i > m_i$ for all $i \in \mathcal{N}$, there exists $\bar{\beta} \in [0, 1)$ such that for all $\beta > \bar{\beta}$, v can be supported as the payoff profile of a SPE.

PROOF. (Sketch) Construct the following punishment strategies for any deviation: the first player to deviate, i , is held down to its minmax payoff m_i (which can be supported as a SPE). Then, the payoff from any deviation $a \in A_i$ is $D_i(a | \beta) \leq d_i + \beta m_i / (1 - \beta)$ where d_i is the highest payoff player i can obtain by deviating, which is finite by the fact that u_i is continuous and bounded and A_i is compact. v_i can be supported if

$$\frac{v_i}{1 - \beta} \geq d_i + \beta \frac{m_i}{1 - \beta}.$$

Since d_i is finite and $v_i > m_i$, there exists $\bar{\beta}_i \in [0, 1)$ such that for all $\beta \geq \bar{\beta}_i$ this inequality is true. Letting $\bar{\beta} = \max_{i \in \mathcal{N}} \bar{\beta}_i$ establishes the desired result. $\square$

THEOREM C.8. (*Unique Markov Perfect Equilibrium in Repeated Games*) Suppose that the stage game has a unique equilibrium a^* . Then, there exists a unique MPE in which a^* is played at every date.

PROOF. The result follows immediately since K is a singleton and the stage payoff has a unique equilibrium. $\square$

This last theorem is natural, but also very important. In repeated games, there is no state vector, so strategies cannot be conditioned on anything. Consequently, in MPE we can only look at the strategies that are best response in the stage game.

EXAMPLE C.1. (Prisoner's Dilemma) Consider the following standard prisoner's dilemma, which, in fact, has many applications in political economy.

	D	C
D	(0, 0)	(4, -1)
C	(-1, 4)	(2, 2)

The stage game has a unique equilibrium, which is (D,D). Now imagine this game being repeated an infinite number of times with both agents having discount factor β . The unique MPE is playing (D,D) at every date.

In contrast, when $\beta \geq 1/2$, then (C,C) at every date can be supported as a SPE. To see this, recall that we only need to consider the minmax punishment, which in this case is (0,0). Playing (C,C) leads to a payoff of $2/(1-\beta)$, whereas the best deviation leads to the payoff of 4 now and a continuation payoff of 0. Therefore, $\beta \geq 1/2$ is sufficient to make sure that the following *grim strategy profile* implements (C,C) at every date: for both players, the strategy is to play C if h^t includes only (C,C) and to play D otherwise.

Why the grim strategy profile is not a MPE is also straightforward to see. This profile ensures cooperation by conditioning on past history, that is, it conditions on whether somebody has defected at any point in the past. This history is not payoff relevant for the future of the game given the action profile of the other player—fixing the action profile of the other player, whether somebody has cheated in the past or not has no effect on future payoffs.

C.4. Exercises

EXERCISE C.1. A simple application of the ideas in this appendix are *common pool games*. Consider a society consisting of $N + 1 < \infty$ players each with payoff function

$$\sum_{s=0}^{\infty} \beta^s \log(c_i(t+s))$$

at time t where $\beta \in (0,1)$ and $c_i(t)$ denotes consumption of individual $i \in \mathcal{N}$ at time t . The society has a common resource, denoted by $K(t)$, which can be thought of as the capital stock at time t . This capital stock follows the nonstochastic law of motion:

$$K(t+1) = AK(t) - \sum_{i \in \mathcal{N}} c_i(t),$$

where $A > 0$, $K(0)$ is given and $K(t) \geq 0$ must be satisfied in every period. The stage game is as follows: at every date all players simultaneously announce $\{c_i(t)\}_{i \in \mathcal{N}}$. If $\sum_{i \in \mathcal{N}} c_i(t) \leq AK(t)$, then each individual consumes $c_i(t)$. If $\sum_{i \in \mathcal{N}} c_i(t) > AK(t)$, then $AK(t)$ is equally allocated among the $N + 1$ players.

- (1) First consider the single-person decision problem corresponding to this game, where $\{c_i(t)\}_{i \in \mathcal{N}}$ is being chosen by a benevolent planner, wishing to maximize the total discounted payoff of all the agents in the society: $\sum_{i \in \mathcal{N}} \sum_{s=0}^{\infty} \beta^s \log(c_i(t+s))$. Set up this problem as a dynamic programming problem and show that the value function of the planner given capital stock K , $V(K)$, is uniquely defined, continuous, concave and also differentiable whenever $S \in (0, AK)$. Also show that the saving level as a function of the capital stock, $\pi(K)$, is a single-valued and continuous function. Moreover, show that $\pi(K) = \beta AK$. Also derive an explicit form equation for the value function, $V(K)$.

- (2) Consider the MPE. First show that there are some uninteresting MPEs, where all individuals announcing $c_i(0) = AK(0)$ or some consumption level close to this. Explain why these are MPEs.
- (3) Next focus on “continuous” and symmetric MPE, where each agent will pursue a strategy of consuming $c^N(K)$ when the capital stock is K . Given symmetry, this implies that when all other agents are pursuing this strategy and agent i chooses consumption c , this will imply a saving level of

$$S = AK - Nc^N(K) - c.$$

Using this observation, write the value function of an individual as

$$(C.5) \quad V^N(K) = \max_{S \leq AK - Nc^N(K)} \{ \log(AK - Nc^N(K) - S) + \beta V^N(S) \}.$$

Explain this expression and provide an intuition.

- (4) Assuming differentiability, derive the first-order condition of the maximization problem in (C.5) and show that there exists a symmetric equilibrium where the equilibrium aggregate saving level in the economy will be given by

$$\pi(K) = \frac{\beta A}{1 + N - \beta N} K.$$

Verify that this is indeed the unique solution to (C.5). Compare this expression to that in Part 2. What is the effect of an increase in N ? Provide an intuition for this.

- (5) Show that if $\beta A > 1 > \beta A / (1 + N - \beta N)$, then the single-person decision problem would involve growth over time, while the MPE would involve the resources shrinking over time.
- (6) Next show that in this game there always exist SPE that implement the single-person solution for any value of $\beta > 0$. Explain this result.
- (7) Now suppose that $\beta A = 1$ and again focus on MPE. Suppose that the game starts with capital stock $K(0)$ and consider the following discontinuous Markovian strategy profile:

$$c_i(K) = \begin{cases} \frac{\beta AK}{1+N} & \text{if } K \geq K(0) \\ K & \text{if } K < K(0) \end{cases}.$$

Show that when all players other than i' pursue this strategy, it is a best response for player i' to play this strategy as well, and along the equilibrium path, the single-person solution is implemented. Carefully provide an intuition for this result. Show that the same result cannot be obtained when $\beta A \neq 1$. Why not?

CHAPTER D

List of Theorems

In this appendix, I list the theorems presented in various different chapters for reference. Many of these theorems refer to mathematical results used in different parts of the book. Some of them are economic results that are more general and more widely applicable than the results I labeled “propositions”. To conserve space, I do not list additional mathematical results given in Lemmas, Corollaries and Facts.

Chapter 2

Theorem 2.1: Euler’s Theorem.

Theorem 2.2: Stability for Systems of Linear Difference Equations.

Theorem 2.3: Stability for Systems of Nonlinear Difference Equations.

Theorem 2.4: Stability for Systems of Linear Differential Equations.

Theorem 2.5: Stability for Systems of Nonlinear Differential Equations.

Chapter 5

Theorem 5.1: Debreu-Mantel-Sonnenschein Theorem.

Theorem 5.2: Gorman’s Aggregation Theorem.

Theorem 5.3: Existence of a Normative Representative Household.

Theorem 5.4: The Representative Firm Theorem.

Theorem 5.5: The First Welfare Theorem for Economies with Finite Number of Commodities.

Theorem 5.6: The First Welfare Theorem for Economies with Infinite Number of Commodities.

Theorem 5.7: The Second Welfare Theorem.

Theorem 5.8: Equivalence of Sequential and Non-Sequential Trading with Arrow Securities.

Chapter 6

Theorem 6.1: Equivalence of Sequential and Recursive Formulations.

Theorem 6.2: Principle of Optimality in Dynamic Programming.

Theorem 6.3: Existence of Solutions in Dynamic Programming.

Theorem 6.4: Concavity of the Value Function.

Theorem 6.5: Monotonicity of the Value Function.

Theorem 6.6: Differentiability of the Value Function.

Theorem 6.7: The Contraction Mapping Theorem.

Theorem 6.8: Applications of the Contraction Mapping Theorem.

Theorem 6.9: Blackwell's Sufficient Conditions for a Contraction.

Theorem 6.10: Sufficiency of Euler Equations and the Transversality Condition.

Chapter 7

Theorem 7.1: Variational Necessary Conditions for an Interior Optimum with Free End Points.

Theorem 7.2: Variational Necessary Conditions for Interior Optimum with Fixed End Points.

Theorem 7.3: Variational Necessary Conditions for Interior Optimum with Inequality Constrained End Points.

Theorem 7.4: Simplified Version of Pontryagin's Maximum Principle.

Theorem 7.5: Mangasarian's Sufficiency Conditions for an Optimum.

Theorem 7.6: Arrow's Sufficiency Conditions for an Optimum.

Theorem 7.7: Pontryagin's Maximum Principle for Multivariate Problems.

Theorem 7.8: Sufficiency Conditions for Multivariate Problems.

Theorem 7.9: Pontryagin's Maximum Principle for Infinite-Horizon Problems.

Theorem 7.10: Hamilton-Jacobi-Bellman Equations.

Theorem 7.11: Sufficiency Conditions for Infinite-Horizon Problems.

Theorem 7.12: General Transversality Condition for Infinite-Horizon Problems.

Theorem 7.13: The Maximum Principle and the Transversality Conditions for Discounted Infinite-Horizon Problems.

Theorem 7.14: Sufficiency Conditions for Discounted Infinite-Horizon Problems.

Theorem 7.15: Existence of Solutions in Optimal Control.

Theorem 7.16: Concavity of the Value Function in Optimal Control.

Theorem 7.17: Differentiability of the Value Function in Optimal Control.

Theorem 7.18: Saddle-Path Stability for Systems of Linear Differential Equations.

Theorem 7.19: Saddle-Path Stability for Systems of Nonlinear Differential Equations.

Chapter 10

Theorem 10.1: Separation Theorem for Investment in Human Capital.

Chapter 16

Theorem 16.1: Equivalence of Sequential and Recursive Formulations.

Theorem 16.2: Principle of Optimality in Stochastic Dynamic Programming.

Theorem 16.3: Existence of Solutions in Stochastic Dynamic Programming.

Theorem 16.4: Concavity of the Value Function.

Theorem 16.5: Monotonicity of the Value Function in State Variables.

Theorem 16.6: Differentiability of the Value Function.

- Theorem 16.7: Monotonicity of the Value Function in Stochastic Variables.
- Theorem 16.8: Sufficiency of Euler Equations and the Transversality Condition.
- Theorem 16.9: Existence of Solutions with Markov Processes.
- Theorem 16.10: Continuity of the Value Function with Markov Processes.
- Theorem 16.11: Concavity of the Value Functions with Markov Processes.
- Theorem 16.12: Monotonicity of the Value Functions with Markov Processes.
- Theorem 16.13: Differentiability of the Value Function with Markov Processes.

Chapter 22

- Theorem 22.1: Arrow's Impossibility Theorem.
- Theorem 22.2: The Median Voter Theorem.
- Theorem 22.3: The Median Voter Theorem with Strategic Voting.
- Theorem 22.4: Downs's Policy Convergence Theorem.
- Theorem 22.5: The Median Voter Theorem without Single Peaked Preferences.
- Theorem 22.6: Downs's Policy Convergence Theorem without Single Peaked References.
- Theorem 22.7: Probabilistic Voting and Preference Aggregation.
- Theorem 22.8: Lobbying and Preference Aggregation.

Appendix Chapter A

- Theorem A.1: Properties of Open and Closed Sets in Metric Spaces.
- Theorem A.2: Continuity and Open Sets in Metric Spaces.
- Theorem A.3: The Intermediate Value Theorem.
- Theorem A.4: Continuity and Open Sets in Topological Spaces.
- Theorem A.5: Convergence of Nets and Continuity in Topological Spaces.
- Theorem A.6: The Heine-Borel Theorem.
- Theorem A.7 The Bolzano-Weierstrass Theorem.
- Theorem A.8: Continuity and Compactness in Topological Spaces.
- Theorem A.9: Weierstrass's Theorem.
- Theorem A.10: Uniform Continuity over Compact Sets.
- Theorem A.11: Continuity of Projection Maps and the Product Topology.
- Theorem A.12: Continuity of Discounted Utilities in the Product Topology.
- Theorem A.13: Tychonoff's Theorem.
- Theorem A.14: Totally Bounded and Compact Spaces.
- Theorem A.15: The Arzela-Ascoli Theorem.
- Theorem A.16: Berge's Maximum Theorem.
- Theorem A.17: Properties of Maximizers under Quasi-Concavity.
- Theorem A.18: Properties of Minimizers under Quasi-Convexity.
- Theorem A.19: Kakutani's Fixed Point Theorem.
- Theorem A.20: Brouwer's Fixed Point Theorem.
- Theorem A.21: Mean Value Theorems.

- Theorem A.22: L'Hospital's Rule.
- Theorem A.23: Taylor's Theorem and Taylor Approximations.
- Theorem A.24: Taylor's Theorem for Functions of Several Variables.
- Theorem A.25: The Inverse Function Theorem.
- Theorem A.26: The Implicit Function Theorem.
- Theorem A.27: Continuity of Linear Functionals in Normed Vector Spaces.
- Theorem A.28: Geometric Form of the Hahn-Banach Theorem.
- Theorem A.29: Separating Hyperplane Theorem.
- Theorem A.30: The Saddle-Point Theorem.
- Theorem A.31: The Kuhn-Tucker Theorem.
- Theorem A.32: The Envelope Theorem.

Appendix Chapter B

- Theorem B.1: Fundamental Theorem of Calculus I.
- Theorem B.2: Fundamental Theorem of Calculus II.
- Theorem B.3: Integration by Parts.
- Theorem B.4: Leibniz's Rule.
- Theorem B.5: Solutions to Systems of Linear Differential Equations with Constant Coefficients.
- Theorem B.6: Solutions to General Systems of Linear Differential Equations.
- Theorem B.7: The Grobman-Hartman Theorem on Stability of Nonlinear Systems of Differential Equations.
- Theorem B.8: Picard's Theorem on Existence and Uniqueness for Differential Equations.
- Theorem B.9: Existence and Uniqueness for Differential Equations on Compact Domain.
- Theorem B.10: Picard's Theorem on Existence and Uniqueness for Systems of Differential Equations
- Theorem B.11: Existence and Uniqueness for Systems of Differential Equations on Compact domain.
- Theorem B.13: Peano's Existence Theorem.
- Theorem B.12: Continuity of Solutions to Differential Equations.
- Theorem B.14: Differentiability of Solutions to Differential Equations.
- Theorem B.15: Solutions to Systems of Linear Difference Equations with Constant Coefficients.
- Theorem B.16: Solutions to Systems of Linear Difference Equations with Constant Coefficients.
- Theorem B.17: Existence and Uniqueness of Solutions to Difference Equations.

Appendix Chapter C

- Theorem C.1: One-Stage Deviation Principle.
- Theorem C.2: Existence of Markov Perfect Equilibria in Finite Dynamic Games.

Theorem C.4: Existence of Subgame Perfect Equilibria in Finite Dynamic Games.

Theorem C.3: Relationship between Markov and Subgame Perfect Equilibria.

Theorem C.5: Punishment with the Worst Equilibrium.

Theorem C.6: Punishment with the Minmax Continuation Values.

Theorem C.7: The Folk Theorem for Repeated Games.

Theorem C.8: Uniqueness of Markov Perfect Equilibria in Repeated Games.

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